



## Chapitre 227

### 227. On the Theory of Reciprocal Surfaces (continuation).

[From the *Proceedings of the Royal Irish Academy*, vol. vii. (1862),  
pp. 20–28.]

[continuation from page 25 ; cf. paper 227, opening.]

and then, since all the quantities entering into the two excepted equations are expressed in terms of the data, these equations are satisfied identically, and it is easy to see that this proves what was before assumed, viz., that in the system of twenty-one equations, the second of the equations marked (\*) is satisfied identically.

Several of the other quantities may be expressed without difficulty in terms of the data  $n, b, c, 4\beta + \gamma, 8k - 18h$  : we in fact have (besides

$a, a', \kappa', \delta'$ , which are originally so expressed) :

$$2\sigma = (n-2)c - (4\beta + \gamma),$$

$$8p = (16n-24)b - (10n-18)c - 2(4b^2 - 9c^2) + 2(8k-18h) - 9(4\beta + \gamma),$$

$$8\kappa = 8n(n-1)(n-2) - (32n-56)b - (17n-46)c \\ + 2(4b^2 - 9c^2) - 2(8k-18h) + 17(4\beta + \gamma),$$

$$2\delta = n(n-1)(n-2)(n-3) - (4n^2 - 20n + 24)b - (6n^2 - 15n + 18)c \\ + 12bc + 18c^2 + (8k-18h) - 9(4\beta + \gamma),$$

$$8n' = 8n(n-1)^2 - (32n-40)b - (21n-30)c + 2(4b^2 - 9c^2) \\ - 2(8k-18h) + 21(4\beta + \gamma),$$

$$c' = 4n(n-1)(n-2) - (16n-28)b - (10n-26)c \\ + (4b^2 - 9c^2) - (8k-18h) + 10(4\beta + \gamma),$$

but the expressions for the remaining quantities, viz.,  $b', p', \sigma', 4\beta' + \gamma', 8k' - 18h'$  would be very complicated. If we suppose that  $b, c, 4\beta + \gamma, 8k - 18h$ , vanish, or, what is the same thing, attend only to the terms which contain  $n$  alone, we have :

$$2\delta' = n(n-1)(n-2)(n^2 - n + 12),$$

$$p' = n(n-2)(n^2 - n + 12),$$

$$\sigma' = 4n(n-2),$$

$$4\beta' + \gamma' = 4n^2(n-2)(n^2 - 3n + 3n - 3),$$

$$8k' - 18h' = n(n-2)\{n^6 - 6n^5 + 16n^4 - 54n^3 + 164n^2 - 288n + 403 \\ - 482n + 348n^2 - 242n + 60\},$$

which agree with the values which Mr Salmon has obtained for  $\beta', \gamma', h', k'$  by means of the twenty-one equations, and the additional equations (peculiar to the case in question, of a surface of the degree  $n$  without singularities, and which are obtained by him from independent considerations),  $i' = 0$ , and  $\beta' = 2n(n-2)(11n-24)$ .

The system of seventeen equations completely accounts for the reduction of the order of the given surface considered as the reciprocal of the reciprocal surface, but the omitted equations are important for other purposes. We may by means of them express  $i, t$  in terms of the data for the system of twenty-one equations, viz.,  $n, b, c, \beta, \gamma, h, k$ ; and, effecting this, and annexing the corresponding values of  $i', t'$ ,

we have the supplementary system :

$$\begin{aligned}
 4t &= (5n - 6)c - 6c^2 + 12h - 5\gamma, \\
 24t &= -(8n - 8)b + (15n - 18)c + 2(4b^2 - 9c^2) - 16k + 36h + 20\beta - 15\gamma, \\
 4i' &= (5n' - 6)c' - 6c'^2 + 12h' - 5\gamma', \\
 24t' &= -(8n' - 8)b' + (15n' - 18)c' + 2(4b'^2 - 9c'^2) - 16k' + 36h' + 20\beta' - 15\gamma',
 \end{aligned}$$

to which I annex also, without transformation, the four equations for  $q, r, q', r'$ , viz. :

$$\begin{aligned}
 q &= b^2 - b - 2k - 3\gamma - 6t, \\
 r &= c^2 - c - 2h - 3\beta, \\
 q' &= b'^2 - b' - 2k' - 3\gamma' - 6t', \\
 r' &= c'^2 - c' - 2h' - 3\beta',
 \end{aligned}$$

the last two of which, neglecting singularities, give

$$q' = 4n(n - 2)(n - 3)(n^2 + 2n - 4), \quad r' = 2n(n - 2)(3n - 4),$$

which are the values given by Mr Salmon. I remark, in conclusion, that there is considerable difficulty in the geometrical conception of the points  $i$  and the planes  $i'$ , and the subject appears to require further examination. In the case of a surface of the order  $n$  without multiple lines, we have not only  $i = 0$  (which is a matter of course), but also  $i' = 0$ . In my paper before referred to, I showed, or attempted to show, by geometrical reasoning, that the common tangent planes of the spinode developpe and the node-couple developpe are stationary planes of the one or the other of the two developes, that is,  $i' = 0$ , and the reasoning seems correct as far as it goes, but it was not shown how the demonstration would (as it ought to do) fail in the case of a surface having a double or cuspidal curve. I showed also that in the case where the common tangent plane is a stationary plane of the spinode developpe (that is for the planes  $\beta'$ ), the spinode curve and the node-couple curve touch instead of simply intersecting; it would seem that the tangent plane at such point is to be counted once, and not twice, in reckoning the number  $\beta'$  of such tangent planes : the like remark applies, of course, also to the points of intersection  $B$  of the double and cuspidal curves.



## Chapitre 228

### 228. Sur l'intégrale $\int \frac{x^{i-1} dx}{(a + bx + cx^2)^i}$

[From the *Journal de Mathématiques pures et appliquées* (Liouville),  
vol. ii. (1857), pp. 49–51.]

*Extrait d'une Lettre adressée à M. Liouville.*

Je suppose que le lecteur ait sous les yeux la Note de M. Liouville (tome I. [1856] page 421). Cela étant, en rétablissant les valeurs de  $g$ ,  $k$  et en écrivant  $i - 1$  au lieu de  $\mu$ , la formule de M. Liouville devient

$$\int_0^1 \frac{t^{i-\frac{1}{2}}(1-t)^{i-\frac{1}{2}} dt}{(a+bt-ct^2)^{i+\frac{1}{2}}} = \frac{\Gamma(\frac{1}{2})\Gamma(i-\frac{1}{2})}{\sqrt{a+b-c}\Gamma(i)[2a+b+2\sqrt{a(a+b-c)}]^{i-\frac{1}{2}}}, \quad (228.1)$$

laquelle, en y posant

$$t = \frac{x}{1+x},$$

se transforme en celle-ci :

$$\int_0^\infty \frac{x^{i-\frac{1}{2}} dx}{[(a+b-c)x^2 + (2a+b)x + a]^{i+\frac{1}{2}}} = \frac{\Gamma(\frac{1}{2})\Gamma(i-\frac{1}{2})}{\sqrt{a+b-c}\Gamma(i)[2a+b+2\sqrt{a(a+b-c)}]^{i-\frac{1}{2}}}, \quad (228.2)$$

et en mettant le dénominateur de l'intégrale sous la forme

$$[(a+b-c)(x+\lambda)(x+\mu)]^{i+\frac{1}{2}},$$

la formule devient

$$\int_0^\infty \frac{x^{i-\frac{1}{2}} dx}{[(x+\lambda)(x+\mu)]^{i+\frac{1}{2}}} = \frac{\Gamma(\frac{1}{2}) \Gamma(i-\frac{1}{2})}{\Gamma(i)} \cdot \frac{1}{(\sqrt{\lambda} + \sqrt{\mu})^{2i-1}}, \quad (228.3)$$

formule qui se trouve dans mon Mémoire : “On certain formulae for differentiation with applications to the evaluation of definite integrals,” *Camb. and Dubl. Math. Journal*, tom. II., page 122, (1847), [41]. La démonstration que j’y ai donnée n’est cependant rigoureuse (à moins que l’on n’admette la théorie de la différentiation à indices quelconques) que pour le cas de  $i + \frac{1}{2}$  égal à un entier positif ; en effet, en écrivant

$$U_{\lambda,\mu,i} = [(x+\lambda)(x+\mu)]^{\frac{1}{2}i} (\sqrt{x+\lambda} - \sqrt{x+\mu})^{2i},$$

je suis parti de la formule

$$\frac{d}{dx} U_{\lambda,\mu,i} = \frac{1}{2}k(\lambda - \mu)^{i-1} U_{\lambda,\mu,i-1} - (k+i) U_{\lambda,\mu,i},$$

pour en déduire l’expression générale de  $\left(\frac{d}{dx}\right)^s U_{\lambda,\mu,i}$ . Cette expression devient très-simple pour le cas  $s = i + 1$  ; on a alors

$$\frac{1}{i} \left(\frac{d}{dx}\right)^{i+1} (\sqrt{x+\lambda} - \sqrt{x+\mu})^{2i} = 2(2i+1)(\lambda - \mu)^i [(x+\lambda)(x+\mu)]^{-i-\frac{1}{2}}, \quad (228.4)$$

formule de différentiation assez singulière. En y écrivant  $\frac{1}{2} - i$  au lieu de  $i$ , et en intégrant  $i + \frac{1}{2}$  fois par la formule

$$\int_x^\infty dx = -\frac{\Gamma(i+\frac{1}{2})}{(-)^{i+\frac{1}{2}}} \left(\int_0^\infty dx\right) / k, \quad s = i + 1,$$

on obtient la formule intégrale ci-devant mentionnée. Il convient d’ajouter que cette formule, représentée sous la forme

$$\int_0^\infty \frac{x^{i-1} dx}{(ax^2 + bx + c)^{i+\frac{1}{2}}} = \frac{\Gamma(i) \Gamma(\frac{1}{2})}{\Gamma(i+\frac{1}{2})} \cdot \frac{1}{(b+2\sqrt{ac})^{i-\frac{1}{2}}} \cdot \frac{1}{c^{\frac{1}{2}}}, \quad (228.5)$$

est comprise dans une formule générale donnée par M. Schlömilch : “Note sur la variation des constantes arbitraires d’une intégrale définie” (*Crelle*, tom. xxxiii., page 268 ; 1847).

Je remarque aussi que M. Donkin, en comparant ses résultats concernant les fonctions de Laplace avec ceux de M. Boole sur le même sujet, a trouvé une identité, laquelle m'a conduit à cette autre formule de différentiation

$$\left\{ \frac{[(x + \lambda)(x + \mu)]^{\frac{1}{2}}}{2x + \lambda + \mu} \frac{d}{dx} \right\}^n \frac{(2x + \lambda + \mu)^{n+i-1}}{[(x + \lambda)(x + \mu)]^{n-\frac{1}{2}i+\frac{1}{2}}} = \frac{(2x + \lambda + \mu)^{n-i+1}}{[(x + \lambda)(x + \mu)]^{\frac{1}{2}}} \left( \frac{d}{dx} \right)^n \frac{(\sqrt{\dots})}{[\dots]}$$

(228.6)

formule que j'ai démontrée au moyen d'une analyse assez compliquée.





## Chapitre 229

# 229. Note sur une formule pour la réversion des séries.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LII. (1856), pp. 276–284.]

Je me propose de développer dans cette note, pour le cas de trois variables (ce qui suffit pour faire voir la loi dans le cas d'un nombre quelconque de variables) une formule qui se trouve dans le mémoire remarquable de Jacobi, “*de resolutione aequationum per series infinitas*,” *Crelle*, t. VI. [1830] p. 257. Voici la formule dont il s'agit.

Soit  $f(x, y, z)$  une fonction rationnelle et entière des variables  $x, y, z$ , et mettons  $u = X, v = Y, w = Z$ , où  $X, Y, Z$  sont des fonctions rationnelles et entières des variables  $x, y, z$ , telles que  $X - x, Y - y, Z - z$  ne contiennent que les puissances et les produits du deuxième ordre et des ordres supérieurs des variables. Cela étant, on aura

$$f(x, y, z) = \left[ f(x, y, z) \frac{\partial(X, Y, Z)}{\partial(x, y, z)} \cdot \frac{1}{(X - u)(Y - v)(Z - w)} \right]_{x^{-1}y^{-1}z^{-1}},$$

où la notation  $\frac{\partial(X, Y, Z)}{\partial(x, y, z)}$  dénote le déterminant fonctionnel ou “Jacobian” de  $X, Y, Z$  par rapport à  $x, y, z$  et la notation  $[ ]_{x^{-1}y^{-1}z^{-1}}$  signifie le coefficient de  $x^{-1}y^{-1}z^{-1}$  dans le développement de la fonction en dedans des  $[ ]$ ; ce développement doit s'effectuer d'une manière déterminée, savoir : il faut d'abord développer les facteurs  $\frac{1}{Y - v}$  selon les puissances descendantes de  $x$ , c'est-à-dire dans la forme

$y^{-1} + y^{-2}v + \text{etc.}$ , et en écrivant  $X = x + P$  (où  $P$  est fonction des trois variables) il faut développer les puissances de  $X$  selon les puissances descendantes du monome  $x$ , c'est-à-dire dans la forme  $X^{-m} = x^{-m} - mx^{-m-1}P + \frac{1}{2}m(m+1)x^{-m-2}P^2 - \text{etc.}$ , ce qui est en effet un développement selon les puissances ascendantes des variables.

La formule donne

$$\text{Coeff. } u^a v^b w^c \text{ dans } f(x, y, z) = \left[ f(x, y, z) \frac{\partial(X, Y, Z)}{\partial(x, y, z)} \cdot X^{-a-1} Y^{-b-1} Z^{-c-1} \right]_{x^{-1}y^{-1}z^{-1}}$$

ou ce qui est la même chose

$$\text{Coeff. } u^a v^b w^c \text{ dans } f(x, y, z) = \left[ f(x, y, z) \frac{\partial(X, Y, Z)}{\partial(x, y, z)} \cdot X^{-a-1} Y^{-b-1} Z^{-c-1} \right]_{x^{-1}y^{-1}z^{-1}}$$

Soit à présent

$$\begin{aligned} X &= x \dots + A_{f,g,h \text{ etc.}} x^f y^g z^h + \text{etc.} = u, \\ Y &= y \dots + B_{i,j,k \text{ etc.}} x^i y^j z^k + \text{etc.} = v, \\ Z &= z \dots + C_{l,m,n \text{ etc.}} x^l y^m z^n + \text{etc.} = w, \\ f(x, y, z) &= \dots + \Theta_{p,q,r \text{ etc.}} x^p y^q z^r + \text{etc.}; \end{aligned}$$

dans ces expressions et partout dans ce qui suit, les etc.'s se rapportent à des termes que l'on obtient en affixant des accents en nombre quelconque aux symboles indéterminés.

En employant la notation de Gauss  $\Pi a = 1 \cdot 2 \cdot 3 \dots a$ , on obtient pour le terme général de  $-\frac{1}{a}X^{-a}$ ,

$$(-)^{r-1} \frac{\Pi(a+r-1)}{\Pi a \Pi f \text{ etc.} \Pi g \text{ etc.} \Pi h \text{ etc.}} A_{f,g,h \text{ etc.}}^a x^{-a+F} y^G z^H,$$

où

$$\alpha + \text{etc.} = r, \quad f\alpha + \text{etc.} = F, \quad g\alpha + \text{etc.} = G, \quad h\alpha + \text{etc.} = H.$$

De même le terme général de  $-\frac{1}{b}Y^{-b}$  est

$$(-)^{s-1} \frac{\Pi(b+s-1)}{\Pi b \Pi i \text{ etc.} \Pi j \text{ etc.} \Pi k \text{ etc.}} B_{i,j,k \text{ etc.}}^b x^I y^{-b+J} z^K,$$

où

$$\beta + \text{etc.} = s, \quad i\beta + \text{etc.} = I, \quad j\beta + \text{etc.} = J, \quad k\beta + \text{etc.} = K,$$

et le terme général de  $-\frac{1}{c}Z^{-c}$  est

$$(-)^{t-1} \frac{\Pi(c+t-1)}{\Pi c \Pi l \text{ etc. } \Pi m \text{ etc. } \Pi n \text{ etc.}} C_{l,m,n \text{ etc.}}^c x^L y^M z^{-c+N},$$

où

$$\gamma + \text{etc.} = t, \quad l\gamma + \text{etc.} = L, \quad m\gamma + \text{etc.} = M, \quad n\gamma + \text{etc.} = N.$$

En formant de là le terme général du Jacobian et en multipliant par le terme général de  $f(x, y, z)$ , on obtient pour le terme général de l'expression en dedans des [], la valeur que voici,

$$\begin{aligned} & (-)^{r+s+t} \frac{\Pi(a+r-1) \Pi(b+s-1) \Pi(c+t-1)}{\Pi a \Pi b \Pi c \Pi f \text{ etc. } \Pi i \text{ etc. } \Pi l \text{ etc.}} \\ & \quad \times A_{f,g,h \text{ etc.}}^a B_{i,j,k \text{ etc.}}^b C_{l,m,n \text{ etc.}}^c \Theta_{p,q,r} \\ & \quad \times \begin{vmatrix} a+r-F, & -G, & -H \\ -I, & b+s-J, & -K \\ -L, & -M, & c+t-N \end{vmatrix} \\ & \quad \times x^{-1-r+F+I+L+p-1} y^{-1-s+G+J+M+q-1} z^{-1-t+H+K+N+r-1}. \end{aligned}$$

et pour trouver le terme qui contient  $x^{-1}y^{-1}z^{-1}$  on n'a qu'à écrire dans cette expression

$$F+I+L+P = a+r, \quad G+J+M+Q = b+s, \quad H+K+N+R = c+t,$$

le coefficient de  $x^{-1}y^{-1}z^{-1}$  sera alors la valeur de l'expression  $[\ ]_{x^{-1}y^{-1}z^{-1}}$  qu'il s'agissait de trouver. En effectuant cela et en récapitulant les formules on obtient le théorème suivant : en posant

$$\begin{aligned} X &= x \dots + A_{f,g,h \text{ etc.}} x^f y^g z^h + \text{etc.} = u, \\ Y &= y \dots + B_{i,j,k \text{ etc.}} x^i y^j z^k + \text{etc.} = v, \\ Z &= z \dots + C_{l,m,n \text{ etc.}} x^l y^m z^n + \text{etc.} = w, \\ f(x, y, z) &= \dots + \Theta_{p,q,r \text{ etc.}} x^p y^q z^r + \text{etc.} \end{aligned}$$

on aura pour le terme général du coeff.  $u^a v^b w^c$  dans  $f(x, y, z)$  la

valeur que voici,

$$(-)^{r+s+t} \frac{\Pi(a+r-1) \Pi(b+s-1) \Pi(c+t-1)}{\Pi a \Pi b \Pi c (\Pi f \Pi g \Pi h)^\alpha \text{ etc. } (\Pi i \Pi j \Pi k)^\beta \text{ etc. } (\Pi l \Pi m \Pi n)^\gamma \text{ etc. } \Pi p \Pi q \Pi r} \\ \times A_{f,g,h \text{ etc.}}^\alpha B_{i,j,k \text{ etc.}}^\beta C_{l,m,n \text{ etc.}}^\gamma \Theta_{p,q,r} \\ \times \begin{vmatrix} P+I+L, & -G, & -H \\ -I, & Q+G+M, & -K \\ -L, & -M, & R+H+K \end{vmatrix}$$

dans laquelle

$$\begin{aligned} \alpha + \text{etc.} &= r, & \beta + \text{etc.} &= s, & \gamma + \text{etc.} &= t, \\ f\alpha + \text{etc.} &= F, & g\alpha + \text{etc.} &= G, & h\alpha + \text{etc.} &= H, \\ i\beta + \text{etc.} &= I, & j\beta + \text{etc.} &= J, & k\beta + \text{etc.} &= K, \\ l\gamma + \text{etc.} &= L, & m\gamma + \text{etc.} &= M, & n\gamma + \text{etc.} &= N, \\ P+F+I+L &= a+r, & Q+G+J+M &= b+s, & R+H+K+N &= c+t. \end{aligned}$$

Les dernières Équations peuvent s'écrire sous la forme

$$\begin{aligned} P + (f-1)\alpha + \text{etc.} + i\beta + \text{etc.} + l\gamma + \text{etc.} &= a, \\ Q + g\alpha + \text{etc.} + (j-1)\beta + \text{etc.} + m\gamma + \text{etc.} &= b, \\ R + h\alpha + \text{etc.} + k\beta + \text{etc.} + (n-1)\gamma + \text{etc.} &= c, \end{aligned}$$

qui sont les conditions auxquelles doivent satisfaire les valeurs de  $P$ ,  $Q$ ,  $R$ ,  $\alpha$ , etc.,  $\beta$ , etc.,  $\gamma$ , etc.,  $f$ ,  $g$ ,  $h$ , etc.,  $i$ ,  $j$ ,  $k$ , etc.,  $l$ ,  $m$ ,  $n$ , etc.; en ajoutant ces Équations on obtient

$$P+Q+R+(f+g+h-1)\alpha+\text{etc.}+(i+j+k-1)\beta+\text{etc.}+(l+m+n-1)\gamma+\text{etc.} = a+b+c,$$

où les nombres  $f+g+h-1$ , etc.  $i+j+k-1$ , etc.  $l+m+n-1$ , etc. sont positifs, il n'y a donc qu'un nombre fini (comme cela doit être) de solutions des Équations indéterminées.

Il y a une manière assez simple pour calculer le déterminant qui entre dans la formule, pour cela je représente les termes  $P$ ,  $I$ , etc. par les notations symboliques  $X_x$ ,  $X_y$ , etc. de manière que le déterminant devient

$$\begin{vmatrix} X_x + X_y + X_z, & -Y_x, & -Z_x \\ -X_y, & Y_y + Y_x + Y_z, & -Z_y \\ -X_z, & -Y_z, & Z_z + Z_x + Z_y \end{vmatrix}$$

ou ce déterminant est ce que devient le produit

$$(X_x + X_y + X_z)(Y_y + Y_z + Y_x)(Z_z + Z_x + Z_y),$$

en omettant du développement tous les termes qui contiennent un cycle tel que  $X_y \cdot Y_x$ , ou  $X_y \cdot Y_z \cdot Z_x$ . Cela donne pour le déterminant la somme des seize termes

$$\begin{aligned} & X_x \cdot Y_y \cdot Z_z + Y_y \cdot Z_z(X_y + X_z) + Z_z \cdot X_x(Y_z + Y_x) + X_x \cdot Y_y(Z_x + Z_y) \\ & + X_x(Y_x \cdot Z_x + Y_x \cdot Z_y + Y_z \cdot Z_x) + Y_y(Z_y \cdot X_y + Z_y \cdot X_z + Z_x \cdot X_y) \\ & + Z_z(X_z \cdot Y_z + X_z \cdot Y_x + X_y \cdot Y_z); \end{aligned}$$

et la même chose est vraie quel que soit l'ordre du déterminant. C'est M. Sylvester qui m'a fait cette remarque.

Les formules de Jacobi s'appliquent aussi au cas où  $u, v, w$ , etc. sont données en termes de  $x, y, z$ , etc. au moyen d'équations et non pas explicitement comme auparavant; mais je ne chercherai pas à présent ce que deviennent les formules pour ce cas plus général.

On peut appliquer la formule au problème de la transformation des variables indépendantes dans le calcul différentiel. En effet, soient  $u, v, w$  des fonctions quelconques de  $x, y, z$ , et prenons  $\xi, \eta, \zeta$  pour les incréments de  $u, v, w$  respectivement; on aura

$$\begin{aligned} \xi &= \left\{ \left( \xi \frac{d}{du} + \eta \frac{d}{dv} + \zeta \frac{d}{dw} \right) + \frac{1}{2} \left( \xi \frac{d}{du} + \eta \frac{d}{dv} + \zeta \frac{d}{dw} \right)^2 + \dots \right\} x, \\ \eta &= \left\{ \left( \xi \frac{d}{du} + \eta \frac{d}{dv} + \zeta \frac{d}{dw} \right) + \frac{1}{2} \left( \xi \frac{d}{du} + \eta \frac{d}{dv} + \zeta \frac{d}{dw} \right)^2 + \dots \right\} y, \\ \zeta &= \left\{ \left( \xi \frac{d}{du} + \eta \frac{d}{dv} + \zeta \frac{d}{dw} \right) + \frac{1}{2} \left( \xi \frac{d}{du} + \eta \frac{d}{dv} + \zeta \frac{d}{dw} \right)^2 + \dots \right\} z. \end{aligned}$$

Soit

$$\nabla = \frac{\partial(x, y, z)}{\partial(u, v, w)}$$

le Jacobian de  $x, y, z$  par rapport à  $u, v, w$ ; et mettons

$$\begin{aligned} \left( \xi \frac{d}{du} + \eta \frac{d}{dv} + \zeta \frac{d}{dw} \right) \nabla &= \nabla', \\ \left( \xi \frac{d}{du} + \eta \frac{d}{dv} + \zeta \frac{d}{dw} \right)^2 \nabla &= \nabla'', \end{aligned}$$

...

$\nabla, \nabla', \nabla''$  seront des fonctions linéaires de  $p, q, r$ , et, en formant avec ces valeurs l'expression de  $\xi \frac{dx}{du} + \eta \frac{dx}{dv} + \zeta \frac{dx}{dw}$ , on trouve

$$\xi \frac{dx}{du} + \eta \frac{dx}{dv} + \zeta \frac{dx}{dw} = \mathfrak{x},$$

où  $\mathfrak{x}, \mathfrak{y}, \mathfrak{z}$  sont des opérations de la forme  $L \frac{d}{du} + M \frac{d}{dv} + N \frac{d}{dw}$ , qui peuvent être représentées symboliquement de cette manière,

$$\mathfrak{x} = \frac{1}{\nabla} \begin{vmatrix} \frac{d}{du}, & \frac{d}{dv}, & \frac{d}{dw} \\ \frac{dy}{du}, & \frac{dy}{dv}, & \frac{dy}{dw} \\ \frac{dz}{du}, & \frac{dz}{dv}, & \frac{dz}{dw} \end{vmatrix}$$

et de même pour  $\mathfrak{y}$  et  $\mathfrak{z}$ ; il faut faire attention, qu'en opérant avec ces symboles, il faut traiter comme des constantes les fonctions de  $u, v, w$  qui entrent dans ces mêmes symboles.

Nous avons évidemment  $\mathfrak{x}x = 1, \mathfrak{y}x = 0, \mathfrak{z}x = 0$ , et de même  $\mathfrak{x}y = 0$ , etc.; on obtient de là

$$\begin{aligned} \xi &= P + \frac{1}{2}(p\mathfrak{x} + q\mathfrak{y} + r\mathfrak{z})^2 x + \text{etc.}, \\ \eta &= Q + \frac{1}{2}(p\mathfrak{x} + q\mathfrak{y} + r\mathfrak{z})^2 y + \text{etc.}, \\ \zeta &= R + \frac{1}{2}(p\mathfrak{x} + q\mathfrak{y} + r\mathfrak{z})^2 z + \text{etc.} \end{aligned}$$

Soit à présent  $\phi$  une fonction quelconque de  $x, y, z$ , ou de  $u, v, w$ ; en envisageant  $\phi$  comme fonction de  $x, y, z$ , on trouve l'incrément de cette fonction en opérant sur  $\phi$  avec le symbole

$$\left( \xi \frac{d}{dx} + \eta \frac{d}{dy} + \zeta \frac{d}{dz} \right) + \frac{1}{2} \left( \xi \frac{d}{dx} + \eta \frac{d}{dy} + \zeta \frac{d}{dz} \right)^2 + \text{etc.},$$

mais en envisageant  $\phi$  comme fonction de  $u, v, w$  et en faisant attention à l'équation

$$\xi \frac{d\phi}{du} + \eta \frac{d\phi}{dv} + \zeta \frac{d\phi}{dw} = p\mathfrak{x} + q\mathfrak{y} + r\mathfrak{z},$$

on trouve ce même incrément en opérant sur  $\phi$  avec le symbole

$$(p\mathfrak{x} + q\mathfrak{y} + r\mathfrak{z}) + \frac{1}{2}(p\mathfrak{x} + q\mathfrak{y} + r\mathfrak{z})^2 + \text{etc.},$$

et les deux résultats deviendront identiques en substituant pour  $p$ ,  $q$ ,  $r$  les valeurs de ces quantités en termes de  $\xi$ ,  $\eta$ ,  $\zeta$ , valeurs qui se trouvent par la réversion des séries qui donnent  $\xi$ ,  $\eta$ ,  $\zeta$  en termes de  $p$ ,  $q$ ,  $r$ . C'est-à-dire nous aurons

$$\left( \xi \frac{d}{dx} + \eta \frac{d}{dy} + \zeta \frac{d}{dz} \right)^n = \Pi \text{ en } p, q, r \text{ coeff. } \xi^a \eta^b \zeta^c \text{ dans } (p\mathfrak{x} + q\mathfrak{y} + r\mathfrak{z})^n + \text{etc.},$$

où

$$\begin{aligned} \xi &= p + \frac{1}{2}(p\mathfrak{x} + q\mathfrak{y} + r\mathfrak{z})^2 x + \text{etc.}, \\ \eta &= q + \frac{1}{2}(p\mathfrak{x} + q\mathfrak{y} + r\mathfrak{z})^2 y + \text{etc.}, \\ \zeta &= r + \frac{1}{2}(p\mathfrak{x} + q\mathfrak{y} + r\mathfrak{z})^2 z + \text{etc.} \end{aligned}$$

C'est là le problème de la réversion des séries qui vient d'être traité ; et en substituant

$$\xi, \quad \eta, \quad \zeta, \quad u, \quad v, \quad w, \quad \mathfrak{x}, \quad \mathfrak{y}, \quad \mathfrak{z}$$

au lieu de

$$u, \quad v, \quad w, \quad x, \quad y, \quad z, \quad P, \quad Q, \quad R,$$

on trouve le théorème suivant :

**Théorème.** Le terme général de  $\left( \frac{d}{d\xi} \right)^a \left( \frac{d}{d\eta} \right)^b \left( \frac{d}{d\zeta} \right)^c$  est

$$K \Omega_{(\mathfrak{x}^f \mathfrak{y}^g \mathfrak{z}^h x)^\alpha \text{ etc. } (\mathfrak{x}^i \mathfrak{y}^j \mathfrak{z}^k y)^\beta \text{ etc. } (\mathfrak{x}^l \mathfrak{y}^m \mathfrak{z}^n z)^\gamma \text{ etc. } \mathfrak{x}^{-p} \mathfrak{y}^{-q} \mathfrak{z}^{-r}},$$

expression dans laquelle

$$\begin{aligned} \alpha + \text{etc.} &= r, \\ f\alpha + \text{etc.} &= F, \\ i\beta + \text{etc.} &= I, \\ l\gamma + \text{etc.} &= L, \end{aligned}$$

$$F + I + L + P = a + r,$$

$$K = (-)^{r+s+t} \frac{\Pi(a+r-1) \Pi(b+s-1) \Pi(c+t-1)}{\Pi a \text{ etc. } \Pi \beta \text{ etc. } \Pi \gamma \text{ etc. } (\Pi f \Pi g \Pi h)^\alpha \text{ etc. } (\Pi i \Pi j \Pi k)^\beta \text{ etc. } (\Pi l \Pi m \Pi n)^\gamma \text{ etc.}},$$

$$\Omega = \begin{vmatrix} P + I + L, & -G, & -H \\ -I, & Q + G + M, & -K \\ -L, & -M, & R + H + K \end{vmatrix},$$



et où les nombres  $f + g + h - 1$ , etc.,  $i + j + k - 1$ , etc.,  $l + m + n - 1$ , etc. sont tous positifs comme auparavant.

La formule contient les symboles  $\mathfrak{x}$ ,  $\mathfrak{y}$ ,  $\mathfrak{z}$  qui sont chacun une fonction linéaire de  $\frac{d}{du}$ ,  $\frac{d}{dv}$ ,  $\frac{d}{dw}$  ; on pourrait donc se proposer la question de trouver le terme général en développant la formule de manière à ne contenir que des puissances et des produits de ces symboles  $\frac{d}{du}$ ,  $\frac{d}{dv}$ ,  $\frac{d}{dw}$  ; c'est à quoi se rapportent les recherches très étendues que vient de faire M. Sylvester sur ce sujet, et qui embrassent aussi bien le cas où les nouvelles variables sont données explicitement que celui où les deux systèmes de variables sont liés par des Équations données.

Il y a une autre manière de traiter cette question de la transformation des variables indépendantes, savoir en écrivant

$$\rho = \hat{\xi} x + \hat{\eta} y + \hat{\zeta} z,$$

on peut exprimer les puissances de  $R$  au moyen de  $\rho$  et de cette autre quantité symbolique

$$\Lambda = \hat{\xi} \mathfrak{x} + \hat{\eta} \mathfrak{y} + \hat{\zeta} \mathfrak{z}.$$

En effet, en mettant  $\rho^2 = p\hat{\xi} + q\hat{\eta} + r\hat{\zeta}$ , on a

$$R + \frac{1}{2}R^2 + \frac{1}{6}R^3 + \text{etc.} = \chi + \frac{1}{2}\chi^2 + \frac{1}{6}\chi^3 + \text{etc.},$$

où

$$\begin{aligned} \xi &= p + \frac{1}{2}\chi^2 x + \frac{1}{6}\chi^3 x + \text{etc.}, \\ \eta &= q + \frac{1}{2}\chi^2 y + \frac{1}{6}\chi^3 y + \text{etc.}, \\ \zeta &= r + \frac{1}{2}\chi^2 z + \frac{1}{6}\chi^3 z + \text{etc.} \end{aligned}$$

En se servant de la méthode des approximations successives, on trouve comme première approximation

$$p = \xi, \quad q = \eta, \quad r = \zeta,$$

et de là

$$\chi = \rho;$$

comme seconde approximation

$$p = \xi - \frac{1}{2}\rho^2 x, \quad q = \eta - \frac{1}{2}\rho^2 y, \quad r = \zeta - \frac{1}{2}\rho^2 z,$$

et de là

$$\chi = \rho - \frac{1}{2}\rho^2\Lambda;$$

comme troisième approximation

$$\begin{aligned} p &= \xi - \frac{1}{2}(\rho^2 - \rho^2\Lambda\rho) x - \frac{1}{6}\rho^3 x, \\ q &= \eta - \frac{1}{2}(\rho^2 - \rho^2\Lambda\rho) y - \frac{1}{6}\rho^3 y, \\ r &= \zeta - \frac{1}{2}(\rho^2 - \rho^2\Lambda\rho) z - \frac{1}{6}\rho^3 z, \end{aligned}$$

et de là

$$\chi = \rho - \frac{1}{2}\rho^2\Lambda + \frac{1}{6}\rho^3\Lambda^2 - \frac{1}{6}\rho^3\Lambda\rho\Lambda + \frac{1}{6}\rho^3\Lambda^2\rho\Lambda,$$

et ainsi de suite ; donc, en substituant,

$$\begin{aligned} R + \frac{1}{2}R^2 + \frac{1}{6}R^3 + \text{etc.} &= \rho - \frac{1}{2}\rho^2\Lambda + \frac{1}{6}\rho^3\Lambda^2 - \frac{1}{6}\rho^3\Lambda\rho\Lambda \\ &\quad + \frac{1}{6}\rho^4 + \text{etc. etc.}, \end{aligned}$$

c'est-à-dire

$$\begin{aligned} R &= \rho, \\ R^2 &= \rho^2 - \rho^2\Lambda, \\ R^3 &= \rho^3 - 3\rho^2\Lambda\rho - (\rho^3 - 3\rho^2\Lambda)\Lambda, \\ &\text{etc.} \end{aligned}$$

formules dans lesquelles un terme tel que  $\rho^2\Lambda$  signifie  $\rho^2 \mathfrak{x} \cdot x + \rho^2 \mathfrak{y} \cdot y + \rho^2 \mathfrak{z} \cdot z$ , c'est-à-dire l'opération du symbole  $\rho^2$  s'arrête aux quantités  $x, y, z$  faisant partie du symbole  $\Lambda$  qui vient immédiatement après le  $\rho^2$  ; la même chose a lieu dans tous les cas semblables. L'équation générale est

$$R^n = \Pi - \Pi\Lambda,$$

mais pour expliquer la forme de la fonction  $\Pi$  il faudrait faire des développements assez longs, dans lesquels je n'entrerai pas à cette occasion ; la découverte de cette équation générale est due à M. Sylvester, qui l'a établie d'une autre manière.

Londres, 2, Stone Buildings, 16 Avril, 1855.



## Chapitre 230

### 230. Note sur la méthode d'élimination de Bezout.

[From the *Journal für die reine und angewandte Mathematik* (Crelle),  
tom. LIII. (1857), pp. 366–367.]

Voici la forme la plus simple sous laquelle on peut présenter cette méthode. Pour éliminer les variables  $x, y$  entre deux équations du  $n^{\text{ième}}$  degré

$$\begin{aligned}(a, \dots | x, y)^n &= 0, \\ (a', \dots | x, y)^n &= 0,\end{aligned}$$

on n'a qu'à former l'équation identique

$$\begin{aligned}& \frac{(a, \dots | x, y)^n (a', \dots | \lambda, \mu)^n - (a', \dots | x, y)^n (a, \dots | \lambda, \mu)^n}{\mu x - \lambda y} \\ &= \begin{vmatrix} a_{0,0} x^{n-1} + a_{1,0} x^{n-2} y \dots + a_{n-1,0} y^{n-1}, & \lambda^{n-1} \\ a_{0,1} x^{n-1} + a_{1,1} x^{n-2} y \dots + a_{n-1,1} y^{n-1}, & \lambda^{n-2} \mu \\ \vdots & \vdots \\ a_{0,n-1} x^{n-1} + a_{1,n-1} x^{n-2} y \dots + a_{n-1,n-1} y^{n-1}, & \mu^{n-1} \end{vmatrix}\end{aligned}$$

où l'expression qui forme le second membre représente la fonction

suivante,

$$\begin{aligned}
 & (a_{0,0} x^{n-1} + a_{1,0} x^{n-2} y \dots + a_{n-1,0} y^{n-1}) \lambda^{n-1} \\
 & + (a_{0,1} x^{n-1} + a_{1,1} x^{n-2} y \dots + a_{n-1,1} y^{n-1}) \lambda^{n-2} \mu \\
 & \quad \quad \quad + \dots \\
 & + (a_{0,n-1} x^{n-1} + a_{1,n-1} x^{n-2} y \dots + a_{n-1,n-1} y^{n-1}) \mu^{n-1};
 \end{aligned}$$

le résultat de l'élimination sera

$$\begin{vmatrix} a_{0,0} & a_{1,0} & \dots & a_{n-1,0} \\ a_{0,1} & a_{1,1} & \dots & a_{n-1,1} \\ \vdots & \vdots & & \vdots \\ a_{0,n-1} & a_{1,n-1} & \dots & a_{n-1,n-1} \end{vmatrix} = 0.$$

Par exemple on trouve

$$\begin{aligned}
 & \frac{(a, b, c | x, y)^2 (a', b', c' | \lambda, \mu)^2 - (a', b', c' | x, y)^2 (a, b, c | \lambda, \mu)^2}{\mu x - \lambda y} \\
 & = \left( \begin{array}{cc} 2(ab' - a'b), & ac' - a'c \\ ac' - a'c, & 2(bc' - b'c) \end{array} | x, y | \lambda, \mu \right);
 \end{aligned}$$

$$\begin{aligned}
 & \frac{(a, b, c, d | x, y)^3 (a', b', c', d' | \lambda, \mu)^3 - (a', b', c', d' | x, y)^3 (a, b, c, d | \lambda, \mu)^3}{\mu x - \lambda y} \\
 & = \left( \begin{array}{ccc} 3(ab' - a'b), & 3(ac' - a'c), & (ad' - a'd) \\ 3(ac' - a'c), & (ad' - a'd) + 9(bc' - b'c), & 3(bd' - b'd) | x^2, xy, y^2 | \lambda^2, \lambda\mu, \mu^2 \\ (ad' - a'd), & 3(bd' - b'd), & 3(cd' - c'd) \end{array} \right);
 \end{aligned}$$

d'où l'on tire immédiatement les résultats de l'élimination entre deux Équations quadratiques ou cubiques.

Londres, 2 Stone Buildings, Avril, 1855.

## Chapitre 231

### 231. Note sur l'équation $x^2 - Dy^2 = \pm 4$ , $D \equiv 5 \pmod{8}$ .

[From the *Journal für die reine und angewandte Mathematik* (Crelle),  
tom. LIII. (1857), pp. 369–371.]

ON sait que pour un déterminant positif  $D \equiv 5 \pmod{8}$  le nombre des formes quadratiques proprement primitives est égal au nombre des formes improprement primitives, quand il existe une solution en nombres impairs de l'équation  $x^2 - Dy^2 = 4$ ; mais que le nombre des formes proprement primitives est égal à trois fois le nombre des formes improprement primitives, quand il n'existe pas de solution en nombres impairs de l'équation  $x^2 - Dy^2 = 4$ .

Les tables de Degen font voir immédiatement, pour les nombres  $D \equiv 5 \pmod{8}$  qui ne sont pas plus grands que 997, s'il existe ou non une solution en nombres impairs de l'équation  $x^2 - Dy^2 = 4$ . À savoir si le nombre 4 se trouve dans la série des dénominateurs des quotients complets (la seconde ligne dans les tables), il existe une solution de l'équation; si non, il n'en existe pas. De plus, si la place où se trouve le nombre 4 est d'ordre pair il existe une solution tant de l'équation  $x^2 - Dy^2 = +4$  que de l'équation  $x^2 - Dy^2 = -4$ , mais si cette place est d'ordre impair il existe seulement une solution de l'équation  $x^2 - Dy^2 = 4$ . On trouve la plus petite solution de ces Équations au moyen de la série des quotients (la première ligne des tables), en s'arrêtant au nombre qui précède le nombre placé au-dessus du nombre 4, et en calculant la fraction continue déterminée

par cette suite; par exemple, pour le nombre 61 on trouve dans les tables

|    |            |     |    |    |          |
|----|------------|-----|----|----|----------|
| 61 | 7,         | 1,  | 4, | 3, | 1        |
|    | 1,         | 12, | 3, | 4, | 9 (5, 5) |
|    | <hr/>      |     |    |    |          |
|    | 226153980  |     |    |    |          |
|    | 1766319049 |     |    |    |          |

Cela fait voir qu'il existe une solution de l'équation  $x^2 - Dy^2 = -4$ , solution que l'on obtient au moyen de la fraction continue  $7 + \frac{1}{1 + \frac{1}{4 + \frac{1}{3}}}$

et faisant  $x = 39$ ,  $y = 5$ . La plus petite solution de l'équation  $x^2 - Dy^2 = 4$  se déduit très facilement de la plus petite solution de l'équation  $\tau^2 - Dv^2 = -4$  au moyen de la formule  $x + y\sqrt{D} = \frac{1}{2}(\tau + v\sqrt{D})^2$ , ce qui donne  $x = \frac{1}{2}\tau^2 + 2$ ,  $y = \frac{1}{2}\tau v$ .

On obtient pareillement la plus petite solution de l'équation  $x^2 - Dy^2 = -1$  au moyen de la formule  $(x + y\sqrt{D}) = \frac{1}{8}(\tau + v\sqrt{D})^3$ , ce qui donne

$$x = \frac{1}{8}(\tau^3 + 3\tau), \quad y = \frac{1}{8}(\tau^2 + 1)v.$$

De même la plus petite solution de l'équation  $x^2 - Dy^2 = 1$  se déduit de la plus petite solution de l'équation  $T^2 - DU^2 = 4$  au moyen de la formule  $x + y\sqrt{D} = \frac{1}{8}(T + U\sqrt{D})^3$ , ce qui donne  $x = \frac{1}{8}(T^3 - 3T)$ ,  $y = \frac{1}{8}(T^2 - 1)U$ .

Je fais observer à cette occasion que suivant une remarque de Göpel ("Notiz über A. Göpel," t. XXXV. [1847] p. 315 de ce Journal) si dans l'équation  $\left(\frac{x + y\sqrt{-n}}{\rho}\right)^m = P + \sqrt{Q}$ , où  $x$ ,  $y$ ,  $\rho$ ,  $n$ ,  $P$ ,  $Q$  sont des entiers, le dénominateur  $\rho$  est plus grand que l'unité et  $x$ ,  $y$ ,  $\rho$  n'ont pas de dénominateur commun, on aura nécessairement  $\rho = 2$ ,  $m = 3$  ou égal à un multiple de 3,  $x$  impair, et  $y$  de la forme  $8n + 5$ .

J'ai calculé au moyen des tables de Degen la table suivante des plus petites solutions en nombres impairs de l'équation  $x^2 - Dy^2 = -4$  ou, si cette équation n'en admet pas, de l'équation  $x^2 - Dy^2 = 4$ ; en d'autres termes, une table des plus petites solutions impaires de l'équation  $x^2 - Dy^2 = \pm 4$ ,  $D \equiv 5 \pmod{8}$ .

Londres, 2 Stone Buildings, 10 Mars, 1855.

**Table des plus petites solutions impaires de l'équation**  
 $x^2 - Dy^2 = \pm 4$ ,  $D \equiv 5 \pmod{8}$ .

| $D$ | $\pm$   | $x$    | $y$  | $D$ | $\pm$   | $x$          | $y$     | $D$ | $\pm$   | $x$    | $y$  |
|-----|---------|--------|------|-----|---------|--------------|---------|-----|---------|--------|------|
| 5   | +       | 3      | 1    | 341 | +       | 277          | 15      | 677 | imposs. |        |      |
| 13  | -       | 1      | 1    | 349 | imposs. |              |         | 685 | -       | 759    | 29   |
| 21  | +       | 5      | 1    | 357 | +       | 19           | 1       | 693 | +       | 2779   | 3    |
| 29  | -       | 5      | 1    | 365 | -       | 19           | 1       | 701 | imposs. |        |      |
| 37  | imposs. |        |      | 373 | imposs. |              |         | 709 | imposs. |        |      |
| 45  | +       | 7      | 1    | 381 | imposs. |              |         | 717 | +       | 241    | 9    |
| 53  | -       | 7      | 1    | 389 | imposs. |              |         | 725 | +       | 27     | 1    |
| 61  | -       | 39     | 5    | 397 | -       | 3447         | 173     | 733 | -       | 27     | 1    |
| 69  | +       | 25     | 3    | 405 | imposs. |              |         | 741 | +       | 245    | 9    |
| 77  | +       | 9      | 1    | 413 | +       | 259          | 3       | 749 | +       | 12945  | 473  |
| 85  | -       | 9      | 1    | 421 | -       | 449134299511 | 21685   | 757 | imposs. |        |      |
| 93  | +       | 29     | 3    | 429 | +       | 145          | 7       | 765 | +       | 139    | 5    |
| 101 | imposs. |        |      | 437 | +       | 21           | 1       | 773 | -       | 139    | 5    |
| 109 | -       | 251    | 11   | 445 | -       | 21           | 1       | 781 | imposs. |        |      |
| 117 | +       | 11     | 1    | 453 | +       | 173          | 1       | 789 | +       | 31825  | 1133 |
| 125 | -       | 11     | 1    | 461 | -       | 1771         | 3       | 797 | -       | 318    | 5    |
| 133 | +       | 173    | 15   | 469 | +       | 65           | 3       | 805 | +       | 1447   | 13   |
| 141 | imposs. |        |      | 477 | +       | 2599         | 119     | 813 | imposs. |        |      |
| 149 | -       | 213    | 5    | 485 | imposs. |              |         | 821 | -       | 16189  | 565  |
| 157 | -       | 213    | 17   | 493 | -       | 111          | 5       | 829 | imposs. |        |      |
| 165 | +       | 13     | 1    | 501 | +       | 23           | 1       | 837 | +       | 29     | 1    |
| 173 | -       | 13     | 1    | 509 | -       | 23           | 1       | 845 | -       | 29     | 1    |
| 181 | +       | 130153 | 9671 | 517 | +       | 23           | 1       | 853 | -       | 27483  | 941  |
| 189 | imposs. |        |      | 525 | +       | 23           | 1       | 861 | +       | 1027   | 35   |
| 197 | imposs. |        |      | 533 | -       | 23           | 1       | 869 | +       | 49377  | 1675 |
| 205 | +       | 43     | 3    | 541 | -       | 13961452253  | 600377  | 877 | imposs. |        |      |
| 213 | +       | 7135   | 1    | 549 | +       | 1523         | 65      | 885 | imposs. |        |      |
| 221 | +       | 15     | 1    | 557 | imposs. |              |         | 893 | +       | 2301   | 77   |
| 229 | -       | 15     | 1    | 565 | -       | 309          | 13      | 901 | imposs. |        |      |
| 237 | +       | 477    | 5    | 573 | imposs. |              |         | 909 | +       | 1181   | 31   |
| 245 | +       | 1177   | 3    | 581 | +       | 43596377     | 179625  | 917 | imposs. |        |      |
| 253 | +       | 1177   | 74   | 589 | +       | 4359377      | 179625  | 925 | imposs. |        |      |
| 261 | +       | 727    | 45   | 597 | +       | 7914293      | 399     | 933 | imposs. |        |      |
| 269 | imposs. |        |      | 605 | +       | 123          | 5       | 941 | -       | 1135   | 37   |
| 277 | -       | 2613   | 157  | 613 | -       | 98763        | 3989    | 949 | -       | 32685  | 1061 |
| 285 | +       | 17     | 1    | 621 | +       | 987622355    | 3989    | 957 | -       | 31     | 1    |
| 293 | -       | 17     | 1    | 629 | -       | 14159        | 561     | 965 | -       | 31     | 1    |
| 301 | +       | 22745  | 1311 | 637 | +       | 14159        | 561     | 973 | imposs. |        |      |
| 309 | +       | 5045   | 287  | 645 | +       | 561          | 8       | 981 | +       | 68123  | 2175 |
| 317 | -       | 89     | 5    | 653 | +       | 1626013      | 6960655 | 989 | +       | 103245 | 3283 |
| 325 | imposs. |        |      | 661 | -       | 1789539      | 69605   | 997 | imposs. |        |      |
| 333 | imposs. |        |      | 669 | +       | 305285       | 11803   |     |         |        |      |

[Note : certain entries above contain values which the OCR could not



2Chapitre 231. Note sur l'équation  $x^2 - Dy^2 = \pm 4$ ,  $D \equiv 5 \pmod{8}$

*fully resolve cleanly from the printed scan; these correspond to large  $x, y$  that the original Crelle table prints in two-line stacks.]*

## Chapitre 232

# 232. Mémoire sur la forme canonique des fonctions binaires.

[From the *Journal für die reine und angewandte Mathematik* (Crelle),  
tom. LIV. (1857), pp. 48–58.]

L'IDÉE de la forme canonique d'une fonction binaire de degré impair et la théorie de la réduction d'une pareille fonction à la forme canonique sont dues à M. Sylvester<sup>(1)</sup>, qui a, en outre, étendu sa théorie aux fonctions binaires des degrés pairs 4 et 6. Mais il n'a établi nulle part l'idée générale de la forme canonique d'une fonction binaire d'un degré pair quelconque. Je me propose de reprendre cette théorie en considérant d'abord les fonctions binaires de degré impair et ensuite les fonctions binaires de degré pair.

<sup>1</sup>Les recherches de M. Sylvester sont contenues dans le mémoire intitulé "Sketch of a Memoir on Elimination, Transformation, and Canonical Forms," *Camb. and Dublin Mathematical Journal*, t. vi. [1851] p. 186, dans un supplément publié conjointement avec ce mémoire et intitulé "Essay on Canonical Forms by J. J. Sylvester," London, Bell, 1851, et dans le mémoire intitulé, "On a remarkable discovery in the Theory of Canonical Forms and of Hyperdeterminants," *Phil. Mag.* Nov. 1851.

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1. <sup>(1)</sup>

Soit donc proposée la fonction de degré impair

$$(a, b, \dots a' | x, y)^{2m-1};$$

en désignant par le symbole  $\Sigma$  une somme de  $m$  termes, on voit qu'on pourra écrire

$$(a, b, \dots a' | x, y)^{2m-1} = \Sigma A (x + \alpha y)^{2m-1},$$

l'expression qui forme le second membre de cette Équation est appelée la forme canonique de la fonction donnée, et il s'agit de trouver les valeurs des quantités  $A, \alpha$ . En supposant que le produit des facteurs linéaires  $x + \alpha y$  soit égal, à un facteur constant près, à

$$(a, b, \dots a'' | x, y)^m,$$

la question peut s'énoncer sous la forme suivante : trouver une fonction de degré  $m$ , telle qu'en représentant  $x + \alpha y$  un facteur linéaire quelconque de cette fonction, la fonction donnée de degré  $2m - 1$  se réduise à la forme  $\Sigma A (x + \alpha y)^{2m-1}$ . La fonction de degré  $m$  contiendra un facteur constant dont la valeur peut être prise à volonté. Mais à ce facteur près la fonction de degré  $m$  sera complètement déterminée.

Pour fixer les idées je suppose que la fonction donnée soit la fonction cubique

$$(a, b, c, d | x, y)^3,$$

dans ce cas il s'agira de trouver une fonction quadratique

$$(a, b, c | x, y)^2$$

telle qu'en posant  $(a, b, c | x, y)^2 = a(x + \alpha y)(x + \beta y)$  on ait identiquement

$$(a, b, c, d | x, y)^3 = A(x + \alpha y)^3 + B(x + \beta y)^3.$$

Cela se fait très-facilement par les méthodes ordinaires. En effet, on a

$$\begin{aligned} A + B &= a, \\ A\alpha + B\beta &= b, \\ A\alpha^2 + B\beta^2 &= c, \\ A\alpha^3 + B\beta^3 &= d, \end{aligned}$$

d'où il suit

$$\begin{aligned} a\xi\eta - b(\xi + \eta) + c &= A(\xi - \alpha)(\eta - \alpha) + B(\xi - \beta)(\eta - \beta), \\ b\xi\eta - c(\xi + \eta) + d &= A\alpha(\xi - \alpha)(\eta - \alpha) + B\beta(\xi - \beta)(\eta - \beta), \end{aligned}$$

$\xi$  et  $\eta$  étant des quantités quelconques ; donc, en prenant  $\xi = \alpha$ ,  $\eta = \beta$ , on obtient

$$\begin{aligned} a\alpha\beta - b(\alpha + \beta) + c &= 0, \\ b\alpha\beta - c(\alpha + \beta) + d &= 0. \end{aligned}$$

Mais l'équation  $(a, b, c|x, y)^2 = a(x + \alpha y)(x + \beta y)$  donne

$$\alpha\beta : \alpha + \beta : 1 = c : 2b : a,$$

on a donc

$$\begin{aligned} a\mathfrak{c} - 2b\mathfrak{b} + c\mathfrak{a} &= 0, \\ b\mathfrak{c} - 2c\mathfrak{b} + d\mathfrak{a} &= 0, \end{aligned}$$

système qui donne les rapports  $\mathfrak{a} : \mathfrak{b} : \mathfrak{c}$ . Mais pour compléter la solution de la manière la plus élégante, il faut ajouter à ces Équations, l'équation  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}|x, y)^2 = 0$ , mise sous la forme  $(\mathfrak{c}, \mathfrak{b}, \mathfrak{a}|y, x)^2 = 0$ , ou ce qui est la même chose

$$y^2\mathfrak{c} + 2yx\mathfrak{b} + x^2\mathfrak{a} = 0;$$

en éliminant  $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}$  on obtient une équation de laquelle on déduit

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}|x, y)^2 = \begin{vmatrix} y^2, & -yx, & x^2 \\ a, & b, & c \\ b, & c, & d \end{vmatrix}.$$

Mais on peut trouver ce résultat d'une manière encore plus facile. Considérant l'équation

$$(a, b, c, d|x, y)^3 = A(x + \alpha y)^3 + B(x + \beta y)^3,$$

on n'a qu'à opérer sur cette équation en se servant du symbole

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}|\partial_y, -\partial_x)^2;$$

le second membre se réduit à zéro à cause des Équations  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}|\alpha, -1)^2 = 0$ ,  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}|\beta, -1)^2 = 0$  que donne l'équation

$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | x, y)^2 = \mathfrak{a}(x + \alpha y)(x + \beta y)$ . Par conséquent on a identiquement

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | \partial_y, -\partial_x)^2 (a, b, c, d | x, y)^3 = 0,$$

c'est-à-dire

$$x(ac - 2bb + ca) + y(bc - 2cb + da) = 0,$$

ou enfin

$$ac - 2bb + ca = 0,$$

$$bc - 2cb + da = 0,$$

et de là on tire comme auparavant la valeur de la fonction  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | x, y)^2$ .

Les valeurs des coefficients  $A, B$  s'expriment alors en fonctions linéaires des coefficients de la fonction cubique, et des quantités  $\alpha, \beta$  qui sont pour ainsi dire les racines de la fonction quadratique  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | x, y)^2$ .

Il est évident que le procédé suivi dans ce qui précède est tout à fait général, et que l'on obtient toujours explicitement la fonction du degré  $m$  : par exemple pour la fonction du cinquième degré

$$(a, b, c, d, e, f | x, y)^5$$

la fonction dont il s'agit sera

$$\begin{vmatrix} y^5, & -y^4x, & y^3x^2, & -y^2x^3, & yx^4, & -x^5 \\ a, & b, & c, & d, & & \\ b, & c, & d, & e, & & \\ c, & d, & e, & f, & & \end{vmatrix}$$

et de même dans tous les autres cas. Cette fonction du degré  $m$  (laquelle est un covariant de la fonction donnée) peut être appelée le *canonisant*. Pour la fonction du cinquième degré le canonisant peut être mis sous cette autre forme

$$\begin{vmatrix} ax + by, & bx + cy, & cx + dy \\ bx + cy, & cx + dy, & dx + ey \\ cx + dy, & dx + ey, & ex + fy \end{vmatrix}$$

et on démontre facilement qu'il existe toujours une transformation semblable.

Considérons maintenant une fonction de degré pair

$$(a, b, \dots a' | x, y)^{2m},$$

et posons

$$(a, b, \dots a' | x, y)^{2m} = \Sigma A (x + \alpha y)^{2m} + \Lambda (\mathfrak{a}, \mathfrak{b}, \dots \mathfrak{a}' | x, y)^m (\mathfrak{a}_1, \mathfrak{b}_1, \dots \mathfrak{a}'_1 | x, y)^m,$$

où l'on suppose  $(\mathfrak{a}, \mathfrak{b}, \dots | x, y)^m = \mathfrak{a}(x + \alpha y) \dots$  et où  $(\mathfrak{a}_1, \mathfrak{b}_1, \dots \mathfrak{a}'_1 | x, y)^m$  représente un certain covariant parfaitement déterminé de  $(a, b, \dots a' | x, y)^{2m}$ . L'expression qui forme le second membre de l'équation proposée s'appellera la forme canonique de la fonction de degré pair.

Pour fixer les idées je suppose que la fonction donnée soit la fonction du quatrième degré

$$(a, b, c, d, e | x, y)^4;$$

dans ce cas, en prenant la fonction quadratique  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | x, y)^2 = \mathfrak{a}(x + \alpha y)(x + \beta y)$  et en représentant par  $(\mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{c}_1 | x, y)^2$  un covariant déterminé de  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | x, y)^2$  on a

$$(a, b, c, d, e | x, y)^4 = A(x + \alpha y)^4 + B(x + \beta y)^4 + \Lambda (\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | x, y)^2 (\mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{c}_1 | x, y)^2.$$

Cela posé, la condition qui sert à déterminer le covariant  $(\mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{c}_1 | x, y)^2$  est la suivante

$$\frac{1}{2}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | \partial_y, -\partial_x)^2 (a, b, c, d, e | x, y)^4 = K (\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | x, y)^2 (\mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{c}_1 | x, y)^2,$$

où  $K$  est un facteur constant dont la valeur est arbitraire.

Le premier membre sera

$$\begin{pmatrix} 7\mathfrak{a}\mathfrak{c} - 6\mathfrak{b}^2, & -2\mathfrak{a}\mathfrak{b}, & \mathfrak{a}^2 \\ -2\mathfrak{b}\mathfrak{c}, & 12\mathfrak{a}\mathfrak{c} - 16\mathfrak{b}^2, & 2\mathfrak{a}\mathfrak{b} \\ \mathfrak{c}^2, & -2\mathfrak{b}\mathfrak{c}, & 7\mathfrak{a}\mathfrak{c} - 6\mathfrak{b}^2 \end{pmatrix} | \mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{c}_1 \rangle (x, y)^2$$

en représentant par cette notation l'expression

$$\begin{aligned} & [(7\mathfrak{a}\mathfrak{c} - 6\mathfrak{b}^2)\mathfrak{a}_1 - 2\mathfrak{a}\mathfrak{b}\mathfrak{b}_1 + \mathfrak{a}^2\mathfrak{c}_1] x^2 \\ & + [-2\mathfrak{b}\mathfrak{c}\mathfrak{a}_1 + (12\mathfrak{a}\mathfrak{c} - 16\mathfrak{b}^2)\mathfrak{b}_1 + 2\mathfrak{a}\mathfrak{b}\mathfrak{c}_1] 2xy \\ & + [\mathfrak{c}^2\mathfrak{a}_1 - 2\mathfrak{b}\mathfrak{c}\mathfrak{b}_1 + (7\mathfrak{a}\mathfrak{c} - 6\mathfrak{b}^2)\mathfrak{c}_1] y^2. \end{aligned}$$

On a donc

$$\begin{aligned}(7ac - 6b^2)a_1 - 2ab\,b_1 + a^2c_1 &= K\,a, \\ -2bc\,a_1 + (12ac - 16b^2)b_1 + 2ab\,c_1 &= K\,2b, \\ c^2a_1 - 2bc\,b_1 + (7ac - 6b^2)c_1 &= K\,c,\end{aligned}$$

et en ajoutant à ces Équations la suivante

$$x^2a_1 + 2xy\,b_1 + y^2c_1 = u,$$

on obtient, en éliminant  $a_1, b_1, c_1$ , l'équation

$$\begin{vmatrix} u, & x^2, & 2xy, & y^2 \\ K\,a, & 7ac - 6b^2, & -2ab, & a^2 \\ 2K\,b, & -2bc, & 12ac - 16b^2, & 2ab \\ K\,c, & c^2, & -2bc, & 7ac - 6b^2 \end{vmatrix} = 0,$$

laquelle peut s'écrire aussi comme il suit

$$\begin{vmatrix} 7ac - 6b^2, & -2ab, & a^2 \\ -2bc, & 12ac - 16b^2, & 2ab \\ c^2, & -2bc, & 7ac - 6b^2 \end{vmatrix} u + K \begin{vmatrix} x^2, & 2xy, & y^2 \\ a, & 7ac - 6b^2, & -2ab, & 2ab \\ 2b, & -2bc, & 12ac - 16b^2 \\ c, & c^2, & -2bc, & 7ac - 6b^2 \end{vmatrix} = 0$$

donc, en rétablissant la valeur de  $u$ , savoir  $u = (a_1, b_1, c_1 | x, y)^2$ , et en prenant

$$K = \begin{vmatrix} 7ac - 6b^2, & -2ab, & a^2 \\ -2bc, & 12ac - 16b^2, & 2ab \\ c^2, & -2bc, & 7ac - 6b^2 \end{vmatrix},$$

on obtient

$$(a_1, b_1, c_1 | x, y)^2 = - \begin{vmatrix} x^2, & 2xy, & y^2 \\ a, & 7ac - 6b^2, & -2ab, & a^2 \\ 2b, & -2bc, & 12ac - 16b^2, & 2ab \\ c, & c^2, & -2bc, & 7ac - 6b^2 \end{vmatrix}.$$

Ces valeurs se réduisent à des expressions très simples. En effet, on a  $K = 576(ac - b^2)^3$ ,  $(a_1, b_1, c_1 | x, y)^2 = -72(ac - b^2)^2(a, b, c | x, y)^2$ , de sorte qu'en supprimant le facteur constant  $-72(ac - b^2)^2$  on peut prendre

$$(a_1, b_1, c_1 | x, y)^2 = (a, b, c | x, y)^2.$$

C'est ce que l'on aurait pu prévoir dès le commencement ; car le procédé par lequel on obtient  $(\mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{c}_1 | x, y)^2$  fait voir que cette fonction est un covariant de  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | x, y)^2$ , du second degré par rapport aux variables et du cinquième degré par rapport aux coefficients ; donc cette fonction sera, à un facteur numérique près, identique avec  $(\mathfrak{a}\mathfrak{c} - \mathfrak{b}^2)^2(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | x, y)^2$ . Une circonstance pareille se présente dans la réduction d'une fonction du sixième degré à la forme canonique. En effet le covariant  $(\mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{c}_1, \mathfrak{d}_1 | x, y)^3$  sera une fonction du troisième degré par rapport aux variables et du septième degré par rapport aux coefficients, donc à un facteur numérique près, cette fonction sera identique avec le discriminant  $(a^2d^2 + \text{etc.})$  multiplié par le cubcovariant  $(a^2d - 3abc + 2b^3)x^3 + \text{etc.}$ , ou, en supprimant le facteur constant, le covariant cherché sera tout simplement le cubcovariant de  $(a, b, c, d | x, y)^3$ . De même pour la fonction du huitième degré on trouve que le covariant  $(\mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{c}_1, \mathfrak{d}_1, \mathfrak{e}_1 | x, y)^4$  sera tout simplement  $(a, b, c, d, e | x, y)^4$ . Mais rien ne nous autorise à supposer qu'une réduction de cette espèce ait lieu dans le cas général, et il paraît qu'en général le covariant  $(\mathfrak{a}_1, \mathfrak{b}_1, \dots, \mathfrak{a}'_1 | x, y)^m$  sera une fonction indécomposable du degré  $2m + 1$  par rapport aux coefficients ; à savoir, en posant

$$(\mathfrak{a}, \mathfrak{b}, \dots, \mathfrak{a}' | \partial_y, -\partial_x)^m (a, b, \dots, a' | x, y)^{2m} (\mathfrak{a}_1, \mathfrak{b}_1, \dots, \mathfrak{a}'_1 | x, y)^m = K (\mathfrak{a}, \mathfrak{b}, \dots, \mathfrak{a}' | x, y)^m,$$

le premier membre sera une fonction de la forme

$$\begin{pmatrix} \mathfrak{A}, & \mathfrak{B}, & \dots \\ \mathfrak{A}', & \mathfrak{B}', & \dots | \mathfrak{a}_1, \mathfrak{b}_1, \dots, \mathfrak{a}'_1 \\ \vdots & & \vdots \end{pmatrix} (x, y)^m$$

où  $\mathfrak{A}, \mathfrak{B}, \dots, \mathfrak{A}', \mathfrak{B}', \dots$  etc. sont des fonctions données du second degré par rapport aux coefficients  $(a, b, \dots, a')$  ; cela étant, on aura

$$K = \begin{vmatrix} \mathfrak{A}, & \mathfrak{B}, & \dots \\ \mathfrak{A}', & \mathfrak{B}', & \dots \\ \vdots & & \vdots \end{vmatrix}$$

et

$$(\mathfrak{a}_1, \mathfrak{b}_1, \dots, \mathfrak{a}'_1 | x, y)^m = - \begin{vmatrix} & x^m, & mx^{m-1}y, & \dots \\ \mathfrak{a}, & \mathfrak{A}, & \mathfrak{B}, & \dots \\ m\mathfrak{b}, & \mathfrak{A}', & \mathfrak{B}', & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix}$$



ce qui fait voir que le covariant dont il s'agit est une fonction du degré  $2m + 1$  par rapport aux coefficients.

Je reviens au cas particulier, le cas général pouvant être traité précisément de la même manière. On a

$$(a, b, c, d, e | x, y)^4 = A(x + \alpha y)^4 + B(x + \beta y)^4 + \Lambda(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | x, y)^2 (\mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{c}_1 | x, y)^2.$$

J'opère sur cette Équation en me servant du symbole  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | \partial_y, -\partial_x)^2$  et j'obtiens

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | \partial_y, -\partial_x)^2 (a, b, c, d, e | x, y)^4 = 2\Lambda K(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | x, y)^2,$$

c'est-à-dire

$$\begin{aligned} & \mathfrak{a}(cx^2 + 2dxy + ey^2) \\ & - 2\mathfrak{b}(bx^2 + 2cxy + dy^2) \\ & + \mathfrak{c}(ax^2 + 2bxy + cy^2) = \frac{1}{6}\Lambda K(\mathfrak{a}x^2 + 2\mathfrak{b}xy + \mathfrak{c}y^2); \end{aligned}$$

en posant, pour abréger,  $\frac{1}{6}\Lambda K = \lambda$ , on aura

$$\begin{aligned} & x^2[\mathfrak{c}\mathfrak{a} - 2b\mathfrak{b} + a(\mathfrak{c} - \lambda)] \\ & + 2xy[d\mathfrak{a} - 2c\mathfrak{b} + b(\mathfrak{c} + \frac{1}{2}\lambda) + a\mathfrak{d}] = 0, \\ & y^2[c(\mathfrak{c} - \lambda) - 2bd + ae] \end{aligned}$$

ce qui donne

$$\begin{aligned} & \mathfrak{c}\mathfrak{a} - 2b\mathfrak{b} + a(\mathfrak{c} - \lambda) = 0, \\ & d\mathfrak{a} - 2c\mathfrak{b} + b(\mathfrak{c} + \frac{1}{2}\lambda) + a\mathfrak{d} = 0, \\ & c(\mathfrak{c} - \lambda) - 2bd + ae = 0, \end{aligned}$$

et de là on tire l'équation

$$\begin{vmatrix} a, & b, & c - \lambda \\ b, & c + \frac{1}{2}\lambda, & d \\ c - \lambda, & d, & e \end{vmatrix} = 0$$

qui servira à la détermination de la quantité  $\lambda$ . Les coefficients des différentes puissances de  $\lambda$  seront des invariants; en effet, si on développe le déterminant, l'équation devient

$$2(ace + 2bcd - ad^2 - b^2e + c^3) + \lambda(ae - 4bd + 3c^2) - \lambda^3 = 0.$$

La quantité  $\lambda$  étant connue, on peut au moyen de deux quelconques des trois Équations trouver les rapports  $\mathfrak{a} : \mathfrak{b} : \mathfrak{c}$ , ou ce qui revient à la même chose, on peut se servir des trois Équations en introduisant les quantités arbitraires  $L, M, N$ , et l'on trouve

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | x, y)^2 = \begin{vmatrix} y^2, & -yx, & x^2 & \\ L, & a, & b, & c - \lambda \\ M, & b, & c + \frac{1}{2}\lambda, & d \\ N, & c - \lambda, & d, & e \end{vmatrix}$$

expression qui, en réalité, ne contient rien d'indéterminé. La quantité  $K$  est une fonction connue de  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c})$ , donc  $\Lambda = \frac{6\lambda}{K}$  sera pareillement connu, et les coefficients  $A, B$  s'expriment en fonctions linéaires des coefficients de la fonction du quatrième degré et des quantités  $\alpha, \beta$  qui sont pour ainsi dire les racines de la fonction quadratique  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} | x, y)^2$ . Comme je l'ai déjà fait observer, tout ce qui précède s'applique également à une fonction de degré pair quelconque. Le déterminant qui contient  $\lambda$  peut être appelé le *Lambdaïque*. On voit aisément quelle est sa forme générale; par exemple pour la fonction  $(a, b, c, d, e, f, g | x, y)^6$  ce déterminant est

$$\begin{vmatrix} a, & b, & c, & d - \lambda \\ b, & c, & d + \frac{1}{2}\lambda, & e \\ c, & d - \frac{1}{2}\lambda, & e, & f \\ d + \lambda, & e, & f, & g \end{vmatrix}$$

expression dont le développement ne contient que les puissances paires de  $\lambda$ ; cela arrive, comme M. Sylvester l'a remarqué depuis longtemps, toutes les fois que  $m$  est impair, c'est-à-dire toutes les fois que le degré de la fonction dont il s'agit est un nombre de la forme  $4p + 2$ .

THE COLLECTED MATHEMATICAL  
PAPERS  
OF  
ARTHUR CAYLEY

Volume IV

Papers 232 (conclusion)–238 (start)

Pages 51–75

Cambridge University Press

1891

## Table des matières

|                                                                                                      |           |
|------------------------------------------------------------------------------------------------------|-----------|
| <b>232. Mémoire sur la Forme Canonique des Fonctions Binaires (conclusion)</b>                       | <b>52</b> |
| <b>233. Addition au Mémoire sur la Forme Canonique des Fonctions Binaires</b>                        | <b>53</b> |
| <b>234. Deuxième Note sur une Formule pour la Réversion des Séries</b>                               | <b>55</b> |
| <b>235. Sur quelques Formules pour la Transformation des Intégrales Elliptiques</b>                  | <b>61</b> |
| <b>236. Note sur la Composition du Nombre 47 par rapport aux Vingt-troisièmes Racines de l'Unité</b> | <b>71</b> |
| <b>237. Théorème sur les Déterminants Gauches</b>                                                    | <b>73</b> |
| <b>238. Note sur les Normales d'une Conique</b>                                                      | <b>75</b> |

# Mémoire sur la Forme Canonique des Fonctions Binaires (conclusion)

## [Continuation from p. 50]

The two displayed covariant identities that occupy the upper half of p. 51 are heavy numerical determinants whose coefficients run, on the left-hand side of each equation, to four or five terms across each of three to five rows. They are stated for the quartic and quintic forms, namely

$$(a, b, c, d, e (\partial_y, -\partial_x, ))^4 \cdot (a, b, c, d, e (x, y, ))^4 \cdot (a_1, b_1, c_1, d_1, e_1 (x, y, ))^4 = \cdots ,$$

and

$$(a, b, c, d, e, f (\partial_y, -\partial_x, ))^5 \cdot (a, b, c, d, e, f (x, y, ))^5 \cdot (a_1, b_1, c_1, d_1, e_1, f_1 (x, y, ))^5 = \cdots ,$$

the right-hand sides being respectively the developments in  $(a_1, b_1, c_1, d_1, e_1)(x, y)^4$  and  $(a_1, b_1, c_1, d_1, e_1, f_1)(x, y)^5$ , with numerical coefficients 24 and 120 prefixed, and entries such as  $71ae - 160bd + 90c^2$ ,  $-124ad + 120bc$ ,  $126ac - 120b^2$ , etc., obtained by the developments. (These large determinantal arrays are reproduced verbatim from the scan; the reader is referred to the original page for the explicit numerical entries.)

The results which have been obtained for the simplest particular cases show that the general theory must be susceptible of further developments. Thus the canonical form of the cubic function  $(a, b, c, d(x, y, ))^3$  will be  $u^3 + v^3$  if we set  $u, v$  in place of  $\sqrt[3]{A}(x + \alpha y)$ ,  $\sqrt[3]{B}(x + \beta y)$ . The general method gives only the value of the product  $uv$  to a constant factor; but one can find the values of  $u$  and  $v$  separately. In fact, recalling the results obtained in the “Note sur les covariants d’une fonction quadratique, cubique, ou biquadratique à deux indéterminées” (t. L. p. 285 of this Journal, [135]), one recognises that the two functions  $\Phi + \sqrt{-D}$ ,  $\Phi - \sqrt{-D}$  are each perfect cubes and one has

$$2\sqrt{-D} = (\Phi + \sqrt{-D}) - (\Phi - \sqrt{-D});$$

one has therefore

$$u = \frac{\Phi + \sqrt{-D}}{2\sqrt{-D}}, \quad v = \frac{-\Phi + \sqrt{-D}}{2\sqrt{-D}}.$$

In like manner the canonical form of the function of the fourth degree  $(a, b, c, d, e(x, y, ))^4$  will be  $u^4 + v^4 + 6\theta u^2 v^2 = (u^2 - v^2)^2 + 2(1 + 3\theta)u^2 v^2$ , if we set  $u, v$  in place of  $\sqrt[4]{A}(x + \alpha y)$ ,  $\sqrt[4]{B}(x + \beta y)$ ; now  $IH - \varpi_1 JU$ ,  $IH - \varpi_2 JU$ ,  $IH - \varpi_3 JU$  being all three squares of quadratic functions, the function  $U$  may be expressed by means of any two of these three quantities as a sum of two squares, and from this one deduces the values of  $u$  and  $v$ ; the development of this solution will be found in a memoir “Sur quelques formules pour la transformation des intégrales elliptiques,” [235], to appear shortly in this Journal. As to the function of the fifth degree, one could write  $u, v, w$  for  $\sqrt[5]{A}(x + \alpha y)$ ,  $\sqrt[5]{B}(x + \beta y)$ ,  $\sqrt[5]{C}(x + \gamma y)$ , which would give the

canonical form  $u^5 + v^5 + w^5$ ; but it is better, on multiplying these values by suitable factors, to write  $u + v + w = 0$  and to take for canonical form  $Au^5 + Bv^5 + Cw^5 = 0$ , as M. Sylvester has done in his memoir cited above. There are also developments on this subject in the memoir of M. Faà de Bruno, "Nota sulla teorica degli invarianti," Tortolini, *Annali di Scienze* etc. 1855.

For the function of the sixth degree one can suppose that the cubic function  $(a, b, c, d(x, y))^3$  has been reduced to the canonical form  $u^3 + v^3$ ; this being so, the cubicovariant will be  $u^3 - v^3$ , and on representing by  $\rho$  one of the imaginary cube roots of unity the canonical form will be

$$A(u + v)^6 + B(u + \rho v)^6 + C(u + \rho^2 v)^6 + \Delta(u^3 - v^3)^2;$$

this also M. Sylvester has shown in his memoir.

*Londres, 9 Avril, 1856.*

## Addition au Mémoire sur la Forme Canonique des Fonctions Binaires

[FROM THE *JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK*, (CRELLE), TOM. LIV. (1857), P. 292.]

Dans le mémoire “Sur la forme canonique des fonctions binaires” (p. 48 of this volume [232]) j’ai dit que M. Sylvester avait en outre étendu sa théorie aux fonctions binaires des degrés pairs 4 et 6 ; j’aurais dû dire, des degrés pairs 4, 6 et 8. J’ai aussi omis de faire observer que les termes *canonisant* et *lambdaïque* appartenaient à M. Sylvester. Enfin en citant dans la note les mémoires de M. Sylvester qui ont rapport à cette théorie j’ai omis de citer le mémoire “On the Calculus of Forms otherwise the Theory of Invariants,” § VIII. (*Camb. and Dublin Math. Journal*, t. IX. [1854], p. 93) section qui porte le titre “On the Reduction of a Sextic Function of Two Variables to the Canonical Form.”

*Londres, 16 Juillet, 1857.*

## Deuxième Note sur une Formule pour la Réversion des Séries

[FROM THE *JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK*, (CRELLE), TOM. LIV. (1857), PP. 156–161 : SEQUEL TO NOTE T. LII. (1856), 229.]

Je me propose de montrer, dans cette deuxième note, de quelle manière le théorème de Jacobi conduit à une formule donnée sans démonstration par M. Sylvester dans son mémoire intitulé “On the Change of Systems of Independent Variables,” *Quarterly Math. Journal*, tom. i. [1857], p. 42 à 56 et 126 à 134. Pour fixer les idées je prends le cas de trois variables, et je suppose que  $f(x, y, z)$  soit une fonction rationnelle et entière de  $x, y, z$ , et que ces variables soient données en fonction de  $u, v, w$  au moyen des équations  $u = X, v = Y, w = Z$ , où  $X, Y, Z$  sont des fonctions rationnelles et entières de  $x, y, z$ . Mais ces fonctions ne sont plus assujetties à la condition (admise dans ma première note) d’être telles que  $X - x, Y - y, Z - z$  ne contiennent que les puissances et les produits du deuxième ordre et des ordres supérieurs des variables, et il s’agit de déterminer dans le cas général le développement de  $f(x, y, z)$  en termes de  $u, v, w$ .

Pour résoudre ce problème j’écris

$$\begin{aligned} X &= A_{100}x + A_{010}y + A_{001}z + \cdots + A_{f,g,h}x^f y^g z^h + \text{etc.}, \\ Y &= B_{100}x + B_{010}y + B_{001}z + \cdots + B_{i,j,k}x^i y^j z^k + \text{etc.}, \\ Z &= C_{100}x + C_{010}y + C_{001}z + \cdots + C_{l,m,n}x^l y^m z^n + \text{etc.}, \\ f(x, y, z) &= \cdots + \Theta_{r,s,t}x^r y^s z^t + \text{etc.}; \end{aligned}$$

dans ces expressions et partout dans la suite les “etc.” représentent des termes qu’on obtient en donnant des accents en nombre quelconque aux symboles indéterminés. Je dois faire observer relativement au coefficient  $A_{f,g,h}$  et aux coefficients semblables, que les termes qui correspondent à  $f + g + h = 1$  sont écrits à part ; on doit donc prendre pour les nombres  $f, g, h$  seulement les valeurs qui rendent  $f + g + h > 1$ .

Je pose

$$x' = A_{100}x + A_{010}y + A_{001}z, \quad y' = B_{100}x + B_{010}y + B_{001}z, \quad z' = C_{100}x + C_{010}y + C_{001}z,$$

et en représentant par

$$\begin{vmatrix} a_{100} & a_{010} & a_{001} \\ b_{100} & b_{010} & b_{001} \\ c_{100} & c_{010} & c_{001} \end{vmatrix}$$

la matrice inverse de

$$\begin{vmatrix} A_{100} & A_{010} & A_{001} \\ B_{100} & B_{010} & B_{001} \\ C_{100} & C_{010} & C_{001} \end{vmatrix},$$



on obtient les équations

$$\begin{aligned}x &= a_{100} x' + b_{100} y' + c_{100} z', \\y &= a_{010} x' + b_{010} y' + c_{010} z', \\z &= a_{001} x' + b_{001} y' + c_{001} z' .\end{aligned}$$

Il est presque superflu de faire observer qu'en représentant par  $\nabla$  le déterminant

$$\nabla = \begin{vmatrix} A_{100}, & A_{010}, & A_{001} \\ B_{100}, & B_{010}, & B_{001} \\ C_{100}, & C_{010}, & C_{001} \end{vmatrix}$$

on a

$$a_{100} = \frac{1}{\nabla} \frac{d\nabla}{dA_{100}}, \quad a_{010} = \frac{1}{\nabla} \frac{d\nabla}{dA_{010}}, \quad \text{etc.}$$

À présent, en supposant chacune des quantités  $u'$ ,  $v'$ ,  $w'$  égale à zéro, on a le système d'équations

$$u - X = 0, \quad v - Y = 0, \quad w - Z = 0, \quad u' - X' = 0, \quad v' - Y' = 0, \quad w' - Z' = 0$$

où

$$\begin{aligned}X &= x' + \dots + A_{f,g,h} x^f y^g z^h + \text{etc.}, \\Y &= y' + \dots + B_{i,j,k} x^i y^j z^k + \text{etc.}, \\Z &= z' + \dots + C_{l,m,n} x^l y^m z^n + \text{etc.}, \\X' &= x - a_{100} x' - b_{100} y' - c_{100} z', \\Y' &= y - a_{010} x' - b_{010} y' - c_{010} z', \\Z' &= z - a_{001} x' - b_{001} y' - c_{001} z',\end{aligned}$$

et comme auparavant

$$f(x, y, z) = \dots + \Theta_{r,s,t} x^r y^s z^t + \text{etc.}$$

Or, en appliquant à ce nouveau système la formule de Jacobi, et en remarquant qu'il est permis de poser tout de suite  $u' = 0$ ,  $v' = 0$ ,  $w' = 0$ , cette formule donne

$$f(x, y, z) = \left[ f(x, y, z) \cdot \frac{\partial(X, Y, Z, X', Y', Z')}{\partial(x, y, z, x', y', z')} \cdot \frac{1}{(X - u)(Y - v)(Z - w)X'Y'Z'} \right]_{x^{-1}y^{-1}z^{-1}x'^{-1}y'^{-1}z'^{-1}}$$

équation dans laquelle on doit d'abord développer le dernier facteur de l'expression renfermée entre crochets suivant les puissances ascendantes de  $u$ ,  $v$ ,  $w$ , et ensuite développer les puissances de  $X$ ,  $Y$ ,  $Z$ ,  $X'$ ,  $Y'$ ,  $Z'$  suivant les puissances descendantes de  $x'$ ,  $y'$ ,  $z'$ ,  $x$ ,  $y$ ,  $z$  respectivement. On obtient ainsi

$$\text{coeff. de } u^a v^b w^c \text{ dans le développement de } f(x, y, z) =$$

$$\left[ f(x, y, z) \cdot \frac{\partial(X, Y, Z, X', Y', Z')}{\partial(x, y, z, x', y', z')} \cdot \frac{X^{-a-1} Y^{-b-1} Z^{-c-1}}{X' Y' Z'} \right]_{x^{-1} y^{-1} z^{-1} x'^{-1} y'^{-1} z'^{-1}},$$

ou ce qui est la même chose

coeff. de  $u^a v^b w^c$  dans le développement de  $f(x, y, z) =$

$$\left[ f(x, y, z) \cdot \frac{\partial\left(-\frac{1}{a}X^{-a}, -\frac{1}{b}Y^{-b}, -\frac{1}{c}Z^{-c}, \log X', \log Y', \log Z'\right)}{\partial(x, y, z, x', y', z')} \right]_{x^{-1} y^{-1} z^{-1} x'^{-1} y'^{-1} z'^{-1}}.$$

Or, en posant  $\Pi a = 1 \cdot 2 \cdot 3 \cdots a$ , etc., le terme général de  $-\frac{1}{a}X^{-a}$  est

$$(-1)^{r-1} \frac{\Pi(a+r-1)}{\Pi a \Pi a \text{ etc.}} A_{f,g,h}^{\alpha} \text{ etc. } x^F y^G z^H x'^{-a-r},$$

où

$$\alpha + \text{etc.} = r, \quad f\alpha + \text{etc.} = F, \quad g\alpha + \text{etc.} = G, \quad h\alpha + \text{etc.} = H,$$

de même le terme général de  $-\frac{1}{b}Y^{-b}$  est

$$(-1)^{s-1} \frac{\Pi(b+s-1)}{\Pi b \Pi \beta \text{ etc.}} B_{i,j,k}^{\beta} \text{ etc. } x^I y^J z^K y'^{-b-s},$$

où

$$\beta + \text{etc.} = s, \quad i\beta + \text{etc.} = I, \quad j\beta + \text{etc.} = J, \quad k\beta + \text{etc.} = K,$$

et le terme général de  $-\frac{1}{c}Z^{-c}$  est

$$(-1)^{t-1} \frac{\Pi(c+t-1)}{\Pi c \Pi \gamma \text{ etc.}} C_{l,m,n}^{\gamma} \text{ etc. } x^L y^M z^N z'^{-c-t},$$

où

$$\gamma + \text{etc.} = t, \quad l\gamma + \text{etc.} = L, \quad m\gamma + \text{etc.} = M, \quad n\gamma + \text{etc.} = N.$$

Le terme général du développement de  $\log X'$  est

$$-\frac{\Pi(r'-1)}{\Pi a' \Pi b' \Pi i'} a_{100}^{a'} b_{100}^{b'} c_{100}^{i'} x'^{r'} y'^{b'} z'^{i'} x^{-r'},$$

où

$$r' = a' + b' + i',$$

et je fais observer que pour tenir compte du terme  $\log x$  que contient  $\log X'$ , il suffit d'attribuer à  $a'$ ,  $b'$ ,  $i'$ ,  $r'$  les valeurs  $a' = b' = i' = r' = 0$ . En effet on n'a besoin que des coefficients différentiels de  $\log X'$ , et, en gardant pour le moment  $r'$  au lieu de zéro, le terme dont il s'agit sera  $-\Pi(r'-1)x^{-r'}$ , ce qui différentié par rapport à  $x$  donne  $\Pi r' x^{-r'-1}$ . En faisant  $r' = 0$  cela devient  $\frac{1}{x}$ , ce qui est en effet la valeur du coefficient différentiel de  $\log x$  par rapport à  $x$ .

De même le terme général de  $\log Y'$  est

$$-\frac{\Pi(s' - 1)}{\Pi\delta' \Pi e' \Pi\kappa'} a_{010}^{\delta'} b_{010}^{e'} c_{010}^{\kappa'} x'^{\delta'} y'^{e'} z'^{\kappa'} y^{-s'},$$

où

$$s' = \delta' + e' + \kappa',$$

et le terme général de  $\log Z'$  est

$$-\frac{\Pi(t' - 1)}{\Pi\iota' \Pi\kappa' \Pi\lambda'} a_{001}^{\iota'} b_{001}^{\kappa'} c_{001}^{\lambda'} x'^{\iota'} y'^{\kappa'} z'^{\lambda'} z^{-t'},$$

où

$$t' = \iota' + \kappa' + \lambda'.$$

Donc, en formant le terme général du Jacobien et en multipliant par le terme général de  $f(x, y, z)$ , on obtient pour le terme général de l'expression placée entre crochets la valeur suivante :

$$\begin{aligned} & (-1)^{r+s+t} \frac{\Pi(a+r-1) \Pi(b+s-1) \Pi(c+t-1) \Pi(r'-1) \Pi(s'-1) \Pi(t'-1)}{\Pi a \Pi b \Pi c \Pi \alpha \text{ etc. } \Pi \beta \text{ etc. } \Pi \gamma \text{ etc. } \Pi a' \Pi b' \Pi i' \Pi \delta' \Pi e' \Pi \kappa' \Pi \iota' \Pi \kappa' \Pi \lambda'} \\ & \times A_{f,g,h}^{\alpha} \text{ etc. } B_{i,j,k}^{\beta} \text{ etc. } C_{l,m,n}^{\gamma} \text{ etc. } a_{100}' b_{100}' c_{100}' a_{010}' b_{010}' c_{010}' a_{001}' b_{001}' c_{001}' \\ & \times \Theta_{p,q,r''} x^{F+I+L+P-r'-1} y^{G+J+M+Q-s'-1} z^{H+K+N+R-t'-1} x'^{-a-r+a'+\delta'+\iota'-1} y'^{-b-s+b'+e'+\kappa'-1} z' \end{aligned}$$

où le facteur  $\Omega$  représente le déterminant numérique suivant

$$\Omega = \begin{vmatrix} F & G & H & -a-r & 0 & 0 \\ I & J & K & 0 & -b-s & 0 \\ L & M & N & 0 & 0 & -c-t \\ -r' & 0 & 0 & a' & \delta' & \iota' \\ 0 & -s' & 0 & b' & e' & \kappa' \\ 0 & 0 & -t' & i' & \kappa' & \lambda' \end{vmatrix}.$$

Pour trouver le terme qui contient  $x^{-1} y^{-1} z^{-1} x'^{-1} y'^{-1} z'^{-1}$  on n'a qu'à poser

$$\begin{aligned} F + I + L + P - r' &= 0, & G + J + M + Q - s' &= 0, & H + K + N + R - t' &= 0, \\ -a - r + a' + \delta' + \iota' &= 0, & -b - s + b' + e' + \kappa' &= 0, & -c - t + i' + \kappa' + \lambda' &= 0. \end{aligned}$$

En faisant cela, et en tenant compte des formules précédentes on obtient le théorème suivant : savoir en posant

$$\begin{aligned} X &= A_{100} x + A_{010} y + A_{001} z + \cdots + A_{f,g,h} x^f y^g z^h + \text{etc.} = u, \\ Y &= B_{100} x + B_{010} y + B_{001} z + \cdots + B_{i,j,k} x^i y^j z^k + \text{etc.} = v, \\ Z &= C_{100} x + C_{010} y + C_{001} z + \cdots + C_{l,m,n} x^l y^m z^n + \text{etc.} = w, \\ f(x, y, z) &= \cdots + \Theta_{r,s,t} x^r y^s z^t + \text{etc.} \end{aligned}$$

on trouve pour le terme général du coefficient de  $u^a v^b w^c$  dans le développement de  $f(x, y, z)$  l'expression

$$(-1)^{r+s+t} \frac{\Pi(a+r-1) \Pi(b+s-1) \Pi(c+t-1) \Pi(r'-1) \Pi(s'-1) \Pi(t'-1)}{\Pi a \Pi b \Pi c \Pi \alpha \text{ etc. } \Pi \beta \text{ etc. } \Pi \gamma \text{ etc. } \Pi a' \Pi b' \Pi i' \Pi \delta' \Pi e' \Pi \kappa' \Pi i' \Pi \kappa' \Pi \lambda'}$$

$$\times A_{f,g,h}^\alpha \text{ etc. } B_{i,j,k}^\beta \text{ etc. } C_{l,m,n}^\gamma \text{ etc. } \Theta_{p,q,r''} a'_{100} b'_{100} c'_{100} a'_{010} b'_{010} c'_{010} a'_{001} b'_{001} c'_{001} \Omega,$$

formule dans laquelle les  $a_{100}$ ,  $a_{010}$ , etc., sont les coefficients inverses des  $A_{100}$ ,  $A_{010}$ , etc., c'est-à-dire

$$\begin{vmatrix} a_{100} & a_{010} & a_{001} \\ b_{100} & b_{010} & b_{001} \\ c_{100} & c_{010} & c_{001} \end{vmatrix} = \begin{vmatrix} A_{100} & A_{010} & A_{001} \\ B_{100} & B_{010} & B_{001} \\ C_{100} & C_{010} & C_{001} \end{vmatrix}^{-1};$$

le déterminant numérique  $\Omega$  a la valeur qui vient d'être donnée, en supposant seulement que les nombres  $\alpha$ ,  $\beta$ , etc., qui y entrent satisfassent aux conditions suivantes

$$\begin{array}{lll} \alpha + \text{etc.} = r, & \beta + \text{etc.} = s, & \gamma + \text{etc.} = t, \\ f\alpha + \text{etc.} = F, & g\alpha + \text{etc.} = G, & h\alpha + \text{etc.} = H, \\ i\beta + \text{etc.} = I, & j\beta + \text{etc.} = J, & k\beta + \text{etc.} = K, \\ l\gamma + \text{etc.} = L, & m\gamma + \text{etc.} = M, & n\gamma + \text{etc.} = N, \\ a' + b' + i' = r', & \delta' + e' + \kappa' = s', & i' + \kappa' + \lambda' = t', \\ P + F + I + L = r', & Q + G + J + M = s', & R + H + K + N = t', \\ a' + \delta' + i' = a + r, & b' + e' + \kappa' = b + s, & i' + \kappa' + \lambda' = c + t. \end{array}$$

De ces équations on tire

$$P + Q + R + F + G + H + I + J + K + L + M + N = a + b + c + r + s + t,$$

ou, en substituant pour  $F$ , etc., leurs valeurs,

$$(f+g+h-1)\alpha + \text{etc.} + (i+j+k-1)\beta + \text{etc.} + (l+m+n-1)\gamma + \text{etc.} = a+b+c-P-Q-R,$$

les nombres  $f+g+h-1$ ,  $i+j+k-1$ ,  $l+m+n-1$ , etc., étant positifs. Il n'y a donc qu'un nombre fini de solutions des équations indéterminées, comme cela doit être.

On peut encore modifier un peu la forme du déterminant  $\Omega$ ; on voit d'abord que l'on peut changer les signes des quantités  $-a-r$ ,  $-b-s$ ,  $-c-t$ ,  $-r'$ ,  $-s'$ ,  $-t'$ ; en faisant cela et en substituant pour ces quantités et aussi pour  $F$ ,  $G$ , etc., les valeurs ci-dessus données, on obtient

$$\Omega = \begin{vmatrix} f\alpha + \text{etc.} & g\alpha + \text{etc.} & h\alpha + \text{etc.} & a' + \delta' + i' & 0 & 0 \\ i\beta + \text{etc.} & j\beta + \text{etc.} & k\beta + \text{etc.} & 0 & b' + e' + \kappa' & 0 \\ l\gamma + \text{etc.} & m\gamma + \text{etc.} & n\gamma + \text{etc.} & 0 & 0 & i' + \kappa' + \lambda' \\ a' + b' + i' & 0 & 0 & \alpha' & \delta' & i' \\ 0 & b' + e' + \kappa' & 0 & b' & e' & \kappa' \\ 0 & 0 & i' + \kappa' + \lambda' & i' & \kappa' & \lambda' \end{vmatrix}.$$

Ainsi se trouve démontré le théorème général de M. Sylvester relatif à la réversion des séries, car il est évident que les formules s'appliquent sans difficulté à un nombre quelconque de variables.

*Londres, le 30 Septembre, 1856.*

# Sur quelques Formules pour la Transformation des Intégrales Elliptiques

[FROM THE *JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK*, (CRELLE), TOM. LV. (1858), PP. 15–24.]

## I.

En posant dans les formules “*Fund. nova* p. 15, Tab. III.”,  $\cos \phi = x$ , on obtient

$$\frac{dy}{\sqrt{(y-\alpha)(y-\beta)(y-\gamma)(y-\delta)}} = \frac{2}{\sqrt{(\alpha-\gamma)(\beta-\delta)} + \sqrt{(\alpha-\beta)(\gamma-\delta)}} \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}}$$

où

$$k^2 = \left( \frac{\sqrt{(\alpha-\gamma)(\beta-\delta)} - \sqrt{(\alpha-\beta)(\gamma-\delta)}}{\sqrt{(\alpha-\gamma)(\beta-\delta)} + \sqrt{(\alpha-\beta)(\gamma-\delta)}} \right)^2,$$

$$\frac{1-x}{1+x} = \frac{(\alpha-\beta)(y-\delta)}{(\alpha-\delta)(y-\beta)} \cdot \frac{(\gamma-\beta)}{(\gamma-\delta)} \cdot \frac{(y-\gamma)}{(y-\beta)}.$$

Maintenant soit

$$(y-\alpha)(y-\beta)(y-\gamma)(y-\delta) = (a, b, c, d, e(y, 1))^4,$$

et proposons-nous d'introduire dans les formules les coefficients  $(a, b, c, d, e)$  au lieu des racines  $\alpha, \beta, \gamma, \delta$ . En renvoyant d'ailleurs à la Note sur les covariants d'une fonction quadratique, cubique ou biquadratique insérée dans ce Journal, t. L. (1855), p. 285, [135], je pose pour abrégé

$$\begin{aligned} (\alpha-\beta)(\gamma-\delta) &= B, \\ (\alpha-\gamma)(\delta-\beta) &= C, \\ (\alpha-\delta)(\beta-\gamma) &= D, \end{aligned}$$

de sorte que l'on a identiquement

$$B + C + D = 0.$$

Les invariants  $I, J$  sont des fonctions rationnelles des quantités  $B, C, D$ ; en effet on a

$$\begin{aligned} B^2 + C^2 + D^2 &= 24I, \\ (B-C)(C-D)(D-B) &= 432J, \\ B^2C^2D^2 &= 256(I^3 - 27J^2). \end{aligned}$$

Cela étant, nous avons

$$k^2 = \left( \frac{i\sqrt{C} - \sqrt{B}}{i\sqrt{C} + \sqrt{B}} \right)^2,$$

ce qui donne d'abord

$$k^2 + 14k + 1 = \frac{B^2 + BC + C^2}{(i\sqrt{C} + \sqrt{B})^4},$$

ou, à cause de  $B + C + D = 0$ , d'où  $B^2 + C^2 + D^2 = 2(B^2 + BC + C^2)$ ,

$$k^2 + 14k + 1 = \frac{8(B^2 + C^2 + D^2)}{(i\sqrt{C} + \sqrt{B})^4} = \frac{24I}{(i\sqrt{C} + \sqrt{B})^4} \cdot 8/8.$$

On trouve ensuite

$$1 - k^2 = \frac{4i\sqrt{BC}}{(i\sqrt{C} + \sqrt{B})^2},$$

et de là

$$k^2(1 - k^2)^2 = \frac{256 B^2 C^2 D^2}{(i\sqrt{C} + \sqrt{B})^8},$$

ou en multipliant le numérateur et le dénominateur par  $(i\sqrt{C} + \sqrt{B})^4$  et posant  $D^2$  au lieu de  $(B + C)^2$ , on obtient

$$k^2(1 - k^2)^2 = \frac{256 B^2 C^2 D^2}{(i\sqrt{C} + \sqrt{B})^{12}}.$$

Ces équations donnent

$$\frac{k^2(1 - k^2)^2}{(k^2 + 14k^2 + 1)^3} = \frac{B^2 C^2 D^2}{2(B^2 + C^2 + D^2)^3},$$

ou, en posant

$$N = \frac{27}{1 - \frac{27J^2}{I^3}},$$

on trouve aussi

$$k^2 + 14k^2 + 1 = \frac{192 I}{(i\sqrt{C} + \sqrt{B})^4}.$$

ce qui donne

$$\frac{2}{i\sqrt{C} + \sqrt{B}} = \sqrt[4]{\frac{k^2 + 14k^2 + 1}{12 I}},$$

et la formule de transformation devient ainsi

$$\frac{dy}{\sqrt{(a, b, c, d, e(y, 1))^4}} = \sqrt[4]{\frac{k^2 + 14k^2 + 1}{12 I}} \frac{dx}{\sqrt{(1 - x^2)(1 - k^2 x^2)}},$$

le module  $k$  étant déterminé par l'équation

$$\frac{(k^2 + 14k^2 + 1)^3}{(k^2 + 14k + 1)^3} = \frac{16 N}{27},$$

où

$$N = \frac{27}{1 - \frac{27J^2}{I^3}}.$$

Cela revient à une formule que j'ai donnée dans le mémoire intitulé, "On the Reduction of  $du \div \sqrt{U}$  when  $U$  is a Function of the Fourth Order," *Camb. and Dublin Math. Journal*, t. i. (1846), p. 70, [33], et de laquelle j'ai déduit des conséquences que je vais reproduire ici. En effet l'équation en  $k$  peut s'écrire sous la forme

$$(k^4 + 14k^2 + 1)^3 - 16Nk^2(k^2 - 1)^4 = 0,$$

c'est-à-dire en écrivant  $\frac{1}{k} + k = \omega$ ,

$$\left(k^2 + \frac{1}{k^2} + 14\right)^3 - 16N\left(k - \frac{1}{k}\right)^4 = 0;$$

on obtient l'équation très-simple

$$\omega^3 = \frac{4}{\sqrt[3]{N-1}},$$

pour  $\omega = k + \frac{1}{k}$ , et on a ensuite

$$k = \frac{\omega + \sqrt{\omega^2 - 4}}{2},$$

ce qui donne aussi

$$k^2 = \frac{(\omega^2 - 1)^2}{4} = -\frac{7 + 8\sqrt{N-1} + N}{2 + \sqrt[3]{N-1}}.$$

Soit  $k = \beta$  l'une des valeurs de  $k$ , l'équation en  $k$  devient

$$(k^4 + 14k^2 + 1)^3 = (\beta^4 + 14\beta^2 + 1)^3$$

équation à laquelle on satisfait, en outre, par la valeur  $k = \frac{1-\beta}{1+\beta}$ . En effet cette valeur donne

$$k^2 + 14k^2 + 1 = \frac{16(\beta^4 + 14\beta^2 + 1)^3}{(1+\beta)^{12}}, \quad k^2 - 1 = -\frac{8\beta(1+\beta^2)}{(1+\beta)^4},$$

expressions qui rendent l'équation identique. Cela fait voir que l'équation peut s'écrire sous la forme

$$(k^4 + 14k^2 + 1)^3 - k^2(k^2 - 1)^4 \frac{(\beta^4 + 14\beta^2 + 1)^3}{\beta^2(\beta^2 - 1)^4} = 0,$$



$$= (k^2 - \beta^2) \left( k^2 - \frac{1}{\beta^2} \right) \left( k - \frac{1 - \beta}{1 + \beta} \right)^2 \left( k - \frac{1 + \beta}{1 - \beta} \right)^2 \left( k^2 - \left( \frac{1 - \beta i}{1 + \beta i} \right)^2 \right) \left( k^2 - \left( \frac{1 + \beta i}{1 - \beta i} \right)^2 \right),$$

et les racines de l'équation en  $k^2$  sont

$$\beta^2, \quad \frac{1}{\beta^2}, \quad \left( \frac{1 - \beta}{1 + \beta} \right)^2, \quad \left( \frac{1 + \beta}{1 - \beta} \right)^2, \quad \left( \frac{1 - \beta i}{1 + \beta i} \right)^2, \quad \left( \frac{1 + \beta i}{1 - \beta i} \right)^2,$$

ce qui s'accorde avec un résultat obtenu par Abel (voir les œuvres d'Abel t. i. p. 310 [Ed. 2, p. 459]).

Je fais observer à présent qu'en écrivant

$$k = \left( \frac{1 - \beta i}{1 + \beta i} \right)^2,$$

on obtient pour  $k$  l'équation  $(27 - 4N)\varpi^3 + 27N(\varpi - 1) = 0$ , qui, en posant  $M = -\frac{27N}{27 - 4N}$ , ou ce qui est la même chose

$$N = \frac{27M}{4M - 27},$$

devient

$$\varpi^3 - M(\varpi - 1) = 0.$$

Cette équation est précisément la même que celle à laquelle je suis parvenu dans la Note sur les covariants etc. ci-dessus citée. La quantité  $\varpi$  est liée au module  $k$  par la relation

$$k = \frac{3\sqrt{\varpi - 1} + 2\sqrt{2\varpi - 3}}{\sqrt{\varpi + 3}}.$$

On peut introduire  $M$  au lieu de  $N$  dans l'équation en  $k$ , et en combinant les formules précédemment obtenues, on trouve

$$\frac{dy}{\sqrt{(a, b, c, d, e(y, 1))^4}} = \sqrt[4]{\frac{k^4 + 14k^2 + 1}{12I}} \frac{dx}{\sqrt{(1 - x^2)(1 - k^2x^2)}} \\ \frac{27(1 + 14k^2 + k^4)^3}{(1 + k^2)^2(1 - 34k^2 + k^4)^2} = \frac{I^3}{J^2},$$

ou ce qui revient à la même chose,

$$k = \frac{3\sqrt{\varpi - 1} + 2\sqrt{2\varpi - 3}}{\sqrt{\varpi + 3}}, \quad \varpi^3 - M(\varpi - 1) = 0.$$

De l'équation entre  $k$  et  $\varpi$  on déduit facilement la relation

$$\frac{k^2 + 14k^2 + 1}{k^4 - 34k^2 + 1} = -\frac{\varpi}{2\varpi - 3},$$

et en substituant dans cette équation pour  $\varpi$  sa valeur on obtient

$$\frac{1 - 34k^2 + k^4}{-3(1 + 14k^2 + k^4)} = \frac{1}{\varpi},$$

ce qui est encore une forme de la relation entre  $k$  et  $\varpi$ .

## II.

On peut obtenir les résultats qui viennent d'être déduits de la formule de Jacobi, en prenant pour point de départ la transformation d'une fonction du quatrième ordre dans sa forme canonique. Je suppose d'abord que l'on a identiquement

$$\begin{aligned}(a, b, c, d, e(x, y))^4 &= (\lambda x + \mu y)^4 + (\lambda' x + \mu' y)^4 + 6\theta(\lambda x + \mu y)^2(\lambda' x + \mu' y)^2, \\ &= x_1^4 + y_1^4 + 6\theta x_1^2 y_1^2.\end{aligned}$$

Cela étant, je pose  $\lambda\mu' - \lambda'\mu = \Lambda$ , et je forme les covariants des deux expressions ; on obtient par la propriété fondamentale de ces fonctions

$$\begin{aligned}I &= \Lambda^2(1 + 3\theta^2), \\ J &= \Lambda^3(\theta - \theta^3), \\ U &= x_1^4 + y_1^4 + 6\theta x_1^2 y_1^2, \\ H &= -\frac{1}{6}\Lambda^2\{\theta x_1^4 + \theta y_1^4 + (1 - 3\theta^2)x_1^2 y_1^2\}, \\ \Phi &= \frac{1}{2}\Lambda^3(9\theta^2 - 1)x_1 y_1(x_1^4 - y_1^4).\end{aligned}$$

De ces relations on tire

$$\frac{J^2}{I^3} = \frac{(\theta - \theta^3)^2}{(1 + 3\theta^2)^3},$$

de sorte qu'en posant

$$N = \frac{1 + 3\theta^2}{4\theta^2 - 4\theta},$$

on aura pour déterminer  $\theta$ , l'équation

$$1 - 9\lambda^2 I - 54\lambda^3 J = 0,$$

et pour déterminer  $x_1, y_1$  les équations

$$U = x_1^4 + y_1^4 + 6\theta x_1^2 y_1^2, \quad H = (\text{the value above}).$$

Je fais observer qu'en désignant par  $\lambda$  un coefficient tel que  $U + 6\lambda H$  soit un carré parfait, on obtient pour  $\lambda$  l'équation

$$1 - 9\lambda^2 I - 54\lambda^3 J = 0.$$

En effet le cubicovariant  $\Phi$  d'une fonction qui est un carré parfait est identiquement égal à zéro, et le cubicovariant de la fonction  $U + 6\lambda H$  est  $(1 - 9\lambda^2 I - 54\lambda^3 J)\Phi$ , on a donc pour  $\lambda$  l'équation qui vient d'être proposée ; cela posé, en observant que

$$\frac{I}{J} \cdot \frac{1 - \theta^2}{1 + 3\theta^2} = \frac{9\theta^2 - 1}{\theta - \theta^3} \cdot \frac{J}{I},$$

on aura pour une des valeurs de  $\lambda$

$$\lambda = -\frac{I}{6J} \cdot \frac{1 - \theta^2}{1 + 3\theta^2},$$

et en effet cette valeur donne

$$\frac{J^2}{I^3} = \frac{(\theta - \theta^3)^2}{(1 + 3\theta^2)^3},$$

ce qui est l'équation en  $\theta$ . Cela étant, je pose

$$\lambda = -\frac{I}{6J} \varpi_1,$$

alors  $\varpi_1$  sera une racine de l'équation

$$\varpi^3 - M(\varpi - 1) = 0,$$

on obtient

$$\varpi_1 = \frac{1 + 3\theta^2}{1 - \theta^2},$$

et de là

$$\theta^2 = \frac{\varpi_1 - 1}{\varpi_1 + 3}.$$

Soient  $\varpi_2, \varpi_3$  les deux autres racines de l'équation en  $\varpi$ , on aura

$$\varpi_1 + \varpi_2 + \varpi_3 = 0, \quad \varpi_2 \varpi_3 = -M, \quad \varpi_2 + \varpi_3 = -\varpi_1,$$

$$\varpi_2 - \varpi_3 = \pm \sqrt{\varpi_1^2 - 4\varpi_2 \varpi_3} = \pm \sqrt{\varpi_1^2 + 4M}.$$

On pourra donc écrire

$$\theta^2 = \frac{\varpi_2(\varpi_3 + 3)\varpi_3}{\varpi_2 - 1},$$

et au moyen des valeurs de  $U, H$  on obtient, par une très-simple réduction, les trois équations suivantes

$$\begin{aligned} IH - \varpi_1 JU &= -(\varpi_2 - \varpi_3) J (x_1^2 + y_1^2)^2, \\ IH - \varpi_2 JU &= -(\varpi_3 - \varpi_1) J (x_1^2 - y_1^2)^2, \\ IH - \varpi_3 JU &= -(\varpi_1 - \varpi_2)(\varpi_3 - \varpi_2) J \cdot 4x_1^2 y_1^2 / (\varpi_2 - \varpi_3), \end{aligned}$$

dont deux quelconques donnent les valeurs de  $x_1, y_1$ ; ainsi on a obtenu la solution complète du problème de la réduction de la fonction  $(a, b, c, d, e(x, y))^4$  à la forme canonique.

Je fais observer que ces équations montrent *a posteriori* que les expressions  $IH - \varpi_1 JU, IH - \varpi_2 JU, IH - \varpi_3 JU$  sont toutes les trois des carrés de fonctions quadratiques. Je fais observer, en outre, que si l'on forme la valeur de l'expression

$$\sqrt{IH - \varpi_1 JU} + i\sqrt{IH - \varpi_2 JU} + i\sqrt{IH - \varpi_3 JU},$$

en posant pour un moment  $\varpi_2 - \varpi_3 = \alpha, \varpi_3 - \varpi_1 = \beta, \varpi_1 - \varpi_2 = \gamma$ , l'expression dont il s'agit sera égale, à un facteur constant près, à

$$\sqrt{\alpha} (x_1^2 + y_1^2) + i\sqrt{\beta} (x_1^2 - y_1^2) + 2i\sqrt{\gamma} x_1 y_1,$$

ou, ce qui est la même chose, à

$$(\sqrt{\alpha} + i\sqrt{\beta})x_1^2 + 2i\sqrt{\gamma}x_1y_1 + (\sqrt{\alpha} - i\sqrt{\beta})y_1^2.$$

Or, cette fonction doit être un carré parfait, ce qu'on reconnaît en effet, en se rappelant que

$$(\sqrt{\alpha} + i\sqrt{\beta})(\sqrt{\alpha} - i\sqrt{\beta}) - (i\sqrt{\gamma})^2 = \alpha + \beta + \gamma = 0.$$

Sa racine carrée sera, à un facteur constant près,  $(\sqrt{\alpha} + i\sqrt{\beta})x_1 + i\sqrt{\gamma}y_1$ , et cette dernière fonction sera un des facteurs linéaires de  $x_1^4 + y_1^4 + 6\theta x_1^2 y_1^2$ . Pour vérifier cela je pose  $x_1^4 + y_1^4 + 6\theta x_1^2 y_1^2 = 0$ , ce qui donne

$$x_1^2 + (3\theta + \sqrt{9\theta^2 - 1})y_1^2 = 0,$$

et la démonstration s'achève par calcul direct, en exprimant chacun des radicaux au moyen des  $\varpi$ .

### III.

Pour obtenir la formule qui sert à la transformation d'une intégrale elliptique, il faut présenter les résultats précédemment obtenus sous une forme un peu différente. J'écris

$$\begin{aligned} (a, b, c, d, e(y, 1, ))^4 &= (\lambda + \mu y)^4 - (1 + k^2)(\lambda + \mu y)^2(\lambda' + \mu' y)^2 + k^2(\lambda' + \mu' y)^4, \\ &= (\lambda + \mu y)^4(1 - x^2)(1 - k^2 x^2), \end{aligned}$$

où

$$x = \frac{\lambda' + \mu' y}{\lambda + \mu y}.$$

Cela posé, en remplaçant comme auparavant  $\lambda\mu' - \lambda'\mu$  par  $\Lambda$ , et en formant les covariants, on obtient

$$I = \frac{1}{12}\Lambda^4(1 + 14k^2 + k^4),$$

$$U = (\lambda + \mu y)^4\{1 - (1 + k^2)x^2 + k^2x^4\},$$

$$H = -\frac{1}{12}\Lambda^2(\lambda + \mu y)^4\{2(1 + k^2) - (1 - 10k^2 + k^4)x^2 + 2k^2(1 + k^2)x^4\},$$

$$\Phi = \frac{1}{2}\Lambda^3(\lambda + \mu y)^6(1 - k^2)x(1 - k^2x^4),$$

$$\frac{27(1 + 14k^2 + k^4)^3}{(1 + k^2)^2(1 - 34k^2 + k^4)^2} = \frac{I^3}{J^2},$$

en posant comme auparavant  $M = \frac{I^3}{12J^2}$ ; on obtient ensuite

$$M = \frac{1 + 14k^2 + k^4}{(1 + k^2) \text{ (simple factor)}},$$

et en remarquant que l'équation entre  $x$  et  $y$  donne

$$\frac{(\lambda' - \lambda x^2) dy}{(\lambda + \mu y)^2} = \frac{\Lambda dy}{(\lambda + \mu y)^2},$$

on trouve

$$\frac{dy}{\sqrt{(a, b, c, d, e(y, 1, ))^4}} = \frac{1}{\Lambda} \frac{dx}{\sqrt{(1 - x^2)(1 - k^2 x^2)}},$$

c'est-à-dire

$$\frac{dy}{\sqrt{(a, b, c, d, e(y, 1, ))^4}} = \sqrt[4]{\frac{k^4 + 14k^2 + 1}{12I}} \frac{dx}{\sqrt{(1 - x^2)(1 - k^2 x^2)}},$$

ce qui s'accorde avec la formule ci-dessus trouvée ; pour compléter la solution, je pose

$$-\frac{1 - 34k^2 + k^4}{3(1 + 14k^2 + k^4)} = \frac{1}{\varpi_1},$$

$\varpi_3$  sera une des racines de l'équation  $\varpi^3 - M(\varpi - 1) = 0$ , en représentant par  $\varpi_1, \varpi_2$  les deux autres racines, on obtient

$$2\varpi_1 = -\varpi_3 \frac{1 + 6k + k^2}{1 + k^2}, \quad 2\varpi_2 = -\varpi_3 \frac{1 - 6k + k^2}{1 + k^2},$$

et ensuite

$$\begin{aligned} IH - \varpi_1 JU &= J(\lambda + \mu y)^4 (\varpi_3 - \varpi_1)(1 + kx^2)^2, \\ IH - \varpi_2 JU &= J(\lambda + \mu y)^4 (\varpi_3 - \varpi_2)(1 - kx^2)^2, \\ IH - \varpi_3 JU &= J \frac{(\varpi_3 - \varpi_1)(\varpi_3 - \varpi_2)}{\varpi_1 - \varpi_2} (\lambda + \mu y)^4 4kx^2. \end{aligned}$$

Donc en posant  $\varpi_2 - \varpi_3 = \alpha$ ,  $\varpi_3 - \varpi_1 = \beta$ ,  $\varpi_1 - \varpi_2 = \gamma$ , on a

$$\sqrt{IH - \varpi_1 JU} + i\sqrt{IH - \varpi_2 JU} + i\sqrt{IH - \varpi_3 JU} = J(\lambda + \mu y)^2 \sqrt{\alpha\beta} (\sqrt{\alpha} + i\sqrt{\beta} + i\sqrt{\gamma} \sqrt{k})^2,$$

de même

$$= J(\lambda + \mu y)^2 \sqrt{\alpha\beta} (\sqrt{\alpha} + i\sqrt{\beta} - i\sqrt{\gamma} \sqrt{k})^2,$$

et de là enfin, en remplaçant  $\alpha, \beta, \gamma$  par leurs valeurs,

$$\frac{(\varpi_2 - \varpi_3)\sqrt{IH - \varpi_1 JU} + (\varpi_3 - \varpi_1)\sqrt{IH - \varpi_2 JU} + (\varpi_1 - \varpi_2)\sqrt{IH - \varpi_3 JU}}{(\varpi_2 - \varpi_3)\sqrt{IH - \varpi_1 JU} + (\varpi_3 - \varpi_1)\sqrt{IH - \varpi_2 JU} - (\varpi_1 - \varpi_2)\sqrt{IH - \varpi_3 JU}} = \left( \frac{\sqrt{\varpi_2 - \varpi_3}}{\sqrt{\varpi_2 - \varpi_3}} \right)$$

équation dont le premier membre est le carré d'une fraction rationnelle de la forme

$$\frac{A + Bx}{C + Dx}.$$

#### IV.

Je terminerai ces recherches en démontrant le théorème de M. Hermite dont j'ai parlé dans la Note sur les covariants etc. ci-dessus citée. L'identité

$$JU^3 - IU^2H + 4H^3 = -\Phi^2,$$

peut être mise sous la forme

$$-J + I \frac{H}{U} - 4 \frac{H^3}{U^3} = \left( \frac{\Phi}{U} \right)^2 \cdot \frac{1}{U}.$$

Posons maintenant

$$z = \frac{H}{U},$$

formule dans laquelle je suppose qu'on ait fait  $y = 1$ , de manière que  $U$ ,  $H$  soient des fonctions rationnelles et entières de la seule variable  $x$ , savoir

$$\begin{aligned} U &= (a, b, c, d, e(x, 1))^4, \\ H &= (ac - b^2, \frac{1}{2}(ad - bc), \frac{1}{6}(ae + 2bd - 3c^2), \frac{1}{2}(be - cd), ce - d^2(x, 1))^4, \end{aligned}$$

alors on aura

$$\sqrt{-J + zI - 4z^3} = \frac{\Phi}{U^2} \sqrt{(a, b, c, d, e(x, 1))^4},$$

et

$$dz = \frac{U dH - H dU}{U^2}.$$

En vertu de la théorie de Jacobi,  $U dH - H dU$  est de la forme  $M\Phi dx$ , où  $M$  est un facteur constant ; on trouve en effet très-facilement  $U dH - H dU = 2\Phi dx$ , d'où l'on déduit la formule

$$\frac{dz}{\sqrt{-J + zI - 4z^3}} = \frac{2 dx}{\sqrt{(a, b, c, d, e(x, 1))^4}},$$

et je fais observer que l'intégrale du premier membre se ramène immédiatement à une forme qui ne contient que la seule constante  $M = \frac{I^3}{12J^2}$ .

*Londres, 9 Avril, 1856.*

## Note sur la Composition du Nombre 47 par rapport aux Vingt-troisièmes Racines de l'Unité

[FROM THE *JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK*, (CRELLE), TOM. LV. (1858), P. 192.]

M. KUMMER a trouvé (*Journal de Liouville*, t. XII. [1847] p. 208) que le nombre 47 peut être décomposé en onze facteurs qui se déduisent du suivant  $\alpha^{10} + \alpha^8 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha^0$ ,  $\alpha$  désignant une racine 23<sup>ième</sup> de l'unité, et on sait par la théorie générale qu'il doit y avoir une puissance  $47^f$  qui se décompose en vingt-deux facteurs. Le nombre  $47^2$  peut se décomposer en deux facteurs formés avec les demi-périodes des racines ; il était donc naturel d'essayer si le facteur  $(\alpha^{10} + \alpha^8 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha^0)^2$  pourrait se décomposer de même en deux facteurs, ce qui donnerait la décomposition de  $47^2$  en vingt-deux facteurs. Mais on démontre très-facilement que cette décomposition n'est pas possible. En effet en posant  $\alpha^\lambda = 1$  ( $\lambda$  étant un nombre premier) et en faisant

$$A + B\alpha + \dots + K\alpha^{\lambda-1} = (a + b\alpha + \dots + k\alpha^{\lambda-1})(a + b\alpha^{\lambda-1} + \dots + k\alpha),$$

on aura  $A = a^2 + b^2 + \dots + k^2$ . Le nombre qui forme le premier membre peut se réduire au moyen de l'équation  $1 + \alpha + \dots + \alpha^{\lambda-1} = 0$  à la forme  $B'\alpha + C'\alpha^2 + \dots + K'\alpha^{\lambda-1}$  et l'on aura

$$B'\alpha + C'\alpha^2 + \dots + K'\alpha^{\lambda-1} = (a + b\alpha + \dots + k\alpha^{\lambda-1})(a + b\alpha^{\lambda-1} + \dots + k\alpha) - (a^2 + b^2 + \dots + k^2)(1 + \alpha + \dots + \alpha^{\lambda-1})$$

équation qui subsiste lorsqu'on y fait  $\alpha = 1$ , ce qui donne

$$B' + C' + \dots + K' = (a + b + \dots + k)^2 - \lambda(a^2 + b^2 + \dots + k^2);$$

or, la fonction qui forme le second membre, prise avec le signe négatif peut se mettre sous la forme  $(a - b)^2 + (a - c)^2 + (b - d)^2 + \text{etc.}$  ; donc la décomposition n'existe pas à moins

que  $B' + C' + \dots + K'$  ne soit négatif. Mais, en réduisant seulement au moyen de l'équation  $\alpha^{23} - 1 = 0$ , on trouve la suivante

$$(\alpha^{10} + \alpha^8 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha^0)^2 =$$

$$6 + 7\alpha + 7\alpha^2 + 3\alpha^3 + 6\alpha^4 + 9\alpha^5 + 6\alpha^6 + 16\alpha^7 + 15\alpha^8 + 9\alpha^9 + 18\alpha^{10} + 9\alpha^{11} \\ + 7\alpha^{12} + 7\alpha^{13} + 3\alpha^{14} + 6\alpha^{15} + 9\alpha^{16} + 6\alpha^{17} + 16\alpha^{18} + 15\alpha^{19} + 9\alpha^{20} + 18\alpha^{21} + 9\alpha^{22},$$

laquelle, en vertu de  $1 + \alpha + \dots + \alpha^{22} = 0$ , se réduit à

$$\alpha + \alpha^2 - 3\alpha^3 + 3\alpha^5 + 10\alpha^7 + 9\alpha^8 + 3\alpha^9 + 12\alpha^{10} + 3\alpha^{11} \\ + \alpha^{12} + \alpha^{13} - 3\alpha^{14} + 3\alpha^{16} + 10\alpha^{18} + 9\alpha^{19} + 3\alpha^{20} + 12\alpha^{21} + 3\alpha^{22},$$

où la somme des coefficients est positive ; donc la décomposition ne peut pas s'effectuer. On pourrait sans beaucoup de peine essayer de la même manière les nombres  $f = 2$  ou  $f = 3$ , mais je ne sais pas si l'on a une idée quelconque de la grandeur du nombre  $f$ .

Londres, le 10 Mai, 1857.

## Théorème sur les Déterminants Gauches

[FROM THE *JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK*, (CRELLE), TOM. LV. (1858), PP. 277–278.]

Dans le mémoire intitulé “Recherches ultérieures sur les déterminants gauches,” ce journal t. L. pp. 299–313 (1855), [137], j’ai donné une formule pour le développement d’un déterminant gauche bordé ; mais j’ai omis de remarquer un cas particulier assez important. La formule générale se rapporte à un déterminant tel que :

$$\alpha 123 | \beta 123 = \begin{vmatrix} \alpha\beta, & \alpha 1, & \alpha 2, & \alpha 3 \\ 1\beta, & 11, & 12, & 13 \\ 2\beta, & 21, & 22, & 23 \\ 3\beta, & 31, & 32, & 33 \end{vmatrix},$$

que l’on obtient en bordant d’une manière quelconque la matrice gauche

$$\begin{vmatrix} 11, & 12, & 13 \\ 21, & 22, & 23 \\ 31, & 32, & 33 \end{vmatrix}.$$

On a par exemple :

$$\begin{aligned} \alpha 123 | \beta 123 &= \alpha\beta \cdot 11 \cdot 22 \cdot 33 \\ &\quad + \alpha\beta \cdot 12 \cdot 12 \cdot 33 \\ &\quad + \alpha\beta \cdot 13 \cdot 13 \cdot 22 \\ &\quad + \alpha\beta \cdot 23 \cdot 23 \cdot 11 \\ &\quad + \alpha 1 \cdot \beta 1 \cdot 22 \cdot 33 \\ &\quad + \alpha 2 \cdot \beta 2 \cdot 11 \cdot 33 \\ &\quad + \alpha 3 \cdot \beta 3 \cdot 11 \cdot 22 \\ &\quad + \alpha 123 \cdot \beta 123. \end{aligned}$$

$$\begin{aligned} \alpha 1234 | \beta 1234 &= \alpha\beta \cdot 11 \cdot 22 \cdot 33 \cdot 44 \\ &\quad + \alpha\beta \cdot 12 \cdot 12 \cdot 33 \cdot 44 \\ &\quad + \cdots \\ &\quad + \alpha\beta 1234 \cdot 1234 \\ &\quad + \alpha 1 \cdot \beta 1 \cdot 22 \cdot 33 \cdot 44 \\ &\quad + \cdots \\ &\quad + \alpha 123 \cdot \beta 123 \cdot 44 \\ &\quad + \cdots \\ &\quad + \alpha 234 \cdot \beta 234 \cdot 11, \end{aligned}$$



où les expressions  $\alpha\beta12$ ,  $\alpha\beta13$ , etc., sont des Pfaffiens. Cela étant, le déterminant formé en bordant d'une manière quelconque une matrice gauche et symétrique peut se nommer déterminant gauche et symétrique bordé, et la formule fait voir qu'un déterminant de cette espèce se réduit toujours au produit de deux Pfaffiens. En effet, en écrivant dans les exemples  $11 = 22 = 33 = 44 = 0$ , on obtient :

$$\alpha123|\beta123 = \alpha123 \cdot \beta123,$$

$$\alpha1234|\beta1234 = \alpha\beta1234 \cdot 1234,$$

et de même pour un déterminant gauche et symétrique bordé quelconque, suivant que l'ordre du déterminant est pair ou impair. Je remarque à propos de cela, que dans le cas d'un déterminant d'ordre pair, le terme  $\alpha\beta$  est multiplié par un mineur qui (comme déterminant gauche et symétrique d'ordre impair) se réduit à zéro ; le déterminant ne contient donc pas ce terme  $\alpha\beta$ , et sera par conséquent réduction linéo-linéaire des quantités  $\alpha1$ ,  $\alpha2$ , etc., et  $1\beta$ ,  $2\beta$ , etc. ; de manière qu'on ne saurait être surpris de voir ce déterminant se présenter sous la forme d'un produit de deux facteurs dont l'un est fonction linéaire de  $\alpha1$ ,  $\alpha2$ , etc., et l'autre fonction linéaire de  $1\beta$ ,  $2\beta$ , etc. Mais pour un déterminant d'ordre impair, le coefficient du terme  $\alpha\beta$  ne se réduit pas à zéro ; en supposant donc que le déterminant puisse s'exprimer comme produit de deux facteurs, il est nécessaire que l'un de ces facteurs soit (comme le déterminant même) fonction linéaire de  $\alpha\beta$  et linéo-linéaire de  $\alpha1$ ,  $\alpha2$ , etc., et  $1\beta$ ,  $2\beta$ , etc. ; de cette manière on se rend compte de la différence de la forme des facteurs, qui a lieu dans les deux cas dont il s'agit.

En écrivant  $\beta = \alpha$  (ce qui implique  $\alpha\alpha = 0$ , car on suppose toujours  $\alpha\beta = -\beta\alpha$ ) le déterminant gauche et symétrique bordé se réduit à un déterminant gauche et symétrique ordinaire, de plus le Pfaffien  $\alpha\beta1234$  se réduit à zéro, et les équations deviennent :

$$\alpha123|\alpha123 = (\alpha123)^2,$$

$$\alpha1234|\alpha1234 = 0;$$

savoir, quand l'ordre est pair, le déterminant se réduit au carré d'un Pfaffien, et quand l'ordre est impair, le déterminant s'évanouit ; ce qui est en effet la propriété fondamentale des déterminants gauches et symétriques.

*2 Stone Buildings, Londres, le 16 Nov. 1857.*

## Note sur les Normales d'une Conique

[FROM THE *JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK*, (CRELLE), TOM. LVI. (1859), PP. 182–185.]

On connaît les recherches très élégantes de M. Joachimsthal sur les normales d'une conique [voir le Mémoire "Ueber die Normalen der Ellipse und des Ellipsoids," *Crelle*, t. XXVI. (1843) pp. 172–180]; en particulier l'auteur a obtenu le théorème suivant : en supposant que 1, 2, 3, 4 soient des points d'une conique tels que les quatre normales se rencontrent dans un même point, on prend le pôle de la droite (1, 2) par rapport à la conique et on mène par ce pôle des perpendiculaires aux diamètres de la conique ; cela étant, en prenant sur chaque diamètre dans le sens opposé un point dont la distance du centre est égale à la distance du pied de la perpendiculaire sur ce même diamètre, la droite menée par ces deux points passe par les deux points 3 et 4. Mais cette propriété peut s'énoncer d'une manière beaucoup plus simple ; la droite dont il s'agit est la polaire (ou autrement dit l'harmonique) – par rapport au triangle formé par les deux diamètres et la droite située à l'infini – du pôle de la droite (1, 2) par rapport à la conique. Or on sait que l'idée de la perpendiculaire peut être généralisée. Savoir en prenant une conique quelconque que nous appelons la *conique absolue*, deux droites harmoniques par rapport à cette conique peuvent être appelées *perpendiculaires* (et de même deux points harmoniques par rapport à la conique absolue peuvent être appelés perpendiculaires). Cela posé, on peut parler dans un sens plus général des normales, etc. d'une courbe quelconque. En effet, que l'on s'imagine comme auparavant (outre la conique absolue) une conique donnée quelconque et quatre points 1, 2, 3 et 4 de cette conique tels que les normales se rencontrent dans un même point. Au lieu du triangle ci-dessus mentionné on a le triangle formé par les trois axes harmoniques (ou autrement dit, conjugués) communs aux deux coniques, et le théorème peut s'énoncer comme suit : En prenant le pôle commun de la droite (1, 2) par rapport à la conique donnée et à la conique absolue, et puis la polaire (l'harmonique) de ce pôle par rapport aux axes conjugués de la conique donnée et de la conique absolue, cette polaire passe par les deux points

3 et 4. Ou (ce qui revient à la même chose) on peut considérer les points 1, 2, 3, 4 comme les angles d'un quadrilatère inscrit dans la conique donnée, les quatre tangentes à cette conique aux points dont il s'agit seront les côtés d'un quadrilatère circonscrit à la conique donnée ; cela étant, les six côtés du quadrilatère inscrit seront les polaires (les harmoniques) – par rapport aux trois axes conjugués de la conique donnée et de la conique absolue – des six sommets du quadrilatère circonscrit.

Pour démontrer cela, je fais observer qu'il est permis de rapporter la conique absolue et la conique donnée aux trois axes conjugués communs, c'est-à-dire de prendre

$$x^2 + y^2 + z^2 = 0,$$

pour équation de la conique absolue, et

$$ax^2 + by^2 + cz^2 = 0,$$

pour équation de la conique donnée : cela posé (et en observant que, d'après la définition, deux droites  $Ax+By+Cz=0$ ,  $A'x+B'y+C'z=0$  seront perpendiculaires si  $AA'+BB'+CC'=0$ ) on obtient sans peine

$$\frac{x(b-c)}{x_1} + \frac{y(c-a)}{y_1} + \frac{z(a-b)}{z_1} = 0,$$

pour équation de la normale au point 1, en désignant par  $(x_1, y_1, z_1)$  les coordonnées de ce point. On a de même, en désignant par  $(x_2, y_2, z_2)$  et  $(x_3, y_3, z_3)$  les coordonnées des points 2 et 3,

$$\begin{aligned} \frac{x(b-c)}{x_2} + \frac{y(c-a)}{y_2} + \frac{z(a-b)}{z_2} &= 0, \\ \frac{x(b-c)}{x_3} + \frac{y(c-a)}{y_3} + \frac{z(a-b)}{z_3} &= 0, \end{aligned}$$

pour les équations des normales aux points 2 et 3 ; et la condition qui exprime que ces trois normales se rencontrent dans un même point sera évidemment

$$\begin{vmatrix} \frac{1}{x_1} & \frac{1}{y_1} & \frac{1}{z_1} \\ \frac{x_1}{1} & \frac{y_1}{1} & \frac{z_1}{1} \\ \frac{x_2}{1} & \frac{y_2}{1} & \frac{z_2}{1} \\ \frac{x_3}{1} & \frac{y_3}{1} & \frac{z_3}{1} \end{vmatrix} = 0.$$

Mais les coordonnées  $(x_1, y_1, z_1)$  etc. satisfont à l'équation  $ax^2+by^2+cz^2=0$ , on a donc aussi

$$\begin{vmatrix} x_1^2 & y_1^2 & z_1^2 \\ x_2^2 & y_2^2 & z_2^2 \\ x_3^2 & y_3^2 & z_3^2 \end{vmatrix} = 0.$$

[Paper 238 continues to page 77.]

THE COLLECTED MATHEMATICAL  
PAPERS  
OF  
ARTHUR CAYLEY

Volume IV

Papers 238 (continuation)–245 (part)

Pages 76–100

Cambridge University Press

1891

## Table des matières

|                                                                                                              |     |
|--------------------------------------------------------------------------------------------------------------|-----|
| 238. Note sur les Normales d'une Conique (suite)                                                             | 77  |
| 1 Addition à la Note sur la Composition du Nombre 47 par rapport aux Vingt-troisièmes Racines de l'Unité     | 79  |
| 2 Note on a Theorem in Spherical Trigonometry                                                                | 80  |
| 3 On Poincot's Four New Regular Solids                                                                       | 83  |
| 4 Second Note on Poincot's Four New Regular Polyhedra                                                        | 87  |
| 5 On the Theory of Groups, as Depending on the Symbolic Equation $\theta^n = 1$ . Third Part                 | 89  |
| 6 On an Analytical Theorem relating to the Distribution of Electricity upon Spherical Surfaces               | 93  |
| 7 On an Analytical Theorem connected with the Distribution of Electricity on Spherical Surfaces. Second Part | 100 |

## Note sur les Normales d'une Conique (suite)

### [Continuation from p. 75]

et de ces deux équations on déduit la suivante

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = 0$$

en désignant par le symbole qui forme le premier membre la fonction

$$x_1y_2z_3 + x_2y_3z_1 + x_3y_1z_2 + x_1y_3z_2 + x_2y_1z_3 + x_3y_2z_1.$$

C'est ce qui résulte de l'identité

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \cdot \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = \begin{vmatrix} x_1^2 & y_1^2 & z_1^2 \\ x_2^2 & y_2^2 & z_2^2 \\ x_3^2 & y_3^2 & z_3^2 \end{vmatrix} + 2x_1y_1z_1 x_2y_2z_2 x_3y_3z_3 \begin{vmatrix} \frac{1}{x_1} & \frac{1}{y_1} & \frac{1}{z_1} \\ \frac{x_2}{1} & \frac{y_2}{1} & \frac{z_2}{1} \\ \frac{x_3}{1} & \frac{y_3}{1} & \frac{z_3}{1} \end{vmatrix};$$

car le déterminant qui forme le second facteur du premier membre de l'équation ne s'évanouissant pas, c'est l'autre facteur qui devra s'évanouir en vertu des deux relations données.

L'équation de la droite (1, 2) sera

$$\begin{vmatrix} x & y & z \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix} = 0;$$

les coordonnées du pôle de cette droite par rapport à la conique donnée  $ax^2 + by^2 + cz^2 = 0$ , seront

$$\frac{1}{a}(y_1z_2 - y_2z_1) : \frac{1}{b}(z_1x_2 - z_2x_1) : \frac{1}{c}(x_1y_2 - x_2y_1);$$

mais les deux équations  $ax_1^2 + by_1^2 + cz_1^2 = 0$ ,  $ax_2^2 + by_2^2 + cz_2^2 = 0$  donnent  $a : b : c = y_1z_2 - y_2z_1 : z_1x_2 - z_2x_1 : x_1y_2 - x_2y_1$ ; par suite de cela les coordonnées du pôle deviennent

$$\frac{1}{y_1z_2 + y_2z_1} : \frac{1}{z_1x_2 + z_2x_1} : \frac{1}{x_1y_2 + x_2y_1},$$

donc l'équation de la polaire (l'harmonique) de ce pôle par rapport aux trois droites ( $x = 0$ ,  $y = 0$ ,  $z = 0$ ) sera

$$x(y_1z_2 + y_2z_1) + y(z_1x_2 + z_2x_1) + z(x_1y_2 + x_2y_1) = 0,$$

laquelle peut être représentée comme suit

$$\begin{Bmatrix} x, & y, & z \\ x_1, & y_1, & z_1 \\ x_2, & y_2, & z_2 \end{Bmatrix} = 0,$$

et, en substituant  $(x_3, y_3, z_3)$  au lieu de  $(x, y, z)$ , on voit que la droite dont il s'agit contient le point 3. De même cette droite contient le point 4, de sorte que le théorème se trouve démontré.

Observons encore que dans la géométrie de la sphère on peut prendre pour la conique absolue la conique imaginaire qui est l'intersection de la surface sphérique avec la surface conique imaginaire  $x^2 + y^2 + z^2 = 0$ . Le mot perpendiculaire aura alors la signification ordinaire, et on aura pour les coniques sphériques ce théorème très simple, savoir que les six côtés du quadrilatère inscrit seront les polaires (les harmonicales) — par rapport aux trois axes de la conique sphérique donnée — des six sommets du quadrilatère circonscrit. C'est ce que l'on reconnaît aussi par l'analyse que je viens de donner, laquelle en effet est précisément celle dont on se servirait naturellement pour les coniques sphériques.

*Londres, le 10 Mars, 1857.*

# 1. Addition à la Note sur la Composition du Nombre 47 par rapport aux Vingt-troisièmes Racines de l'Unité

[FROM THE JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK (CRELLE), T. LVI. (1859), PP. 186–187: SEQUEL TO 236.]

M. Kummer a bien voulu m'écrire une lettre où il remarque que le cube du facteur complexe du nombre 47, multiplié par l'unité complexe convenable, peut effectivement se décomposer en deux facteurs réciproques; c'est-à-dire qu'il existe une unité complexe  $E(\alpha)$  telle que

$$E(\alpha)(\alpha^{11} + \alpha^{10} + \alpha^8 + \alpha^7 + \alpha^4 + \alpha^2)^3 = F(\alpha)F(\alpha^{-1}).$$

Pour  $F(\alpha)$  M. Kummer a trouvé la valeur

$$F(\alpha) = \alpha^{16} + \alpha^{13} + \alpha^9 + \alpha^8 + \alpha^6 - \alpha^4 + \alpha^2,$$

fonction qui par conséquent est l'un des 22 facteurs complexes de  $47^3$ . Dans un postscriptum M. Kronecker m'a communiqué l'expression suivante de cette unité complexe

$$E(\alpha) = \frac{(\alpha^6 + \alpha^{17})(\alpha^8 + \alpha^{15})^2(\alpha^{10} + \alpha^{13})^2(\alpha^9 + \alpha^{14})}{(\alpha^5 + \alpha^{18})^2}$$

équivalente à l'expression en fonction entière :

$$E(\alpha) = \begin{cases} -23\alpha + 2\alpha^2 - 20\alpha^3 - 21\alpha^4 - 3\alpha^5 - 17\alpha^6 - 4\alpha^7 - 8\alpha^8 - 12\alpha^9 \\ + \alpha^{10} + \alpha^{11} - 3\alpha^{12} - 3\alpha^{13} - 17\alpha^{14} - 4\alpha^{15} - 14\alpha^{16} - 8\alpha^{17} - 12\alpha^{18}. \end{cases}$$

En supposant que cette valeur de  $E(\alpha)$  soit connue, on trouve sans peine une condition à laquelle  $F(\alpha)$  doit satisfaire. La valeur que j'ai donnée pour  $(\alpha^{11} + \alpha^{10} + \alpha^8 + \alpha^7 + \alpha^4 + \alpha^2)^2$

contient le terme constant +6; en y ajoutant la quantité évanouissante  $-6(1 + \alpha + \dots + \alpha^{22})$  elle se réduit à

$$\begin{aligned} &\alpha + \alpha^2 - 3\alpha^3 + 3\alpha^4 + 10\alpha^5 + 9\alpha^6 + 3\alpha^7 + 12\alpha^8 + 3\alpha^9 \\ &+ \alpha^{12} + \alpha^{13} - 3\alpha^{14} + 3\alpha^{15} + 10\alpha^{16} + 9\alpha^{17} + 3\alpha^{18} + 12\alpha^{19} + 3\alpha^{20}, \end{aligned}$$

et en multipliant cette valeur par  $E(\alpha)$ , le terme constant sera +808; donc en y ajoutant la quantité évanouissante  $-808(1 + \alpha + \dots + \alpha^{22})$  on obtient

$$\begin{aligned} &E(\alpha)(\alpha^{11} + \alpha^{10} + \alpha^8 + \alpha^7 + \alpha^4 + \alpha^2)^3 \\ &= -5\alpha - 8\alpha^2 - 7\alpha^3 - 7\alpha^4 - 6\alpha^5 - 3\alpha^6 - 7\alpha^7 - 7\alpha^8 - 5\alpha^9 \\ &\quad - 5\alpha^{10} - 8\alpha^{11} - 7\alpha^{12} - 7\alpha^{13} - 6\alpha^{14} - 3\alpha^{15} - 7\alpha^{16} - 7\alpha^{17} - 5\alpha^{18}. \end{aligned}$$

En représentant cette expression par  $B\alpha + C\alpha^2 + \dots + K\alpha^{18}$ , j'écris

$$B\alpha + C\alpha^2 + \dots + K\alpha^{18} = (\alpha + b\alpha + \dots + k\alpha^{18})(\alpha + b'\alpha^{-1} + \dots + k'\alpha^{-18})(1 + \alpha + \dots + \alpha^{22}),$$



équation qui subsiste pour  $\alpha = 1$ . On a donc

$$B + C + \dots + K = (a + b + \dots + k)(a' + b' + \dots + k') \cdot 23(a + b + \dots + k'),$$

ou, d'après les valeurs de  $B, C, \dots, K$ ,

$$-136 = (a + b + \dots + k)(a' + b' + \dots + k'),$$

ce qui donne

$$(a + b + \dots + k)^2 = -136 \pmod{23}.$$

On peut ajouter à  $(a + b + \dots + k)^2$  un multiple quelconque de  $(1 + \alpha + \dots + \alpha^{22})$  et changer le signe ; il est donc permis de prendre  $(a + b + \dots + k)$  positif et plus petit que  $\frac{23}{2}$ . Cela étant la congruence donne

$$a + b + \dots + k = 7.$$

Au moyen de cette valeur l'équation à laquelle il faut satisfaire devient

$$\begin{aligned} 7 + 2a &= -a^2 + 3a^2 + 4a^3 - a^4 + 2a^5 \\ &\quad + 2a^6 - a^7 + 3a^8 + 4a^9 - a^{10} + 2a^{11} \\ &= (a + ba \dots + ka^{18})(a + ba^{-1} \dots + ka^{-18}). \end{aligned}$$

À cause des coefficients numériques négatifs, les coefficients cherchés  $a, b, \dots k$  ne peuvent pas être tous à la fois positifs ; et cela étant il y a quelque manière de satisfaire aux deux conditions ci-dessus écrites, savoir il faut donner à sept des coefficients  $a, b, \dots k$  les valeurs 1, 1, 1, 1, 1, 1, -1, et aux autres coefficients la valeur zéro ; l'expression

$$F(\alpha) = \alpha^2 + \alpha^4 + \alpha^6 + \alpha^8 + \alpha^{13} - \alpha^{14} + \alpha^{16}$$

s'accorde en effet avec cette conclusion.

*Londres, le 6 Octobre, 1858.*

## 2. Note on a Theorem in Spherical Trigonometry

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVII. (1859), P. 151.]

I am not aware that the following theorem has been noticed: viz., in any spherical triangle, if as usual  $a, b, c$  are the sides, and  $A, B, C$  the opposite angles, then

$$\begin{aligned}\sin b \sin c + \cos b \cos c \cos A &= \sin B \sin C - \cos B \cos C \cos a, \\ \sin c \sin a + \cos c \cos a \cos B &= \sin C \sin A - \cos C \cos A \cos b, \\ \sin a \sin b + \cos a \cos b \cos C &= \sin A \sin B - \cos A \cos B \cos c.\end{aligned}$$

The demonstration is very simple; in fact we have

$$\begin{aligned}\sin b \sin c + \cos b \cos c \cos A &= \sin b \sin c (\sin^2 A + \cos^2 A) + \cos b \cos c \cos A \\ &= \sin b \sin c \sin^2 A + \cos A (\cos b \cos c + \sin b \sin c \cos A) \\ &= \sin B \sin C \sin^2 a + \cos A \cos a \\ &= \sin B \sin C (1 - \cos^2 a) + \cos A \cos a \\ &= \sin B \sin C + \cos a (\cos A - \sin B \sin C \cos a) \\ &= \sin B \sin C - \cos B \cos C \cos a,\end{aligned}$$

which proves the theorem.

*2, Stone Buildings, W.C., January 5, 1859.*

A geometrical proof and interpretation are given, G. B. Airy, "Remarks on Mr Cayley's Trigonometrical Theorem, etc.," *Phil. Mag.* same volume, p. 176. I transfer to this place the concluding sentence of the subsequent paper 243. "I take the opportunity of noticing that the theorem in spherical trigonometry, which I gave in the February Number, is not new, but, as pointed out by Prof. Chauvenet in the Mathematical Monthly (Cambridge, U.S.), is to be found in Cagnoli's 'Trigonometry' (1808)."

### 3. On Poinso't's Four New Regular Solids

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVII. (1859), PP. 123–128.]

It is shown by Poinso't, in the “Mémoire sur les Polygones et les Polyèdres,” *Journ. École Polyt.* vol. IV. pp. 16 to 48 (1810), that, besides the regular polyhedrons of ordinary geometry, there are (of course in an extended signification of the term) four new regular polyhedrons, viz. an icosahedron, which I will call the great icosahedron (No. 33 of the Memoir), and three dodecahedrons, which I will call the great dodecahedron (No. 37), the great stellated dodecahedron (No. 38), and the small stellated dodecahedron (No. 39). The nature of Poinso't's generalization will be best understood by conceiving, as he does, that the polyhedron is projected on a concentric sphere, so that the faces become spherical polygons. Then for the ordinary polyhedrons of geometry, the sum of the angles at a vertex = 4 right angles; but it may, according to the more general notion, be =  $s$  times 4 right angles. In like manner for the ordinary polyhedrons, the sides of a face subtend at the centre angles the sum of which is =  $e'$  times 4 right angles. And finally, the sum of the spherical polygons is ordinarily equal to the entire spherical surface; but according to the more general notion, it may be =  $D$  times the spherical surface ( $e$  is Poinso't's  $s$ ;  $e'$  does not occur in Poinso't; and, for a reason which will appear, I have written  $D$  for Poinso't's  $E$ .)

The new polyhedra are constructed, as follows:

#### 1. The great Icosahedron.

Each face is made up of seven faces, or rather four faces and six half faces of the ordinary icosahedron, in the manner shown by fig. 1. There are six right angles in each vertex, and together, not four, but eight right angles, or  $s = 2$ ; but, as in the ordinary polyhedra,  $e' = 1$ ; and the sum of all the faces, or eight times the spherical surface, or  $D = 7$ . (Also  $E = 7$ .)

#### 2. The great Dodecahedron.

Each face is made up of five faces of the ordinary icosahedron in the manner shown by the figure 2. There are five angles at each vertex, and these make up together eight right angles, or  $s = 2$ ; but, as in ordinary polyhedra,  $e' = 1$ ; and the sum of all the faces is obviously  $12 \times \frac{5}{12}$ , that is three times the spherical surface, or  $D = 3$ . (Also  $E = 3$ .)

#### 3. The great stellated Dodecahedron.

Each face is formed by stellating a face of the great dodecahedron in the manner shown by fig. 3. There are, as in the ordinary

dodecahedron, three angles at each vertex, and the sum of these is simply four right angles, or  $s = 1$ . On account of the stellation,  $e' = 2$ . Each of the projecting parts of the face is equal  $\frac{1}{6}$  of the face of the ordinary icosahedron; and if we reckon the area of the stellated pentagon as twice the interior pentagon plus the projecting

parts, then the area of the stellated face will be four times the spherical surface, and accordingly Poinot writes  $E = 4$ . If, however, what seems preferable, we reckon the area of the stellated pentagon as a side (or what is the same thing, reckon the stellated face as the centre of the face and standing on the sides.

#### 4. The small stellated Dodecahedron.

Each face is twelve that of the face of the ordinary dodecahedron, and the sum of the faces is twice the spherical surface, and accordingly Poinot writes  $E = 2$ . But according to the second mode of measurement, the area of the stellated face is three times the spherical surface; and the sum of the faces is three times the spherical surface, and the sum is  $D = 3$ .

$E$ , viz. the faces make together  $E$  times the spherical surface, the area of a stellated face being reckoned as the sum of the triangles having their vertices at the centre of the face and standing on the sides.

$D$ , viz. the faces make together  $D$  times the spherical surface, the area of a stellated face reckoned as the sum of the triangles having their vertices at the centre of the face and standing on the sides.

The Table is

| Designation                  | $H$ | $S$ | $A$ | $a$ | $a'$ | $e$ | $e'$ | $D$ |
|------------------------------|-----|-----|-----|-----|------|-----|------|-----|
| Tetrahedron                  | 4   | 4   | 6   | 3   | 3    | 1   | 1    | 1   |
| Hexahedron                   | 6   | 8   | 12  | 4   | 3    | 1   | 1    | 1   |
| Octahedron                   | 8   | 6   | 12  | 3   | 4    | 1   | 1    | 1   |
| Dodecahedron                 | 12  | 20  | 30  | 5   | 3    | 1   | 1    | 1   |
| Icosahedron                  | 20  | 12  | 30  | 3   | 5    | 1   | 1    | 1   |
| Great stellated dodecahedron | 12  | 20  | 30  | 5   | 3    | 1   | 2    | 7   |
| Great icosahedron            | 20  | 12  | 30  | 3   | 5    | 2   | 1    | 7   |
| Small stellated dodecahedron | 12  | 12  | 30  | 5   | 5    | 1   | 2    | 3   |
| Great dodecahedron           | 12  | 12  | 30  | 5   | 5    | 2   | 1    | 3   |

where the figures which are polar reciprocals of each other are written in pairs: viz., as is well known, the tetrahedron is its own reciprocal, the hexahedron and octahedron are reciprocals, and the dodecahedron and icosahedron are reciprocals; moreover the great stellated dodecahedron and the great dodecahedron are reciprocals. I have not been able to define  $E$  in such a manner as to enable me to form the definition of a reciprocal number  $E'$ ; this may be possible, but in the mean time it seems better to discard  $E$  altogether, and use instead of it the number  $D$ .

Euler's well-known relation applying to ordinary polyhedra is

$$S + H = A + 2.$$

Poinot in his memoir has (by an extension of Legendre's demonstration of Euler's theorem) obtained the more general relation,

$$eS + H = A + 2E,$$

which, however, does not apply to the two stellated figures where  $e'$  is different from unity; the general form is

$$eS + e'H = A + 2D,$$

which applies to all the nine figures. This applies to all polyhedra, regular or not, which are such that  $e$  has the same value for each vertex, and  $e'$  the same value for each face. If for any other polyhedra, regular or not,  $e$ , then, according to the foregoing mode of reckoning, the area of the face (measured in the ordinary manner) is to be reckoned as  $e'$  times the same value  $a$ , then, according to the foregoing mode of reckoning, the area of the face will be  $E + e'H = A + 2D$ .

I remark that the small stellated dodecahedron and the great dodecahedron are connected by the following Table:

$$\begin{array}{ll} a b c d e = P, & A C E R D = p, \\ p b i k e = A, & P I E B H = a, \\ p e j i a = B, & P J A C I = b, \\ p d f j b = C, & P F B D J = c, \\ p c g f c = D, & P G C E F = d, \\ p a h g d = E, & P H D A G = e, \\ j c d g q = F, & J D Q C G = f, \\ f d e h q = G, & F E Q D H = g, \\ g e a i q = H, & G A Q E I = h, \\ h a b j q = I, & H B Q A J = i, \\ i b c f q = J, & I C Q B F = j, \\ f g h i j = Q, & F H J G I = q, \end{array}$$

where it is to be noticed that in either Table each non-consecutive triad of any printed column occurs once, and only once, as a non-consecutive triad of another printed. The restriction that a non-consecutive triad of any multiplet is not to occur as a triad, consecutive or non-consecutive, of any other multiplet (see my note appended to Mr Kirkman's paper "On Autopolar Polyhedra," *Phil. Trans.* 1857, p. 183 [259]), applies only to ordinary polyhedra, and not to the class here considered.

2, *Stone Buildings*, W.C., January 13, 1859.

## 4. Second Note on Poinot's Four New Regular Polyhedra

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVII. (1859), PP. 209–210.]

The Note on Poinot's four new regular Polyhedra (Feb. Number, p. 123), [241] was written without my being acquainted with Cauchy's first memoir, "Recherches sur les Polyèdres" (*Journ. Polyt.* vol. IX. pp. 68–86, 1813), the former part of which (pp. 68–76) relates to Poinot's polyhedra. Cauchy considers the polyhedra, not as projected on the sphere, but as solids, and he shows, very elegantly, that all such polyhedra are obtained by stellating the regular dodecahedron and the regular icosahedron. The reciprocal method would lead to two figures, and explained and developed in my former Note. Cauchy does not at all consider Poinot's generalized equation,  $eS + H = A + 2E$ ; but the latter part of his memoir employed by Poinot, and explained and developed in my former Note. Cauchy does not at all consider Poinot's generalized equation,  $eS + H = A + 2E$ ; viz. Cauchy's theorem is, for any polyhedron, regular or not, partitioned into any number of polyhedra (including those of the original polyhedron), and  $A$  the total number of edges (including those of the original polyhedron), and  $A$  the total number of vertices (including those of the original polyhedron), then  $S + H = A + P + 1$ ; that is, the sum of the number of vertices and the number of faces exceeds by unity the sum of the number of edges and the number of polyhedra.

For  $P = 1$ , we have Euler's equation  $S + H = A + 2$ ; and we have a theorem relating to the partition of a polygon; viz. if the polygon is divided into  $R$  polygons, and if  $S$  be the number of vertices, and  $A$  the number of edges, then  $S + H = A + 1$ ;

polyhedra. I remark that, in the equation  $S + H = A + 1$ ,  $H$  should, in analogy with Cauchy's notation for polyhedra, be replaced by  $P$ ; so that we have for a single polygon,

$$A = S;$$

and for the partitions of a polygon,

$$A = S + P - 1;$$

corresponding respectively to Euler's theorem for a single polyhedron, viz.

$$S + H = A + 2;$$

and to Cauchy's theorem for the partitions of a polyhedron, viz.

$$S + H = A + 2 + (P - 1).$$

Cauchy's second memoir (pp. 87–98) contains a very beautiful demonstration of the theorem implied in the ninth definition of the eleventh book of Euclid, viz. that two convex polyhedra are equal when they are bounded by the same number of faces equal each to each.

2, *Stone Buildings*, W.C., February 1, 1859.

## 5. On the Theory of Groups, as Depending on the Symbolic Equation $\theta^n = 1$ . Third Part

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVIII. (1859), PP. 34–37: SEQUEL TO 125 AND 126.]

The following is, I believe, a complete enumeration of the groups of 8:

- I.  $1, \alpha, \alpha^2, \alpha^3, \alpha^4, \alpha^5, \alpha^6, \alpha^7$  ( $\alpha^8 = 1$ ).
- II.  $1, \alpha, \alpha^2, \alpha^3, \beta, \beta\alpha, \beta\alpha^2, \beta\alpha^3$  ( $\alpha^4 = 1, \beta^2 = 1, \alpha\beta = \beta\alpha$ ).
- III.  $1, \alpha, \alpha^2, \alpha^3, \beta, \beta\alpha, \beta\alpha^2, \beta\alpha^3$  ( $\alpha^4 = 1, \beta^2 = 1, \alpha\beta = \beta\alpha^3$ ).
- IV.  $1, \alpha, \alpha^2, \alpha^3, \beta, \beta\alpha, \beta\alpha^2, \beta\alpha^3$  ( $\alpha^4 = 1, \beta^2 = \alpha^2, \alpha\beta = \beta\alpha^3$ ).
- V.  $1, \alpha, \beta, \beta\alpha, \gamma, \gamma\alpha, \gamma\beta, \gamma\beta\alpha$  ( $\alpha^2 = 1, \beta^2 = 1, \gamma^2 = 1, \alpha\beta = \beta\alpha, \alpha\gamma = \gamma\alpha, \beta\gamma = \gamma\beta$ ).

That the groups are really distinct is perhaps most readily seen by writing down the indices of the different terms of each group; these are

- I. 1, 8, 4, 8, 2, 8, 4, 8.
- II. 1, 4, 2, 4, 2, 4, 2, 4.
- III. 1, 4, 2, 4, 2, 2, 2, 2.
- IV. 1, 4, 2, 4, 4, 4, 4, 4.
- V. 1, 2, 2, 2, 2, 2, 2, 2.

It will be presently seen why there is no group where the symbols  $\alpha, \beta$  are such that  $\alpha^4 = 1, \beta^2 = 1, \alpha\beta = \beta\alpha^2$ . A group which presents itself for consideration is

$$1, \alpha, \alpha^2, \alpha^3, \beta, \beta\alpha, \beta\alpha^2, \beta\alpha^3 \quad (\alpha^4 = 1, \beta^2 = \alpha^2, \alpha\beta = \beta\alpha);$$

but the indices of the different terms of this group are

$$1, 4, 2, 4, 2, 4, 2, 4,$$

and if we write  $\beta\alpha = \gamma$ , then we find  $\gamma^2 = \beta\alpha\beta\alpha = \beta\beta\alpha\alpha = \alpha^4 = 1, \alpha\gamma = \alpha\beta\alpha = \beta\alpha\alpha = \gamma\alpha$ ; and the group is

$$1, \alpha, \alpha^2, \alpha^3, \gamma, \gamma\alpha, \gamma\alpha^2, \gamma\alpha^3 \quad (\alpha^4 = 1, \gamma^2 = 1, \alpha\gamma = \gamma\alpha),$$

which is the group II.

The group IV is a remarkable one; it appears to arise from the circumstance that the factors 2 and 4 of the number 8 are not prime to each other; this can only happen when the number which denotes the order of the group contains a square factor. But the nature of the group in question will be better understood by presenting it under a different form. In fact, if we write  $\beta\alpha^3 = \gamma, \alpha^2 = \beta^2 = \delta$ , then we find  $\alpha = \delta\alpha, \beta\alpha^2 = \delta\beta, \beta\alpha = \delta\gamma$ , and the group will be

$$1, \alpha, \beta, \gamma, \delta, \delta\alpha, \delta\beta, \delta\gamma,$$

where the laws of combination are

$$\begin{aligned}\delta^2 &= 1, & \alpha^2 &= \beta^2 = \gamma^2 = \delta, & \alpha\beta &= \gamma, \\ \beta\gamma &= \alpha, & \gamma\alpha &= \beta, & \beta\alpha &= \gamma\delta = \delta\gamma, \\ \gamma\beta &= \alpha\delta = \delta\alpha, & \alpha\gamma &= \beta\delta = \delta\beta.\end{aligned}$$

Observe that  $\delta$  is a symbol of operation such that  $\delta^2 = 1$ , and that  $\delta$  is convertible with each of the other symbols  $\alpha, \beta, \gamma$ . It will be not so much a restrictive assumption in regard to  $\delta$ , as a definition of  $-1$  considered as a symbol of operation if we write  $\delta = -1$ ; the group thus becomes

$$1, \alpha, \beta, \gamma, -1, -\alpha, -\beta, -\gamma,$$

where

$$\alpha = \beta\gamma = -\gamma\beta, \quad \beta = \gamma\alpha = -\alpha\gamma, \quad \gamma = \alpha\beta = -\beta\alpha.$$

Hence  $\alpha, \beta, \gamma$  combine according to the laws of the quaternion symbols  $i, j, k$ ; and it is only the point of view from which the question is here considered which obliges us to consider the symbols as belonging to a group of 8, instead of (as in the theory of quaternions) a group of 4.

Suppose in general that the symbols  $\alpha, \beta$  are such that

$$\alpha^m = 1, \quad \beta^n = 1, \quad \alpha\beta = \beta\alpha^s,$$

then we find

$$\alpha^u \beta^v = \beta^v \alpha^{us^v};$$

and therefore if  $v = n$ ,  $\alpha^u = \alpha^{us^n}$  or  $\alpha^{u(s^n-1)} = 1$ , whence  $u(s^n - 1) \equiv 0 \pmod{m}$ ; or since  $u$  is arbitrary,  $s^n - 1 \equiv 0 \pmod{m}$ , an equation which, if  $m, n$  are given, determines the admissible values of  $s$ ; thus, for example, if  $n = 2$ , and  $m$  is a prime number, then  $s = 1$  or  $s = m - 1$ . The equation  $\alpha^u \beta^v = \beta^v \alpha^{us^v}$  shows that any combination whatever of the symbols  $\alpha, \beta$  can be expressed in the form  $\beta^q \alpha^p$  (or, if we please, in the form  $\alpha^p \beta^q$ ). It is proper to show that the assumed law is consistent with the associative law, viz. that the expression

$$\beta^{v_1} \alpha^{u_1} \cdot \beta^{v_2} \alpha^{u_2} \cdot \beta^{v_3} \alpha^{u_3}$$

can be transformed in one way only into the form  $\beta^v \alpha^u$ . We in fact have

$$\beta^{v_1} \alpha^{u_1} \cdot \beta^{v_2} \alpha^{u_2} = \beta^{v_1+v_2} \alpha^{u_1 s^{v_2} + u_2},$$

and multiplying this by the remaining factor  $\beta^{v_3} \alpha^{u_3}$ , we have

$$\beta^{v_1+v_2+v_3} \alpha^{(u_1 s^{v_2} + u_2) s^{v_3} + u_3},$$

which is equal to

$$\beta^{v_1+v_2+v_3} \alpha^{u_1 s^{v_2+v_3} + u_2 s^{v_3} + u_3},$$



and the result would have been precisely the same if, instead of thus combining together the first and second factors and the product with the third factor, we had combined the first factor with the product of the second and third factors, so that the associative law is satisfied.

It is now easy to see that if, as before,

$$\alpha^m = 1, \quad \beta^n = 1, \quad \alpha\beta = \beta\alpha^s,$$

conditions which it has been shown imply  $s^n \equiv 1 \pmod{m}$ , then the symbols  $\beta^q\alpha^p$  (or, if we please,  $\alpha^p\beta^q$ ), where  $p$  has the values  $0, 1, 2, \dots, m-1$ , and  $q$  the values  $0, 1, 2, \dots, n-1$ , form a group of  $nm$  terms. In particular, as already noticed, if  $n = 2$  and  $m$  is prime, then  $s = 1$  or  $s = m-1$ ; the two groups so obtained are essentially distinct from each other. If  $n = 2$ , but  $m$  is not prime, then  $s$  has in general more than two values: thus for  $m = 12$ ,  $s^2 \equiv 1 \pmod{12}$ , which is satisfied by  $s = 1, 5, 7$  and  $11$ ; the group corresponding to  $s = 1$  is distinct from that for any other value of  $s$ , but I have not ascertained whether the values other than unity do, or do not, give groups distinct from each other.

For the sake of an observation to which it gives rise, I write down an example of a group corresponding to  $n = 2$ ,  $s = m-1$ , say  $m = 5$ , and therefore  $s = 4$ , so that we have

$$\alpha^5 = 1, \quad \beta^2 = 1, \quad \alpha\beta = \beta\alpha^4,$$

and the group is

$$1, \alpha, \alpha^2, \alpha^3, \alpha^4, \beta, \beta\alpha, \beta\alpha^2, \beta\alpha^3, \beta\alpha^4,$$

the indices of the several terms being

$$1, 5, 5, 5, 5, 2, 2, 2, 2, 2.$$

The group is here expressed by means of the symbols  $\alpha, \beta$ , having the indices 5 and 2 respectively, but it may be expressed by means of two symbols having each of them the index 2. Thus putting  $\beta\alpha = \gamma$ , we find  $\beta^2 = 1, \gamma^2 = 1, (\beta\gamma)^5 = 1$ , which is equivalent to  $(\gamma\beta)^5 = 1$ , and the group may be represented in the form

$$1, \beta, \gamma, \beta\gamma, \gamma\beta, \beta\gamma\beta, \gamma\beta\gamma, \beta\gamma\beta\gamma, \gamma\beta\gamma\beta = \gamma\beta\gamma\beta\gamma,$$

the equality of the last two symbols being an obvious consequence of the equation  $(\beta\gamma)^5 = 1$ . It is clear that for any even number  $2p$  whatever, there is always a group which can be expressed in this form.

*2, Stone Buildings, W.C., June 9, 1859.*

## 6. On an Analytical Theorem relating to the Distribution of Electricity upon Spherical Surfaces

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVIII. (1859), PP. 119–127.]

There is contained in Plana's "Mémoire sur la distribution de l'électricité à la surface de deux sphères conductrices complètement isolées" (*Mém. de Turin*, vol. VII. 1845), an identical relation which is remarkable, as well in itself as because by means of it the author corrects an error into which Poisson had fallen in his researches on the same subject. The development of a certain definite integral is obtained in the form (equation 165)

$$y = -2 \cdot 3 \frac{M_1 h}{b^2} \sin^4 \frac{1}{2} \theta + \frac{3 \cdot 4 \cdot 5}{1 \cdot 2} \frac{M_2 h}{b^3} \sin^6 \frac{1}{2} \theta + \&c.$$

Poisson had in effect shown that  $M_1 = 0$ ; and he thence inferred that,  $\theta$  being small, the function in question  $\propto \sin^4 \frac{1}{2} \theta$ , or what is the same thing,  $\propto (1 - \cos \theta)^2$ . In the former part of the memoir, Plana shows that this is not the true form of the development; the foregoing development must therefore be illusory; and Plana in fact shows, by a laborious induction carried as far as  $M_7$ , that all the coefficients  $M$  vanish identically. The identical equation  $M_i = 0$ , where  $i$  is any positive integer whatever, constitutes the analytical theorem above referred to. Plana's expression for the function  $M_i$  is as follows:

$B_1, B_2, B_3, \&c.$  denote Bernoulli's numbers as given by the equation

$$\frac{t}{e^t - 1} = 1 - \frac{1}{2}t - B_1 \frac{t^2}{1 \cdot 2} + B_2 \frac{t^4}{1 \cdot 2 \cdot 3 \cdot 4} + \&c.$$

( $B_1 = \frac{1}{6}$ ,  $B_2 = \frac{1}{30}$ ,  $B_3 = \frac{1}{42}$ ,  $B_4 = \frac{1}{30}$ ,  $\&c.$  I have, in conformity with the usual practice, written the equations so as to make these numbers all positive; with Plana they are

alternately positive and negative). And in the equation 162, writing for  $k$  its value  $\frac{1}{1+b}$ , we have,  $\lambda$  being any positive integer,

$$\begin{aligned} G_\lambda &= \frac{1}{\lambda + 1} - \frac{1}{2}(1 + b) + B_1 \frac{\lambda}{1 \cdot 2} \frac{(1 + b)^2}{\lambda - 1} \\ &\quad - B_2 \frac{\lambda \cdot \lambda - 1 \cdot \lambda - 2}{1 \cdot 2 \cdot 3 \cdot 4} \frac{(1 + b)^4}{\lambda - 3} \\ &\quad + B_3 \frac{\lambda \cdot \lambda - 1 \cdot \lambda - 2 \cdot \lambda - 3 \cdot \lambda - 4}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} \frac{(1 + b)^6}{\lambda - 5} \\ &\quad + \&c., \end{aligned}$$

where the series is continued for so long as the factor in the denominator is positive. It should be observed that this factor really divides out, and that the rule just mentioned amounts to this, viz. that when  $\lambda$  is odd, the finite series on the right-hand side is to be continued to its last term; but when  $\lambda$  is even, the series is to be

continued only to the last term but one. And  $G_\lambda$  being thus defined, the expression for  $M_i$  (see equation 164, in which I have written for  $k$  its value  $\frac{1}{1+b}$ ) is

$$M_i = (1+b)^{i-1} \left\{ G_i + (1+2) \frac{i \cdot i - 1}{1 \cdot 2} \left( \frac{1}{b} \right)^2 G_{i+2} + (1+3) \frac{i \cdot i - 1 \cdot i - 2 \cdot i - 3}{1 \cdot 2 \cdot 3 \cdot 4} \left( \frac{1}{b} \right)^4 G_{i+4} + \&$$

where on the right-hand side the finite series is continued up to its last term, the value of which is obviously

$$\frac{1}{i}(1+2i-1) \left( \frac{1}{b} \right)^{2i-1} G_{3i-1}, \quad \text{that is,} \quad \frac{2}{i} \left( \frac{1}{b} \right)^{2i-1} G_{3i-1}.$$

But the form of this equation may be somewhat simplified. We in fact have

$$G_i = \frac{1}{i+1} - \frac{1}{2}(1+b) + B_1 \frac{i}{1 \cdot 2} \frac{(1+b)^2}{i-1} - B_2 \frac{i \cdot i - 1 \cdot i - 2}{1 \cdot 2 \cdot 3 \cdot 4} \frac{(1+b)^4}{i-3} + \&c.,$$

which is at once changed into

$$(1+b)M_i = \theta_i + \frac{i+1}{i} \frac{i}{1 \cdot 2} \theta_{i+1} + \&c.,$$

where

$$\theta_i = \frac{1}{i+1} - \frac{1}{2}i(1+b) + B_1 \frac{i \cdot i - 1}{1 \cdot 2} (1+b)^2 - B_2 \frac{i \cdot i - 1 \cdot i - 2 \cdot i - 3}{1 \cdot 2 \cdot 3 \cdot 4} (1+b)^4 + \&c.$$

or multiplying by  $i+1$ , and then putting  $i-1$  in the place of  $i$ , we have

$$i(1+b)M_{i-1} = \theta_i + \frac{i \cdot 1}{1 \cdot 2} \theta_{i+1} + \frac{i \cdot i - 1}{1 \cdot 2 \cdot 3} \theta_{i+2} + \&c. = 0,$$

where

$$\theta_i = 1 - \frac{1}{2}i(1+b) + B_1 \frac{i \cdot i - 1}{1 \cdot 2} (1+b)^2 - B_2 \frac{i \cdot i - 1 \cdot i - 2 \cdot i - 3}{1 \cdot 2 \cdot 3 \cdot 4} (1+b)^4 - \&c.$$

In this last equation the finite series on the right-hand side is, when  $i$  is even, to be continued up to its last term, but when  $i$  is odd, then only up to the last term but one. The equation to be proved is

$$0 = \theta_i + \frac{i \cdot 1}{1 \cdot 2} \theta_{i+1} + \frac{i \cdot i - 1}{1 \cdot 2 \cdot 3} \theta_{i+2} + \&c.,$$

where on the right-hand side the finite series is to be continued up to its last term; and the equation holds for any integer value of  $i$  which is  $\geq 2$ . This is the simplest form of Plana's theorem.

We have

$$\theta_i = i(1+b)^{i-1} G_{i-1};$$

or writing this equation under the form  $\theta_i = i(1+b)\frac{G_{i-1}}{(1+b)^{2-i}}$ , and comparing with Plana's developed expressions for  $G$  (which are continued by him as far as  $G_{11}$ ), we find

$$\begin{aligned}\theta_2 &= -\frac{1}{6}b, \\ \theta_3 &= \frac{1}{6}b^2, \\ \theta_4 &= \frac{1}{6}b + \frac{1}{2}b^2 - \frac{1}{6}b^3 - \frac{1}{6}b^4, \\ \theta_5 &= -\frac{2}{15}b^2 - 2b^3 - \frac{5}{6}b^5, \\ \theta_6 &= -\frac{1}{6}b - \frac{1}{2}b^2 - \frac{2}{3}b^3 + \frac{4}{3}b^4 + b^5 + \frac{1}{6}b^6, \\ \theta_7 &= \frac{2}{15}b^2 + 4b^3 + \frac{15}{2}b^4 + 4b^5 + \frac{2}{15}b^6, \\ \theta_8 &= \frac{2}{15}b + \frac{1}{2}b^2 + \frac{5}{3}b^3 + \frac{1}{6}b^4 + \frac{15}{6}b^5 + b^6 + \frac{1}{4}b^7 + \frac{1}{30}b^8, \\ \&c. &= -\frac{2}{15}b^2 - 12b^3 - 37b^4 - 54b^5 - 37b^6 - 12b^7 - \frac{2}{15}b^8,\end{aligned}$$

which are of course the results obtained by developing the foregoing expression for  $\theta_i$ , in powers of  $b$ , and collecting the terms. The formulae put in evidence a remarkable symmetry which does not exist in the original expression in powers of  $1+b$ .

It would be now easy to verify, for moderately small values of the suffix, the equations

$$\theta_2 + \frac{2}{1.2}\theta_3 + \frac{2.1}{1.2.3}\theta_4 = 0,$$

&c.

This is, in fact, Plana's process, which, however, as the suffixes increase, becomes a very laborious one, and the law of the terms which destroy each other is not in anywise exhibited thereby.

I have succeeded in obtaining a complete demonstration, founded on Herschel's theorem for the development of a function of  $e^t$ , and the expression thereby given for Bernoulli's numbers. The theorem in question [See Herschel's *Collection of Examples in the Calculus of Finite Differences*, Cambridge, 1820, p. 70, where the theorem is given in the form  $f\{(1+\Delta)^x\}0^x = u^x f(1+\Delta)0^x]$  is, that for any function of  $e^t$  which admits of development in positive integer powers of  $t$ ,

$$f(e^t) = f(1+\Delta)e^{t.0},$$

where the right-hand side denotes the series the general term whereof is

$$\frac{t^n}{1 \cdot 2 \cdot 3 \dots n} f(1+\Delta)0^n;$$

and  $f(1+\Delta)$  is of course to be developed in powers of  $\Delta$ ,  $\Delta^2$ ,  $\Delta^3$ , &c., applied to the symbol  $0^n$  (viz.  $\Delta 0^n = 1^n - 0^n$ ;  $\Delta^2 0^n = 2^n - 2 \cdot 1^n + 0^n$ , &c.).

This gives

$$\frac{t}{e^t - 1} = \frac{\log(1+\Delta)}{\Delta} e^{t.0},$$

and comparing the development of the right-hand side with the development

$$\frac{t}{e^t - 1} = 1 - \frac{1}{2}t + B_1 \frac{t^2}{1 \cdot 2} - B_2 \frac{t^4}{1 \cdot 2 \cdot 3 \cdot 4} + \&c.,$$

we find

$$\begin{aligned} \frac{\log(1 + \Delta)}{\Delta} 0^0 &= 1, \\ \frac{\log(1 + \Delta)}{\Delta} 0 &= -\frac{1}{2}, \\ \frac{\log(1 + \Delta)}{\Delta} 0^{2\sigma} &= (-)^{\sigma-1} B_\sigma, \\ \frac{\log(1 + \Delta)}{\Delta} 0^{2\sigma+1} &= 0, \quad (\sigma \geq 1). \end{aligned}$$

It is now easy to obtain the equation

$$\theta_i = \frac{\log(1 + \Delta)}{\Delta} \{(1 + 0(1 + b))^i - (-0(1 + b))^i\}.$$

In fact, the first two terms of the development of the expression on the right-hand side agree with those of the foregoing expression for  $\theta_i$ . For any even power  $2x$  (except, when  $i$  is even, the power  $2x = i$ ) the term is

$$\frac{[i]^{2x}}{[2x]^{2x}} (1 + b)^{2x} \frac{\log(1 + \Delta)}{\Delta} 0^{2x},$$

which agrees; and when  $i$  is even, then for the power  $2x = i$  there are two equal and opposite terms which destroy each other, and the whole term in  $\theta_i$  is, as it ought to be, zero. For any odd power  $2x - 1$ , ( $x > 1$ ), (including, when  $i$  is odd, the power  $2x - 1 = i$ ), the term vanishes as containing an evanescent factor. The expression for  $\theta_i$  is thus shown to be true.

I write for shortness,

$$X = 1 + 0(1 + b), \quad Y = -0(1 + b).$$

Forming the expression for  $i(1 + b)M_{i-1}$ ,

$$\theta_i + \frac{i \cdot 1}{1 \cdot 2} \theta_{i+1} + \frac{i \cdot i + 1}{1 \cdot 2 \cdot 3} \theta_{i+2} + \&c.,$$

this is

$$i(1 + b)M_{i-1} = \frac{\log(1 + \Delta)}{\Delta} \left\{ \left( X \left( 1 + \frac{X}{b} \right) \right)^i - \left( Y \left( 1 + \frac{Y}{b} \right) \right)^i \right\},$$

and we have

$$X = (1 + 0)(1 + b) - b,$$

and therefore

$$1 + \frac{X}{b} = \frac{1 + 0}{b} (1 + b),$$

$$\begin{aligned} \left(X \left(1 + \frac{X}{b}\right)\right)^i &= \frac{(1+0)^i(1+b)^i}{b^i} (1+0) \{(1+0)(1+b) - b\}^i, \\ 1 + \frac{Y}{b} &= -\frac{0(1+b) - b}{b} = -\frac{1}{b} \{0(1+b) - b\}. \end{aligned}$$

We see that the expression for  $\left(X \left(1 + \frac{X}{b}\right)\right)^i$  is deduced from that of  $\left(Y \left(1 + \frac{Y}{b}\right)\right)^i$  by writing therein  $1+0$  in the place of  $0$ ; we have therefore

$$\left(X \left(1 + \frac{X}{b}\right)\right)^i = \Delta_0 \left(Y \left(1 + \frac{Y}{b}\right)\right)^i$$

and consequently

$$\left(X \left(1 + \frac{X}{b}\right)\right)^i - \left(Y \left(1 + \frac{Y}{b}\right)\right)^i = (\Delta_0 - 1) \left(Y \left(1 + \frac{Y}{b}\right)\right)^i = \log(1 + \Delta_0) (0\{0(1+b) - b\})^i;$$

whence also

$$i(1+b)M_{i-1} = \left(\frac{1}{b}\right)^i \log(1 + \Delta) (0\{0(1+b) - b\})^i.$$

We have by the general theorem,

$$t = \log e^t = \log(1 + \Delta)e^{t \cdot 0};$$

and consequently whenever  $n \geq 2$ ,

$$\log(1 + \Delta) 0^n = 0.$$

But if  $i \geq 2$ , the reduced function  $(0\{0(1+b) - b\})^i$  contains only powers of  $0$ , and the superior powers; it is therefore reduced to zero by the operation  $\log(1 + \Delta)$ , and we have

$$i(1+b)M_{i-1} = \theta_i + \frac{i \cdot 1}{1 \cdot 2} \theta_{i+1} + \frac{i \cdot i - 1}{1 \cdot 2 \cdot 3} \theta_{i+2} + \&c. = 0;$$

and the theorem in question is thus proved. The foregoing expressions for  $\theta_2$ ,  $\theta_3$ , &c. show that these functions all divide by  $b$ , and moreover that when  $i$  is even and greater than 2, then that  $\theta_i$  divides by  $b^2$ . The equation

$$\theta_i = \frac{\log(1 + \Delta)}{\Delta} \{(1 + 0(1+b))^i - (-0(1+b))^i\}$$

gives generally for the term in  $\theta_i$  involving  $b^\alpha$ , the expression

$$\frac{[i]^\alpha}{[\alpha]^\alpha} \frac{\log(1 + \Delta)}{\Delta} \{(1+0)^{i-\alpha} 0^\alpha - (-0)^i\};$$

and it is to be shown, first, that the coefficient vanishes for  $\alpha = 0$ ; and next, that when  $i$  is even and  $> 2$ , the coefficient also vanishes for  $\alpha = 1$ . Putting  $\alpha = 0$ , the coefficient is

$$\frac{\log(1 + \Delta)}{\Delta} \{(1+0)^i - (-0)^i\},$$

or to

$$\log(1 + \Delta) 0^{i-1} + \{1 - (-)^i\} \frac{\log(1 + \Delta)}{\Delta} 0^i,$$

where, since  $i \geq 2$ , the former term vanishes, as above remarked; and the latter term, when  $i$  is even, vanishes on account of the factor  $1 - (-)^i$ ; and when  $i$  is odd, on account of the other factor. Hence the coefficient vanishes for  $\alpha = 0$ .

Next, if  $i$  is even, and  $\alpha = 1$ , the coefficient becomes

$$\frac{\log(1 + \Delta)}{\Delta} \{(1 + 0)^{i-1} 0 - (-0)^i\},$$

which, writing  $(1 + 0) - 1$  for 0, becomes

$$\frac{\log(1 + \Delta)}{\Delta} \{(1 + 0)^i - (1 + 0)^{i-1} - 0^i\},$$

which, since  $(1 + 0)^i - 0^i = \Delta 0^i$ ,  $(1 + 0)^{i-1} = (1 + \Delta) 0^{i-1}$ , is equal to

$$\log(1 + \Delta) 0^{i-1} - \frac{(1 + \Delta) \log(1 + \Delta)}{\Delta} 0^{i-1},$$

or since the first term vanishes, to

$$- \frac{(1 + \Delta) \log(1 + \Delta)}{\Delta} 0^{i-1}.$$

But this function is to a numerical factor  $\frac{1}{i}$  près the coefficient of  $t^{i-1}$  in  $\frac{e^t}{e^t - 1}$ , or (what is the same thing) in  $\frac{t}{e^t - 1}$ ; and if in the expression for  $\frac{t}{e^t - 1}$  we write  $-t$  in the place of  $t$ , we find

$$\frac{t}{e^{-t} - 1} = -1 - \frac{1}{2}t + B_1 \frac{t^2}{1 \cdot 2} - B_2 \frac{t^4}{1 \cdot 2 \cdot 3 \cdot 4} + \&c.,$$

so that,  $i$  being even and greater than 2, the function in question vanishes. Hence in the case under consideration the coefficient vanishes for  $\alpha = 1$ .

Writing  $\beta$  for  $i - \alpha$ , or assuming  $\alpha + \beta = i$ , the symmetry of the foregoing expressions for  $\theta_2, \theta_3$ , &c. shows that we ought to have

$$\frac{\log(1 + \Delta)}{\Delta} \{(1 + 0)^\alpha 0^\beta - (-0)^{\alpha+\beta}\} = \pm \frac{\log(1 + \Delta)}{\Delta} \{(1 + 0)^\beta 0^\alpha - (-0)^{\alpha+\beta}\},$$

where the upper or under sign is to be taken according as  $\alpha + \beta$  is even or odd. Or separating the two cases, we find

$$\frac{\log(1 + \Delta)}{\Delta} \{(1 + 0)^\alpha 0^\beta - (1 + 0)^\beta 0^\alpha\} = 0, \quad \alpha + \beta \text{ even},$$

$$\frac{\log(1 + \Delta)}{\Delta} \{(1 + 0)^\alpha 0^\beta + (1 + 0)^\beta 0^\alpha - 2 \cdot 0^{\alpha+\beta}\} = 0, \quad \alpha + \beta \text{ odd}.$$

I have not attempted to verify *à posteriori* these elegant formulae.

2, *Stone Buildings, W.C., June 18, 1859.*

## 7. On an Analytical Theorem connected with the Distribution of Electricity on Spherical Surfaces. Second Part

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVIII. (1859), PP. 193–202: CONTINUATION OF 244.]

The theorem is certainly true; but its existence gives rise to a difficulty to which I shall advert in the sequel. I propose, in the first instance, to give a demonstration which starts from the expression for  $fx$  given by Plana's equation (115), instead of the deduced equation which was the basis of my former proof. It will be proper to explain the origin and meaning of the formulae. We have two conducting spherical surfaces, radii 1 and  $b$ , in contact with each other (so that the distance between the centres is  $1 + b$ ): and then, if  $x$  is the distance from the centre of the sphere, radius 1, of an exterior point, and  $\mu (= \cos \theta)$  the cosine of the inclination of this distance to the line from the centre to the centre of the other sphere, the potential  $\phi(\mu, x)$  of the sphere, radius 1, at the point whose coordinates are  $(x, \mu)$  is deduced from the potential  $fx$  of a point in the axis; that is, if

$$fx = A_0 + A_1 x + A_2 x^2 + \&c.,$$

then

$$\phi(\mu, x) = A_0 P_0 + A_1 P_1 x + A_2 P_2 x^2 + \&c.,$$

where  $P_0, P_1, P_2, \&c.$  are Legendre's functions, viz. the functions of  $\mu$  which are the coefficients of the successive powers of  $x$  in the development of  $(1 - 2\mu x + x^2)^{-\frac{1}{2}}$  in ascending powers of  $x$ . And the electrical thickness  $y$  at any point of the surface of the sphere, radius 1, is given by the formula

$$y = x \frac{d\phi(\mu, x)}{dx} + \frac{1}{2}\phi(\mu, x),$$

where, after the differentiation,  $x = 1$ .

The problem consequently depends on the determination of the potential  $fx$  for a point on the axis; and this is determined by the functional equation

$$fx = \frac{h}{1 + 2b - (1 + b)x} F\left(\frac{1 + b - x}{1 + 2b - (1 + b)x}\right) + \frac{hb}{1 + b - x},$$

(Plana's equation (G), in which I have written for  $\beta, \gamma, H$  their values, and substituted also for  $g$  its value  $= h$ ). The solution of this equation is (equation (H), writing therein  $g = h$ )

$$fx = \sum_0^{\infty} \frac{h}{n + n(1 + b) - n(1 + b)x} - \sum_0^{\infty} \frac{Ph}{(n + 1)(1 + b) - (1 + n(1 + b))x},$$

where  $P$  is an arbitrary constant *quoad* the functional equation, viz. it is any function whatever which has the property of remaining unaltered when  $x$  is changed into



$\frac{1+b-x}{1+2b-(1+b)x}$ . Poisson, and Plana after him, arrive at the conclusion that in the physical problem  $P = 0$ . It appears to me that there is ground for holding that this is only true *sub modo*, and that  $\frac{P}{1+b-x}$  for  $x = 1$  (which, if  $P$  were retained, would be a term occurring in the expression for the thickness at the point of contact) is not of necessity zero. But the term, if it exists, can be replaced at the conclusion; and I write therefore

$$fx = \sum_0^{\infty} \frac{h}{n + n(1+b) - n(1+b)x} - \sum_0^{\infty} \frac{Ph}{(n+1)(1+b) - (1+n(1+b))x}.$$

According to the process by which the solution of the functional equation was obtained, this is the true form of the solution; for although the series are non-convergent, and the two sums are in fact each of them infinite, there is nothing to show a relation between the number of terms which must be taken in each series. However, nothing immediately turns upon this, as the expression is only used for obtaining an expression for  $fx$  in the form of a definite integral, viz., equation (36),

$$fx = \frac{b}{1+2b-(1+b)x} \int_0^1 \frac{dt(t^{-\frac{b}{1+b}} - 1)t^{\frac{bx}{(1+b)(1-x)}}}{1-t},$$

or, equation (39),

$$fx = \frac{b}{1+b} \int_0^1 \frac{1}{1-t(1+b)(1-x)}.$$

THE COLLECTED MATHEMATICAL  
PAPERS  
OF  
ARTHUR CAYLEY

Volume IV

Papers 245 (conclusion)–252 (part)

Pages 101–150

Cambridge University Press

1891

## Table des matières

|                                                                                                                                                    |     |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 245. On an Analytical Theorem connected with the Distribution of<br>Electricity upon Spherical Surfaces (conclusion)                               | 102 |
| 246. On Contour and Slope Lines                                                                                                                    | 109 |
| 247. On the Analytical Forms called Trees. Second Part.                                                                                            | 113 |
| 248. Sketch of a Proof of the Theorem that every Algebraic Equation<br>has a Root                                                                  | 117 |
| 249. Note on Cones of the Third Order                                                                                                              | 122 |
| 250. Sur la Surface qui est l'Enveloppe des Plans Conduits par les<br>Points d'un Ellipsoïde Perpendiculairement aux Rayons Menés par<br>le Centre | 123 |
| 251. Sur quelques Formules pour la Différentiation                                                                                                 | 135 |
| 252. Note sur l'Équation des Différences pour une Équation donnée de<br>Degré quelconque                                                           | 151 |

## On an Analytical Theorem connected with the Distribution of Electricity upon Spherical Surfaces (conclusion)

### [Continuation from p. 100]

This is the starting-point of the present investigation ; and attending to the equations

$$\begin{aligned}\frac{\log(1+\Delta)}{\Delta} \sigma &= -\frac{1}{p}, \\ \frac{\log(1+\Delta)}{\Delta} \sigma^{n-1} &= 0, \quad (n > 1), \\ \frac{\log(1+\Delta)}{\Delta} \sigma^n &= B_{p,n},\end{aligned}$$

we see that the development of  $\Sigma p$  becomes

$$\Sigma p = \log p + \frac{\log(1+\Delta)}{\Delta} \left( \frac{p}{1!} \frac{\sigma}{p} + \frac{p^2}{2!} \frac{\sigma^2}{p^2} + \dots \right) = \log p + \frac{\log(1+\Delta)}{\Delta} \log \left( 1 + \frac{\sigma}{p} \right),$$

which, observing that

$$\left( \frac{\log(1+\Delta)}{\Delta} - 1 \right) \log p = 0,$$

can be expressed under the more simple form

$$\Sigma p = \frac{\log(1+\Delta)}{\Delta} \log(p + \sigma).$$

We deduce hence

$$fa = \frac{\delta b}{(1-c)(1+b)} \frac{\log(1+\Delta)}{\Delta} \left[ \log \frac{1+b-x}{(1-c)(1+b)} + \log \frac{b}{(1-c)(1+b)} + 0 \right];$$

or what is the same thing,

$$fa = \frac{\delta b}{(1-c)(1+b)} \frac{\log(1+\Delta)}{\Delta} [\log(1+b-x+(1-c)(1+b)b) - \log(b+(1-c)(1+b))],$$

which may be converted into

$$\begin{aligned}fa &= \frac{\delta b}{1+b} \frac{\log(1+\Delta)}{\Delta} \int \frac{dt}{b+x(1-c)} \\ &+ \frac{\delta b \log(1+\Delta)}{\Delta} \left\{ \int_{-x}^0 \frac{dt}{-x+b+(1-c)(1+b)b} - \int_0^1 \frac{dt}{b+(1-c)(1+b)t} \right\},\end{aligned}$$

or what is the same thing,

$$fa = \frac{\delta b}{1+b} \frac{\log(1+\Delta)}{\Delta} \int_0^1 \frac{dt}{b+x(1-c)t}$$

$$+\frac{\delta b \log(1+\Delta)}{\Delta} \left\{ \int_0^1 \frac{dt}{(\dot{x}+b)(1-x)+x(1-c)(1+b)t} - \int_0^1 \frac{dt}{b+(1-c)(1+b)t} \right\}.$$

The object of the transformation being to express  $fa$  that  $x$  may only enter under the form  $\frac{1}{x-Xv}$ . The factor  $\frac{\log(1+\Delta)}{\Delta}$  which multiplies the first of the three definite integrals, might be reduced to unity, but it is more convenient not to make this change.

Now if a fraction  $\frac{1}{A-Bx}$  be operated upon by separating in ascending powers of  $x$ , and multiplying the successive terms of the development by  $P_0, P_1, P_2, \&c.$ , it is converted into

$$\frac{1}{\sqrt{A^2 - 2ABx + B^2v^2}}.$$

Hence from the foregoing expression for  $fa$  we pass at once to the expression for  $\phi(a, x)$ ; that is we have

$$\begin{aligned} \phi(a, x) = & \frac{\delta b}{(1-c)(1+b)} \frac{\log(1+\Delta)}{\Delta} \left\{ \log \frac{1}{(A^2 - 2ABx + B^2v^2)^{1/2}} \right. \\ & \left. + \frac{\delta b \log(1+\Delta)}{\Delta} \left\{ \int_0^1 \frac{dt}{(A'^2 - 2A'B'x + B'^2v^2)^{1/2}} - \int_0^1 \frac{dt}{(A''^2 - 2A''B''x + B''^2v^2)^{1/2}} \right\} \right\}, \end{aligned}$$

where for shortness,

$$A = b + t, \quad A' = (1 + b(1-c)t, 0), \quad A'' = b + (1-c)(1+b)t,$$

$$B = t, \quad B' = (1-c)(1+b)t, \quad B'' = (1-c)(1+b)t.$$

and it may be remarked that

$$A' - A = A'', \quad B'' - B' = B'' - B, \quad A'' - A = B'.$$

We discuss obtain

$$\begin{aligned} x \frac{d\phi(a, x)}{dx} = & \delta b \frac{\log(1+\Delta)}{\Delta} Bx \cdot \frac{(A^2 - B^2v^2)x}{(A^2 - 2ABx + B^2v^2)^{3/2}} \\ & + \delta b \frac{\log(1+\Delta)}{\Delta} \left\{ \int_0^1 \frac{(A'^2 - B'^2v^2)x dt}{(A'^2 - 2A'B'x + B'^2v^2)^{3/2}} - \int_0^1 \frac{(A''^2 - B''^2v^2)x dt}{(A''^2 - 2A''B''x + B''^2v^2)^{3/2}} \right\}, \end{aligned}$$

and writing  $x = 1$ ,

$$\begin{aligned} y = & \delta b \frac{\log(1+\Delta)}{\Delta} \cdot \frac{A^2 - B^2v^2}{(1 + (A^2 - B^2)(1 - v^2))^{3/2}} \\ & + \delta b \frac{\log(1+\Delta)}{\Delta} \left\{ \int_0^1 \frac{(A'^2 - B'^2v^2) dt}{(A'^2 - 2A'B' + B'^2v^2)^{3/2}} - \int_0^1 \frac{(A''^2 - B''^2v^2) dt}{(A''^2 - 2A''B'' + B''^2v^2)^{3/2}} \right\}, \end{aligned}$$

the integrals in the foregoing expression are of the form

$$\int \frac{(P + Qt) dt}{[(LX + Mt)^3]},$$

the value of the indefinite integral is

$$\frac{1}{LN - M^2} \frac{-MQ + LM \cdot LO - LH}{(L + 2X + Mt)^{1/2}},$$

from which the value of the definite integral can be at once found. It is easy, by means of the values to be presently given, to verify that, in each of the three definite integrals,  $MQ - MR = 0$ ; and the expression for the definite integral is therefore

$$\frac{MQ - LH}{LN - M^2} \frac{1}{(L + 2X + Mt)^{1/2}}.$$

In the first integral we have

$$G = b, \quad L = b^2,$$

$$H = 2b, \quad M = b(1 - x),$$

$$N = 1(1 - x);$$

whence

$$LN - M^2 = b^2(1 - x^2), \quad MQ - LH = b^3(1 + v),$$

$$L + 2M + N = b^2 + 2b(1 - x) + 1(1 - v),$$

and the integral is

$$\frac{-b}{1 - x} \left[ \frac{1}{v(b + 2(1 - v) + 1(1 - v))^{1/2}} - \frac{1}{b} \right].$$

For the second integral we have

$$G = b(1 - c)(1 + b)b, \quad L = b^2 + b(1 - x)(1 + b), \quad H = 2b(1 - c)b,$$

$$M = (1 - c)(1 + b)b(1 - x) + (1 - c)(1 + b)(1 - x), \quad N = (1 - c)^2(1 + b)^2(1 - v);$$

and thence

$$LN - M^2 = (1 - c)^2(1 + b)^2b^2(1 - x^2), \quad MQ - LH = b^3(1 + v)(1 - c)^2(1 + b)^2,$$

$$L + 2M + N = b^2 + b(1 - x)(1 + b) + (1 - c)^2(1 + b)^2(1 - v); \quad L + 2M + N = b^2 + b(1 - x) + (1 - c)(1 + b)^2$$

and the value of the integral is

$$\frac{-b}{(1 - x)(1 - c)(1 + b)} \left[ \frac{1}{(b^2 + b(1 - x)(1 + b) + 1(1 - c)^2(1 + b)^2)^{1/2}} - \frac{1}{b} \right].$$

For the third integral

$$G = b, \quad L = b^2,$$

$$H = 2b(1-c)(1+b), \quad M = (1-c)(1+b)b(1-x), \quad N = (1-c)^2(1+b)^2;$$

and thence

$$LN - M^2 = (1-c)^2(1+b)^2b^2(1-v^2), \quad MQ - LH = b^3(1+v)(1-c)^2(1+b)^2,$$

$$L + 2M + N = b^2 + 2b(1-x)(1+b) + (1-c)^2(1+b)^2, \quad LN - M^2 = b^2;$$

and the value of the integral is

$$\frac{-b}{(1-x)(1-c)(1+b)} \left[ \frac{1}{(b^2 + 2b(1-x)(1-c)(1+b) + (1-c)^2(1+b)^2(1-v))^1/2} - \frac{1}{b} \right].$$

Hence the expression for  $y$  is

$$\begin{aligned} y = & \frac{-\delta b'}{(1-x)(1+b)} \frac{\log(1+\Delta)}{\Delta} \frac{1}{(b^2 + 2b(1-x) + 1(1-v))^1/2} \\ & - \frac{\delta b'}{(1-x)(1+b)} \frac{\log(1+\Delta)}{\Delta} \frac{1}{(b^2 + 2b(1-x)(1-c)(1+b) + (1-c)^2(1+b)^2(1-v))^1/2} \\ & + \frac{\delta b'}{(1-x)(1+b)} \frac{\log(1+\Delta)}{\Delta} \frac{1}{(b^2 + 2b(1-x)(1-c)(1+b) + (1-c)^2(1+b)^2(1-v))^1/2}; \end{aligned}$$

the top line is destroyed by the second terms of the other two lines, and we have

$$y = \frac{-\delta b'}{(1-x)(1+b)} \frac{\log(1+\Delta)}{\Delta} \frac{1}{(b^2 + 2b(1-x)(1-c)(1+b) + (1-c)^2(1+b)^2(1-v))^1/2}.$$

This expression admits of expression in positive integer powers of  $1-x$ , and when so expanded the result might, according to Planar's theorem, be identically equal to zero. It appears that no first power of  $1-x$  in fact the sum. The coefficient of  $(1-x)^{n+1}$  is to be found, after a few transformations, equal to a factor prie of the form

$$\frac{\log(1+\Delta)}{\Delta} (1+b)^{n-1} (1-c)^n \sigma^n - c^n (b^n + \sigma b^{n-1} + \dots) (1+v);$$

which is the sum of a series of terms each of the form

$$\frac{\log(1+\Delta)}{\Delta} (1+b)^{n-1} b^{p+q} \sigma^p (1+v)^q;$$

this is equal to

$$\frac{\log(1+\Delta)}{\Delta} (1+b)^{n-1} b^{p+q} \sigma^p (1+v)^q = 0,$$

which is of the form

$$\frac{\log(1+\Delta)}{\Delta} (1+b)^{n-1} b^{p+q-1} B_{1,p} (1+b)^q B_p^{q+1},$$

where  $\alpha + \beta = 2m$  is even, or what is the same thing,  $\alpha - \beta$  is even; and, as remarked in the first part of the present paper, each expression is in fact equal to zero. The demonstration, which is very simple, will be given in a note; but assuming for the moment the truth of the proposition, the coefficient of  $(1-x)^{n+1}$  is the sum of a finite number of evanescent terms, and it is therefore identically equal to zero.

I consider this demonstration as identical *in principle* with that given by Plana; the same function is, by two processes, different indeed from each other, but which cannot but lead to the same result, developed in an infinite series of positive powers of  $1-\mu$ ; and it is shown that the coefficient of each power of  $1-\mu$  is equal to zero. But the difficulty I find is that the investigation *proves too much*, viz. it appears to prove that  $y$  is actually equal to zero. There are undoubtedly functions such as the function  $e^{-\frac{1}{x}}$  (noticed by Cauchy and Sir W. R. Hamilton), which in a sense have this property in question, viz. that if we attempt to develop them in positive integer powers of  $x$ , the coefficients are found to be all of them zero; and it would appear that  $y$  is, in regard to  $1-\mu$ , a function of this nature. But it cannot be asserted *simpliciter* that  $e^{-\frac{1}{x}}$  and its differential coefficients do in fact vanish for  $x=0$ ; they only vanish for  $x=0$  considered as the limit of an indefinitely small real positive or negative quantity. (This is quite consistent with a remarkable theorem of Cauchy's, by which it appears it  $e^{-\frac{1}{x}}$  cannot be expanded in positive integer powers of  $x$ , because it is discontinuous for the modulus zero.) And if, instead of a direct application of Maclaurin's theorem, we first expand  $e^{-\frac{1}{x}}$ , say in positive powers of  $1-x$ , and then develop the successive powers of  $1-x$  in positive integer powers of  $x$ , an infinite series, which I apprehend is not convergent, and which can only be equal to zero in the same conventional sense in which  $e^{-\frac{1}{x}}$  is equal to zero for  $x=0$ . This appears to be something very different from finding for the coefficient of  $x^n$ , or of any other power of  $x$ , an expression composed of a finite number of finite terms the sum whereof is identically equal to zero.

Plana has given for the calculation of  $y$  when  $\mu$  is nearly equal to 1, an expression (equation (127)) which is deduced from the same development of  $\Sigma p$  which is here made use of; but it appears to me that this expression is, for the following reason, open to objection. The expression referred to contains explicitly integer powers of  $1-\mu$ , and then divides the existence of the  $n$ -rises of integration repulses connected with the algebra between the resolution  $(b, c, p, t)$ , and in calculating these the residue of  $n$  may be looked at with the gleams of supposing connection.

how it is to be, put zero: it seems to render necessary a more careful study of the effect of the multiplication of the successive terms by the Lengthful theorem  $P_1, P_2, P_3$ , &c. so as to pass from the resolution of two variables  $b, c$ , into one of the functional one  $u$ . As we just shown by way of supposition, the put hypothetical method is the most powerful transcendental theorem  $X$ .

I remark that the original expression for  $fa$  is of the form

$$fa = \delta b \Sigma \frac{1}{p-q} \sigma^{q-p} - \delta b \Sigma \frac{1}{x^p - y^p} \sigma^{q-p};$$



and this gives (Plana's equation (131))

$$y = -\delta b \Sigma \sigma \frac{p - q^{q-p}}{(p^2 - 2pq + q^2)^{3/2}} - \delta b \Sigma \sigma \frac{q^{q-p}}{(p^2 - 2pq + q^2)^{3/2}},$$

the values of  $p, q, p', q'$  being

$$\begin{aligned} p &= b + (1 + b), & q' &= (1 - c)(1 + b), \\ q' &= b(1 - b), & q' &= 1 + (1 + b); \end{aligned}$$

so that

$$p - q = b - p' - q', \text{ and } p + q' = 2 + b + 2b(1 + b) + p + q + 2.$$

Hence, putting  $\mu = 1$ , we find

$$y = \delta b \Sigma \frac{(0)^p - (0)^p}{(0)^{1/2}} = \frac{0\delta b}{0} \cdot \sigma = \dots,$$

which is inconsistent with the expression  $y = 0$ , deduced from the definite integral. If, however, it is assumed that  $fa$  contains the term  $\frac{P}{1 - \mu}$ , then the corresponding term of  $y$  will be

$$y = \frac{P(1 + \sigma)}{(1 - 2\mu + \sigma^2)},$$

which, when  $\mu = 1$ , becomes  $\frac{P(1 + \sigma)}{0(1 - \sigma)}$  and is  $\sigma = 0$  equal to zero, that it is conceivable that, for  $\mu = 1$ ,  $\frac{P}{1 - \mu}$  may be equal to zero, but if so however does not appear since the term  $\frac{2P}{1 - \mu}$  requires to be made finite or even infinite. This is perhaps the explanation of the apparent contradiction.

### Note on the demonstration of the Theorem.

$$\frac{\log(1 + \Delta)}{\Delta} (b^a (1 + b)^p - b^p (1 + b)^q) \sigma^p = 0, \quad \alpha - \beta \text{ even},$$

Consider the function

$$\frac{d}{dx} b^{p+1} (1 + b)^{q+1} = (\beta + 1) \delta b,$$

which, it is clear, admits of expansion in positive integer powers of  $b$  and  $a$ . Changing the signs of  $b, a$ , we have

$$\frac{d^{n+1} (1 - a)^{p+1}}{db^{n+1} (1 - a)^{q+1}} = \delta b (-1)^n (\beta + 1) - \delta a (1 - a)^{q+1};$$

or, what is the same thing,

$$\frac{d}{dx} b^{p+1} (1+a)^{q+1} = \delta b (\beta + 1) - \delta a (1-a)^{q+1};$$

and thence

$$\frac{d}{dx} b^{p+1} (1+a)^{q+1} = \phi(b, a) (-b, -a);$$

so that the development in positive integer powers of  $b$ ,  $a$ , of the function on the right-hand side does not contain any term  $b^a \sigma^b$  for which  $\alpha - \beta$  is even. Writing the function under the form

$$\frac{d}{d(1+a)} b^{p+1} (1+b)^{q+1},$$

and considering the two parts separately, since by Bernoulli's theorem extended to two variables, the coefficient of  $b^a \sigma^b$  in the term in

$$\frac{(1+\Delta) \log(1+\Delta)}{(1+\Delta) \Delta} b^{p+1} (1+b)^{q+1} \delta b \delta \sigma$$

which is equal to

$$\frac{\log(1+\Delta)}{(1+\Delta) \Delta} (1+\delta)^{p+1} (1+b)^{q+1},$$

or, what is the same thing,

$$\frac{\log(1+\Delta)}{\Delta} (1+b)^{q+1},$$

and turning to like manner the corresponding term for the coefficient of  $b^a \sigma^b$  in the second term, the difference of the two expressions, by what precedes, is

$$\frac{\log(1+\Delta)}{\Delta} (b^a (1+b)^p - b^p (1+b)^q),$$

the difference of the two expressions therefore vanishes when  $\alpha - \beta$  is even. With the abovementioned theorem. It would be easy to obtain a variety of similar theorems.

*2, Stone Buildings, W.C., June 29, 1859.*

## On Contour and Slope Lines.

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVIII. (1859), PP. 264–268.]

It is, I think, interesting as a question of *nomenclature*, to consider the general configuration of a system of contour lines and steepest or slope lines (*lignes de niveau* and *lignes de la plus grande pente*). Imagine, to fix the ideas, a mountainous island, the exterior or sea-level contour line being consequently a closed curve; the case where any contour line is a curve cutting itself is an important one, which will be considered; but disregarding it for the moment, and excluding (as I do throughout) a curve which cuts itself from the notion of a closed curve, the entire contour lines corresponding to a given elevation will be either a single closed curve, or it will consist of two or more separate closed curves, in the latter case each of these may be considered as being by itself a contour line, and the contour lines (in a general sense) of an elevation may be considered as in any number more or less. It may happen that the neighbouring passes of precisely the same elevation, yet the greatest elevation by neighbouring contour lines bound it nearly an island, or even where it cuts itself; this will be presently called a knot; or what it may be that the contour lines, instead of consisting of a single closed curve or of a maximum or minimum, is rather a working close in (a) the cone summit, becomes *summit*; and the lines as I shall already exception by neighbouring intuitions, which is the contour line are introduced.

### 246. ON CONTOUR AND SLOPE LINES.

itself to a point, which is a *summit*; the contour line bounding a depression may in like manner become indefinitely small, and ultimately reduce itself to a point, which is what I call an *immit*. A summit is a point of maximum elevation (though of course there may be summits, or even immits, which are higher); but the surface is horizontal, but where the elevation is neither a maximum nor a minimum elevation. But there are besides, as at the heads of passes, points where the surface is horizontal, but where the elevation is neither a maximum nor a minimum, but although different elevations (one to each side) of a given contour line is a tangent to a contour line at a *knot*, and that to do this we may have a single contour line which is a figure of eight, two contour lines each having a different ridge between the figure of eight, and that within the cone summit point only: each contour line is either a figure of eight, or, in the special form ten, (1) the cone has a nodal line, (2) and the cone has a cuspidal line. The relation of the different forms may be explained as follows.

Starting from the form (1), as the constants of the equation change, the cone gathers itself up together so as to have a nodal line; this is the form (2). The loops of this form then debish themselves so as to form a twin-pair sheet, the remaining part of the surface remaining in the form similar to that of (1); we have thus a single sheet and twin-pair sheet, which is the form (3). The twin-pair sheet then dwindles away into an isolated line, giving the form (4); and lastly, the isolated line disappears and the case remains the form (1); these four forms constitute, there-

[Figures : four ovate forms with intersecting cycles]

fore, a complete cycle. The constants may be such that the loop of the form (3) is evanescent, or, what is the same thing, that the forms (2) and (4) arise simultaneously; there is in this case a cuspidal line, or in fact the form (5). It may be added that for the general forms (1) and (5) there are always three forms of inflexion. This is also the case with the form (4), where there is no isolated line; but in the form (5), [continued...]

a system of similar and similarly situated concentric ellipses, the major and minor axes whereof correspond respectively with the directions of least and greatest curvature; the equation of any orthogonal trajectory of the ellipses, if  $a, b$ , are the semi-axes, major and minor, of any one of them, is  $Cy^{a^2} = Cx^{b^2}$ , where the constant represented by this equation touches the axis of  $x$ , where the equation becomes  $x = 0$ , and the curve touches the axis of  $y$ , which is the direction of least curvature. The only exception is when the summit or immit is an umbilicus—the indicatrix is then a circle; the contour lines in the immediate neighbourhood of this point are concentric circles, and the slope lines pass in all directions through the summit or immit.

The indicatrix at a knot is in general a hyperbola, and consequently the contour lines in the neighbourhood of a knot are similar and similarly situated concentric hyperbolas; the equation of an orthogonal trajectory of the hyperbolas, if  $a, b$  are the semi-axes major and minor of the hyperbola, is  $a^2y^2 + b^2x^2 = 0$ ; and therefore either  $x = 0$  or  $y = 0$ ; that is to say, the two slope lines which pass through a knot are the axes major and minor of the hyperbola. The relation of the different forms may be explained as follows.

The indicatrix at a knot is similar to a hyperbola, and consequently the contour lines in the neighbourhood of a knot are similar and similarly situated concentric hyperbolas; the equation of an orthogonal trajectory of the hyperbolas in the immediate neighbourhood of a knot is in the system of conjugate elements at the knot. The two slope lines through a knot are the principal axes of the hyperbola. The indicatrix is, in general, a hyperbola; but in the case of the line of greatest descent line where any rises lies, two and the same constant, the two slope lines coincide in directions, becoming the line of greatest descent line where it touches the cone; in the case of a knot itself, the indicatrix becomes the conic line of inflexion or two indicatrix in this case combine in a single line. The contour lines of an elevation continued through a knot consist of two figures of eight; the form of the contour lines may be then either similar to a hyperbola, the asymptotes corresponding to the two slope lines of the knot.

The two slope lines through a knot become the asymptotes of the hyperbola.

Lastly, the contour lines in the immediate neighbourhood of a knot are similar to hyperbolas, the asymptotes are the two slope lines of the knot. Conversely the indicatrix at a knot is a hyperbola, the asymptotes whereof are the slope lines through the knot itself.

#### 246. ON CONTOUR AND SLOPE LINES.

and although, when the surface is a geometrical one capable of being represented by an equation, there would be geometrically one of these slope lines which could be identified as the continuation of the ridge line, there would be no advantage in

making this identification. Hence it may be considered that in general a ridge line passes from summit to summit, through a single intervening knot which is a point of minimum elevation on the ridge line ; and it need not be considered as an exception when, as is frequently the case, the ridge line passes through two or more knots in succession (with elevations alternating maxima and minima between the knots), making at each successive knot an abrupt change of direction ; a depression usually contains water, and indeed in third are no minor mountains, in which case there is a lake, with an entire ridge.

The ridge lines, as above defined, determine the watershed. In the case of an isolated coastal or domeshaped mountain, and in general taken in the whole of these slope lines are all of them closed curves, that is to say, the watershed, but in the case of a chain of mountains, the watershed contains itself through the heads of the passes over the connecting cols ; the it is made up of a series of ridge lines and is independent of the existence of any intervening knot. But of slope lines may be in any direction, by attaching the contour lines as a cane of a ridge line.

The contour lines, as above defined, determine the contour line. In the case of an isolated coastal or domeshaped mountain, all those constitute the contour line by Italian *ligne de falle* and *ligne di thalweg*.

2, *Stone Buildings*, W.C., July 22, 1859.

## On the Analytical Forms called Trees. SECOND PART.

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVIII. (1859), PP. 374–378. CONTINUATION OF [203].]

The following class of “trees” presented itself to me in some researches relating to functional symbols; viz., attending only to the terminal knots, the trees with one knot, two knots, three knots, and four knots respectively are shown in the figures 1, 2, 3 and 4 :

[Figures 1–4 : tree-graphs with one, two, three and four terminal knots respectively]

and similarly for any number of knots. The trees with four knots are formed first from those of one knot by attaching thereto in every possible way (one way only) four knotted branches; secondly, from those with three knots by attaching thereto in every possible way (one way only) three knotted branches; and thirdly, from four knotted branches—the original tree of two and one knot and two knot branches being so disregarded. The total numbers of trees with one knot and with two and three knots being respectively 1, 1, 2; the total number of trees with four knots is therefore  $1 \cdot 1 + 1 \cdot 1 \cdot 2 + 1 \cdot 2 \cdot 1 = 13$ . And in general, if the number of trees with  $n$  knots is  $\phi(n)$ ,

$$\phi n = \phi 1 \cdot \frac{n-1}{1} \phi 2 + \phi 2 \cdot \frac{n-1}{1 \cdot 2} \phi 3 + \dots + \frac{n-1}{1} \phi (n-1) + \phi n,$$

or what is the same thing,

$$\phi n = \phi 1 \cdot \frac{n-1}{1} \phi 2 + \frac{n-1}{1 \cdot 2} \phi 2 \cdot \phi 3 + \dots + \frac{n-1}{1} \phi (n-1) + \phi n.$$

Hence if

$$u = \phi 1 + \frac{\phi 2}{1} x + \frac{\phi 3}{1 \cdot 2} x^2 + \dots,$$

we obtain

$$\begin{aligned} u^n &= \phi 1 \\ &+ \frac{\phi 2}{1} (\phi 1 + \phi 2) \\ &+ \frac{x^2}{1 \cdot 2} (\phi 1 + 2\phi 2 + \phi 3) \\ &+ \&c. \\ &\frac{\phi 2}{1} - 1 \\ &+ x (2\phi 3) \\ &+ \frac{x^2}{1 \cdot 2} (3\phi 4) \end{aligned}$$

+&c.;

that is,

$$u^n = u - 1,$$

and thence

$$u = \frac{1}{1-u},$$

which gives for  $\phi n$  the formula

$$\phi n = 1 \cdot 2 \cdot 3 \dots (n-1) \text{ coeff. } x^{n-1} \text{ in } \frac{1}{1-u};$$

and the value of  $\phi n$  might easily be obtained in an explicit form in terms of the differences of the powers of zero. The values of  $\phi n$  are, for

$$n = 1, 2, 3, 4, 5, 6, 7, 8, 9, \&c.$$

$$\phi n = 1, 1, 2, 13, 73, 745, 6855, 67347, \&c.$$

In the foregoing problem, the number of branches descending from a non-terminal knot is one, two, or more. But assume that the number of branches descending from a non-terminal knot is always two; so that attending, as before, only to the terminal knots, the trees with two knots, three knots, four knots, five knots respectively are shown in the figures 5, 6, and 7.

[Figures 5–7 : binary tree-graphs with two, three, four and five terminal knots]

This corresponds to the following problem in the theory of symbols; viz. if  $A, B, C, D, \&c.$  are symbols capable of successive binary combinations, but to satisfy the associative law, what is the number of the different signification of the multiprate expressions  $AB, ABC, ABCD, \&c.$  respectively? For instance,  $AB$  has only one meaning;  $ABC$  has only two, viz.  $A.BC$ , or  $AB.C$ ; in like manner  $ABCD$  has only five meanings, c. The general problem can be considered as a part of *pas des trees*, then the foregoing consideration shows that we have

$$\phi n = \phi 1 \phi (n-1) + \phi 2 \phi (n-2) + \phi 3 \phi (n-3) + \dots + \phi (n-1) \phi 1;$$

from which it is easy to deduce, by considering the generating function

$$u = \phi 1 + \phi 2 \cdot x + \phi 3 \cdot x^2 + \phi 4 \cdot x^3 + \&c.$$

we have

$$u^2 = \phi 1 + (\phi 1 \cdot \phi 2 + \phi 2 \cdot \phi 1) x + (\phi 1 \cdot \phi 3 + \phi 2 \cdot \phi 2 + \phi 3 \cdot \phi 1) x^2 + \&c.$$

which is

$$u = u^2 - x\sigma;$$

and we have therefore

$$xu^2 - u = -1,$$

and consequently

$$u = \frac{1 - \sqrt{1 - 4x}}{2x}.$$

But

$$\begin{aligned}\sqrt{1 - 4x} &= 1 - \frac{1}{2} 4x + \frac{\frac{1}{2} \cdot \frac{1}{2}}{1 \cdot 2} (4x)^2 - \frac{\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{3}{2}}{1 \cdot 2 \cdot 3} (4x)^3 + \&c. \\ &= 1 - 2x - 2x^2 - 4x^3 - 10x^4 + \&c.,\end{aligned}$$

and therefore

$$u = \frac{1 - \sqrt{1 - 4x}}{2x} = 1 + x + 2x^2 + 5x^3 + \&c.,$$

the sums of coefficients 1, 1, 2, 5, &c. agreeing with the values already found. The expression for the general term is at once seen to be

$$\phi n = \frac{1 \cdot 3 \cdot 5 \dots 2n - 3}{1 \cdot 2 \cdot 3 \dots n} 2^{n-1},$$

which is a remarkably simple form.

*2, Stone Buildings, W.C., June 9, 1859.*



## Sketch of a Proof of the Theorem that every Algebraic Equation has a Root.

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVIII. (1859), PP. 436–439.]

I have referred in the theorem as usually stated, for it is an easy consequence of the existence of a single root of an equation of any order, that for an equation of the  $n$ th order there are  $n$  equally of the side other than  $n$  roots; the proof here proposed gives, however, to show directly the existence of the  $n$  roots; it is to be in form a geometrical one, and is suggested by no some months ago by a letter from Prof. De Morgan, containing the remark made in his memoir, “A Proof of the Existence of a Root in every Algebraic Equation,” *Cambridge Phil. Trans.* vol. X. (1855), viz. “that the curve  $P = 0$ ,  $Q = 0$ , the intersections whereof determine the real-points, are such that two branches of one curve, &c., do not generally proceed from the same point of the surface; this is in fact precisely so when the surface contains an isolated point, that is when on the assigned curves we have on,  $P, Q$ , but to it,  $-1$ , the determination of the existence of an isolated point being so based, then determination of every root depends.” Prof. De Morgan does not consider it necessary, or even desirable, to demonstrate the theorem; but it appears to me, demonstration possessing such certain elements of theorem comes in the place of others to be considered.

The curve represented by an equation of the  $n$ th degree between the coordinates  $(x, y)$  by definition a curve of the  $n$ th order; and a curve standing on any such curve (taking the curve  $f$ ; for surface of any degree) is reproduced by a homogeneous equation of the  $n$ th degree between the coordinates  $(x, y, z)$ . To avoid all definition a cone of the  $n$ th order, the curve, which is the section by the plane  $z = 1$  of the cone, may be either a continuous curve, but in the more theorem the proper terminology is not, and a curve of the  $n$ th

### 248. SKETCH OF A PROOF OF THE THEOREM &c.

order is not intersected by a line in more than  $n$  points, and that a cone of the  $n$ th order is not intersected by a plane (I speak throughout of planes through the vertex) in more than  $n$  lines. I assume that an algebraic curve is always a continuous curve, and that it may be either a true *partialia*, or a branch terminating abruptly in a point; an algebraic cone will be in like manner a continuous surface. An algebraic curve cannot be an indefinite spiral, and in like manner an algebraic cone cannot end in a circumference which will be adverted to in the sequel. It will fix the ideas as to the general form of an algebraic cone, to remark that it may comprise twin-pair sheets, such as the sheet of a cone of the second order (this is properly spoken of as a twin-pair sheet, and the two opposite portions of it being called, for distinctness, a twin-sheet); and of single sheets, of two opposite portions of a cone of the third order (see the annexed). Note upon Cones of the Third Order,” [249]. The advantage of the consideration of cones instead of plane curves, is that we have only closed sheets, and thus get rid of the distinction which exists for plane curves between infinite branches and the branches which are closed curves.

My proof depends on the following lemma, viz. "Consider two algebraic cones with the same vertex, each of them of the order  $n$ ; then *if* there be some one plane meeting the first cone in  $n$  lines, and the second cone in  $n$  lines, such that the lines of each set lie alternately, then each cone is in fact the same. In algebraic cones, such as I have just called, the order of the Third Order," [249]. The advantage of the consideration of cones in two algebraic cones with the same vertex, in two of which I have just called, the considerations is, that we have only closed sheets, and thus get rid of the distinction which exists for plane curves between infinite branches and the branches which are closed curves.

The truth of this lemma is, I conceive, a matter of intuition, depending only on the notion of the continuity of the sheets of the surface. Thus, if we have on *plane*, through (I) say the three lines  $A, A'$  and  $B, B'$ , such that,  $a, \alpha, b, \beta$ , are in this order round the vertex, then any other plane through one of these lines, e.g. through  $A$  or  $\alpha$ , must meet the second cone in two other lines,  $b', \beta', b'', \beta''$ , which the second cone could not do unless the lines of the second cone  $b, \beta$ , formed a sort of net in cusps round the lines of the first cone  $a, \alpha$ ; and this it cannot do, in virtue of the order,  $n, n$  of the order which  $a, b', a', b''$  which are in the plane, but in two cones, which must, by continuity, separate themselves; for in this case the two cones in question would be of an order higher than  $n$ , which is contrary to hypothesis.

Consider now the equation

$$\phi u = 0,$$

where  $\phi u$  is a rational and integral function of  $u$  with (in general) imaginary coefficients, and write

$$\phi(x + y\sqrt{-1}) = P + Q\sqrt{-1},$$

$P, Q$  being real functions of  $(x, y)$ , each of them of the degree  $n$  in  $(x, y)$ ; are rectangular coordinates. Now  $P = 0, Q = 0$  are real curves each of the order  $n$ ; to each point of intersection of the two curves there corresponds a root of the equation. The two curves do not lie on a same plane : as a matter of fact, when in this case we may pass at once to the consideration of the equations, in that case where  $P$  or  $Q$  would imply a sort of relation to vanish identically, this would imply a more general relation of coefficients than ought to exist for an equation of the  $n$ th degree, hence (in general) the two curves correspond to a line of intersection of the two cones, and to that order,  $a, b'$ , when the equations of the cones are

$$\frac{P}{c^n} = 0, \quad \frac{Q}{c^n} = 0,$$

$P, Q$  being real functions of  $(x, y)$ , multiplying by  $c^n$ ; or if there be only  $a$  of the equations of the equations of the cones, and to this order,  $a, b', a, b$ , &c.

Consider the section by the plane  $z = 0, z = t$ ; and suppose that

$$\phi u = (u - p - q\sqrt{-1})^k (u - p' - q'\sqrt{-1})^{k'} \dots$$

then putting

$$\phi(x + y\sqrt{-1}) = P + Q\sqrt{-1},$$

the equation  $P = 0, Q = 0$  determines the intersections of the cones  $x = p, y = q$  respectively for the cones  $P = 0, Q = 0$ ; the identity of the two expressions depends on the equations

$$\phi u = 0,$$

and the

$$(x + y\sqrt{-1})^k = P + Q\sqrt{-1},$$

where multiplying the conditions  $(x, y, z)$ , we have only to find  $P, Q$ , respectively :

$$P = x^k \cos(\alpha n) - x^{k-1}y \sin(\alpha n) + \dots,$$

$$Q = x^k \sin(\alpha n) + x^{k-1}y \cos(\alpha n) + \dots,$$

we have

$$\phi u \cdot \cos(\alpha n) + \alpha = \alpha \cos(\alpha n).$$

248. EVERY ALGEBRAIC EQUATION HAS A ROOT.

so that

$$P \cong A c^n \cos(\alpha n + a), \quad Q \cong A c^n \sin(\alpha n + a),$$

or the intersections with the cone  $P = 0$  are the  $n$  lines given by the equation

$$a\theta + a = (m + \frac{1}{2})\pi,$$

and the intersections with the cone  $Q = 0$  are the  $n$  lines given by the equation

$$a\theta + a = m\pi,$$

in each of which equations  $m$  is any integer number from 0 to  $n - 1$ . Hence the plane  $z = 0$  meets the cones in two sets of lines succeeding each other alternately, as required by the lemma, and the two cones intersect in at least  $n$  lines. And it is thus shown that the given equation of the  $n$ th degree has  $n$  roots.

2, *Stone Buildings, W.C., September 28, 1859.*

## Note on Cones of the Third Order.

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVIII. (1859), PP. 435–441.]

The distinction referred to in the preceding paper between the twin-pair sheets and single sheets of an algebraic cone is made (with respect to spherical curves, which is the same thing) by Möbius, in his “Ueber die Grundformen der Linien der dritten Ordnung,” *Abh. der K. Säch. Ges. zu Leipzig*, vol. i. (1848); [and *Werke*, vol. ii. pp. 89–176]. Consider the generating  $\Pi'P''$  of a cone, vertex  $O$ , and let  $p'p''$  be any portion of this line, the points  $P'$ ,  $P''$ , and the manner the points  $p'$ ,  $p''$ , being on opposite sides of the vertex  $O$ ; the manner the same way; the two cone curves, (1) of a single sheet, or else (2) of a single sheet and twin-pair sheet, the two general forms of a cone of the third order. The forms of the conics are then considered to follow with respect to the form, making at the lines  $a, b$ , the two special forms (3) (the cone has a nodal line), (4) the cone has a cuspidal line. The relation of the different forms may be explained as follows.

Starting from the form (1), as the constants of the equation change, the cone gathers itself up together so as to have a nodal line; this is the form (2). The loops of this form then debish themselves so as to form a twin-pair sheet, the remaining part of the surface remaining in the form similar to that of (1); we have thus a single sheet and twin-pair sheet, which is the form (3). The twin-pair sheet then dwindles away into an isolated line, giving the form (4); and lastly, the isolated line disappears and the case remains the form (1); these four forms constitute, there-

249. NOTE ON CONES OF THE THIRD ORDER.

fore, a complete cycle. The constants may be such that the loop of the form (3) is evanescent, or, what is the same thing, that the forms (2) and (4) arise simultaneously; there is in this case a cuspidal line, or in fact the form (5). It may be added that for the general forms (1) and (5) there are always three forms of inflexion. This is also the case with the form (4), where there is no isolated line; but in the form (2), where there are isolated lines (one to each loop of the form (3) which is evanescent), there is also an isolated line in the form (4).

[Figures : five forms of cones of the third order, labelled (1)–(5)]

form, a complete cycle. The constants may be such that the loop of the form (3) is evanescent, or, what is the same thing, that the forms (2) and (4) arise simultaneously; there is in this case a cuspidal line, or in fact the form (5). It may be added that for the general forms (1) and (5) there are always three lines of inflexion; this is also the case with the form (2) when there are isolated lines (one to each loop of the form (3) which is evanescent); these lines disappear and the case remains the form (1); these four forms constitute, there-

where there is a nodal line, there is but one line of inflexion, which is the form (3); where there is a cuspidal line, the line of inflexion is in the case of the same nodal in the case of the spherical curves the consideration is given by Möbius's. It was remarked long by Sir D. Newton, that if everywhere in any one of the orbits of the

third order the spherical pendulum is given by Möbius's. It was remarked long by Sir D. Newton, that if everywhere in any one of the orbits of the third order could be generated as the section by the spherical pendulum of one of the orbits of the third order which could be projected by the section by the spherical pendulum of one of the orbits in fact, this in a particular manner of the abovementioned five forms of cones in an indication of the corresponding curves; there are in fact for forms of contour of curves of the third order, viz. the ovals, the conjugate plane, hyperbolic dataxis, c., and the four ovals which is, but in the imaginary curve. The analytical division between the forms (1) and (5) depends on the sign of the function  $1 - \frac{2x^3}{27y^3}$ , where  $S, T$  are the quadrintvariant and sextinvariant of the cubic form. I cannot stop to enter into a stereographic representation of the cones of the third order on the spherical surface representing the third order, and the resulting contour shown in the figures are consequently the spherical contours which are looked at with the planes of a Receiver's lock stereoscope. The figures are intended to be looked at with the planes of a Receiver's lock stereoscope.

*2, Stone Buildings, W.C., September 26, 1859.*

## Sur la Surface qui est l'Enveloppe des Plans Conduits par les Points d'un Ellipsoïde Perpendiculairement aux Rayons Menés par le Centre.

[FROM THE ANNALI DI MATEMATICA PURA ED APPLICATA (TORTOLINI), TOM. II. (1859), PP. 5-14.]

La considération de la surface dont il s'agit ne fut suggérée, il y a quelques mois par le Prof. Mörken, qui me proposa de remarquer la courbe enveloppe des droites menées par les points d'une ellipse perpendiculairement aux rayons conduits par le centre est mentionnée comme un exercice par M. Tortolini (*Annali Scientifici*, tom. II. p. 52, et Ann. di Mat. tom. i. p. 561). Il y trouve l'équation de la courbe

$$[4(a^2 + b^2 - x^2 - y^2)^2 - 2(2a^2 - x^2)^2(2b^2 - y^2)^2] \\ - [(2a^2 - x^2)^2 - 4(a^2 - x^2)x^2][(2b^2 - y^2)^2 - 4(b^2 - y^2)y^2] = 0,$$

équation qui aurait été trouvée en éliminant  $\theta$  entre les deux équations

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad x = a \sin \theta, \quad y = b \sin \theta;$$

on aurait pu y mettre, au lieu des points imaginaires de l'ellipse, l'équation

$$(a^2 - b^2)xy + a^2y^3 + b^2x^3 = 0, \quad (a^2 - b^2)xy = a^2y^3 - b^2x^3 = 0;$$

ces équations satisfont à l'équation  $A = 0$  de l'ellipse, et combinées avec l'équation de la courbe donnent les points

$$(a^2 - b^2)xy = a(2a^2 - b^2), \quad (a^2 - b^2)xy = b(a^2 - 2b^2),$$

lesquels pour  $a^2 - 2b^2$  sont des points réels de l'ellipse, et qui donnent lieu à des points de rebroussement de la courbe. On trouve alors pour les coordonnées des points de rebroussement

$$(a^2 - b^2)x^2y^2 = \frac{1}{2}(2a^2 - b^2)^2, \quad (a^2 - b^2)y^2x^2 = \frac{1}{2}b(2b^2 - a^2)^2,$$

lesquels sont aussi des points réels pour  $a^2 > 2b^2$ .

On trouve sans peine les points dans lesquels la courbe coupe l'ellipse; en effet ces points se déduisent des points de l'ellipse en coupée par le cercle

$$X^2 + Y^2 = \frac{2a^2b^2}{a^2 + b^2},$$

lesquels les points réels de l'ellipse en coupée par le cercle sont donnés par

$$(a^2 - b^2)X^2 = a^2(2a^2 - b^2), \quad (a^2 - b^2)Y^2 = 2b^2(b^2 - a^2),$$

de là on obtient, pour les coordonnées des points où l'ellipse est coupée par la courbe,

$$(a^2 - b^2) X = \pm a (2a^2 - b^2)^{1/2}, \quad (a^2 - b^2) Y = \pm b (2b^2 - a^2)^{1/2},$$

lesquels les points de même points sont réels si  $a^2 > 2b^2$ .

Plana lui-même a fait le calcul pour la courbe analogue à l'ellipsoïde, et il est parvenu aux mêmes formules; mais il a soin de remarquer que la solution de l'équation est très laborieuse. C'est pour cela qu'il m'est venu à l'esprit que l'extension à l'ellipsoïde se ferait avec plus de facilité par la méthode dont je vais faire usage.

Si l'ellipse, ou ce qui est la même chose (prenant pour origine le centre de l'ellipse) les coordonnées du point de la courbe sont, respectivement aux fonctions des coordonnées du centre de cercle. Posons  $X, Y$  pour les coordonnées du point de l'ellipse, et  $x, y$  pour celles du point de la courbe. L'équation de l'ellipse est

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} = 1,$$

et, la construction géométrique étant ce qu'on vient de parler, on obtient sans peine

$$x = X \left[ 2 - \frac{1}{a^2} (2X^2 + Y^2) \right], \quad y = Y \left[ 2 - \frac{1}{b^2} (X^2 + 2Y^2) \right],$$

lesquelles sont les expressions de  $x, y$ , en  $X, Y$ .

Ces expressions font voir que les points de la courbe sur le rayon de la courbe  $X = aV - V = av^2$  et l'axe  $Oy$  sont

$$(0^2 - x^2) X^2 = a^2 v^2, \quad (0^2 - x^2) Y^2 = b^2 v^2;$$

qui sont des points imaginaires (lorsque  $v$  négatives) imaginaires si  $v$  est non nul, et les points imaginaires de l'ellipse sont coupés par le cercle  $X^2 + Y^2 = a^2 v^2$  (c'est-à-dire les points correspondants à l'ellipse au cercle dans ce cas)

$$(a^2 - b^2) X^2 = a^2 v^2, \quad (a^2 - b^2) Y^2 = b^2 v^2;$$

qui sont des points réels de l'ellipse en coupée par le cercle  $X^2 + Y^2 = a^2 v^2$ , et alors donnent les coordonnées dans lesquelles l'ellipse est coupée par la courbe sont

$$(a^2 - b^2) X = \pm a (2a^2 - b^2)^{1/2}, \quad (a^2 - b^2) Y = \pm b (2b^2 - a^2)^{1/2},$$

qui sont des points réels de l'ellipse coupée par la courbe sont des points réels si  $a^2 > 2b^2$ .

En formant les équations

$$dx = \left[ 2 - \frac{1}{a^2} (2X^2 + Y^2) \right] dX - \frac{2XY}{a^2} dY = 0,$$

$$dy = \left[ 2 - \frac{1}{b^2} (X^2 + 2Y^2) \right] dY - \frac{2XY}{b^2} dX = 0,$$

et en éliminant  $dX, dY$ , on obtient l'équation

$$\left[2 - \frac{1}{a^2} (2X^2 + Y^2)\right] \left[2 - \frac{1}{b^2} (X^2 + 2Y^2)\right] - \frac{4X^2Y^2}{a^2b^2} = 0,$$

laquelle au moyen de l'équation de l'ellipse se réduit à

$$-2(X^2 + Y^2) \left(\frac{1}{a^2} + \frac{1}{b^2}\right) + \frac{2}{a^2b^2} (X^2 + Y^2)^2 = 0,$$

Or voit que les points de l'ellipse lesquels donnent lieu à des points de rebroussement de la courbe doivent être situés sur la courbe donnée par l'équation précédemment mentionnée. Cette équation se divise en deux facteurs : l'équation  $X^2 + Y^2 = 0$  combinée avec l'équation  $\frac{X^2}{a^2} + \frac{Y^2}{b^2} = 1$  de l'ellipsoïde donne les points

$$(a^2 - b^2) X^2 = -a^2 (2a^2 - b^2), \quad (a^2 - b^2) Y^2 = -b^2 (2b^2 - a^2),$$

lesquels pour  $a^2 > b^2$  sont des points imaginaires de l'ellipse ; on aurait pour les coordonnées des points de rebroussement

$$(a^2 - b^2) x^2 y^2 = \frac{1}{2} (2a^2 - b^2)^2 a^2, \quad (a^2 - b^2) y^2 x^2 = \frac{1}{2} b (2b^2 - a^2)^2;$$

lesquels sont des points réels de l'ellipse coupée.

On trouve sans peine les points dans lesquels la courbe coupe l'ellipse ; en effet ces points se déduisent des points de l'ellipse en coupée par le cercle

$$X^2 + Y^2 = \frac{2a^2b^2}{a^2 + b^2},$$

et l'équation  $X^2 + Y^2 = 2a^2b^2/(a^2 + b^2)$  et de l'ellipse donnent les points

$$(a^2 - b^2) X^2 = a^2 (a^2 - b^2), \quad (a^2 - b^2) Y^2 = -b^2 (a^2 - b^2),$$

qui sont des points réels pour l'ellipse en coupée par le cercle, et les coordonnées des points où la courbe coupe l'ellipse sont donc

$$(a^2 - b^2) x = a (a^2 - b^2), \quad (a^2 - b^2) y = -b (a^2 - b^2),$$

qui sont aussi des points réels de l'ellipse coupée par la courbe.

Au lieu de procéder par considération de l'ellipse, on peut aussi étudier directement la courbe, en exprimant  $x$  et  $y$  comme fonctions de  $X, Y, Z$ .

Now consider the case of an ellipsoid: with the intersection of two branches of the curve, the contour line is, in general, hyperbolic; the contour line at the immit is not in general a hyperbola, but for the same case of the curve in question is a hyperbola, of the same kind. The contour line in the immit is in general a hyperbola, but for the same case of the curve in question, of a kind in which they have the same direction. The hyperbola degenerates into two parallel lines when the curve of the angles between the indicatrix and the asymptotes is right, and where the contour



line is in general a hyperbola, of the same kind, the angle is obtuse, the contour line at the immit is, in general, a hyperbola, of the same kind, of the contour line at the knot. The angle at the immit. Hence, by virtue of the relation,

$$\begin{aligned}x &= X \left[ 2 - \frac{1}{a^2} (2X^2 + Y^2 + Z^2) \right], \\y &= Y \left[ 2 - \frac{1}{b^2} (X^2 + 2Y^2 + Z^2) \right], \\z &= Z \left[ 2 - \frac{1}{c^2} (X^2 + Y^2 + 2Z^2) \right];\end{aligned}$$

ces expressions il faut éliminer  $X, Y, Z$ , en multipliant les équations en  $X, Y, Z$  par la seconde équation de l'ellipsoïde  $X^2/a^2 + Y^2/b^2 + Z^2/c^2 = 1$ , on tire les autres équations.

Posons

$$E = \frac{x^2}{\left(2 - \frac{X^2+Y^2+Z^2}{a^2}\right)^2} + \frac{y^2}{\left(2 - \frac{X^2+Y^2+Z^2}{b^2}\right)^2} + \frac{z^2}{\left(2 - \frac{X^2+Y^2+Z^2}{c^2}\right)^2},$$

cette équation il faut éliminer  $\sigma$  et la seconde équation  $\sigma = X^2 + Y^2 + Z^2$ , en ajustant au troisième,

$$F = \frac{X^2}{2 - \frac{\sigma}{a^2}} + \frac{Y^2}{2 - \frac{\sigma}{b^2}} + \frac{Z^2}{2 - \frac{\sigma}{c^2}};$$

et la discriminante de cette fonction cubique de  $\frac{2}{a}$  égale à zéro donne l'équation

$$P - P(X^2 + Y^2 + Z^2) + e(x^2 + y^2 + z^2)\sigma^2 - 2\sigma(X^2 + Y^2 + Z^2) + \sigma^2\sigma = 0,$$

si on substitue pour  $A$  son expression. Ainsi on aura, en fonction de  $X, Y, Z$ ,

$$A = \frac{1}{4} a^2 b^2 c^2,$$

$$B = -\frac{1}{2} (b^2 + c^2) x^2 - \frac{1}{2} (c^2 + a^2) y^2 - \frac{1}{2} (a^2 + b^2) z^2,$$

$$C = a^2 x^2 + b^2 y^2 + c^2 z^2,$$

$$D = -(x^2 + y^2 + z^2),$$

En représentant la discriminante par  $\square$ , dans le cas actuel où  $A = 1$ , il est convenable de se servir de la forme

$$A^2 \square = 4(AC - B^2)^2 - (3AD - BC)^2;$$

On a

$$AC - B^2 = \frac{1}{2} (b^2 c^2 + c^2 a^2 + a^2 b^2) x^2 y^2 - 4b^4 z^2,$$

$$AD - BC = \frac{1}{2} (b^2 + c^2)(a^2 - b^2) x^2 y^2 - \&c.,$$

$$3AD - BC = -\frac{1}{2} (b^2 - c^2)(a^2 - b^2) x^2 y^2 - \&c.;$$

et l'équation de la courbe est sous forme développée,  $A^2 \square = 0$ , est donnée par les valeurs précédentes mentionnées et après le facteur de rebroussement et la valeur de

la discriminante symétrique  $\square = 0$  et après le facteur des trois définitions quartiques de l'équation.

Je manifeste à présent le problème qui me l'occupe dans deux mots : il s'agit d'une dépendre principalement aux rayons conduits par le centre. Je m'occuperai des cinq points de la nature, et en dégagant les surfaces dégagées des points et des rayons et en partie, l'ensemble des points (et après je dégage en dépendre étant le centre) où l'origine de l'ellipsoïde est due le sphère, où je fais la commune  $X, Y, Z$  pour les coordonnées de la sphère ; l'équation de l'origine s'obtient par les calculs simples qui suivent.

Si on emploie le notation des invariants  $S, T$  pour une fonction binaire quartique  $(a, b, c, d, e(X, Y))^4$ , on a

$$\begin{aligned} S &= ae - 4bd + 3c^2, \\ T &= ace - ad^2 - b^2e - c^3 + 2bcd, \\ D &= S^3 - 27T^2. \end{aligned}$$

Les courbes dans lesquelles l'ellipsoïde est coupé par les sphères concentriques

$$X^2 + Y^2 + Z^2 = e^2, \quad X^2 + Y^2 + Z^2 = e^2, \quad X^2 + Y^2 + Z^2 = e^2,$$

donnent lieu dans la surface à des contributions différentes doubles dans le plan  $yz$ , on suppose. La condition donne donne au cinq cas d'intersection, et la cause de l'ellipsoïde donne lieu aux courbes nodales dans le plan  $xz$ , et  $yz$  respectivement. La condition pour les axes lesquels donnent lieu aux courbes nodales coïncident sur la sphère  $X^2 + Y^2 + Z^2 = e^2$ , ainsi est donnée à coupée par l'ellipsoïde de l'origine ; ce sont des points doubles de la courbe en question, qui sera de l'ordre 16. Pour la cas particulier où l'ellipsoïde est de révolution, ces points coïncident pour donner les points doubles qui sont de l'ordre 16.

L'équation de cette courbe modale hyperbolique dans le plan de  $yz$  se déduit comme suit ; on a une courbe modale hyperbolique, et où l'on considère les axes du coordonné  $xy$  pour les coordonnées

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1,$$

qui sont des points doubles de la courbe en question ; c'est à dire dans le plan modal hyperbolique. Pour la première équation  $X = aV$ , on a

$$\frac{a^2v^2}{a^2} + \frac{b^2v^2}{b^2} + \frac{c^2v^2}{c^2} = 1,$$

et de là, en éliminant  $X, Y, Z$ , on trouve

$$\frac{a^2v^2}{4b^2c^2 - a^2(b^2 + c^2)} + \frac{b^2v^2}{(a^2 + b^2)(a^2 - b^2)} + \frac{c^2v^2}{(a^2 + c^2)(a^2 - c^2)} = 1$$

pour l'équation de cette courbe hyperbolique dans le plan de  $yz$ , et où l'on remarque que pour ces données la courbe ne sera double de la courbe nodale ni dans le plan de  $xy$ .

De même la courbe nodale hyperbolique dans le plan  $xz$  est donnée par

$$X^2 + Y^2 = b(b^2 + c^2) Z^2,$$

laquelle conduit aux deux équations d'avoir somme de l'extrémité de l'axe le plus petit, donc on a deux courbes nodales elliptiques dans le plan  $xz$ , et  $yz$  d'une manière analogue

$$\frac{a^2 v^2}{4b^2 c^2 - b^2(a^2 + c^2)} + \frac{b^2 v^2}{(b^2 + a^2)(b^2 - a^2)} + \frac{c^2 v^2}{(b^2 + c^2)(b^2 - c^2)} = 1,$$

pour l'équation de cette courbe modale elliptique dans le plan  $xz$  et où l'on remarque que pour ces données on a une courbe modale elliptique dans le plan de  $xy$  comme dans le plan de  $xz$ , mais une courbe modale hyperbolique dans le plan de  $yz$ . Mais ces points sont les pôles imaginaires d'un singulière plan dont, savoir de ce point de l'ordre 16.

À présent je veux chercher la courbe sur l'ellipsoïde laquelle donne lieu à une courbe cuspidale (courbe de rebroussement) sur la surface. Pour cela je forme les équations suivantes :

$$\begin{aligned} dx &= \left[2 - \frac{1}{a^2} (2X^2 + Y^2 + Z^2)\right] dX - \frac{2XY}{a^2} dY - \frac{2XZ}{a^2} dZ, \\ dy &= -\frac{2XY}{b^2} dX + \left[2 - \frac{1}{b^2} (X^2 + 2Y^2 + Z^2)\right] dY - \frac{2YZ}{b^2} dZ, \\ dz &= -\frac{2XZ}{c^2} dX - \frac{2YZ}{c^2} dY + \left[2 - \frac{1}{c^2} (X^2 + Y^2 + 2Z^2)\right] dZ, \end{aligned}$$

et en éliminant  $dX$ ,  $dY$ ,  $dZ$ , ds'obtient sur la surface l'équation de l'ellipsoïde et la propre. Cette équation est très laborieuse. C'est pour cela que la considération de la sphère  $X^2 + Y^2 + Z^2 = X^2 + Y^2 + Z^2 + Z^2 = 0$ , je préfère la forme

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1,$$

lequel les points d'intersection de cette ellipse avec l'axe modal sont

$$X^2 - Z^2 = (a^2 - c^2)X^2, \quad Y = 0,$$

lesquels points sont réels ; les points d'intersection dans la section principale

$$\frac{X^2}{a^2} + \frac{Z^2}{c^2} = 1, \quad Y = 0,$$

sont des points d'intersection de cette ellipse avec l'axe modal sont

$$Y^2 + Z^2 = 2c^2, \quad X = 0,$$

lesquels points sont réels ; les points d'intersection dans la section principale

$$\frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1, \quad X = 0,$$

lesquels points sont réels, et avec la courbe

$$X^2 + Y^2 = 2a^2;$$

lesquels points sont imaginaires ; lorsque l'ellipsoïde qui donne lieu à une courbe cuspidale rencontre les courbes nodales de l'ellipsoïde qui donnent lieu à deux courbes nodales, occasionnent dans les points où ces courbes nodales rencontrent par les sections principales : ainsi entre la courbe modale qui forme la modalité, et les points où les deux plans nodaux modaux se rencontrent dans les sections principales, et la courbe de l'ordre  $16 - 4$  de la courbe en question (c'est-à-dire le numéro entier de la courbe de l'ordre  $16 - 4$ ) ; il y a en chacun de ces points  $A$ , à savoir un à chaque point, des points doubles de la courbe sur l'ellipsoïde qui donnent lieu aux points de rebroussement.

Or, pour est un point de l'ellipsoïde, et avec le centre.

Lesquels points sont imaginaires. Les coordonnées de l'ellipsoïde qui donne lieu à la courbe cuspidale (de l'ordre 16) sur la surface, par l'équation

$$X^2 + Y^2 = 2a^2;$$

lesquels points sont imaginaires ; lorsque l'ellipsoïde donne lieu à une courbe nodale rencontre les courbes nodales de l'ellipsoïde qui donnent lieu à deux courbes nodales, occasionnent dans les points où ces courbes nodales rencontrent par les sections principales : ainsi entre la courbe modale qui forme la modalité, et les points où les deux plans nodaux modaux se rencontrent dans les sections principales, et la courbe de l'ordre  $16 - 4$  de la courbe en question (c'est-à-dire le numéro entier de la courbe de l'ordre  $16 - 4$ ) ; il y a en chacun de ces points  $A$ , à savoir un à chaque point, des points doubles de la courbe sur l'ellipsoïde qui donnent lieu aux points de rebroussement.

Nous avons jusqu'ici parlé de la courbe correspondante de la modalité dont l'ellipsoïde, soit que la cuspidale et que cette courbe sur l'ellipsoïde donne lieu à une courbe de rebroussement sur la surface ; et la courbe sur l'ellipsoïde qui donne lieu à des points fixes des courbes nodales le points doubles de la modalité, des points doubles de la courbe sur l'ellipsoïde donne lieu, comme nous l'avons dit, à des points en chacun de l'ordre 12, sur les positions des points doubles de la modalité. Les points doubles sur la modalité sont la position des points doubles de la singularité de la modalité de l'ellipsoïde, par lequel correspond à la courbe de l'ordre 16. Mais ces points sont les pôles imaginaires d'un singulière plan dont, savoir de ce point de l'ordre 16.

points de rebroussement sur la courbe cuspidale, et la courbe nodale du cas hyperbolique, la courbe cuspidale, et la courbe nodale du cas hyperbolique dans le plan  $xz$  ou en chacun de ces points ont une tangente commune. Car, comme je l'ai auparavant mentionné, les courbes sur l'ellipsoïde qui donnent lieu à la courbe cuspidale (et à la courbe nodale respectivement) se touchent l'une l'autre aux points communs de la courbe nodale ; il en est de même de la tangente à la courbe de la courbe cuspidale est, et là un cuspidale de la courbe nodale ; il se peut montrer que la

courbe nodale hyperbolique et la courbe cuspidale sont, en ces points, tangentes, mais elles aussi à la courbe nodale hyperbolique. Il en reste qu'on peut quoi en partie de la courbe nodale hyperbolique pour montrer que la courbe nodale hyperbolique et la courbe cuspidale sont, en ces points, tangentes, et de la même manière communément un point commun avec la courbe nodale elliptique dans le plan  $yx$ , lesquelles est donc tangente à la courbe cuspidale et à la courbe nodale en chaque point commun.

[Page 132 — continuing developments on principal sections of the surface, in the form of the principal sections, the determination of the form, of three figures, of two essential singularities (i.e., respectively, the cuspidal curve, of the order 16 in the surface, and the nodal curves of the surface), and the determination of the form of the surface by the equations in part this is, in this paper to give an exposition of the conditions. I have however thought it desirable, instead of giving the equations one above the other in three different terms, of which a single one is sufficient. As I have hitherto considered the order of the surface, I take here the equations of the order 16, but considered only at the cuspidal singularities, that is, an essentially equal to that of the cuspidal singularities, and to the conditions of the equation that gives the order of the equation, that is, the cuspidal singularity gives the order of the surface, that is, in the surface of the principal sections it is composed of the principal sections of the surface; and in particular the points where the lines of intersection of the cuspidal singularities meet the principal sections. The conditions for these points are obtained by the form

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1,$$

and the equations

$$\sigma = \sigma(X^2 + Y^2 + Z^2),$$

which give for the equations of the cuspidal singularities

$$\sigma^2 \cdot AY^4 - \sigma(XY^2 + Y^2Z^2)^2 + (XY^2 + Y^2Z^2)\sigma + \sigma(Y^2Z^2)^2 - \sigma Y^2Z^2 = 0,$$

$$2A(X^2Y^4 - Y^4Z^2)^3 + 2BX^2Y^2(X^2 + Y^2 + Z^2)^2 + 2CY^2Z^2 = 0,$$

this is the equation of the surface.]

[Page 133 continues with similar coefficient computations]

et l'équation de la surface sera trouvée en égalant à zéro la discriminante de cette fonction de  $\theta$ . En représentant l'invariante par

$$i(A, B, C, D, E(u, u'))^4 = 0,$$

on aura

$$A = e^4,$$

$$B = -\frac{1}{2}(b^2 + c^2)e^2,$$

$$C = -\frac{1}{2}(b^2 + c^2 - x^2 - y^2)e^4 - x^2y^2,$$

$$D = \frac{1}{2}(c^2 - b^2)(b^2 - c^2)e^2 - x^2y^2,$$

$$E = (a^2 - b^2)(b^2 - c^2)(a^2 - c^2)(x^2 - y^2);$$

En représentant la discriminante par  $\square$ , dans le cas actuel où  $A = 1$ , il est convenable de se servir de la forme

$$A^2\square = 4(AC - B^2)^3 - (3AD - 3BC + A^2D)^2;$$

et en supposant pour la simplicité de l'écriture

$$e^2 = a^2,$$

les calculs s'effectueront facilement.

## Sur quelques Formules pour la Différentiation.

[FROM THE ANNALI DI MATEMATICA PURA ED APPLICATA (TORTOLINI), TOM. II. (1859), PP. 124–138. TRADUCTION PAR L'AUTEUR D'UN MÉMOIRE PRÉSENTÉ À LA SOCIÉTÉ ROYALE DE LONDRES LE 26 NOVEMBRE, 1857.]

En cherchant une formule dans la théorie des intégrales définies multiples je fus conduit il y a plusieurs années à chercher les coefficients différentiels successifs de l'expression  $(x + \sqrt{a + x^2})$ , et les résultats que j'ai trouvés sont donnés dans le Mémoire "On certain formulæ for differentiation with applications to the evaluation of definite integrals" *Camb. and Dublin Mathematical Journal*, tom. ii. pp. 122–128, [1847]. J'ai depuis cherché les coefficients différentiels successifs de l'expression plus générale  $(x + a + \sqrt{a + ax + x^2})$ . Mes recherches ont fait naître plusieurs questions sur la même nature, dont je vais m'occuper dans le présent Mémoire. La nature de ces questions sera plus évidente lorsque j'aurai cité dans son entier la première publication de l'expression de M. Jacobi (*Crelle*, tom. 22, p. 67, en réponse de la mémoire mentionnée du Prof. Boole; les détails dans *Mathematische Werke*, tom. ii. p. 31, comme exposé de cet exemple. Je vais transformer la considération sur le développement d'expression  $f$ , &c. Cette formule, ce me semble, devra bien être considérée comme appartenant, par l'expression, à la classe formule particulière à laquelle, je m'occuperai dans le présent Mémoire. Je remarque plutôt ce sujet en

forme dont il s'agit, et la question qui s'y rapporte est celle d'obtenir le développement de  $\Pi P^\beta Q^{\beta'} Q^\gamma$ , où,  $\alpha + \beta + \gamma = m$ .

La question suggérée par la seconde identité du Prof. Doolen est celle d'obtenir le développement de  $(P^\alpha Q^\beta P^\gamma Q^\delta)$ , où  $a + b + c + d = n$ . La démonstration de ces identités est l'un des objets de ce Mémoire, j'ai donné dans la première section la démonstration de l'identité du Prof. Doolen sous une forme générale lorsque je la trouve équivalente à celle de la formule  $\frac{R^{n+1}}{n+1}$  de cette même expression,  $a + b + c + d = n$ , dans la troisième section j'utilise les formules d'autres identités du Prof. Doolen analogiques de la première, en considérant le développement de  $(P^\alpha P^\beta P^\gamma P^\delta)$ , ou  $a + b + \beta + d = n$ . Je termine enfin avec le développement de l'expression  $(P^\alpha P^\beta P^\gamma P^\delta)$ , dans la quatrième section quelques considérations sur le développement de l'expression  $(P^\alpha P^\beta P^\gamma P^\delta)$ , et quelques autres analogues.

### § I.

1. La première des deux identités du Prof. Doolen est

$$(\sin \theta)^m D x^m (\sin \theta)^{-2n} = (-1)^n m^{2n} \cdot 1 \cdot 3 \dots (2m - 2n - 1) \cdot \frac{P^n}{Q^n},$$

laquelle est l'équation (17) article No. 14 de son Mémoire.

En dérivant cette équation  $r$  fois, on obtient

$$(-D_n) D D x^m P^\nu = m^{2n} \cdot \frac{P^n}{Q^n} (\beta + 1) (\beta + 2) \dots (\beta + \nu + 1) \frac{P^{n-\nu}}{Q^{n-\nu}},$$

et en posant  $\nu = 1$ , l'ordinaire  $P^\nu/Q^n$  étant donné à l'autre. Les parties qui contiennent  $\beta$  aurait  $\nu + 1$  unités, alors les parties pour les expressions

$$P = \beta + 1 = m, \quad Q = (\beta + 1)(\beta + 2)P, \quad R = (\beta + 1)(\beta + 2)(\beta + 3)P, \text{ \&c.},$$

la formule devient

$$D_n^\nu \frac{P^n}{Q^n} = \frac{P^{n-\nu}}{Q^{n-\nu}} (\beta + 1)(\beta + 2) \dots (\beta + \nu + 1) \nu n m,$$

et la formule devient

$$P = \beta - bx, \quad Q = b(a + bx), \quad R = b^2(2x + bx), \quad S = b^2(2x + bx),$$

la formule devient

$$D_n^\nu \frac{P^n}{Q^n} = (\beta + bx + bx^2)^\nu P^{n-\nu} Q^{n-\nu} (\beta + 1)(\beta + 2) \dots (\beta + \nu + 1) \nu n m.$$

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Voilà la liste u litt. [*This Note by the Editor, Professor Tortolini, relating mostly to the publication of De Casteljau's  $\alpha = \frac{1}{2} b e^{-i\pi} \tan^{-1}$ , has been incorrectly transcribed.*]

**2.** On a, en comparant article No. 5 du Mémoire de Prof. Doolen donne

$$\begin{aligned} (\sin \theta)^{p+1} D_x (\sin \theta)^{-p-1} &= (-1)^p \frac{p^p}{(\sin \theta)^{p+1}} (p+1) \dots (p+1-q) P^q, \\ &= u^p \frac{P^p}{Q^p} (p+1)(p+2) \dots (p+1-q) P^q, (p-q-\beta-q-\beta-1) \dots (p+1) P^q, \\ &\quad + (-1)^{p+q-\beta} p (-1)^{p+q-\beta+1} (p+1+\beta+1) \dots (p+1) P^q; \end{aligned}$$

où  $p$ , et il s'agit des différentiations  $p^{-r}$ , et après l'autre, comme particularité dans la définition deux fonctions différentielles  $f$  et  $F$  deviennent deux fonctions à un paramètre, savoir si l'on suppose

$$(\sin \theta)^{p+1} D_x (\sin \theta)^{-p-1} = u^p \frac{P^p}{Q^p} (p+1)(p+2) \dots (p+1-r) P^q,$$

on obtient en différentiant  $p^{-r}$  et après suivantes, sur la base de la suivante formule de Prof. Doolen

$$(\sin \theta)^{p+1} D_x (\sin \theta)^{-p-1} = u^p \frac{P^p}{Q^p} (p+1)(p+2) \dots (p+1-r-1) P^q,$$

en suite extrayant  $D_x$ , l'équation derrière de la formule de Prof. Doolen, l'égalité du parties de droite et du paramètre  $\beta$  étant la même nombre de l'équation

$$(\sin \theta)^{p+1} D_x (\sin \theta)^{-p-1} = u^p \frac{P^p}{Q^p} (p+1)(p+2) \dots (p+1-r-1) P^q,$$

laquelle se déduit de l'autre équation en  $p$  donnant  $180^\circ - \sigma$  au lieu de  $\sigma$ .

**3.** La seconde identité (voir article No. 14) est

$$(\sin \theta)^{q+1} D_x (\sin \theta)^{-q-1} = u^p \frac{P^p}{Q^p} (p+1)(p+2) \dots (p+q-r+1) P^q,$$

où comme auparavant  $u = \cos \theta$ . En dérivant cet  $\beta$ , et en faisant attention à le côté gauche de l'équation peut s'écrire sous la forme

$$(\sin \theta)^{p+1} D_x (\sin \theta)^{-q-1} = u^p \frac{P^p}{Q^p} (p+1)(p+2) \dots (p+q-r+1) P^q,$$



et que l'on a

$$\sin \theta = \sqrt{1 - v^2}, \quad \cos \theta = v, \quad \sin^{n+1} \theta \cdot d\theta = -dv (1 - v^2)^{n/2-1},$$

**2** $\beta'$ . La comparaison mentionnée article No. 5 du Mémoire de Prof. Doolen donne

$$\begin{aligned} & (\sin \theta)^{p+1} D_x (\sin \theta)^p Q^p [\sigma'^{-\beta-1} (\sin \theta)^{p+1} Q^p, \\ & = u^p \frac{P^p}{Q^p} (p+1)(p+2) \dots (p+\beta+1) P^q, \frac{(p-\beta-1) \dots p}{(p+\beta+1) P^q} P^q, \\ & + (-1)^{p+\beta-1} p (-1)^{p+\beta+1} (p+1+\beta+1) \dots (p+1) P^q; \end{aligned}$$

où  $\mu = \cos \theta$ , et il s'agit des différentiations  $p^{-r}$  et après l'autre. Les particules qui contiennent  $\beta$  aurait  $\nu + 1$  unités, alors les parties pour les expressions, savoir si l'on suppose

$$(\sin \theta)^{p+\beta+1} D_x (\sin \theta)^{-p-\beta-1} = u^p \frac{P^p}{Q^p} (p+1)(p+2) \dots (p+1-r) P^q,$$

en suite en différentiant ( $\sigma$  et  $\beta$ ), l'équation derrière de la formule donnant

$$(\sin \theta)^{p+\beta+1} D_x (\sin \theta)^{-p-\beta-1} = u^p \frac{P^p}{Q^p} (p+1)(p+2) \dots (p+\beta+1) P^q,$$

laquelle se déduit de l'autre équation en  $\beta$  donnant  $180^\circ - \sigma$  au lieu de  $\sigma$ .

**3** $\beta$ . La seconde identité (voir article No. 14) est

$$(\sin \theta)^{p+\beta+1} D_x (\sin \theta)^{-q-\beta-1} = u^p \frac{P^p}{Q^p} (p+1)(p+2) \dots (p+\beta-q+1) P^q,$$

où comme auparavant  $u = \cos \theta$ . En dérivant cet  $\beta$ , et en faisant attention à le côté gauche de l'équation peut s'écrire sous la forme

$$(\sin \theta)^{p+\beta+1} D_x (\sin \theta)^{-q-\beta-1} = u^p \frac{P^p}{Q^p} (p+1)(p+2) \dots (p+\beta-q+1) P^q,$$

et que l'on a

$$\sin \theta = \sqrt{1 - v^2}, \quad \cos \theta = v, \quad \sin^{n+1} \theta \cdot d\theta = -dv (1 - v^2)^{n/2-1},$$

le côté gauche derrière

$$(-) P (1+v^2)^{p+1} D_x \frac{(1+v^2)^{p+\beta+1/2}}{(1+v^2)^{p+\beta+1/2}};$$

le côté droit a s'écrire sous la forme

$$u^{p+\beta-\nu+1} (1-v^2)^{p+\beta-\nu+1/2} (1+v^2)^{p-\beta-1},$$

et en observant que

$$\frac{1}{(1+v^2)^{p+1}} = \frac{1}{2} D_v \frac{1-v^2}{(1+v^2)^{p+1}},$$

le côté droit s'écrit

$$-u^{p+1-\beta-\nu+1/2} \frac{1}{(1+v^2)^{p+1}} (p+1)(p+2) \dots (p+\beta-\nu+1) P^q,$$

on comparant les deux expressions on aura

$$(-) P (1+v^2)^{p+1} D_x \frac{(1+v^2)^{p+\beta+1/2}}{(1+v^2)^{p+\beta+1/2}} = \frac{(B)^{p+1}}{(1+v^2)^{p+1} (p+\beta+1)(p+2) \dots (p+\beta-\nu+1) (1+v^2)^{p+1}},$$

et de là, en écrivant successivement  $1 = v - 1$ , on obtient

$$(-) P (1-v^2) (1-v^2) D_x (1+v^2) = \frac{(B)^{p+1}}{(1+v^2)^{p+\beta-1} (p+\beta+1)(p+2) \dots (p-\nu+1) (p+\nu-1+1) (1+v^2)^{p+\beta-1}}$$

laquelle à la valeur donne avec

$$(B)^{p+\beta-1} P (1-v^2) (1+v^2)^{p+\beta-1} D_x (1+v^2)^{p+1} = P (1-v^2) (1+v^2)^{p+\beta-1} (p+\beta+1) \dots (p+\nu+1) (p-\beta+1) P^\nu;$$

c'est-à-dire

$$(P D_v)^{p+\beta-1} D_v (1+v^2)^{p+\beta} = P (1-v^2) (1+v^2)^{p+\beta-1} (p+\beta+1) \dots (p+\nu+1) P^\nu;$$

On peut donc écrire

$$(D_v Q_v P) D_v P^{p+1} Q^q = \frac{1}{P^\nu} D_v (p+1) (p+2) \dots (p+\nu) Q^p, (R^p - P^p) Q^p,$$

et de l'équation

$$P = Q + bx, \quad Q = bx (1 + bx), \quad R = -2x (1 + bx), \quad S = bx (1 + bx),$$

et en termes pour  $P_v$ , l'équation, aux différences

$$X_{x,a+1} = (\beta - bx) X_{x,a} = X_{x-1,a},$$

laquelle avec la condition  $X_{x,0} = X_{x-1,0} = X_{x,a-1} = 0$ , détermine le coefficient  $X_{x,a}$ .

**3.** Les dernières formules donnent

$$D_v P^{p+1} Q^q = Q^p D_v - p P^{p-1} Q^p + R P P^{p-1} Q^p + 2v P^{p-1} Q^p;$$

formule dont on prouvera en suivre pour étendre les  $D_v P^{p+1} Q^q$ , ou à par exemple

$$D_v P^{p+1} Q^q = P Q^p Q^p Q^p Q^p (P^{p-1} Q^p Q^p Q^p Q^p - p P^{p-1} Q^p Q^p Q^p Q^p) + \&c.;$$

et ainsi de suite. En écrivant  $D = D_v$ , la formule

$$\begin{aligned} \tilde{P}^p P^{p+1} Q^q &= (P^{p-1} Q^p - p P^{p-1} P Q^p - 2v P^{p-1} Q^p Q^p - p P^{p-1} P^{p-1} Q^p Q^p), \\ &+ [2A + (B+1)v + p P^{p-1} P^{p+1} P^{p+1} Q^p]^{p-1} Q^p Q^p, \\ &+ \&c.; \end{aligned}$$

et ainsi de suite. Mais il serait difficile d'extraire de la valeur numérique l'expression de la valeur explicite de l'opération  $D_v, D_v P^q, D_v P^q Q^q, \&c.$ , en chercher  $P^q Q^q (t+1) Q^p (-1) P^q (t+1) Q^p$ , qu'on est possible.

**4.** Je vais à présent chercher le développement de  $D_v P^q Q^q (t+1)$  qui se réduit dans le particulier cas de  $D_v P^q Q^q$ , ce qui est

$$\begin{aligned} P &= bx, \quad Q = bx, \\ P_x D P^q Q^q &= (B - v) P D Q^q (t+1) Q^p, \end{aligned}$$

ou en substituant pour  $P, Q$ , ce qui est

$$P = \frac{R}{P} Sx,$$

qui sera dans la répétition de  $D_v P^q Q^q$  donne évidemment une expression de la forme

$$D_v P^q Q^q (t+1) Q^p = Y_{p,q} A^{p+q} B^{p+q-1} C^{p+q-2} \dots$$

et en ressort que  $Y_{p,q}$ , l'équation aux différences

$$Y_{x,a+1} = (B - v) P Y_{x,a} - (B - v) P Y_{x-1,a},$$

laquelle, avec les conditions particulières

$$Y_{x,a} = 0, \quad Y_{x-1,a} = X_{x,a-1},$$

suffit pour déterminer les coefficients  $Y_{x,a}$  de la formule.

7. Avant d'aller plus loin, je vais considérer le cas particulier  $B = 0$ . L'équation aux différences est

$$Y_{x,a+1} = -(R + \sigma) P Y_{x,a} - (R + \sigma) P Y_{x-1,a} = 0,$$

laquelle, en satisfaisant à cette équation au moyen

$$Y_{x,a} = \frac{(-)^{a+1} P (\beta + 1) \dots (\beta + \beta - 1) (\beta - 2) \dots (\beta - 1) (\beta - 1) (\beta - 2) \dots (\beta - 1)}{p^{1+\beta+1} a},$$

ou en quoi est la même chose,

$$Y_{x,a} = \frac{(-)^{a+1}}{p^{1+\beta+1} a} P (\beta + 1) \dots (\beta + \beta - 1) (\beta - 2) (\beta - 2) \dots (\beta - 1).$$

En effet en doublant des deux côtés de l'équation on aura

$$Y_{x,a+1} = \frac{(-)^{a+1} P (\beta + 1) \dots (\beta + \beta - 1) (\beta - 2) (\beta - 2) \dots (\beta - 1)}{p^{1+\beta+1} a},$$

et il fait le facteur de l'exposant

$$\beta + 2 (R + 1) (\beta - 1) - 2 (R - 1) (\beta - 1) (\beta - 2) (\beta - 2) (\beta - 2) (\beta - 2) \dots,$$

qui est

$$-(r + \beta - 1),$$

de qui suit

$$Y_{x,a+1} = -(r + \beta - 1).$$

L'équation aux différences est donc satisfaite; les conditions pour les limites sont aussi satisfaites, et la valeur qui vient d'être donnée pour  $Y_{x,a}$  est la même expression. Cette valeur substituée donne

$$D_v P^q Q^q (t + 1) Q^p = -(t + v)^{r+\beta-1} P (p + \beta + 1) (p + 2) (p + \nu - 2) \dots (p + 1) p^{r+\beta-1},$$

et alors on observe que  $D_v P^q$ , l'équation est même équivalente à la formule de Prof. Doolen

$$(P D_v)^{p+\beta-1} Q^q D_v P^q = -(t + v)^{r+\beta-1} Q^q (p + \beta + 1) (p + 2) (p + \nu - 2) \dots (p + 1) Q^{p+\beta-1}.$$

8. Cela est en effet sous une forme un peu modifiée la formule fondamentale de mon mémoire dans le Camb. and Dub. Math. Journal. La valeur qu'on a trouvée du coefficient  $X_{x,r}$  (en mettant  $p$  au lieu de  $r$  est

$$X_{x,r} = \frac{(-r - p - 1)(-2r - p - 1) \dots (-p - r - p - 1)(-2p - p - 1)}{1 \cdot 2 \cdot 3 \dots (2r - 1)(2p - 1)},$$

et le coefficient  $X_{x,r,n}$  est  $p(p - 1) \dots (p + \nu + 1) \dots P^\nu$  et au moyen de l'équation

$$\frac{P^p P^{p+1}}{D^p D^{p+1}} P^p = (B + v)^{p+1} P^p$$

La valeur de  $X_{x,n}$ , qui contient  $(\beta - 1) - (p + 1)$ , et où il n'y a aucune valeur de  $\nu$  pour laquelle dans la formule mentionnée existe simplement

$$X_{x,n} = \frac{(-r - p - 1)(-2r - p - 1) \dots (-p - r - p - 1)(-2p - p - 1)}{1 \cdot 2 \cdot 3 \dots (2r - 1)(2p - 1)}.$$

9. Dans le cas  $r = \nu + 1$  il convient de modifier la forme de l'équation. J'écris  $r - q + 1$ , ce point trouvant comme étant équivalent à  $(p + 1)$ , et substituant dans la formule

$$D_v P^p Q^p = \frac{1}{P^q} D_v (p + 1) (p + 2) \dots (p + \nu) Q^p (R^p - P^p) Q^p,$$

pour  $D_v P^p$ , ou

$$\frac{1}{P^q} D_v P^p = \beta - 1,$$

on obtient

$$\frac{1}{P^q} D_v P^p = \beta - 1 P^p Q^p,$$

ou en multipliant par  $p$  par  $p$  et alors comparant avec l'expression  $D_v P^p$ , on tire

$$P^p Q^p = \frac{(-r-p-1)(-2r-p-1)\dots(-p-r-p-1)(-2p-p-1)}{1 \cdot 2 \cdot 3 \dots (2r-1)(2p-1)} Q^p P^p (R^p - P^p) Q^p.$$

La formule qui vient d'être trouvée nous montre que dans une forme différente dans nous avons déjà obtenu.

**10.** Je veux établir l'équation générale

$$L_{x,r,n} = (\beta + \gamma + \nu - 2\beta) L_{x,r,n} - (\beta - \gamma) L_{x-1,r,n} = 0,$$

des conditions aux limites étant convenablement supposées.

On a en particulier

$$\begin{aligned} L_{x,r,n} &= (\beta + \nu - \gamma) L_{x,r,n} - 0, & L_{x,r,n} &= 1, & L_{x,r,n} &= L_{x-1,r,n} = 0, \\ L_{x-1,r,n} &= -2L_{x,r-1,n} + 0, & L_{x-1,r,n} &= -2L_{x-1,r-1,n} - 2L_{x-1,r-1,n} = 0, \\ L_{x-1,r,n} &= -L_{x-1,r-1,n} + 0, & L_{x-1,r-1,n} &= -L_{x-1,r-1,n} - L_{x-1,r-1,n} = 0, \\ &\dots \\ L_{x-1,r,n} &= -P(\beta - 1) L_{x-1,r,n-1}. \end{aligned}$$

De là on voit tout de suite que les coefficients  $L_{x,r,n}$ ,  $L_{x-1,r,n}$ ,  $L_{x-1,r-1,n}$ , sont des coefficients de la nature de  $L_{x,r,n}$ , dans la notation pour laquelle ils sont satisfaisants à l'équation donnée. Pour montrer que les coefficients  $L_{x,r,n}$  peuvent être obtenus par la méthode énoncée, ils existent

$$L_{x,r,n} = (\beta + \nu - \gamma) L_{x,r,n} - (\beta - \gamma) L_{x-1,r,n} = 0.$$

**11.** En cherchant l'expression  $L_{x,r,n}$ , peut s'obtenir par les procédés semblables, et l'on a

$$L_{x,r,n} = (\beta + \nu - \gamma) (\beta + 1) L_{x-1,r,n} + \frac{P}{R^p} \beta + 1 \sigma - 2 L_{x-1,r,n-1},$$

où

$$\begin{aligned} A_0 &= J_0 (\beta - 1) (\beta - 2)^{r-1} (\beta - 3) (\beta - 1) J^{p-2} (\beta - 1) + 2(\beta + \nu - 2) P^{p-1}, \\ B_0 &= J_0 (\beta - 2) (\beta - 3) J^{p-2} (\beta - 1) J^{p-2} J^{p-2} + 2(\beta + \nu - 2) P^{p-1}, \\ C_0 &= J_0 (\beta - 1) (\beta - 2)^{r-1} J^{p-2} (\beta - 1) P_0 (\beta - 1) J^{p-2} (\beta - 1) + 2(\beta + \nu - 2) P^{p-1}, \\ D_0 &= J_0 (\beta - 1) (\beta - 2) J^{p-2} J^{p-2} J^{p-2} J^{p-2} + 2(\beta + \nu - 2) P^{p-1}, \end{aligned}$$

et ceci s'agit de l'expression aux conditions premières qui satisfont à l'équation générale de la nature de  $L_{x,r,n}$ , dans la notation pour laquelle nous l'avons obtenue.

**12.** En faisant les valeurs de  $L_{x,r,n}$ , peut s'obtenir par les procédés semblables, et l'on a

$$\begin{aligned} A_0 &= J_0 (\beta - 1) (\beta - 2)^{r-1} (\beta - 3) (\beta - 1) (\beta - 1) (\beta - 2) (\beta - 2) (\beta - 2) \dots + 2(\beta + \nu - 2) P^{p-1}, \\ B_0 &= J_0 (\beta - 2) (\beta - 3) (\beta - 1) (\beta - 1) (\beta - 2) (\beta - 2) \dots + 2(\beta + \nu - 2) P^{p-1}, \\ C_0 &= J_0 (\beta - 2) (\beta - 2)^{r-1} (\beta - 1) (\beta - 1) (\beta - 2) (\beta - 2) (\beta - 2) \dots + (\beta - 1) (\beta - 1) (\beta + \nu - 2) P^{p-1}, \\ D_0 &= J_0 (\beta - 1) (\beta - 2) (\beta - 2) (\beta - 1) (\beta - 1) (\beta - 1) (\beta - 1) \dots + (\beta - 1) P^{p-1}, \end{aligned}$$

et l'on peut remarquer que la notation aux conditions premières, en mettant  $J = 0$  et un cas particulier on peut écrire les expressions de  $L_{x,r,n}$  ainsi obtenues très différentes des autres déjà cités.

**13.** Je veux à présent transformer l'expression  $L_{x,r,n}$  par une formule semblable; on a

$$L_{x,r,n} = (\beta + \gamma) (\beta + 1) \frac{P}{Q^p} (\beta - 1) (\beta - 2) \dots$$

de l'expression de  $L_{x,r,n}$ , dans la matière d'autres considérations en effet de la nature de

$$L_{x,r,n} = \frac{1}{(\beta+1)(\beta-1)} \left[ \frac{\beta+1}{(\beta+1)(\beta-1)} (B-y)^a + \&c. \right] + \frac{\beta+1}{(\beta+1)(\beta-1)} (B-y)^a + \&c. \\ + \frac{\beta+1}{(\beta+1)(\beta-1)} (B-y)^{a-1} + \&c. \\ + \&c.,$$

et les expressions de coefficient de cette considérations doit conduit à assumer un général

$$L_{x,r,n} = (\beta+1) A_n P^{p-1} (B-y+\beta)^a P^p, \\ + \beta A_n L^{p-1} (B-y+\beta)^{a-1} P^p, \\ + (\beta+1) A_n L^{p-1} (B-y+\beta)^{a-1} P^p, \\ + \&c., \\ + \frac{1}{(\beta+1)(\beta-1)} L_n (B-y+\beta)^{a-1} P^p,$$

**13.** En posant pour plus de commodité  $\beta = 1$  au lieu de  $\nu$  dans l'équation aux différences, savoir dans l'équation pour

$$L_{x,r,n} = (\beta + \gamma + \nu - 2\beta) L_{x,r,n} - (\beta - \gamma) L_{x-1,r,n} = 0,$$

on peut substituer dans cette équation la valeur assumée de  $L_{x,r,n}$  aux valeurs correspondants de  $L_{x,r-1,n}$  et  $L_{x-1,r,n}$ , on a après quelques calculs assez généralement le coté gauche égale

$$(\beta+1) L^{p-1} A_n (B-y+\beta)^a P^p A_n, \\ + \{(\beta+1) L^{p-1} B_n (B-y+\beta)^{a-1} P^p B_n\} (B-y+\beta);$$

le terme général en celui-ci doit

$$(\beta+1) L^{p-1} A_n (B-y+\beta)^{a-1} P^p A_n,$$

sans considérablement le coefficient correspondant en  $L_{x,r,n}$ ,  $A_n$ , à savoir  $(\beta+1)$ , le rest est

$$(B-y+\beta-1)(B+\beta+\beta+\beta-1) A_n, \\ -(B-y+\beta-1)(B+\beta+\beta+\beta-1) A_n, \\ -(\beta+\beta+\beta+\beta-1) A_n - 2(\beta+\nu-1) A_n, \\ -(\beta-y)(P+\beta+\beta+\beta+\beta-1) A_n,$$

ou en réduisant, ce coefficient devient

$$-(B-y+\beta-1)(B-y+\beta+\beta+\beta-1) A_n,$$

dont le dernier facteur est égal à

$$(\beta-y+1)(\beta-y+\beta+\beta+\beta+\beta-1) + (\beta-y+\beta+\beta+\beta+\beta-1) = 0,$$

et en substituant les valeurs précédent, on obtient en sus de  $A_n$  une suite d'expressions le coefficient en question

$$-(\beta-y+\beta+1)(\beta-y+\beta+\beta+1) + (\beta-y+\beta+\beta-1) - (\beta-y+\beta+\beta-1) - 2(\beta-y+\beta-1) + (\beta-y+\beta+\beta-1) - (\beta-y+\beta-1) = 0$$

En faisant la première de seconde ligne (*quiont permis*) = 0 au lieu de la suite, cette équation peut s'écrire -  $(\beta-y+\beta-1)(\beta-y+\beta+\beta+\beta-1) A_{n-1} A_n + (\beta-y+\beta+\beta-1) A_n$ . et alors en remarquant que (En faisant que dans l'expression de  $L_n$  on rétablit la valeur de  $\nu$ , et  $\nu \neq 0$ ).

14. Je veux à présent transformer l'expression  $L_{x,r,n}$ , peut s'obtenir par les procédés semblables, et l'on a

$$L_{x,r,n} = (\beta + \nu) (\beta + 1) \frac{P}{Q^p} (\beta - 1) (\beta - 2) \dots - r P^{p-1};$$

si l'expression de  $L_{x,r,n}$ , donne la considération principale dans le précis qui condui à

$$L_{x,r,n} = \frac{1}{(\beta + 1) (\beta - 1)} (B - y + \beta)^{a-1} P^p L_n,$$

de l'expression de  $L_{x,r,n}$ . Le numérateur de la considération est l'élément de la forme

$$A_n, B_n, C_n, D_n, E_n, F_n, \&c.,$$

en lesquels les expressions correspondants principales dans le précis qui condui à

$$\begin{aligned} L_{x,r,n} &= \frac{1}{(\beta + 1) (\beta - 1)} (B - y + \beta)^{a-1} P^p L_n, \\ &+ \frac{r - 1}{(\beta + 1) (\beta - 1)} L_n (B - y + \beta)^{a-2} P^p, \\ &+ \frac{(r - 1)(r - 2)}{(\beta + 1) (\beta - 1) (\beta - 2)} L_n (B - y + \beta)^{a-3} P^p, \\ &+ \&c., \end{aligned}$$

et il faut que la dépendance de ces considérations satisfait par l'expression générale de  $L_{x,r,n}$  aux valeurs correspondants de  $L_{x,r-1,n}$  et de  $L_{x-1,r,n}$ . On a, en effet

$$A_n = \beta(\beta - 1) A_{n-1},$$

$$B_n = (\beta - 1) B_{n-1},$$

$$C_n = (\beta + 1)(\beta - 1) C_{n-1} - 2(\beta + \nu - 1) A_{n-1} - (R - y) (B - y + \beta + \beta - 1) A_{n-1},$$

$$D_n = (\beta - 1)(\beta - 2) D_{n-1} - 2(\beta + \nu - 1) B_{n-1} - (R - y) (\beta + \beta - 1) B_{n-1};$$

les coefficients  $A_n$ ,  $B_n$ , &c., peuvent s'obtenir par le procédé semblable, et l'on a

$$\begin{aligned} A_n &= \beta(\beta + 1) L^{p-1} B^{n-1} (B^{n-1} - y)^{a-1}, \\ B_n &= (\beta + \beta + 1) P^{p-1} B^{n-1} (B^{n-1} - y)^{a-1}, \\ C_n &= (\beta + 1) P^{p-1} (B^{n-1} - y)^{a-1} (R - y)^a, \\ D_n &= (\beta + 1) P^{p-1} (B^{n-1} - y)^{a-1} (R - y)^a, \\ &\dots \end{aligned}$$

est qui en l'expression de  $L_{x,r,n}$ , dans la formule

$$\begin{aligned} D_v P^p Q^q &= \frac{1}{P} L_{x,r,n} Q^p, \\ &= (-)^{p+1} L_{p,n} Q^p P^p, \\ &+ \&c., \\ &+ \frac{1}{P} L_{p,n} Q^p P^p Q^p, \end{aligned}$$

15. Je vais à présent au développement de  $L_{x,r,n}$  (sur  $\nu = \beta$ ).

On a, ce qui fait la difficile,

$$(P D_v) P^{p+\nu} Q^q = P_n P^q Q^q Q^p,$$

16. On a, en particulier

$$X_{p,1,n,n} = (\beta + \nu - 2 P) X_{p,1,n,n} + N_{p-1,n,n} X_{p-1,1,n,n},$$

$$X_{p,1,n+1,n} = (\beta + \nu - 2 P) X_{p,1,n+1,n} + N_{p-1,n+1,n} X_{p-1,1,n+1,n},$$

$$X_{p,1,n+1,n+1} = (\beta + \nu - 2P) X_{p,1,n+1,n+1} + N_{p-1,n+1,n+1} X_{p-1,1,n+1,n+1},$$

...

$$X_{p,1,n,r} = (\beta + \nu - 2P) X_{p,1,n,r-1} + N_{p-1,n,r-1} X_{p-1,1,n,r-1},$$

lesquelles donnent tour de suite

$$X_{p,1,n,n} = X_{p-1,1,n,n} + (\beta + \nu - 2P) N_{p-1,n,n} \frac{P-1}{P+1} N_{p-1,n,n-1}.$$

**18.** L'équation aux différences en effet aux déterminations  $L_{x,r,n}$  peut être trouvée par la même manière, et l'on aura la formule générale

$$L_n = -(r + P + 1) P^{r+P-1} Y_{n,r} V^q, W^p,$$

on obtient

$$\frac{(-)^p}{N^p} Y_{p,r} V^q W^p = -(t - v)^{r+P-1} V^q (p + \beta + 1) (p + 2) \dots (p + \nu - 1) P^{r+\nu-1};$$

et en supposant que  $\nu - r = 1$ , l'équation devient  $A = -P - Q P = R P$ , où en faisant ses coefficients numériques  $V_{p+\nu} = \beta = 0$ . L'équation peut s'écrire encore dans la forme

$$A^{p+r} P^{p+r-1} P^{p+r-1} = X (P - V) (V - y)^{a-1} (V - y)^{a-1} (V - y)^{a-1}.$$

**19.** Je veux au présent au développement de  $L_{x,r,n}$  ( $n = \nu - \beta$ ).

On a, ce qui fait la difficile,

$$(P D_v) P^{p+\nu} Q^q = P_n P^q Q^q Q^p,$$

ou ce qui est la même chose,

$$(P D_v) P^{p+\nu} Q^q P^{p-1} Q^p,$$

et de là,

$$P_n P^q Q^q = P_n P^q Q^q, \quad P^p Q^p = P^p Q^p, \quad B = (e + v) P^q (t + v) P^q;$$

et il s'agit de chercher  $P^p Q^p$  comme fonction de  $X, Y, Z$ .

L'équation est

$$P_{n,p} = (\beta + \beta + 1) (\nu + \beta + 1) P_{n,p-1} + 2v P_{n-1,p-1} - R P_{n-1,p-1} = 0,$$

laquelle avec les conditions  $P^p P_{n,p} = 0, P^p P_{n+1,p} = 0, P^p P_{n,p-1} = 0$ , détermine la coefficient  $P_{n,p}$ .

**20.** La première des identités du Prof. Doolen est

$$D_n^p P^p = (-1)^n P (p + 1) (p + 2) \dots (p + \nu - 1) Q^p,$$

mais par mode déjà bornée trouvée No. 9 on y arrivera de même par les lecteurs lieux et  $\nu$  déjà.

et en remarquant que  $D_n P^p, P^p$ , on voit que la suite donne formules sont identiques, et la relation des deux donnent *deux* démontrer.

La seconde identité du Prof. Doolen est

$$P^p Q^p = (-1)^p P (p + 1) (p + 2) \dots (p + \nu - 1) V^q,$$

On a par la formule (No. 16) on tire

$$\begin{aligned} \frac{(-)^p}{N^p} Q^p Q^p &= L_{n,n} V^q (B - v + \beta)^{a-1}, \\ &+ \frac{(R - y)^{a-1} P^p (B - v + \beta)^{a-1} P^p P^q (R - v)^{a-1}}{1 P (B - v + \beta - 1) (B - v + \beta - 2)} \\ &+ \frac{(R - v)^{a-2} P^p (B - v + \beta - 1) (B - v + \beta - 2) L_n (R - v)^{a-2}}{1 P P (B - v + \beta - 1) (B - v + \beta - 2)} \end{aligned}$$

+&c.,

où le coefficient  $L_n$  a une valeur déterminée par

$$L_n = X_{n,n,r} P^p,$$

le coefficient  $X_{n,n,r}$ , étant celui qui se déduit en mettant  $\nu = r$  de la formule

$$X_{n,n,r} P^p P^q = X (\beta - 1) (\beta - 2) (\beta - 3) \dots$$

et on développant les binomiales on obtient pour  $Q^p (-)^{p+1} Q^p$  une expression  $(P_p^q)^p P_p^q (P - (P^p)^q)^p$  de

$$(P^p)^q = P^p Q^p L_n (R^p - V^q) P^p \dots (P^p)^q,$$

$$+ (P)^q L_n V^q (B - v + \beta)^{a-1} P^p L_n V^q P_n^q,$$

Cela dernière donc égale

$$(P^p)^q Q^p P^p P^p P^p Q^p Q^p Q^p,$$

et le développement de cette expression se déduit de celui de  $L_n$  (No. 13) en y substituant les valeurs précédent. On obtient ainsi

$$(P^p)^q Q^p P^p P^p Q^p P^p Q^p P^p Q^p P^p Q^p P^p Q^p,$$

laquelle s'accorde avec les démonstrations par l'expression générale de  $L_{x,r,n}$  vis pour avoir comme constituant l'expression pour le coefficient  $X_{n,n,r}$ , et a couvent supposée; mais la formule différents-fais est dans cet ouvrage déjà cité.

## § IV.

21. La première des identités du Prof. Doolen est

$$D_x^q B^p = (-1)^q P (p+1) \dots (p+q-1) Q^p;$$

on, ce qui est la même chose,

$$(P^q D_x)^p P^p = P^p Q^p (-1)^p P (p+1) (p+2) \dots (p+q-1) Q^p,$$

formule dont on peut s'occuper de manière de  $D_x P^p Q^p$ , on a tout simplement

$$P^q D_x P^p Q^p = P_n P^q (-1)^p P (p+1) \dots (p+q-1) Q^p,$$

$$+ \frac{1}{1 \cdot P} P^q P^q Q^p P^q Q^p P^p Q^p, + \frac{1}{1 \cdot 2 \cdot P^q} P^q P^q Q^p P^q P^q Q^p P^p P^p Q^p, + \&c.,$$

et en termes pour  $P_n$ , l'équation, aux différences

$$X_{x,a+1} = (\beta - bx) X_{x,a} - X_{x-1,a},$$

laquelle avec les conditions  $X_{x,0} = X_{x-1,0} = X_{x,a-1} = 0$ , détermine le coefficient  $X_{x,a}$ .



## Note sur l'Équation des Différences pour une Équation donnée de Degré quelconque.

[FROM THE ANNALI DI MATEMATICA PURA ED APPLICATA (TORTOLINI), TOM. II. (1859), PP. 305, 306.]

Il s'agit de trouver l'équation qui a pour racines les carrés des différences des racines d'une équation donnée

$$(\ast(x, 1))^n = 0.$$

En représentant cette équation par  $\phi x = 0$ , soient  $y$  deux racines différentes quelconques, on a non seulement  $\phi x = 0$ ,  $\phi y = 0$ , mais aussi

$$\frac{\phi x - \phi y}{x - y} = 0,$$

et en éliminant entre ces équations

$$\frac{d\phi y}{dy} = 0,$$

on aura l'équation aux racines  $x - y$ . Pour effet, la première équation est de degré  $\nu$ , la seconde de degré  $n - 1$  pour rapport à  $y$ ; on obtient donc deux équations rationnelles en  $x$ ,  $y$ , et en éliminant l'équation en  $x - y$  aura la forme

$$u(x - y) = (x - y)^n + (x - y)^{n-1} \phi y - (x - y)^{n-1} \phi x = 0.$$

Il convient de changer ces je la forme des équations : en effet la première équation est du degré  $n$ , la seconde du degré  $n - 1$  par rapport à  $y$ ; on obtient donc deux équations en  $x$  aura la forme

$$u(x - y) = (x - y) \frac{d\phi y}{dy} - \frac{d\phi x}{dx} = 0,$$

$$\frac{d\phi x}{dx} - \frac{d\phi y}{dy} = 0,$$

THE COLLECTED MATHEMATICAL PAPERS  
of  
ARTHUR CAYLEY

Volume IV — pages 131–153

Papers 250 (conclusion), 251, 252, 253 (start)

[250]

L'ENVELOPPE DES PLANS &amp;C.

131

points de rebroussement sur la courbe cuspidale, et de plus la courbe nodale hyperbolique, la courbe cuspidale, et la courbe sextique dans le plan  $zx$  ont à chacun de ces points une tangente commune. Car, comme je l'ai auparavant mentionné, les courbes sur l'ellipsoïde qui donnent lieu à la courbe nodale hyperbolique et la courbe cuspidale respectivement, se touchent l'une l'autre aux points qui donnent lieu aux points critiques ; de là il suit que la courbe nodale hyperbolique et la courbe cuspidale se touchent l'une l'autre aux points critiques, et par cela seulement que la tangente de la courbe cuspidale est, à l'un quelconque des points critiques, dans le plan de  $zx$ , (puisque la courbe cuspidale est symétrique par rapport à ce plan) on voit que le point critique sera un point de rebroussement sur la courbe cuspidale. Il ne reste qu'à montrer que la courbe nodale hyperbolique et la courbe sextique ont une tangente commune au point critique. La tangente de la courbe nodale hyperbolique est donnée par l'équation

$$\frac{a^2 x dx}{a^2 - b^2} + \frac{c^2 z dz}{c^2 - b^2} = 0.$$

ou

$$x = X \left[ 2 - \frac{1}{a^2} (X^2 + Y^2 + Z^2) \right], \quad z = Z \left[ 2 - \frac{1}{c^2} (X^2 + Y^2 + Z^2) \right],$$

et

$$X^2 + Z^2 = b^2, \quad X^2 + Y^2 + Z^2 = 2b^2, \quad Y = 0,$$

c'est-à-dire que l'on a

$$\frac{x}{a^2} = \frac{X}{a^2}, \quad \frac{z}{c^2} = \frac{Z}{c^2},$$

et l'équation devient ainsi  $X dx + Z dz = 0$ , ce qui fait voir que la tangente est perpendiculaire au rayon mené par le centre de l'ellipsoïde au point qui donne lieu au point critique, et ainsi la tangente de la courbe nodale hyperbolique coïncide avec celle de la courbe sextique dans le plan de  $zx$ . Et l'on peut à présent expliquer plus particulièrement la forme de la courbe cuspidale (le cas dont il s'agit est celui où l'on a non seulement  $a^2 > 2b^2$ ,  $b^2 > 2c^2$  mais aussi  $a^2 + c^2 > 3b^2$ ) en effet la courbe cuspidale est composée en premier lieu de deux ovales en forme d'œil situés symétriquement de chaque côté du plan de  $xy$ , et qui passent par les points critiques, et les points de rebroussement de la courbe sextique dans le plan de  $yz$ , et en second lieu de deux ovales ordinaires situés symétriquement de chaque côté du plan de  $yz$ , et qui passent par les points de rebroussement des courbes sextiques dans les plans de  $xy$  et  $zx$  respectivement : tous les ovales dont il s'agit étant, il va sans dire, des courbes à double courbure. Les relations des différentes courbes sur l'ellipsoïde seront mieux comprises au moyen de la figure 1<sup>re</sup>, où les sections principales de l'ellipsoïde sont tracées, non pas en perspective, mais développées sur le plan de la figure ; et les courbes qui donnent lieu aux courbes nodales et cuspidales respectivement sont seulement marquées par des courbes menées par les points dans les sections principales par lesquels points passent respectivement les courbes dont il s'agit. On aura une idée de la forme de la surface au moyen de la figure 2<sup>e</sup> (laquelle est dessinée en perspective orthogonale au moyen de la première figure) et qui fait voir la forme des

sections principales de la surface (y compris les courbes nodales) et la forme de la courbe cuspidale. Les numéros et lettres dans les deux figures font voir quels points des différentes courbes sur l'ellipsoïde donnent lieu respectivement aux points des différentes courbes sur la surface. En particulier il convient de remarquer que dans la figure 1<sup>re</sup> le point 4 qui donne lieu à un point critique est situé entre les points 3 et 5 qui donnent lieu respectivement au point double et au point de rebroussement de la courbe sextique dans le plan de  $zx$ , et cela fait voir très évidemment dans quelle partie de la courbe sextique est située le point critique, où cette courbe se touche elle-même par la courbe nodale hyperbolique. Je n'ai pas cherché à dessiner d'autres sections de la surface, cela donnerait trop de complication à la figure, et il serait encore plus difficile d'apercevoir la forme de la surface, laquelle peut à peine être représentée sinon par un modèle ou au moins une stéréographie. On peut trouver l'équation de la surface de la même manière que celle de la courbe sextique, à savoir il faut éliminer  $X, Y, Z$  entre l'équation

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1,$$

et les équations

$$x = X \left[ 2 - \frac{1}{a^2} (X^2 + Y^2 + Z^2) \right], \quad y = Y \left[ 2 - \frac{1}{b^2} (X^2 + Y^2 + Z^2) \right], \quad z = Z \left[ 2 - \frac{1}{c^2} (X^2 + Y^2 + Z^2) \right],$$

qui donnent les points de la surface.

En écrivant  $\theta = X^2 + Y^2 + Z^2$  on trouve

$$\frac{x^2}{\left(2 - \frac{\theta}{a^2}\right)^2} + \frac{y^2}{\left(2 - \frac{\theta}{b^2}\right)^2} + \frac{z^2}{\left(2 - \frac{\theta}{c^2}\right)^2} = \theta,$$

$$\frac{x^2}{a^2 \left(2 - \frac{\theta}{a^2}\right)^3} + \frac{y^2}{b^2 \left(2 - \frac{\theta}{b^2}\right)^3} + \frac{z^2}{c^2 \left(2 - \frac{\theta}{c^2}\right)^3} = 1,$$

et en multipliant la seconde équation par 2, et la première équation par  $-\theta$  et en ajoutant on trouve

$$\theta = \frac{x^2}{2 - \frac{\theta}{a^2}} + \frac{y^2}{2 - \frac{\theta}{b^2}} + \frac{z^2}{2 - \frac{\theta}{c^2}},$$

laquelle a pour dérivée par rapport à  $\theta$  la seconde équation. Cette équation peut s'écrire

$$\begin{aligned} & \theta \cdot 8a^2b^2c^2 + \theta^2[-2(a^2b^2 + b^2c^2 + c^2a^2)] \\ & + \theta^3[4(b^2c^2 + c^2a^2 + a^2b^2) + a^2x^2 + b^2y^2 + c^2z^2] \\ & + \theta^4[-8a^2b^2c^2 - 2(b^2 + c^2)a^2x^2 - 2(c^2 + a^2)b^2y^2 - 2(a^2 + b^2)c^2z^2] \\ & + 4a^2b^2c^2(a^2x^2 + b^2y^2 + c^2z^2)\theta = 0, \end{aligned}$$

et l'équation de la surface sera trouvée en égalant à zéro le discriminant de cette fonction de  $\theta$ . En représentant l'équation par

$$\frac{1}{4}(A, B, C, D, E \mid \theta, 1)^4 = 0,$$

on aura

$$\begin{aligned} A &= 6, \\ B &= -3(a^2 + b^2 + c^2), \\ C &= 4(b^2c^2 + c^2a^2 + a^2b^2) + a^2x^2 + b^2y^2 + c^2z^2, \\ D &= -12a^2b^2c^2 - 3(b^2 + c^2)a^2x^2 - 3(c^2 + a^2)b^2y^2 - 3(a^2 + b^2)c^2z^2, \\ E &= 24a^2b^2c^2(x^2 + y^2 + z^2), \end{aligned}$$

et l'équation de la surface sera

$$(AE - 4BD + 3C^2)^3 - 27(ACE + 2BCD - AD^2 - B^2E - C^3)^2 = 0.$$

Il convient de remarquer que  $C$ , qui serait du douzième ordre en  $x, y, z$  n'entre pas dans l'équation, et que tous les autres termes sont au plus de l'ordre 10, la surface est donc (comme j'ai déjà remarqué) de l'ordre 10. Il ne m'a pas paru nécessaire de développer plus complètement l'équation<sup>1</sup>.

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1. Mi sia lecito di richiamare qui per un' istante in seguito alla bella ed elegante Memoria dell' illustre Geometra, come più volte questa superficie di decimo ordine derivata dall' ellissoide sia stato fin da molti anni per me il soggetto di ricerche relative al calcolo integrale per ciò che riguarda specialmente la sua quadratura dipendente da trascendenti ellittici di prima e seconda specie; così in un breve articolo inserito nella Raccolta Scientifica di Roma Anno 2° No. 9 in data del 2 Aprile 1846 ritrovai le coordinate della nuova superficie in funzione delle coordinate corrispondenti dell' ellissoide, e determinai ancora l' integrale definito duplicato di forma razionale per la quadratura della stessa superficie: più estesamente mi occupai dello stesso argomento in una Memoria inserita nel tom. 34° del Sig. Crelle in data 15 Aprile 1846: più completamente poi risolsi la stessa questione in altro articolo pubblicato nella citata Raccolta Scientifica dello stesso Anno 2° No. 21 con la data del 23 Ottobre 1846. Infine ripresi a risolvere lo stesso problema con altri metodi in una Memoria *Sulla riduzione di alcuni integrali definiti ai trascendenti ellittici* inserita pel mese di Agosto e Settembre 1848 nel giornale arcadico di Roma, e precisamente nel No. 18 e seguenti. Sovente ancora nei privati miei studii tentai di giungere all' equazione algebrica della superficie, il che dipendeva da un' eliminazione, ma proponendomi la questione sotto un' aspetto complicato, mi trovava arrestato dalla lunghezza dei calcoli. Il meraviglioso ripiego analitico ritrovato dal Sig. Cayley riduce quest' eliminazione ad una semplicità inaspettata, e rende inutile qualunque altra ricerca, che si volesse intraprendere sullo stesso soggetto. Termino questa breve nota col fare i miei distinti ringraziamenti al chiarissimo Autore, che si è compiaciuto dietro un mio suggerimento comunicatogli dal Sig. D. Hirst fare la traduzione della Memoria per essere inserita negli *Annali*.—B. T.

*[Page 134 blank in the original.]*

[Figure plate, inserted between pp. 133 and 135.]

*Fig. 1.*

Sections principales de l'ellipsoïde

$$a^2 = 84, \quad b^2 = 28, \quad c^2 = 12$$

et aussi

$$a^2 = 25, \quad b^2 = 5, \quad c^2 = 1, \quad a^2 + c^2 = 26.$$

- (a) Génératrice de la courbe nodale elliptique dans le plan de  $xy$
- (b) Génératrice d'une partie de la courbe Cuspidale
- (c) Génératrice de la courbe nodale hyperbolique dans le plan de  $zx$
- (d) Génératrice d'une partie de la courbe cuspidale

*Fig. 2.*

*[Vue de la surface en perspective orthogonale.]*

## 251.

## SUR QUELQUES FORMULES POUR LA DIFFÉRENTIATION.

[FROM THE ANNALI DI MATEMATICA PURA ED APPLICATA (TORTOLINI), TOM. II. (1859), PP. 214–230. TRADUCTION PAR L'AUTEUR D'UN MÉMOIRE PRÉSENTÉ À LA SOCIÉTÉ ROYALE DE LONDRES LE 26 NOVEMBRE, 1857.]

En cherchant une formule dans la théorie des intégrales définies multiples je fus conduit il y a plusieurs années à chercher les coefficients différentiels successifs de l'expression  $(\sqrt{x+\lambda} - \sqrt{x+\mu})^{2s}$ , et les résultats que j'ai trouvés sont donnés dans le Mémoire "On certain formulae for differentiation with applications to the evaluation of definite integrals," *Camb. and Dublin Mathematical Journal*, tom. II. pp. 122–128 (1847), [41]. J'ai depuis cherché les coefficients différentiels successifs de l'expression plus générale  $[(x+\lambda)(x+\mu)]^\alpha (\sqrt{x+\lambda} - \sqrt{x+\mu})^{2s}$ , mais l'investigation n'était pas achevée. Mon attention fut rappelée à ce sujet par deux identités remarquables trouvées par le Prof. Donkin dans son Mémoire "On the equation of Laplace's functions etc." *Phil. Trans.* tom. CXLVII. (1857), pp. 43–57, au moyen de la comparaison de ses résultats avec ceux du Prof. Boole; identités qui appartenaient, je m'aperçus, à la classe des formules ci-dessus mentionnées: la première de ces deux identités se déduit en effet assez facilement d'une formule exposée dans mon mémoire; la démonstration de la seconde identité est beaucoup plus difficile, et je n'ai réussi à l'établir qu'en la faisant dépendre de l'établissement de l'égalité des coefficients numériques de deux expressions de la même forme. Je suis depuis revenu aux investigations incomplètes dont j'ai parlé ci-dessus et les résultats que j'ai trouvés sont donnés dans le présent mémoire. Je remarque qu'en écrivant pour abréger

$$P = 2x + \lambda + \mu, \quad Q = \sqrt{(x+\lambda)(x+\mu)}, \quad R = (\sqrt{x+\lambda} - \sqrt{x+\mu})^2,$$

le sujet auquel appartiennent tous les résultats est la différentiation de l'expression  $P^\alpha Q^\beta R^\gamma$ : l'expression ci-dessus mentionnée  $[(x+\lambda)(x+\mu)]^\alpha (\sqrt{x+\lambda} - \sqrt{x+\mu})^{2s}$  est de la



forme dont il s'agit, et la question qui s'y rapporte est celle d'obtenir le développement de  $D_x^n P^\alpha Q^\beta R^\gamma$ , où  $\alpha = 0$ .

La question suggérée par la seconde identité du Prof. Donkin est celle d'obtenir le développement de  $(P^{-1} Q^2 D_x)^n P^\alpha Q^\beta R^\gamma$ , où  $\alpha = \gamma - \beta$ . Comme la démonstration de ces identités est l'un des objets de ce Mémoire, j'ai donné dans la première section la réduction des identités à la forme sous laquelle je les ai depuis considérées. La seconde section se rapporte au développement de l'expression  $D_x^n P^\alpha Q^\beta R^\gamma$ , où  $\alpha = 0$ ; la troisième section à celui de l'expression  $(P^{-1} Q^2 D_x)^n P^\alpha Q^\beta R^\gamma$ , où  $\alpha = \gamma - \beta$ ; enfin la quatrième section contient l'application des deux identités, et quelques autres applications des formules.

## § I.

1. La première des deux identités du Prof. Donkin est

$$(\sin \theta D_\theta \sin \theta)^n (\tan \frac{1}{2} \theta)^n = 1 \cdot 3 \cdot 5 \dots (2n-1) (\sin \theta)^n,$$

laquelle est l'équation (27) article No. 14, de son Mémoire<sup>2</sup>.

En écrivant  $\cot \theta = t$ , l'équation devient

$$\left(-\frac{1}{t}\right)^n D_t^n \frac{(\sqrt{1+t^2}-t)^n}{\sqrt{1+t^2}} = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{(1+t^2)^{n+\frac{1}{2}}},$$

et en posant comme à l'ordinaire  $i = \sqrt{-1}$  on obtient

$$\sqrt{1+t^2}-t = -\frac{i}{2} (\sqrt{t+i}-\sqrt{t-i})^2,$$

et en écrivant aussi

$$1 \cdot 3 \cdot 5 \dots (2n-1) = 2^n \cdot \frac{1}{2} \cdot \frac{3}{2} \dots (n - \frac{1}{2}) = 2^n [n - \frac{1}{2}]^n,$$

la formule devient

$$D_t^n \frac{(\sqrt{t+i}-\sqrt{t-i})^{2n}}{\sqrt{1+t^2}} = 2^n [n - \frac{1}{2}]^n \frac{1}{(1+t^2)^{n+\frac{1}{2}}}.$$

Ceci est un cas particulier de

$$D_x^n \frac{(\sqrt{x+\lambda}-\sqrt{x+\mu})^{2n}}{\sqrt{(x+\lambda)(x+\mu)}} = (-)^n (\lambda-\mu)^n [n - \frac{1}{2}]^n \frac{1}{[(x+\lambda)(x+\mu)]^{n+\frac{1}{2}}},$$

ou, en écrivant comme auparavant

$$P = 2x + \lambda + \mu, \quad Q = \sqrt{(x+\lambda)(x+\mu)}, \quad R = (\sqrt{x+\lambda}-\sqrt{x+\mu})^2,$$

la formule est

$$D_x^n Q^{-1} R^n = (-)^n (\lambda-\mu)^n [n - \frac{1}{2}]^n Q^{-2n-1}.$$

2. Vedi la nota in fine. [This Note by the Editor, Professor Tortolini, relating merely to the transformation  $(\sin \theta D_\theta \sin \theta)^n = (\sin \theta)^{n+1} (\sin^{-1} \theta D_\theta)^n \sin \theta$ , is not here reproduced.]

[251

SUR QUELQUES FORMULES POUR LA DIFFÉRENTIATION.

137

2.<sup>3</sup>

3. La seconde identité (voir article No. 16) est

$$(\sin \theta)^{-n} (\sin \theta D_\theta \sin \theta)^n (\tan \frac{1}{2} \theta)^s = \mu^{s-n} (1 - \mu^2)^{\frac{1}{2}n} (D_{\mu^{-1}})^n \mu^{n+s} (1 + \mu)^{-s},$$

où comme auparavant  $\mu = \cos \theta$ . En écrivant  $\cot \theta = t$ , et en faisant attention que le côté gauche de l'équation peut s'écrire sous la forme

$$(\sin \theta)^{-n-1} (\sin^2 \theta D_\theta)^n \sin \theta (\tan \frac{1}{2} \theta)^s,$$

et que l'on a

$$\sin \theta = \frac{1}{\sqrt{1+t^2}}, \quad \tan \frac{1}{2} \theta = \sqrt{1+t^2} - t, \quad \sin^2 \theta D_\theta = -D_t,$$

4

C. IV.

18

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3. La comparaison mentionnée article No. 5 du Mémoire du Prof. Donkin donne

$$(\sin \theta)^{-n} (\sin \theta D_\theta \sin \theta)^n f$$

où  $\mu = \cos \theta$ , et après les différentiations  $\mu' = \cos \theta$ . Les parties qui contiennent les fonctions indéterminées  $f$  et  $F$  doivent être égales, chacune prise à part de l'autre. Les parties qui contiennent  $f$  seront égales, si cette égalité subsiste pour  $\mu = \mu'^s$ , où  $s$  est un indice quelconque; c'est-à-dire l'égalité subsistera si

$$\begin{aligned} & (\sin \theta)^{-n} (\sin \theta D_\theta \sin \theta)^n (\tan \frac{1}{2} \theta)^s \\ &= \mu^{s-n} (1 - \mu^2)^{\frac{1}{2}n} (D_{\mu^{-1}})^n \mu^{n+s} (1 + \mu)^{-s}, \end{aligned}$$

ou enfin en écrivant  $\mu$  au lieu de  $\mu'$ , si

$$(\sin \theta)^{-n} (\sin \theta D_\theta \sin \theta)^n (\tan \frac{1}{2} \theta)^s = \mu^{s-n} (1 - \mu^2)^{\frac{1}{2}n} (D_{\mu^{-1}})^n \mu^{n+s} (1 + \mu)^{-s},$$

où  $\mu = \cos \theta$ ; c'est en effet la seconde des deux identités du Prof. Donkin. L'égalité des parties qui contiennent  $F$  dépend de la même manière de l'équation

$$(\sin \theta)^{-n} (\sin \theta D_\theta \sin \theta)^n (\tan \frac{1}{2} \theta)^s = (-)^n (1 - \mu^2)^{\frac{1}{2}n} (D_{\mu^{-1}})^n \mu^{n+s} (1 - \mu)^{-s},$$

laquelle se déduit de l'autre équation en y écrivant  $180^\circ - \theta$  au lieu de  $\theta$ .

4. Le lecteur pourrait omettre cet article, qui ne fait que montrer qu'une certaine formule du Professeur Donkin se réduit à son identité seconde.

le côté gauche devient

$$(-)^n (1+t^2)^{\frac{1}{2}(n+s+\frac{1}{2})} D_t^n \frac{(\sqrt{1+t^2}-t)^s}{\sqrt{1+t^2}}.$$

Le côté droit peut s'écrire sous la forme

$$\mu^{s-n+1} (1-\mu^2)^{\frac{1}{2}n} \left( \frac{1}{\mu} D_\mu \right)^n \mu^{n+s} (1+\mu)^{-s},$$

et en observant que

$$\mu = \frac{t}{\sqrt{1+t^2}}, \quad \frac{1}{\mu} D_\mu = \frac{(1+t^2)^2}{t} D_t,$$

le côté droit devient

$$\frac{t^{s-n+1}}{(1+t^2)^{\frac{1}{2}(s-n+1)}} \left( \frac{(1+t^2)^2}{t} D_t \right)^n \frac{t^{n+s}}{(1+t^2)^{\frac{1}{2}(n+s)} (\sqrt{1+t^2}+t)^{-s}};$$

en comparant les deux expressions on obtient

$$\left( \frac{(1+t^2)^2}{t} D_t \right)^n \frac{t^{n+s}}{(1+t^2)^{\frac{1}{2}(n+s)} (\sqrt{1+t^2}+t)^{-s}} = \frac{(1+t^2)^{n+\frac{1}{2}}}{t^{n-s+1}} D_t^n \frac{(\sqrt{1+t^2}-t)^s}{\sqrt{1+t^2}},$$

et de là, en écrivant comme auparavant  $i = \sqrt{-1}$ , et

$$\sqrt{1+t^2}-t = -\frac{i}{2} (\sqrt{t+i}-\sqrt{t-i})^2,$$

la formule devient

$$\left( \frac{(1+t^2)^2}{t} D_t \right)^n \frac{t^{n+s} (\sqrt{t+i}-\sqrt{t-i})^{-2s}}{(1+t^2)^{\frac{1}{2}(n+s)}} = \frac{(1+t^2)^{n+\frac{1}{2}}}{t^{n-s+1}} D_t^n \frac{(\sqrt{t+i}-\sqrt{t-i})^{2s}}{\sqrt{1+t^2}},$$

laquelle est un cas particulier de

$$\begin{aligned} & \left( \frac{[(x+\lambda)(x+\mu)]^2}{2x+\lambda+\mu} D_x \right)^n \frac{(2x+\lambda+\mu)^{n+s-1}}{[(x+\lambda)(x+\mu)]^{n-s}} (\sqrt{x+\lambda}-\sqrt{x+\mu})^{-2s} \\ &= \frac{[(x+\lambda)(x+\mu)]^{n+s-1}}{(2x+\lambda+\mu)^{-n+s-1}} D_x^n \frac{(\sqrt{x+\lambda}-\sqrt{x+\mu})^{2s}}{\sqrt{(x+\lambda)(x+\mu)}}, \end{aligned}$$

c'est-à-dire de

$$(P^{-1} Q^2 D_x)^n (P^{n+s-1} Q^{-2n+1} R^{-n+s}) = P^{-n+s-1} Q^{2n+2} D_x^n (Q^{-1} R^s).$$

4. En écrivant comme auparavant

$$P = 2x + \lambda + \mu, \quad Q = \sqrt{(x+\lambda)(x+\mu)}, \quad R = (\sqrt{x+\lambda}-\sqrt{x+\mu})^2,$$

on a  $R = P - 2Q$ , et en écrivant pour abréger  $(\lambda - \mu)^2 = \Delta$ ,

[251

SUR QUELQUES FORMULES POUR LA DIFFÉRENTIATION.

139

on obtient aussi

$$\Delta = P^2 - 4Q^2, \quad \frac{1}{2}R = P + 2Q;$$

on trouve de plus

$$D_x P = 2, \quad D_x Q = \frac{1}{2} \frac{P}{Q}, \quad D_x R = -\frac{R}{Q}.$$

5. Les dernières formules donnent

$$D_x P^\alpha Q^\beta R^\gamma = \frac{1}{2}\beta P^{\alpha+1} Q^{\beta-2} R^\gamma - \gamma P^\alpha Q^{\beta-1} R^\gamma + 2\alpha P^{\alpha-1} Q^\beta R^\gamma,$$

formule dont on pourrait se servir pour chercher  $D_x^r P^\alpha Q^\beta R^\gamma$ ; on a par exemple

$$\begin{aligned} D_x^2 P^\alpha Q^\beta R^\gamma &= \frac{1}{4}\beta(\beta-2) P^{\alpha+2} Q^{\beta-4} R^\gamma - \frac{1}{2}\gamma(2\beta-1) P^{\alpha+1} Q^{\beta-3} R^\gamma \\ &\quad + [\beta(2\alpha+1) + \gamma^2] P^\alpha Q^{\beta-2} R^\gamma - 4\alpha\gamma P^{\alpha-1} Q^{\beta-1} R^\gamma + 4(\alpha-1)\gamma P^{\alpha-2} Q^\beta R^\gamma, \end{aligned}$$

et ainsi de suite. Et de même

$$P^{-1} Q^2 D_x P^\alpha Q^\beta R^\gamma = \frac{1}{2}\beta P^\alpha Q^\beta R^\gamma - \gamma P^{\alpha-1} Q^{\beta+1} R^\gamma + 2\alpha P^{\alpha-2} Q^{\beta+2} R^\gamma,$$

et de là

$$\begin{aligned} (P^{-1} Q^2 D_x)^2 P^\alpha Q^\beta R^\gamma &= \frac{1}{4}\beta(\beta+2) P^\alpha Q^{\beta+2} R^\gamma - \frac{1}{2}(2\beta+3)\gamma P^{\alpha-1} Q^{\beta+1} R^\gamma \\ &\quad + [2\alpha(\beta+2) + \gamma^2] P^{\alpha-2} Q^{\beta+2} R^\gamma - 2(2\alpha-1)\gamma P^{\alpha-3} Q^{\beta+3} R^\gamma \\ &\quad + 4\alpha(\alpha-2) P^{\alpha-4} Q^{\beta+4} R^\gamma, \end{aligned}$$

et ainsi de suite. Mais il serait difficile d'obtenir de cette manière l'expression de la  $r$ -ième répétition de l'opération  $D_x$ , ou  $P^{-1} Q^2 D_x$ , sur  $P^\alpha Q^\beta R^\gamma$  et je ne poursuis pas la question.

6. Je vais à présent chercher le développement de  $D_x^r P^\alpha Q^\beta R^\gamma$  ( $\alpha = 0$ ) ou ce qui est la même chose  $D_x^r Q^\beta R^\gamma$ . On a

$$D_x Q^\beta R^\gamma = \frac{1}{2}\beta P Q^{\beta-2} R^\gamma - \gamma Q^{\beta-1} R^\gamma,$$

ou en substituant pour  $P$  la valeur

$$P = \frac{\Delta}{R} - 2Q,$$

on a

$$D_x Q^\beta R^\gamma = -(\beta + \gamma) Q^{\beta-1} R^\gamma - \frac{1}{2}\beta \Delta Q^{\beta-2} R^{\gamma-1}.$$

La répétition de l'opération  $D_x$  donne évidemment une expression de la forme

$$D_x^r Q^\beta R^\gamma = (-)^r L_{r,0} Q^{\beta-r} R^\gamma + (-)^r \frac{1}{2^r} L_{r,1} \Delta Q^{\beta-r-2} R^{\gamma-1} + \dots + (-)^r \frac{1}{2^r} L_{r,r} \Delta^r Q^{\beta-3r} R^{\gamma-r},$$

et on obtient pour  $L_{r,\theta}$  l'équation aux différences

$$L_{r+1,\theta+1} - (\beta + \gamma - r - 2\theta - 2) L_{r,\theta+1} - (\beta - r - \theta) L_{r,\theta} = 0,$$

laquelle, avec les conditions particulières

$$L_{r,-1} = 0, \quad L_{r,r+1} = 0, \quad L_{0,0} = 1,$$

suffit pour déterminer les coefficients  $L_{r,\theta}$  de la formule.

7. Avant d'aller plus loin, je vais considérer le cas particulier  $\beta = 0$ . L'équation aux différences est

$$L_{r+1,\theta+1} - (\gamma - r - 2\theta - 2) L_{r,\theta+1} + (r + \theta) L_{r,\theta} = 0,$$

et l'on satisfait à cette équation en posant

$$L_{r,\theta} = \frac{(-)^r [r + \theta - 1]^\theta [\gamma - \theta - 1]^{r-\theta-1}}{2^\theta [\theta]^\theta},$$

ou, ce qui est la même chose,

$$L_{r,\theta} = \frac{(-)^r 2^\theta \gamma [\theta - \frac{1}{2}]^\theta [r + \theta - 1]^{r-\theta-1} [\gamma - \theta - 1]^{r-\theta-1}}{(r - \theta - 1) [\theta]^\theta}.$$

En effet on déduit de la première expression

$$\begin{aligned} & L_{r+1,\theta+1} - (\gamma - r - 2\theta - 2) L_{r,\theta+1} \\ &= 2^{\theta+1} [\theta + 1]^{\theta+1} (r + \theta) [r + \theta - 1]^{r-\theta-1} [\gamma - \theta - 2]^{r-\theta-1} \\ & \quad \times [(r + \theta + 1)(\gamma - r) - (\gamma - r - 2\theta - 2)(r - \theta - 1)], \end{aligned}$$

et le facteur en [ ] est  $2(\theta + 1)(\gamma - \theta - 1)$ , de manière que l'expression devient

$$2^{\theta+1} [\theta + 1]^{\theta+1} (r + \theta) [r + \theta - 1]^{r-\theta-1} [\gamma - \theta - 1]^{r-\theta-1},$$

ce qui est

$$-(r + \theta) [\gamma - \theta - 1]^{r-\theta-1} \cdot L_{r,\theta}.$$

L'équation aux différences est donc satisfaite; les conditions pour les limites sont aussi satisfaites, et la valeur qui vient d'être donnée est donc celle de  $L_{r,\theta}$  dans le cas particulier dont il s'agit où  $\beta = 0$ .

8. Il sera convenable d'écrire

$$\frac{(-)^r}{2^r \gamma} \frac{[r + \theta - 1]^{r-\theta-1} [\gamma - \theta - 1]^{r-\theta-1}}{[\theta - \frac{1}{2}]^\theta} \cdot L_{r,\theta} = \frac{[\theta - \frac{1}{2}]^\theta [r + \theta - 1]^{r-\theta-1} [\gamma - \theta - 1]^{r-\theta-1}}{[r - \theta - 1]^{r-\theta-1}} = L'_{r,\theta},$$

et alors en observant que  $L_{r,\theta}$  ou  $L'_{r,\theta}$  se réduit à zéro pour  $\theta = r$ , on obtient

[251

SUR QUELQUES FORMULES POUR LA DIFFÉRENTIATION.

141

9. Cela est en effet sous une forme un peu modifiée la formule fondamentale de mon mémoire dans le *Camb. and Dub. Math. Journal* ; la valeur qu'on y trouve du coefficient  $K_{r,\theta}$  (en mettant  $\gamma$  au lieu de  $i$  est)

$$K_{r,\theta} = \frac{\Gamma(r - \frac{1}{2} + \theta) \Gamma(2r - 1 - \theta) \Gamma(\gamma - r - \theta - 1)}{\Gamma(\theta + 1) \Gamma(r - \frac{1}{2}) \Gamma(2r - 1 - 2\theta) \Gamma(\gamma - r + 1)},$$

et le coefficient  $L_{r,\theta}$  ( $\beta = 0$ ) du présent mémoire est lié avec  $K_{r,\theta}$  par l'équation

$$\frac{1}{2^\theta \gamma} L_{r,\theta} = L'_{r,\theta} = K_{r,r-1-\theta}.$$

La valeur de  $L'_{r,r-1}$  est  $[r - \frac{1}{2}]^{r-1}$ , et pour  $r = \gamma + 1$  le coefficient devient  $[\gamma - \frac{1}{2}]^\gamma$ , et tous les autres coefficients se réduisent à zéro : la formule devient donc tout simplement

$$D_x^{\gamma+1} R^\gamma = [\gamma - \frac{1}{2}]^\gamma \Delta^\gamma Q^{-2\gamma-1}.$$

10. Dans le cas  $r > \gamma + 1$  il convient de modifier la forme de l'équation. J'écris  $r = \gamma + 1 + s$ , puis en faisant attention que les coefficients  $L'_{\gamma+1+s,\theta}$  se réduisent à zéro pour  $\theta < \gamma$ , on peut écrire  $\gamma + \theta$  au lieu de  $\theta$ , et dans la formule nouvelle étendre  $\theta$  depuis  $\theta = 0$  jusqu'à  $\theta = s$ . On a ainsi

$$\sum_{\theta=0}^s (-)^{\gamma+\theta} L'_{\gamma+1+s,\gamma+\theta} \Delta^{\gamma+\theta} Q^{-2\gamma-1-2\theta-\gamma+\theta} R^{-\theta},$$

et dans cette équation le côté gauche peut s'exprimer sous la forme

$$D_x^{\gamma+1+s} R^\gamma.$$

On obtient ainsi

$$\begin{aligned} D_x^{\gamma+1+s} R^\gamma &= L''_{s,0} \Delta^\gamma Q^{-2\gamma-1-s} R^0 \\ &\quad - L''_{s,1} \Delta^{\gamma+1} Q^{-2\gamma-2-s} R^{-1} \\ &\quad + \dots + (-)^s L''_{s,s} \Delta^{\gamma+s} Q^{-2\gamma-3s-1} R^{-s}, \end{aligned}$$

où  $L''_{s,\theta} = (-)^{s-\theta} L'_{\gamma+1+s,\gamma+\theta} = L'_{\gamma+1+s,\gamma+\theta}$ , et en substituant pour les coefficients  $L'$  leurs valeurs, et en faisant attention que  $[-\theta - 1]^{s-\theta} = (-)^{s-\theta} [s]^{s-\theta}$ , on trouve

$$L''_{s,\theta} = \frac{(-)^s [2\gamma + s + \theta]^{\gamma+\theta} [s]^{s-\theta}}{2^\theta [2\gamma]^{2\theta} [\gamma + \theta]^\gamma} \cdot \frac{(-)^s [\gamma + \theta - \frac{1}{2}]^\theta [2\gamma + s + \theta]^{s-\theta} [s]^{s-\theta}}{[s - \theta]^{s-\theta}},$$

la formule qui vient d'être trouvée est donnée sous une forme différente dans mon mémoire déjà cité.

11. Je résume l'équation générale

$$L_{r+1,\theta+1} - (\beta + \gamma - r - 2\theta - 2) L_{r,\theta+1} - (\beta - r - \theta) L_{r,\theta} = 0,$$

les conditions aux limites étant comme auparavant

$$L_{r,-1} = 0, \quad L_{r,r+1} = 0, \quad L_{0,0} = 1.$$

On a en particulier

$$\begin{aligned} L_{r+1,0} - (\beta + \gamma - r) L_{r,0} &= 0, \\ L_{r+1,1} - (\beta + \gamma - r - 2) L_{r,1} - (\beta - r) L_{r,0} &= 0, \\ L_{r+1,2} - (\beta + \gamma - r - 4) L_{r,2} - (\beta - r - 1) L_{r,1} &= 0, \\ &\vdots \\ L_{r+1,r} - (\beta - 2r) L_{r,r} &= 0. \end{aligned}$$

De là on voit tout de suite que les valeurs de  $L_{r,0}$ ,  $L_{r,r}$  sont

$$L_{r,0} = [\beta + \gamma]^r, \quad L_{r,r} = [\beta, -2]^r,$$

où dans la dernière équation la notation au côté droit dénote une factorielle à différence  $-2$ .

12. Les autres coefficients  $L_{r,1}$ ,  $L_{r,2}$ , etc. peuvent s'obtenir successivement par une intégration directe des équations; et quoiqu'on ne les obtienne pas de cette manière sous la forme la plus commode, cependant il convient de donner l'investigation. Pour  $L_{r,1}$  on peut écrire

$$L_{r,1} = [\beta + \gamma - 2]^r M_{r,1},$$

l'équation pour  $M_{r,1}$  sera

$$M_{r+1,1} - M_{r,1} = \frac{(\beta - r) [\beta + \gamma]^r}{[\beta + \gamma - 2]^{r+1}} = \frac{(\beta - r)(\beta + \gamma)(\beta + \gamma - 1)}{(\beta + \gamma - r)(\beta + \gamma - r - 1)(\beta + \gamma - r - 2)},$$

laquelle peut aussi s'écrire

$$M_{r+1,1} - M_{r,1} = -\frac{\frac{1}{2}(r+2)(r+1)(\gamma-2)}{\beta + \gamma - r - 2} + \frac{(r+1)r(\gamma-1)}{\beta + \gamma - r - 1} - \frac{\frac{1}{2}r(r-1)\gamma}{\beta + \gamma - r}.$$

Cette équation a une solution

$$M_{r,1} = \frac{A_r}{\beta + \gamma - r - 1} + \frac{B_r}{\beta + \gamma - r};$$

cela donne en effet

$$M_{r+1,1} - M_{r,1} = \frac{A_{r+1} - A_r}{\beta + \gamma - r - 1} + \frac{B_{r+1} - B_r}{\beta + \gamma - r},$$

et l'on doit ainsi avoir

$$A_{r+1} = -\frac{1}{2}(r+2)(r+1)(\gamma-2), \quad B_{r+1} - A_r = (r+1)r(\gamma-1), \quad -B_r = -\frac{1}{2}r(r-1)\gamma,$$

équations qui sont satisfaites par

$$A_r = -\frac{1}{2}(r+1)r(\gamma-2), \quad B_r = \frac{1}{2}r(r-1)\gamma.$$

Les conditions aux limites sont aussi satisfaites, et en formant l'expression de  $L_{r,1}$  on obtient

13. La valeur de  $L_{r,2}$  peut s'obtenir par un procédé semblable, et l'on a

$$L_{r,2} = [\beta + \gamma - 4]^r \left\{ \frac{A_r}{\beta + \gamma - r - 3} + \frac{B_r}{\beta + \gamma - r - 2} + \frac{C_r}{\beta + \gamma - r - 1} + \frac{D_r}{\beta + \gamma - r} \right\},$$

où

$$\begin{aligned} A_r &= \frac{1}{24} (r+1) r ((r^2 + 5r - 6) \gamma^2 + (-11r^2 - 31r + 42) \gamma + (24r^2 + 48r - 72)), \\ B_r &= -\frac{1}{6} r (r-1) ((r^2 + 3r - 2) \gamma^2 + (-7r^2 - 13r + 6) \gamma + 8r^2 + 16r), \\ C_r &= \frac{1}{4} r (r-1)(r-2) ((r^2 + r) \gamma^2 + (-3r^2 - 3r) \gamma), \\ D_r &= -\frac{1}{6} r (r-2)(r-3) ((r^2 - r) \gamma^2 + (r^2 - r) \gamma), \end{aligned}$$

et l'on peut remarquer que les suites des coefficients numériques  $(1, 1, 1, 1)$ ,  $(5, 3, 1, -1)$ ,  $(-6, -2, 0, 0)$  ont respectivement les premières, secondes, et troisièmes différences égales à zéro, les suites  $(-11, -7, -3, 1)$ ,  $(-31, -13, -3, -1)$  ont respectivement les secondes, et troisièmes différences égales à zéro, et la suite  $(24, 8, 0, 0)$  a les troisièmes différences égales à zéro. Je remarque que pour  $r = 1$ ,  $r = 2$  les expressions de  $L_{r,1}$ ,  $L_{r,2}$  donnent

$$L_{1,1} = \frac{(\beta + \gamma - 2)(1 - \frac{1}{2}\gamma)}{1} = \beta,$$

lesquelles s'accordent avec les résultats donnés par l'expression générale de  $L_{r,r}$ .

14. L'expression de  $L_{r,1}$  peut se transformer en

$$L_{r,1} = r\beta [\beta + \gamma - 2]^{r-1} - \frac{1}{2}r(r-1) [\beta + \gamma - 2]^{r-2},$$

et l'expression de  $L_{r,2}$  peut de même, avec beaucoup plus de peine, se transformer en

$$L_{r,2} = \frac{1}{2}r(r-1) \left( \beta(\beta-2) [\beta+\gamma-4]^{r-2} - (r-2)\beta(\beta-1) [\beta+\gamma-4]^{r-3} + \frac{1}{4}(r-2)(r-3)(\beta+1)\beta [\beta+\gamma-4]^{r-4} \right),$$

et la forme de ces équations et d'autres considérations m'ont conduit à assumer en général

$$\begin{aligned} L_{r,\theta} &= \frac{1}{[\theta]^\theta} \left\{ [\beta]^{2\theta} A_{0,\theta} [r]^{\theta+1} [\beta + \gamma - 2\theta]^{r-2\theta} \right. \\ &+ \frac{(-)^1}{[\theta-1]^1} [\beta]^{2\theta-1} A_{1,\theta} [r]^{\theta+1} [\beta + \gamma - 2\theta]^{r-2\theta-1} \\ &\left. + \dots + \frac{(-)^\theta}{[\theta]^\theta} [\beta]^\theta A_{\theta,\theta} [r]^{2\theta} [\beta + \gamma - 2\theta]^{r-3\theta} \right\}. \end{aligned}$$



**15.** En posant pour plus de commodité  $\theta - 1$  au lieu de  $\theta$  dans l'équation aux différences, cette équation devient

$$L_{r+1,\theta} - (\beta + \gamma - r - 2\theta) L_{r,\theta} = (\beta - r - \theta + 1) L_{r,\theta-1},$$

et puis substituant dans cette équation la valeur assumée de  $L_{r,\theta}$ , et les valeurs correspondantes de  $L_{r+1,\theta}$  et  $L_{r,\theta-1}$ , le terme général au côté gauche sera

$$[\sigma]^\sigma [\theta - \sigma]^{\theta-\sigma} \frac{1}{2^\sigma} [\beta]^{2\theta-\sigma-1} [\beta + \gamma - 2\theta]^{r-2\theta-\sigma-1} A_{\sigma,\theta} [(r+1) [r]^{\theta+\sigma-1} + 1] \\ \times [(r+1)(\beta + \gamma - r - \theta - \sigma) - (\beta + \gamma - r - 2\theta)(r - \theta - \sigma + 1)];$$

l'expression en [ ] est  $(\theta + \sigma)(\beta + \sigma - 2\theta + 1)$ , et en substituant cette valeur le terme général au côté gauche devient

$$[\sigma]^\sigma [\theta - \sigma]^{\theta-\sigma} \frac{1}{2^\sigma} (\theta + \sigma) [r]^{\theta+\sigma} [\beta + \gamma - 2\theta + 1]^{r-2\theta-\sigma-1} A_{\sigma,\theta}.$$

Le terme général au côté droit est

$$[\sigma]^\sigma [\theta - 1 - \sigma]^{\theta-1-\sigma} \frac{1}{2^\sigma} [\beta]^{2\theta-1-\sigma} [\beta + \gamma - 2\theta + 1]^{r-2\theta-\sigma+1} A_{\sigma,\theta-1} \\ \times (\beta - r - \theta + 1)(\beta + \gamma - 2\theta + 2),$$

dont le dernier facteur est égal à

$$(\beta + \gamma - r - \theta + 1 + \sigma)(\beta - 2\theta + 2 - \sigma) - (r - \theta + 1 - \sigma)(\gamma + \sigma);$$

et en substituant cette valeur le terme devient

$$-[\sigma]^\sigma [\theta - 1 - \sigma]^{\theta-1-\sigma} \frac{1}{2^\sigma} [\beta]^{2\theta-1-\sigma+1} [\beta + \gamma - 2\theta + 1]^{r-2\theta-\sigma} (\gamma + \sigma) A_{\sigma,\theta-1}.$$

En écrivant dans la seconde ligne (ce qui est permis)  $\sigma - 1$  au lieu de  $\sigma$ , cette ligne devient

$$-[\sigma - 1]^{\sigma-1} [\theta - \sigma]^{\theta-\sigma} \frac{1}{2^{\sigma-1}} [\beta]^{2\theta-\sigma+1} [\beta + \gamma - 2\theta + 1]^{r-2\theta-\sigma+1} (\gamma + \sigma - 1) A_{\sigma-1,\theta-1},$$

et les deux lignes ensemble seront

$$\frac{(-)^\sigma}{[\sigma]^\sigma [\theta - \sigma]^{\theta-\sigma}} \frac{1}{2^\sigma} [\beta]^{2\theta-\sigma+1} [\beta + \gamma - 2\theta + 1]^{r-2\theta-\sigma+1} \\ \times [(\theta - \sigma)(\beta - 2\theta + 2 - \sigma) A_{\theta-1,\sigma} + 2\sigma(\gamma + \sigma - 1) A_{\theta-1,\sigma-1}],$$

**[251**

SUR QUELQUES FORMULES POUR LA DIFFÉRENTIATION.

**145**

ce qui est le terme général au côté droit. En comparant cela avec l'expression ci-dessus trouvée pour le terme général au côté gauche on obtient

$$(\theta + \sigma) A_{\theta, \sigma} = (\theta - \sigma)(\beta - 2\theta + 2 - \sigma) A_{\theta-1, \sigma} + 2\sigma(\gamma + \sigma - 1) A_{\theta-1, \sigma-1},$$

pour l'équation aux différences à laquelle doit satisfaire le coefficient  $A_{\theta, \sigma}$ ; en y écrivant

$$A_{\theta, \sigma} = [\gamma + \sigma - 1]^\sigma B_{\theta, \sigma},$$

on aura pour  $B_{\theta, \sigma}$  l'équation

$$(\theta + \sigma) B_{\theta, \sigma} = (\theta - \sigma)(\beta - 2\theta + 2 - \sigma) B_{\theta-1, \sigma} + 2\sigma B_{\theta-1, \sigma-1},$$

et on voit sans peine que les conditions aux limites sont

$$B_{\theta, -1} = 0, \quad B_{\theta, \theta+1} = 0, \quad B_{0,0} = 1.$$

**16.** On a en particulier

$$\beta_{\theta,0} = (\beta - 2\theta + 2) B_{\theta-1,0}, \quad B_{\theta,\theta} = B_{\theta-1,\theta-1},$$

et de là

$$B_{\theta,0} = [\beta, -2]^\theta, \quad B_{\theta,\theta} = 1.$$

Les autres fonctions  $B_{\theta, \sigma}$  sont des fonctions rationnelles et entières de  $\beta$ , du degré  $\theta - \sigma$ , données par l'équation aux différences; mais les coefficients numériques des différentes puissances de  $\beta$  pourraient être, je l'admets, une espèce de transcendentes rationnelles propres pour exprimer la valeur de  $L_{r,\theta}$  et il sera suffisant de tabuler les valeurs de  $B_{\theta, \sigma}$  sans pousser plus loin la recherche de la loi de ces valeurs. On a en effet pour  $B_{0,0}$ ,  $B_{1,0}$ , etc. ...  $B_{\theta-1,\theta-1}$  les valeurs

$$1, \beta, \beta(\beta - 2), \beta(\beta - 2)(\beta - 4), \beta(\beta - 2)(\beta - 4)(\beta - 6), \text{ etc. } \dots$$

$$1, \beta - 1, \beta^2 - \frac{4}{3}\beta + 1, \beta^3 - 6\beta^2 + \frac{43}{3}\beta - 2,$$

$$1, \beta - 2, \beta^2 - 6\beta + 7,$$

$$1, \beta - 3,$$

$$1.$$

**17.** Puis en substituant pour  $A_{\theta, \sigma}$  la valeur  $[\gamma + \sigma - 1]^\sigma B_{\theta, \sigma}$  on trouve

$$\begin{aligned} L_{r,\theta} = & \frac{1}{[\theta]^\theta} \left\{ B_{\theta,0} [\beta]^{2\theta} [r]^{\theta+1} [\beta + \gamma - 2\theta]^{r-2\theta} \right. \\ & + \frac{(-)^1}{[\theta-1]^1} \frac{1}{2^1} B_{\theta,1} [\gamma]^1 [\beta]^{2\theta-1} [r]^{\theta+1} [\beta + \gamma - 2\theta]^{r-2\theta-1} \\ & + \dots + \frac{(-)^\theta}{[\theta]^\theta 2^\theta} B_{\theta,\theta} [\gamma + \theta - 1]^\theta [\gamma]^\theta [r]^{2\theta} [\beta + \gamma - 2\theta]^{r-3\theta} \left. \right\} \end{aligned}$$

ce qui est l'expression de  $L_{r,\theta}$  dans la formule

$$\begin{aligned} D_x^r Q^\beta R^\gamma &= (-)^r L_{r,0} Q^{\beta-r} R^\gamma \\ &+ (-)^r \frac{1}{2^r} L_{r,1} \Delta Q^{\beta-r-2} R^{\gamma-1} \\ &+ \dots + \frac{1}{2^r} L_{r,r} \Delta^r Q^{\beta-3r} R^{\gamma-r}. \end{aligned}$$

### § III.

18. Je passe à présent au développement de

$$(P^{-1} Q^2 D_x)^n P^\alpha Q^\beta R^\gamma, \quad (\alpha = \gamma - \beta),$$

ou, ce qui est la même chose,

$$(P^{-1} Q^2 D_x)^n P^{\gamma-\beta} Q^\beta R^\gamma.$$

On a comme auparavant

$$P^{-1} Q^2 D_x P^\alpha Q^\beta R^\gamma = \frac{1}{2} \beta P^\alpha Q^\beta R^\gamma - \gamma P^{\alpha-1} Q^{\beta+1} R^\gamma + 2\alpha P^{\alpha-2} Q^{\beta+2} R^\gamma,$$

et de là

$$\begin{aligned} P^{-1} Q^2 D_x P^{\gamma-\beta} Q^\beta R^\gamma &= \frac{1}{2} \beta P^{\gamma-\beta} Q^\beta R^\gamma - \gamma P^{\gamma-\beta-1} Q^{\beta+1} R^\gamma + 2(\gamma - \beta) P^{\gamma-\beta-2} Q^{\beta+2} R^\gamma \\ &= \frac{1}{2} \beta (P - 2Q) P^{\gamma-\beta-1} Q^\beta R^\gamma - (\gamma - \beta) (P - 2Q) P^{\gamma-\beta-2} Q^{\beta+1} R^\gamma, \end{aligned}$$

c'est-à-dire

$$P^{-1} Q^2 D_x P^{\gamma-\beta} Q^\beta R^\gamma = \frac{1}{2} \beta P^{\gamma-\beta-1} Q^\beta R^{\gamma+1} - (\gamma - \beta) P^{\gamma-\beta-2} Q^{\beta+1} R^{\gamma+1}.$$

On peut donc écrire

$$\begin{aligned} (P^{-1} Q^2 D_x)^n P^{\gamma-\beta} Q^\beta R^\gamma &= \frac{1}{2^n} N_{n,0} P^{\gamma-\beta-n} Q^\beta R^{\gamma+n} \\ &+ \dots + (-)^n N_{n,n} P^{\gamma-\beta-2n} Q^{\beta+n} R^{\gamma+n}, \end{aligned}$$

et on trouve pour  $N_{r,\theta}$  l'équation aux différences

$$N_{r+1,\theta+1} - (\beta + 2r + \theta + 1) N_{r,\theta+1} - (\gamma - \beta - r - \theta) N_{r,\theta} = 0,$$

laquelle avec les conditions aux limites

$$N_{r,-1} = 0, \quad N_{r,r+1} = 0, \quad N_{0,0} = 1,$$

détermine le coefficient  $N_{r,\theta}$ .

On a, en particulier

$$\begin{aligned} N_{r+1,0} &= (\beta + 2r) N_{r,0}, \\ N_{r+1,1} &= (\beta + 2r + 1) N_{r,1} + (\gamma - \beta - r) N_{r,0}, \\ N_{r+1,2} &= (\beta + 2r + 2) N_{r,2} + (\gamma - \beta - r - 1) N_{r,1}, \\ &\vdots \\ N_{r+1,r+1} &= (\gamma - \beta - 2r) N_{r,r}, \end{aligned}$$

lesquelles donnent tout de suite

$$N_{r,0} = [\beta + 2r - 2, -2]^r, \quad N_{r,r} = [\gamma - \beta, -2]^r.$$

**19.** L'équation ressemble à celle pour  $L_{r,\theta}$ , et l'on pourrait croire que l'intégration s'accomplirait d'une manière semblable, mais cela n'est pas ainsi ; car en considérant la seconde équation de la suite, c'est-à-dire

$$N_{r+1,1} - (\beta + 2r + 1) N_{r,1} = (\gamma - \beta - r) [\beta + 2r - 2, -2]^r,$$

en y mettant

$$N_{r,1} = [\beta + 2r - 1, -2]^r M_{r,1},$$

on obtient

$$[\beta + 2r + 1, -2]^{r+1} (M_{r+1,1} - M_{r,1}) = (\gamma - \beta - r) [\beta + 2r - 2, -2]^r,$$

et de là

$$M_{r+1,1} - M_{r,1} = \frac{(\gamma - \beta - r) [\beta + 2r - 2, -2]^r}{[\beta + 2r + 1, -2]^{r+1}};$$

mais les facteurs du numérateur, et du dénominateur ne sont pas ici (ce qui arrivait dans l'équation pour la quantité dénotée auparavant par le même symbole  $M_{r,1}$ ) de la même forme, et il n'y a pas de simplification dans la forme de la fraction. En écrivant successivement  $r - 1, r - 2, \dots, 2, 1, 0$  pour  $r$  on trouve

$$M_{r,1} = \frac{(\gamma - \beta - r + 1)(\beta + 2r - 4) \dots (\beta + 2) \beta}{(\beta + 2r - 1) \dots (\beta + 5)(\beta + 3)},$$

et de là

$$\begin{aligned} N_{r,1} &= [\beta + 2r - 1, -2]^r \left\{ \frac{(\gamma - \beta - 1) \beta}{\beta + 3} + \frac{(\gamma - \beta - 2)(\beta + 2) \beta}{(\beta + 5)(\beta + 3)} \right. \\ &\quad \left. + \dots + \frac{(\gamma - \beta - r + 1)(\beta + 2r - 4) \dots (\beta + 2) \beta}{(\beta + 2r - 1) \dots (\beta + 5)(\beta + 3)} \right\}. \end{aligned}$$

Il ne paraît pas que la suite en [ ] puisse être additionnée, et cela m'empêche de pousser plus loin la recherche de la forme des coefficients  $N_{r,\theta}$  ; la solution, telle que je l'ai trouvée, est donc donnée par l'expression de  $(P^{-1} Q^2 D_x)^n P^{\gamma-\beta} Q^\beta R^\gamma$  en termes des coefficients numériques  $N_{r,\theta}$ , et par l'équation aux différences et conditions aux limites qui déterminent les coefficients  $N_{r,\theta}$ .

20. La première des identités du Prof. Donkin est

#### § IV.

$$D_x^n Q^{-1} R^n = (-)^n \Delta^n [n - \frac{1}{2}]^n Q^{-2n-1};$$

mais par une forme ci-dessus trouvée No. 9 en y écrivant  $n$  au lieu de  $\gamma$  on a

$$D_x^{n+1} R^n = [n - \frac{1}{2}]^n \Delta^n Q^{-2n-1},$$

et en remarquant que  $D_x R^n = -nQ^{-1} \Delta R^n$ , on voit que les deux formules sont identiques, et la vérité du théorème est ainsi démontrée.

21. La seconde des deux identités est

$$P^{-n+s-1} Q^{2n+2} D_x^n Q^{-1} R^s = (P^{-1} Q^2 D_x)^n P^{n+s-1} Q^{-2n+1} R^{-n+s}.$$

On a par la formule (No. 10) en y mettant  $s$  au lieu de  $\gamma$  et  $m+1$  au lieu de  $r$ ,

$$D_x^{s+1+m} R^s = (-)^{m+s+1} \sum_{\theta=0}^m L''_{m,\theta} \Delta^{s+\theta} Q^{-2s-1-2\theta-m+\theta} R^{-\theta},$$

où

$$L''_{m,\theta} = \frac{[s-\theta]^s [s+\theta]^{s-\theta} [s-\theta-1]^{s-\theta-1}}{2^\theta [s]^{2\theta} [m-\theta]^{m-\theta}}.$$

22. En renversant l'ordre des termes et mettant aussi  $L''_{m+1,\theta} = V_{m-\theta}$ , de manière que

$$V_\theta = \frac{[s-1-\theta]^{s-\theta} [2m-\theta]^\theta [s-m+\theta-1]^\theta}{[\theta]^\theta},$$

on obtient

$$\begin{aligned} & \sum_{\theta=0}^m (-)^{m+1} V_\theta \Delta^{s+m-\theta} Q^{-2s-m-1+\theta} R^{\theta-m} \\ & + V_1 \Delta^s Q^{-2s-1+\theta} R^{s-m+\theta} + \dots + V_\theta \Delta^{s+\theta} Q^{-2s-1-\theta} R^{-\theta}, \end{aligned}$$

et en remarquant que  $D_x R^s = -sQ^{-1} R^s$ , et que l'on a aussi  $\Delta = P^2 - 4Q^2$  et  $R = P - 2Q$ , de manière que  $\Delta R^{-1} = -(P + 2Q)$ , l'équation peut s'écrire sous la forme

$$\begin{aligned} Q^{2n+2} P^{-n+s-1} D_x^n Q^{-1} R^s &= (-)^n \{ V_0 (P + 2Q)^n P^{-n+s-1} Q^{-1} \\ &+ V_1 (P + 2Q)^{n-1} P^{-n+s-1} Q^{1+1} R^0 \dots \\ &+ V_\theta (P + 2Q)^{n-\theta} P^{-n+s-1} Q^{1+\theta} R^0 \} \end{aligned}$$

[251

SUR QUELQUES FORMULES POUR LA DIFFÉRENTIATION.

149

et en développant les binômes on obtient pour  $Q^{2n+2} P^{-n+s-1} D_x^n Q^{-1} R^s$  la valeur  $(-)^n \times$

$$(V_0) P^{s-1} Q R^s + \left( \binom{n}{1} 2V_0 + V_1 \right) P^{s-2} Q^2 R^s + \left( \binom{n}{2} 2^2 V_0 + \binom{n-1}{1} 2V_1 + V_2 \right) P^{s-3} Q^3 R^s \\ \dots + (2^n V_0 + 2^{n-1} V_1 \dots + 2V_{n-1} + V_n) P^{s-n-1} Q^{n+1} R^s.$$

Cela devrait être égal à

$$(P^{-1} Q^2 D_x)^n P^{n+s-1} Q^{-2n+1} R^{-n+s}$$

et le développement de cette dernière expression se déduit de celui (No. 18) de

$$(P^{-1} Q^2 D_x)^n P^{\gamma-\beta} Q^\beta R^\gamma$$

en y écrivant  $n, -2n+1, -n+s$  au lieu de  $r, \beta, \gamma$ ; la valeur sera ainsi

$$= \frac{1}{2^n} N_{n,0} P^{s-1} Q R^s - \frac{1}{2^{n-1}} N_{n,1} P^{s-2} Q^2 R^s + \frac{1}{2^{n-2}} N_{n,2} P^{s-3} Q^3 R^s \dots + (-)^n N_{n,n} P^{s-n-1} Q^{n+1} R^s,$$

laquelle a la même forme que le développement de la fonction au côté gauche de l'équation, et l'identité des deux expressions dépend des équations

$$\begin{aligned} \frac{1}{2^n} N_{n,0} &= (-)^n V_0, \\ \frac{-1}{2^{n-1}} N_{n,1} &= (-)^{n-1} \left( \binom{n}{1} 2V_0 + V_1 \right), \\ \frac{1}{2^{n-2}} N_{n,2} &= (-)^{n-2} \left( \binom{n}{2} 2^2 V_0 + \binom{n-1}{1} 2V_1 + V_2 \right), \\ &\vdots \\ N_{n,n} &= (2^n V_0 + 2^{n-1} V_1 \dots + 2V_{n-1} + V_n); \end{aligned}$$

mais comme on ne sait pas la forme générale des coefficients  $N_{n,\theta}$  je n'ai pas pu vérifier complètement ces équations. On a cependant en mettant  $n, -2n+1, -n+s$  pour  $r, \beta, \gamma$

$$\begin{aligned} N_{n,0} &= [-1, +2]^n = (-)^n [2n-1, -2]^n = (-)^n 2^n [n - \frac{1}{2}]^n, \\ N_{n,1} &= s[-3, +2]^{n-1} = (-)^{n-1} s[2n-1, -2]^{n-1} = (-)^{n-1} 2^{n-1} s[n - \frac{1}{2}]^{n-1}, \\ &\vdots \\ V_0 &= [s-1]^s \cdot \frac{1}{2^0 [m]^m} \cdot [2m]^0 [s-m-1]^0 = [s-1]^s, \\ V_1 &= s[s]^{s-1} [2m-1]^1 [s-m]^1 / [1]^1 = s(2m-1)(s-m) [s-1]^{s-1} / 1. \end{aligned}$$

et les deux premières équations seront

$$[n - \frac{1}{2}]^n = [n - \frac{1}{2}]^n, \quad s[n - \frac{1}{2}]^{n-1} = \frac{1}{2} \{ 2[n - \frac{1}{2}]^n + [n - \frac{1}{2}]^{n-1} (2n-1)(s-n) \},$$

où dans la seconde équation l'expression au côté droit est

$$[2n+2(s-n)][n - \frac{1}{2}]^{n-1} = 2s[n - \frac{1}{2}]^{n-1} \div 2 = s[n - \frac{1}{2}]^{n-1},$$

comme cela devrait être. La dernière équation de la suite est

$$N_{n,n} = 2^n V_0 + 2^{n-1} V_1 \dots + 2V_{n-1} + V_n$$

et comme on a  $N_{n,n} = [n+s-1, -2]^n$ , cette dernière équation peut aussi se vérifier.

## 252.

NOTE SUR L'ÉQUATION DES DIFFÉRENCES POUR UNE  
ÉQUATION DONNÉE DE DEGRÉ QUELCONQUE.

[FROM THE ANNALI DI MATEMATICA PURA ED APPLICATA (TORTOLINI), TOM. II. (1859), PP. 365, 366.]

Il s'agit de trouver l'équation qui a pour racines les carrés des différences des racines d'une équation donnée

$$(*| \ v, 1)^n = 0.$$

En représentant cette équation par  $\phi v = 0$ , soient  $x, y$  deux racines différentes quelconques, on a non seulement  $\phi x = 0, \phi y = 0$ ; mais aussi

$$\frac{\phi x - \phi y}{x - y} = 0,$$

et en écrivant dans ces équations

$$x + y = 2s, \quad (x - y)^2 = 4\theta,$$

(ou ce qui est la même chose  $x = s + \sqrt{\theta}, y = s - \sqrt{\theta}$ ) on obtient deux équations rationnelles en  $s$ , et  $\theta$ , et en éliminant  $s$  on obtient l'équation qui donne

$$\theta = \frac{1}{4}(x - y)^2.$$

Il convient de changer un peu la forme des équations; en effet la première équation est du degré  $n$ , la seconde du degré  $n - 1$  par rapport à  $s$ , mais en écrivant les deux équations sous la forme

$$\frac{\phi x + \phi y}{2} = 0, \quad \frac{\phi x - \phi y}{x - y} = 0,$$

[252

NOTE SUR L'ÉQUATION DES DIFFÉRENCES &amp;C.

151

l'une et l'autre équation sera du degré  $n - 1$  par rapport à  $s$ . La forme sous laquelle j'ai présenté la méthode de Bézout s'applique au problème ; en représentant les deux équations par  $Fs = 0$ ,  $Gs = 0$  j'écris pour le moment

$$\phi(s + \sqrt{\theta}) = A, \quad \phi(s - \sqrt{\theta}) = B,$$

on a alors

$$Fs = \frac{1}{2} (A + B), \quad Gs = \frac{A - B}{\sqrt{\theta}},$$

et en écrivant

$$\phi(s' + \sqrt{\theta}) = A', \quad \phi(s' - \sqrt{\theta}) = B',$$

on a de même

$$Fs' = \frac{1}{2} (A' + B'), \quad Gs' = \frac{A' - B'}{\sqrt{\theta}}.$$

On obtient de là

$$\begin{aligned} \Phi(s, s') &= \frac{Fs \cdot Gs' - Fs' \cdot Gs}{s - s'} \\ &= \frac{\frac{1}{2} (A + B) \cdot \frac{A' - B'}{\sqrt{\theta}} - \frac{1}{2} (A' + B') \cdot \frac{A - B}{\sqrt{\theta}}}{s - s'} \\ &= \frac{2 (AB' - A'B) (A - B) (A' - B')}{(s - s') \sqrt{\theta}} \quad (\text{en réduisant}). \end{aligned}$$

Donc, en rétablissant les valeurs de  $A, B, A', B'$ , on a la fonction

$$\begin{aligned} &\frac{2n [\phi(s + \sqrt{\theta}) \phi(s' - \sqrt{\theta}) - \phi(s' + \sqrt{\theta}) \phi(s - \sqrt{\theta})]}{(s - s') \sqrt{\theta}} \\ &+ \frac{[\phi(s + \sqrt{\theta}) - \phi(s - \sqrt{\theta})] [\phi(s' + \sqrt{\theta}) - \phi(s' - \sqrt{\theta})]}{\theta}, \end{aligned}$$

laquelle sera de la forme

$$(a_{0,0}, a_{0,1}, \dots a_{n-1,n-1}) (s, 1)^{n-2} (s', 1)^{n-2},$$

où les coefficients  $a$  sont des fonctions rationnelles de  $\theta$ , et en égalant à zéro le déterminant formé avec ces coefficients on a l'équation qu'il s'agissait de trouver. Quoique cette solution soit analytiquement la plus simple, j'ai une autre méthode nouvelle plus adaptée au calcul, laquelle j'ai appliquée à trouver l'équation des différences pour l'équation quintique

$$(a, b, c, d, e, f \mid v, 1)^5 = 0.$$

Londres, 4 Fév. 1860.



## 253.

## SUR LA COURBE PARALLÈLE À L'ELLIPSE.

[FROM THE ANNALI DI MATEMATICA PURA ED APPLICATA (TORTOLINI), TOM. III. (1860), PP. 311–316.]

Il fut remarqué par Cauchy (*Comptes Rendus*, tom. XIII. [1841], p. 1063) que l'équation de cette courbe pourrait se trouver en éliminant  $\theta$  entre les équations

$$\frac{x^2}{(\theta + a^2)^2} + \frac{y^2}{(\theta + b^2)^2} = 1, \quad \frac{\theta x^2}{(\theta + a^2)^3} + \frac{\theta y^2}{(\theta + b^2)^3} = k^2,$$

et cette élimination fut effectuée, et le résultat trouvé sous la forme la plus simple par M. Catalan (*Terquem*, tom. III. 1844, p. 553); mais pour faire l'élimination de la manière la plus facile je remarque que ces deux équations donnent

$$\frac{x^2}{\theta + a^2} + \frac{y^2}{\theta + b^2} = 1 + \frac{k^2}{\theta}, \quad \frac{x^2}{(\theta + a^2)^2} + \frac{y^2}{(\theta + b^2)^2} = \frac{k^2}{\theta^2},$$

dont la seconde est la dérivée, par rapport à  $\theta$ , de la première; or cette première équation est

$$(\theta + k^2)(\theta + a^2)(\theta + b^2) - x^2 \theta (\theta + b^2) - y^2 \theta (\theta + a^2) = 0,$$

et, en égalant à zéro le discriminant de la fonction cubique, on aura l'équation de la courbe; en posant

$$A = x^2 + y^2 - k^2 - a^2 - b^2, \quad B = b^2 x^2 + a^2 y^2 - a^2 k^2 - b^4 - a^2 b^2, \quad C = a^2 b^2 k^2,$$

la fonction cubique, multipliée par 3, sera

$$(3, -A, -B, 3C | \theta, 1)^3;$$

on aura donc

$$4(A^2 + 3B)(B^2 + 3AC) - (AB - 9C)^2 = 0,$$

[253

SUR LA COURBE PARALLÈLE À L'ELLIPSE.

153

ou ce qui est la même chose

$$A^2B^2 + 4B^3 + 4A^3C + 18ABC - 27C^2 = 0,$$

c'est-à-dire en substituant les valeurs de  $A, B, C$ , l'équation de la courbe sera

$$\begin{aligned} & (x^2 + y^2 - k^2 - a^2 - b^2)^2 (b^2x^2 + a^2y^2 - a^2k^2 - b^4 - a^2b^2)^2 \\ & + 4(b^2x^2 + a^2y^2 - a^2k^2 - b^4 - a^2b^2)^3 \\ & + 4a^2b^2k^2(x^2 + y^2 - k^2 - a^2 - b^2)^3 \\ & + 18a^2b^2k^2(x^2 + y^2 - k^2 - a^2 - b^2)(b^2x^2 + a^2y^2 - a^2k^2 - b^2k^2 - a^2b^2) \\ & - 27a^4b^4k^4 = 0, \end{aligned}$$

laquelle est en effet l'équation trouvée par M. Catalan.

En écrivant  $k = 0$  l'équation devient

$$0 = (b^2x^2 + a^2y^2 - a^2b^2)^2 [(x^2 + y^2)^2 - 2(a^2 - b^2)(x^2 + y^2) + (a^2 - b^2)^2],$$

ce qui équivaut à l'équation de l'ellipse

$$b^2x^2 + a^2y^2 - a^2b^2 = 0$$

deux fois répétée, et aux équations

$$(x \pm ae)^2 + y^2 = 0,$$

(où comme à l'ordinaire  $a^2e^2 = a^2 - b^2$ ) lesquelles appartiennent aux droites menées chacune par un foyer réel et un foyer imaginaire de l'ellipse ; ces droites sont aussi les tangentes menées à l'ellipse par les quatre foyers. En effet en prenant sur l'ellipse le point imaginaire dont les coordonnées sont

$$x = \frac{a}{e}, \quad y = ia\sqrt{\frac{1}{e^2} - e^2} \quad (i = \sqrt{-1}),$$

l'équation du cercle, rayon 0, ayant pour centre le point dont il s'agit, sera

$$\left(x - \frac{a}{e}\right)^2 + \left[y - ia\sqrt{\frac{1}{e^2} - e^2}\right]^2 = 0,$$

ce cercle se réduit donc à deux droites, savoir la droite

$$\left(x - \frac{a}{e}\right) + i \left[y - ia\sqrt{\frac{1}{e^2} - e^2}\right] = 0,$$

ou, ce qui est la même chose,

$$x - ae + iy = 0,$$

laquelle est tangente à l'ellipse, et la droite

C. IV.

THE COLLECTED MATHEMATICAL PAPERS  
of  
ARTHUR CAYLEY

Volume IV — pages 154–178

Papers 253 (conclusion), 254, 255, 256, 257

ou, ce qui est la même chose,

$$x - a \left( \frac{1}{e} - e \right) - iy = 0,$$

laquelle est la droite menée par le point de contact à l'autre point circulaire à l'infini (celui qui n'est pas situé sur la tangente  $x - ae + iy = 0$ ). Le cercle, comme courbe composée d'une droite tangente à l'ellipse et d'une autre droite menée par le point de contact, a même avec l'ellipse une intersection à trois points réunis. Ces considérations font voir pourquoi au cas  $k = 0$  les quatre droites  $(x \pm ae)^2 + y^2 = 0$  font partie de la courbe.

En supposant que  $k$  est quelconque, mais que l'on a  $a = b$ , l'équation devient

$$a^4 (x^2 + y^2)^2 [(x^2 + y^2 - a^2)^2 - 2k^2 (x^2 + y^2 + a^2) + k^4] = 0,$$

où, en écartant le facteur  $(x^2 + y^2)^2$  et le facteur constant  $a^4$ , l'équation se réduit à

$$(x^2 + y^2 - a^2)^2 - 2k^2 (x^2 + y^2 + a^2) + k^4 = 0,$$

c'est-à-dire

$$[x^2 + y^2 - (a + k)^2] [x^2 + y^2 - (a - k)^2] = 0;$$

ainsi la courbe se réduit à l'ensemble des deux droites  $(x^2 + y^2) = 0$  (chacune deux fois répétée), et des deux cercles

$$x^2 + y^2 - (a + k)^2 = 0, \quad x^2 + y^2 - (a - k)^2 = 0$$

comme évidemment cela doit être.

Par rapport aux singularités de la courbe, la forme de l'équation montre à premier coup d'œil qu'il y a quatre points doubles à l'infini ; savoir les deux points circulaires à l'infini (points où l'infini est rencontré par les droites  $x^2 + y^2 = 0$ ), et les deux points où l'infini est rencontré par les droites  $b^2 x^2 + a^2 y^2 = 0$ . Il y a de plus deux points doubles sur chacun des axes : en effet en écrivant dans l'équation de la courbe  $y = 0$  l'équation résultante, toute réduction faite, devient

$$[(x - a)^2 - k^2] [(x + a)^2 - k^2] [b^2 x^2 - (a^2 - b^2)(k^2 - b^2)]^2 = 0,$$

ce qui fait voir qu'il y a sur l'axe de  $x$  les deux points doubles dont les coordonnées sont données par l'équation

$$b^2 x^2 - (a^2 - b^2)(k^2 - b^2) = 0.$$

Mais pour parvenir à cette conclusion il convient de considérer que l'axe de  $x$  rencontre la courbe dans des points tels que pour chacun la distance normale à l'ellipse est égale à  $k$  : il y a quatre distances normales  $k$ , mesurées le long de l'axe ; et quatre distances normales  $k$  mesurées à des points situés symétriquement par rapport à l'axe, lesquels se rencontrent deux à deux sur l'axe aux points dont les coordonnées sont  $b^2 x^2 = (a^2 - b^2)(b^2 - k^2)$  ; c'est là l'origine des deux points doubles sur l'axe de  $x$ . On doit

aussi remarquer que les coordonnées des points sur l'ellipse qui correspondent à ces points doubles sont données par les équations

$$x^2 = \frac{a^4}{b^2} \frac{b^2 - k^2}{a^2 - b^2}, \quad y^2 = \frac{a^2}{a^2 - b^2} \frac{k^2 - b^4}{b^2};$$

on voit de là que les points sur l'ellipse qui donnent lieu aux deux points doubles sur l'axe de  $x$ , ne seront réels à moins que  $k > \frac{b^2}{a}$ ,  $< b$ ; les deux points doubles seront cependant réels pour toute valeur  $k > 0 < b$ . J'ajoute que pour les valeurs  $k \geq 0$ ,  $< \frac{b^2}{a}$ , les deux points doubles seront des points conjugués ou isolés; pour  $k = \frac{b^2}{a}$ , chacun de ces points se réunit à la courbe, et a deux points de rebroussement, mais il n'y a pas de singularité visible dans la forme de la courbe; pour  $k > \frac{b^2}{a}$ ,  $< b$  il y a deux points doubles avec des branches réelles; et pour  $k = b$ , ces deux points viennent se réunir au centre de l'ellipse et il y a deux branches ayant pour tangente commune à ce point l'axe de  $x$ ; enfin pour  $k > b$ , les deux points doubles sur l'axe de  $x$  deviennent imaginaires.

Des considérations pareilles s'appliquent aux points doubles sur l'axe de  $y$ ; pour  $x = 0$  l'équation de la courbe se transforme en

$$[(y - b)^2 - k^2][(y + b)^2 - k^2][a^2 y^2 - (a^2 - b^2)(k^2 - a^2)]^2 = 0;$$

il y a donc sur l'axe de  $y$  deux points doubles dont les coordonnées sont données par

$$a^2 y^2 - (a^2 - b^2)(k^2 - a^2) = 0,$$

ces points correspondent à des points sur l'ellipse dont les coordonnées sont données par

$$x^2 = \frac{a^2 - b^2}{a^2 - b^2} \frac{k^2}{a^2}, \quad y^2 = \frac{b^4}{a^2} \frac{k^2 - a^2}{a^2 - b^2};$$

les points sur l'ellipse ne seront donc réels à moins que  $k > \frac{a^2}{b}$ ; les deux points doubles seront cependant réels pour toute valeur  $k > a$ . Pour  $k < a$  les deux points doubles seront imaginaires. Pour  $k = a$  le centre est point double commun aux deux branches symétriques de l'axe de  $y$ ; pour  $k = \frac{a^2}{b}$ , chacun des deux points doubles se réunit à la courbe et a deux points de rebroussement, mais il n'y a pas de singularité visible dans la forme de la courbe; enfin pour  $k > \frac{a^2}{b}$ , les deux points doubles se détachent de la courbe et deviennent des points conjugués ou isolés. En résumé, il y a huit points doubles, savoir quatre points doubles à l'infini, et quatre points doubles sur les deux axes.

Il y a de plus 12 points de rebroussement; ces points sont en effet les centres de courbure aux points de l'ellipse pour lesquels le rayon de courbure est égal à  $k$ ;

les douze points seront tous imaginaires à moins que  $k$  n'ait une valeur entre les limites  $\frac{b^2}{a}$ ,  $\frac{a^2}{b}$  : pour une telle valeur de  $k$  les douze points seront 4 réels, et 8 imaginaires. Pour  $k = \frac{b^2}{a}$  les quatre points réels se réunissent deux à deux aux deux points doubles sur l'axe de  $x$  ; pour  $k = \frac{a^2}{b}$ , ils se réunissent deux à deux aux deux points doubles sur l'axe de  $y$  ; cela s'accorde avec ce qui est dit ci-dessus par rapport aux points doubles. On peut encore trouver que le nombre des points de rebroussement est 12 à moyen des équations

$$A^2 + 3B = 0, \quad B^2 + 3AC = 0, \quad AB - 9C = 0,$$

lesquelles appartiennent chacune à une courbe du quatrième ordre qui passe par les points de rebroussement ; la forme des équations fait voir que ces courbes ont en commun 12 points, et seulement 12 points, d'intersection. C'est là en effet la manière la plus simple de trouver les coordonnées des points de rebroussement ; car ces équations donnent tout de suite

$$A = -3C^{\frac{1}{3}}, \quad B = -3C^{\frac{2}{3}},$$

c'est-à-dire on a les équations

$$x^2 + y^2 = a^2 + b^2 + k^2 - 3(abk)^{\frac{2}{3}},$$

$$b^2x^2 + a^2y^2 = a^2b^2 + (a^2 + b^2)k^2 - 3(abk)^{\frac{4}{3}},$$

qui donnent les coordonnées des douze points de rebroussement.

Pour vérifier que ces points correspondent aux points de l'ellipse pour lesquels le rayon de courbure est égal à  $k$  ; je prends  $a \cos \theta$ ,  $b \sin \theta$  pour les coordonnées d'un point sur l'ellipse, les coordonnées du centre de courbure seront données par

$$ax = (a^2 - b^2) \cos^3 \theta, \quad by = -(a^2 - b^2) \sin^3 \theta,$$

et en supposant que le rayon de courbure soit égal à  $k$ , on a

$$k = \frac{1}{ab} (a^2 \sin^2 \theta + b^2 \cos^2 \theta)^{\frac{3}{2}}.$$

Cela donne

$$a^2 \sin^2 \theta + b^2 \cos^2 \theta = (abk)^{\frac{2}{3}},$$

et de là on déduit

$$(a^2 - b^2) \cos^2 \theta = a^2 - (abk)^{\frac{2}{3}}, \quad (a^2 - b^2) \sin^2 \theta = (abk)^{\frac{2}{3}} - b^2,$$

et de là

$$a^2x^2 = \frac{1}{a^2 - b^2} \left[ a^2 - (abk)^{\frac{2}{3}} \right]^3, \quad b^2y^2 = \frac{1}{a^2 - b^2} \left[ (abk)^{\frac{2}{3}} - b^2 \right]^3,$$

valeurs qui s'accordent en effet avec les valeurs ci-dessus trouvées.

La courbe parallèle à l'ellipse est de l'ordre 8 et il y a 8 points doubles, et 12 points de rebroussement ; elle sera donc de la classe  $56 - 16 - 36 = 4$ . En effet pour mener d'un point donné une tangente à la courbe parallèle on peut décrire avec ce point comme centre un cercle rayon  $k$ , pour mener une tangente commune au cercle et à l'ellipse ; la droite, menée par le point donné, parallèle à la tangente commune sera la tangente cherchée : on peut donc par un point donné mener 4 tangentes à la courbe, ou la courbe parallèle est de la classe 4. Au reste on peut trouver très facilement l'équation de la courbe parallèle en coordonnées tangentielles. Car en représentant par  $\theta$  l'inclinaison d'une tangente quelconque de l'ellipse à l'axe de  $x$ , l'équation de cette tangente sera

$$x \cos \theta + y \sin \theta - \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} = 0,$$

et l'équation de la tangente parallèle de la courbe sera ainsi

$$x \cos \theta + y \sin \theta - k - \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} = 0;$$

donc en représentant cette équation par  $\xi x + \eta y + \zeta = 0$  on aura

$$\xi : \eta : \zeta = \cos \theta : \sin \theta : -k - \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta},$$

et de là

$$\zeta + k\sqrt{\xi^2 + \eta^2} + \sqrt{a^2 \xi^2 + b^2 \eta^2} = 0,$$

laquelle est l'équation dont il s'agit ; la forme rationnelle est

$$(a^2 - k^2)^2 \xi^4 + (b^2 - k^2)^2 \eta^4 + 2(a^2 - k^2)(b^2 - k^2) \xi^2 \eta^2 - 2(a^2 + k^2) \xi^2 \zeta^2 - 2(b^2 + k^2) \eta^2 \zeta^2 + \zeta^4 = 0,$$

équation du quatrième ordre comme cela devait être.

J'ai cru qu'il n'était pas nécessaire de faire voir comment on pourrait à moyen de l'équation tracer la courbe ; en effet on trouve les différentes formes assez facilement par des considérations géométriques, en considérant la courbe comme la développée (ou révolue) de l'ellipse : on se rend compte à ce moyen de ce qui est déjà dit par rapport aux points doubles et aux points de rebroussement.

Je remarque enfin que l'équation d'une normale quelconque de l'ellipse est

$$ax \sin \theta - by \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

(où  $\theta$  est un paramètre arbitraire) : donc en considérant la trajectoire orthogonale de ce système de droites, on obtient

$$(x dx + y dy) \sqrt{b^2 dx^2 + a^2 dy^2} + (a^2 - b^2) dx dy = 0$$

comme équation différentielle de la courbe parallèle à l'ellipse. L'intégrale de cette équation est donc l'équation du huitième ordre (contenant  $k$  comme constante arbitraire) qu'on a ci-dessus trouvée.

Londres, 2 Stone Buildings, 22 Oct. 1860.

## 254.

## SUR LA SURFACE PARALLÈLE À L'ELLIPSOÏDE.

[FROM THE ANNALI DI MATEMATICA PURA ED APPLICATA, (TORTOLINI), TOM. III. (1860), PP. 345–352.]

Il y a pour la surface parallèle à l'ellipsoïde une solution tout à fait semblable à celle pour la courbe parallèle à l'ellipse. En effet soit

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1$$

l'équation de l'ellipsoïde ;  $k$  la distance de la surface parallèle, en prenant cette distance sur la normale extérieure au point  $(X, Y, Z)$  et en écrivant pour abréger

$$\sqrt{\frac{X^2}{a^4} + \frac{Y^2}{b^4} + \frac{Z^2}{c^4}} = \frac{k}{\lambda},$$

on trouve pour les coordonnées  $(x, y, z)$  de l'extrémité de la normale

$$x = X \left(1 + \frac{\lambda}{a^2}\right), \quad y = Y \left(1 + \frac{\lambda}{b^2}\right), \quad z = Z \left(1 + \frac{\lambda}{c^2}\right),$$

et, en substituant pour  $X, Y, Z$  les valeurs données par ces équations, on obtient

$$\begin{aligned} \frac{a^2 x^2}{(a^2 + \lambda)^2} + \frac{b^2 y^2}{(b^2 + \lambda)^2} + \frac{c^2 z^2}{(c^2 + \lambda)^2} &= 1, \\ \frac{x^2}{(a^2 + \lambda)^2} + \frac{y^2}{(b^2 + \lambda)^2} + \frac{z^2}{(c^2 + \lambda)^2} &= \frac{k^2}{\lambda^2}. \end{aligned}$$

Or ces équations donnent

$$\begin{aligned} \frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} &= 1 + \frac{k^2}{\lambda}, \\ \frac{x^2}{(a^2 + \lambda)^2} + \frac{y^2}{(b^2 + \lambda)^2} + \frac{z^2}{(c^2 + \lambda)^2} &= \frac{k^2}{\lambda^2}, \end{aligned}$$



et la seconde équation est la dérivée par rapport à  $\lambda$  de la première. C'est-à-dire on obtient l'équation de la surface parallèle en égalant à zéro le discriminant par rapport à  $\lambda$  de l'équation

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} = 1 + \frac{k^2}{\lambda},$$

ou autrement dit, la surface parallèle est l'enveloppe par rapport à  $\lambda$  de cette équation. En multipliant par  $\lambda(a^2 + \lambda)(b^2 + \lambda)(c^2 + \lambda)$ , on aura une équation du quatrième ordre, dont le discriminant égalé à zéro donne l'équation de la surface : on voit sans peine que cette équation sera de l'ordre 10.

J'avais à peine trouvé cette méthode, quand j'ai appris que le problème était déjà résolu par MM. Salmon et W. Roberts. M. Salmon considère la surface comme lieu des centres des sphères, rayon  $k$ , qui touchent à l'ellipsoïde  $\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1$ , ce qui lui donne tout de suite le théorème que voici ; savoir on obtient l'équation de la surface parallèle, en égalant à zéro le discriminant par rapport à  $\lambda$  du discriminant par rapport à  $(X, Y, Z)$  de l'équation

$$\lambda \left( \frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} - 1 \right) + (X - x)^2 + (Y - y)^2 + (Z - z)^2 - k^2 = 0;$$

cette solution est donnée en passant dans une Note publiée dans le No. de Septembre 1860 des Nouvelles Annales (Terquem et Gerono). En formant le discriminant par rapport à  $X, Y, Z$  on trouve l'équation en  $\lambda$  ci-dessus donnée.

La solution de M. W. Roberts n'avait pas été publiée : il a bien voulu me permettre d'en faire part aux géomètres. En considérant une section circulaire de l'ellipsoïde, l'enveloppe des sphères, rayon  $k$ , ayant leurs centres sur cette section, sera une surface annulaire, et l'enveloppe des surfaces annulaires qui correspondent aux différentes sections circulaires de l'ellipsoïde sera la surface parallèle. Dans la figure (qui est censée située dans le plan des axes le plus grand et le plus petit) soit  $Of$  la trace de la section circulaire centrale,  $Og$  le diamètre perpendiculaire,  $AB$  la trace d'une section quelconque parallèle à la section circulaire centrale : soit de plus  $AB = 2h$ , et  $(\alpha, \gamma)$

les coordonnées du point  $C$  qui est le centre de la section  $AB$ . En prenant pour un moment ce point  $C$  pour origine, et l'axe de  $\eta$  parallèle à l'axe moyen, la surface annulaire sera l'enveloppe de la sphère

$$(\xi - \delta \cos \phi)^2 + (\eta - \delta \sin \phi)^2 + \zeta^2 = k^2,$$

(où  $\phi$  est un paramètre arbitraire) ; donc l'équation de la surface annulaire sera

$$4\delta^2 (\xi^2 + \eta^2) = (\xi^2 + \eta^2 + \zeta^2 + \delta^2 - k^2)^2,$$

ou, en prenant le centre de l'ellipsoïde pour origine, cette équation sera

$$4\delta^2 (\xi^2 + \eta^2 - 2\alpha\xi + \alpha^2) = (\xi^2 + \eta^2 + \zeta^2 + \delta^2 - k^2 - 2\alpha\xi - 2\eta\gamma - \alpha^2 - \gamma^2)^2.$$

Mais par les propriétés de l'ellipse on a  $\gamma = l\alpha$  et  $\delta^2 = m + n\alpha^2$ , où  $l, m, n$  sont des quantités constantes, fonctions des axes  $a, b, c$  de l'ellipsoïde. L'équation de la surface annulaire deviendrait ainsi une équation qui contient le paramètre variable  $\alpha$  au quatrième degré, et en égalant à zéro le discriminant par rapport à  $\alpha$  de cette équation, on obtiendrait l'équation de la surface parallèle. A propos de cette solution je remarque qu'en considérant la surface annulaire, enveloppe des sphères, rayon  $k$ , ayant leurs centres sur l'ellipse

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1, \quad Z = 0,$$

on obtient sans peine le système d'équations

$$\frac{a^2x}{a^2 + \lambda} = X, \quad \frac{b^2y}{b^2 + \lambda} = Y,$$

et de là l'équation de la surface annulaire s'obtient en égalant à zéro le discriminant par rapport à  $\lambda$  de l'équation

$$\frac{a^2x^2}{a^2 + \lambda} + \frac{b^2y^2}{b^2 + \lambda} = 1 + \frac{k^2 - z^2}{\lambda},$$

c'est-à-dire en égalant à zéro le discriminant d'une fonction cubique de  $\lambda$ . Mais en supposant que l'ellipse devient un cercle, l'on n'aura qu'une fonction quadratique de  $\lambda$  ; c'est là pourquoi la surface annulaire considérée par M. Roberts s'exprime sous une forme assez simple pour qu'on puisse s'en servir pour trouver l'équation de la surface parallèle. Soit à présent l'ellipse, section de l'ellipsoïde

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1,$$

par un plan quelconque

$$lX + mY + nZ = p;$$

[254

SUR LA SURFACE PARALLÈLE À L'ELLIPSOÏDE.

161

il est clair que l'équation de la surface annulaire doit se trouver de même à moyen d'une fonction cubique de  $\lambda$ . Pour vérifier cela, je remarque que l'on a à trouver l'enveloppe de la sphère  $(X - x)^2 + (Y - y)^2 + (Z - z)^2 = k^2$ , où les paramètres  $X, Y, Z$  sont liés par les deux équations qui viennent d'être mentionnées : cela donne tout de suite pour la surface annulaire le système des équations

$$\begin{aligned} x - X - \lambda \frac{X}{a^2} - \mu l &= 0, \\ y - Y - \lambda \frac{Y}{b^2} - \mu m &= 0, \\ z - Z - \lambda \frac{Z}{c^2} - \mu n &= 0, \end{aligned}$$

où  $\lambda, \mu$  sont des paramètres arbitraires : ces équations donnent

$$X = \frac{a^2(x - \mu l)}{a^2 + \lambda}, \quad Y = \frac{b^2(y - \mu m)}{b^2 + \lambda}, \quad Z = \frac{c^2(z - \mu n)}{c^2 + \lambda},$$

et de là, en mettant pour abrégier

$$\begin{aligned} \frac{la^2x}{a^2 + \lambda} + \frac{mb^2y}{b^2 + \lambda} + \frac{nc^2z}{c^2 + \lambda} &= P, \\ \frac{la^2}{a^2 + \lambda} + \frac{mb^2}{b^2 + \lambda} + \frac{nc^2}{c^2 + \lambda} &= Q, \end{aligned}$$

on trouve à moyen de l'équation linéaire

$$Q\mu - P = 0, \quad \text{ou} \quad \mu = \frac{P}{Q},$$

et les deux autres équations donnent alors

$$\begin{aligned} \frac{a^2(x - \mu l)^2}{(a^2 + \lambda)^2} + \frac{b^2(y - \mu m)^2}{(b^2 + \lambda)^2} + \frac{c^2(z - \mu n)^2}{(c^2 + \lambda)^2} &= 1, \\ \frac{(\lambda x + a^2\mu l)^2}{(a^2 + \lambda)^2} + \frac{(\lambda y + b^2\mu m)^2}{(b^2 + \lambda)^2} + \frac{(\lambda z + c^2\mu n)^2}{(c^2 + \lambda)^2} &= k^2, \end{aligned}$$

où  $\mu$  est censé dénoter la valeur  $\frac{P}{Q}$ . Je déduis de là les équations

$$\begin{aligned} \frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} - 1 &= \frac{\mu P}{\lambda}, \\ \frac{x^2}{(a^2 + \lambda)^2} + \frac{y^2}{(b^2 + \lambda)^2} + \frac{z^2}{(c^2 + \lambda)^2} \\ - \frac{2\mu}{\lambda} \left( \frac{a^2lx}{(a^2 + \lambda)^2} + \frac{b^2my}{(b^2 + \lambda)^2} + \frac{c^2nz}{(c^2 + \lambda)^2} \right) + \frac{\mu^2}{\lambda^2} \left( \frac{a^4l^2}{(a^2 + \lambda)^2} + \frac{b^4m^2}{(b^2 + \lambda)^2} + \frac{c^4n^2}{(c^2 + \lambda)^2} \right) &= \frac{k^2}{\lambda^2}. \end{aligned}$$

Soient  $P', Q'$  les dérivées de  $P, Q$  par rapport à  $\lambda$ , on a

$$-P' = \frac{la^2x}{(a^2 + \lambda)^2} + \frac{mb^2y}{(b^2 + \lambda)^2} + \frac{nc^2z}{(c^2 + \lambda)^2},$$

$$-Q' = \frac{la^2}{(a^2 + \lambda)^2} + \frac{mb^2}{(b^2 + \lambda)^2} + \frac{nc^2}{(c^2 + \lambda)^2},$$

et de là aussi

$$Q + \lambda Q' = \frac{a^2l^2}{(a^2 + \lambda)^2} + \frac{b^2m^2}{(b^2 + \lambda)^2} + \frac{c^2n^2}{(c^2 + \lambda)^2}.$$

À moyen de ces équations, et en substituant pour  $\mu$  la valeur  $\frac{P}{Q}$ , les deux équations ci-dessus données deviennent

$$\left( \frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} - 1 - \frac{k^2}{\lambda} \right) Q + \frac{P^2}{\lambda} = 0,$$

$$\left( \frac{x^2}{(a^2 + \lambda)^2} + \frac{y^2}{(b^2 + \lambda)^2} + \frac{z^2}{(c^2 + \lambda)^2} - \frac{k^2}{\lambda^2} \right) Q - 2 \frac{PP'}{\lambda} + (Q + \lambda Q') \frac{P^2}{Q\lambda^2} = 0;$$

en différentiant la première équation par rapport à  $\lambda$  on obtient

$$\begin{aligned} & - \left( \frac{x^2}{(a^2 + \lambda)^2} + \frac{y^2}{(b^2 + \lambda)^2} + \frac{z^2}{(c^2 + \lambda)^2} - \frac{k^2}{\lambda^2} \right) Q \\ & + \left( \frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} - 1 - \frac{k^2}{\lambda} \right) Q' + \frac{2PP'}{\lambda} - \frac{P^2}{\lambda^2} = 0, \end{aligned}$$

et cette équation se réduit à une identité à moyen des deux équations : donc l'équation de la surface annulaire s'obtient en égalant à zéro le discriminant par rapport à  $\lambda$  de l'équation

$$\left( \frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} - 1 - \frac{k^2}{\lambda} \right) Q + \frac{P^2}{\lambda} = 0,$$

c'est-à-dire, en substituant pour  $P, Q$  leurs valeurs, l'équation sera

$$\begin{aligned} & \left( \frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} - 1 - \frac{k^2}{\lambda} \right) \left( \frac{la^2}{a^2 + \lambda} + \frac{mb^2}{b^2 + \lambda} + \frac{nc^2}{c^2 + \lambda} \right) \\ & + \frac{1}{\lambda} \left( \frac{a^2lx}{a^2 + \lambda} + \frac{b^2my}{b^2 + \lambda} + \frac{c^2nz}{c^2 + \lambda} - p \right)^2 = 0. \end{aligned}$$

Dans cette équation, en réunissant les termes

$$\frac{a^2}{a^2 + \lambda} - \frac{a^2l^2}{a^2 + \lambda} - \frac{1}{\lambda} \frac{a^4l^2x^2}{(a^2 + \lambda)^2},$$

on voit que ces termes se réduisent à

$$\frac{a^2}{(a^2 + \lambda)} - \frac{a^4}{\lambda(a^2 + \lambda)^2},$$

[254

SUR LA SURFACE PARALLÈLE À L'ELLIPSOÏDE.

163

et de même pour les termes semblables en  $y, z$ . Donc en multipliant par

$$\lambda(a^2 + \lambda)(b^2 + \lambda)(c^2 + \lambda)$$

les dénominateurs disparaissent, et l'équation est réellement de l'ordre 3 par rapport à  $\lambda$ .

La section sera circulaire en supposant

$$l = \sqrt{\frac{1}{b^2} - \frac{1}{a^2}}, \quad m = 0, \quad n = \sqrt{\frac{1}{c^2} - \frac{1}{b^2}};$$

cela donne

$$\frac{a^2 l^2}{a^2 + \lambda} + \frac{b^2 m^2}{b^2 + \lambda} + \frac{c^2 n^2}{c^2 + \lambda} = \frac{\frac{a^2}{b^2} - 1}{a^2 + \lambda} + \frac{1 - \frac{c^2}{b^2}}{c^2 + \lambda} = \frac{(a^2 - c^2)(b^2 + \lambda)}{b^2(a^2 + \lambda)(c^2 + \lambda)},$$

l'équation en  $\lambda$  devient ainsi

$$\begin{aligned} & \frac{(a^2 - c^2)(b^2 + \lambda)}{b^2(a^2 + \lambda)(c^2 + \lambda)} \left[ \frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} - 1 - \frac{k^2}{\lambda} \right] \\ & + \frac{1}{\lambda} \left[ \frac{a^2 l x}{a^2 + \lambda} + \frac{c^2 n z}{c^2 + \lambda} - p \right]^2 = 0, \end{aligned}$$

laquelle se réduit à

$$\begin{aligned} & x^2 [\lambda(a^2 - b^2) + c^2(a^2 - b^2)] + y^2 \lambda(a^2 - c^2) + z^2 [\lambda(a^2 - c^2) + a^2(b^2 - c^2)] \\ & + 2acxz \sqrt{a^2 - b^2} \sqrt{b^2 - c^2} \\ & - 2bp [ax(c^2 + \lambda)\sqrt{a^2 - b^2} + cz(a^2 + \lambda)\sqrt{b^2 - c^2}] \\ & - 2(a^2 - c^2)(b^2 + \lambda)(c^2 + \lambda) + b^2 y^2 (a^2 + \lambda)(c^2 + \lambda) = 0, \end{aligned}$$

équation de l'ordre 2 par rapport à  $(x, y, z)$ , par rapport à  $p$ , et par rapport à  $\lambda$ . En égalant à zéro le discriminant par rapport à  $\lambda$ , on obtiendrait une équation du quatrième ordre par rapport aux coordonnées  $(x, y, z)$ , et par rapport à  $p$ ; on aurait ainsi l'équation de la surface annulaire de M. Roberts, rapportée aux axes de l'ellipsoïde, et en termes des quantités  $a, b, c, k$  et du paramètre  $p$  qui donne la position du plan de la section circulaire.

M. Roberts a donné (*Comptes Rendus* Nov. 14, 1859) le théorème que voici qui se rapporte à une surface primitive quelconque :—“Soit  $D$  la surface semblable à la primitive en multipliant par 2 ses rayons vecteurs, et soit  $P$  la surface parallèle ou équidistante de  $D$  par la longueur constante  $k$ , l'équation qui résulte de la substitution de  $\sqrt{x^2 + y^2 + z^2}$  au lieu de  $k$  dans l'équation de  $P$ , coïncide avec l'équation

de la première dérivée négative de la surface primitive”—où je rappelle que dans la théorie des surfaces dérivées de M. Hirst la première dérivée négative est la surface enveloppe des plans conduits par les points de la surface primitive perpendiculairement aux rayons menés par un point quelconque. Pour démontrer le théorème, prenons  $f(X, Y, Z) = 0$  pour l'équation de la surface primitive : en supposant que  $(X, Y, Z)$  soient les coordonnées d'un point de cette surface,  $(2X, 2Y, 2Z)$  seront les coordonnées d'un point de la surface  $D$ , et la surface parallèle à  $D$  sera l'enveloppe des sphères

$$(x - 2X)^2 + (y - 2Y)^2 + (z - 2Z)^2 = k^2,$$

avec la relation  $f(X, Y, Z) = 0$  entre les paramètres. Or on peut avant d'effectuer l'élimination écrire  $\sqrt{x^2 + y^2 + z^2}$  au lieu de  $k$ , l'équation devient ainsi

$$X(x - X) + Y(y - Y) + Z(z - Z) = 0,$$

avec cette même relation  $f(X, Y, Z) = 0$  entre les paramètres ; or cette équation est celle d'un plan conduit par le point  $(X, Y, Z)$  perpendiculairement au rayon mené par l'origine des coordonnées : et ce dernier système donne ainsi l'équation de la surface dérivée.

L'équation de la surface parallèle de l'ellipsoïde est trouvée en égalant à zéro le discriminant par rapport à  $\lambda$  de l'équation

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} = 1 + \frac{k^2}{\lambda};$$

donc, en écrivant dans cette équation  $x^2 + y^2 + z^2$  au lieu de  $k^2$ , et  $4a^2, 4b^2, 4c^2$  au lieu de  $a^2, b^2, c^2$ , on doit obtenir l'équation de la dérivée de l'ellipsoïde, les rayons étant menés par le centre. Pour me conformer à la notation de mon Mémoire “Sur la surface qui est l'enveloppe des plans conduits par les points d'un ellipsoïde perpendiculairement aux rayons menés par le centre” (*Journal*, tom. II. 1859 [250]) j'écris  $-2\theta$  au lieu de  $\lambda$  : l'équation en  $\lambda$  devient ainsi

$$\frac{x^2}{4a^2 - 2\theta} + \frac{y^2}{4b^2 - 2\theta} + \frac{z^2}{4c^2 - 2\theta} = 1 - \frac{x^2 + y^2 + z^2}{2\theta},$$

laquelle se réduit tout de suite à

$$\frac{x^2}{2 - \frac{\theta}{a^2}} + \frac{y^2}{2 - \frac{\theta}{b^2}} + \frac{z^2}{2 - \frac{\theta}{c^2}} = \theta,$$

et la surface dérivée est l'enveloppe de cette équation en  $\theta$ , résultat trouvé dans le mémoire que je viens de mentionner.

Mais je dois à M. Roberts la remarque que réciproquement l'équation en  $\lambda$  pour la surface parallèle peut se déduire des formules de ce mémoire. En effet en prenant  $a \cos \alpha, b \cos \beta, c \cos \gamma$  pour les coordonnées d'un point de l'ellipsoïde (on a comme à l'ordinaire  $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$ ) l'équation du plan perpendiculaire au rayon par le centre sera

$$ax \cos \alpha + by \cos \beta + cz \cos \gamma = a^2 \cos^2 \alpha + b^2 \cos^2 \beta + c^2 \cos^2 \gamma,$$

[254

SUR LA SURFACE PARALLÈLE À L'ELLIPSOÏDE.

165

et l'enveloppe de cette équation est l'enveloppe par rapport à  $\theta$  de l'équation

$$\frac{x^2}{2 - \frac{\theta}{a^2}} + \frac{y^2}{2 - \frac{\theta}{b^2}} + \frac{z^2}{2 - \frac{\theta}{c^2}} = \theta.$$

Donc en général l'enveloppe de

$$l \cos \alpha + m \cos \beta + n \cos \gamma = p \cos^2 \alpha + q \cos^2 \beta + r \cos^2 \gamma,$$

(où comme auparavant  $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$ ) sera l'enveloppe par rapport à  $\theta$  de l'équation

$$\frac{l^2}{2p - \theta} + \frac{m^2}{2q - \theta} + \frac{n^2}{2r - \theta} = \theta.$$

Or la surface parallèle à l'ellipsoïde est l'enveloppe de

$$(X - x)^2 + (Y - y)^2 + (Z - z)^2 = k^2,$$

ou, en écrivant  $a \cos \alpha$ ,  $b \cos \beta$ ,  $c \cos \gamma$  au lieu de  $X, Y, Z$ , cette surface est l'enveloppe de

$$l \cos \alpha + m \cos \beta + n \cos \gamma = p \cos^2 \alpha + q \cos^2 \beta + r \cos^2 \gamma,$$

en posant  $\rho = x^2 + y^2 + z^2 - k^2$  et

$$l = 2ax, \quad m = 2by, \quad n = 2cz, \quad p = a^2, \quad q = b^2, \quad r = c^2.$$

Donc cette surface sera l'enveloppe par rapport à  $\theta$  de l'équation

$$\frac{4a^2x^2}{2a^2 - \theta} + \frac{4b^2y^2}{2b^2 - \theta} + \frac{4c^2z^2}{2c^2 - \theta} = \rho + \theta,$$

et en écrivant  $(2\rho - k^2) - \theta = 2\lambda$  cette équation devient

$$\frac{a^2x^2}{a^2 + \lambda} + \frac{b^2y^2}{b^2 + \lambda} + \frac{c^2z^2}{c^2 + \lambda} = 1,$$

c'est-à-dire

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} = 1 + \frac{k^2}{\lambda},$$

ou enfin

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} = 1 + \frac{k^2}{\lambda},$$

ce qu'il s'agissait de faire voir. Je réserve à une autre occasion la discussion de la forme, et des singularités de la forme, et des singularités de la surface parallèle à l'ellipsoïde.

2, Stone Buildings, W. C., Londres, 7 Nov. 1860.

## 255.

## ON A PROBLEM OF DOUBLE PARTITIONS.

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XX. (1860), PP. 337–341.]

IF  $a + b + c + \dots = m$ ,  $\alpha + \beta + \gamma + \dots = \mu$  (the quantities being all positive integer numbers, not excluding zero), then  $(a, \alpha) + (b, \beta) + (c, \gamma) + \dots$  is considered as a partition of  $(m, \mu)$ : and the partible quantity  $(m, \mu)$ , and parts  $(a, \alpha)$ ,  $(b, \beta)$ , &c. being each of them composed of two elements, such partition is said to belong to the theory of Double Partitions. The subject (so far as I am aware) has hardly been considered except by Professor Sylvester, and it is greatly to be regretted that only an outline of his valuable researches has been published: the present paper contains the demonstration of a theorem, due to him, by which (subject to certain restrictions) the question of Double Partitions is made to depend upon the ordinary theory of Single Partitions.

Let the question be proposed, “In how many ways can  $(m, \mu)$  be made up of the given parts  $(a, \alpha)$ ,  $(b, \beta)$ ,  $(c, \gamma)$ , &c.” under the following conditions (which are, it will be seen, necessary in the demonstration of the theorem constituting the solution), viz.

$\frac{a}{\alpha}$ ,  $\frac{b}{\beta}$ ,  $\frac{c}{\gamma}$ , &c. are unequal fractions, each in its least terms,

and

$\alpha$ ,  $\beta$ ,  $\gamma$ , &c., are each less than  $\mu + 2$ .

The number of partitions is

$$= \text{coeff. } x^m y^\mu \text{ in } \frac{1}{(1 - x^a y^\alpha)(1 - x^b y^\beta)(1 - x^c y^\gamma) \dots},$$

the fraction being developed in ascending powers of  $x, y$ .



[255

ON A PROBLEM OF DOUBLE PARTITIONS.

167

Considering the fraction as a function of  $y$ , it may be expressed as a sum of partial fractions in the form

$$\frac{A(x, y)}{1 - x^a y^\alpha} + \frac{B(x, y)}{1 - x^b y^\beta} + \frac{C(x, y)}{1 - x^c y^\gamma} + \dots,$$

where

$$\begin{array}{ll} A(x, y) \text{ is rational in } x, \text{ rational and integral of degree } \alpha - 1 \text{ in } y, \\ B(x, y) & \text{,,} & \beta - 1 \text{ ,,} \\ C(x, y) & \text{,,} & \gamma - 1 \text{ ,,} \\ & \&c. \end{array}$$

To find  $A(x, y)$  we have, when  $y = x^{-\frac{a}{\alpha}}$ ,

$$A(x, y) = \frac{1}{(1 - x^b y^\beta)(1 - x^c y^\gamma) \dots},$$

or what is the same thing, we have

$$A(x, x^{-\frac{a}{\alpha}}) = \frac{1}{(1 - x^{b-\frac{a\beta}{\alpha}})(1 - x^{c-\frac{a\gamma}{\alpha}}) \dots}.$$

This in fact determines  $A(x, y)$ ; for the right-hand side of the equation may be reduced to the form

$$A(x, x^{-\frac{a}{\alpha}}) = \frac{A_0 + A_1 x^{\frac{a}{\alpha}} + \dots + A_{\alpha-1} x^{\frac{(\alpha-1)a}{\alpha}}}{(1 - x^{ab-a\beta})(1 - x^{ac-a\gamma}) \dots},$$

where  $A_0, A_1, \dots, A_{\alpha-1}$  are rational functions of  $x$ ; to do this, it is only necessary (taking  $\omega$  an imaginary  $\alpha$ -th root of unity) to multiply the numerator and denominator by

$$\Pi(1 - \omega x^{b-\frac{a\beta}{\alpha}}) \Pi(1 - \omega x^{c-\frac{a\gamma}{\alpha}}) \dots,$$

where  $\Pi$  denotes the product of the factors corresponding to the  $\alpha - 1$  values of  $\omega$ ; the denominator is thus converted into

$$(1 - x^{ab-a\beta})(1 - x^{ac-a\gamma}) \dots,$$

which is of the form in question; and the numerator becomes a rational function of  $x$  and  $x^{\frac{a}{\alpha}}$ , integral as regards  $x^{\frac{a}{\alpha}}$ , and therefore at once expressible in the form in question. And the equation, viz.

$$A(x, x^{-\frac{a}{\alpha}}) = \frac{A_0 + A_1 x^{\frac{a}{\alpha}} + \dots + A_{\alpha-1} x^{\frac{(\alpha-1)a}{\alpha}}}{(1 - x^{ab-a\beta})(1 - x^{ac-a\gamma}) \dots},$$

remains true if instead of  $x^{-\frac{a}{\alpha}}$  we write  $\omega x^{-\frac{a}{\alpha}}$ ; in fact, instead of writing in the first instance  $y = x^{-\frac{a}{\alpha}}$ , it would have been allowable to write  $y = \omega x^{-\frac{a}{\alpha}}$ ,  $\omega$  being any  $\alpha$ -th

root, real or imaginary, of unity. Hence recollecting that  $A(x, y)$  is a rational and integral function of the degree  $\alpha - 1$  in  $y$ , the equation

$$A(x, y) = A_0 + A_1 y + \dots + A_{\alpha-1} y^{\alpha-1}$$

which is true for the  $\alpha$  values  $\omega x^{-\frac{a}{\alpha}}$  of  $y$ , must be true identically; or this equation gives the value of  $A(x, y)$ . And the values of  $B(x, y)$ ,  $C(x, y)$ , &c. are of course of the like form.

Now consider the term

$$\frac{A(x, y)}{1 - x^a y^\alpha},$$

where  $A(x, y)$  is a rational and integral function of the degree  $\alpha - 1$  in  $y$ , and  $\frac{a}{\alpha}$  is by hypothesis a fraction in its least terms. The coefficient therein of  $x^m y^\mu$  (the fraction being developed in ascending powers of  $x, y$ ) is

$$= \text{coeff. } \frac{1}{x^m y^\mu} \text{ in } \frac{A(x, y)}{1 - x^a y^\alpha},$$

(the fraction being developed in ascending powers of  $x, y$ ). In fact the two fractions only differ by a wholly irrational function of  $x$ , as is at once obvious by developing

$$\frac{1}{1 - x^a y^\alpha} \text{ in ascending powers of } y.$$

We have, separating the integral part,

$$\frac{A(x, y)}{1 - x^a y^\alpha} = U + \frac{A(x, x^{-\frac{a}{\alpha}})}{1 - x^a y^\alpha},$$

where  $U$  is a rational and integral function of the degree  $\alpha - 2$  in  $y$ . But  $\alpha$  being by hypothesis  $< \mu + 2$ , or what is the same thing,  $\alpha - 2 < \mu$ ,  $U$  does not contain any term of the form  $x^m y^\mu$ , and therefore

$$\begin{aligned} & \text{coeff. } x^m y^\mu \text{ in } \frac{A(x, y)}{1 - x^a y^\alpha} \\ &= \text{do. in } \frac{A(x, x^{-\frac{a}{\alpha}})}{1 - x^a y^\alpha}, \end{aligned}$$

and this last is

$$\begin{aligned} &= \text{coeff. } x^m \text{ in } x^{a\mu} A(x, x^{-\frac{a}{\alpha}}), \\ &= \text{coeff. } x^{m-a\mu} \text{ in } A(x, x^{-\frac{a}{\alpha}}), \\ &= \text{coeff. } x^{\alpha m - a\mu} \text{ in } A(x^\alpha, x^{-a}); \end{aligned}$$

and from the foregoing equation

$$A(x, x^{-\frac{a}{\alpha}}) = \frac{1}{(1 - x^{b-\frac{a\beta}{\alpha}})(1 - x^{c-\frac{a\gamma}{\alpha}}) \dots},$$

this is

$$= \text{coeff. } x^{\alpha m - a\mu} \text{ in } \frac{1}{(1 - x^{\alpha b - a\beta})(1 - x^{\alpha c - a\gamma}) \dots}.$$

The last-mentioned expression is thus the value of

$$\text{coeff. } x^m y^\mu \text{ in } \frac{A(x, y)}{1 - x^a y^\alpha};$$

and hence, THEOREM,

$$\begin{aligned} & \text{coeff. } x^m y^\mu \text{ in } \frac{1}{(1 - x^a y^\alpha)(1 - x^b y^\beta)(1 - x^c y^\gamma) \dots} \\ &= \text{coeff. } x^{\alpha m - a\mu} \text{ in } \frac{1}{(1 - x^{\alpha b - a\beta})(1 - x^{\alpha c - a\gamma}) \dots} \\ &+ \text{coeff. } x^{\beta m - b\mu} \text{ in } \frac{1}{(1 - x^{\beta a - b\alpha})(1 - x^{\beta c - b\gamma}) \dots} \\ &+ \text{coeff. } x^{\gamma m - c\mu} \text{ in } \frac{1}{(1 - x^{\gamma a - c\alpha})(1 - x^{\gamma b - c\beta}) \dots} \\ &+ \&c., \end{aligned}$$

the fraction on the left-hand side being expanded in ascending powers of  $x, y$ , and those on the right-hand side being expanded in ascending powers of  $x$ , and the data satisfying the above-mentioned conditions. The number of partitions of  $(m, \mu)$  is thus found to be equal to the expression on the right-hand side. It is to be noticed that on the right-hand side, when any of the indices  $\alpha m - a\mu, \beta m - b\mu, \dots$  is negative, the corresponding coefficient vanishes; and that when the index of the power of  $x$  in any factor of a denominator is negative, e.g. if  $\alpha b - a\beta = -p$ , then (in order to develop in ascending powers of  $x$ ) we must in the place of  $\frac{1}{1 - x^{-p}} = \frac{1}{-x^{-p}}$  write  $\frac{1 \cdot x^p}{x^p - 1}, = -\frac{x^p}{1 - x^p}$ , and develop in the form  $-(x^p + x^{2p} + x^{3p} + \dots)$ . The right-hand side is thus seen to be the sum of a series of positive or negative numbers, each of which taken positively denotes the number of the single partitions of a given partible number into given parts.

If, using a term of Professor Sylvester's, we say that

$$\text{coeff. } x^m \text{ in } \frac{1}{(1 - x^a)(1 - x^b) \dots}$$

(where  $m, a, b, c, \dots$  are positive or negative integers, and the fraction is developed in ascending powers of  $x$ ) is

$$\begin{aligned} &= \text{Denumerant of } m \text{ in respect to the elements } (a, b, c, \dots), \text{ say} \\ &= \text{Denumerant } (m; a, b, c, \dots), \end{aligned}$$

then when  $m, a, b, c, \dots$  are positive, but not otherwise, Denumerant  $(m; a, b, c, \dots)$  denotes the number of ways in which  $m$  can be made up of the parts  $a, b, c, \dots$ ; and the foregoing result shows that the number of ways in which  $(m, \mu)$  can be made up of the parts  $(a, \alpha), (b, \beta), (c, \gamma), \&c.$  is equal to the sum

$$\begin{aligned} &\text{Denumerant } (\alpha m - a\mu; \alpha b - a\beta, \alpha c - a\gamma, \dots) \\ &+ \text{Denumerant } (\beta m - b\mu; \beta a - b\alpha, \beta c - b\gamma, \dots) \\ &+ \text{Denumerant } (\gamma m - c\mu; \gamma a - c\alpha, \gamma b - c\beta, \dots) \\ &+ \&c. \end{aligned}$$

But, as appears from what precedes, a denumerant may be equal to zero, or may denote a number of partitions taken negatively; and it is not allowable, in the place, e.g., of the first denumerant, to write *simpliciter*, number of partitions of  $\alpha m - a\mu$  in respect of  $\alpha b - a\beta, \alpha c - a\gamma, \&c.$  The notion of a Denumerant is, in fact, an important generalization of the notion of a number of partitions.

2, Stone Buildings, W. C., October 4, 1860.

## 256.

## ON A SYSTEM OF ALGEBRAIC EQUATIONS.

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XX. (1860), PP. 341, 342.]

THE determination of  $a, b, c$  from the system of equations

$$a^2 + bc = \lambda,$$

$$b^2 + ca = \mu,$$

$$c^2 + ab = \nu,$$

in the case where  $\lambda, \mu, \nu$  have the values 16, 17, and 18 respectively, is the problem known as Colonel Titus's Arithmetical Problem. See Maseres' *Tracts on the Resolution of affected Algebraic Equations*, Lond. 1800. If for shortness we put

$$\sigma = \nu - c^2,$$

then the third equation gives  $b = \frac{\sigma}{a}$ ; and substituting this value of  $b$  in the two other equations, we have

$$a^2 + \frac{c\sigma}{a} = \lambda,$$

$$\frac{\sigma^2}{a^2} + ca = \mu;$$

or what is the same thing,

$$a^3 - \lambda a + \sigma c = 0,$$

$$ca^3 - \mu a^2 + \sigma^2 = 0;$$

and from these equations, eliminating  $a$ , we have

$$\sigma^4 - 3c^2\sigma^3 + (3c^4 - 2\lambda\mu)\sigma^2 - c^2(c^4 - \lambda\mu)\sigma + c^4\lambda\mu - c^2(\lambda^3 + \mu^3) + \lambda^2\mu^2 = 0,$$

where  $\sigma = \nu - c^2$ . The equation in  $c^2$  is thus of the fourth order; and in like manner, if instead of  $c^2$  we take  $a$  as the unknown quantity, and substitute therefore for  $c^2$  its value  $\nu - \sigma$ , the equation in  $\sigma$  will be also of the fourth order: and effecting the reduction, this equation is

$$8\sigma^4 - 12\nu\sigma^3 + (6\nu^2 - 2\lambda\mu)\sigma^2 + (\nu^3 + \lambda\nu - \mu^3 - \nu^3)\sigma + (\nu\lambda - \mu^2)(\nu\mu - \lambda^2) = 0.$$

It may be remarked that if  $\sigma = 0$ , then  $a$  or  $b$  vanishes; and therefore, from the original equations,  $\nu\lambda - \mu^2 = 0$ , or  $\nu\mu - \lambda^2 = 0$ , which agrees with the result afforded by the foregoing equation in  $\sigma$ . Again, if  $\sigma = \nu$ , then  $c = 0$ ; and therefore, from the original equations,  $\nu^2 - \lambda\mu = 0$ . The left-hand side of the equation in  $\sigma$ , writing therein  $\sigma = \nu$ , should therefore contain the factor  $\nu^2 - \lambda\mu$ ; its value in fact is  $\nu^4 - 2\lambda\mu\nu^2 + \lambda^2\mu^2$ , or  $(\nu^2 - \lambda\mu)^2$ .

If in the original equations we write  $a = \frac{x}{z}$ ,  $b = \frac{y}{z}$ , the equations become

$$x^2 + cyz - \lambda z^2 = 0,$$

$$y^2 + czx - \mu z^2 = 0,$$

$$(c^2 - \nu)z^2 + xy = 0,$$

which are three homogeneous equations of the second order; from which, if the variables  $x, y, z$  are eliminated, we have the required equation in  $c$ . And it would not, I think, be difficult, from the known formula for the general case, to deduce the foregoing result corresponding to the very particular case which is here in question.

2, Stone Buildings, W. C., September 25, 1860.

## 257.

ON THE CUBIC CENTRES OF A LINE WITH RESPECT TO THREE LINES  
AND A LINE.

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XX. (1860), PP. 418–423.]

CONSIDER a line  $L$  in relation to the three lines  $X, Y, Z$  and the line  $I$ : through the point of intersection of the lines  $X, L$ , draw any line meeting the lines  $I, Y, Z$ , and let the harmonic of the intersection with  $I$ , in relation to the intersections with  $Y, Z$ , be  $\xi$ ; then the locus of the point  $\xi$  is a conic passing through the points  $YI, ZI, YZ$ .

If, in like manner, through the point of intersection of the lines  $Y, L$ , there is drawn any line meeting the lines  $I, Z, X$ , and the harmonic of the intersection with  $I$ , in relation to the intersections with  $Z, X$ , is called  $\eta$ , the locus of the point  $\eta$  is a conic passing through the points  $ZI, XI, ZX$ .

And so, if through the point of intersection of the lines  $Z, L$  there is drawn any line meeting the lines  $I, X, Y$ , and the harmonic of the intersection with  $I$ , in relation to the intersections with  $X, Y$ , is called  $\zeta$ , then the locus of  $\zeta$  is a conic passing through the points  $XI, YI, XY$ .

The pairs of conics, viz. the second and third, third and first, first and second conics, have obviously in common the points  $XI, YI, ZI$  respectively. They besides intersect all three of them in three points, which may be termed the cubic centres of the line  $L$  in relation to the lines  $X, Y, Z$  and the line  $I$ .

The line  $L$  may be such that two of the three cubic centres coincide; the locus of the coincident centres is in this case a conic which touches the lines  $X, Y, Z$  harmonically in regard to the line  $I$ ; that is, it touches each of the three lines in the point which is the harmonic of its intersection with  $I$  in relation to its intersections with the other two lines.

Except that the line  $I$  is there taken to be infinity, the foregoing theorems occur in Plücker's *System der analytischen Geometrie* (Berlin, 1835), p. 177 et seq.; and they play an important part in his classification of curves of the third order (see p. 220 et seq.). It is, I think, an omission that he has not sought for the curve which is the envelope of the line  $L$  in the above-mentioned case of the two coincident centres: I find that the envelope is a curve of the fourth order, having four-pointic contact with the lines  $X, Y, Z$  harmonically in regard to the line  $I$ ; viz., if the equations of the lines  $X, Y, Z$  are  $x = 0, y = 0, z = 0$  respectively, and the equation of the line  $I$  is  $x + y + z = 0$ , then the equation of the envelope in question is

$$\sqrt{x} + \sqrt{y} + \sqrt{z} = 0,$$

a result which is also interesting as exhibiting a geometrical construction of the curve represented by this equation.

The investigation of the series of theorems is as follows; taking

$$\begin{array}{lll} x = 0 & \text{for the equation of } X, \\ y = 0 & \text{„} & Y, \\ z = 0 & \text{„} & Z, \\ x + y + z = 0 & \text{„} & I, \\ \lambda x + \mu y + \nu z = 0 & \text{„} & L, \end{array}$$

then, first, in order to find the curve which is the locus of  $\xi$ , the coordinates of the point  $XL$  are given by  $x : y : z = 0 : \nu : -\mu$ ; or if, as it is convenient to do, we take  $X, Y, Z$  (instead of  $x, y, z$ ) for current coordinates, by  $X : Y : Z = 0 : \nu : -\mu$ . Hence taking  $x, y, z$  as the coordinates of  $\xi$ , the equation of the line through  $XL, \xi$  is

$$\begin{vmatrix} X, & Y, & Z \\ x, & y, & z \\ 0, & \nu, & -\mu \end{vmatrix} = 0,$$

viz.

$$X(\mu y + \nu z) - x(\mu Y + \nu Z) = 0;$$

and at the point where this line meets the line  $I$ , the equation whereof is

$$X + Y + Z = 0,$$

we have

$$(Y + Z)(\mu y + \nu z) + x(\mu Y + \nu Z) = 0;$$

that is

$$Y(\mu x + \mu y + \nu z) + Z(\nu x + \mu y + \nu z) = 0.$$

Hence this line, and the line

$$Yz - Zy = 0,$$



[257

WITH RESPECT TO THREE LINES AND A LINE.

175

with the lines

$$Y = 0, \quad Z = 0,$$

are the lines which pass through the point  $YZ$  and the four harmonic points, and they form therefore a harmonic pencil; or we have

$$y(\mu x + \mu y + \nu z) - z(\nu x + \mu y + \nu z) = 0,$$

or, what is the same thing,

$$(\mu y - \nu z)(x + y + z) + 2yz(\nu - \mu) = 0,$$

as the locus of the point  $\xi$ : the locus is therefore a conic passing through the points  $YI, ZI, YZ$ .

The equations of the conics which are the loci of  $\xi, \eta, \zeta$  respectively, are therefore

$$\begin{aligned} U &= (\mu y - \nu z)(x + y + z) + 2yz(\nu - \mu) = 0, \\ V &= (\nu z - \lambda x)(x + y + z) + 2zx(\lambda - \nu) = 0, \\ W &= (\lambda x - \mu y)(x + y + z) + 2xy(\mu - \lambda) = 0; \end{aligned}$$

and the identical equation,

$$U\lambda x + V\mu y + W\nu z = 0,$$

shows that these conics have three points of intersection in common. The three equations, and a fourth one to which they give rise, may be written

$$\begin{aligned} \frac{z}{y} - \frac{\nu}{\mu} + \frac{2(\mu - \nu)}{x + y + z} &= 0, \\ \frac{x}{z} - \frac{\lambda}{\nu} + \frac{2(\nu - \lambda)}{x + y + z} &= 0, \\ \frac{y}{x} - \frac{\mu}{\lambda} + \frac{2(\lambda - \mu)}{x + y + z} &= 0, \\ \frac{\nu - \mu}{x} + \frac{\lambda - \nu}{y} + \frac{\mu - \lambda}{z} &= 0, \end{aligned}$$

and each of these is the equation of a conic passing through the three cubic centres.

If two of the three centres coincide, then the conics all touch at the coincident centres. Consider the first and second conics: these intersect at the point  $z = 0, x + y + z = 0$ ; and the line  $x + y + z = 2kz$ , if  $k$  be properly determined, or what is the same thing, the line  $x + y + z = \frac{2(\nu - \mu)}{\theta + \nu} z$ , if  $\theta$  is properly determined, will be a line passing through the last-mentioned point and one of the other points of intersection:  $k$  or  $\theta$  will of course be determined by a cubic equation; and if this has a pair of equal

roots, the conics will touch. But the equation of the line, combined with those of the two conics, gives

$$x : y : z = \frac{1}{\theta + \lambda} : \frac{1}{\theta + \mu} : \frac{1}{\theta + \nu},$$

and substituting these values in the equation of the line, we have

$$\frac{1}{\theta + \lambda} + \frac{1}{\theta + \mu} + \frac{1}{\theta + \nu} - \frac{2}{\theta} = 0,$$

which is (as it should be) a cubic equation in  $\theta$ .

If the equation in  $\theta$  has equal roots, then

$$\frac{1}{(\theta + \lambda)^2} + \frac{1}{(\theta + \mu)^2} + \frac{1}{(\theta + \nu)^2} - \frac{2}{\theta^2} = 0;$$

and putting in these two equations,

$$x = \frac{m}{\theta + \lambda}, \quad y = \frac{m}{\theta + \mu}, \quad z = \frac{m}{\theta + \nu},$$

we have

$$\begin{aligned} x + y + z - \frac{2m}{\theta} &= 0, \\ x^2 + y^2 + z^2 - \frac{2m^2}{\theta^2} &= 0; \end{aligned}$$

whence eliminating  $m$ ,

$$(x + y + z)^2 = 2(x^2 + y^2 + z^2);$$

that is

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0;$$

or, what is the same thing,

$$\sqrt{x} + \sqrt{y} + \sqrt{z} = 0,$$

for the equation of the locus of the coincident centres: such locus is therefore a conic touching the lines  $x = 0$ ,  $y = 0$ ,  $z = 0$ , in the points of intersection with the lines  $y - z = 0$ ,  $z - x = 0$ ,  $x - y = 0$  respectively; it is a conic touching the lines  $X, Y, Z$  harmonically in regard to the line  $I$ .

To find the envelope of the line  $L$ , the most convenient course is to take the equation in  $\theta$  in the reduced form

$$\theta^3 - \theta(\mu\nu + \nu\lambda + \lambda\mu) - 2\lambda\mu\nu = 0;$$

this will have a pair of equal roots if

$$(\mu\nu + \nu\lambda + \lambda\mu)^3 - 27\lambda^2\mu^2\nu^2 = 0;$$

[257

WITH RESPECT TO THREE LINES AND A LINE.

177

that is, if

$$\mu\nu + \nu\lambda + \lambda\mu - 3(\lambda\mu\nu)^{\frac{2}{3}} = 0;$$

or finally if

$$\lambda^{-\frac{1}{2}} + \mu^{-\frac{1}{2}} + \nu^{-\frac{1}{2}} = 0,$$

which is the relation between  $\lambda, \mu, \nu$  in order that the line

$$\lambda x + \mu y + \nu z = 0$$

may have two coincident centres; this gives at once for the equation of the envelope

$$\sqrt{x} + \sqrt{y} + \sqrt{z} = 0,$$

which is the equation of a curve of the fourth order having four-pointic contact with the lines  $x = 0, y = 0, z = 0$ , at the points of intersection with the lines  $y - z = 0, z - x = 0, x - y = 0$  respectively, i. e. it has four-pointic contact with the lines  $X, Y, Z$  harmonically in regard to the line  $I$ .

It may be noticed that the rationalized form of the equation  $\sqrt{x} + \sqrt{y} + \sqrt{z} = 0$  is

$$x^4 + y^4 + z^4 - 4(yx^3 + y^3z + zx^3 + z^3x + xy^3 + x^3y) + 6(y^2z^2 + z^2x^2 + x^2y^2) - 124(x^2yz + y^2zx + z^2xy) = 0.$$

If, to fix the ideas, the signs of the coordinates  $x, y, z$  are so determined that a point within the triangle  $x = 0, y = 0, z = 0$  has its coordinates positive (in which case the line  $x + y + z = 0$  will cut the three sides produced), the curve  $\sqrt{x} + \sqrt{y} + \sqrt{z} = 0$  will lie wholly within the triangle, and will be of the form shown by the annexed figure. This is, in fact, the form of the curve in the case considered by Plücker, where the line  $I$  is at infinity, the points of contact being the middle points of the sides. And his five groups of curves,  $\alpha, \beta, \gamma, \delta, \epsilon$ , and two subdivisions of the group

$\beta$  (see pp. 221–224), correspond to the following positions of the line in regard to the triangle and curve, viz.

- $\alpha$ . The line cuts the three sides produced.
- $\beta$ . It passes through an angle, (a) cutting, or (b) not cutting the curve.
- $\gamma$ . It cuts two sides and a side produced, but does not cut or touch the curve.
- $\delta$ . It cuts two sides and a side produced, and touches the curve.
- $\epsilon$ . It cuts two sides and a side produced, and cuts the curve.

It is hardly necessary to remark that, in the general case, the tangential equation of the curve is

$$\xi^{-\frac{1}{2}} + \eta^{-\frac{1}{2}} + \zeta^{-\frac{1}{2}} = 0,$$

or what is the same thing,

$$(\eta\zeta + \zeta\xi + \xi\eta)^3 - 27\xi^2\eta^2\zeta^2 = 0,$$

and that the curve is therefore of the sixth class.

2, Stone Buildings, W. C., October 16, 1860.

THE COLLECTED MATHEMATICAL PAPERS  
of  
ARTHUR CAYLEY

Volume IV — pages 179–203

Papers 258, 259, 260

## 258.

## ON A RELATION BETWEEN TWO TERNARY CUBIC FORMS.

[*From the Philosophical Magazine*, vol. xx. 1860, pp. 512–514.]

The cubic form

$$x^3 + y^3 + z^3 + 6lxyz$$

is in general linearly transformable into the form

$$(X + Y + Z)^3 + 27kXYZ;$$

in fact, writing

$$X = 2lx - y - z,$$

$$Y = 2ly - z - x,$$

$$Z = 2lz - x - y,$$

we have identically

$$(1 - 2l + 4l^2)(X + Y + Z)^3 + 24(1 - l)^3XYZ = 8(2l + 1)^3(1 - l)^3(x^3 + y^3 + z^3 + 6lxyz);$$

and the value of  $k$  consequently is

$$k = -\frac{8(l - 1)^3}{9(1 - 2l + 4l^2)}.$$

If, however,  $l = \frac{1}{2}$  or  $l = -\frac{1}{2}$ , the transformation fails. In the former case, viz. for  $l = \frac{1}{2}$ , the equations for the linear transformation become

$$X = 2x - y - z,$$

$$Y = 2y - z - x,$$

$$Z = 2z - x - y,$$

which give  $X + Y + Z = 0$ , so that  $X, Y, Z$  are no longer independent; and the formula of transformation becomes

$$(X + Y + Z)^3 = 0.$$

It may be noticed that the invariant  $S$  of the form

$$x^3 + y^3 + z^3 + 6lxyz$$

is  $S = -l + l^4$ , so that  $l = 1$  is one of the values which make  $S$  vanish. And the above transformation is not applicable to the cubic form  $x^3 + y^3 + z^3 + 6xyz$ , which is a form for which  $S$  vanishes. The transformation, however, holds good for  $l = 0$ , which is another value which makes  $S$  vanish; or it does apply to the form  $x^3 + y^3 + z^3$ , for which  $S$  vanishes. The transformation, in fact, is

$$(X + Y + Z)^3 + 24XYZ = -8(x^3 + y^3 + z^3),$$

with the linear equations

$$X = -y - z,$$

$$Y = -z - x,$$

$$Z = -x - y.$$

The above two forms for which  $S$  vanishes, viz.

$$x^3 + y^3 + z^3 + 6xyz, \quad x^3 + y^3 + z^3,$$

are, notwithstanding, equivalent to each other, as appears by the identical equation

$$(x + y + z)^3 + (x + \omega y + \omega^2 z)^3 + (x + \omega^2 y + \omega z)^3 = 3(x^3 + y^3 + z^3 + 6xyz),$$

where  $\omega$  is an imaginary cube root of unity. In the latter of the two cases of failure, viz. for  $l = -\frac{1}{2}$ , the equations for the linear transformations are

$$X = Y = Z = -x - y - z;$$

so that  $X, Y, Z$  are not only not independent, but they are connected by two linear relations. And the formula of transformation becomes

$$(X + Y + Z)^3 - 27XYZ = 0,$$

which is, in fact, true in virtue of the equations  $X = Y = Z$ .

The two forms of equation,

$$x^3 + y^3 + z^3 + 6lxyz = 0, \quad (x + y + z)^3 + 27kxyz = 0,$$

represent each of them equally well a curve of the third order without a double point. In the first form the three real points of inflexion are given by

$$(x = 0, y + z = 0), \quad (y = 0, z + x = 0), \quad (z = 0, x + y = 0);$$

or what is the same thing, the points in question are the intersections of the lines  $x = 0$ ,  $y = 0$ ,  $z = 0$  with the line  $x + y + z = 0$ ; or we have  $x + y + z = 0$  for the equation of the line through the three points of inflexion; and the equations of the tangents at the points of inflexion are

$$2lx - y - z = 0, \quad 2ly - z - x = 0, \quad 2lz - x - y = 0.$$

For the second form it is obvious that the points of inflexion are the intersections of the lines  $x = 0$ ,  $y = 0$ ,  $z = 0$  with the line  $x + y + z = 0$ ; and, moreover, that the lines  $x = 0$ ,  $y = 0$ ,  $z = 0$  are the tangents at the point of inflexion.

The first of the above-mentioned forms, however, cannot represent a curve with a double point. In fact the condition for its doing so would be  $1 + 8l^3 = 0$ ; but when this condition is satisfied, the left-hand side breaks up into linear factors, and the equation represents, not a proper curve of the third order, but a system of three lines. The second form can represent a curve having a double point; viz. if  $k = -1$ , the curve will have a conjugate or isolated point at the point  $x = y = z$ . It is clear *a priori* that ( $x = 0$ ,  $y = 0$ ,  $z = 0$  being real lines) neither of the forms can represent a curve of the third order having a double point with two real branches through it, since in this case the curve has only one real point of inflexion.

I have elsewhere used the word "node" to denote a double point, and I take the opportunity of suggesting the employment of the words "crunode" (*crus*) and "acnode" (*acus*) to denote respectively a double point with two real branches through it, and a conjugate or isolated point.

2, *Stone Buildings, W.C., October 19, 1860.*



## 259.

## THE PROBLEM OF POLYHEDRA.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLVII. (for the year 1857), pp. 183–185: printed as a Note to Mr Kirkman's Memoir "On Autopolar Polyhedra," pp. 183–215.]

Let  $a, b, c, d, e, f, g, h$ , &c. represent the vertices of a polyhedron, then a face will be represented, e.g. by  $abcde$ , where the contiguous duads, viz.  $ab, bc, cd, de, ea$  are the edges of the face; and calling the face  $K$ , we may write

$$K = abcde. \quad (1)$$

It is to be noticed that the letters of a face-symbol may be taken forwards or backwards from any letter without altering the meaning of the symbol. Thus,  $abcde, bcdea$ , &c.,  $edcba$ , &c. might any of them be taken to denote the face  $K$ . The diagonal of a face cannot be either an edge or a diagonal of any other face, i.e. a non-contiguous duad such as  $ac$  in a face-symbol  $K$  cannot be a duad, contiguous or non-contiguous, of any other face-symbol. But each edge of a face must be an edge of one and only one other face, i.e. each contiguous duad such as  $ab$  in the face-symbol  $K$  must be a contiguous duad of one and only one other face-symbol  $L$ . And moreover two faces cannot have more than a single edge in common, i.e. two face-symbols cannot contain more than a single contiguous duad, the same in each symbol.

The face  $K$  contains the edges  $ab, ae$ , i.e. the edge  $ab$  is contained in the face  $K$ ; it will also be contained in one and only one other face, suppose  $L$ ; this face will contain another edge through the vertex  $a$ , suppose the edge  $af$ , and so on, until we arrive at a face containing the edge  $ae$ ; we have, for example,

$$\begin{aligned}
K &= eabcd, \\
L &= baf \dots, \\
M &= fag \dots, \\
N &= gah \dots, \\
P &= hai \dots, \\
Q &= iae \dots;
\end{aligned}$$

and we thence derive the vertex-symbol

$$a = KLMNPQ, \quad (2)$$

where the contiguous duads  $KL, LM, MN, NP, PQ, QK$  represent in order the edges through the vertex  $a$ . The remarks before made with respect to the face-symbols apply to the vertex-symbols. A non-contiguous duad such as  $KN$  of the vertex-symbol  $a$  cannot be a duad, contiguous or non-contiguous, of any other vertex-symbol; but each contiguous duad such as  $KL$  of the vertex-symbol  $a$  must be a contiguous duad of one and only one other vertex-symbol  $b$ . And the symbols of two vertices cannot contain more than one contiguous duad, the same in each symbol.

Any edge of the polyhedron admits of a double representation; it is the junction of two vertices, or the intersection of two faces. Thus  $ab$  and  $KL$  will represent the same edge, or we may write

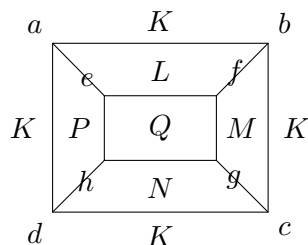
$$ab = KL. \quad (3)$$

It is to be remarked that in this system, to each equation  $ab = KL$  there corresponds one and only one equation of the form  $ae = KQ$ , i.e. to an edge considered as drawn from a given vertex in a given face there corresponds one and only one other edge from the same vertex in the same face.

It has been shown how the system of face-symbols (1) leads to the system of vertex-symbols (2), and the system of edge symbols (3); and generally, any one of the three systems leads to the other two; and the three systems conjointly, or each system by itself, is a complete representation of the polyhedron. As an example, take the hexaedron: the three systems are:

$$\begin{aligned}
K &= abcd, & a &= LPK, \\
L &= abfe, & b &= LMK, \\
M &= bfgc, & c &= MNK, \\
N &= gcdh, & d &= NPK, \\
P &= dhe a, & e &= LPQ, \\
Q &= hefg, & f &= LMQ, \\
& & g &= MNQ, \\
& & h &= NPQ;
\end{aligned} \quad (2)$$

$$\begin{aligned}
 ab &= KL, & ae &= PL, & ef &= LQ, & (3) \\
 bc &= KM, & bf &= LM, & fg &= MQ, \\
 cd &= KN, & cg &= MN, & gh &= NQ, \\
 de &= KP, & dh &= NP, & he &= PQ.
 \end{aligned}$$



Consider, now, two polyhedra having the same number of vertices and also the same number of faces. And let the vertices and faces of the first polyhedron taken in any order be represented by

$$abcde \dots KLM \dots,$$

and the vertices and faces of the second polyhedron taken in a certain order be represented by

$$a'b'c'd'e' \dots K'L'M' \dots;$$

then, forming the substitution symbol

$$a'b'c'd'e' \dots K'L'M' \dots abcde \dots KLM \dots,$$

which denotes that  $a'$  is to be written for  $a$ ,  $b'$  for  $b$ ,  $\dots$   $K'$  for  $K$ , &c., if operating with this upon the symbol system of the first polyhedron, we obtain the symbol system of the second polyhedron, the second polyhedron will be syntypic with the first. It should be noticed, that there may be several modes of arrangement of the vertices and faces of the second polyhedron, which will render it syntypic according to the foregoing definition with the first polyhedron, i.e. the second polyhedron may be syntypic in several different ways with the first polyhedron. This is, in fact, the same as saying that a polyhedron may be syntypic with itself in several different ways. Suppose, next, that the number of vertices of the second polyhedron is equal to the number of faces of the first polyhedron, and the number of faces of the second polyhedron is equal to the number of vertices of the first polyhedron: and let the vertices and faces of the first polyhedron in any order be represented by

$$abcde \dots KLM \dots,$$

and the faces and vertices of the second polyhedron in a certain order be represented by

$$A'B'C'D'E' \dots k'l'm' \dots$$

Then, forming the substitution symbol

$$A'B'C'D'E' \dots k'l'm' \dots abcde \dots KLM \dots,$$

if, operating with this upon the symbol system of the first polyhedron, we obtain the symbol system of the second polyhedron, the second polyhedron is said to be polar-syntypic with the first; and, as in the case of syntypicism, this may happen in several different ways.

Lastly, if there be a polyhedron having the same number of vertices and faces, and if the vertices and faces in any order be represented by

$$abcd \dots KLMN \dots,$$

and the faces and vertices in a certain order be represented by

$$ABCD \dots klmn \dots;$$

then, forming the substitution symbol

$$ABCD \dots klmn \dots abcd \dots KLMN \dots,$$

if, operating with this upon the symbol system of the polyhedron, we reproduce such symbol system, i.e. in fact, if the polyhedron be polar-syntypic with itself, the polyhedron is said to be autopolar; and in accordance with a preceding remark, this may happen in several different ways. It is clear that the substitution symbol, operating on the symbol system of the vertices, must give the symbol system of the faces, and conversely; but operating on the symbol system of the edges, it must reproduce such symbol system of the edges: and this last condition will by itself suffice to make the polyhedron autopolar, i.e. the polyhedron will be autopolar if the substitution symbol, operating on the symbol system of the edges, reproduces such symbol system.

## 260.

## ON THE DOUBLE TANGENTS OF A PLANE CURVE.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLIX, for the year 1859, pp. 193–212. Received March 17,—Read April 14, 1859.]

It was first shown by Plücker on geometrical principles, that the number of the double tangents of a plane curve of the order  $m$  was  $\frac{1}{2}m(m-2)(m^2-9)$ : see the note, “Solution d’une question fondamentale concernant la théorie générale des Courbes,” *Crelle*, t. XII. pp. 105–108 (1834), and the “Theorie der algebraischen Curven” (1839). The memoir by Hesse, “Ueber die Wendepuncte der Curven dritter Ordnung,” *Crelle*, t. XXVIII. pp. 97–107 (1844), contains the analytical solution of the allied easier problem of the determination of the points of inflexion of a plane curve. In the memoir, “Recherches sur l’élimination et sur la théorie des Courbes,” *Crelle*, t. XXXIV. (1847), pp. 30–45, [53], I showed how the problem of double tangents admitted of an analytical solution, viz. if  $U = 0$  is the equation of the curve,  $L$ ,  $M$ ,  $N$  the first derived functions of  $U$ , and

$$D = \alpha(M\partial_z - N\partial_y) + \beta(N\partial_y - L\partial_z) + \gamma(L\partial_y - M\partial_x)$$

(where  $\alpha$ ,  $\beta$ ,  $\gamma$  are arbitrary), then the points of contact of the double tangents are given as the intersections of the curve  $U = 0$ , with a curve the equation whereof is in the first instance obtained under the form  $[F] = 0$ ;  $[F]$  being a given function of

$$DU, D^2U, \dots D^{m-1}U,$$

of the degree  $m^2 - m - 6$  in respect of  $(\alpha, \beta, \gamma)$ , the degree  $m^3 - 2m^2 - 10m + 12$  in respect of  $(x, y, z)$ , and the degree  $m^2 + m - 12$  in respect of the coefficients of  $U$ . It was necessary, in order that the points of intersection should be independent of the arbitrary quantities  $(\alpha, \beta, \gamma)$ , that we should have identically

$$[F] = \Lambda \cdot U + N \cdot \Pi U,$$

$N$  being of the degree  $m^2 - m - 6$  in  $(\alpha, \beta, \gamma)$ , and consequently  $\Pi U$  a function of  $(x, y, z)$  without  $(\alpha, \beta, \gamma)$ . Guided by Hesse's investigation for the points of inflexion, I asserted that it was probable that  $N$  was of the form  $(\alpha x + \beta y + \gamma z)^{m^2-m-6}$ ; which being so,  $\Pi U$  would be of the degree  $(m-2)(m^2-9)$  in respect of  $(x, y, z)$ , and the degree  $m^2 + m - 12$  in respect of the coefficients, and I was thus led to the theorem, "On trouve les points de contact des tangentes doubles en combinant avec l'équation de la courbe une équation  $\Pi U = 0$ , de l'ordre  $(m-2)(m^2-9)$  par rapport aux variables et de l'ordre  $m^2 + m - 12$  par rapport aux coefficients—c'est-à-dire, puisqu'il correspond deux points de contact à une tangente double, le nombre de ces tangentes est égal à  $\frac{1}{2}m(m-2)(m^2-9)$ : théorème démontré indirectement par M. Plücker."

Hesse, in the memoir "Ueber Curven dritter Ordnung u.s.w.," *Crelle*, t. XXXVI. pp. 143–176 (1848), showed how the components  $D^i U, D^2 U, \dots D^{m-1} U$  of  $[F]$  could each of them be expressed in a simplified form, and he thus effected the actual reduction of  $[F]$  to the form  $\Lambda \cdot U + (\alpha x + \beta y + \gamma z)^{m-2} \cdot R$ , where  $R$  still contained the arbitrary quantities  $(\alpha, \beta, \gamma)$  in the degree  $(m-2)(m-3)$ . In particular for a quartic curve, the equation  $R = 0$  was shown to be

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \alpha, \beta, \gamma)^2 \cdot H = 0,$$

where the left-hand side is of the degree 2 in  $(\alpha, \beta, \gamma)$  and the degree 16 in  $(x, y, z)$ ; and which should therefore by means of the equation  $U = 0$  be reducible so as to contain the factor  $(\alpha x + \beta y + \gamma z)^2$ .

Jacobi's paper, "Beweis des Satzes, dass eine Curve  $n$ -ten Grades im allgemeinen  $\frac{1}{2}n(n-2)(n^2-9)$  Doppeltangenten hat," *Crelle*, t. XL. pp. 237–260 (1850), did not, I think, materially advance the solution of the question. In a letter to Jacobi, dated the 30th December, 1849, published at the conclusion of the last-mentioned paper, Hesse gave the equation of the curve of the 14th order for the points of contact of the double tangents of a quartic, viz. in my notation,

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 \cdot H - H (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H = 0,$$

and the demonstration is given in Hesse's paper, "Ueber die ganzen homogenen Functionen von der dritten und vierten Ordnung zwischen drei Variabeln," *Crelle*, t. XLI. pp. 285–292 (1851), and is reproduced in Mr Salmon's *Treatise on the Higher Plane Curves* (1852). Two very interesting memoirs by Hesse and Steiner, *Crelle*, t. XLIX. (1855), relate to the geometrical theory of the double tangents of a quartic, and it is not necessary to refer to them more particularly. It is to be observed that the curve which determines the points of contact of the double tangents is not absolutely determinate; for we may, it is clear, in the place of  $\Pi U = 0$ , write  $\Pi U + M \cdot U = 0$ , where  $M$  is an arbitrary function of the proper degree: a very elegant transformation in the case of the quartic is given in Hesse's paper, "Transformation der Gleichung der Curven 14ten Grades, welche eine gegebene Curve 4ten Grades in den Berührungspuncten ihrer Doppeltangenten schneiden," *Crelle*, t. LII. pp. 97–103 (1856).

Mr Salmon's work above referred to, contains the fundamental theorem of the tangential of a cubic, viz. a tangent to a cubic meets the cubic in a third point

which lies on the second or line polar of the point of contact with respect to the Hessian. In my "Memoir on Curves of the Third Order," *Phil. Trans.* vol. CXLVII. (1857), pp. 415–446, art. No. 37, [146], I gave an identical equation relating to the tangential of a cubic, but which is not there exhibited in its proper form; this was afterwards effected by Mr Salmon, in the paper "On Curves of the Third Order," *Phil. Trans.* vol. CXLVIII. (1858), pp. 535–541. The equation, as given by Mr Salmon, is in the notation of the present memoir,

$$-\mathfrak{H} \cdot U + \frac{1}{3}\mathfrak{D}\mathfrak{H} \cdot DU - \frac{1}{3}DH \cdot \mathfrak{D}\mathbf{T} + H \cdot \mathbf{T} = 0,$$

an equation which in fact puts in evidence the last-mentioned theorem for the tangential of a cubic.

The idea occurred to me of considering, in the case of the higher plane curves, the tangentials of a given point of the curve, viz. the points in which the tangent again meets the curve; for by expressing that two of these tangentials were coincident, we should have the condition that the given point is the point of contact of a double tangent. But I was not able to complete the solution.

Finally, Mr Salmon discovered the equation of a curve of the order  $n - 2$ , which by its intersections with the tangent at the given point determines the tangentials, and by expressing that the curve in question is touched by the tangent, he was led to a complete solution of the Double-tangent problem. Mr Salmon's result is given in the note, "On the Double Tangents to Plane Curves," in the *Philosophical Magazine* for October 1858. The discovery just referred to led me to the investigations of the present memoir, in which it will be seen that I obtain, for a curve of any order whatever, the identical equation corresponding to the before-mentioned equation obtained by Mr Salmon in the case of a cubic; which identical equation puts in evidence the theorem as to the tangentials of the curve, and may thus be considered as containing in itself the solution of the Double-tangent problem: the identical equation is besides interesting for its own sake, as a part of the theory of ternary quantics.

1. Mr Salmon's solution of the problem of double tangents is based upon the following analytical determination of the tangentials of any point of the curve.

Let

$$T = (* \mid X, Y, Z)^n = 0$$

be the equation of the given curve,  $(X, Y, Z)$  being current coordinates; and let  $(x, y, z)$  be the coordinates of a point on the curve, so that we have

$$U = (* \mid x, y, z)^n = 0,$$

a condition satisfied by the coordinates of the point in question.

Then the tangent

$$V = (X\partial_x + Y\partial_y + Z\partial_z)U = 0$$

at the point  $(x, y, z)$ , meets the curve besides in  $(n - 2)$  points, which are the tangentials of the given point  $(x, y, z)$ , and which are determined as the intersections of the tangent  $V = 0$  with a certain curve,

$$\Omega = (* \mid x, y, z)^{n-2} = 0.$$

2. To express the equation of this curve, let  $U_1, U_2, \dots$  be the successive emanants of  $U$ , taken with the facients of emanation  $(x_1, y_1, z_1)$ , viz.

$$U_1 = \frac{1}{n} \left[ \text{should be } \frac{1}{n-2} \right] (x_1 \partial_x + y_1 \partial_y + z_1 \partial_z) U,$$

$$U_2 = \frac{1}{n(n-1)} \left[ \text{should be } \frac{1}{(n-2)(n-3)} \right] (x_1 \partial_x + y_1 \partial_y + z_1 \partial_z)^2 U,$$

$$\dots$$

where it should be noticed that the numerical determination is such, that putting  $(x, y, z)$  for  $(x_1, y_1, z_1)$ , then  $U_1, U_2, \dots$  become respectively equal to  $U$ . [The numerical determination should have been and in the latter part of the memoir is assumed to be such as to render  $H_1, H_2$ , &c. equal to  $H$ , on making the substitution in question: the correction was made in a later memoir "On the double tangents of a curve of the fourth order."] Suppose also that  $H, H_1, H_2, \dots$  are the Hessians of  $U, U_1, U_2, \dots$ , viz.  $H$  is the determinant formed with the second derived functions of  $U$  with respect to  $(x, y, z)$ ,  $H_1$  the like determinant with the second derived functions of  $U_1$  with respect to the same quantities  $(x, y, z)$ ; and so on. Moreover let  $D^{n-2}H_r = (X\partial_x + Y\partial_y + Z\partial_z)^{n-2}H$ , denote the  $(n-2)$ th emanant of  $H$  with respect to the current coordinates  $(X, Y, Z)$  as facients of emanation; and similarly let  $D^{n-2}H_1, D^{n-2}H_2, \dots$  denote the  $(n-2)$ th emanants of  $H_1, H_2, \dots$  in respect to the same facient of emanation—it being understood that in all these functions,  $(x_1, y_1, z_1)$  are after the differentiations to be replaced by  $(x, y, z)$ . It is to be observed that  $U_r$  is of the degree  $(n-r)$  in  $(x, y, z)$ , and consequently  $H_r$  of the degree  $3(n-2-r)$ ; hence  $D^{n-2}H_r$  is of the degree  $3(n-2-r) - (n-2)$ ,  $= 2(n-2) - 3r$ , which implies that  $r \not\geq \frac{1}{3}(n-2)$ , for otherwise  $D^{n-2}H_r$  would be identically equal to zero. Upon replacing  $(x_1, y_1, z_1)$  by  $(x, y, z)$ ,  $D^{n-2}H_r$  ( $r$  satisfying the above condition) becomes of the degree  $2(n-2)$  in  $(x, y, z)$ , and it is obviously of the degree 3 in the coefficients of  $U$ , and of the degree  $(n-2)$  in the current coordinates  $(X, Y, Z)$ .

3. This being premised, we have

$$\Omega = D^{n-2}H - \frac{n-1}{1} D^{n-2}H_1 + \&c. = 0,$$

for the equation of the curve of the order  $(n-2)$ , which by its intersection with the tangent gives the tangentials of the given point; the numerical coefficients are the binomial coefficients of the order  $(n-1)$  taken with the signs  $+$  and  $-$  alternately, and the series is continued as long as the terms do not vanish, that is, if as before  $r$  denote the suffix of  $H$ , for so long as  $r \not\geq \frac{1}{3}(n-2)$ ; but of course the value will not be altered by continuing the series to  $r = n-1$ . In particular, for the quartic we have

$$\Omega = D^2H - 3D^2H_1,$$

for the quintic

$$\Omega = D^3H - 4D^3H_1 + 6D^3H_2,$$



and so on. The function  $\Omega$ , like the several component terms, is of course of the degree 3 in the coefficients of  $U$ , and of the degree  $2(n-2)$  in  $(x, y, z)$ .

4. It is to be remarked that the formula applies to a cubic; we have here simply  $\Omega = DH$ , which agrees with a result already mentioned. It may be noticed also that in the general case the formula gives at once the condition for the points of inflexion; in fact, if the point  $(x, y, z)$  be a point of inflexion, then one of the tangentials must coincide with this point, or the equation  $\Omega = 0$  will be satisfied by writing therein  $(x, y, z)$  for  $(X, Y, Z)$ ; but when this is done  $D^{n-2}H$ ,  $D^{n-2}H_1$ , &c. reduce themselves (to numerical factors *près*) to  $H$ , and the equation becomes simply  $H = 0$ , which is the well-known condition for the points of inflexion.

5. If two of the tangentials coincide, or what is the same thing, if the tangent  $V = 0$  touches the curve  $\Omega = 0$ , then the point  $(x, y, z)$  will be the point of contact of a double tangent. The equation which expresses the condition in question, treating therein  $(x, y, z)$  as current coordinates, is consequently that of a curve, intersecting the given curve (now represented by  $U = 0$ ) in the points of contact of the double tangents. The process leads to a determinate form  $\Pi U = 0$ , of the curve in question, but of course any curve whatever,  $\Pi U + M \cdot U = 0$ , will intersect the curve  $U = 0$  in the points of contact of the double tangents.

6. I write for the moment

$$\Omega = (A, \dots | X, Y, Z)^{n-2} = 0,$$

$$V = \xi X + \eta Y + \zeta Z = 0,$$

for the two equations: the coefficients  $(A, \dots)$ , as already mentioned, are of the degree  $2(n-2)$  in  $(x, y, z)$  and of the degree 3 in the coefficients of  $U$ ; or as we may express it,

$$A, \dots = (a, \dots)^3 (x, y, z)^{2(n-2)}.$$

In like manner  $\xi, \eta, \zeta$  are of the degree  $(n-1)$  in  $(x, y, z)$ , and the degree 1 in the coefficients of  $U$ , or we may write

$$\xi, \eta, \zeta = (a, \dots)^1 (x, y, z)^{n-1}.$$

7. The equation which expresses that the line  $V = 0$  touches the curve  $\Omega = 0$ , is  $\nabla \Omega = 0$ , where the facients of the Reciprocat  $\nabla \Omega$  are the coefficients  $(\xi, \eta, \zeta)$  of the linear function. This equation is of the form

$$(A, \dots)^{2(n-3)} (\xi, \eta, \zeta)^{(n-2)(n-3)} = 0;$$

or attending to the forms of  $(A, \dots)$  and  $(\xi, \eta, \zeta)$ , it is of the form

$$(a, \dots)^{n+4(n-3)} (x, y, z)^{(n-2) \cdot 2(n-3) + (n-1)(n-2)(n-3)} = 0,$$

or what is the same thing, the form

$$(a, \dots)^{n+4(n-3)} (x, y, z)^{(n-2)(n^2-9)} = 0,$$

viz. the curve through the points of contact of the double tangents is a curve of the order  $(n-2)(n^2-9)$ , and its equation contains the coefficients of the equation  $U=0$  of the given curve in the degree  $(n+4)(n-3)$ . And since each double tangent corresponds to two points of contact, the number of double tangents is  $\frac{1}{2}n(n-2)(n^2-9)$ . This agrees with the before-mentioned results.

8. The whole problem is thus reduced to the demonstration of Mr Salmon's expression for the curve  $\Omega=0$ . To fix the ideas, consider the case of a quartic curve  $T = (* \mid X, Y, Z)^4 = 0$ , and let the function  $U = (* \mid x, y, z)^4$  (or as for shortness we may write it,  $U = (x, y, z)^4$ ) and certain of its emanants be represented as follows, viz.—

$$\begin{aligned} a &= (x, y, z)^4, \\ b &= (x, y, z)^3(X, Y, Z), \\ c &= (x, y, z)^2(X, Y, Z)^2, \\ d &= (x, y, z)(X, Y, Z)^3, \\ e &= (X, Y, Z)^4, \\ a' &= (x, y, z)^3(X', Y', Z'), \\ b' &= (x, y, z)^2(X, Y, Z)(X', Y', Z'), \\ c' &= (x, y, z)(X, Y, Z)^2(X', Y', Z'), \\ d' &= (X, Y, Z)^3(X', Y', Z'), \\ a'' &= (x, y, z)^2(X', Y', Z')^2, \\ b'' &= (x, y, z)(X, Y, Z)(X', Y', Z')^2, \\ c'' &= (X, Y, Z)^2(X', Y', Z')^2, \end{aligned}$$

where  $(X', Y', Z')$  are new arbitrary facients; but, as before,  $(X, Y, Z)$  are taken to be current coordinates, and  $(x, y, z)$  the coordinates of the given point on the curve:

$e=0$  is the equation of the curve;

$d=0$ , the equation of the first or cubic polar of the point  $(x, y, z)$ ;

$b=0$ , the equation of the last or line polar of the point  $(x, y, z)$ , or what is the same thing (the point being on the curve), the tangent of the curve at this point;

$a=0$ , the condition which expresses that the point is on the curve.

9. Imagine now an identical equation,

$$a\text{I} + b\text{II} + d\text{III} + e\text{IV} = 0;$$

then, since  $a=0$ , we have

$$b\text{II} + d\text{III} + e\text{IV} = 0;$$

and if in this equation we write  $b = 0$ ,  $e = 0$ , it becomes  $\text{III } d = 0$ , that is, the points of intersection of the curve  $e = 0$  and the tangent  $b = 0$  lie on one or other of the curves  $d = 0$ ,  $\text{III} = 0$ . But the points in question do not lie on the curve  $d = 0$ , consequently they lie on the curve  $\text{III} = 0$ .

10. To explain the law of formation of the multipliers I, II, III, IV, I form the matrix

$$\begin{pmatrix} a, & b, & c, & d, & e; & a', & b', & c', & d'; \\ b, & c, & d, & e, & f; & b', & c', & d', & e'; \\ a', & b', & c', & d', & e'; & a'', & b'', & c'', & d'' \end{pmatrix}$$

(matrix layout reconstructed from scan; the relevant rows for the quartic are the top two and the accented row, with the appropriate truncation.)

and then we have

$$\begin{aligned} \text{I} &= - \begin{vmatrix} d, & e, & b' \\ e, & f, & c' \\ d', & e', & b'' \end{vmatrix}, \quad \text{II} = + \begin{vmatrix} d, & e, & a' \\ e, & f, & b' \\ d', & e', & a'' \end{vmatrix} - \begin{vmatrix} c, & e, & b' \\ d, & f, & c' \\ c', & e', & b'' \end{vmatrix}, \\ \text{III} &= - \begin{vmatrix} b, & c, & b' \\ c, & d, & c' \\ b', & c', & b'' \end{vmatrix} - \begin{vmatrix} a, & c, & a' \\ b, & d, & b' \\ a', & c', & a'' \end{vmatrix}, \quad \text{IV} = + \begin{vmatrix} a, & b, & a' \\ b, & c, & b' \\ a', & b', & a'' \end{vmatrix}, \end{aligned}$$

values which, as I proceed to show, satisfy the identical equation

$$a\text{I} + b\text{II} + d\text{III} + e\text{IV} = 0.$$

11. We have in fact

$$\begin{aligned} \text{I} &= d(db'' - c'^2 + cc'' - b'd') \\ &\quad + e(b'c' - b''c + b'c' - bc'') \\ &\quad + d'(cc' - db' + bd' - cc'), \end{aligned}$$

where the last line is  $= bd'^2 - db'd'$ ;

$$\begin{aligned} \text{II} &= d(b'c' - a''d + b'c' - b''c + a'd' - bc'') \\ &\quad + e(ca'' - a'c' + bb'' - b'^2 + ac'' - a'c') \\ &\quad + d'(a'd - b'c + b'c - bc' + bc' - ad'), \end{aligned}$$

where the last line is  $da'd' - ad'^2$ ;

$$\begin{aligned} \text{III} = & a(-b'd' + cc'' + c'^2 - b''d + b'd' - ea'') \\ & + b(bc'' - b'c' + b''c - b'c' + a''d - ad') \\ & + a'(cc' - bd' + b'd - cc' + ae - b'd), \end{aligned}$$

where the last line is  $-ba'd' + ea'^2$ ; and

$$\begin{aligned} \text{IV} = & a(b''c - b'c' + da'' - b'c') \\ & + b(b'^2 - bb'' + a'c' - a''c) \\ & + a'(bc' - b'c + cb' - a'd), \end{aligned}$$

where the last line is  $ba'c - a'^2d$ .

These values may be expressed as follows:

$$\begin{aligned} \text{I} = & a(0) \\ & + b(d'^2 - ) \quad (12) \\ & + d(b''d + c''c - b'd' - cc' - d'b') \quad (13) \\ & + e(-b''c - c''b + b'c' + c'b') \quad (14) \end{aligned}$$

$$\begin{aligned} \text{II} = & a(-d'^2) \quad (21) \\ & + b(0) \\ & + d(-a''d - b''c - c''b + a'd' + b'c' + c'b' + d'a') \quad (23) \\ & + e(a''c + b''b + c''a - a'c' - b'b' - c'a') \quad (24) \end{aligned}$$

$$\begin{aligned} \text{III} = & a(-b''d - c''c + b'd' + c'c' + d'b') \quad (31) \\ & + b(a''d + b''c + c''b - a'd' - b'c' - c'b' - d'a') \quad (32) \\ & + d(0) \\ & + e(-a''a + a'a') \quad (34) \end{aligned}$$

$$\begin{aligned} \text{IV} = & a(b''c + c''b - b'c' - c'b') \quad (41) \\ & + b(-a''c - b''b - c''a + a'c' + b'b' + c'a') \quad (42) \\ & + d(a''a - a'a') \quad (43) \\ & + e(0). \end{aligned}$$

which are of the form

$$\begin{aligned} \text{I} &= a \cdot 0 + b(12) + d(13) + e(14), \\ \text{II} &= a(21) + b \cdot 0 + d(23) + e(24), \\ \text{III} &= a(31) + b(32) + d \cdot 0 + e(34), \\ \text{IV} &= a(41) + b(42) + d(43) + e \cdot 0, \end{aligned}$$

where  $(12) = -(21)$ , &c., and which therefore satisfy the equation

$$a\text{I} + b\text{II} + d\text{III} + e\text{IV} = 0.$$

12. The equation of the curve which by its intersection with the tangent gives the tangentials, is

$$\text{III} = - \begin{vmatrix} b, & c, & b' \\ c, & d, & c' \\ b', & c', & b'' \end{vmatrix} - \begin{vmatrix} a, & c, & a' \\ b, & d, & b' \\ a', & c', & a'' \end{vmatrix},$$

the degrees of which are

$$\begin{aligned} &\text{in the coefficients of } U, \quad 3, \\ &\text{in } (x, y, z), \quad 6, \\ &\text{in } (X, Y, Z), \quad 4, \\ &\text{in } (X', Y', Z'), \quad 2; \end{aligned}$$

and it only remains to divest this equation of a factor which it contains,

$$\begin{vmatrix} x, & y, & z \\ X, & Y, & Z \\ X', & Y', & Z' \end{vmatrix}^2,$$

which being thrown out, the equation will be independent of  $(X', Y', Z')$  and will be of the degrees

$$\begin{aligned} &\text{in the coefficients of } U, \quad 3, \\ &\text{in } (x, y, z), \quad 4, \\ &\text{in } (X, Y, Z), \quad 2. \end{aligned}$$

and will in fact be the before-mentioned equation  $\Omega = D^2H - 3D^2H_1 = 0$ .

13. Write for shortness,

$$\Delta = \begin{vmatrix} x, & y, & z \\ X, & Y, & Z \\ X', & Y', & Z' \end{vmatrix};$$

it is to be shown that

$$\text{III} = -\frac{1}{6}\Delta^2(D^2H - 3D^2H_1).$$

260]

ON THE DOUBLE TANGENTS OF A PLANE CURVE.

195

14. To effect this I remark that we have identically

$$\begin{vmatrix} a, & b, & a' \\ b, & c, & b' \\ a', & b', & a'' \end{vmatrix} = \Delta^2 H;$$

and I proceed to operate upon this equation with  $D = X\partial_x + Y\partial_y + Z\partial_z$ .

I notice that

$$a, b, c, d, e; \quad a', b', c', d'; \quad a'', b'', c''$$

are in regard to  $(x, y, z)$  of the degrees

$$4, 3, 2, 1, 0; \quad 3, 2, 1, 0; \quad 2, 1, 0;$$

or what is the same thing, since for the case in hand  $n = 4$ , of the degrees

$$n, n-1, \dots; \quad n-1, n-2, \dots; \quad n-2, n-3, \dots$$

and we have

$$Da = nb, \quad Db = (n-1)c, \dots \quad Da' = (n-1)b', \quad Db' = (n-2)c', \dots \quad Da'' = (n-2)b'', \dots$$

15. In the determinant

$$\begin{vmatrix} a, & b, & a' \\ b, & c, & b' \\ a', & b', & a'' \end{vmatrix} = \Delta^2 H,$$

the degrees of the terms (other than each top term, the degree of which is higher by unity) in the several columns are  $n-1, n-2, n-2$ ; if then we operate on the determinant with  $D$ , and as regards the top terms we write

$$\begin{aligned} Da &= b + (n-1)b, \\ Db &= c + (n-2)c, \\ Da' &= b' + (n-2)b', \end{aligned}$$

we have in the first place a term

$$\begin{vmatrix} b, & b, & b' \\ c, & c, & c' \\ b', & b', & a'' \end{vmatrix} + \begin{vmatrix} a, & c, & a' \\ b, & d, & b' \\ a', & c', & a'' \end{vmatrix} + \begin{vmatrix} a, & b, & b' \\ b, & c, & c' \\ a', & b', & b'' \end{vmatrix}$$

next the terms

$$(n-1) \begin{vmatrix} b, & b, & a' \\ c, & c, & b' \\ b', & b', & a'' \end{vmatrix} + (n-2) \begin{vmatrix} a, & c, & a' \\ b, & d, & b' \\ a', & c', & a'' \end{vmatrix} + (n-2) \begin{vmatrix} a, & b, & b' \\ b, & c, & c' \\ a', & b', & b'' \end{vmatrix},$$

the first of which vanishes. On the right-hand side  $D\Delta = 0$  identically, and therefore  $D \cdot \Delta^2 H = \Delta^2 DH$ , or we have

$$(n-2) \begin{vmatrix} a, & c, & a' \\ b, & d, & b' \\ a', & c', & a'' \end{vmatrix} + (n-2) \begin{vmatrix} a, & b, & b' \\ b, & c, & c' \\ a', & b', & b'' \end{vmatrix} = \Delta^2 DH.$$

16. I repeat the operation  $D$ : we have

$$\begin{aligned} & (n-2)(n-1) \begin{vmatrix} a, & d, & a' \\ b, & e, & b' \\ a', & d', & a'' \end{vmatrix} + (n-2)(n-1) \begin{vmatrix} b, & c, & a' \\ c, & d, & b' \\ b', & c', & a'' \end{vmatrix} \\ & + (n-2)(n-3) \begin{vmatrix} a, & c, & b' \\ b, & d, & c' \\ a', & c', & b'' \end{vmatrix} + (n-2)(n-2) \begin{vmatrix} a, & b, & c' \\ b, & c, & d' \\ a', & b', & c'' \end{vmatrix} \\ & + (n-2)(n-2) \begin{vmatrix} b, & b, & b' \\ c, & c, & c' \\ b', & b', & b'' \end{vmatrix} + (n-2)(n-3) \begin{vmatrix} a, & b, & b' \\ b, & c, & c' \\ a', & b', & b'' \end{vmatrix} = \Delta^2 D^2 H, \end{aligned}$$

or, collecting the different terms,

$$(n-2)(n-3) \begin{vmatrix} a, & c, & b' \\ b, & d, & c' \\ a', & c', & b'' \end{vmatrix} + 2(n-2)^2 \begin{vmatrix} a, & b, & c' \\ b, & c, & d' \\ a', & b', & c'' \end{vmatrix} + (n-2)(n-3) \begin{vmatrix} a, & b, & b' \\ b, & c, & c' \\ a', & b', & b'' \end{vmatrix} + (n-2)(n-1) \begin{vmatrix} a, & d, & a' \\ b, & e, & b' \\ a', & d', & a'' \end{vmatrix} = \Delta^2 D^2 H.$$

17. A little consideration will show that in this equation we may write  $n-1$  for  $n$ , and  $H_1$  for  $H$ . In fact, putting for a moment  $\delta = x_1 \partial_x + y_1 \partial_y + z_1 \partial_z$ , corresponding to the equation

$$\begin{vmatrix} a, & b, & a' \\ b, & c, & b' \\ a', & b', & a'' \end{vmatrix} = \Delta^2 H,$$

this other equation,

$$\begin{vmatrix} \delta a, & \delta b, & \delta a' \\ \delta b, & \delta c, & \delta b' \\ \delta a', & \delta b', & \delta a'' \end{vmatrix} = \Delta^2 H_1,$$

where ultimately  $(x_1, y_1, z_1)$  are to be replaced by  $(x, y, z)$ . We may operate upon this equation with  $D, D^2, \dots$  as before, the only difference being that in the first

instance  $\delta a$ ,  $\delta b$ , &c. are as regards  $(x, y, z)$  of degrees lower by unity than  $a$ ,  $b$ , &c., that is  $n-1$  must be substituted throughout in the place of  $n$ ; and when at the end of the process  $(x_1, y_1, z_1)$  are replaced by  $(x, y, z)$ , then  $\delta a$ ,  $\delta b$ , &c. become equal to  $a$ ,  $b$ , &c., from which the truth of the asserted proposition is manifest.

18. Hence writing  $n = 4$ , we have

$$2 \begin{vmatrix} a, & b, & c' \\ b, & c, & d' \\ a', & b', & c'' \end{vmatrix} + 8 \begin{vmatrix} a, & d, & a' \\ b, & e, & b' \\ a', & d', & a'' \end{vmatrix} + 2 \begin{vmatrix} a, & c, & b' \\ b, & d, & c' \\ a', & c', & b'' \end{vmatrix} + 6 \begin{vmatrix} b, & c, & a' \\ c, & d, & b' \\ b', & c', & a'' \end{vmatrix} = \Delta^2 D^2 H,$$

$$2 \begin{vmatrix} a, & d, & a' \\ b, & e, & b' \\ a', & d', & a'' \end{vmatrix} + 2 \begin{vmatrix} b, & c, & a' \\ c, & d, & b' \\ b', & c', & a'' \end{vmatrix} = \Delta^2 D^2 H_1,$$

and hence

$$2 \begin{vmatrix} a, & b, & c' \\ b, & c, & d' \\ a', & b', & c'' \end{vmatrix} + 2 \begin{vmatrix} a, & c, & b' \\ b, & d, & c' \\ a', & c', & b'' \end{vmatrix} + 2 \begin{vmatrix} b, & c, & a' \\ c, & d, & b' \\ b', & c', & a'' \end{vmatrix} = \Delta^2 (D^2 H - 3D^2 H_1),$$

which is the required equation

$$\text{III} = -\frac{1}{6}\Delta^2(D^2 H - 3D^2 H_1).$$

19. It is to be added, that the equation for  $\Delta^2 DH$  gives  $\text{IV} = \frac{1}{2}\Delta^2 DH$ ; the values of II and I are at once obtained from those of III and IV by interchanging  $(x, y, z)$  and  $(X, Y, Z)$ . Hence if we represent by  $\mathfrak{H}$ ,  $\mathfrak{D}$ , &c. the values which  $H$ ,  $D$ , &c. assume by this interchange, we may write

$$\begin{aligned} \text{I} &= -\frac{1}{2}\Delta^2 \mathfrak{D}\mathfrak{H}, \\ \text{II} &= +\frac{1}{6}\Delta^2(\mathfrak{D}^2 \mathfrak{H} - 3\mathfrak{D}^2 \mathfrak{H}_1), \\ \text{III} &= -\frac{1}{6}\Delta^2(D^2 H - 3D^2 H_1), \\ \text{IV} &= +\frac{1}{2}\Delta^2 DH; \end{aligned}$$

and the identical equation,

$$a \text{I} + b \text{II} + d \text{III} + e \text{IV} = 0,$$

gives therefore

$$-\mathfrak{D}\mathfrak{H} \cdot U + \frac{1}{3}(\mathfrak{D}^2 \mathfrak{H} - 3\mathfrak{D}^2 \mathfrak{H}_1) DU - \frac{1}{3}(D^2 H - 3D^2 H_1) \mathfrak{D}\mathbf{T} + DH \cdot \mathbf{T} = 0,$$

which is of itself sufficient to put in evidence the property that the curve  $D^2 H - 3D^2 H_1 = 0$  gives by its intersections with the tangent  $DU = 0$ , the tangentials of the point  $(x, y, z)$ . The last-mentioned equation is the equation for a quartic corresponding to Mr Salmon's equation

$$-\mathfrak{H} \cdot U + \frac{1}{3}\mathfrak{D}\mathfrak{H} \cdot DU - \frac{1}{3}DH \cdot \mathfrak{D}\mathbf{T} + H \cdot \mathbf{T} = 0$$

for the cubic  $U = 0$ .



20. It is worth while to give the investigation of the equation for the cubic; the matrix is

$$\begin{pmatrix} a, & b, & c, & d; & a', & b' \\ b, & c, & d, & e; & b', & c' \\ a', & b', & c', & d'; & a'', & b'' \end{pmatrix}$$

and the identical equation is

$$a\text{I} + b\text{II} + c\text{III} + d\text{IV} = 0,$$

where

$$\text{I} = \begin{vmatrix} c, & b, & b' \\ d, & c, & c' \\ c', & b', & b'' \end{vmatrix} - \begin{vmatrix} a, & c, & a' \\ b, & d, & b' \\ a', & c', & a'' \end{vmatrix},$$

$$\text{II} = - \begin{vmatrix} c, & b, & a' \\ d, & c, & b' \\ c', & b', & a'' \end{vmatrix},$$

$$\text{III} = - \begin{vmatrix} a, & b, & a' \\ b, & c, & b' \\ a', & b', & a'' \end{vmatrix},$$

$$\text{IV} = \begin{vmatrix} a, & b, & a' \\ b, & c, & b' \\ a', & b', & a'' \end{vmatrix},$$

(matrix variants reconstructed from scan: the III and IV expressions follow the same pattern with the appropriate column substitutions.)

or, as we may express them,

$$\begin{aligned} \text{I} = & a(0) \\ & + b(c'^2) \quad (12) \\ & + c(b''c - b'c' - c'b') \quad (13) \\ & + d(-b''b + b'b') \quad (14) \end{aligned}$$

$$\begin{aligned} \text{II} = & a(-c'^2) \quad (21) \\ & + b(0) \\ & + c(-a''c - b''b + a'c' + b'b' + c'a') \quad (23) \\ & + d(a''b + b''a - a'b' - b'a') \quad (24) \end{aligned}$$

$$\begin{aligned} \text{III} = & a(-b''c + b'c' + c'b') \quad (31) \\ & + b(a''c + b''b - a'c' - b'b' - c'a') \quad (32) \\ & + c(0) \\ & + d(-a''a + a'a') \quad (34) \end{aligned}$$

$$\begin{aligned} \text{IV} = & a(b''b - b'b') \quad (41) \\ & + b(-a''b - b''a + a'b' + b'a') \quad (42) \\ & + c(a''a - a'a') \quad (43) \\ & + d(0) \end{aligned}$$

260]

ON THE DOUBLE TANGENTS OF A PLANE CURVE.

199

which verify the identical equation. We have  $\text{III} = -\Delta^2 DH$ ,  $\text{IV} = \Delta^2 H$ , and thence  $\text{II} = +\Delta^2 \mathfrak{D}\mathfrak{H}$ ,  $\text{I} = -\Delta^2 \mathfrak{H}$ ; hence the equation in question

$$-\mathfrak{H} \cdot U + \frac{1}{3} \mathfrak{D}\mathfrak{H} \cdot DU - \frac{1}{3} DH \cdot \mathfrak{D}\mathbf{T} + H \cdot \mathbf{T} = 0.$$

21. One other example will be sufficient to render manifest the law of the formation of the multipliers I, II, III, IV.

In the case of a sextic curve we have the matrix

$$\begin{pmatrix} a, & b, & c, & d, & e, & f; & a', & b', & c', & d', & e' \\ b, & c, & d, & e, & f, & g; & b', & c', & d', & e', & f' \\ a', & b', & c', & d', & e', & f'; & a'', & b'', & c'', & d'', & e'' \end{pmatrix}$$

the identical equation is

$$a\text{I} + b\text{II} + f\text{III} + g\text{IV} = 0;$$

and the expressions for the multipliers I, II, III, IV are:

$$\begin{aligned} \text{I} &= \begin{vmatrix} f, & e, & b' \\ g, & f, & c' \\ f', & e', & b'' \end{vmatrix} + \begin{vmatrix} f, & d, & c' \\ g, & e, & d' \\ f', & d', & c'' \end{vmatrix} + \begin{vmatrix} f, & c, & d' \\ g, & d, & e' \\ f', & c', & d'' \end{vmatrix} + \begin{vmatrix} f, & b, & e' \\ g, & c, & f' \\ f', & b', & e'' \end{vmatrix}, \\ \text{II} &= - \begin{vmatrix} f, & e, & a' \\ g, & f, & b' \\ f', & e', & a'' \end{vmatrix} - \begin{vmatrix} f, & d, & b' \\ g, & e, & c' \\ f', & d', & b'' \end{vmatrix} - \begin{vmatrix} f, & c, & c' \\ g, & d, & d' \\ f', & c', & c'' \end{vmatrix} - \begin{vmatrix} f, & b, & d' \\ g, & c, & e' \\ f', & b', & d'' \end{vmatrix} - \begin{vmatrix} f, & a, & e' \\ g, & b, & f' \\ f', & a', & e'' \end{vmatrix}, \\ \text{III} &= - \begin{vmatrix} a, & b, & e' \\ b, & c, & f' \\ a', & b', & e'' \end{vmatrix} - \begin{vmatrix} a, & c, & d' \\ b, & d, & e' \\ a', & c', & d'' \end{vmatrix} - \begin{vmatrix} a, & d, & c' \\ b, & e, & d' \\ a', & d', & c'' \end{vmatrix} - \begin{vmatrix} a, & e, & b' \\ b, & f, & c' \\ a', & e', & b'' \end{vmatrix} - \begin{vmatrix} a, & f, & a' \\ b, & g, & b' \\ a', & f', & a'' \end{vmatrix}, \\ \text{IV} &= \begin{vmatrix} a, & b, & d' \\ b, & c, & e' \\ a', & b', & d'' \end{vmatrix} + \begin{vmatrix} a, & c, & c' \\ b, & d, & d' \\ a', & c', & c'' \end{vmatrix} + \begin{vmatrix} a, & d, & b' \\ b, & e, & c' \\ a', & d', & b'' \end{vmatrix} + \begin{vmatrix} a, & e, & a' \\ b, & f, & b' \\ a', & e', & a'' \end{vmatrix}. \end{aligned}$$

22. We have in fact

$$\begin{aligned} \text{I} &= a(0) \\ &+ b(-ge'' + ff') \quad (12) \\ &+ f(b''f + c''e + d''d + e''c - b'f' - c'e' - d'd' - e'c' - f'b') \quad (13) \\ &+ g(-b''e - c''d - d''c - e''b + b'e' + c'd' + d'c' + e'b') \quad (14) \end{aligned}$$

$$\begin{aligned} \text{II} &= a(e''g) \quad (21) \\ &+ b(0) \\ &+ f(-a''f - b''e - c''d - d''c - e''b + a'f' + b'e' + c'd' + d'c' + e'b' + f'a') \quad (23) \\ &+ g(a''e + b''d + c''c + d''b - a'e' - b'd' - c'c' - d'b' - e'a') \quad (24) \end{aligned}$$

$$\text{III} = a(-b''f - c''e - d''d - e''c + b'f' + c'e' + d'd' + e'c' + f'b') \quad (31)$$

$$+ b(a''f + b''e + c''d + d''c + e''b - a'f' - b'e' - c'd' - d'c' - e'b' - f'a') \quad (32)$$

$$+ f(0) \quad (34)$$

$$\text{IV} = a(b''e + c''d + d''c + e''b - b'e' - c'd' - d'c' - e'b') \quad (41)$$

$$+ b(-a''e - b''d - c''c - d''b + a'e' + b'd' + c'c' + d'b' + e'a') \quad (42)$$

$$+ f(a''a - a'a') \quad (43)$$

$$+ g(0)$$

which are of the form

$$\text{I} = a \cdot 0 + b(12) + f(13) + g(14),$$

$$\text{II} = a(21) + b \cdot 0 + f(23) + g(24),$$

$$\text{III} = a(31) + b(32) + f \cdot 0 + g(34),$$

$$\text{IV} = a(41) + b(42) + f(43) + g \cdot 0,$$

where (12) = -(21) &c., and the equation

$$a\text{I} + b\text{II} + f\text{III} + g\text{IV} = 0$$

is consequently satisfied.

23. The expression

$$\text{III} = - \begin{vmatrix} a & b & e' \\ b & c & f' \\ a' & b' & e'' \end{vmatrix} - \begin{vmatrix} a & c & d' \\ b & d & e' \\ a' & c' & d'' \end{vmatrix} - \begin{vmatrix} a & d & c' \\ b & e & d' \\ a' & d' & c'' \end{vmatrix} - \begin{vmatrix} a & e & b' \\ b & f & c' \\ a' & e' & b'' \end{vmatrix} - \begin{vmatrix} a & f & a' \\ b & g & b' \\ a' & f' & a'' \end{vmatrix}$$

leads to

$$\text{III} = -\Delta^2(D^4H - 5D^4H_1 + 10D^4H_2),$$

and consequently the equation of the curve which by its intersections with the tangent determines the tangentials of a point of a sextic, is

$$D^4H - 5D^4H_1 + 10D^4H_2 = 0.$$

24. In the general case of a curve of the order  $n$  the matrix is

$$\begin{pmatrix} a_0 & a_1 & a_2 & \dots & a_{n-1}; & a'_0 & a'_1 & \dots & a'_{n-2} \\ a_1 & a_2 & a_3 & \dots & a_n; & a'_1 & a'_2 & \dots & a'_{n-1} \\ a'_0 & a'_1 & a'_2 & \dots & a'_{n-1}; & a''_0 & a''_1 & \dots & a''_{n-2} \end{pmatrix}$$

where, in analogy with what precedes,

$$a_0 = (X, Y, Z)^n,$$

$$a_1 = (X, Y, Z)^{n-1}(x, y, z),$$

$$\vdots$$

$$a_{n-1} = (X, Y, Z)(x, y, z)^{n-1},$$

$$a_n = (x, y, z)^n,$$

260]

ON THE DOUBLE TANGENTS OF A PLANE CURVE.

201

and similarly for the accented letters, so that

$a_0 = 0$  is the equation of the curve;

$a_1 = 0$  is the equation of the first or  $(n-1)$ thic polar;

$a_{n-1} = 0$  is the equation of the last or line-polar, or what is the same thing, since  $(x, y, z)$  is a point on the curve, the tangent at this point;

$a_n = 0$ , the condition which expresses that  $(x, y, z)$  is a point of the curve;

and we have to form the identical equation

$$a_0 \text{ I} + a_1 \text{ II} + a_{n-1} \text{ III} + a_n \text{ IV} = 0.$$

25. If, for shortness, the columns of the last-mentioned matrix are represented by

$$1, 2, 3 \dots n, (1), (2) \dots (n-1),$$

and the determinants formed with these columns respectively by a corresponding notation  $\{1, 2, (1)\}$ ,  $\{1, 2, (2)\}$ , &c., then the expressions for the multipliers I, II, III, IV are as follows, viz.

$$\text{I} = \{n, n-1, (2)\} + \{n, n-2, (3)\} \dots + \{n, 2, (n-1)\},$$

$$\text{II} = -\{n, n-1, (1)\} - \{n, n-2, (2)\} \dots - \{n, 2, (n-2)\} - \{n, 1, (n-1)\},$$

$$\text{III} = -\{1, 2, (n-1)\} - \{1, 3, (n-2)\} \dots - \{1, n-1, (2)\} - \{1, n, (1)\},$$

$$\text{IV} = \{1, 2, (n-2)\} + \{1, 3, (n-1)\} \dots + \{1, n-1, (1)\};$$

the truth of the identical equation being shown, as in the foregoing special cases, by the transformation of the multipliers into the form

$$\text{I} = a_0 \cdot 0 + a_1(12) + a_{n-1}(13) + a_n(14),$$

$$\text{II} = a_0(21) + a_1 \cdot 0 + a_{n-1}(23) + a_n(24),$$

$$\text{III} = a_0(31) + a_1(32) + a_{n-1} \cdot 0 + a_n(34),$$

$$\text{IV} = a_0(41) + a_1(42) + a_{n-1}(43) + a_n \cdot 0,$$

where  $(12) = -(21)$ , &c.: the required expressions may be written down without difficulty.

26. Proceeding then to reduce the equation

$$\text{III} = -\{1, 2, (n-1)\} - \{1, 3, (n-2)\} \dots - \{1, n-1, (2)\} - \{1, n, (1)\},$$

we have the equation

$$\{1, 2, (1)\} = \Delta^2 H,$$

which is to be successively operated on with  $D$ . The degrees (less unity) of the columns

$$1, 2, \dots n-1, n, (1) (2), \dots n-1,$$

are

$$n-1, n-2, \dots 1, 0, n-2, n-3, \dots 0;$$

and the rule is to operate on each column of the determinant, multiplying by the degree less unity, and increasing the symbolical number by unity. Thus

$$\begin{aligned} D\{1, 2, (1)\} &= (n-1)\{2, 2, (1)\} + (n-2)\{1, 3, (1)\} + (n-2)\{1, 2, (2)\}, \\ &= (n-2)\{1, 3, (1)\} + (n-2)\{1, 2, (2)\}, \end{aligned}$$

since  $\{2, 2, (1)\}$  vanishes identically. The following Table shows the mode of effecting the operations:

*[The original page contains an extensive table headed by  $H, DH, D^2H, D^3H, D^4H, D^5H, D^6H, D^7H$ , etc., showing the numerical coefficients (with factorials of  $[n-2]$  and  $[n-1]$ ) which multiply the symbolic determinants  $\{1\,2(r)\}, \{1\,3(r)\}, \{1\,4(r)\}$ , &c. in the expansion of  $\Delta^2 D^r H$ . The general law of the table is summarised in art. 27 below.]*

The first three columns of the table show the numbers which give, by the addition of the numbers in the same horizontal line, the numerical coefficients of the factorials which multiply the different terms of  $H, DH$ , &c., and where in the last column  $12(1)$ , &c. are written for shortness in the place of  $\{1, 2, (1)\}$ , &c.

260]

ON THE DOUBLE TANGENTS OF A PLANE CURVE.

203

where the first three columns show the numbers which give, by the addition of the numbers in the same horizontal line, the numerical coefficients of the factorials which multiply the different terms of  $H$ ,  $DH$ , &c., and where in the last column  $12(1)$ , &c. are written for shortness in the place of  $\{1, 2, (1)\}$ , &c.

27. It is clear that we have in general

$$\begin{aligned}
 \Delta^2 D^r H = & \frac{1}{1} [n-2]^r [n-2]^r \{1, 2, (r+1)\} \\
 & + \frac{r}{1} [n-2]^r [n-2]^{r-1} \{1, 3, (r)\} \\
 & + \frac{r \cdot r - 1}{1 \cdot 2} [n-2]^r [n-2]^{r-2} \{1, 4, (r-1)\} \\
 & + \vdots \\
 & + \frac{1}{1} [n-2]^r [n-2]^0 \{1, r+2, (1)\} \\
 & + [n-1]^1 \left\{ \frac{r}{1} + R'_{r-1} \right\} [n-2]^{r-1} [n-2]^{r-2} \{2, 3, (r-1)\} \\
 & + \dots \\
 & + R'^{r-1}_{r-1} [n-2]^{r-1} [n-2]^0 \{2, r+1, (1)\} \\
 & + [n-1]^2 \left\{ \frac{r}{1} + R''_{r-2} \right\} [n-2]^{r-2} [n-2]^{r-3} \{3, 4, (r-3)\} \\
 & + \dots \\
 & + R''^{r-2}_{r-2} [n-2]^{r-2} [n-2]^0 \{3, r, (1)\} \\
 & + \dots \\
 & + [n-1]^{[(r-1)/2]} [r \text{ even or odd}] [n-2]^{[r/2]} [\dots] \left\{ \frac{1}{2}r+1, \frac{1}{2}r+2, (1) \right\} \\
 & + [n-1]^{[r/2]} [n-2]^{r/2} [n-2]^0 \left\{ \frac{1}{2}(r+1), \frac{1}{2}(r+3), (2) \right\} \\
 & + [n-1]^{[r/2]+1} [n-2]^{r/2+1} \left\{ \frac{1}{2}(r+1), \frac{1}{2}(r+5), (1) \right\}.
 \end{aligned}$$

and the general term is

$$[n-1]^\delta R_r^\delta [n-2]^{\beta+1} [n-2]^{r-2\delta-s} \{\beta+1, \beta+2+s, (r-2\delta+1-s)\},$$

where  $s$  extends from  $s=0$  to  $s=r-2\delta$ , and  $\delta$  from  $\delta=0$  to  $\delta=\frac{1}{2}r$  or  $\frac{1}{2}(r-1)$ , according as  $r$  is even or odd. The expression for the coefficients  $R_r^\delta$  is

$$R_r^\delta = \frac{[r]^\delta}{[\delta]^\delta},$$

and that of the other coefficients  $R_{r-s}^\delta$  ( $\delta = \text{or} < 1$ ) is not required for the present purpose.

28. According to a remark already made, the expressions for  $D^{n-2}H_1$ ,  $D^{n-2}H_2$ , &c. are at once obtained from that for  $D^rH$  by merely writing  $n-1$ ,  $n-2$ , &c. in the place of  $n$ : it is however to be noticed, that the quantity within the [ ] must not be negative, and that on its becoming so, the factorial is to be omitted.

THE COLLECTED MATHEMATICAL PAPERS  
of  
ARTHUR CAYLEY

Volume IV — pages 204–228

Papers 260 (conclusion), 261

29. I write now

$$\alpha + \beta = \sigma, \quad r - 2\delta - \sigma = \beta,$$

and I consider the expression

$$D^r H - \frac{n-1}{1} D^r H_1 + \&c.;$$

the general term of which is

$$R_r^\delta \{ \delta + 1, \delta + 2 + s, (r - 2\delta + 1 - s) \} \\ \times \left\{ \begin{array}{l} \frac{[n-1]^\delta}{1} [n-2]^\sigma [n-2]^\beta \\ - \frac{n-1}{1} [n-2]^\sigma [n-2]^\beta \\ + \frac{n-1 \cdot n-2}{1 \cdot 2} [n-3]^\sigma [n-3]^\beta \\ + \&c. \end{array} \right\},$$

or, as this may be written, putting  $q = n - \delta - 1$ ,

$$R_r^\delta \{ \delta + 1, \delta + 2 + s, (r - 2\delta + 1 - s) \} \\ \times [n-1]^\delta \left\{ \begin{array}{l} [n-2]^\sigma [n-2]^\beta \\ - \frac{q}{1} [n-3]^\sigma [n-3]^\beta \\ + \&c. \end{array} \right\}.$$

30. I assume  $r \not\geq n - 2$ , we have then  $\alpha + \beta = r - \delta \not\geq n - \delta - 2$ , and therefore  $\alpha + \beta < q$ . The general term of the series in  $\{ \}$  is

$$(-)^\xi \frac{[q]^\xi}{[\xi]^\xi} [n-2-\xi]^\sigma [n-2-\xi]^\beta,$$

where the terms for which  $n - 2 - \xi$  is negative are to be excluded, or what is the same thing, the series is not to be continued beyond  $\xi = n - 2$ . But observing that  $[q]^\xi$  vanishes for  $\xi > q$ , that is,  $\xi > n - \delta - 1$ , it is in fact the same thing whether the series is continued indefinitely or only to the term for which  $\xi = n - \delta - 1$ , and we may consistently with the condition  $\xi \not\geq n - 2$ , continue the series as far as  $\xi = n - \delta - 1$ , except in the case  $\delta = 0$ , when by doing so we include the term corresponding to  $\xi = n - 1$ , which in virtue of the condition ought to be excluded. The expression for the term in question is  $(-)^{n-1} [-1]^\sigma [-1]^\beta$ ; hence if the sum of the series continued to the proper point is  $S$ , the sum continued indefinitely (in the particular case  $\delta = 0$ ) is  $S + (-)^{n-1} [-1]^\sigma [-1]^\beta$ , but in every other case the sum continued indefinitely is simply  $S$ . And by a well-known theorem in finite differences, the sum continued indefinitely is in fact zero. That is, except in the case  $\delta = 0$ , we have  $S = 0$ , but in the excepted case

$$S + (-)^{n-1} [-1]^\sigma [-1]^\beta = 0;$$

or observing that  $\alpha + \beta (= r - \delta)$  is in this case  $= r$ , and transforming the factorials, we have

$$S = (-)^{n-r} [\alpha]^\alpha [\beta]^\beta,$$

or substituting for  $\alpha$  and  $\beta$  their values,

$$S = (-)^{n-r} [s]^s [r-s]^{r-s}.$$



31. Hence the general term of

$$D^r H - \frac{n-1}{1} D^r H_1 + \&c.$$

vanishes except for  $\delta = 0$ , but when  $\delta = 0$ , its value is

$$R_r^s \{1, 2+s, (r+1-s)\} \times (-)^{n-r} [s]^s [r-s]^{r-s};$$

or observing that  $R_r^s$  is equal to  $[r]^r \div [s]^s [r-s]^{r-s}$ , the value is simply

$$(-)^{n-r} [r]^r \{1, 2+s, (r+1-s)\},$$

that is, we have

$$\Delta^2 \left( D^r H - \frac{n-1}{1} D^r H_1 + \&c. \dots \right) = (-)^{n-r} [r]^r \Sigma \{1, 2+s, (r+1-s)\},$$

the summation in respect to  $s$  extending from  $s = 0$  to  $s = r$ . In particular, giving to  $r$  the values  $n-2$  and  $n-1$ , and attending to the expressions for III and IV, we find

$$\Delta^2 \left( D^{n-2} H - \frac{n-1}{1} D^{n-2} H_1 + \&c. \dots \right) = -[n-2]^{n-2} \text{III},$$

$$\Delta^2 \left( D^{n-1} H - \frac{n-1}{1} D^{n-1} H_1 + \&c. \dots \right) = -[n-3]^{n-1} \text{IV}.$$

32. The equation  $\text{III} = 0$  belongs to the curve which by its intersections with the tangent, gives the tangentials of a point of the curve  $U = 0$ . Hence the equation of the curve in question is

$$D^{n-2} H - \frac{n-1}{1} D^{n-2} H_1 + \&c. = 0,$$

which is Mr Salmon's theorem, leading to the solution of the problem of double tangents.

33. The expressions for I and II are obtained from those of IV and III by interchanging  $(X, Y, Z)$  and  $(x, y, z)$ , and reversing the sign. Hence if, as before,  $\mathfrak{H}$ ,  $\mathfrak{D}$ , &c. denote the values which  $H$ ,  $D$ , &c. assume by this interchange, we have

$$\Delta^2 \left( \mathfrak{D}^{n-2} \mathfrak{H} - \frac{n-1}{1} \mathfrak{D}^{n-2} \mathfrak{H}_1 + \&c. \dots \right) = [n-2]^{n-2} \text{II},$$

$$\Delta^2 \left( \mathfrak{D}^{n-1} \mathfrak{H} - \frac{n-1}{1} \mathfrak{D}^{n-1} \mathfrak{H}_1 + \&c. \dots \right) = [n-3]^{n-1} \text{I},$$

and the identical equation

$$a_0 \text{I} + a_1 \text{II} + a_{n-1} \text{III} + a_n \text{IV} = 0$$

becomes therefore

$$\left\{ \begin{array}{l} \left( \mathfrak{D}^{n-1}\mathfrak{H} - \frac{n-1}{1} \mathfrak{D}^{n-1}\mathfrak{H}_1 + \&c. \right) U \\ + \frac{1}{n(n-2)} \left( \mathfrak{D}^{n-2}\mathfrak{H} - \frac{n-1}{1} \mathfrak{D}^{n-2}\mathfrak{H}_1 + \&c. \right) DU \\ - \frac{1}{n(n-2)} \left( D^{n-2}H - \frac{n-1}{1} D^{n-2}H_1 + \&c. \right) \mathfrak{D}\mathbf{T} \\ - \left( D^{n-1}H - \frac{n-1}{1} D^{n-1}H_1 + \&c. \right) \mathbf{T} \end{array} \right\} = 0,$$

which is the general identical equation referred to in the introduction to the present memoir.

34. It is to be noticed that for  $n = 3$ , the equation is

$$(\mathfrak{D} - 2\mathfrak{H}_1)U + \frac{1}{3}\mathfrak{D}\mathfrak{H} \cdot DU - \frac{1}{3}DH \cdot \mathfrak{D}\mathbf{T} - (H - 2H_1)\mathbf{T} = 0.$$

But we have  $H_1 = H$ , and in like manner  $\mathfrak{H}_1 = \mathfrak{H}$ , and the equation thus becomes

$$-\mathfrak{H}U + \frac{1}{3}\mathfrak{D}\mathfrak{H} \cdot DU - \frac{1}{3}DH \cdot \mathfrak{D}\mathbf{T} + H\mathbf{T} = 0.$$

And so also for  $n = 4$ , the equation is

$$(\mathfrak{D}^2\mathfrak{H} - 3\mathfrak{D}^2\mathfrak{H}_1)U + \frac{1}{8}(\mathfrak{D}^2\mathfrak{H} - 3\mathfrak{D}^2\mathfrak{H}_1)DU - \frac{1}{8}(D^2H - 3D^2H_1)\mathfrak{D}\mathbf{T} - (DH - 3DH_1)\mathbf{T} = 0.$$

But we have in general  $DH_1 = \frac{n-3}{n-2}DH$ , and therefore in the present case  $DH_1 = \frac{1}{2}DH$ , and consequently  $\mathfrak{D}\mathfrak{H}_1 = \frac{1}{2}\mathfrak{D}\mathfrak{H}$ , and the equation thus becomes

$$-\mathfrak{D}\mathfrak{H} \cdot U + \frac{1}{8}(\mathfrak{D}^2\mathfrak{H} - 3\mathfrak{D}^2\mathfrak{H}_1)DU - \frac{1}{8}(D^2H - 3D^2H_1)\mathfrak{D}\mathbf{T} + DH \cdot \mathbf{T} = 0,$$

which agree with the results previously obtained for the two particular cases.

## 261.

## ON THE CONIC OF FIVE-POINTIC CONTACT AT ANY POINT OF A PLANE CURVE.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLIX. (for the year 1859), pp. 371–400. Received March 1,—Read March 24, 1859.]

The tangent is a line passing through two consecutive points of a plane curve, and we may in like manner consider the conic which passes through five consecutive points of a plane curve; and as there are certain singular points, viz. the points of inflexion where three consecutive points of the curve lie in a line, so there are singular points where six consecutive points of the curve lie in a conic. In the particular case where the given curve is a cubic, the last-mentioned species of singular points have been considered by Plücker and Steiner, and in the same particular case, the theory of the conic of five-pointic contact has recently been established by Mr Salmon. But the general case, where the curve is of any order whatever, has not so far as I am aware been hitherto considered;—the establishment of this theory is the object of the present memoir.

## I. INVESTIGATION OF THE EQUATION OF THE CONIC OF FIVE-POINTIC CONTACT.

1. I take  $(X, Y, Z)$  as current coordinates, and I represent the equation of the given curve by

$$\mathbf{T} = (* \mid X, Y, Z)^m = 0.$$

Let  $(x, y, z)$  be the coordinates of a given point on the curve, and let  $U = (* \mid x, y, z)^m$  be what  $\mathbf{T}$  becomes when  $(x, y, z)$  are written in the place of  $(X, Y, Z)$ ; we have therefore  $U = 0$  as a condition satisfied by the coordinates of the point in question.

2. Write for shortness

$$DU = (X\partial_x + Y\partial_y + Z\partial_z)U,$$

$$D^2U = (X\partial_x + Y\partial_y + Z\partial_z)^2U,$$

and let  $\Pi = aX + bY + cZ = 0$  be the equation of a line. It is easy to see that

$$D^2U - \Pi \cdot DU = 0$$

will be the equation of a conic having an ordinary (two-pointic) contact with the curve at the point  $(x, y, z)$ . In fact the equation  $DU = 0$  is that of the tangent at the point in question, and the equation  $D^2U = 0$  is that of the penultimate polar (or polar conic) of the point, which conic is touched by the tangent; the assumed equation represents therefore a conic having an ordinary (two-pointic) contact with the polar conic, and therefore with the curve. It may be added that the two conics intersect besides in a pair of points, and that the line joining these, or common chord of the two conics, is the line represented by the equation  $\Pi = 0$ ; and this being so, the constants  $(a, b, c)$  of the line  $\Pi = 0$  can be so determined as to give rise to a five-pointic contact.

3. Consider the coordinates of a point of the curve as functions of a single variable parameter; then for the present purpose the coordinates of a point consecutive to  $(x, y, z)$  may be taken to be

$$\begin{aligned} x + dx + \frac{1}{2}d^2x + \frac{1}{6}d^3x + \frac{1}{24}d^4x, \\ y + dy + \frac{1}{2}d^2y + \frac{1}{6}d^3y + \frac{1}{24}d^4y, \\ z + dz + \frac{1}{2}d^2z + \frac{1}{6}d^3z + \frac{1}{24}d^4z, \end{aligned}$$

values which, substituted for  $X, Y, Z$ , must satisfy the equations

$$\mathbf{T} = 0, \quad D^2U - \Pi \cdot DU = 0.$$

4. I write for shortness

$$\begin{aligned} \partial_1 &= dx \partial_x + dy \partial_y + dz \partial_z, \\ \partial_2 &= d^2x \partial_x + d^2y \partial_y + d^2z \partial_z, \\ \partial_3 &= d^3x \partial_x + d^3y \partial_y + d^3z \partial_z, \\ \partial_4 &= d^4x \partial_x + d^4y \partial_y + d^4z \partial_z, \end{aligned}$$

then the consecutive value of  $\mathbf{T}$  is

$$\exp\left(\partial_1 + \frac{1}{2}\partial_2 + \frac{1}{6}\partial_3 + \frac{1}{24}\partial_4\right) U$$

(Read  $\exp. z$ , exponential of  $z$ ,  $= e^z$ ), which is

$$\begin{aligned} &\times \left(1 + \partial_1 + \frac{1}{2}\partial_1^2 + \frac{1}{6}\partial_1^3 + \frac{1}{24}\partial_1^4\right) \\ &\times \left(1 + \frac{1}{2}\partial_2 + \frac{1}{4}\partial_1\partial_2 + \frac{1}{8}\partial_2^2\right) \\ &\times \left(1 + \frac{1}{6}\partial_3 + \frac{1}{6}\partial_1\partial_3\right) \\ &\times \left(1 + \frac{1}{24}\partial_4\right), \\ &= U \\ &+ \partial_1 U \\ &+ \frac{1}{2}(\partial_1^2 + \partial_2)U \\ &+ \frac{1}{6}(\partial_1^3 + 3\partial_1\partial_2 + \partial_3)U \\ &+ \frac{1}{24}(\partial_1^4 + 6\partial_1^2\partial_2 + 4\partial_1\partial_3 + 3\partial_2^2 + \partial_4)U, \end{aligned}$$

261]

CONTACT AT ANY POINT OF A PLANE CURVE.

209

the several terms of which must respectively vanish, and we have therefore

$$\begin{aligned}
 U &= 0, \\
 \partial_1 U &= 0, \\
 \partial_2 U &= -\partial_1^2 U, \\
 \partial_3 U &= -(\partial_1^3 + 3\partial_1 \partial_2)U, \\
 \partial_4 U &= -(\partial_1^4 + 6\partial_1^2 \partial_2 + 4\partial_1 \partial_3 + 3\partial_2^2)U.
 \end{aligned}$$

5. Next, preparing to substitute in the equation

$$D^2 U - \Pi \cdot DU = 0,$$

the consecutive value of  $DU$  is

$$(x + dx + \frac{1}{2}d^2x + \frac{1}{6}d^3x + \frac{1}{24}d^4x) \partial_x U + \&c.$$

where

$$\partial_x U = (x\partial_x + y\partial_y + z\partial_z)U = mU.$$

Reducing by the above results, the consecutive value of  $DU$  is

$$= -\frac{1}{2}\partial_1^2 U - \frac{1}{6}(\partial_1^3 + 3\partial_1 \partial_2)U - \frac{1}{24}(\partial_1^4 + 6\partial_1^2 \partial_2 + 4\partial_1 \partial_3 + 3\partial_2^2)U.$$

6. Hence also writing

$$\begin{aligned}
 P &= ax + by + cz, \\
 \partial_1 P &= a dx + b dy + c dz, \\
 \partial_2 P &= a d^2x + b d^2y + c d^2z,
 \end{aligned}$$

the consecutive value of  $-\Pi DU$  is  $-(P + \partial_1 P + \frac{1}{2}\partial_2 P)$  multiplied into the consecutive value of  $DU$ , and the product is

$$\begin{aligned}
 &= P \cdot \frac{1}{2}\partial_1^2 U \\
 &\quad + P \cdot \frac{1}{6}(\partial_1^3 + 3\partial_1 \partial_2)U \quad + \partial_1 P \cdot \frac{1}{2}\partial_1^2 U \\
 &\quad + P \cdot \frac{1}{24}(\partial_1^4 + 6\partial_1^2 \partial_2 + 4\partial_1 \partial_3 + 3\partial_2^2)U + \partial_1 P \cdot \frac{1}{6}(\partial_1^3 + 3\partial_1 \partial_2)U + \frac{1}{2}\partial_2 P \cdot \frac{1}{2}\partial_1^2 U.
 \end{aligned}$$

7. The consecutive value of  $D^2 U$  is

$$\begin{aligned}
 &= (x + dx + \frac{1}{2}d^2x + \frac{1}{6}d^3x + \frac{1}{24}d^4x)^2 \partial_x^2 U + \&c. \\
 &\quad \left. \begin{aligned}
 &= x^2 \\
 &+ 2x dx \\
 &+ x^2 dx + (dx)^2 \\
 &+ \frac{1}{3}x d^3x + dx d^2x \\
 &+ \frac{1}{12}x d^4x + \frac{1}{3}dx d^3x + \frac{1}{4}(d^2x)^2
 \end{aligned} \right\} \partial_x^2 U + \&c.,
 \end{aligned}$$

C. IV.

27

which is

$$\left. \begin{aligned} & \partial_1^2 \\ & + 2\partial_1\partial_2 \\ & + \partial_2^2 + \frac{2}{3}\partial_1\partial_3 \\ & + \frac{1}{6}\partial_1\partial_4 + \frac{2}{3}\partial_2\partial_3 + \frac{1}{4}\partial_2^2 \end{aligned} \right\} U;$$

and observing that

$$\begin{aligned} \partial_x^2 U &= m(m-1)U, \\ \partial_x \partial_1 U &= (m-1)\partial_1 U, \\ \partial_x \partial_2 U &= (m-1)\partial_2 U, \\ \partial_x \partial_3 U &= (m-1)\partial_3 U, \\ \partial_x \partial_4 U &= (m-1)\partial_4 U, \end{aligned}$$

and reducing as before, the consecutive value of  $D^2U$  is

$$\begin{aligned} &= -(m-2)\partial_1^2 U \\ & - \frac{1}{3}[(m-1)\partial_1^3 + 3(m-2)\partial_1\partial_2]U \\ & - \frac{1}{12}[(m-1)(\partial_1^4 + 6\partial_1^2\partial_2) + (m-2)(4\partial_1\partial_3 + 3\partial_2^2)]U. \end{aligned}$$

8. Substituting in the equation  $D^2U - \Pi \cdot DU = 0$ , we obtain as the conditions of a five-pointic contact

$$\begin{aligned} & -(m-2)\partial_1^2 U + P \cdot \frac{1}{2}\partial_1^2 U = 0, \\ & -\frac{1}{3}[(m-1)\partial_1^3 + 3(m-2)\partial_1\partial_2]U + P \cdot \frac{1}{6}(\partial_1^3 + 3\partial_1\partial_2)U + \partial_1 P \cdot \frac{1}{2}\partial_1^2 U = 0, \\ & -\frac{1}{12}[(m-1)(\partial_1^4 + 6\partial_1^2\partial_2) + (m-2)(4\partial_1\partial_3 + 3\partial_2^2)]U \\ & + P \cdot \frac{1}{24}[\partial_1^4 + 6\partial_1^2\partial_2 + 4\partial_1\partial_3 + 3\partial_2^2]U + \partial_1 P \cdot \frac{1}{6}(\partial_1^3 + 3\partial_1\partial_2)U + \frac{1}{2}\partial_2 P \cdot \frac{1}{2}\partial_1^2 U = 0; \end{aligned}$$

or reducing

$$\begin{aligned} P &= 2(m-2), \\ \partial_1 P &= \frac{2}{3} \frac{\partial_1^3 U}{\partial_1^2 U}, \\ \partial_2 P &= \frac{1}{2} \frac{[\partial_1^4 + 6\partial_1^2\partial_2]U}{\partial_1^2 U} - \frac{4}{3} \frac{\partial_1^3 U}{\partial_1^2 U} \cdot \frac{[\partial_1^3 + 3\partial_1\partial_2]U}{\partial_1^2 U}, \end{aligned}$$

which are the conditions of a five-pointic contact: it is to be remarked that if only the first and second conditions are satisfied, we have a four-pointic contact, and if only the first condition is satisfied, a three-pointic contact.

9. We have to reduce the last-mentioned equations; suppose that  $A, B, C$  are the first derived functions of  $U$ , then the equation  $\partial_1 U = 0$  may be written

$$A dx + B dy + C dz = 0,$$

261]

211

and this will be satisfied identically if

$$\begin{aligned} dx &= B\nu - C\mu, \\ dy &= C\lambda - A\nu, \\ dz &= A\mu - B\lambda, \end{aligned}$$

where  $\lambda, \mu, \nu$  are arbitrary multipliers, which may be taken to be constants. We have therefore  $\partial_1 = \mathbf{D}$ , where

$$\mathbf{D} = (B\nu - C\mu)\partial_x + (C\lambda - A\nu)\partial_y + (A\mu - B\lambda)\partial_z.$$

10. The resulting expressions for  $\partial_1^2 U$ ,  $\partial_1^3 U$ ,  $\partial_1^4 U$  may be exhibited in the reduced forms given by Hesse, viz. if  $\varpi = \lambda x + \mu y + \nu z$ , we have

$$\begin{aligned} \partial_1^2 U &= P_2 U - Q_2 \varpi^2, \\ \partial_1^3 U &= P_3 U - Q_3 \varpi^3, \\ \partial_1^4 U &= P_4 U - Q_4 \varpi^4, \end{aligned}$$

where the values of  $P_2, P_3, P_4; Q_2, Q_3, Q_4$  are as follows, viz. if  $(a, b, c, f, g, h)$  are the second derived functions of  $U$ , and if

$$H = \begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix}$$

be the Hessian; if, moreover,

$$\Phi = - \begin{vmatrix} \cdot, & \lambda, & \mu, & \nu \\ \lambda, & a, & h, & g \\ \mu, & h, & b, & f \\ \nu, & g, & f, & c \end{vmatrix}$$

be the bordered Hessian (we may also write  $\Phi = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \lambda, \mu, \nu)^2$ , where  $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$  are the inverse coefficients of  $(a, b, c, f, g, h)$ , viz.  $\mathfrak{A} = (bc - f^2)$ , &c.); and finally, if for shortness we write

$$\Omega = \partial_x \Phi \cdot \partial_x \Phi + \partial_y \Phi \cdot \partial_y \Phi + \partial_z \Phi \cdot \partial_z \Phi,$$

then we have

$$\begin{aligned} P_2 &= \frac{m}{m-1} \Phi, & Q_2 &= \frac{1}{(m-1)^2} H, \\ P_3 &= \frac{m}{m-1} \mathbf{D}\Phi, & Q_3 &= \frac{1}{(m-1)^2} \mathbf{D}H, \\ P_4 &= \frac{m}{m-1} \mathbf{D}^2\Phi - \frac{m\Omega}{(m-1)^2} H, & Q_4 &= \frac{1}{(m-1)^2} \mathbf{D}^2H - \frac{\Omega}{(m-1)^2} \Phi + \frac{3(m-2)}{(m-1)^3} H\Phi. \end{aligned}$$

In the present case  $U = 0$ , and we have

$$\begin{aligned}\partial_1^2 U &= -Q_2 \varpi^2, \\ \partial_1^3 U &= -Q_3 \varpi^3, \\ \partial_1^4 U &= -Q_4 \varpi^4.\end{aligned}$$

11. Hence, substituting for  $Q_2$  and  $Q_3$  their values, the first and second of the equations for the five-pointic contact give

$$\begin{aligned}P &= 2(m-2), \\ \partial_1 P &= \frac{2}{3} \frac{\mathbf{D}H}{H} \varpi,\end{aligned}$$

and observing that  $\Pi$  is a linear function of  $(X, Y, Z)$ , and consequently that  $P, \partial_1 P$  denote simply the values which  $\Pi$  assumes when  $(x, y, z), (dx, dy, dz)$  are respectively substituted for  $(X, Y, Z)$ , we see at once that these two conditions will be satisfied if we put

$$\Pi = \frac{1}{3} \frac{1}{H} DH + \Lambda \cdot DU,$$

where  $\Lambda$  is an arbitrary constant, or, what is the same thing, an arbitrary function of  $(x, y, z)$ . We have thus the general equation of a conic of four-pointic contact.

12. The above value of  $\Pi$  gives

$$\partial_2 P = \frac{1}{3} \frac{1}{H} \partial_2 H + \Lambda \partial_1 U,$$

and the third equation of the system of conditions for a five-pointic contact is therefore

$$\frac{1}{3} \frac{1}{H} \partial_2 H + \Lambda \partial_1 U = \frac{1}{2} \frac{[\partial_1^4 + 6\partial_1^2 \partial_2]U}{\partial_1^2 U} - \frac{4}{3} \frac{\partial_1^3 U}{\partial_1^2 U} \cdot \frac{[\partial_1^3 + 3\partial_1 \partial_2]U}{\partial_1^2 U},$$

which leads to the value of  $\Lambda$ .

13. We have in general

$$\begin{aligned}\partial_1^2 U &= \frac{m}{m-1} \Phi U - \frac{1}{(m-1)^2} H \varpi^2, \\ \partial_1^3 U &= \frac{m}{m-1} \mathbf{D}\Phi \cdot U - \frac{1}{(m-1)^2} \mathbf{D}H \cdot \varpi^3, \\ (\partial_1^3 + 3\partial_1 \partial_2) U &= \frac{m}{m-1} \mathbf{D}\Phi \cdot U - \frac{1}{(m-1)^2} \mathbf{D}H \cdot \varpi^3, \\ (\partial_1^4 + 6\partial_1^2 \partial_2) U &= \frac{m}{m-1} \left( 2\partial_2 \Phi + \mathbf{D}^2 \Phi + \frac{3}{m-1} \Omega \right) U - \frac{1}{(m-1)^2} \left( \mathbf{D}^2 H + \frac{3\Omega}{m-1} \Phi - \frac{3(m-2)}{m-1} H \Phi \right) \varpi^4,\end{aligned}$$



261]

CONTACT AT ANY POINT OF A PLANE CURVE.

213

the last two of which have not yet been demonstrated. The value of  $\partial_2 \partial_1^2 U$  (which, however, is not required for the present purpose) is

$$= \varpi^2 \frac{-3(m-3)}{m-1} \Omega - \frac{4\Omega}{m-1} \varpi^2,$$

which also is not yet demonstrated.

14. Putting  $U = 0$ , we have

$$\partial_1^2 U = -\frac{1}{(m-1)^2} H \varpi^2,$$

$$\partial_1^3 U = -\frac{1}{(m-1)^2} \mathbf{D}H \cdot \varpi^3,$$

$$(\partial_1^3 + 3\partial_1 \partial_2) U = -\frac{1}{(m-1)^2} \mathbf{D}H \cdot \varpi^3,$$

$$(\partial_1^4 + 6\partial_1^2 \partial_2) U = -\frac{1}{(m-1)^2} \left( \mathbf{D}^2 H + \mathbf{D}^2 H + \frac{3\Omega}{m-1} \Phi - \frac{3(m-2)}{m-1} H \Phi \right) \varpi^4,$$

and substituting,

$$\frac{1}{3} \frac{1}{H} \partial_2 H + \frac{1}{(m-1)^2} H \varpi^2 \Lambda = \frac{1}{2} \frac{1}{H} \left( 2\partial_2 H + \mathbf{D}^2 H + \frac{3\Omega}{m-1} \Phi - \frac{3(m-2)}{m-1} H \Phi \right) - \frac{4}{3} \left( \frac{\mathbf{D}H}{H} \right)^2,$$

where the term involving  $\partial_2 H$  disappears; the equation may be written

$$\frac{9H \varpi^2}{(m-1)^2} \Lambda = 3H \left( \mathbf{D}^2 H + \frac{3\Omega}{m-1} \Phi - \frac{3(m-2)}{m-1} H \Phi \right) - 4(\mathbf{D}H)^2,$$

which I will represent by

$$\frac{9H \varpi^2}{(m-1)^2} \Lambda = 3R_2 R_4 - 4R_3^2,$$

the values of  $R_2, R_3, R_4$  being

$$\begin{cases} R_2 = H, \\ R_3 = \mathbf{D}H, \\ R_4 = \mathbf{D}^2 H + \frac{3\Omega}{m-1} \Phi - \frac{3(m-2)}{m-1} H \Phi. \end{cases}$$

15. We have  $R_2 = H$ , and it will be shown that

$$(m-1)^2 R_3^2 = -9(m-2)^2 H \Psi + 3(m-2) H \Omega \Phi - \Psi \varpi^2,$$

$$(m-1)^2 R_4 = -12(m-2)^2 H \Psi + 4(m-2) \Omega \Phi - \Omega \varpi^2,$$

where for shortness

$$\begin{aligned}\Psi &= (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2, \\ \Omega &= (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H,\end{aligned}$$

and hence, writing

$$9H \varpi^2 \Lambda = 3R_2(m-1)^2 R_4 - 4(m-1)^2 R_3^2,$$

we have

$$9H \varpi^2 \Lambda = -3\Omega\Psi + 4\Psi;$$

or replacing  $H, \Psi$  by their values,

$$9H \varpi^2 \Lambda = -3H (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H + 4(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2,$$

and  $\Lambda$  having this value, the equation of the five-pointic conic is

$$D^2U - \left( \frac{1}{3} \frac{1}{H} DH + \Lambda DU \right) DU = 0,$$

where it will be recollected that the current coordinates are  $(X, Y, Z)$ , and that  $D$  denotes  $X\partial_x + Y\partial_y + Z\partial_z$ .

16. I remark, in passing, that the problem of finding the circle of curvature at a given point of a plane curve, is in fact that of determining the conic having with the curve at the given point a three-pointic contact, and besides passing through two given points. The equation of a conic having an ordinary contact, is

$$D^2U - \Pi \cdot DU = 0,$$

where

$$\Pi = aX + bY + cZ,$$

and the condition of a three-pointic contact is

$$ax + by + cz = 2(m-2).$$

Let the coordinates of the two given points be

$$(x_1, y_1, z_1), \quad (x_2, y_2, z_2),$$

and let  $(D^2U)_1$  &c. be the corresponding values of  $D^2U$ , &c., then we have

$$aX + bY + cZ = \frac{D^2U}{DU},$$

$$ax_1 + by_1 + cz_1 = \left( \frac{D^2U}{DU} \right)_1,$$

$$ax_2 + by_2 + cz_2 = \left( \frac{D^2U}{DU} \right)_2;$$

and if from the four equations we eliminate  $a, b, c$ , we find

$$\begin{vmatrix} XDU, & YDU, & ZDU, & D^2U \\ x, & y, & z, & 2(m-2) \\ x_1, & y_1, & z_1, & \left(\frac{D^2U}{DU}\right)_1 \\ x_2, & y_2, & z_2, & \left(\frac{D^2U}{DU}\right)_2 \end{vmatrix} = 0,$$

for the equation of the conic in question;  $x, y, z$  being the coordinates of the point of contact, and  $X, Y, Z$  current coordinates.

## II. DEMONSTRATION OF IDENTITIES ASSUMED IN THE PRECEDING SECTION.

*Proof of the expressions for  $(\partial_1^3 + 3\partial_1\partial_2)U$  and  $(\partial_1^4 + 6\partial_1^2\partial_2)U$ :*

17. It will be remembered that  $\partial_1, \partial_2$  stand originally for

$$\begin{aligned} dx \partial_x + dy \partial_y + dz \partial_z, \\ d^2x \partial_x + d^2y \partial_y + d^2z \partial_z, \end{aligned}$$

and that  $A, B, C$  being the first derived functions of  $U$ ,  $dx, dy, dz$  are changed into

$$B\nu - C\mu, \quad C\lambda - A\nu, \quad A\mu - B\lambda,$$

and that the resulting value of  $\partial_1$ , viz.

$$(B\nu - C\mu)\partial_x + (C\lambda - A\nu)\partial_y + (A\mu - B\lambda)\partial_z,$$

is also represented by  $\mathbf{D}$ , so that  $\partial_1 = \mathbf{D}$ . The corresponding values of  $d^2x, d^2y, d^2z$  are

$$\begin{aligned} d^2x &= \nu dB - \mu dC, \\ d^2y &= \lambda dC - \nu dA, \\ d^2z &= \mu dA - \lambda dB, \end{aligned}$$

where we have

$$dA = a dx + h dy + g dz, \quad \&c.,$$

in which  $dx, dy, dz$  are to be replaced by the values

$$B\nu - C\mu, \quad C\lambda - A\nu, \quad A\mu - B\lambda;$$

and  $\partial_2$  really denotes what

$$d^2x \partial_x + d^2y \partial_y + d^2z \partial_z$$

becomes when the above values are substituted for  $d^2x, d^2y, d^2z$ . But in the expressions  $\partial_1^2 U, \partial_1\partial_2 U$  &c., the symbols  $\partial_x, \partial_y, \partial_z$  contained in  $\partial_1$  and  $\partial_2$  operate only on  $U$ , and not on the variable quantities  $A, B, C$ , &c. contained in  $\partial_1$  and  $\partial_2$ .

18. If now we treat  $\partial_1$  as an operand, that is, perform the differentiations on the variable quantities  $A, B, C$  which enter into  $\partial_1$ , we obtain

$$\partial_1 \cdot \partial_1 = \partial_1^2 + \partial_2,$$

or, what is the same thing, operating on  $\partial_1 U$  with  $\partial_1$ , the result is

$$\partial_1^2 U + \partial_2 U,$$

and in like manner

$$\partial_1 \cdot \partial_1^2 U = \partial_1^3 U + 2\partial_1 \partial_2 U.$$

It is, in fact, upon these principles that Hesse's values of  $\partial_1^2 U, \partial_1^3 U$ , &c. were obtained, and we may by means of them obtain the other expressions assumed in the preceding section.

19. In fact, starting from Hesse's equation,

$$\partial_1^2 U = \frac{m}{m-1} \Phi U - \frac{1}{(m-1)^2} H \varpi^2,$$

we have

$$(\partial_1^3 + 2\partial_1 \partial_2) U = \frac{m}{m-1} (U \mathbf{D} \Phi + \Phi \mathbf{D} U) - \frac{1}{(m-1)^2} (\mathbf{D} H \cdot \varpi^2 + H \cdot 2\varpi \mathbf{D} \varpi).$$

But we have identically  $\mathbf{D} U = 0$ ,  $\mathbf{D} \varpi = 0$ , and this equation becomes therefore

$$(\partial_1^3 + 2\partial_1 \partial_2) U = \frac{m}{m-1} U \mathbf{D} \Phi - \frac{1}{(m-1)^2} \mathbf{D} H \cdot \varpi^2;$$

this is precisely Hesse's value of  $\partial_1^3 U$ , and thus we have  $\partial_1 \partial_2 U = 0$ , and therefore

$$\partial_1^3 U = \frac{m}{m-1} U \mathbf{D} \Phi - \frac{1}{(m-1)^2} \mathbf{D} H \cdot \varpi^3.$$

20. In like manner, starting from the expression of  $\partial_1^3 U$ , we have

$$(\partial_1^4 + 3\partial_1^2 \partial_2) U = \frac{m}{m-1} (U \mathbf{D} \cdot \mathbf{D} \Phi + \mathbf{D} U \cdot \mathbf{D} \Phi) - \frac{1}{(m-1)^2} (\mathbf{D} \cdot \mathbf{D} H \cdot \varpi^3 + 3\mathbf{D} H \cdot \varpi^2 \mathbf{D} \varpi),$$

or since  $\mathbf{D} U$  and  $\mathbf{D} \varpi$  vanish identically, and the values of  $\mathbf{D} \cdot \mathbf{D} \Phi$ ,  $\mathbf{D} \cdot \mathbf{D} H$  are  $\partial_1^2 \Phi + \mathbf{D}^2 \Phi$ ,  $\partial_2 H + \mathbf{D}^2 H$ , we have

$$(\partial_1^4 + 3\partial_1^2 \partial_2) U = \frac{m}{m-1} U (\partial_2 \Phi + \mathbf{D}^2 \Phi) - \frac{1}{(m-1)^2} (\partial_2 H + \mathbf{D}^2 H) \varpi^3,$$

and if from the double of this equation we subtract Hesse's equation,

$$\partial_1^4 U = \frac{m}{m-1} \left( U \mathbf{D}^2 \Phi + \frac{3\Omega}{m-1} U \Phi \right) - \frac{1}{(m-1)^2} \left( \mathbf{D}^2 H \cdot \varpi^4 + \frac{3\Omega}{m-1} \Phi \varpi^4 - \frac{3(m-2)}{m-1} H \Phi \varpi^4 \right),$$

261]

CONTACT AT ANY POINT OF A PLANE CURVE.

217

we find the required relation,

$$6\partial_1^2\partial_2U = \frac{m}{m-1}\left(2U\partial_2\Phi + \frac{3\Omega}{m-1}U\Phi\right) - \frac{1}{(m-1)^2}\left(2\partial_2H + \frac{3\Omega}{m-1}\Phi - \frac{3(m-2)}{m-1}H\Phi\right)\varpi^2.$$

*Proof of the expression for  $\partial_2H$ :*

21. We have

$$\partial_2H = \partial_xH(\nu dB - \mu dC) + \partial_yH(\lambda dC - \nu dA) + \partial_zH(\mu dA - \lambda dB),$$

where

$$\begin{aligned} dA &= a dx + h dy + g dz, \\ dB &= h dx + b dy + f dz, \\ dC &= g dx + f dy + c dz, \end{aligned}$$

in which  $dx, dy, dz$  are to be replaced by their values: we have therefore

$$\partial_2H = \partial_xH \left\{ \nu[(bC-fB)\lambda + (fA-hC)\mu + (hB-bA)\nu] - \mu[(fC-cB)\lambda + (cA-gC)\mu + (gB-fA)\nu] \right\} + \&c.,$$

where the coefficient of  $\partial_xH$  is

$$b\lambda^2 + (gC - cA)\mu^2 + (hB - bA)\nu^2 - (2fA - gB - hC)\mu\nu + (bC - fB)\nu\lambda + (cB - fC)\lambda\mu;$$

or, since we have

$$\begin{aligned} (m-1)A &= ax + hy + gz, \\ (m-1)B &= hx + by + fz, \\ (m-1)C &= gx + fy + cz, \end{aligned}$$

the coefficient, omitting the factor  $\frac{1}{m-1}$ , which will be afterwards restored, is

$$\begin{aligned} &+ [g(gx + fy + cz) - c(ax + hy + gz)]\mu^2 \\ &+ [h(hx + by + fz) - b(ax + hy + gz)]\nu^2 \\ &+ [2f(ax + hy + gz) - g(hx + by + fz) - h(gx + fy + cz)]\mu\nu \\ &+ [b(gx + fy + cz) - f(hx + by + fz)]\nu\lambda \\ &+ [c(hx + fy + cz) - f(gx + fy + cz)]\lambda\mu \\ &= 0\lambda^2 \\ &+ (-\mathfrak{C}x + \mathfrak{F}y)\mu^2 \\ &+ (-\mathfrak{C}x + \mathfrak{G}z)\nu^2 \\ &+ (-2\mathfrak{F}x + \mathfrak{G}y + \mathfrak{H}z)\mu\nu \\ &+ (-\mathfrak{G}x + \mathfrak{B}z)\nu\lambda \\ &+ (-\mathfrak{H}x + \mathfrak{F}y + \mathfrak{B}z)\lambda\mu, \end{aligned}$$

C. IV.

28

which is equal to

$$-(\mathfrak{A}\lambda^2 + \mathfrak{B}\mu^2 + \mathfrak{C}\nu^2 + 2\mathfrak{F}\mu\nu + 2\mathfrak{G}\nu\lambda + 2\mathfrak{H}\lambda\mu)x + (\mathfrak{A}\lambda + \mathfrak{H}\mu + \mathfrak{G}\nu)(\lambda x + \mu y + \nu z).$$

The coefficients of  $\partial_y H$ ,  $\partial_z H$  have a similar form; and uniting the three terms, observing that  $x\partial_x H + y\partial_y H + z\partial_z H$  is equal to  $3(m-2)H$ , and attending to the values of  $\Phi$ ,  $\Omega$ ,  $\varpi$ , we have, restoring the omitted factor,

$$\partial_2 H = -\frac{3(m-2)}{m-1} H\Phi + \frac{\Omega}{m-1} \varpi.$$

*Proof of the expressions for  $(DH)^2$  and  $D^2\Phi$ :*

22. These are obtained (for the particular case  $m=4$ , which makes but little difference) in Mr Salmon's *Higher Plane Curves*, pp. 88 and 89, and I merely reproduce his investigation; we have

$$(DH)^2 = \{(B\nu - C\mu)\partial_x H + (C\lambda - A\nu)\partial_y H + (A\mu - B\lambda)\partial_z H\}^2,$$

or, what is the same thing,

$$(DH)^2 = \{\lambda(C\partial_y - B\partial_z)H + \mu(A\partial_z - C\partial_x)H + \nu(B\partial_x - A\partial_y)H\}^2;$$

and if we consider first the term which contains  $\lambda^2$ , the coefficient is

$$[(C\partial_y - B\partial_z)H]^2.$$

Now making use of the equations

$$\begin{aligned} (m-1)A &= ax + hy + gz, \\ (m-1)B &= hx + by + fz, \\ (m-1)C &= gx + fy + cz, \end{aligned}$$

and

$$m(m-1)U = ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0,$$

we have

$$\begin{aligned} (m-1)^2 C^2 &= (gx + fy + cz)^2 - c(ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy) \\ &= -\mathfrak{B}x^2 + 2\mathfrak{F}xy - \mathfrak{A}y^2, \\ (m-1)^2 BC &= (hx + by + fz)(gx + fy + cz) - f(ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy) \\ &= \mathfrak{F}x^2 - \mathfrak{G}xy - \mathfrak{H}xz + \mathfrak{A}yz, \\ (m-1)^2 B^2 &= (hx + by + fz)^2 - b(ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy) \\ &= -\mathfrak{C}x^2 + 2\mathfrak{G}xz - \mathfrak{A}z^2; \end{aligned}$$

and hence

$$\begin{aligned} (m-1)^2 [(C\partial_y - B\partial_z)H]^2 &= (-\mathfrak{B}x^2 + 2\mathfrak{F}xy - \mathfrak{A}y^2)(\partial_y H)^2 \\ &\quad + 2(-\mathfrak{F}x^2 + \mathfrak{G}xy + \mathfrak{H}xz - \mathfrak{A}yz)\partial_y H \cdot \partial_z H \\ &\quad + (-\mathfrak{C}x^2 + 2\mathfrak{G}xz - \mathfrak{A}z^2)(\partial_z H)^2 \\ &= -x^2(\mathfrak{B}, \mathfrak{C}, \mathfrak{F} \mid \partial_y H, \partial_z H)^2 \\ &\quad + 2x(y\partial_y H + z\partial_z H)(\mathfrak{H}\partial_y H + \mathfrak{G}\partial_z H) \\ &\quad - \mathfrak{A}(y\partial_y H + z\partial_z H)^2, \end{aligned}$$

261]

CONTACT AT ANY POINT OF A PLANE CURVE.

219

and

$$y\partial_y H + z\partial_z H = 3(m-2)H - x\partial_x H,$$

so that the term is

$$\begin{aligned} &= -x^2(\mathfrak{B}, \mathfrak{C}, \mathfrak{F} \mid \partial_y H, \partial_z H)^2 \\ &\quad + 2x(3(m-2)H - x\partial_x H)(\mathfrak{H}\partial_y H + \mathfrak{G}\partial_z H) \\ &\quad - \mathfrak{A}(3(m-2)H - x\partial_x H)^2; \end{aligned}$$

or, reducing, we have

$$\begin{aligned} (m-1)^2[(C\partial_y - B\partial_z)H]^2 &= -x^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 \\ &\quad + 6(m-2)H x (\mathfrak{A}\partial_x H + \mathfrak{H}\partial_y H + \mathfrak{G}\partial_z H) \\ &\quad - 9(m-2)^2 \mathfrak{A} H^2. \end{aligned}$$

23. The other terms may be obtained in a similar manner; and it is easy to see that, collecting all the terms, the sum will be

$$\begin{aligned} &-(\lambda x + \mu y + \nu z)^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 \\ &+ 6(m-2)H(\lambda x + \mu y + \nu z)[(\mathfrak{A}\lambda + \mathfrak{H}\mu + \mathfrak{G}\nu)\partial_x H + (\mathfrak{H}\lambda + \mathfrak{B}\mu + \mathfrak{F}\nu)\partial_y H + (\mathfrak{G}\lambda + \mathfrak{F}\mu + \mathfrak{C}\nu)\partial_z H] \\ &- 9(m-2)^2 H^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \lambda, \mu, \nu)^2; \end{aligned}$$

or, attending to the signification of the symbols  $\Psi, \Phi, \Omega, \varpi$ , we have

$$(m-1)^2(DH)^2 = -9(m-2)^2 H^2 \Phi + 3(m-2)H \varpi \Omega - \varpi^2 \Psi.$$

24. Next

$$D^2 H = \{(B\nu - C\mu)\partial_x + (C\lambda - A\nu)\partial_y + (A\mu - B\lambda)\partial_z\}^2 H,$$

or, what is the same thing,

$$D^2 H = \{\lambda(C\partial_y - B\partial_z) + \mu(A\partial_z - C\partial_x) + \nu(B\partial_x - A\partial_y)\}^2 H;$$

and if we attend first to the term which contains  $\lambda^2$ , the coefficient is

$$(C\partial_y - B\partial_z)^2 H.$$

Now substituting for  $C^2, CB, B^2$  as before, we have

$$\begin{aligned} (m-1)^2(C\partial_y - B\partial_z)^2 H &= (-\mathfrak{B}x^2 + 2\mathfrak{F}xy - \mathfrak{A}y^2)\partial_y^2 H \\ &\quad + 2(-\mathfrak{F}x^2 + \mathfrak{G}xy + \mathfrak{H}xz - \mathfrak{A}yz)\partial_y\partial_z H \\ &\quad + (-\mathfrak{C}x^2 + 2\mathfrak{G}xz - \mathfrak{A}z^2)\partial_z^2 H \\ &= -x^2(\mathfrak{B}, \mathfrak{C}, \mathfrak{F} \mid \partial_y, \partial_z)^2 H \\ &\quad + 2x(y\partial_y + z\partial_z)(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H \\ &\quad - \mathfrak{A}(y\partial_y + z\partial_z)^2 H, \end{aligned}$$

where it is to be observed that the symbols of differentiation affect  $H$  only. We have

$$\begin{aligned} (y\partial_y + z\partial_z)^2 H &= (x\partial_x + y\partial_y + z\partial_z)^2 H \\ &\quad - 2(x\partial_x + y\partial_y + z\partial_z) x\partial_x H \\ &\quad + (x\partial_x)^2 H, \end{aligned}$$

where  $H$  is a homogeneous function of the degree  $3m - 6$ ;  $x\partial_x H$ , since the  $x$  is not affected by the differentiation, must be treated as of the degree  $3m - 7$ , and  $(x\partial_x)^2 H$ , for the like reason, stands for  $x^2\partial_x^2 H$ ; we have therefore

$$\begin{aligned} (y\partial_y + z\partial_z)^2 H &= (3m - 6)(3m - 7)H \\ &\quad - 2(3m - 7)x\partial_x H \\ &\quad + x^2\partial_x^2 H. \end{aligned}$$

In like manner,

$$\begin{aligned} (y\partial_y + z\partial_z)(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H &= (x\partial_x + y\partial_y + z\partial_z)(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H - x\partial_x(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H \\ &= (3m - 7)(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H - x\partial_x(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H; \end{aligned}$$

and hence

$$\begin{aligned} (m - 1)^2(C\partial_y - B\partial_z)^2 H &= -x^2(\mathfrak{B}, \mathfrak{C}, \mathfrak{F} \mid \partial_y, \partial_z)^2 H \\ &\quad + 2x[(3m - 7)(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H - x\partial_x(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H] \\ &\quad - \mathfrak{A}[(3m - 6)(3m - 7)H - 2(3m - 7)x\partial_x H + x^2\partial_x^2 H] \\ &= -x^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H \\ &\quad + 2(3m - 7)x(\mathfrak{A}\partial_x + \mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H \\ &\quad - (3m - 6)(3m - 7)\mathfrak{A}H. \end{aligned}$$

25. The other terms are formed in a similar manner; and collecting all the terms, we have

$$\begin{aligned} (m - 1)^2 D^2 H &= -(3m - 6)(3m - 7)(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \lambda, \mu, \nu)^2 H \\ &\quad + 2(3m - 7)(\lambda x + \mu y + \nu z)[(\mathfrak{A}\lambda + \mathfrak{H}\mu + \mathfrak{G}\nu)\partial_x H + (\mathfrak{H}\lambda + \mathfrak{B}\mu + \mathfrak{F}\nu)\partial_y H + (\mathfrak{G}\lambda + \mathfrak{F}\mu + \mathfrak{C}\nu)\partial_z H] \\ &\quad - (\lambda x + \mu y + \nu z)^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H, \end{aligned}$$

or, attending to the signification of the symbols  $\Psi$ ,  $\Phi$ ,  $\Omega$ ,  $\varpi$ , this is

$$(m - 1)^2 D^2 H = -(3m - 6)(3m - 7)H\Phi + (3m - 7)\varpi\Omega - \varpi^2\Psi.$$



261]

CONTACT AT ANY POINT OF A PLANE CURVE.

221

*Proof of the expressions for  $(m-1)^2 R_3^2$ ,  $(m-1)^2 R_4$ :*

26. We have

$$\begin{aligned}
 (m-1)^2 R_3^2 &= (m-1)^2 (DH)^2 \\
 &= -9(m-2)^2 H^2 \Phi \\
 &\quad + 3(m-2)H \Omega \varpi \\
 &\quad - \Psi \varpi^2, \\
 (m-1)^2 R_4 &= (m-1)^2 D^2 H + (m-1) \Omega \Phi - (3m-6)(m-1) \Phi H \\
 &= -12(m-2)^2 H \Phi \\
 &\quad + 4(m-2) \Omega \varpi \\
 &\quad - \Omega \varpi^2,
 \end{aligned}$$

which are the expressions required.

*Proof of corresponding expressions for  $(m-1)^2 Q_3^2$ ,  $(m-1)^2 Q_4$ :*

27. We have

$$\begin{aligned}
 (m-1)^2 Q_3^2 &= (m-1)^2 (\mathbf{D}H)^2 \\
 &= -9(m-2)^2 H^2 \Phi \\
 &\quad + 3(m-2)H \Omega \Phi \\
 &\quad - \Psi \Phi, \\
 (m-1)^2 Q_4 &= (m-1)^2 \mathbf{D}^2 H - (m-1) \Omega \Phi + (3m-6)(m-1) H \Phi \\
 &= -6(m-2)(m-3) H \Phi \\
 &\quad + 2(m-3) \Omega \Phi \\
 &\quad - \Omega \Phi;
 \end{aligned}$$

to which I join

$$(m-1)^2 Q_2 = H.$$

28. We have consequently

$$\begin{aligned}
 (m-1)^4 (3Q_2 Q_4 - Q_3^2) &= -9(m-2)(m-4) H^2 \Phi \\
 &\quad + 3(m-4) H \Omega \Phi \\
 &\quad + (-3\Omega H + \Psi) \Phi;
 \end{aligned}$$

and

$$(m-1)^6 (3(m-2)Q_2 Q_4 - 2(m-3)Q_3^2) = (-3(m-2)H \Omega + 2(m-3)\Psi) \Phi^2,$$

which for  $m=4$  becomes

$$729 (3Q_2 Q_4 - Q_3^2) = (-3\Omega H + \Psi) \Phi^2.$$

29. In the case  $m=4$ , we have Hesse's theorem, that the equation  $3Q_2 Q_4 - Q_3^2 = 0$  gives a curve of the 14th order, which passes through the points of contact of the double tangents, viz. substituting for  $\Omega$ ,  $\Psi$  their values, the equation of this curve is

$$\begin{aligned}
 &-3H (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H \\
 &+ (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 = 0.
 \end{aligned}$$

I have added these remarks for the sake of pointing out the striking resemblance of the expressions which occur in the double-tangent problem for  $m = 4$ , and in the present theory of the five-pointic conic for any value whatever of  $m$ . It has not hitherto been shown what the expressions  $3(m-2)Q_2Q_4 - 2(m-3)Q_3^2$  and  $-3(m-2)H\Omega + 2(m-3)\Psi$  respectively denote, except in the particular case  $m = 4$ .

### III. APPLICATION OF THE FORMULÆ TO THE CUBIC.

30. I shall apply the formula for the five-pointic conic to a cubic; to avoid confusion as to a numerical factor, I write  $U'$ ,  $H'$  in the place of  $U$ ,  $H$ , so that we have

$$\Lambda = \frac{1}{9H'} \left\{ -3(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H' \cdot H' \right. \\ \left. + 4(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H', \partial_y H', \partial_z H')^2 \right\},$$

and then the equation of the five-pointic conic is

$$D^2U' - \left( \frac{1}{3} \frac{1}{H'} DH' + \Lambda DU' \right) DU' = 0.$$

I take as the equation of the cubic,

$$U = x^3 + y^3 + z^3 + 6lxyz = 0;$$

the formulae Table No. 70 of my Third Memoir on Quantics, *Phil. Trans.*, vol. CXLVI. (1856), pp. 627–647, [144], putting  $H$  for  $HU$ , give

$$H = l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz.$$

Hence writing

$$U' = \frac{1}{6}(x^3 + y^3 + z^3 + 6lxyz),$$

the first derived functions are

$$\frac{1}{2}(x^2 + 2lyz), \quad \frac{1}{2}(y^2 + 2lzx), \quad \frac{1}{2}(z^2 + 2lxy);$$

the second derived functions, or  $(a, b, c, f, g, h)$ , are  $(x, y, z, lx, ly, lz)$ , whence  $H' = -U$ ; the inverse coefficients  $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$  are

$$(yz - l^2x^2, \quad zx - l^2y^2, \quad xy - l^2z^2, \quad l^2yz - lx^2, \quad l^2zx - ly^2, \quad l^2xy - lz^2);$$

and putting  $U' = \frac{1}{6}U$  and  $H' = -H$ , we have

$$\Lambda = -\frac{1}{9H} \left\{ -3(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H \cdot H \right. \\ \left. + 4(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 \right\},$$

and the equation of the five-pointic conic is

$$D^2U - U \left( \frac{1}{3} DH + \frac{1}{6} \Lambda DU \right) DU = 0.$$

31. We have

$$\begin{aligned}
 (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H &= (yz - l^2 x^2) \cdot 6lx \\
 &\quad + (zx - l^2 y^2) \cdot 6ly \\
 &\quad + (xy - l^2 z^2) \cdot 6lz \\
 &\quad + 2(l^2 yz - lx^2) \cdot -(1 + 2l^3)x \\
 &\quad + 2(l^2 zx - ly^2) \cdot -(1 + 2l^3)y \\
 &\quad + 2(l^2 xy - lz^2) \cdot -(1 + 2l^3)z,
 \end{aligned}$$

which is

$$\begin{aligned}
 &= 18l^2 xyz - 6l^4(x^3 + y^3 + z^3) \\
 &\quad - 6l^2(1 + 2l^3)xyz + 2l(1 + 2l^3)(x^3 + y^3 + z^3), \\
 &= (12l^2 - 12l^5)xyz + (2l - 2l^4)(x^3 + y^3 + z^3), \\
 &= 2l(1 - l^3)(x^3 + y^3 + z^3 + 6lxyz),
 \end{aligned}$$

or we have

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H = -2S \cdot U,$$

where  $S$  is the quartinvariant (see the Table No. 70). For the present purpose  $U = 0$ , and consequently

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H = 0.$$

32. Next,

$$\begin{aligned}
 (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 &= (yz - l^2 x^2)[3l^2 x^2 - (1 + 2l^3)yz]^2 \\
 &\quad + (zx - l^2 y^2)[3l^2 y^2 - (1 + 2l^3)zx]^2 \\
 &\quad + (xy - l^2 z^2)[3l^2 z^2 - (1 + 2l^3)xy]^2 \\
 &\quad + 2(l^2 yz - lx^2)[3l^2 y^2 - (1 + 2l^3)zx][3l^2 z^2 - (1 + 2l^3)xy] \\
 &\quad + 2(l^2 zx - ly^2)[3l^2 z^2 - (1 + 2l^3)xy][3l^2 x^2 - (1 + 2l^3)yz] \\
 &\quad + 2(l^2 xy - lz^2)[3l^2 x^2 - (1 + 2l^3)yz][3l^2 y^2 - (1 + 2l^3)zx],
 \end{aligned}$$

the first three lines of which are

$$\begin{aligned}
 &9l^4 xyz(x^3 + y^3 + z^3) - 18l^2(1 + 2l^3)x^2 y^2 z^2 + (1 + 2l^3)^2(y^3 z^3 + z^3 x^3 + x^3 y^3) \\
 &\quad - 9l^6(x^5 + y^5 + z^5) + 6l^3(1 + 2l^3)xyz(x^3 + y^3 + z^3) - 3l^2(1 + 2l^3)^2 x^2 y^2 z^2;
 \end{aligned}$$

or collecting and reducing,

$$\begin{aligned}
 &(-9l^6) (x^6 + y^6 + z^6) \\
 &\quad + (1 + 4l^3 + 4l^6)(y^3 z^3 + z^3 x^3 + x^3 y^3) \\
 &\quad + (-15l^4 + 12l^7)(x^3 + y^3 + z^3)xyz \\
 &\quad + (-21l^2 - 48l^5 - 12l^8)x^2 y^2 z^2;
 \end{aligned}$$

the second three lines are

$$18l^4(y^3z^3 + z^3x^3 + x^3y^3) - 12l^2(1 + 2l^3)xyz(x^3 + y^3 + z^3) + 6l^2(1 + 2l^3)^2x^2y^2z^2 \\ - 6l^6x^2y^2z^2 + 12l^3(1 + 2l^3)(y^3z^3 + z^3x^3 + x^3y^3) - 2l(1 + 2l^3)xyz(x^3 + y^3 + z^3);$$

or collecting and reducing,

$$(12l^4 + 42l^6)(y^3z^3 + z^3x^3 + x^3y^3) \\ + (-2l - 20l^4 - 32l^7)(x^3 + y^3 + z^3)xyz \\ + (6l^2 - 30l^5 + 24l^8)x^2y^2z^2.$$

Hence in the first part replacing the top line by

$$-9l^6(x^6 + y^6 + z^6) + 18l^6(y^3z^3 + z^3x^3 + x^3y^3),$$

and uniting the two parts, we find

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 = (1 + 8l^3)^2 (y^3z^3 + z^3x^3 + x^3y^3) \\ + (-9l^6) (x^6 + y^6 + z^6) \\ + (-2l - 5l^4 - 20l^7) (x^3 + y^3 + z^3) xyz \\ + (-15l^2 - 78l^5 + 12l^8) x^2y^2z^2;$$

and referring to the Table No. 70, and writing  $\Theta$  for  $\Theta U$ , we have

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 = \Theta,$$

where  $\Theta$  is the first of the three functions which may be chosen to represent the octicovariant of the cubic.

33. We have thus

$$\Lambda = -\frac{4}{9} \frac{\Theta}{H},$$

and thence

$$D^2U - \left( \frac{1}{3} \frac{1}{H} DH - \frac{4}{9} \frac{\Theta}{H} DU \right) DU = 0$$

as the equation of the five-pointic conic: the investigation has been conducted by means of the canonical form of the equation of the cubic, but the form of the result shows that it applies to the equation of the cubic in any form whatever.

34. If, however, we continue to represent the cubic by the canonical equation

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

the result may be further reduced. We have, putting  $U = 0$ , or writing  $x^3 + y^3 + z^3 = -6lxyz$ ,

$$H = -(1 + 8l^3)xyz;$$

261]

CONTACT AT ANY POINT OF A PLANE CURVE.

225

moreover, putting  $U = 0$ , the Table No. 70 gives

$$\Theta = (1 + 8l^3)^2(y^3z^3 + z^3x^3 + x^3y^3) - 3l^2x^2y^2z^2;$$

or substituting for  $H$  the last-mentioned value, and putting for shortness

$$Q = y^3z^3 + z^3x^3 + x^3y^3 - 3l^2x^2y^2z^2,$$

we have

$$\Theta = (1 + 8l^3)^2 Q,$$

and with these values of  $H$  and  $Q$ , the equation of the five-pointic conic is

$$D^2U + \left( \frac{1}{3} \frac{DH}{(1 + 8l^3)xyz} + \frac{4}{9} \frac{(1 + 8l^3)Q}{x^2y^2z^2} DU \right) DU = 0,$$

$$DU = 3\{(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z\},$$

$$D^2U = 6\{(X^2 + 2lYZ)x + (Y^2 + 2lZX)y + (Z^2 + 2lXY)z\},$$

or, as it will be convenient to write it,

$$D^2U = 6(x, y, z, lx, ly, lz \mid X, Y, Z)^2,$$

$$DH = \{3l^2x^2 - (1 + 2l^3)yz\}X + \{3l^2y^2 - (1 + 2l^3)zx\}Y + \{3l^2z^2 - (1 + 2l^3)xy\}Z;$$

whence, finally, the equation of the five-pointic conic of the cubic

$$X^3 + Y^3 + Z^3 + 6lXYZ = 0$$

at the point  $(x, y, z)$  is

$$\begin{aligned} & 6(1 + 8l^3)x^2y^2z^2(x, y, z, lx, ly, lz \mid X, Y, Z)^2 \\ & + \{(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z\} \\ & \times \left\{ x^2y^2z^2[(3l^2x^2 - (1 + 2l^3)yz)X + (3l^2y^2 - (1 + 2l^3)zx)Y + (3l^2z^2 - (1 + 2l^3)xy)Z] \right. \\ & \left. - Q[(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z] \right\} = 0, \end{aligned}$$

a result which I had previously obtained by a special method.

35. But the expression may be exhibited in a different form by a transformation suggested by a geometrical theorem of Mr Salmon's. In fact the tangent at the point  $(x, y, z)$  meets the cubic in the tangential of this point, and the coordinates of the tangential are  $x(y^3 - z^3)$ ,  $y(z^3 - x^3)$ ,  $z(x^3 - y^3)$ . Calling these  $\xi$ ,  $\eta$ ,  $\zeta$ , the equation of the tangent to the cubic at the tangential is

$$(\xi^2 + 2l\eta\zeta)X + (\eta^2 + 2l\zeta\xi)Y + (\zeta^2 + 2l\xi\eta)Z = 0.$$

Now we have identically,

$$\begin{aligned} & x^2y^2z^2[(3l^2x^2 - (1 + 2l^3)yz)X + (3l^2y^2 - (1 + 2l^3)zx)Y + (3l^2z^2 - (1 + 2l^3)xy)Z] \\ & - Q[(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z] \\ & = [(\xi^2 + 2l\eta\zeta)X + (\eta^2 + 2l\zeta\xi)Y + (\zeta^2 + 2l\xi\eta)Z] \\ & + U[x^2(-y^3 - z^3 + 2lyz)X + y^2(-z^3 - x^3 + 2lzx)Y + z^2(-x^3 - y^3 + 2lxy)Z]. \end{aligned}$$

In fact this relation will be true if only

$$3x^2y^2z^2(3l^2x^2 - (1 + 2l^3)yz) - Q(x^2 + 2lyz) - (\xi^2 + 2l\eta\zeta) = Ux^2(-y^3 - z^3 + 2lyz);$$

and substituting for  $\xi$ ,  $\eta$ ,  $\zeta$  and  $Q$  their values, the left-hand side is

$$\begin{aligned} & 3x^2y^2z^2(3l^2x^2 - (1 + 2l^3)yz) \\ & - (x^2 + 2lyz)(y^3z^3 + z^3x^3 + x^3y^3 - 3l^2x^2y^2z^2) \\ & - x^2(y^3 - z^3)^2 \\ & - 2lyz(z^3 - x^3)(x^3 - y^3); \end{aligned}$$

and expanding and reducing, the result is

$$x^2(-(y^3 + z^3)x^3 - x^3(y^3 + z^3)) + 2lyz(-2x^3(y^3 + z^3) + x^6) + 12l^2x^2y^2z^2 \cdot x^2;$$

whence, dividing by  $x^2$ , the equation becomes

$$-(y^3 + z^3)(x^3 + y^3 + z^3) + 2lxyz(x^3 - 2(y^3 + z^3)) + 12l^2x^2y^2z^2 = (-y^3 - z^3 + 2lyz)(x^3 + y^3 + z^3 + 6lxyz),$$

which is identically true.

36. Hence in the identical equation putting  $U = 0$ , we see that the equation of the five-pointic conic may be written

$$\begin{aligned} & 9(1 + 8l^3)x^2y^2z^2(x, y, z, lx, ly, lz \mid X, Y, Z)^2 \\ & + [(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z] \\ & \times [(\xi^2 + 2l\eta\zeta)X + (\eta^2 + 2l\zeta\xi)Y + (\zeta^2 + 2l\xi\eta)Z] = 0, \end{aligned}$$

where  $\xi$ ,  $\eta$ ,  $\zeta$  stand for  $x(y^3 - z^3)$ ,  $y(z^3 - x^3)$ ,  $z(x^3 - y^3)$ , the coordinates of the tangential of the given point, and which puts in evidence the geometrical theorem above referred to, viz.

**THEOREM.**—*The common chord of the five-pointic conic and the polar conic is the tangent to the cubic at the tangential of the given point.*

37. The five-pointic conic meets the cubic in the point of contact, considered as five coincident points, and in a remaining sixth point or point of simple intersection. The process by which I originally obtained the equation of the five-pointic conic, led also to the equation of the line joining the point of contact with the point of simple intersection: the equation of this line is

$$\begin{aligned} & 9x^2y^2z^2 \left\{ \begin{aligned} & Xx((1 + 8l^3)x^4 + (4l + 4l^4)x^2yz + (-2l^2 + 2l^5)y^2z^2) \\ & + Yy((1 + 8l^3)y^4 + (4l + 4l^4)y^2zx + (-2l^2 + 2l^5)z^2x^2) \\ & + Zz((1 + 8l^3)z^4 + (4l + 4l^4)z^2xy + (-2l^2 + 2l^5)x^2y^2) \end{aligned} \right\} \\ & - 9Qx^2y^2z^2 \{ X(3l^2x^2 - (1 + 2l^3)yz) + Y(3l^2y^2 - (1 + 2l^3)zx) + Z(3l^2z^2 - (1 + 2l^3)xy) \} \\ & + Q^2 \{ X(x^2 + 2lyz) + Y(y^2 + 2lzx) + Z(z^2 + 2lxy) \} = 0. \end{aligned}$$

261]

CONTACT AT ANY POINT OF A PLANE CURVE.

227

38. If the conic meet the cubic in six coincident points, that is, if the point of contact be a singular point of the kind already spoken of, or, as we may term it, a sextactic point, then the last-mentioned line must coincide with the tangent at the point. Represent for a moment the equation of the line by

$$AX + BY + CZ = 0,$$

then this line is to coincide with the line

$$(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z = 0,$$

or we must have

$$B(z^2 + 2lxy) - C(y^2 + 2lzx) = 0,$$

$$C(x^2 + 2lyz) - A(z^2 + 2lxy) = 0,$$

$$A(y^2 + 2lzx) - B(x^2 + 2lyz) = 0,$$

which must be equivalent to a single condition. The terms  $A, B, C$ , which contain

$$x^2 + 2lyz, \quad y^2 + 2lzx, \quad z^2 + 2lxy$$

respectively, may, it is clear, be omitted, and omitting also a factor  $3x^2y^2z^2$ , we may write

$$A = Xx^2yz((1 + 8l^3)x^4 + (4l + 4l^4)x^2yz + (-2l^2 + 2l^5)y^2z^2) - 2Q(3l^2x^2 - (1 + 2l^3)yz),$$

$$B = Xxy^2z((1 + 8l^3)y^4 + (4l + 4l^4)y^2zx + (-2l^2 + 2l^5)z^2x^2) - 2Q(3l^2y^2 - (1 + 2l^3)zx),$$

and the like value for  $C$ . The last of the three equations is

$$\begin{aligned} & \{(y^2 + 2lzx)((1 + 8l^3)x^4 + (4l + 4l^4)x^2yz + (-2l^2 + 2l^5)y^2z^2) \\ & - (x^2 + 2lyz)((1 + 8l^3)y^4 + (4l + 4l^4)y^2zx + (-2l^2 + 2l^5)yz^2x^2)\} \\ & - 2Q\{(y^2 + 2lzx)(3l^2x^2 - (1 + 2l^3)yz) - (x^2 + 2lyz)(3l^2y^2 - (1 + 2l^3)zx)\} = 0, \end{aligned}$$

where the function on the left hand is

$$\begin{aligned} & = 2xyz \left\{ \begin{aligned} & (1 + 8l^3)(x^3y^3 - x^3y^3 + 2l(x^4z - y^4z)) \\ & + (4l + 4l^4)2l(x^3yz^2 - x^3yz^2) \\ & + (-2l^2 + 2l^5)(xy^2z^3 - x^2yz^3) \end{aligned} \right\} \\ & - 2Q\{(1 + 2l^3)(y^3z - x^3z) + 6l^2(x^2z - y^2z)\}, \end{aligned}$$

or, what is the same thing, throwing out the factor  $z$ , it is

$$\begin{aligned} & = \{3(1 + 8l^3)x^2y^2 + 6l(1 + 8l^3)xyz(x^2 + y^2) + 30l^2(1 + 8l^3)x^2y^2z^2\}(x^2 - y^2) \\ & - 2Q(1 + 8l^3)(x^3 - y^3); \end{aligned}$$

or throwing out the factor  $(1 + 8l^3)$  and substituting for  $Q$  its value, it is

$$= \{3x^2y^2 + 6lxyz(x^2 + y^2) + 30l^2x^2y^2z^2 - 2(y^3z^3 + z^3x^3 + x^3y^3 - 3l^2x^2y^2z^2)\}(x^2 - y^2).$$

The first factor, reducing by the equation  $x^3 + y^3 + z^3 + 6lxyz = 0$ , is

$$\begin{aligned} &= 3x^2y^2 + z^3(x^3 + y^3 + z^3) - 2(y^3z^3 + z^3x^3 + x^3y^3) \\ &= (z^3 - x^3)(z^3 - y^3). \end{aligned}$$

39. Hence putting for the moment

$$M = (y^3 - z^3)(z^3 - x^3)(x^3 - y^3),$$

it appears that the last of the three equations is  $Mz = 0$ ; the first and second are of course  $Mx = 0$  and  $My = 0$ , and the required condition is  $M = 0$ , that is,

$$(y^3 - z^3)(z^3 - x^3)(x^3 - y^3) = 0,$$

the equation which, combined with the equation of the curve

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

gives the sextactic points. There are consequently twenty-seven such points, and it is at once seen that these are the points of contact of the tangents to the cubic from the points of inflexion, or, what is the same thing, that the twenty-seven sextactic points form nine groups of three each, such that the three points of a group have for their common tangential one of the nine points of inflexion. In fact, let  $\omega$  be a cube root (real or imaginary) of unity, the three sextactic points of one of the groups will be given by

$$\begin{cases} x - \omega y = 0, \\ x^3 + y^3 + z^3 + 6lxyz = 0. \end{cases}$$

Now consider the tangential of any one of these points, its coordinates are

$$\begin{aligned} x_1 &= x(y^3 - z^3), \\ y_1 &= y(z^3 - x^3), \\ z_1 &= z(x^3 - y^3); \end{aligned}$$

or, reducing by the equation  $x - \omega y = 0$ ,  $x_1 = \omega y(x^3 - z^3)$ ,  $y_1 = -y(x^3 - z^3)$ ,  $z_1 = 0$ , or, what is the same thing,  $x_1 + \omega y_1 = 0$ ,  $z_1 = 0$ ; that is, the point  $(x_1, y_1, z_1)$  is one of the points of inflexion. This is the construction of the sextactic points obtained by Plücker and Steiner.

40. Reverting to the equation of the line joining the point of contact of the five-pointic conic with the point of simple intersection, this meets the cubic in a third point, and Mr Salmon has shown that this third point is in fact the second tangential (tangential of the tangential) of the point of contact, or, what is the same thing:

**THEOREM.** *The point of simple intersection of the cubic and the five-pointic conic is the third point of intersection with the cubic, of the line joining the point of contact with the second tangential of this point.*



THE COLLECTED MATHEMATICAL PAPERS  
of  
ARTHUR CAYLEY

Volume IV — pages 229–253

Papers 261 (conclusion), 262

41. I have not sought to verify this theorem by my formulæ. I remark, that combining it with the before-mentioned theorem, the five-pointic conic is completely determined as follows; viz.

THEOREM. The five-pointic conic touches the cubic at the point of contact (two conditions); it passes through the two points in which the polar conic is intersected by the tangent to the cubic at the tangential of the point of contact (two conditions); and it passes through the point which is the third point of intersection with the cubic of the line joining the point of contact with its second tangential.

42. The construction for the point of simple intersection leads at once to that for the sextactic points; in fact, consider a point having for its tangential a point of inflexion: a point of inflexion is its own tangential, and the second tangential of the first-mentioned point is therefore the point of inflexion: the line joining the point with the second tangent is therefore the tangent at the point, and the point of simple intersection coincides with the point itself, that is, the point in question is a sextactic point.

43. I represent the equation of the five-pointic conic by

$$(a, b, c, f, g, h \mid X, Y, Z)^2 = 0;$$

the value of  $a$  is

$$\begin{aligned} a = & 9(1 + 8l^3) x^4 y^2 z^2 \\ & + 3x^2 y^2 z^2 (3l^2 y^2 - (1 + 2l^3) yz)(x^2 + 2lyz) \\ & - Q(x^2 + 2lyz)^2, \end{aligned}$$

in which equation

$$Q = y^3 z^3 + z^3 x^3 + x^3 y^3 - l^3 x^2 y^2 z^2;$$

or, reducing by the equation  $x^3 + y^3 + z^3 + 6lxyz = 0$ ,

$$Q = y^3 z^3 + x^3(-x^3 - 6lxyz) - 3l^2 x^2 y^2 z^2,$$

that is,

$$-Q = x^6 + 6lx^4 yz + 3l^2 x^2 y^2 z^2 - y^3 z^3,$$

and we have

$$\begin{aligned} a = & 9(1 + 8l^3) x^4 y^2 z^2 \\ & + 3x^2 y^2 z^2 (3l^2 y^2 - (1 + 2l^3) yz)(x^2 + 2lyz) \\ & + (x^6 + 6lx^4 yz + 3l^2 x^2 y^2 z^2 - y^3 z^3)(x^2 + 4xl^2 yz + 4l^2 y^2 z^2). \end{aligned}$$

We have in like manner

$$\begin{aligned} 2f = & 18(1 + 8l^3) b^2 y^2 z^2 \\ & + 3x^2 y^2 z^2 [(3l^2 y^2 - (1 + 2l^3) zx)(z^2 + 2lxy) + (3l^2 z^2 - (1 + 2l^3) xy)(y^2 + 2lzx)] \\ & - 2Q(y^2 + 2lzx)(z^2 + 2lxy); \end{aligned}$$

the coefficient of  $3x^2 y^2 z^2$  in the second line is

$$= 6lyz^3 + (-1 + 4l^3)x(y^3 + z^3) - 4l(1 + 2l^3)x^2 yz;$$

or reducing by the equation of the curve,

$$= (1 - 4l^3)x^4 + (2l - 32l^4)x^2 yz + 6l^2 xy^2 z^2;$$

and the coefficient of  $-2Q$  is

$$= y^2 z^2 + 2lx(y^3 + z^3) + 4l^2 x^2 yz;$$

or reducing by the equation of the curve,

$$= -2lx^4 - 8l^2 x^2 yz + y^2 z^2.$$

We thus have

$$\begin{aligned} 2f &= 18(1 + 8l^3) b^2 y^2 z^2 \\ &\quad + 3x^2 y^2 z^2 ((1 - 4l^3) x^4 + (2l - 32l^4) x^2 yz + 6l^2 xy^2 z^2) \\ &\quad + 2(x^6 + 6lx^4 yz + 3l^2 x^2 y^2 z^2 - y^3 z^3)(-2lx^4 - 8l^2 x^2 yz + y^2 z^2). \end{aligned}$$

Reducing the expressions of  $a$  and  $2f$ , we find for the coefficients  $(a, b, c, f, g, h)$ ,

$$\begin{aligned} a &= x^{10} + 10lx^7 yz + 40l^2 x^4 y^2 z^2 + (5 + 120l^3) x^4 y^3 z^3 - 10lx^2 y^4 z^4 - y^5 z^5, \\ 2f &= -4lx^{10} - 40l^2 x^7 yz + (5 - 120l^3) x^4 y^2 z^2 + 40lx^4 y^3 z^3 + 8l^2 x^2 y^4 z^4 - 2y^5 z^5, \end{aligned}$$

which gives the completely developed form of the equation of the five-pointic conic.

44. I investigate the coordinates of the point of simple intersection of the cubic and the five-pointic conic as follows: the equations of the two curves are

$$\begin{aligned} X^3 + Y^3 + Z^3 + 6lXYZ &= 0, \\ (a, b, c, f, g, h \mid X, Y, Z)^2 &= 0; \end{aligned}$$

or if we write

$$\begin{aligned} \alpha &= 1, & A &= c, \\ \beta &= 6lXY, & B &= 2(gX + fY), \\ \gamma &= X^3 + Y^3, & C &= aX^2 + 2hXY + bY^2, \end{aligned}$$

then the two equations are

$$\begin{aligned} \alpha Z^3 + \beta Z + \gamma &= 0, \\ AZ^2 + BZ + C &= 0, \end{aligned}$$

and the result of the elimination of  $Z$  will be

$$(\alpha Z_1^3 + \beta Z_1 + \gamma)(\alpha Z_2^3 + \beta Z_2 + \gamma) = 0,$$

where  $Z_1, Z_2$  are the roots of the equation  $AZ^2 + BZ + C = 0$ ; that is, we have

$$\begin{aligned} &\alpha^2 \frac{C}{A} (B^2 - 2AC) \\ &+ \alpha\beta (-BC + 2ABC) \\ &\quad \alpha\gamma \\ &+ \gamma^2 CA^2 \\ &- \gamma\beta BA^2 \\ &+ \beta^2 A^2 = 0; \end{aligned}$$

261]

CONTACT AT ANY POINT OF A PLANE CURVE.

231

and substituting for  $A, B, C$  their values, but attending only to the terms which involve  $X^7$  and  $Y^7$ , the result is

$$(a^3 + c^3 - 8f^3 + 6acf) X^7 + \dots + (b^3 + c^3 - 8f^3 + 6bcf) Y^7 = 0.$$

45. But the result of the elimination must obviously be

$$(Xy_1 - Yx_1)(Xy_2 - Yx_2) = 0,$$

if  $(x_1, y_1, z_1)$  are the coordinates of the point of simple intersection. Comparing the two results, and forming the analogous third equation, we may write

$$\begin{aligned} x^7 x_1 &= b^3 + c^3 - 8f^3 + 6bcf, \\ y^7 y_1 &= c^3 + a^3 - 8g^3 + 6cag, \\ z^7 z_1 &= a^3 + b^3 - 8h^3 + 6abh, \end{aligned}$$

where the value of  $x^7 x_1$  may also be written

$$(b + c - 2f)(b + c\omega - 2\omega^2 f)(b + c\omega^2 - 2\omega f),$$

$\omega$  being an imaginary cube root of unity, and so for the other two terms. The factors of  $x^7 x_1$  might be calculated from the identical equation

$$\begin{aligned} bY^2 + 2fYZ + cZ^2 &= 9(1 + 8l^3) x^2 y^2 z^2 (y^2 Y^2 + z^2 Z^2 + 2lxYZ) \\ &+ [(y^2 + 2lzx)Y + (z^2 + 2lxy)Z]x \\ &\times \left\{ 3x^2 y^2 z^2 [(3l^2 y^2 - (1 + 2l^3)zx)Y + (3l^2 z^2 - (1 + 2l^3)xy)Z] \right. \\ &\quad \left. - Q[(y^2 + 2lzx)Y + (z^2 + 2lxy)Z] \right\}. \end{aligned}$$

I remark, that putting  $x = 0$ , we have

$$bY^2 + 2fYZ + cZ^2 = -y^2 z^2 (yY + zZ)^2,$$

and hence writing 1 for  $Y$ , and  $-1, -\omega, -\omega^2$  for  $Z$ , we have

$$\begin{aligned} b + c - 2f &= -y^2 z^2 (y - z)^2, \\ b + c\omega^2 - 2\omega f &= -y^2 z^2 (y - \omega z)^2, \\ b + c\omega - 2\omega^2 f &= -y^2 z^2 (y - \omega^2 z)^2; \end{aligned}$$

and hence the product of the three factors is

$$-y^6 z^6 (y - z)^2 (y - \omega z)^2 (y - \omega^2 z)^2,$$

which is equal to  $-y^6 z^6 (y^3 - z^3)^2$ , or  $-y^6 z^6 (y^3 + z^3)^2$ , which vanishes in virtue of the assumed equation  $x = 0$ . This shows that the function  $b^3 + c^3 - 8f^3 + 6bcf$  contains the factor  $x$ . I have not verified *à posteriori*, but I assume it to be true, that it contains in fact the factor  $x^7$ , and consequently that the expressions for  $x_1, y_1, z_1$  are rational and integral functions of  $(x, y, z)$  of the degree 25, and containing respectively the factors  $x, y, z$ .

46. In the theory of the cubic, a point which depends linearly upon a given point may be termed a derivative of such point. According to a very beautiful theorem of Professor Sylvester's, the coordinates of a derivative point are necessarily rational and integral functions of a square degree of the coordinates  $(x, y, z)$  of the given point; and moreover, there is but one derivative point having its coordinates of any given square degree  $m^2$ , or, as we may express it, only one derivative point of the degree  $m^2$ . The successive tangentials are derivative points of the degrees 4, 16, 64, &c.; the third point of intersection with the cubic, of the line joining two

derivative points of the degrees  $m^2$  and  $n^2$  respectively, is a derivative point of the degree  $(m+n)^2$ . Thus the third point of intersection with the cubic, of the line joining the given point with its second tangential, is a derivative point of the degree  $(4+1)^2$ , and it is easy to see that the degree is not 9; it is therefore 25. The point of simple intersection of the five-pointic conic is a derivative point of the degree 25; it is therefore, according to Professor Sylvester's general theory, identical with the point given by the former construction; this agrees with the before-mentioned theorem of Mr Salmon.

#### IV. INDEPENDENT INVESTIGATION FOR THE CUBIC.

47. The following is, in substance, the method by which I first obtained the equation of the five-pointic conic, for the cubic

$$X^3 + Y^3 + Z^3 + 6lXYZ = 0.$$

Write for shortness

$$\begin{aligned} U &= x^3 + y^3 + z^3 + 6lxyz, \\ V &= (x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z, \\ W &= (X^2 + 2lYZ)x + (Y^2 + 2lXZ)y + (Z^2 + 2lXY)z, \\ T &= X^3 + Y^3 + Z^3 + 6lXYZ, \\ P &= ax + by + cz, \\ \Pi &= aX + bY + cZ; \end{aligned}$$

then,  $X, Y, Z$  being current coordinates, and  $x, y, z$  the coordinates of a point of the cubic (so that  $U = 0$ ), the equation of the cubic will be

$$T = 0,$$

and the equation of a conic having with it an ordinary (two-pointic) contact at the point  $(x, y, z)$ , will be<sup>1</sup>

$$2W - \Pi V = 0.$$

48. Now imagine from the point of contact lines drawn to the other four intersections of the two curves; in the case of the five-pointic conic, three of these lines will coincide with the tangent  $V = 0$ , and the remaining line will be the line joining the point of contact with the point of simple intersection. The equations of the lines in question can be found by Joachimsthal's theorem, viz. if  $(x, y, z)$  be the coordinates of a given point, and  $(X, Y, Z)$  current coordinates, then if in the equations of any two curves we substitute for the coordinates,  $\lambda x + \mu X$ ,  $\lambda y + \mu Y$ ,  $\lambda z + \mu Z$ , and between the equations so obtained eliminate  $\lambda, \mu$ , the resulting equation will be that

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<sup>1</sup>I have introduced the factor 2, to make this correspond with the form  $DW - \Pi DU = 0$ , in the case in question,  $m = 3$ .

261]

CONTACT AT ANY POINT OF A PLANE CURVE.

233

of the lines drawn from the point  $(x, y, z)$  to the points of intersection of the two curves. The point  $(x, y, z)$  is any point whatever, and it may therefore be a point of intersection, or, as in the present instance, a point of contact of the two curves; the only difference is, that in either case the degree of each equation as regards  $(\lambda, \mu)$  is reduced by unity, and the degree of the resulting equation in  $X, Y, Z$  is also reduced by unity: in the case of a point of simple intersection this is the only reduction; but in the case of a point of contact, the resulting equation contains the equation of the tangent as a factor, and rejecting this factor, the reduction in degree is two units.

49. Applying the method to the two equations,  $T = 0$ ,  $2W - \Pi V = 0$ , and substituting therein for the original current coordinates  $X, Y, Z$  the values  $\lambda x + \mu X$ ,  $\lambda y + \mu Y$ ,  $\lambda z + \mu Z$ , the equations become

$$\begin{aligned}\lambda^3 U + 3\lambda^2 \mu V + 3\lambda \mu^2 W + \mu^3 T &= 0, \\ 2(\lambda^2 U + 2\lambda \mu V + \mu^2 W) - (\lambda P + \mu \Pi)(\lambda V + \mu V) &= 0;\end{aligned}$$

or writing  $U = 0$ , and omitting from each equation the factor  $\mu$ , the equations become

$$\begin{aligned}3\lambda^2 V + 3\lambda \mu W + \mu^2 T &= 0, \\ \lambda(4 - P)V + \mu(2W - \Pi V) &= 0;\end{aligned}$$

and putting in the first equation  $\lambda = 2W - \Pi V$ ,  $\mu = -(4 - P)V$ , the result of the elimination contains the factor  $V$ , rejecting which it becomes

$$3(2W - \Pi V)^2 - 3(4 - P)(2W - \Pi V)W + (4 - P)^2 VT = 0,$$

which is of the fourth degree in  $(X, Y, Z)$ , as it should be, and represents therefore the lines drawn from the point of contact to the other four points of intersection of the conic and cubic.

50. The equation may be written

$$-3(2W - \Pi V)((2 - P)W + \Pi V) + (4 - P)^2 VT = 0,$$

and we obtain at once the condition that this may contain the factor  $V$ , viz. this condition is

$$P = 2;$$

and if this be satisfied the conic will have a three-pointic contact, and there will be three other points of intersection. And writing  $P = 2$ , and throwing out the factor  $V$ , we find

$$3\Pi^2 V - 6\Pi W + 4T = 0$$

for the equation of the lines from the point of contact to the three points of intersection. And we have now to determine  $\Pi$  so that the function on the left hand may divide by  $V^2$ .

51. I simplify my original method by the use of a theorem of Mr Salmon's, viz. writing

$$\begin{aligned} T &= X^3 + Y^3 + Z^3 + 6lXYZ, & \mathfrak{H} &= l^2(X^3 + Y^3 + Z^3) - (1 + 2l^3)XYZ, \\ W &= (X^2 + 2lYZ)x + \dots, & \mathfrak{H}_1 &= (3l^2X^2 - (1 + 2l^3)YZ)x + \dots, \\ & & H_1 &= (3l^2x^2 - (1 + 2l^3)yz)X + \dots, \\ U &= x^3 + y^3 + z^3 + 6lxyz, & H &= l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz, \end{aligned}$$

we have identically

$$TH - U\mathfrak{H} = WH_1 - V\mathfrak{H}_1;$$

and in the present case, since  $U = 0$ ,

$$TH = WH_1 - V\mathfrak{H}_1.$$

Hence, multiplying by  $H$  and substituting this value of  $TH$ , the equation becomes

$$3H\Pi^2V - 6H\Pi W + 4WH_1 - 4V\mathfrak{H}_1 = 0,$$

or, as we may write it,

$$V(3H\Pi^2 - 4\mathfrak{H}_1) + 2W(2H_1 - 3H\Pi) = 0;$$

and we can at once make the equation divide by  $V$ , viz. by assuming

$$\Pi = \frac{2}{3} \frac{H_1}{H} + AV,$$

where  $A$  is arbitrary: we have thus a four-pointic contact. And substituting for  $\Pi$ , and throwing out the factor  $V$ , the equation becomes

$$3H\left(\frac{2}{3} \frac{H_1}{H} + AV\right)^2 - 4\mathfrak{H}_1 + 2W\left(\frac{2}{3} A\right) = 0;$$

or reducing,

$$16(H_1^2 - 3H\mathfrak{H}_1) - 36H^2AW + 24HH_1AV + 9H^2A^2V^2 = 0,$$

which is the equation of the lines drawn from the point of contact to the remaining two points of intersection.

52. I write for greater convenience

$$\Theta = \frac{16}{9} \mathfrak{H},$$

( $\Theta$  being as yet indeterminate); the equation is thus reduced to

$$9H^2\{H(H_1^2 - 3H\mathfrak{H}_1) + \Theta W\} - 6H^2H_1\Theta V + \Theta^2V^2 = 0,$$

and we have then to determine  $\Theta$  so that the left-hand side may divide by  $V$ ; or, what is the same thing, we must determine  $\Theta$  so that

$$H(H_1^2 - 3H\mathfrak{H}_1) + \Theta W$$

261]

CONTACT AT ANY POINT OF A PLANE CURVE.

235

may divide by  $V$ . This implies the existence of an identical equation,

$$H(H_1^2 - 3H\mathfrak{H}_1) + \Theta W = MU + NV,$$

which for  $U = 0$  would give the decomposition in question; but I have not investigated the values of  $M$  and  $N$ . I assume at the outset  $U = 0$ , and putting as before

$$-Q = x^6 + 6lx^4yz + 3l^2x^2y^2z^2 - y^3z^3,$$

and writing also

$$\begin{aligned} S = & Xx[(1 + 8l^3)x^4 + (4l + 4ll^4)x^2yz + (-2l^2 + 2l^5)y^2z^2] \\ & + Yy[(1 + 8l^3)y^4 + (4l + 4ll^4)y^2zx + (-2l^2 + 2l^5)z^2x^2] \\ & + Zz[(1 + 8l^3)z^4 + (4l + 4ll^4)z^2xy + (-2l^2 + 2l^5)x^2y^2], \end{aligned}$$

I remark that for  $U = 0$  we have

$$-xyzH_1^2 + VS + (1 + 8l^3)(QW - 3x^2y^2z^2\mathfrak{H}_1) = 0,$$

an equation which, observing that

$$H = -(1 + 8l^3)xyz,$$

and assuming also

$$\Theta = (1 + 8l^3)^2Q,$$

may be written

$$H(H_1^2 - 3H\mathfrak{H}_1) + \Theta W = -(1 + 8l^3)VS,$$

which gives the required decomposition, viz.,  $\Theta$  having this value, the conic will have a five-pointic contact. Reducing by the last equation, and throwing out the factor  $V$ , we find

$$-9H^2(1 + 8l^3)S - 6H^2H_1\Theta + \Theta^2V = 0$$

for the equation of the line joining the point of contact with the point of simple intersection. And if in this equation we write  $H = -(1 + 8l^3)xyz$ , and  $\Theta = (1 + 8l^3)^2Q$ , we obtain, finally, for the equation of the line in question,

$$9x^2y^2z^2S - 6x^2y^2z^2QH_1 + Q^2V = 0,$$

which is the before-mentioned result.

53. It only remains to verify the assumed equation

$$-xyzH_1^2 + VS + (1 + 8l^3)(QW - 3x^2y^2z^2\mathfrak{H}_1) = 0.$$

We may write

$$-Q = x^6 + 6lx^4yz + 3l^2x^2y^2z^2 - y^3z^3,$$

and then observing that

$$\begin{aligned} W &= xX^2 + \dots + 2lXYZ + \dots, \\ \mathfrak{H}_1 &= 3l^2xX^2 + \dots - (1 + 2l^3)XYZ - \dots, \end{aligned}$$



we find at once

$$(1 + 8l^3)(QW - 3x^2y^2z^2\mathfrak{H}_1) = -(1 + 8l^3)\left\{(x^7 + 6lx^5yz + \dots)X^2 + (2lx^7 + 12l^2x^5yz - 3x^2y^3z^2 - 2lx^4y^2z^2)YZ + \dots\right\}.$$

Next writing

$$\begin{aligned} H_1 &= (3l^2x^2 - (1 + 2l^3)yz)X + \dots, \\ V &= (x^2 + 2lyz)X + \dots, \\ S &= x\{(1 + 8l^3)x^4 + (4l + 4ll^4)x^2yz + (-2l^2 + 2l^5)y^2z^2\}X + \dots, \end{aligned}$$

and forming the expression for

$$-xyzH_1^2 + VS,$$

the coefficient of  $X^2$  is

$$\begin{aligned} &-xyz[3l^2x^2 - (1 + 2l^3)yz]^2 \\ &+ (x^2 + 2lyz)x\{(1 + 8l^3)x^4 + (4l + 4ll^4)x^2yz + (-2l^2 + 2l^5)y^2z^2\}, \end{aligned}$$

which is

$$= (1 + 8l^3)(x^7 + 6lx^5yz + 12l^2x^3y^2z^2 - xy^3z^3);$$

the coefficient of  $YZ$  is

$$\begin{aligned} &-2xyz(3l^2y^2 - (1 + 2l^3)zx)(3l^2z^2 - (1 + 2l^3)xy) \\ &+ (y^2 + 2lzx)z[(1 + 8l^3)z^4 + (4l + 4ll^4)z^2xy + (-2l^2 + 2l^5)x^2y^2] \\ &+ (z^2 + 2lxy)y\{(1 + 8l^3)y^4 + (4l + 4ll^4)y^2zx + (-2l^2 + 2l^5)x^2y^2\}, \end{aligned}$$

which is

$$= (1 + 8l^3)\{y^2z^2(y^3 + z^3) + 8lxy^3z^3 + 12l^2x^2yz(y^3 + z^3) - 2lx(y^3 + z^3) - 2xy^2z^2\};$$

or substituting for  $y^3 + z^3$  and  $y^3z^3$  the values  $-x^3 - 6lxyz$  and  $(x^3 + 6lxyz)^2 - 2y^3z^3$  respectively, this is

$$= (1 + 8l^3)(2lx^7 + 12l^2x^5yz - 3x^2y^3z^2 - 2lx^4y^2z^2).$$

We have thus

$$-xyzH_1^2 + VS = (1 + 8l^3)\left\{(x^7 + 6lx^5yz + 12l^2x^3y^2z^2 - xy^3z^3)X^2 + (2lx^7 + 12l^2x^5yz - 3x^2y^3z^2 - 2lx^4y^2z^2)YZ + \dots\right\}$$

and the equation

$$-xyzH_1^2 + VS + (1 + 8l^3)(QW - 3x^2y^2z^2\mathfrak{H}_1) = 0$$

is thus verified.

*Addition.*—The foregoing memoir was communicated to Mr Salmon, and I am indebted to him for two notes, containing the extension to a curve of any order, of the preceding investigation for the case of a cubic; I reproduce this extension in the following section.

## V. EXTENSION OF THE LAST PRECEDING METHOD TO A CURVE OF ANY ORDER.

54. Consider the curve of the  $m$ -th order  $T = 0$ , and in the place of the coordinates write  $\lambda x + \mu X$ ,  $\lambda y + \mu Y$ ,  $\lambda z + \mu Z$ , where, as before,  $(x, y, z)$  are the coordinates of the point of the curve, and  $X, Y, Z$  current coordinates; the term involving  $\lambda^m$  vanishes, and dividing out the factor  $\mu$  the equation becomes

$$\lambda^{m-1} DU + \frac{m-1}{1} \lambda^{m-2} \mu D^2 U + \frac{m-1 \cdot m-2}{1 \cdot 2} \lambda^{m-3} \mu^2 D^3 U + \frac{1}{4} \lambda^{m-4} \mu^3 D^4 U + \&c. = 0.$$

Making the like substitution in  $D^2 U - \Pi DU = 0$ , the assumed equation of the five-pointic conic, the factor  $\mu$  divides out and the equation becomes

$$\lambda(2DU - P DU) + \mu(D^2 U - \Pi DU) = 0,$$

or, what is the same thing,

$$\lambda DU(2(m-1) - P) + \mu(D^2 U - \Pi DU) = 0;$$

and if from the two equations we eliminate  $\lambda, \mu$ , the result, throwing out the factor  $DU$ , is

$$(D^2 U - \Pi DU)^{m-1} - \frac{1}{4}(2(m-1) - P) DU (D^2 U - \Pi DU)^{m-2} + \&c. = 0,$$

where all the terms after the second contain the factor  $DU$ ; the condition in order that the equation may divide by  $DU$ , is consequently  $2(m-1) - P = 2$ , or  $P = 2(m-2)$ , the condition of a three-pointic contact. Substituting this value, and dividing by  $DU$ , the equation becomes

$$-\Pi(D^2 U - \Pi DU)^{m-2} + \frac{1}{3} D^3 U (D^2 U - \Pi DU)^{m-3} - \frac{1}{4} D^2 U \cdot DU (D^2 U - \Pi DU)^{m-4} + \&c. = 0,$$

which will be divisible by  $DU$  if  $-\Pi D^2 U + \frac{1}{3} D^3 U$  is divisible by  $DU$ , and the condition for this is found to be, as before,

$$\Pi = \frac{1}{3} \frac{DH}{H} + A DU,$$

where  $A$  is arbitrary; we have thus the conditions of a four-pointic contact.

55. Substituting this value of  $\Pi$ , we see that  $D^3 U - \frac{1}{H} DH \cdot D^2 U$  divides by  $DU$ , viz. there exists an identical equation,

$$D^3 U - \frac{1}{H} DH \cdot D^2 U = IU + J DU;$$

and hence if  $U = 0$ ,

$$\frac{1}{DU} (D^3 U - \frac{1}{H} DH \cdot D^2 U) = J,$$

where  $J$  is a quadric function of  $(X, Y, Z)$ . I do not know the general form of this function, but Mr Salmon has obtained a result which may be generalised as

follows, viz. writing for  $X, Y, Z$  the values  $B\nu - C\mu, C\lambda - A\nu, A\mu - B\lambda$  (where, as before,  $A, B, C$  are the first derived functions of  $U$  and  $\lambda, \mu, \nu$  are arbitrary), the expression for  $J$  is

$$J = \frac{1}{(m-1)^2} \left( D^2U - \frac{1}{H} DH \cdot D^2U \right) = \frac{2(m-2)}{(m-1)} \Omega_3 - \frac{(m-4)H}{(m-1)H} \Omega_2,$$

a formula which will be presently useful.

56. The foregoing equation may be written

$$(D^2U)^{m-2} \left( -\Pi D^2U + \frac{1}{3} D^3U \right) + (D^2U)^{m-3} DU \left\{ (m-2) \Pi^2 D^2U - \frac{1}{3} (m-3) \Pi D^3U - \frac{1}{4} D^4U \right\} + \&c. (DU)^2 \dots = 0;$$

and the term  $-\Pi D^2U + \frac{1}{3} D^3U$  is equal to

$$\frac{1}{3} \left( D^3U - \frac{1}{H} DH \cdot D^2U \right) - A DU \cdot D^2U, \quad \frac{1}{3} J DU - A DU \cdot D^2U.$$

Substituting this value the equation divides by  $DU$ , and throwing out this factor it becomes

$$(D^2U)^{m-2} \left( \frac{1}{3} J - A D^2U \right) + (D^2U)^{m-3} \left\{ (m-2) \Pi^2 DU - \frac{1}{3} (m-3) \Pi D^3U - \frac{1}{4} D^4U \right\} + \&c. DU = 0;$$

or observing that  $\frac{1}{3} \Pi D^3U = \Pi^2 D^2U + \text{term containing } DU$ , this may be written

$$(D^2U)^{m-2} \left( \frac{1}{3} J - A D^2U \right) + (D^2U)^{m-3} \left( \Pi^2 D^2U - \frac{1}{4} D^4U \right) + \&c. DU = 0.$$

57. If the equation divides by  $DU$  we shall have a five-pointic contact; the condition for this is that

$$-A(D^2U)^2 + \Pi^2 D^2U + \Pi^2 D^2U - \frac{1}{4} D^4U$$

may divide by  $DU$ , or more simply that

$$-A(D^2U)^2 + \frac{1}{3} J D^2U + \frac{1}{9} \left( \frac{DH}{H} \right)^2 D^2U - \frac{1}{4} D^4U$$

may divide by  $DU$ , or, what is the same thing, the function in question must vanish in virtue of the substitution of the values  $B\nu - C\mu, C\lambda - A\nu, A\mu - B\lambda$  in the place of  $X, Y, Z$ . The expression for  $J$  has just been given; we have besides

$$\left( \frac{DH}{H} \right)^2 = \frac{m^2}{(m-1)^2} \Omega_1, \quad D^2U = \mu^2 \Omega_2, \quad D^4U = -\mu^4 \Omega_4,$$

where the values of  $\Omega_1, \Omega_2$  are given (ante, No. 27); we have thus

261]

CONTACT AT ANY POINT OF A PLANE CURVE.

239

and

$$D^2U = \frac{1}{(m-1)^2} \mu^2 \Omega_2,$$

whence

$$\begin{aligned} \Lambda &= \frac{1}{\mu^2} \left[ \frac{3(m-3)}{m-1} \Omega_4 - \frac{1}{(m-1)^2} \Omega_2^2 \right. \\ &\quad - \frac{1}{(m-1)^2} \Omega_2 \mu^2 \left\{ \frac{4(m-2)}{m-1} \Phi + \frac{(m-2)}{(m-1)H} \Omega_2 - \frac{1}{(m-1)^2} \frac{1}{H} \Phi \mu^2 \right\} \\ &\quad \left. + \mu^4 \left\{ -\frac{2(m-2)(m-3)}{(m-1)^2} \Phi + \Omega_4 \frac{(m-5)}{(m-1)H} \Omega_2 - \frac{1}{(m-1)^2} \frac{1}{H} \Omega_2^2 \right\} \right] \\ &= \frac{1}{9(m-1)^2 H^2} (4W - 3\Omega H), \end{aligned}$$

and consequently

$$\Lambda = \frac{1}{9H^2} (4W - 3\Omega H),$$

which agrees with the result before obtained, and thus the present method gives the complete solution of the problem.

## 262.

ON THE EQUATION OF DIFFERENCES FOR AN EQUATION OF ANY ORDER, AND IN PARTICULAR FOR THE EQUATIONS OF THE ORDERS TWO, THREE, FOUR, AND FIVE.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CL. (for the year 1860), pp. 93–112. Received March 2,—Read March 29, 1860.]

The term, *equation of differences*, denotes the equation for the squared differences of the roots of a given equation; the equation of differences afforded a means of determining the number of real roots, and also limits for the real roots, of a given numerical equation, and was upon this account long ago sought for by geometers. In the *Philosophical Transactions* for 1763, Waring gives, but without demonstration or indication of the mode of obtaining it, the equation of differences for an equation of the fifth order wanting the second term: the result was probably obtained by the method of symmetric functions. This method is employed in the *Meditationes Algebraicæ* (1782), where the equation of differences is given for the equations of the third and fourth orders wanting the second terms; and in p. 85 the before-mentioned result for the equation of the fifth order wanting the second term, is reproduced. The formulæ for obtaining by this method the equation of differences, are fully developed by Lagrange in the *Traité des Equations Numériques* (1808); and he finds by means of them the equation of differences for the equations of the orders two and three, and for the equation of the fourth order wanting the second term; and in Note III. he gives, after Waring, the result for the equation of the fifth order wanting the second term. It occurred to me that the equation of differences could be most easily calculated by the following method. The coefficients of the equation of differences, *qua* functions of the differences of the roots of the given equation, are leading

coefficients of covariants, or (to use a shorter expression) they are “Seminvariants<sup>2</sup>,” that is, each of them is a function of the coefficients which is reduced to zero by one of the two operators which reduce an invariant to zero. In virtue of this property they can be calculated, when their values are known for the particular case in which one of the coefficients of the given equation is zero. To fix the ideas, let the given equation be  $(* | v, 1)^n = 0$ ; then, when the last coefficient or constant term vanishes, the equation breaks up into  $v = 0$  and into an equation of the degree  $n - 1$ , which I call the reduced equation; the equation of differences will break up into two equations, one of which is the equation of differences for the reduced equation, the other is the equation for the squares of the roots of the same reduced equation. This hardly requires a proof; let the roots of the given equation be  $\alpha, \beta, \gamma, \delta$ , &c., those of the equation of differences are  $(\alpha - \beta)^2, (\alpha - \gamma)^2, (\alpha - \delta)^2$ , &c.,  $(\beta - \gamma)^2, (\beta - \delta)^2, (\gamma - \delta)^2$ , &c.; but in putting the constant term equal to zero, we in effect put one of the roots, say  $\alpha$ , equal to zero; the roots of the equation of differences thus become  $\beta^2, \gamma^2, \delta^2$ , &c.,  $(\beta - \gamma)^2, (\beta - \delta)^2, (\gamma - \delta)^2$ , &c. The equation for the squares of the roots can be found without the slightest difficulty; hence if the equation of differences for the reduced equation of the order  $n - 1$  is known, we can, by combining it with the equation for the squares of the roots, form the equation of differences for the given equation with the constant term put equal to zero, and thence by the above-mentioned property of the Seminvariance of the coefficients, find the equation of differences for the given equation. The present memoir shows the application of the process to equations of the orders two, three, four, and five: part of the calculation for the equation of the fifth order was kindly performed for me by the Rev. R. Harley. It is to be noticed that the best course is to apply the method in the first instance to the forms  $(a, b, \dots | v, 1)^n = 0$ , without numerical coefficients (or, as they may be termed, the *denumerate forms*), and to pass from the results so obtained to those which belong to the forms  $(a, b, \dots | v, 1)^n = 0$ , or *standard forms*. The equation of differences, for  $(\alpha - \beta)^2$ , &c., the coefficients of which are seminvariants, naturally leads to the consideration of a more general equation having for its roots  $(\alpha - \beta)^2(x - \gamma y)(x - \delta y) \dots$ , &c., the coefficients of which are covariants; and in fact, when, as for equations of the orders two, three, and four, all the covariants are known, such covariant equation can be at once formed from the equation of differences; for equations of the fifth order, however, where the covariants are not calculated beyond a certain degree, [they are now all calculated, see 141 and 143], only a few of the coefficients of the covariant equation can be thus at once formed. At the conclusion of the memoir, I show how the equation of differences for an equation of the order  $n$  can be obtained by the elimination of a single quantity from two equations each of the order  $n - 1$ ; and by applying to these two equations the simplification which I have made in Bézout’s abridged method of elimination, I exhibit the equation of differences for the given equation of the order  $n$ , in a compendious form by means of a determinant; the first-mentioned method is, however, that which is best adapted for the actual development of the equation of differences for the equation of a given order.

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<sup>2</sup>The term “Seminvariant” seems to me preferable to M. Brioschi’s term “Peninvariant.”

The equations successively considered are

$$\begin{aligned}(a, b, c \mid v, 1)^2 &= 0, \\ (a, b, c, d \mid v, 1)^3 &= 0, \\ (a, b, c, d, e \mid v, 1)^4 &= 0, \\ (a, b, c, d, e, f \mid v, 1)^5 &= 0.\end{aligned}$$

The equation of differences for the quadric, and that for the squares of the roots, are considered to be known, and the other results are derived from them: it will be convenient to write down in the first instance the results for the quadric, the cubic, and the quartic equations, and then explain the process of obtaining them.

For the quadric equation,

$$\text{Equation of differences is, } 0 = a^2(ac - b^2) + 4(\theta, 1)^1.$$

That is, the equation of differences for  $(a, b, c \mid v, 1)^2 = 0$  has the form

$$a^2(ac - b^2) + 4\theta = 0,$$

and the equation for the squares of the roots is

$$(a^2, ac + 2b^2, c^2, \ldots \mid \theta, 1)^2 = 0.$$

**Equation for the squares of the roots is,  $0 =$**

$$\begin{aligned}&a^2 \quad (b^2 + 1) \\ &+ 1 \quad (ac + 2b^2) \quad (c^2 + 1) \quad (\theta, 1)^2.\end{aligned}$$

For the cubic equation,

$$\text{Equation of differences is, } 0 = a^4(\theta, 1)^3,$$

having coefficients (denumerate form)

$$\begin{aligned}&+ 1, \\ &a^2c + 6, \quad a^2c + 27, \quad ab^2c + 96, \\ &b^2 - 2, \quad b^2 + 1, \quad a^2c^2 + 96, \\ &abcd + 18, \\ &ac^2 + 4, \\ &b^2d + 4, \\ &b^2c^2 - 1.\end{aligned}$$

The equation for the squares of the roots is

$$\begin{aligned}&a^2 \times \\ &+ 1, \quad ac + 2, \quad c^2 + 1, \quad d^2 + 1, \quad bd - 2, \quad (\theta, 1)^3. \\ &b^2 - 1.\end{aligned}$$

**262]** ON THE EQUATION OF DIFFERENCES FOR AN EQUATION OF ANY ORDER. **243**

For the quartic equation,

$$\text{Equation of differences is, } 0 = a^4 \times \dots (\theta, 1)^6,$$

with denumerate-form coefficients listed in successive columns (each column is the coefficient of  $\theta^k$  for  $k = 0, 1, \dots, 6$ ):

Column  $a^4 \times$ :

$$+1, \quad ac + 8, \quad b^2 - 3.$$

Column  $a^4 \theta \times$ :

$$a^2e + 8, \quad a^2ce + 16, \quad a^2bd - 2, \quad a^2c^2 + 22, \quad ac^2 - 16, \quad b^4 + 3.$$

Column  $a^4 \theta^2 \times$ :

$$\begin{aligned} &a^2ce + 16, \quad a^2c^2 + 26, \\ &a^2b^2e - 6, \quad a^2bcd - 30, \quad a^2c^2 - 16, \\ &a^2b^3d + 288, \quad a^2b^2c^2 - 24, \quad ab^3c + 8, \quad b^6 - 1. \end{aligned}$$

Column  $a^4 \theta^3 \times$ :

$$\begin{aligned} &a^2bc^2e + 48, \quad a^2b^2de + 24, \quad a^2cd^2 + 32, \quad a^2b^2c^2 + 17, \\ &a^2b^3de + 54, \quad a^2bcd^2 + 32, \quad ab^3cd^2 + 25, \quad ab^2e + 38, \quad ab^3d^2 + 6, \quad ab^2d^2 + 26. \end{aligned}$$

Column  $a^4 \theta^4 \times$ :

$$\begin{aligned} &a^2e^2 + 256, \quad a^2b^2cd - 192, \\ &a^2b^2c^2 + 216, \quad a^2bc^2de - 144, \quad a^2b^2d^2 - 128, \\ &a^2bcde - 54, \quad a^2bd^3 - 120, \quad a^2d^4 - 27, \\ &ab^3cd - 144, \quad ab^3d^2 - 6, \quad ab^2cd - 80, \quad abcd^3 + 18, \\ &ac^4d^2 + 16, \quad ac^3d^2 - 4, \quad b^4d^2 + 49, \quad b^4c^2 - 27, \\ &b^3cd^2 + 18, \quad b^3d^4 - 4, \quad b^2c^4 - 1, \quad b^2cd^2 + 1, \quad b^2c^3e - 4. \end{aligned}$$

(The full table is laid out in six columns on the printed page; the above lists the entries column by column, in the order of the printed table.)

**Equation for squares of the roots is,  $0 =$**

$$\begin{aligned} &a^2 \times \\ &+ 1, \quad ac + 2, \quad ae + 2, \quad ce + 2, \quad e^2 + 1, \\ &b^2 - 1, \quad c^2 + 1, \quad bd - 2, \quad d^2 - 1, \quad (\theta, 1)^4. \end{aligned}$$

The multiplication of the equation of differences and the equation of the squares of the roots of the quadric equation, gives the equation,  $0 = (\theta, 1)^3$ ,

$$\begin{aligned} &a^4 \times \\ &+ 1, \quad ac + 6, \quad a^2c^2 + 9, \quad ac^3 + 4, \\ &b^2 - 4, \quad ab^2c + 16, \quad b^2c^2 - 1, \quad b^4 + 4, \end{aligned}$$

where all the coefficients except the last are reduced to zero by the operator  $(\theta, 1)^3$ ,

$$3a\partial_b + 2b\partial_c + c\partial_d,$$



and they are consequently (without any alteration) coefficients of the equation of differences of the cubic equation: the last coefficient is not reduced to zero by the operator, and requires therefore to be completed by the adjunction of the terms in  $d$  (the series, here and in every other case, is of course a finite one, the number of terms might easily be calculated *à priori*). Let the value be  $L_0 + L_1d + L_2d^2 + \&c.$ , we have  $L_0 = +4ac^3 - b^2c^2$ ; and putting for shortness  $\nabla = 3a\partial_b + 2b\partial_c$ , the operator which reduces this to zero is  $\nabla + c\partial_d$ ; we ought therefore to have

$$0 = \nabla L_0 + d(\nabla L_1 + \dots) + d^2(\nabla L_2 + \dots) \\ + cL_1 + 2cL_2 + 3cL_3,$$

and consequently

$$L_1 = -\frac{1}{c}\nabla L_0, \quad L_2 = -\frac{1}{2c}\nabla L_1, \quad L_3 = -\frac{1}{3c}\nabla L_2, \quad \&c.,$$

giving

$$L_1 = -\frac{1}{c}\{(-6 + 24c^2)18abc^2 - 4b^3c\} = -18abc + b^3, \\ L_2 = -\frac{1}{2c}\{-54a^2c + (36 - 36c^2)0ab^2\} = +27a^2, \\ L_3 = 0, \quad \&c.,$$

and consequently for the last coefficient the value above written down; it will be presently seen how in more complicated cases the calculations should be arranged.

Again, multiplying together the equation of differences and the equation for the squares of the roots of the cubic equation, we obtain an equation which it is not necessary to write down, as it can be at once formed by putting  $e = 0$  in the equation of differences for the quartic equation. And from the equation so obtained, by the adjunction of the terms in  $e$ , we find the equation of differences for the quartic equation, viz. each coefficient is of the form  $L_0 + L_1e + L_2e^2 + \&c.$ , where  $L_0$  is known, and such coefficient is reduced to zero by the operator

$$4a\partial_b + 3b\partial_c + 2c\partial_d + d\partial_e,$$

or putting for shortness  $\nabla = 4a\partial_b + 3b\partial_c + 2c\partial_d$ , the operator  $\nabla + d\partial_e$ . We have therefore

$$L_1 = -\frac{1}{d}\nabla L_0, \quad L_2 = -\frac{1}{2d}\nabla L_1, \quad L_3 = -\frac{1}{3d}\nabla L_2, \quad \&c.$$

It is to be observed that the last coefficient of the equation of differences is the discriminant, and that the above method of calculating the coefficients of the equation of differences, as applied to the last coefficient, is nothing else than the method of calculating the discriminant given in my Fourth Memoir on Quantics, [155].

**262]**      ON THE EQUATION OF DIFFERENCES FOR AN EQUATION OF ANY ORDER.      **245**

The multiplication of the equation of differences, and the equation for the squares of the roots of the quartic equation, gives, in like manner, the equation of differences for the quintic equation, except as to the terms involving  $f$ ; and these are obtained as above, viz. each coefficient is of the form  $L_0 + L_1f + L_2f^2 + \&c.$ , where  $L_0$  is known; such coefficient is reduced to zero by the operator

$$5a\partial_b + 4b\partial_c + 3c\partial_d + 2d\partial_e + e\partial_f,$$

or putting for shortness  $\nabla' = 5a\partial_b + 4b\partial_c + 3c\partial_d + 2d\partial_e$ , the operator  $\nabla' + e\partial_f$ . We therefore have

$$L_1 = -\frac{1}{e}\nabla' L_0, \quad L_2 = -\frac{1}{2e}\nabla' L_1, \quad L_3 = -\frac{1}{3e}\nabla' L_2, \quad \&c.,$$

which give  $L_1, L_2, \&c.$  by means of  $L_0$ . For the calculation of  $\nabla L$  (where  $L$  is any one of the coefficients  $L_0, L_1, \&c.$ ), it is proper to separate the terms involving the different powers of  $e$ , and write  $L = M_0 + M_1e + M_2e^2 + \&c.$ ,  $\nabla = \nabla'' + 2d\partial_e$ , where  $\nabla'' = 5a\partial_b + 4b\partial_c + 3c\partial_d$ . We have then

$$\nabla L = \nabla'' M_0 + e(\nabla'' M_1 + 2dM_2) + \&c. + 2dM_1,$$

or, what is the same thing,

$$\frac{1}{1} \nabla L_n + \frac{4dM_1}{\dots} + \dots,$$

and as an equation which should be satisfied identically, and which would therefore serve as a verification,

$$\nabla'' M_n + 2dM_1 = 0;$$

but, since a verification was obtained by other means, the equations of this kind were not used for the purpose. It may be interesting to give the actual calculation of one of the coefficients, say of coefficient  $\theta^4$  (which, with coefficient  $\theta^5$ , was calculated by Mr Harley).

*Calculation of coefficient  $\theta^4$  in equation of differences for the quintic equation*

$$(a, b, c, d, e, f \mid v, 1)^5 = 0.$$

*Calculation of  $L_0$ .*

[The original page presents a large computational table giving the coefficients of  $L_0$  as a sum

$$L_0 = A + Be + Ce^2 + De^3 + Ee^4,$$

with each of  $-A$ ,  $-B$ ,  $-C$ ,  $-D$ ,  $-E$  expanded as a sum of monomial terms in  $a$ ,  $b$ ,  $c$ ,  $d$ ,  $e$ ,  $f$  with integer numerical coefficients. The principal entries, transcribed column by column from the original tabular layout, are:]

$-E =$

$$\begin{aligned} & a^3 cd^2 \cdot (-960), \quad a^3 bd^2 \cdot (+840), \quad a^4 \cdot (-400), \\ & a^2 d^3 \cdot (+270), \\ & a^2 bcd \cdot (+240), \quad a^2 b^2 d^2 \cdot (-275), \\ & a^2 bc^2 d \cdot (+390), \quad a^2 c^3 \cdot (+360), \quad ab^3 c \cdot (-400), \\ & a^2 b^2 d^2 \cdot (-140), \quad ab^4 \cdot (+48), \\ & ab^2 cd^2 \cdot (+14), \quad a^2 c^3 \cdot (+15), \\ & ab^3 cd \cdot (+178), \\ & ab^2 c^2 d^2 \cdot (-246), \\ & abc^4 d \cdot (+164), \quad ab^2 c^3 \cdot (-112), \\ & ac^5 \cdot (-24), \\ & b^3 d^3 \cdot (-48), \\ & b^2 cd^3 \cdot (+54), \quad b^4 c^2 \cdot (+25), \\ & b^2 c^2 d \cdot (-38), \\ & b^2 c^4 \cdot (+6). \end{aligned}$$

(The full layout of the original table involves further columns headed  $-D$ ,  $-C$ ,  $-B$ ,  $-A$ , each enumerating numerous monomials in  $a$ ,  $b$ ,  $c$ ,  $d$ ,  $e$ ,  $f$ . Owing to the very large number of entries (several hundreds of monomials in five columns) and the dense matrix-style typesetting in the original, the table is reproduced here only in summary form; the reader is referred to the original page for the complete listing.)

*Calculation of  $L_1$ .*

Terms of  $L_1f$  not involving  $e$ :

$$-f(\nabla''B + 4dC), \quad \text{viz.} \quad (\text{table of coefficients}).$$

[The original page gives a large rectangular table tabulating, for each monomial in  $a, b, c, d$  multiplied by  $f$ , the contributions  $5a\partial_b B, 4b\partial_c B, 3c\partial_d B, 4dC$ , and the resulting coefficient of the monomial in  $L_1$ . Principal monomials listed include  $a^2cd^2f, a^2b^2d^2f, a^2bc^2d^2f, ab^3cd^2f, abc^4df, ac^5f, b^4d^2f, b^2cd^3f, b^2cdf, b^2c^2df$  with respective totals reported in the rightmost column ( $-420, +600, +5070, +24, +40, +130, +32, +24, +6$ , etc.).]

Terms of  $L_1f$  involving  $e$ :

$$-ef(\nabla''C + 6dD), \quad \text{viz.} \quad (\text{table of coefficients}).$$

[Likewise tabulated; principal monomials include  $a^2cd^2ef, a^2b^2def, a^2bcdef, a^2bc^2ef, ab^3def, ab^2c^2ef, b^4cef$  with totals  $-1550, +1740, -200, +290, +310, +288, +56$ , etc.]

Terms of  $L_1f$  involving  $e^2$ :

$$-e^2f(\nabla''D + 8dE), \quad \text{viz.} \quad (\text{table of coefficients}).$$

[Principal monomials include  $a^2d^3f, a^2bcd f$  with totals  $+1000, +1400$ , etc.]

*Calculation of  $L_2$ .*

$L_2 = A + Be + Ce^2$ , suppose, where

$$\begin{aligned}
 -A : & \quad a^2bd^2 \cdot (+775), \quad a^4d \cdot (-500), \quad a^3bc \cdot (-700), \quad a^3b^2 \cdot (+320), \\
 & \quad a^2b^2cd \cdot (-870), \\
 & \quad a^2bc^2 \cdot (-145), \\
 & \quad ab^4d \cdot (+100), \\
 & \quad ab^3c \cdot (+155), \\
 & \quad b^5 \cdot (-28), \\
 -B : & \quad a^2bcd f^2 \cdot (\dots).
 \end{aligned}$$

Terms of  $L_2 f^2$  not involving  $e$  are given by

$$-f^2(\nabla'' B + 4d^2 C),$$

tabulated as above. Principal contributions yield (for example)

$$a^3bcd f^2 \cdot (+1875), \quad a^2cd^2 f^2 \cdot (+200), \quad a^2b^2df^2 \cdot (-2025), \quad a^2c^3 f^2 \cdot (+1750), \quad ab^4df^2 \cdot (-112), \quad ab^3cdf^2 \cdot (+840).$$

Terms of  $L_2 f^2$  involving  $e$  are given by

$$-ef^2(\nabla'' C + 6dD),$$

with principal contributions  $a^2cef^2 \cdot (-5000)$ ,  $a^2def^2 \cdot (+2000)$ , etc.

**262]** ON THE EQUATION OF DIFFERENCES FOR AN EQUATION OF ANY ORDER. **249**

*Calculation of  $L_3$  (gives  $L_3 = 0$ ).*

$L_3 = A + Be$ , where

$$-A : \quad a^3c \cdot (+1666\frac{2}{3}), \quad a^2b^2 \cdot (-666\frac{2}{3});$$

and

$-B :$  is determined from  $-\frac{1}{3}\nabla''B$  via the table; the resulting coefficient is 0.

Terms of  $L_3f^3$  are given by

$$-\frac{1}{3}f^3\nabla''B \quad \text{viz.} \quad a^3bf^3 \cdot (-6666\frac{2}{3}) + (+6666\frac{2}{3}) = 0.$$

and the required coefficient of  $\theta^4$  is

$$L_0 + L_1f + L_2f^2.$$

All the coefficients were calculated in this manner, except the last coefficient, which was deduced from the known value of the discriminant for the standard form. And we have thus the complete expression of the equation of differences for the general quintic equation under the denumerate form  $(a, b, c, d, e, f \mid v, 1)^5 = 0$ , viz.

250–251

[Two-page tabular display]

[262

*Equation of differences for  $(a, b, c, d, e, f \mid v, 1)^5 = 0$  is*

[The original printed pages 250–251 present, across a double-page spread, the complete equation of differences for the general quintic in denumerate form. The result is a polynomial in  $\theta$  of degree 6, in which each coefficient is itself a large polynomial in  $a, b, c, d, e, f$  (degree  $2k$  in the entries for the coefficient of  $\theta^{6-k}$ , by virtue of the seminvariancy property). The seven columns are headed by the successive powers  $\theta^0, \theta^1, \theta^2, \theta^3, \theta^4, \theta^5, \theta^6$  (with the appropriate multiplier  $a^{2k}$  prefixed). Each column is a long list of monomials with their integer coefficients; principal entries include (in successive columns):]

**Coefficient of  $\theta^0$ :**  $a^4(+1, ac + 10, a^2d - 4, b^2 - 4, \text{ldots})$ , with further entries  $a^2e + 30, ac^2 + 50, a^2dc + 25, a^2be - 20, ab^2c + 30, a^3bd + 81, b^6 + \dots$ , etc.

**Coefficient of  $\theta^1$ :**  $a^4\theta(a^2f + \dots, abef + 100, a^2bde - 110, \dots)$ .

**Coefficient of  $\theta^2$ :**  $a^4\theta^2(a^3bef - 250, a^2d^2e - 360, a^3e^2 + 950, a^4bdf + 625, \dots)$  together with very many further monomials.

**Coefficient of  $\theta^3$ :**  $a^4\theta^3(\dots)$ .

**Coefficient of  $\theta^4$ :**  $a^4\theta^4(\dots)$ , the result of the calculation given in the preceding section.

**Coefficient of  $\theta^5$ :**  $a^4\theta^5(\dots)$ .

**Coefficient of  $\theta^6$ :**  $a^4\theta^6(\dots)$ , equal to the discriminant of the quintic up to the standard normalisation.

(Owing to the very large extent of this table—more than four hundred monomials in six variables—the entries are not reproduced individually in the present transcription. The reader is referred to the original double-page spread of the printed memoir for the full list of monomials and coefficients.)

**252** ON THE EQUATION OF DIFFERENCES FOR AN EQUATION OF ANY ORDER. **[262]**

It may be remarked, that if  $\omega$  is an imaginary cube root of unity, then the roots of the equation  $(1, 1, 1, 1, 1, 1 \mid v, 1)^5 = 0$  are  $-1, \omega, \omega^2, -\omega, -\omega^2$ ; the differences of the roots are

$$\begin{aligned} & -1 - \omega, -1 - \omega^2, -1 + \omega, -1 + \omega^2, \omega - \omega^2, 2\omega, \omega + \omega^2, \omega^2 + \omega, 2\omega^2, -\omega + \omega^2, \\ & = \omega^2, \omega, -1 + \omega, -1 + \omega^2, \omega - \omega^2, 2\omega, -1, -1, 2\omega^2, -\omega + \omega^2, \end{aligned}$$

and the squares of the differences are

$$\omega, \omega^2, -3\omega, -3\omega^2, -3, 4\omega^2, 1, 1, 4\omega, -3,$$

from which the equation of differences is found to be

$$(\theta^2 + \theta + 1)(\theta^2 - 3\theta + 9)(\theta^2 + 3\theta + 9)(\theta^2 - 2\theta + 1)(\theta^2 + 6\theta + 9) = 0;$$

or multiplying out, it is

$$(1, 6, 21, 46, 108, 546, 493, -1410, -567, -540, +12960 \mid \theta, 1)^{10} = 0;$$

which is what the preceding expression of the equation of differences becomes upon writing therein  $a = b = c = d = e = f = 1$ . Moreover, upon passing (as will presently be done) to the standard form, and then writing  $a = b = c = d = e = f = 1$ , all the coefficients (except the first coefficient, which is equal to unity) should become equal to zero; these two tests afford a complete verification of the result.

The following corrections have to be made in Waring's result, as given by himself and Lagrange (Waring, *Phil. Trans.* 1762).

Waring, *Meditationes Algebraicæ*, p. 85:

$$\text{for } +169q^2 \text{ read } +196q^2 \quad (\text{in coefficient } w^2).$$

Lagrange, *Equations Numériques*, p. 108:

$$\text{for } +1200GE \text{ read } +200GE \quad (\text{in } d),$$

$$\text{for } -169B^2D \text{ read } -196B^2D \quad (\text{in } e),$$

$$\text{for } -25B^4 \text{ read } +25B^4 \quad (\text{in } f),$$

$$\text{for } +27G^2B^2 \text{ read } -27G^2D^2 \quad (\text{in } k).$$

It may be noticed, that if in the coefficients of the several powers of  $\theta$  (as they are written down in the columns, without regarding the power of  $a$  which multiplies the entire column), we attend only to the terms independent of  $a$ , we have the series

$$\begin{aligned} & 1, b^2 - 4, b^4 + 6, b^6 - 4, b^8 + 1, b^5d + 8, b^6f + 8, \&c., \\ & \quad b^4ce + 22, \\ & \quad b^5c^2 - 3, \quad b^4d^2 - 2, \\ & \quad b^4c^2d - 16, \\ & \quad b^4c^4 + 3, \end{aligned}$$

the law of the first terms of which, up to the term  $+b^8$ , is obvious; but the term  $+b^8$ , which is the last term of this initial series, is also the first term of a terminal series, the terms of which are deduced from the coefficients in the equation of differences for the quartic equation  $(a, b, c, d, e \mid v, 1)^4 = 0$ , viz. these coefficients are



$$\begin{aligned}
& a^4 \times \\
& + 1, \quad ac + 8, \quad a^2e + 8, \\
& b^2 - 3, \quad a^2bd - 2, \\
& a^2c^2 + 22, \\
& abc - 16, \\
& b^4 + 3;
\end{aligned}$$

and by writing  $b, c, d, e, f$  in the place of  $a, b, c, d, e$  respectively, and multiplying by  $b^4$ , we have the above-mentioned series,

$$b^4 + 1, \quad b^5d + 8, \quad \&c., \quad b^5c^2 - 3.$$

It is easy to see, *à priori*, in the case of an equation of any order, that this property holds good. Passing now to the standard forms:

For the quadric  $(a, b, c \mid v, 1)^2 = 0$ , the equation of differences is,  $0 =$

$$\begin{aligned}
& a^2 \times \\
& + 1 \quad \text{and} \quad 4 \times (ac + 1) (\theta, 1)^1.
\end{aligned}$$

For the cubic equation  $(a, b, c, d \mid v, 1)^3 = 0$ , the equation of differences is,  $0 =$

$$\begin{aligned}
& a^4 \times \\
& + 1, \quad 18a^2 \times (ac + 1, b^2 - 1), \\
& 81 \times (a^2c^2 + 1, b^2abd + 21), \\
& 27 \times (a^2d^2 + 1, abcd - 6, ac^3 + 4, b^3d + 4, b^2c^2 - 3) \quad (\theta, 1)^3.
\end{aligned}$$

For the quartic equation  $(a, b, c, d, e \mid v, 1)^4 = 0$ , the equation of differences is,  $0 =$

(The standard-form expression has the same monomial structure as the denumerate form above, but with the entries multiplied by the appropriate binomial-coefficient powers — specifically  $48a^4 \times, 8a^4 \times, 32 \times, 16 \times, 1152 \times, 256 \times$  as overall numerical factors for the successive columns. The full standard-form quartic expression is set out as a six-column table on page 253 of the

original, with principal entries including:

$$\begin{aligned}
& a^4 \times +1, \quad ac + 1, \quad a^2e + 1, \quad a^2ce + 3, \quad a^2d^2 + 13, \\
& b^2 - 1, \quad a^2bd - 4, \quad a^2b^2e + 56, \quad a^3b^2cde - 7, \quad a^3e^2 - 1, \\
& a^2c^2 + 99, \quad a^2cd^2 + 288, \quad a^2d^2e + 3, \quad a^2c^2e^2 + 1, \\
& a^2b^2ce - 192, \quad a^2b^2d^2 - 400, \quad a^2bcde - 10, \quad a^2cd^2e + 54, \\
& a^2bc^2 - 192, \quad a^2bc^2d - 1944, \quad a^2bd^3 + 1269, \quad a^2c^3d^2 + 54, \\
& b^4 + 96, \quad a^2c^4 + 1377, \quad ab^3d^2 - 6, \quad ab^2cd^2 + 27, \\
& a^2b^3d + 64, \quad ab^3cd - 144, \quad ab^3d^2e - 6, \quad ab^2c^3e + 54, \\
& a^2b^4c - 432, \quad ab^2cd - 80, \quad ab^2c^2d - 78, \quad abcd^3 + 108, \\
& ab^3c + 384, \quad ac^4 + 1377, \quad ab^2c^3 - 78, \quad ac^3e + 81, \\
& b^6 - 128, \quad ab^4e + 96, \quad ac^4 + 27, \quad ac^2d^2 - 54, \\
& ab^3cd + 3648, \quad ab^2cd^3 + 4, \quad b^4ce + 108, \\
& ab^4c - 2592, \quad abcd^2 - 180, \quad b^4d^2 + 27, \\
& b^4d - 1536, \quad ab^3c^2 + 108, \quad b^3cd^2 + 18, \\
& b^6 + 1152, \quad ac^4d + 81, \quad b^2cd^4 + 36; \\
& \text{etc., } (\theta, 1)^4.
\end{aligned}$$

Owing to the very dense layout of the original printed table, only the principal entries are reproduced above; the reader is referred to the original page 253 of the memoir for the complete six-column table of monomials and coefficients.)

[The paper continues on the following pages 254 onwards with the standard-form expression for the quintic and the alternative determinant-elimination method; these pages fall outside the present extract.]

THE COLLECTED MATHEMATICAL PAPERS  
of  
ARTHUR CAYLEY

Volume IV — pages 254–278

Papers 262 (conclusion), 263, 264, 265, 266 (opening)

FOR THE QUINTIC EQUATION  $(a, b, c, d, e, f \mid v, 1)^5 = 0$ , *the equation of differences is,*

[Pages 254–255 of the printed text carry, across the two facing pages, the colossal six-column table giving the coefficients of the equation in  $\theta$  for the quintic. The columns correspond to the successive powers

$$\theta^{10}, \quad \theta^9, \quad \theta^8, \quad \theta^7, \quad \theta^6, \quad \theta^5, \quad \theta^4, \quad \theta^3, \quad \theta^2, \quad \theta^1, \quad \theta^0,$$

and the leading numerical multipliers are

$$100a^4\times, \quad 50a^4\times, \quad 2500a^4\times, \quad 125\times, \quad 625\times, \quad 2500\times, \quad 6250\times, \quad \dots$$

each column being headed by these factors. The body of each column is a long list of distinct monomial seminvariants of degree 5 in the coefficients  $a, b, c, d, e, f$  together with the associated integer coefficient (the sign  $\pm$  being indicated against each monomial); under each column the sum of the positive coefficients is recorded, and equals the sum of the negative coefficients — so that the row sums  $\pm 117, \pm 154, \pm 46627, \pm 91258, \pm 125515, \pm 159884, \pm 23570, \pm 18666, \pm 128505$  appear at the foot of the successive columns.

The leading monomials in column 1 ( $\theta^{10}$  coefficient) are

$$a^4 \cdot ac + 1, \quad a^4 \cdot b^2 - 1;$$

the column total being  $\pm 1$ . In column 2 ( $\theta^9$ , factor  $50a^4\times$ ) the leading entries are

$$a^4 \cdot ad + 1, \quad a^4 \cdot bc - 1;$$

total  $\pm 1$ . The third column ( $\theta^8$ , factor  $2500a^4\times$ ) contains

$$a^4 \cdot ae + 1, \quad a^4 \cdot ac^2 + 1, \quad a^4 \cdot abd - 1, \quad a^4 \cdot b^2c + 1, \quad \dots$$

totalling  $\pm 154$ . The remaining columns develop in like manner, the typical monomials of column 6 ( $\theta^5$ ) including

$$a^3b^2cde - 793860, \quad a^3bc^4 - 1736680000, \quad a^4b^2e^2 + 244000000, \quad \dots$$

the column total being  $\pm 159884$ ; and so on through the last column ( $\theta^0$ ,  $3125\times$ ), which has the bottom total  $\pm 128505$ .

The exact body of these two pages is too dense and too pictorial in its typographic layout to be set in line; the reader is referred to the original printed pages 254–255 for the complete enumeration of every monomial and coefficient.]

[where in each column after the first the sum of the positive coefficients is equal to that of the negative coefficients, and I have inserted at the foot this sum with the sign +. A single coefficient  $-7500$  in place of  $-3750$  has been corrected.]

The coefficients in the preceding equations of differences are functions of the seminvariants of the quantics to which they belong; for instance, in the case of the quartic, the coefficient of  $\theta^4$  is

$$8a^2[a^2(ae - 4bd + 3c^2) + 96(ac - b^2)],$$

that of  $\theta^3$  is

$$32[-13a^2(ace - ad^2 - b^2e + 2bcd - c^3) + 16a^2(ac - b^2)(ae - 4bd + 3c^2) + 128(ac - b^2)^3],$$

and so for the other coefficients; and by replacing each seminvariant by the covariant to which it belongs, we pass from the solution of the original problem of finding the equation for  $\theta = (\alpha - \beta)^2$ , to that of the problem of finding the equation for

$$\theta = \frac{(x - \alpha y)^2(x - \beta y)^2}{(x - \gamma y)^2 \dots}.$$

The results are as follows: —

For the quadric  $(a, b, c \mid x, y)^2$ , the equation in  $\theta$  is,  $0 =$

$$(U^2, 4\Box)(\theta, 1)^1,$$

where  $U$  is the quadric,  $\Box$  the discriminant.

For the cubic  $(a, b, c, d \mid x, y)^3$ , the equation in  $\theta$  is,  $0 =$

$$(U^4, 18U^2H, 81H^2, 27\Box U^2)(\theta, 1)^3,$$

where  $U$  is the cubic,  $H$  the Hessian,  $\Box$  the discriminant.

For the quartic  $(a, b, c, d, e \mid x, y)^4$ , the equation in  $\theta$  is,  $0 =$

$$\left\{ \begin{array}{l} U^7, \\ 48U^5H, \\ 32(-13U^4J + 16U^3HI + 128H^3), \\ 16(-7U^3I^2 - 288UHIJ + 384H^2I), \\ 1152(-3UIJ + 2HI^2), \\ 256(I^3 - 27J^2) \end{array} \right\} (\theta, 1)^4,$$

where  $U$  is the quartic,  $H$  the Hessian,  $I$  and  $J$  the quadrinvariant and the cubinvariant respectively.

**262]** ON THE EQUATION OF DIFFERENCES FOR AN EQUATION OF ANY ORDER. **257**

For the quintic  $(a, b, c, d, e, f \mid x, y)^5$ , the equation in  $\theta$ , as far as it can be expressed in terms of known covariants, is,  $0 =$

$$\left\{ \begin{array}{l} U^8, \\ 100 U^6 (\text{Tab. No. 15}), \\ 50 U^4 [U^2 (\text{Tab. No. 14}) + 75 (\text{Tab. No. 15})^2], \\ \vdots \\ 3125 \text{ Discriminant (= Tab. No. 26)} \end{array} \right\} (\theta, 1)^{10},$$

where the Tables referred to are those in my Second Memoir on Quantics, [141]; for the completed equation see post p. 261].

The form of the preceding results may be modified by writing  $\theta = \frac{1}{a^2} U^2 \vartheta$ ; we have thus the equations for

$$\vartheta = a^2 (\alpha - \beta)^2 (x - \gamma y)^2 \dots;$$

thus for example, in the case of the cubic  $(a, b, c, d \mid x, y)^3$ , the equation for  $\vartheta [= a^2 (\alpha - \beta)^2 (x - \gamma y)^2]$  is

$$0 = (1, 18H, 81H^2, 27 \square U^{-2}) (\vartheta, 1)^3;$$

this equation may be written

$$(\vartheta + 9H)^3 + 27 \square U^{-2} = 0;$$

or putting  $v = U^{2/3} \vartheta$ , we have

$$v^3 + 9Hv + U^2 \sqrt{-27 \square} = 0,$$

an equation the roots of which are

$$a(\alpha - \beta)(x - \gamma y), \quad a(\beta - \gamma)(x - \alpha y), \quad a(\gamma - \alpha)(x - \beta y),$$

and which leads to the formula, given in my fifth Memoir on Quantics, [156], for the solution of a cubic equation. But this decomposition of the equation in  $\vartheta$  is peculiar to the cubic.

The equation of differences for an equation of any order may be found by the following entirely distinct method. Let  $\varphi v = (* \mid v, 1)^n = 0$  be for shortness represented by  $\varphi v = 0$ , and let  $x, y$  be any two distinct roots; we have not only  $\varphi x = 0$ ,  $\varphi y = 0$ , but also  $\varphi x + \varphi y = 0$ ,  $\frac{\varphi x - \varphi y}{x - y} = 0$ .

Writing  $\theta = (x - y)^2$ ,  $s = x + y$ , we have

$$x = \frac{1}{2}(s + \sqrt{\theta}), \quad y = \frac{1}{2}(s - \sqrt{\theta}),$$

values which are to be substituted for  $x, y$  in the equations

$$\varphi x + \varphi y = 0, \quad \frac{\varphi x - \varphi y}{x - y} = 0.$$

We have thus two equations rational in  $s$  and  $\theta$ , and the elimination between them of the quantity  $s$  leads to the required equation in  $\theta$ . But it is proper to modify the form of the system; in fact the two equations are, as regards  $s$ , the first of them of the degree  $n$ , the second of the degree  $n - 1$ ; but if we write

$$\varphi x + \varphi y = 0, \quad \frac{\varphi x - \varphi y}{x - y} = 0,$$

then each of the equations will be of the same degree  $n - 1$  in  $s$ .

For instance, let  $\varphi v = (a, b, c, d \mid v, 1)^3$ ; then  $x = \frac{1}{2}(s + \sqrt{\theta})$ ,  $y = \frac{1}{2}(s - \sqrt{\theta})$ ; the equations  $\varphi x + \varphi y = 0$ ,  $\frac{\varphi x - \varphi y}{x - y} = 0$ , are

$$\begin{aligned} s^3.a + 3s^2.2b + 3s(4c + a\theta) + 8d + 6b\theta &= 0, \\ 3s^2.a + 3s.4b + 12c + a\theta &= 0; \end{aligned}$$

and multiplying the first equation by 3 and the second by  $-s$ , adding and dividing by 2, we have an equation

$$s^2.3b + s(12c + 4a\theta) + 12d + 9b\theta = 0.$$

The second equation and this equation may be written

$$\begin{aligned} (3a, 12b, 12c + a\theta \mid s, 1)^2 &= 0, \\ (3b, 12c + 4a\theta, 12d + 9b\theta \mid s, 1)^2 &= 0, \end{aligned}$$

and the elimination of  $s$  from these equations gives the required equation in  $\theta$ . The result may be obtained under either of the two forms,

$$\{a^2\theta^2 + (15ac - 27b^2)\theta - 36(bd - c^2)\}\{a^2\theta + 3(ac - b^2)\} + 3\{2ab\theta + 3(ad - bc)\}^2 = 0$$

and

$$\{4a^2\theta + (24ac - 27b^2)\theta - 36(bd - c^2)\}[a^2\theta + 12(ac - b^2)] + 3\{ab\theta + 6(ad - bc)\}^2 = 0,$$

the expansions of which respectively coincide with the before-mentioned result.

In the case of the quartic equation  $\varphi v = (a, b, c, d, e \mid v, 1)^4 = 0$ , we have

$$\begin{aligned} s^4.a + 4s^3.2b + 6s^2(4c + a\theta) + 4s(8d + 6b\theta) + 16e + 24c\theta + a\theta^2 &= 0, \\ 4s^3.a + 6s^2.4b + 4s(12c + a\theta) + 32d + 8b\theta &= 0, \end{aligned}$$

from which we derive another cubic equation; the two cubic equations thus are

$$\begin{aligned} (4a, 24b, 48c + 4a\theta, 32d + 8b\theta \mid s, 1)^3 &= 0, \\ (4b, 24c + 10a\theta, 48d + 44b\theta, 32e + 48c\theta + 2a\theta^2 \mid s, 1)^3 &= 0, \end{aligned}$$

from which, if  $s$  be eliminated, we have the equation in  $\theta$ .

Similarly, for the quintic equation  $\varphi v = (a, b, c, d, e, f \mid v, 1)^5 = 0$ , we have

$$\begin{aligned} s^5 \cdot a + 5s^4 \cdot 2b + 10s^3 (4c + a\theta) + 10s^2 (8d + 6b\theta) \\ + 5s (16e + 24c\theta + a\theta^2) + 32f + 80d\theta + 10b\theta^2 = 0, \\ 5s^4 \cdot a + 10s^3 \cdot 4b + 10s^2 (12c + a\theta) + 5s (32d + 8b\theta) + 80e + 40c\theta + a\theta^2 = 0, \end{aligned}$$

from which we derive another quartic equation; the two equations are

$$\begin{aligned} (5a, 40b, 120c + 10a\theta, 160d + 40b\theta, 80e + 40c\theta + a\theta^2 \mid s, 1)^4 = 0, \\ (5b, 40c + 20a\theta, 120d + 110b\theta, 160e + 280c\theta + 12a\theta^2, 80f + 200d\theta + 25b\theta^2 \mid s, 1)^4 = 0, \end{aligned}$$

from which, if  $s$  be eliminated, we have the equation in  $\theta$ .

But we may apply Bézout's method to the two equations each of the order  $n - 1$ , which results from the equation of the order  $n$ , in the present case the equation  $\varphi v = (* \mid v, 1)^n = 0$ . The process is as follows: —Suppose, in general, that  $s$  is to be eliminated from the two equations

$$Fs = 0, \quad Gs = 0,$$

each of the order  $n - 1$ ; it is only necessary to form the expression

$$\frac{FsGs' - Fs'Gs}{s - s'}$$

which will be a function of  $s, s'$  of the form

$$(a_{1,1}, a_{1,2}, \dots, a_{n-1,n-1} \mid s, 1)^{n-2} (s', 1)^{n-2}$$

where the coefficients are such that  $a_{i,j} = a_{j,i}$ ; and by equating to zero the determinant formed with these coefficients, we have the result of the elimination.

In the present case, writing for a moment  $\frac{1}{2}(s + \sqrt{\theta}) = A$ ,  $\frac{1}{2}(s - \sqrt{\theta}) = B$ , and in like manner  $\frac{1}{2}(s' + \sqrt{\theta}) = A'$ ,  $\frac{1}{2}(s' - \sqrt{\theta}) = B'$ , we have

$$\begin{aligned} Fs = \varphi(A + B) - \frac{\varphi(A - B)}{\sqrt{\theta}}, \quad Gs = \frac{A - B}{\sqrt{\theta}}, \\ Fs' = \varphi(A' + B') - \frac{\varphi(A' - B')}{\sqrt{\theta}}, \quad Gs' = \frac{A' - B'}{\sqrt{\theta}}, \end{aligned}$$

and therefore

$$FsGs' - Fs'Gs = \frac{(A + B)(A' - B') - (A' + B')(A - B)}{\sqrt{\theta}} \dots,$$

or reducing and dividing by  $s - s'$ ,

$$\frac{FsGs' - Fs'Gs}{s - s'} = \frac{1}{\sqrt{\theta}(s - s')} \cdot \frac{AB' - A'B}{1} \cdot \frac{(A - B)(A' - B')}{\theta} \dots$$



Hence, substituting for  $A, B, A', B'$  these values, we have the expression

$$\frac{2m}{\sqrt{\theta}} \cdot \frac{\dots}{\theta}$$

which is of the form

$$(a_{1,1}, \dots, a_{n-1,n-1} \mid s, 1)^{n-2} (s', 1)^{n-2},$$

and equating to zero the determinant formed with the coefficients, we have an equation in  $\theta$  which is the equation of differences of the given equation  $\varphi v = 0$ . For instance, if the given equation is  $\varphi v = (a, b, c, d \mid v, 1)^3 = 0$ , then we have

$$\begin{aligned} \varphi\left(\frac{1}{2}(s + \sqrt{\theta})\right) &= (a, 2b + a\sqrt{\theta}, 4c + 4b\sqrt{\theta} + a\theta, 8d + 12c\sqrt{\theta} + 6b\theta + a\theta\sqrt{\theta} \mid s, 1)^3 \\ &= (A, B, C, D \mid s, 1)^3, \\ \varphi\left(\frac{1}{2}(s - \sqrt{\theta})\right) &= (a, 2b - a\sqrt{\theta}, 4c - 4b\sqrt{\theta} + a\theta, 8d - 12c\sqrt{\theta} + 6b\theta - a\theta\sqrt{\theta} \mid s, 1)^3 \\ &= (A', B', C', D' \mid s, 1)^3; \end{aligned}$$

and the function in  $s, s'$  is

$$\begin{aligned} &3(AB' - A'B)ss'^2 \\ &+ 3(AC' - A'C)ss'(s + s') \\ &+ [(AD' - A'D) + 9(BC' - B'C)](s^2 + ss' + s'^2) \\ &+ 3(BD' - B'D)(s + s') \\ &+ 3(CD' - C'D) \\ &+ \frac{1}{\theta}(A - A', B - B', C - C', D - D' \mid s, 1)^3 (A - A', B - B', C - C', D - D' \mid s', 1)^3, \end{aligned}$$

which is equal to

$$\begin{aligned} &- 12abss'(s + s') \\ &+ 12(-12ac - a^2\theta)(s^2 + ss' + s'^2) \\ &+ (36ac - 72b^2 + 9a^2\theta)ss' \\ &+ (24ad - 72bc + 12ab\theta)(s + s') \\ &+ 96bd - 144c^2 + (-48ac + 72b^2)\theta - 3a^2\theta^2 \\ &+ 4(3as^2 + 12bs + 12c + a\theta)(3as'^2 + 12bs' + 12c + a\theta); \end{aligned}$$

**262]** ON THE EQUATION OF DIFFERENCES FOR AN EQUATION OF ANY ORDER. **261**

or reducing and dividing by 32, this is

$$\{9(ac - b^2) + 3a^2\theta\} ss' + \{9(ad - bc) + 6ab\theta\}(s + s') + 36(bd - c^2) + (-15ac + 27b^2)\theta - a^2\theta^2,$$

the terms in  $s^2$ ,  $s'^2$  disappearing, as they should do. Writing this under the form

$$\begin{pmatrix} 9(ac - b^2) + 3a^2\theta, & 9(ad - bc) + 6ab\theta \\ 9(ad - bc) + 6ab\theta, & 36(bd - c^2) + (-15ac + 27b^2)\theta - a^2\theta^2 \end{pmatrix} (s, 1)(s', 1)$$

and equating the determinant to zero, we have the required equation in  $\theta$ : the form is that obtained by the ordinary process of applying Bézout's method to the two equations  $(3a, 12b, 12c + a\theta \mid s, 1)^2 = 0$ ,  $(3b, 12c + 4a\theta, 12d + 9b\theta \mid s, 1)^2 = 0$ , being in fact the before-mentioned equation

$$\{a^2\theta^2 + (15ac - 27b^2)\theta - 36(bd - c^2)\}\{a^2\theta + 3(ac - b^2)\} + 3\{2ab\theta + 3(ad - bc)\}^2 = 0.$$

But, as already remarked, this elimination process is less convenient for the complete development of the result, than the method first explained in the present memoir.

[The equation p. 257, changing the notation and inserting the omitted coefficients, becomes  $0 =$

$$\begin{aligned} &\theta^{10} \cdot 100 \times (A^2B + 1, A^2C + 1); \\ &\theta^9 \cdot 50 \times (A^2B + 1, A^2CD + 75); \\ &\theta^7 \cdot 2500 \times (A^2BC + 1, A^2F^2 + 1, A^2CD + 29); \\ &\theta^6 \cdot 125 \times (A^2BI + 1169, A^2BC^2 + 450, CF^2 + 800, C^4 + 6323); \\ &\theta^5 \cdot 625 \times (A^2G + 1601, AFHD + 60, BF^2 + 112480, BACD + 3620); \\ &\theta^4 \cdot 12500 \times (A^2BH - 1545, A^2J, AECI + 230, AECJ + 275, BEF + 1225, \\ &\quad + 11650, CAHB^2 + 1125); \\ &\theta^3 \cdot 6250 \times (A^2BMG - 33267, -21450200, AEBGCJD + 23820000, BCH + 1625, C^2G); \\ &\theta^2 \cdot 62500 \times (ABDJG - 20, +24, BCG + 19, B^2H - 36); \\ &\theta^1 \cdot 62500 \times (B^2JG - 162, GH, DJ, CM - 28); \\ &\theta^0 \cdot 3125 \times (Q), \end{aligned}$$

where the capital letters denote the covariants of the quantic as explained 141 and 143.]

*Note.* The exact column layout of the printed page 261 is here transposed into a single sequential listing for clarity. The detailed monomial structure and signs follow the original.

## 263.

## DEMONSTRATION OF A THEOREM IN FINITE DIFFERENCES.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CL. (for the year 1860), pp. 321–323: printed as a note to Sir J. W. F. Herschel's Memoir "On the Formulae investigated by Dr Brinkley for the general Term in the Development of Lagrange's Expression for the Summation of Series and for Successive Integrations," pp. 319–321.]

The formula (B) [of Sir J. W. F. Herschel's Memoir], substituting therein for  $\Delta x$  the value  $([x])\Delta\nabla^x 0^n$ , becomes

$$\frac{1}{[n-1]} \left\{ \frac{1}{[x-1]} [n+1]\nabla^x - \frac{1}{[x-2]} [n+2]\nabla^x \&c. \right\} 0^n,$$

or, as this may be written,

$$\frac{1}{[n-1]} \left\{ \frac{[x]}{1 \cdot 2} \nabla^x 0^n - \frac{[x]}{2 \cdot 3} \nabla^x 0^n + \dots \right\},$$

or, inserting a first term which vanishes except in the case  $x = 0$ , and which is required in order that the formula may hold good for this particular value,

$$\frac{1}{[n-1]} \{ [0][x] 0^n - [1][x-1] (n+1)^x + [2][x-2] (n+2)^x - \&c. \},$$

where the series on the right-hand side need only be continued up to the term containing  $\nabla^x 0^n$ , since the subsequent terms vanish. [In these formulæ and throughout the present paper  $[x]$  is written to denote the factorial  $[x]^x$  or  $\Pi(x)$ , and so in other cases.]

Now  $\nabla^x 0^n$ , or  $\left(\frac{e^t - 1}{e^t}\right)^x 0^n$ , is equal to  $[x] \times \text{coef. } t^n$  in  $\left(\frac{e^t - 1}{e^t}\right)^x$ , and so  $\nabla^x 0^n$ , or  $\left(\frac{e^t - 1}{e^t}\right)^x 0^n$ , is equal to  $[x] \times \text{coef. } t^n$  in  $\left(\frac{e^t - 1}{e^t}\right)^x$ .

Hence, putting  $R = \frac{e^t - 1}{t}$ , the last-mentioned formula will be true if, as regards the term which contains  $t^n$ , we have

$$R^{-n} = \frac{1}{[n-1]} \left\{ \frac{1}{[0][x]} n - \frac{1}{[1][x-1]} (n+1) R + \frac{1}{[2][x-2]} (n+2) R^2 - \&c. \right\},$$

the series on the right-hand side being continued up to the term in  $R^x$ . This formula is, in fact, true if  $R$ , instead of being restricted to denote  $\frac{e^t - 1}{t}$ , denotes any function whatever of the form  $1 + bt + ct^2 + \&c.$ , and it is true not only for the term in  $t^n$ , but for all the powers of  $t$  not higher than  $t^x$ . And, moreover,  $R^{-n}$  may denote any positive or negative integral or fractional power of  $R$ . In fact, the formula (assuming for a moment the truth of it) shows that the expansion of any power whatever of a series of the form in question, can be obtained by means of the expansions of the successive positive integer powers of the same series: the existence of such a formula (at least for negative powers) was indicated by Eisenstein, *Crelle*, t. XXXIX. p. 181 (1850), and the formula itself, in a slightly different form, was obtained in a very simple manner by Professor Sylvester in his paper, "Development of an idea of Eisenstein," *Quart. Math. Journ.* t. I. p. 199 (1855); the demonstration was in fact as follows, viz. writing

$$R^n = (1 + (R-1))^n = 1 + n(R-1) + \frac{n(n-1)}{1 \cdot 2} (R-1)^2 + \&c. :$$

if we attend only to the terms involving powers of  $t$  not higher than  $t^x$ , the series on the right-hand side needs only to be continued up to the term involving  $(R-1)^x$ , and the right side being thus converted into a rational and integral function of  $R$ , it may be developed in a series of powers of  $R$  (the highest power being of course  $R^x$ ), and the coefficients of the several powers are finite series which admit of summation; this gives the required formula. But there is an easier method; the process shows that the series on the right-hand side, continued as above up to the term involving  $t^x$ , is, as regards  $n$ , a rational and integral function of the degree  $x$ ; and by Lagrange's interpolation formula, any rational and integral function of  $n$  of the degree  $x$ , can be at once expressed in terms of the values corresponding to  $x+1$  particular values of  $n$ . The investigation will be as follows: — Let  $R$  denote a series of the form  $1 + bt + ct^2 + \&c.$ , and let  $R^n$  denote the development of the  $n$ th power of  $R$ , continued up to the term containing  $t^x$ , the terms involving higher powers of  $t$  being rejected:  $R^0, R^1, R^2, \&c.$ , and generally  $R^s$ , will in like manner denote the developments of these powers up to the term involving  $t^x$ , or what is the same thing, they will be the values of  $R^n$ , corresponding to  $n = 0, 1, 2, \dots x$ . By what precedes  $R^n$  is a rational and integral function of  $n$  of the degree  $x$ , and it can therefore be expressed in terms of the values  $R^0, R^1, R^2, \dots R^x$ , which correspond to  $n = 0, 1, 2, \dots x$ . Let  $s$  have any one of the last-mentioned values, then the expression

$$\frac{n \cdot n - 1 \cdot n - 2 \cdots n - x}{n - s} \cdot \frac{1}{s \cdot s - 1 \cdots 2 \cdot 1 \cdot - 1 \cdot - 2 \cdots - (x - s)} R^s,$$

which, as regards  $n$ , is a rational and integral function of the degree  $x$  (the factor

264

DEMONSTRATION OF A THEOREM IN FINITE DIFFERENCES.

[263

$n - s$ , which occurs in the numerator and in the denominator being of course omitted), vanishes for each of the values  $n = 0, 1, 2, \dots, x$ , except only for the value  $n = s$ , in which case it becomes equal to unity. The required formula is thus seen to be

$$R^n = \sum \left\{ \frac{n \cdot n - 1 \cdot n - 2 \cdots n - x}{n - s} \cdot \frac{1}{s \cdot s - 1 \cdots 2 \cdot 1 \cdot -1 \cdot -2 \cdots -(x - s)} R^s \right\},$$

where the summation extends to the several values  $s = 0, 1, 2, \dots, x$ ; or, what is the same thing, it is

$$R^n = \sum \left\{ \frac{n \cdot n - 1 \cdot n - 2 \cdots n - x}{n - s} \cdot \frac{(-)^{x-s}}{1 \cdot 2 \cdots s \cdot 1 \cdot 2 \cdots (x - s)} R^s \right\},$$

or, as this may be written,

$$R^n = \sum \left\{ \frac{n \cdot n - 1 \cdot n - 2 \cdots n - x}{n + s} \cdot \frac{(-)^{x-s}}{1 \cdot 2 \cdots s \cdot 1 \cdot 2 \cdots (x - s)} R^s \right\},$$

or substituting for  $s$  the values  $0, 1, 2, \dots, x$ , the formula is

$$R^n = \frac{n \cdot n - 1 \cdot n - 2 \cdots n - x}{[n - 1]} \left\{ \frac{1}{[0][x]n} - \frac{1}{[1][x - 1](n + 1)} R + \frac{1}{[2][x - 2](n + 2)} R^2 - \cdots \right\},$$

continued up to the term involving  $R^x$ , which is the theorem in question.

## 264.

## ON AN EXTENSION OF ARBOGAST'S METHOD OF DERIVATIONS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLI. (for the year 1861), pp. 37–43, Received October 18,—Read December 13, 1860.]

ARBOGAST'S Method of Derivations was devised by him with a view to the development of a function  $\varphi(a+bx+cx^2+\dots)$ , but it is at least as useful for the formation of only the literal parts of the coefficients, or, what is the same thing, the combinations of a given degree and weight in the letters ( $a, b, c, d, \dots$ ), the weights of the successive letters being 0, 1, 2, 3, &c. Thus instead of applying the method to finding the coefficients

$$a^4, \quad 4a^3b, \quad 4a^3c + 6a^2b^2, \quad \&c.,$$

we may apply it merely to finding the sets of terms

$$a^4, \quad a^3b, \quad a^3c, \quad a^2b^2, \quad \&c.$$

To derive any column from the one which immediately precedes it, we operate on a letter by changing it into its immediate successor in the alphabet, and we must in each term operate on the last letter, and also, when the last but one letter in the term is the immediate antecessor in the alphabet of the last letter (but in this case only), operate on the last but one letter. Thus  $a^3c$  gives  $a^3d$ , but  $a^2b^2$  gives  $a^2bc$  and  $a^3c$ , and the next succeeding column is therefore

$$a^3d, \quad a^2bc.$$

If the series of letters is finite, and the last letter of the term is also the last letter of the series, then it is impossible to operate on the last letter of the term, but the last but one letter (when the foregoing rule applies to it) is still to be operated on; and if the rule does not apply, then the term does not give rise to a term in the succeeding column; the operations will at length terminate, and a

complete series of columns be obtained. Thus, if the letters are  $(a, b, c, d)$ , and the operations are (as before) performed upon  $a^4$ , the entire series of columns is of the form (with successive columns laid out in alphabetical order):

$$\begin{array}{c|c|c|c|c|c|c|c|c|c|c}
 a^4 & a^3b & a^3c & a^3d & a^2bd & a^2bcd & a^2c^2d & ab^2d^2 & abcd^2 & abd^3 & \dots \\
 & a^2b^2 & a^2bc & a^2c^2 & a^2b^2c & a^2bd^2 & a^2cd^2 & a^2c^2d^2 & b^3cd & b^2cd^2 & d^4 \\
 & & ab^3 & a^2b^3 & ab^3d & ab^2cd & ab^2c^2 & b^4 & bcd^3 & c^2d^3 & \\
 & & & & b^4 & ab^3c & b^3c^2 & b^3d^2 & b^2d^3 & & 
 \end{array}$$

Nothing can be more convenient than the process when the entire series of columns is required; but it is very desirable to have a process for the formation of any column apart from the others; and the object of the present memoir is to investigate a rule for the purpose. But as showing how this rule is to be demonstrated, by very similar principles, I will commence with Arbogast's rule, applied as above, depends upon

If we take any combination  $b^2d$  and operate backwards on the last letter (viz. by changing it into its immediate antecessor in the alphabet), we obtain  $b^2c$ , which is a term in the next preceding column;  $b^2d$  is therefore obtainable from a term in the next preceding column, viz.  $b^2c$ , and the process is to operate on the last letter. If, instead of  $b^2d$ , the term is  $abc^2$  (the last letter here entering as a power), the operation backwards on the last letter gives  $ab^2c$ , which is also a term of the preceding column; and it is to be noticed that the last but one letter  $b$  is here the immediate antecessor of the last letter  $c$  (and would have been so even if  $b$  had not entered into the given term  $abc^2$ , thus  $ac^2$  operated on backwards would have given  $abc$ ). Hence  $abc^2$  is obtained by operating on a term in the next preceding column, viz. the term  $a^2bc$ , but in this case the operation is performed on the last but one letter. Every term is thus obtained from the next preceding column, viz. the terms are obtained by operating on the last letter, and (when the last but one letter is the immediate antecessor in the alphabet of the last letter) then also on the last but one letter, of each term of the next preceding column, and the correctness of the rule is thus demonstrated. It is to be observed that the terms are operated upon in order, the operation on the last but one letter (when it is operated on) being made immediately after that upon the last letter of the same term, and that the terms of a column are thus obtained in the proper alphabetical order.

I pass now to the above mentioned question of the formation of a single column by itself; it will be convenient, by way of illustration, to write down the columns

$$\begin{array}{lll}
 a^2bd^2, & ac^2d^2, & c^2de^2, \\
 ab^2d^2, & ab^2cd^2, & cd^2e^2, \\
 a^2bce, & bc^2de, & bce^3, \\
 a^2c^2e, & c^2de^2, & ad^4, \\
 b^2ce^2, & ad^4, & bcd^3, \\
 bc^2de, & bcd^3, & bd^4, \\
 c^4e^2, & bd^4, & d^5;
 \end{array}$$

which belong to the set  $(a, b, c, d, e)$ , and are of the degree 5 and the weights 12 and 15 respectively.

Some definitions and explanations are required. I speak of the first and last letters of the set, simply as the first and the last letter: there is frequent occasion to speak of the last letter; and to avoid confusion, the last letter of a term will be spoken of as the *ultimate letter*; it is necessary also to consider the *penultimate*, *antepenultimate*, and *pro-antepenultimate letters* of the term. It will be convenient to distinguish between the ultimate letter and the *ultimate*, which may be either the ultimate letter or a power of such letter; and similarly for the penultimate, &c. Thus in the term  $bcd^2$ , the ultimate letter is  $d$  and the ultimate is  $d^2$ , the penultimate and the penultimate letter are each of them  $c$ ; of course the ultimate, penultimate, &c. letters are always distinct from each other. We have also to consider the *pairs* of letters contained in a term;  $cd^2e$  contains the pairs  $(c, d)$ ,  $(c, e)$ ,  $(d, d)$ ,  $(d, e)$ , and so in other cases; the letters of a pair are taken in the natural order. A pair not containing either the first letter or the last letter is *expansible*; thus the set being as before  $(a, b, c, d, e)$ , the pairs  $(b, c)$ ,  $(d, d)$  are expansible: they are *expanded* by retreating the prior and advancing the posterior letter each one step; thus the just mentioned pairs  $(b, c)$ ,  $(d, d)$  are expanded into  $(a, d)$  and  $(c, e)$  respectively.

A pair composed of two distinct non-contiguous letters is *contractible*; it is contracted by advancing the prior and retreating the posterior letter each one step: thus  $(a, d)$ ,  $(c, e)$  are contractible pairs, and they give by contraction the pairs  $(b, c)$ ,  $(d, d)$  respectively: the processes of expansion and contraction are obviously converse to each other.

The expression *the last expansible pair* of a term hardly requires explanation;  $(d, d)$  is the last expansible pair of the term  $bcd^2$ , or of the term  $b^2d^2e^2$ ,  $(c, d)$  the last expansible pair of the term  $c^2de$  (the set being always  $(a, b, c, d, e)$ ), and so in all other cases. The expression *the last expansion*, in regard to any term, means the expansion of the last expansible pair of such term.

The expansion or contraction of any pair of a term leaves unaltered as well the weight as the degree, the resulting term belongs therefore to the same column as the original one. But the effect of an expansion is to diminish, and that of a contraction to increase the alphabetical rank, or *rank in the column*, the ranks being reckoned as the first or lowest, second, third, and so on, up to the last or highest rank. In particular, by performing upon any term the last expansion, we diminish the rank in the column, and by a succession of last expansions we bring the term up to the head of the column. Such term is not susceptible of any further expansion; it may therefore contain the first and last letters or either of them, and it may also contain a single intermediate letter in the first power only; thus the first and last letters being  $a, e$ , the head term of a column is of the form  $a^m e^l$  or  $a^m c e^l$ ,  $c$  being some intermediate letter, and the powers  $a^m$ ,  $e^l$  being both or either of them omittable. In like manner, by a succession of contractions of any term we obtain the term at the foot of the column; such term is not susceptible of any further contraction, and it must therefore be composed either of a single letter or of two contiguous



letters, that is it must be of the form  $c^m$ , or of the form  $c^m d^l$ , where  $c, d$  are any two contiguous letters, not excluding the case where the single letter or each or either of the contiguous letters is a first or a last letter.

It is to be observed that the passage up from any term to the term at the head of the column (or *top term*) by means of a succession of last expansions, is a perfectly unique one; but as no selection has been made of a like unique process of contraction, this is not the case with respect to the passage down from any term to the term at the foot of the column (or *bottom term*) by a succession of contractions.

Every term gives by the last expansion a term above it; it can therefore be obtained from such term above it by means of a contraction. But the contraction of the upper term is by hypothesis such that, performing upon the contracted term the last expansion, we obtain the upper term; a contraction, which, performed on any term, gives a lower term which by performing upon it the last expansion reproduces the first mentioned or upper term, may be called a *reversible contraction*. And it is clear that if we perform on the top term all the reversible contractions, and on each of the resulting terms all the reversible contractions, and so on as long as the process is possible, we obtain without repetition all the terms of the column. The column is in fact similar to a genealogical tree in the male line, each lower term issuing from a single upper term, while each upper term generates a lower term or terms, or does not generate any such term, and the top term being the common origin of the entire series. It may be added that when the order of the reversible contractions of the same term is duly fixed, the alphabetical arrangement of the terms in the column agrees with the order as of primogeniture (an elder son and his issue male preceding all the younger sons and their respective issue male) in the genealogical tree.

It only remains then to inquire under what conditions a contraction is reversible. Now as regards any term, in order that a contraction performed on it may be reversible, it is necessary and sufficient that the pair produced by the contraction should be the last expansible pair of the contracted term. There are several cases to be considered. First, if the contraction affects the ultimate and penultimate letters of the term: this implies that the ultimate and penultimate letters are not contiguous. Let the term terminate in  $e^m h$ , the contracted term will terminate in  $e^{m-1} f g$ , and  $(f, g)$ , the pair produced by the contraction, is the last expansible pair of the contracted term; the contraction is in this case reversible. If, however, the term terminate in  $e^m h^p$  ( $p > 1$ ), the contracted term will terminate in  $e^{m-1} f g h^{p-1}$ , and the last expansible pair is not as before  $(f, g)$ , but it is (according as  $p = 2$  or  $p > 2$ )  $(g, h)$  or  $(h, h)$ : unless indeed  $h$  is the last letter, in which case  $(f, g)$  remains the last expansible pair of the contracted term. In the example  $e^m$  has been written, but the case  $m = 1$  is not excluded; moreover, the penultimate letter  $e$  is removed three steps from the ultimate letter  $h$ , but the result would have been the same if instead of  $e$  we had had any preceding letter, or had had the letter  $f$ ; by hypothesis it is not  $g$ , the letter contiguous to the ultimate letter  $h$ . The conclusion is that a contraction on the ultimate and penultimate letters (these being non-contiguous) is reversible if the ultimate is a simple letter, or if, being a power, it is a power of the last letter.

Next let the contraction affect the ultimate and antepenultimate letters. The two letters are here separated by the penultimate letter, and are therefore not contiguous. Suppose that the termination is  $f^l g^m k$  ( $f$  and  $g$  contiguous), the contracted term terminates in  $f^{l-1} g g^m j$ , and  $(g, j)$  the pair arising from the contraction is the last expansible pair of the contracted term; the contraction is therefore reversible. In the example  $f^l$  has been written, but the case  $l = 1$  is not excluded; moreover the ultimate letter  $k$  is taken non-contiguous to the penultimate letter  $g$ ; but if the ultimate letter had been the contiguous letter  $h$ , the only difference is that the pair would be  $(g, g)$ , and the conclusion is not altered. But suppose the termination of the term is  $e^l g^m k$  ( $e, g$  non-contiguous), then the contracted term terminates in  $e^{l-1} f g^m j$ , where the pair arising from the contraction is  $(f, j)$ , but the last expansible pair is  $(g, j)$ ; the contraction therefore is not reversible. The case  $l = 1$  is not excluded; nor is it necessary that the ultimate letter should be non-contiguous to the penultimate; if the ultimate letter had been  $h$ , the pair arising from the contraction would have been  $(f, g)$ , but the last expansible pair  $(g, g)$ , and the contraction is still not reversible. In each of the cases considered the ultimate has been a simple letter; if in the first case the ultimate had been  $k^p$  ( $p > 1$ ), then the contracted pair would terminate in  $j k^p$ , and (according as  $p = 2$  or  $p > 2$ ) the last expansible pair would be  $(j, k)$  or  $(k, k)$ , which is not the pair  $(g, j)$  produced by the contraction; the contraction is therefore not reversible, unless indeed  $k$  is the last letter, in which case it continues reversible. In the second case the contraction, which is not reversible when the ultimate is the simple letter  $k$ , remains not reversible when the ultimate is a power of such letter. The conclusion is that a contraction on the ultimate and antepenultimate letters is reversible, if the penultimate and antepenultimate letters are contiguous, and the ultimate is a simple letter, or if, being a power, it is a power of the last letter.

A contraction on the ultimate and pro-antepenultimate letters, or on the ultimate letter and any letter preceding the pro-antepenultimate letter, is never reversible. To show this, it will be sufficient to consider the case where the term terminates in  $efgh$ , the contracted term terminates in  $ff'gg$ , the pair arising from the contraction being  $(f, g)$ , and the last expansible pair being  $(g, g)$ ; and *à fortiori*, if the letters or any of them occur as powers, or are non-contiguous.

Consider, next, a contraction on the penultimate and antepenultimate letters; this assumes that these letters are non-contiguous. Such a contraction may be reversible if only the ultimate is the last letter or a power thereof; and the condition is then similar to that in the case of the ultimate and penultimate letters; only as the penultimate cannot be a power of the last letter, it must be a simple letter. And the conditions in order that the contraction may be reversible then are that the ultimate is the last letter or a power thereof, and the penultimate a simple letter.

The next case is that of a contraction on the penultimate and pro-antepenultimate letters. Such contraction may be reversible if the ultimate is the last letter or a power thereof; and the condition is then similar to that for the case of the ultimate and antepenultimate letters; only as the penultimate cannot be a power of the last

letter, it must be a simple letter. The conditions, in order that the contraction may be reversible, are that the ultimate may be the last letter or a power thereof, the penultimate a simple letter, and the antepenultimate and pro-antepenultimate letters contiguous.

A contraction on the penultimate letter and on any letter preceding the pro-antepenultimate letter, or upon any two letters preceding the penultimate letter, is never reversible. If the ultimate, penultimate, antepenultimate and pro-antepenultimate letters are denoted by  $U$ ,  $P$ ,  $A$ ,  $P'$  respectively, then by what precedes, the following contractions, viz.  $UP$ ,  $UA$ ,  $PA$ ,  $PP'$ , may be reversible, and the conditions for them are shown in the annexed Table. It is to be noticed that the conditions for  $UA$ ,  $PA$  are mutually exclusive, and consequently that the number of reversible contractions to be performed upon any term is at most 3. The Table is

|     | $A$                                                                                                                  | $P'$                                                                                                              |
|-----|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| $U$ | $U$ , $P$ , non-contiguous letters.<br>The ultimate a simple letter, or a power of the last letter.                  | $P$ , $A$ , contiguous letters.<br>The ultimate a simple letter, or a power of the last letter.                   |
| $P$ | $P$ , $A$ , non-contiguous letters.<br>Penultimate a simple letter.<br>Ultimate the last letter, or a power thereof. | $A$ , $P'$ , contiguous letters.<br>Penultimate a simple letter.<br>Ultimate the last letter, or a power thereof. |

The contractions are to be applied in the order  $UP$ ,  $UA$ ,  $PA$ ,  $PP'$ , but all the contracted terms originating in a prior contraction of a given term are to be exhausted before proceeding to a posterior contraction of the same term. As an example of the process, the set being  $(a, b, c, d, e)$ , I will take the column

$$\begin{aligned}
 &abe^3, \\
 &acde^2, \\
 &ad^3e, \\
 &b^2de^2, \\
 &bc^2e^2, \\
 &bcd^2e, \\
 &bd^4, \\
 &c^2d^2e, \\
 &c^4e.
 \end{aligned}$$

The top term  $abe^3$  is given. The contractions applicable to it are  $UP$ ,  $UA$ . And  $UP$  gives  $acde^2$ . The only contraction applicable to this is  $UA$ , giving  $ad^3e$ . And

there is not any contraction applicable to  $ad^3e$ . We revert therefore to  $UA$  on  $abe^3$ , this gives  $b^2de^2$ . The only applicable contraction is  $PA$ , giving  $bc^2e^2$ . The contractions applicable to this are  $UP$ ,  $UA$ . And  $UP$  gives  $bcd^2e$ . The only contraction applicable to this is  $UA$ , giving  $bd^4$ , and there is not any contraction applicable to  $bd^4$ . We revert therefore to  $UA$  on  $bc^2e^2$ , this gives  $c^2d^2e$ , and the only contraction applicable to this is  $UA$ , giving  $c^4e$ . There is not any contraction applicable to this, and the process is therefore complete. It may be remarked that the example presents no instance of the contraction  $PP'$ ; in fact the only terms having a pro-antepenultimate are  $acde^2$ , in which  $A$ ,  $P'$  are not contiguous letters, and  $bcd^2e$ , in which the penultimate is not a simple letter, so that the contraction  $PP'$  is in each case excluded. It must be confessed that a considerable amount of practice would be required before the process could be readily made use of.

## 265.

ADDITION TO THE MEMOIR ON AN EXTENSION OF ARBOGAST'S  
METHOD OF DERIVATIONS.

[Now first published, 1891.]

The process explained in the foregoing Memoir is not easily workable, and I have in fact never used it. I have devised a new process theoretically less complete, and of a somewhat mixed character consisting, as it does, in the analytical reduction of the problem to the same problem for a smaller number of letters. I have however found it very convenient for obtaining the literal terms in the theory of the sextic, viz. where we have the seven letters  $(a, b, c, d, e, f, g)$ ; and I propose to explain the process by applying it to the determination of the terms of the discriminant, degree = 6, weight = 18.

Here treating  $a, b, c, d, e, f, g$  to denote the indices of these letters respectively in a term such as  $a^a b^b c^c \dots$  (that is, writing for convenience  $a, b, c, \dots$  instead of  $\alpha, \beta, \gamma, \dots$ ) we have

$$a + b + c + d + e + f + g = 6,$$

and thence

$$b + 2c + 3d + 4e + 5f + 6g = 18,$$

$$6a + 5b + 4c + 3d + 2e + f = 18.$$

I separate off from the others the first three letters  $a, b, c$ . The equation gives  $a = 3$  at most. And then

|              |                                       |                  |
|--------------|---------------------------------------|------------------|
| if $a = 3$ , | then $5b + 4c + 3d + 2e + f = 0$ ,    | $b = 0, c = 0$ ; |
| if $a = 2$ , | $5b + 4c + 3d + 2e + f = 6$ ,         | $b = 1$ at most; |
|              | if $b = 1$ , $4c + 3d + 2e + f = 1$ , | $c = 0$ ;        |
|              | if $b = 0$ , $4c + 3d + 2e + f = 6$ , | $c = 1$ at most. |

|              |                                        |                  |
|--------------|----------------------------------------|------------------|
| if $a = 1$ , | $5b + 4c + 3d + 2e + f = 12$ ,         | $b = 2$ at most; |
|              | if $b = 2$ , $4c + 3d + 2e + f = 2$ ,  | $c = 0$ ;        |
|              | if $b = 1$ , $4c + 3d + 2e + f = 7$ ,  | $c = 1$ at most; |
|              | if $b = 0$ , $4c + 3d + 2e + f = 12$ , | $c = 3$ at most; |
| if $a = 0$ , | $5b + 4c + 3d + 2e + f = 18$ ,         |                  |
|              | if $b = 3$ , $4c + 3d + 2e + f = 3$ ,  | $c = 0$ ;        |
|              | if $b = 2$ , $4c + 3d + 2e + f = 8$ ,  | $c = 2$ at most; |
|              | if $b = 1$ , $4c + 3d + 2e + f = 13$ , | $c = 3$ at most; |
|              | if $b = 0$ , $4c + 3d + 2e + f = 18$ , | $c = 4$ at most. |

Hence considering the several terms as arranged in alphabetical order and writing down only the factors in  $a, b, c$ , we obtain col. 1 of the following diagram;

| Col. 1.  | Col. 2.   | Col. 3. |
|----------|-----------|---------|
| $a^3$    | $(d^3)^9$ | 1       |
| $a^2b$   | $(d^3)^8$ | 1       |
| $a^2c$   | $(d^3)^7$ | 2       |
| $a^2$    | $(d^4)^6$ | 5       |
| $ab^2$   | $(d^3)^7$ | 2       |
| $abc$    | $(d^3)^6$ | 3       |
| $ab$     | $(d^4)^5$ | 4       |
| $ac^3$   | $(d^2)^6$ | 1       |
| $ac^2$   | $(d^3)^5$ | 3       |
| $ac$     | $(d^4)^4$ | 4       |
| $a$      | $(d^5)^3$ | 3       |
| $b^3$    | $(d^3)^6$ | 3       |
| $b^2c^2$ | $(d^2)^6$ | 1       |
| $b^2c$   | $(d^3)^5$ | 3       |
| $b^2$    | $(d^4)^4$ | 4       |
| $bc^3$   | $(d^2)^5$ | 1       |
| $bc^2$   | $(d^3)^4$ | 3       |
| $bc$     | $(d^4)^3$ | 3       |
| $b$      | $(d^5)^2$ | 2       |
| $c^4$    | $(d^2)^4$ | 2       |
| $c^3$    | $(d^3)^3$ | 3       |
| $c^2$    | $(d^4)^2$ | 2       |
| $c$      | $(d^5)^1$ | 1       |
| 1        | $(d^6)^0$ | 1       |
|          |           | 58      |

We then form col. 2 by annexing to each term a term  $(d^\theta)^\varphi$  which denotes the derivative  $\varphi$  of  $d^\theta$ ; the value of  $\theta$  being such that the whole term may be of the proper degree 6, and the value

of  $\varphi$  being such that the whole term may be of the proper weight 18. Thus

$$\begin{aligned} a^3(d^3)^9, \quad & \text{degree is } 3 + 3, = 6; \quad \text{weight is } 0 + 9 + 9, = 18, \\ a^2bc(d^3)^7, \quad & \text{„ } 1 + 2 + 3, = 6; \quad \text{„ „ } 0 + 2 + 9 + 7, = 18, \end{aligned}$$

viz.  $d^3$  being of weight 9, then  $(d^3)^9$  is of weight  $9 + 9$ , and  $(d^3)^7$  of weight  $9 + 7$ .

C. IV.

35

The derivatives of any power of  $d$  are given by those of the indefinite power  $d^m$ , which, omitting therein the several powers of  $d$ , may be tabulated thus. I remark that the table is carried up to the column 21 in order that it may be applicable to the calculation of the literal terms of the degree 15 and weight 45 which belong to the highest invariant of the sextic.

SUBSIDIARY TABLE OF THE DERIVATIVES OF  $d^m$ .

[Page 274 of the original carries an extensive triangular table giving, in 22 columns numbered 0, 1, 2, ... 21 and in 46 rows numbered 1, 2, ... 46, the derivative monomials in the letters ( $e, f, g$ ) which arise from successive derivations of  $d^m$ . The body of the table is read as follows: the column heading is the weight (or order of derivation), and each entry in the body is the monomial in  $e, f, g$  which is obtained by performing the corresponding derivation on  $d^m$  and stripping off the appropriate power of  $d$ . For instance the column 0 contains only 1 (the trivial derivative); col. 1 contains  $e$ ; col. 2 contains  $e^2$  and  $f$ ; col. 3 contains  $e^3, ef, g$ ; col. 4 contains  $e^4, e^2f, f^2, eg$ ; col. 5 contains  $e^5, e^3f, ef^2, e^2g, fg$ ; col. 6 contains  $e^6, e^4f, e^2f^2, f^3, e^3g, efg, g^2$ ; and so on, the higher columns giving the full enumeration of monomials of degree  $\leq$  the column index in the letters  $e, f, g$ . The reader is referred to the original page 274 for the complete tabular display.]



This means for instance that the derivatives of  $d^2$  are

| 0     | 1    | 2      | 3       | 4     | 5     | 6      |
|-------|------|--------|---------|-------|-------|--------|
| $d^2$ | $de$ | $d^2f$ | $df^2$  | $deg$ | $dfg$ | $dg^2$ |
|       |      | $d^2g$ | $deg$   |       |       |        |
|       |      | $def$  | $de^2g$ |       |       |        |

those of  $d^3$  are

| 0     | 1      | 2      | 3      | 4     | 5      | 6      | 7       | 8      | 9      |
|-------|--------|--------|--------|-------|--------|--------|---------|--------|--------|
| $d^3$ | $d^2e$ | $d^2f$ | $d^2g$ | $deg$ | $df^2$ | $dg^2$ | $deg^2$ | $f^2g$ | $e^2f$ |

and so on; viz. to form any column we affix to the terms of the corresponding column of the subsidiary table the proper power of  $d$ , so that the degree in all the letters may be 2, 3, &c. as the case may be, using from each column of the subsidiary table only the terms which are not of too high a degree; for instance for col. 4 of the derivatives of  $d^2$ , only the terms  $eg$ ,  $f^2$ ,  $e^2f$  of the complete column  $eg$ ,  $f^2$ ,  $e^2f$ ,  $e^4$ .

It is to be observed that these tables of the derivatives of  $d^2$ ,  $d^3$ , &c. do not need to be actually formed; any column which is wanted can be written down at once, *currente calamo*, from the column of the subsidiary table. And if we require only the number of terms, then these numbers are at once taken out from the Subsidiary Table. Thus for the numerical column of the diagram, the Subsidiary Table, col. 9, contains but 1 term  $g^3$  of degree 3, and thus the number set opposite to  $(d^3)^9$  is 1; so col. 8 contains but 1 term  $fg^2$  of the degree 3, and so the number opposite to  $(d^3)^8$  is 1; col. 7 contains 2 terms  $eg^2$ ,  $f^2g$  of degree 3, and so the number opposite to  $(d^3)^7$  is 2; and so on: we have in this way the numbers 1, 1, 2, 5, 2, 3, 4, &c. the partial sums of which are 1, 1, 7, 9, 11, 3, 8, 9, 9 giving the total sum 58: viz. this is the number of the terms degree 6, weight 18, which can be formed with the seven letters  $(a, b, c, d, e, f, g)$ .

When the literal terms are required, it is proper in the first instance as a safeguard against accidental omissions in copying, to take the numbers in this manner; and we can then take out the terms themselves, viz. these are

$$\begin{aligned} & a^3b d^3 f g^2, \quad a^3c d^2 e f g, \quad \dots, \\ & a^2b^2c d^2 e^2 f, \quad \dots, \\ & \dots \end{aligned}$$

and so on, the whole number being = 58 as just mentioned.

## 266.

ON THE EQUATION FOR THE PRODUCT OF THE DIFFERENCES  
OF ALL BUT ONE OF THE ROOTS OF A GIVEN EQUATION.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLI. for the year 1861, pp. 45–59. Received November 30, 1860,—Read January 10, 1861.]

IT IS easy to see that for an equation of the order  $n$ , the product of the differences of all but one of the roots will be determined by an equation of the order  $n$ , the coefficients of which are alternately rational functions of the coefficients of the original equation, and rational functions multiplied by the square root of the discriminant. In fact, if the equation be  $\varphi v = (a, \dots | v, 1)^n = a(v - \alpha)(v - \beta) \cdots$ , then putting for the moment  $a = 1$ , and disregarding numerical factors,  $\sqrt{D}$ , the square root of the discriminant, is equal to the product of the differences of the roots, and  $\varphi' \alpha$  is equal to  $(\alpha - \beta)(\alpha - \gamma) \cdots$ , consequently the product of the differences of the roots, all but  $\alpha$ , is equal to  $\sqrt{D} \div \varphi' \alpha$ , and the expression  $\frac{\sqrt{D}}{\varphi' \alpha}$  is the root of an equation of the order  $n$ , the coefficients of which are rational functions of the coefficients of the original equation. I propose in the present memoir to determine the equation in question for equations of the orders three, four, and five: the process employed is similar to that in my memoir “On the Equation of Differences for an Equation of any Order, and in particular for Equations of the Orders Two, Three, Four, and Five,” *Phil. Trans.*, vol. CL. (1860), [262], viz. the last coefficient of the given equation is put equal to zero, so that the given equation breaks up into  $v = 0$  and into an equation of the order  $n - 1$  called the reduced equation; and this being so, the required equation breaks up into an equation of the order  $n - 1$  (which however is not, as for the equation of differences, that which corresponds to the reduced equation) and into a linear equation; the equation of the order  $n - 1$  is calculated by the method

of symmetric functions; and combining it with the linear equation, which is known, we have the required equation, except as regards the terms involving the last coefficient, which terms are found by the consideration that the coefficients of the required equation are seminvariants. The solution leads immediately to that of a more general question; for if the product of the differences of all the roots except  $\alpha$ , of the given equation

$$\varphi v = (* | v, 1)^n = a(v - \alpha)(v - \beta) \cdots = 0$$

(which product is a function of the degree  $n - 2$  in regard to each of the roots  $\beta, \gamma, \delta, \dots$ ), is multiplied by  $(x - \alpha y)^{n-2}$ , the function so obtained will be the root of an equation of the order  $n$ , the coefficients of which are covariants of the quantic  $(* | x, y)^n$ , and these coefficients can be at once obtained by writing, in the place of the seminvariants of the former result, the covariants to which they respectively belong. In the case of the quintic equation, one of these covariants is, in regard to the coefficients, of the degree 6, which exceeds the limit of the tabulated covariants, [the covariants are all tabulated, 141 and 143], the covariant in question has therefore to be now first calculated. The covariant equations for the cubic and the quartic might be deduced from the formulæ Nos. 119 and 142 of my Fifth memoir on Quantics, *Phil. Trans.*, vol. CXLVIII. (1858), [156]; they are in fact the bases of the methods which are there given for the solution of the cubic and the quartic equations respectively; and it was in this way that I was led to consider the problem which is here treated of.

1. The notation  $\zeta^2(\alpha, \beta, \gamma, \dots)$  is used (after Professor Sylvester) to denote the product of the squared differences of  $(\alpha, \beta, \gamma, \dots)$ , and the notation  $\zeta^4(\alpha, \beta, \gamma, \dots)$  to denote the product of the differences taken in a determinate order, viz.

$$\begin{aligned} \zeta^4(\alpha, \beta, \gamma, \delta, \dots) &= (\alpha - \beta)(\alpha - \gamma)(\alpha - \delta) \cdots \\ &\quad (\beta - \gamma)(\beta - \delta) \cdots \\ &\quad (\gamma - \delta) \cdots \end{aligned}$$

2. The product of the differences of the roots of an equation depends, as already noticed, on the square root of the discriminant; and in order to fix the numerical factors and signs, it will be convenient, in regard to the equations

$$\begin{aligned} (a, b, c | v, 1)^2 &= 0, \\ (a, b, c, d | v, 1)^3 &= 0, \\ (a, b, c, d, e | v, 1)^4 &= 0, \\ (a, b, c, d, e, f | v, 1)^5 &= 0, \end{aligned}$$

to write as follows:

$$\begin{aligned}\zeta^4(\alpha, \beta) &= \frac{1}{a} \sqrt{-(4ac - b^2)} = \frac{1}{a} \sqrt{-\square}, \\ \zeta^4(\alpha, \beta, \gamma) &= \frac{1}{a^2} \sqrt{-(27a^2d^2 - \dots)} = \frac{1}{a^2} \sqrt{-\square}, \\ \zeta^4(\alpha, \beta, \gamma, \delta) &= -\frac{1}{a^3} \sqrt{256a^3e^3 - 27a^2d^4 + \dots} = -\frac{1}{a^3} \sqrt{\square}, \\ \zeta^4(\alpha, \beta, \gamma, \delta, \varepsilon) &= -\frac{1}{a^4} \sqrt{3125a^4f^4 + 256a^4e^5 + \dots} = -\frac{1}{a^4} \sqrt{\square},\end{aligned}$$

where it is to be observed, for example, that writing in the last equation  $e = 0$ , and therefore  $f = 0$ , we have  $\zeta^4(\alpha, \beta, \gamma, \delta, 0) = \frac{1}{a^4} \sqrt{256a^4e^5 + \dots}$ , which agrees with the equation  $\zeta^4(\alpha, \beta, \gamma, \delta, 0) = \alpha\beta\gamma\delta\zeta^4(\alpha, \beta, \gamma, \delta) = \frac{e}{a}\zeta^4(\alpha, \beta, \gamma, \delta)$ , if for  $\zeta^4(\alpha, \beta, \gamma, \delta)$  we substitute the value given by the last equation but one.

For the cubic equation  $(a, b, c, d \mid v, 1)^3 = 0$ ;

3. We have to find the equation for  $\theta = \zeta^2(\alpha, \beta) = (\alpha - \beta)^2$ ; the roots are

$$\theta_1 = \beta - \gamma, \quad \theta_2 = \gamma - \alpha, \quad \theta_3 = \alpha - \beta.$$

To apply the method above explained, write  $\gamma = 0$ , and therefore also  $d = 0$ ; the roots thus become

$$\theta_1 = \beta, \quad \theta_2 = -\alpha, \quad \theta_3 = \alpha - \beta,$$

and we have the quadric and linear equations

$$(\theta + \alpha)(\theta - \beta) = 0, \quad \theta - (\alpha - \beta) = 0,$$

where  $(\alpha, \beta)$  are the roots of the equation

$$(a, b, c \mid v, 1)^2 = 0.$$

Hence, writing

$$\square = 4ac - b^2,$$

we have

$$\alpha - \beta = -\sqrt{-\square} / a,$$

and the two equations become

$$\theta^2 - \theta[(\alpha + \beta) + (\alpha - \beta)] + (-\alpha\beta) - \alpha(\alpha - \beta) = 0, \quad \theta - (\alpha - \beta) = 0,$$

or multiplying the two equations together,

$$\theta^3 a^2 + \theta^2 \cdot 0 + \theta(3ac - b^2) + c\sqrt{-\square} = 0,$$

THE COLLECTED MATHEMATICAL PAPERS  
of  
ARTHUR CAYLEY

Volume IV — pages 279–303  
Paper 266 (conclusion), Paper 267

which is what the required equation becomes, on putting therein  $d = 0$ ; the coefficients of the complete equation are seminvariants, and the terms in  $d$  are to be inserted by means of this property. The coefficient  $3ac - b^2$  is reduced to zero by the operator

$$3a \partial_b + 2b \partial_c + c \partial_d;$$

it is therefore a seminvariant, and remains unaltered. The coefficient  $c\sqrt{-\square}$  is what  $\sqrt{-\square}$  becomes ( $\square$  being the discriminant of the cubic equation) on putting therein  $d = 0$ , it is therefore to be changed into  $\sqrt{-\square}$ . Hence

4. For the cubic equation  $(a, b, c, d \mid v, 1)^3$  the equation for  $\theta (= \zeta^2(\alpha, \beta))$  is,  $0 =$

$$(a^2, 0, 3ac - b^2, \sqrt{-\square})(\theta, 1)^3, \quad \text{where columns are headed } a^2X, \sqrt{-\square}X.$$

5. For the quartic equation  $(a, b, c, d, e \mid v, 1)^4 = 0$ ;

$$\theta = \zeta^2(\alpha, \beta, \gamma) = -(\alpha - \beta)(\alpha - \gamma)(\beta - \gamma).$$

The roots are

$$\theta_1 = \zeta^4(\beta, \gamma, \delta),$$

$$\theta_2 = -\zeta^4(\gamma, \delta, \alpha),$$

$$\theta_3 = \zeta^4(\delta, \alpha, \beta),$$

$$\theta_4 = -\zeta^4(\alpha, \beta, \gamma),$$

the signs being in this case (and indeed for an equation of any even order) alternately positive and negative; in fact, if the equation is represented by  $\varphi v = 0$ , then the roots divided by  $\zeta^2(\alpha, \beta, \gamma, \delta)$  should respectively be  $\varphi'\alpha$ ,  $\varphi'\beta$ ,  $\varphi'\gamma$ ,  $\varphi'\delta$ , and this will be the case if the signs are taken as above.

6. Putting now  $\delta = 0$  (and therefore  $e = 0$ ) the roots become

$$\theta_1 = \beta\gamma(\beta - \gamma),$$

$$\theta_2 = \gamma\alpha(\gamma - \alpha),$$

$$\theta_3 = \zeta^4(\delta, \alpha, \beta) \text{ (see text),}$$

$$\theta_4 = -\zeta^4(\alpha, \beta, \gamma),$$

where  $(\alpha, \beta, \gamma)$  are the roots of  $(a, b, c, d \mid v, 1)^3 = 0$ . Let  $\square$  denote the discriminant of the cubic function, then  $\zeta^4(\alpha, \beta, \gamma) = -\frac{1}{a^2}\sqrt{-\square}$ , and we have thus the linear equation; the cubic equation is

$$\Pi_1(\theta - \beta\gamma(\beta - \gamma)) = 0,$$

the coefficients of which can be calculated by the method of symmetric functions (see Annex No. 1).

7. The cubic equation being thus obtained, we have the two equations

$$\begin{aligned}\theta^3 a^2 + \theta^2 \cdot 0 + \theta(3ac - b^2) + c\sqrt{-\square} &= 0, \\ \theta^3 a^2 - \theta^2(3ac - b^2)d\sqrt{-\square}/\square + \dots &= 0,\end{aligned}$$

[the second factor being the cubic equation derived in Annex No. 1]; and multiplying these together, the resultant equation is, schematically,

$$\theta^6 \cdot a^4 + \theta^5 \cdot 0 + \theta^4 a^2(-9a^2d^2 + 4abcd - b^2d + \square) + \theta^3 \cdot a^2(-8a^2d + 4abc - b^3)d\sqrt{-\square} + \dots = 0,$$

where the coefficients have to be completed by adding the terms which contain  $e$ .

We have  $\sqrt{-\square}$  in the place of  $d\sqrt{-\square}$ , and  $\square$  in the place of  $-d^2\square$ . The coefficient  $-8a^2d + 4abc - b^3$  is a seminvariant, and requires no alteration. The coefficient

$$-9a^2d^2 + 4abcd - b^3d + \square$$

that is,

$$\begin{aligned}&-9a^2d^2 + 4abcd - b^3d \\ &+ 27a^2d^2 - 18abcd + 4ac^3 + 4b^3d - b^2c^2;\end{aligned}$$

collects (with the contributions from  $\square$ ) to the following table of monomials and integer coefficients:

|          |     |
|----------|-----|
| $a^2ce$  | -16 |
| $a^2d^2$ | +18 |
| $ab^2e$  | +6  |
| $abcd$   | -14 |
| $ac^3$   | +4  |
| $b^3d$   | +3  |
| $b^2c^2$ | -1  |

and the terms in  $e$  to be added to this, in order to make it a seminvariant, are easily found as displayed.

8. Hence, for the quartic equation  $(a, b, c, d, e \mid v, 1)^4$ , the equation for  $\theta (= \zeta^2(\alpha, \beta, \gamma))$  is  $0 =$

$$(a^6, e^2\sqrt{-\square}X, a^4X, a^2\sqrt{-\square}X, \square X)(\theta, 1)^4$$

where, expanding column by column,

|                                |                                                                                                                       |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| column $a^6$ :                 | +1,                                                                                                                   |
| column $\sqrt{-\square}X$ :    | 0,                                                                                                                    |
| column $a^4X$ :                | $a^2ce - 16, \quad a^2d^2 + 18, \quad ab^2e + 6,$<br>$abcd + 144, \quad ac^3 - 184, \quad b^4 - 7, \quad b^2c^2 + 3,$ |
| column $a^2\sqrt{-\square}X$ : | (the seminvariant $-8a^2d + 4abc - b^3$ ),                                                                            |
| column $\square X$ :           | +1 · $\square$ ,                                                                                                      |

the whole giving the desired equation in  $\theta$  for the quintic equation

$$(a, b, c, d, e, f \mid v, 1)^5 = 0;$$

266]

OF THE DIFFERENCES OF THE ROOTS OF A GIVEN EQUATION.

281

9. We have  $\theta = \zeta^2(\alpha, \beta, \gamma, \delta)$ , the roots being

$$\begin{aligned}\theta_1 &= \zeta^4(\beta, \gamma, \delta, \varepsilon), & \text{which for } \varepsilon = 0 \text{ becomes } \beta\gamma\delta\zeta^4(\beta, \gamma, \delta), \\ \theta_2 &= \zeta^4(\gamma, \delta, \varepsilon, \alpha), & -\gamma\delta\alpha\zeta^4(\gamma, \delta, \alpha), \\ \theta_3 &= \zeta^4(\delta, \varepsilon, \alpha, \beta), & \delta\alpha\beta\zeta^4(\delta, \alpha, \beta), \\ \theta_4 &= \zeta^4(\varepsilon, \alpha, \beta, \gamma), & -\alpha\beta\gamma\zeta^4(\alpha, \beta, \gamma), \\ \theta_5 &= \zeta^4(\alpha, \beta, \gamma, \delta), & \zeta^4(\alpha, \beta, \gamma, \delta).\end{aligned}$$

10. The linear equation is  $\theta a^4 + e\sqrt{\square} = 0$ ; the quartic equation may be written  $\Pi_1(\theta - \theta_i) = 0$ , for the determination of which see Annex No. 2. The two equations are

$$\begin{aligned}\theta^4 \cdot a^6 + \theta^3 \cdot 0 + \theta^2 \cdot a^2(Me - \square) + \theta \cdot a^2 Ne\sqrt{\square} + e^2\square &= 0, \\ \theta \cdot a^4 + e\sqrt{\square} &= 0;\end{aligned}$$

and multiplying these together, the resulting equation is

$$\theta^5 \cdot a^{10} + \theta^4 \cdot 0 + \theta^3 \cdot a^6 (Me - \square) + \theta^2 \cdot a^4 (Ne + M) e\sqrt{\square} + \theta \cdot a^2 (N + a^2 e) e\sqrt{\square} \cdot (\cdot) + e^3 \square \sqrt{\square} = 0,$$

where the coefficients have to be completed by the addition of the terms in  $f$ . We have  $\sqrt{\square}$  in the place of  $e\sqrt{\square}$ , and therefore  $\square$  in the place of  $e^2\square$ .

11. The value of  $Me - \square$  is

$$\begin{aligned}a^3e^3 &-160 \\ a^3bde &+132 \\ a^2ce^2 &+88 \\ a^2cd^2e &-177 \\ a^2d^4 &+27 \\ ab^2ce^2 &-97 \\ ab^2d^2e &+6 \\ abc^2de &+62 \\ abcd^3 &-18 \\ ac^4e &-12 \\ ac^3d^2 &+4 \\ b^4e^2 &+18 \\ b^3cde &-14 \\ b^3d^3 &+4 \\ b^2c^2e &+3 \\ b^2cd^2 &-1\end{aligned}$$

and the terms in  $f$  are found to be

$$\begin{aligned}a^3def &+300 \\ a^3bcef &-130 \\ a^3bd^2f &-120 \\ a^2c^2df &+40 \\ ab^2ef &+28 \\ ab^2cdf &+66 \\ abc^3f &-24 \\ b^4df &-16 \\ b^2c^2f &+6 \\ a^3cf^2 &-125 \\ a^2b^2f^2 &+50\end{aligned}$$



12. The value of  $Ne + M$  is

$$\begin{array}{rcl}
 a^3e^3 & + & 80 \\
 a^2bde & - & 54 \\
 a^2ce^2 & - & 36 \\
 a^2cd^2 & + & 27 \\
 ab^2ce & + & 42 \\
 abc^2d & - & 18 \\
 ac^4 & + & 4 \\
 b^4e & - & 8 \\
 b^3cd & + & 4 \\
 b^2c^3 & - & 1
 \end{array}$$

and the terms in  $f$  to be added thereto are found to be

$$\begin{array}{rcl}
 a^3df & + & 50 \\
 a^2bcf & + & 30 \\
 ab^3f & - & 8
 \end{array}$$

13. The value of  $N + a^2e$  is

$$\begin{array}{rcl}
 a^3e & - & 15 \\
 a^2bd & + & 6 \\
 a^2c^2 & + & 4 \\
 ab^2c & - & 5 \\
 b^4 & + & 1
 \end{array}$$

which is a seminvariant, and requires no addition.

14. Hence, for the quintic equation

$$(a, b, c, d, e, f \mid v, 1)^5 = 0,$$

the equation for  $\theta (= \zeta^2(\alpha, \beta, \gamma, \delta))$  is,  $0 =$

$$(a^{12}X, 0, a^6X, a^3\sqrt{\square}X, \square X, \square\sqrt{\square}X)(\theta, 1)^5,$$

with column entries

| $a^6X$      |      | $a^3\sqrt{\square}X$ |     | $\square X$ |     |
|-------------|------|----------------------|-----|-------------|-----|
| $a^3cf^2$   | -125 | $a^3df$              | -50 | $a^3e$      | -15 |
| $a^3def$    | +300 | $a^3e^2$             | +80 | $a^2bd$     | +6  |
| $a^3e^3$    | -160 | $a^2bcf$             | +30 | $a^2c^2$    | +4  |
| $a^2b^2f^2$ | +50  | $a^2bde$             | -54 | $ab^2c$     | -5  |
| $a^2bde^2$  | +132 | $a^2ce^2$            | -36 | $b^4$       | +1  |
| $a^2bcef$   | -130 | $a^2cd^2$            | +27 |             |     |
| $a^2bd^2f$  | -120 | $ab^3f$              | -8  |             |     |
| $a^2c^2df$  | +40  | $ab^2ce$             | +42 |             |     |
| $a^2c^2e^2$ | +88  | $abc^2d$             | -18 |             |     |
| $a^2cd^3$   | -117 | $ac^4$               | +4  |             |     |
| $a^2d^4$    | +27  | $b^4e$               | -8  |             |     |
| $ab^3ef$    | +28  | $b^3cd$              | +4  |             |     |
| $ab^2ce^2$  | -97  | $b^2c^3$             | -1  |             |     |
| $ab^2d^2e$  | +6   |                      |     |             |     |
| $abc^2de$   | +62  |                      |     |             |     |
| $ab^2cdf$   | +66  |                      |     |             |     |
| $abcd^3$    | -18  |                      |     |             |     |
| $abc^3f$    | -24  |                      |     |             |     |
| $ac^4e$     | -12  |                      |     |             |     |
| $ac^3d^2$   | +4   |                      |     |             |     |
| $b^4df$     | -16  |                      |     |             |     |
| $b^4e^2$    | +18  |                      |     |             |     |
| $b^3cde$    | -14  |                      |     |             |     |
| $b^3d^3$    | +4   |                      |     |             |     |
| $b^2c^2f$   | +6   |                      |     |             |     |
| $b^2c^3e$   | +3   |                      |     |             |     |
| $b^2cd^2$   | -1   |                      |     |             |     |

and the columns headed  $a^{12}X$ ,  $0$ , and  $\square\sqrt{\square}X$  have entries  $+1$ ,  $0$ , and  $+1$ , respectively.

**266]**

OF THE DIFFERENCES OF THE ROOTS OF A GIVEN EQUATION.

**283**

15. As a verification of this result, I remark that, taking for the quintic equation  $v^5 + v^4 + v^3 + v^2 + v + 1 = 0$ , the roots of this equation are  $-1, \omega, \omega^2, -\omega, -\omega^2$ , where  $\omega$  is an imaginary cube root of unity ( $\omega^2 + \omega + 1 = 0$ ). We ought to have  $\zeta^4(\alpha, \beta, \gamma, \delta, \varepsilon) = -\sqrt{\square} = -36$ ; and this will be the case if, for instance,  $\alpha, \beta, \gamma, \delta, \varepsilon$  are respectively  $-1, \omega, \omega^2, -\omega^2, -\omega$ . We have then

$$\zeta^4(\alpha, \beta, \gamma, \delta) = -1 - \omega \cdot -1 - \omega^2 \cdot -1 + \omega^2 \cdot \omega - \omega^2 \cdot \omega + \omega^2 \cdot 2\omega^2 = 6,$$

$$\zeta^4(\beta, \gamma, \delta, \varepsilon) = \omega - \omega^2 \cdot \omega + \omega^2 \cdot 2\omega \cdot 2\omega^2 \cdot \omega^2 + \omega \cdot -\omega^2 + \omega = -12,$$

$$\zeta^4(\gamma, \delta, \varepsilon, \alpha) = 2\omega^2 \cdot \omega + \omega \cdot \omega^2 + 1 \cdot -\omega^2 - \omega \cdot -\omega^2 + 1 \cdot -\omega + 1 = +6(\omega - \omega^2),$$

$$\zeta^4(\delta, \varepsilon, \alpha, \beta) = -\omega^2 + \omega \cdot -\omega^2 + 1 \cdot -\omega^2 - \omega \cdot -\omega + 1 \cdot -2\omega \cdot -1 - \omega = -6(\omega - \omega^2),$$

$$\zeta^4(\varepsilon, \alpha, \beta, \gamma) = -\omega + 1 \cdot -2\omega \cdot -\omega - \omega^2 \cdot -1 - \omega \cdot -1 - \omega \cdot \omega - \omega^2 = 6.$$

The equation in  $\theta$  is thus  $(\theta - 6)^2(\theta + 12)(\theta^2 + 108) = 0$ , or multiplying out it is

$$(1, 0, 0, +432, -11664, +46656 \mid \theta, 1)^5 = 0,$$

which in fact (observing that  $\sqrt{\square} = 36$ ) is what the preceding formula becomes for the equation  $(1, 1, 1, 1, 1, 1 \mid v, 1)^5 = 0$ .

The analogous verifications for the cubic and the quartic equations are as follows:

16. For the cubic, if the assumed equation is  $v^3 + v^2 + v + 1 = 0$ , the roots whereof are  $-1, i, -i$  ( $i^2 = -1$ ), then we should have  $\zeta^4(\alpha, \beta, \gamma) = \sqrt{-\square} = 4i$ , which will be the case if  $\alpha, \beta, \gamma = -1, i, -i$  respectively, and the roots  $\beta - \gamma, \gamma - \alpha, \alpha - \beta$  of the equation in  $\theta$  then are  $2i, -i + 1, -i - 1$ , so that the equation in  $\theta$  is  $(\theta^2 + 2i\theta - 2)(\theta - 2i) = 0$ , or

$$(1, 0, 2, 4i \mid \theta, 1)^3 = 0,$$

which (observing that  $\sqrt{-\square} = 4i$ ) is what the formula for the equation in  $\theta$  becomes for the equation  $(1, 1, 1, 1 \mid v, 1)^3 = 0$ .

17. For the quartic equation, taking this to be  $v^4 + v^3 + v^2 + v + 1 = 0$ , the roots are  $\omega, \omega^2, \omega^3, \omega^4$ , where  $\omega$  is an imaginary fifth root of unity ( $\omega^4 + \omega^3 + \omega^2 + \omega + 1 = 0$ ), and putting  $\alpha, \beta, \gamma, \delta$  equal to  $\omega, \omega^2, \omega^3, \omega^4$  respectively, we have

$$-\sqrt{\square} = \zeta^4(\alpha, \beta, \gamma, \delta) = -5(\omega + \omega^4 - \omega^2 - \omega^3),$$

giving, as it should do,  $\square = 125$ . The equation in  $\theta$  is therefore by the formula

$$(1, 0, 0, -25(\omega + \omega^4 - \omega^2 - \omega^3), 125 \mid \theta, 1)^4 = 0.$$

But the roots are

$$\begin{aligned}\theta_1 &= \zeta^4(\beta, \gamma, \delta) = -\omega^2 - \omega^3 \cdot \omega^2 - \omega^4 \cdot \omega^3 - \omega^4 = 2 - \omega + \omega^2 - 2\omega^3 = 2 - X, \\ \theta_2 &= -\zeta^4(\gamma, \delta, \alpha) = -\omega^3 - \omega^4 \cdot \omega^3 - \omega \cdot \omega^4 - \omega = -1 + 3\omega + 2\omega^2 + \omega^3 = -1 + Y, \\ \theta_3 &= \zeta^4(\delta, \alpha, \beta) = \omega^4 - \omega \cdot \omega^4 - \omega^2 \cdot \omega - \omega^2 = -4 - 3\omega - 2\omega^2 - \omega^3 = -4 - Y, \\ \theta_4 &= -\zeta^4(\alpha, \beta, \gamma) = -\omega - \omega^2 \cdot \omega - \omega^3 \cdot \omega^2 - \omega^3 = 3 + \omega - \omega^2 + 2\omega^3 = 3 + X,\end{aligned}$$

if, for shortness,

$$X = \omega - \omega^2 + 2\omega^3, \quad Y = 3\omega + 2\omega^2 + \omega^3.$$

The equation in  $\theta$  is therefore

$$(\theta - 2 + X)(\theta + 1 - Y)(\theta + 4 + Y)(\theta - 3 - X) = 0,$$

where the left-hand side is the product of the factors, and

$$\begin{aligned}(\theta - 2 + X)(\theta - 3 - X) &= \theta^2 - 5\theta + 6 - X - X^2 = \theta^2 - 5\theta + 10 - 5(\omega + \omega^4), \\ (\theta + 1 - Y)(\theta + 4 + Y) &= \theta^2 + 5\theta + 4 - 3Y - Y^2 = \theta^2 + 5\theta + 10 - 5(\omega^2 + \omega^3),\end{aligned}$$

and the equation in  $\theta$  is, therefore, as it should be,

$$(1, 0, 0, -25(\omega + \omega^4 - \omega^2 - \omega^3), 125 \mid \theta, 1)^4 = 0.$$

Passing from the denumerate to the standard forms:

266]

OF THE DIFFERENCES OF THE ROOTS OF A GIVEN EQUATION.

285

18. For the cubic equation  $(a, b, c, d \mid v, 1)^3 = 0$ , the equation for  $\theta (= \zeta^2(\alpha, \beta))$  is  $0 =$

$$(a^2X, 0, 9HX, \sqrt{-27\Box}X)(\theta, 1)^3,$$

with column entries

$$\begin{array}{ll} a^4X : & +1, \\ \text{column 2} : & 0, \\ 9HX : & ba^2c + 1, \\ \sqrt{-27\Box}X : & +1. \end{array}$$

19. For the quartic equation  $(a, b, c, d, e \mid v, 1)^4 = 0$ , the equation for  $\theta (= \zeta^2(\alpha, \beta, \gamma))$  is  $0 =$

$$(a^6X, 0, 96a^2X, 512\sqrt{\Box}X, 256\Box X)(\theta, 1)^4,$$

with the entries (collected column by column) including

$$\begin{array}{ll} 96a^4HX \text{ column} & \\ a^2ce & -1 \\ abc & +3 \\ b^2 & -2 \\ 512\sqrt{\Box}X \text{ column} & \\ a^2d^2 & +3 \\ ab^2e & +1 \\ abcd & +149 \\ b^4d & +8 \\ b^2c^2 & -6 \end{array}$$

and a final column  $+1 \cdot \Box$  in  $256\Box X$ .

20. For the quintic equation  $(a, b, c, d, e, f \mid v, 1)^5 = 0$ , the equation for  $\theta (= \zeta^2(\alpha, \beta, \gamma, \delta))$  is,  $0 =$

$$(a^{12}X, 0, 625a^6X, 12500\sqrt{\Box}a^3X, 15625\Box X, 76125\Box\sqrt{\Box})(\theta, 1)^5,$$

*[The long coefficient table printed below this display is not yet fully collated in this slice; the coefficient header above has been repaired from the source scan.]*

21. I remark, with respect to the equation in  $\theta$ , for the cubic, that it leads at once to the equation of differences. In fact we have

$$a^2\theta^3 + 9(ac - b^2)\theta + \sqrt{-27\Box} = \Pi_1\{\theta - (\alpha - \beta)\},$$

whence changing the sign of  $d$ ,

$$a^2\theta^3 + 9(ac - b^2)\theta - \sqrt{-27\Box} = \Pi_1\{\theta + (\alpha - \beta)\};$$

or multiplying the two equations and putting  $u$  for  $\theta^2$ ,

$$u\{a^2u + 9(ac - b^2)\}^2 + 27\Box = \Pi_1\{u - (\alpha - \beta)^2\},$$

that is, the equation of differences is

$$a^4u^3 + 18(ac - b^2)a^2u^2 + 81(ac - b^2)^2u + 27\Box = 0;$$

but this mode of composition is peculiar to the case of the cubic.

If in the several equations in  $\theta$  we substitute for the seminvariants the covariants to which they respectively belong, we obtain as follows:

22. For the cubic equation  $(a, b, c, d \mid x, y)^3 = 0$ , the equation for

$$\theta [= (\beta - \gamma)^2(x - \alpha y)^2]$$

is,  $0 =$

$$(U^3, 0, 9H, \sqrt{-27\Box}X)(\theta, 1)^3 = 0.$$

23. For the quartic equation  $(a, b, c, d, e \mid x, y)^4 = 0$ , the equation for

$$\theta [= (\beta - \gamma)(\gamma - \delta)(\delta - \beta)(x - \alpha y)^2]$$

is

$$(U^6, 0, 96U^3(-3JU + 21H), 512\sqrt{\Box}(-\Phi), 256\Box)(\theta, 1)^4 = 0.$$

24. And for the quintic equation  $(a, b, c, d, e, f \mid x, y)^5 = 0$ , denoting the covariants of the quintic as in 141,  $A$  the quintic itself, &c., and completing the expression for the coefficient of  $\vartheta^3$ , the equation for

$$\vartheta [= (\beta - \gamma)(\beta - \delta)(\beta - \varepsilon)(\gamma - \delta)(\gamma - \varepsilon)(\delta - \varepsilon)(x - \alpha y)^3]$$

is

$$\left\{ \begin{array}{l} A^{12}, \\ 0, \\ 625 A^6(48AJ - 80B^3 + 120BH - 50CG), \\ 12500\sqrt{\Box} A^3(4AB^3 - AH - 50CD), \\ 15625\Box(3A^2B + 25C^2), \\ 76125\Box\sqrt{\Box} \end{array} \right\} (\vartheta, 1)^5 = 0,$$

where the covariant which enters into the coefficient of  $\vartheta^3$ , being of the sixth degree in the coefficients, is not given in the Tables.

Its value (completed for me, from the first term, by Mr Davis) is

[a large table of monomial covariants of degree 6, occupying the body of

printed page 287, with monomials such as

|             |         |
|-------------|---------|
| $a^4c^4ef$  | -2      |
| $a^2c^4d^2$ | +18     |
| $a^2b^3c^4$ | -30     |
| $a^3b^4e^2$ | +3100   |
| $a^3b^4e^3$ | -30     |
| $a^4b^4c$   | +18     |
| $a^5c^3d$   | +2      |
| $a^4b^4f$   | -32     |
| $a^2b^2d^2$ | +2      |
| $a^4bcd^2$  | +624    |
| $a^2bcde^2$ | -540    |
| $ab^3cd^2e$ | +1196   |
| $abc^2d^3$  | -840    |
| $ab^3cd^3$  | +1080   |
| $ab^4cd^3$  | +672240 |

and so on,

...

[the full table being too dense to set in line;

the reader is referred to the original printed page 287].

[viz. this is  $= 48 AJ - 80 HI + 120 BI - 50 CO$  as above].

In the following two Annexes, the notation of the symmetric functions is the same as in my "Memoir on the Symmetric Functions of the Roots of an Equation," *Phil. Trans.*, vol. CXLVII. (1857), [147] and the values of the symmetric functions are taken from that memoir, the powers of  $a$  being restored by the principle of homogeneity. The suffixes of the  $\Sigma$  indicate the number of terms in the sum; thus in the first Annex

$$\Sigma_3 \beta \gamma (\beta - \gamma) (\gamma - \alpha) = \Sigma_3 (\beta \gamma^2 - \alpha \beta \gamma - \gamma^3 + \alpha \gamma^2);$$

the terms  $\Sigma_3 (\beta \gamma + \alpha \beta)$  are equal to  $\Sigma_3 \Sigma \beta$ , the complete symmetric function; the correct result will be obtained (though of course neither of these equations is true) by writing  $\Sigma_3 \beta \gamma = \frac{1}{2} \Sigma_3 \alpha^2 \beta$ ,  $\Sigma_3 \alpha \gamma^2 = \frac{1}{2} \Sigma_3 \alpha^2 \beta$ , and so in similar cases; the insertion of the suffix to the  $\Sigma$  very much facilitates the calculation, and is a check on its accuracy.

Annex No. 1, containing the calculation of the equation  $\Pi_1(\theta - \theta_i) = 0$ , where

$$\theta_1 = \beta \gamma (\beta - \gamma), \quad \theta_2 = \gamma \alpha (\gamma - \alpha), \quad \theta_3 = \alpha \beta (\alpha - \beta),$$

$\alpha, \beta, \gamma$  being the roots of the cubic equation  $(a, b, c, d \mid v, 1)^3 = 0$ .

We have

$$\Sigma \theta_i = \Sigma \beta \gamma (\beta - \gamma) = -(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha) = \zeta^4(\alpha, \beta, \gamma) = a^{-2} \sqrt{-\square},$$

where  $\square = 27a^2d^2 + \&c.$  is the discriminant of the cubic.

$$\Sigma\theta_1\theta_2 = \Sigma_3 \beta\gamma(\beta - \gamma) \cdot \gamma\alpha(\gamma - \alpha) = \alpha\beta\gamma \Sigma_3 \gamma(\beta - \gamma)(\gamma - \alpha),$$

where  $\alpha\beta\gamma = -a^{-1}d$  and

$$\begin{aligned} \Sigma_3 \gamma(\beta - \gamma)(\gamma - \alpha) &= \Sigma_3 (\beta\gamma^2 - \alpha\beta\gamma - \gamma^3 + \alpha\gamma^2) \\ &= 3\Sigma_3 \alpha^2\beta - 3(\alpha\beta\gamma) - \Sigma_3 \alpha^3 \\ &= 3(21) - 3(13) - (3), \end{aligned}$$

which, expressed in terms of the power-sums, gives

$$\Sigma_3 \gamma(\beta - \gamma)(\gamma - \alpha) = a^{-3} (9a^2d - 4abc + b^3) \text{ (after reduction),}$$

so that

$$\Sigma\theta_1\theta_2 = -a^{-4} (9a^2d^2 - 4abcd + b^3d).$$

And lastly,

$$\begin{aligned} \theta_1\theta_2\theta_3 &= \alpha^2\beta^2\gamma^2(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha) \\ &= -\alpha^2\beta^2\gamma^2\zeta^4(\alpha, \beta, \gamma) \\ &= -a^{-4}d^2\sqrt{-\square}, \end{aligned}$$

so that the equation is the one given above, No. 7.

Annex No. 2, containing the calculation of the equation  $\Pi_1(\theta - \theta_i) = 0$ , where

$$\theta_1 = \beta\gamma\delta\zeta^4(\beta, \gamma, \delta), \quad \theta_2 = -\gamma\delta\alpha\zeta^4(\gamma, \delta, \alpha), \quad \theta_3 = \delta\alpha\beta\zeta^4(\delta, \alpha, \beta), \quad \theta_4 = -\alpha\beta\gamma\zeta^4(\alpha, \beta, \gamma),$$

$\alpha, \beta, \gamma, \delta$  being the roots of the quartic equation  $(a, b, c, d, e \mid v, 1)^4 = 0$ .

$$\begin{aligned} \Sigma\theta_i &= (\alpha\beta\gamma\delta) \Sigma_4 \beta\gamma\delta\zeta^4(\beta, \gamma, \delta) \\ &= -(\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta) \\ &= -\zeta^4(\alpha, \beta, \gamma, \delta), \end{aligned}$$

where  $\square = 256a^3e^3 + \&c.$  is the discriminant of the quartic.

$$\begin{aligned} \Sigma\theta_1\theta_2 &= \Sigma_4 \theta_i\theta_j = \Sigma_4 \beta\gamma\alpha\zeta^4(\delta, \alpha, \beta) \cdot (-\alpha\beta\gamma)\zeta^4(\alpha, \beta, \gamma) \\ &= \Sigma_4 \beta\gamma(\delta - \alpha)(\delta - \beta)(\alpha - \beta) \cdot -\alpha\beta\gamma(\alpha - \beta)(\alpha - \gamma)(\beta - \gamma) \\ &= -\alpha\beta\gamma\delta \Sigma_4 \alpha\beta(\alpha - \beta)^2(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta), \end{aligned}$$

where  $\alpha\beta\gamma\delta = -a^{-1}e$ , and\*

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\* The signs of  $\theta_1, \theta_2, \theta_3, \theta_4$  are taken account of implicitly.



266]

OF THE DIFFERENCES OF THE ROOTS OF A GIVEN EQUATION.

289

(continuing)

$$= \Sigma_4 \alpha \beta (\alpha - \beta)^2 \{ \alpha^2 \beta^2 - \alpha \beta (\alpha + \beta)(\gamma + \delta) + (\alpha^2 + \beta^2) \gamma \delta \\ + \alpha \beta (\gamma + \delta)^2 - \gamma \delta (\alpha + \beta)(\gamma + \delta) + \gamma^2 \delta^2 \}$$

viz., expanding term by term,

$$\begin{aligned} & \Sigma_4 \alpha^3 \beta^3 (\alpha - \beta)^2, \\ & - \Sigma_4 \alpha^2 \beta^2 (\alpha - \beta)^2 (\alpha + \beta) (\gamma + \delta), \\ & + \alpha \beta \gamma \delta \Sigma_4 (\alpha - \beta)^2 (\alpha^2 + \beta^2), \\ & + \Sigma_4 \alpha^2 \beta^2 (\alpha - \beta)^2 (\gamma + \delta)^2, \\ & - \alpha \beta \gamma \delta \Sigma_4 (\alpha - \beta)^2 (\alpha + \beta)(\gamma + \delta), \\ & + \alpha^2 \beta^2 \gamma^2 \delta^2 \Sigma_4 (\alpha - \beta)^2, \end{aligned}$$

which, expressed in the notation of the memoir [147] (the suffixes indicating term-counts), reduce respectively to the elementary symmetric functions

$$\begin{aligned} & \Sigma_2 \alpha^4 \beta^4 \quad (15), \\ & - \Sigma_3 \alpha^3 \beta^3 \gamma \quad (53), \\ & + \Sigma_4 \alpha^2 \beta^2 \gamma \quad (431), \\ & + \Sigma_5 \alpha^2 \beta^2 \gamma^2 \delta \quad (322), \\ & - \Sigma_6 \alpha^2 \beta \gamma \delta \quad (3211), \\ & + \Sigma_7 \alpha \beta \gamma \delta \quad (21^3), \end{aligned}$$

where for a moment  $a$  is put equal to unity. The value of the last-mentioned expression is then calculated as follows:

[A long column of intermediate sums is set down in the original; the principal contributing monomials, with their integer multipliers, are

$$\begin{array}{ll} bde & -4 - 12, \\ b^4e & +1 + 16, \\ b^2ce & +8 + 8, \\ b^2c^2e^2 & +73 - 8, \\ cb^4ce^2 & +73, \\ b^2c^2de & +311, \\ b^3c^2e^2d & +48, \\ \&c., & \&c.; \end{array}$$

after collecting, the final result equals

$$\Sigma_4 \alpha \beta (\alpha - \beta)^2 (\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta) = a^{-3} M e,$$

say.]

and restoring the powers of  $a$  by the principle of homogeneity, and putting

$$\begin{array}{rcl} a^3 e^2 & + & 4096, \\ a^2 ced & + & 27, \\ ab^2 ed^2 & + & 4670, \\ ac^4 & + & 18, \\ b^2 ce^2 & - & 9, \\ bcd^2 e & - & 4, \\ b^2 c^2 d & + & 1, \end{array}$$

we have

$$\Sigma \theta_i \theta_j = a^{-3} M e.$$

Next,

$$\Sigma \theta_1 \theta_2 \theta_3 = \Sigma \beta \gamma \delta \zeta^4(\beta, \gamma, \delta) \cdot -\gamma \delta \alpha \zeta^4(\gamma, \delta, \alpha) \cdot \delta \alpha \beta \zeta^4(\delta, \alpha, \beta) \quad [\text{four terms in all}],$$

giving, after collecting,

$$\begin{aligned} &= \alpha^2 \beta^2 \gamma^2 \delta^2 (\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta) \Sigma_4 \alpha (\alpha - \beta)(\alpha - \gamma)(\alpha - \delta) \\ &= -a^{-6} e^2 \sqrt{\square} \Sigma_4 \alpha (\alpha - \beta)(\alpha - \gamma)(\alpha - \delta), \end{aligned}$$

and observing that

$$\begin{aligned} (a, b, c, d, e \mid v, 1)^4 &= a(v - \alpha)(v - \beta)(v - \gamma)(v - \delta), \\ 4av^3 + 3bv^2 + 2cv + d &= a(v - \beta)(v - \gamma)(v - \delta) + \&c., \end{aligned}$$

which, putting  $v = \alpha$ , gives

$$4a\alpha^3 + 3b\alpha^2 + 2c\alpha + d = a(\alpha - \beta)(\alpha - \gamma)(\alpha - \delta),$$

we have

$$\begin{aligned} \Sigma_4 \alpha (\alpha - \beta)(\alpha - \gamma)(\alpha - \delta) &= \frac{1}{a} (4a \Sigma \alpha^4 + 3b \Sigma \alpha^3 + 2c \Sigma \alpha^2 + d \Sigma \alpha) \\ &= 4(4) + 3b(3) + 2c(2) + d(1), \end{aligned}$$

where for a moment  $a$  is put equal to 1. This is calculated by the array

$$\begin{array}{rclcl} e & -16 & & & -16, \\ bd & +16 & -9 & -1 & +6, \\ c^2 & +8 & & -4 & +4, \\ b^2 c & -16 & +9 & +2 & -5, \\ b^4 & +4 & -3 & & +1, \end{array}$$

giving the collected terms displayed on the following page.

**266]**

OF THE DIFFERENCES OF THE ROOTS OF A GIVEN EQUATION.

**291**

or restoring the powers of  $a$ , and putting

$$\begin{array}{rcl} a^3e & -16, \\ a^2bd & +6, \\ a^2c^2 & +4, \\ ab^2c & -5, \\ b^4 & +1, \end{array}$$

we have

$$\Sigma_4\theta_1\theta_2\theta_3 = a^{-2} Ne\sqrt{\square}.$$

Lastly,

$$\begin{aligned} \theta_1\theta_2\theta_3\theta_4 \quad (\text{or } \theta_1\theta_2\theta_3\theta_4) &= \alpha^2\beta^2\gamma^2\delta^2 \zeta^4(\beta, \gamma, \delta) \zeta^4(\gamma, \delta, \alpha) \zeta^4(\delta, \alpha, \beta) \zeta^4(\alpha, \beta, \gamma) \\ &= \alpha^2\beta^2\gamma^2\delta^2 \zeta^4(\alpha, \beta, \gamma, \delta) \\ &= a^{-2} e^2 \square, \end{aligned}$$

and the equation  $\Pi_1(\theta - \theta_i) = 0$  is thus found to be the one given above, No. 10.

## 267.

ON THE PORISM OF THE IN-AND-CIRCUMSCRIBED POLYGON.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLI. (for the year 1861), pp. 225–239. Received February 20,—Read March 7, 1861.]

The Porism referred to is as follows, viz. two conics may be so related to each other, that a polygon may be inscribed in the one, and circumscribed about the other conic, in such manner that any point whatever of the circumscribing conic may be taken for a vertex of the polygon. I gave in the year 1853, in the *Philosophical Magazine*<sup>1</sup>, a general formula for the relation between the two conics, viz. if  $U = 0$  is the equation of the inscribed conic,  $V = 0$  that of the circumscribed conic, and if  $\text{disct.}(U + \xi V)$ , where  $\xi$  is an arbitrary multiplier, denotes the discriminant of  $U + \xi V$  in regard to the coordinates  $(x, y, z)$  (such discriminant being of course a cubic function of  $\xi$ , in regard to  $\xi$ , and also in regard to the coefficients of the two conics,  $U$ ,  $V$ , jointly), then if we write

$$\sqrt{\text{disct.}(U + \xi V)} = A + B\xi + C\xi^2 + D\xi^3 + E\xi^4 + F\xi^5 + G\xi^6 + \&c.,$$

the relations for the cases of the triangle, pentagon, heptagon, &c. are

$$C = 0, \quad \begin{vmatrix} C & D \\ D & E \end{vmatrix} = 0, \quad \begin{vmatrix} C & D & E \\ D & E & F \\ E & F & G \end{vmatrix} = 0, \quad \&c.$$

<sup>1</sup>See the papers — “On the Geometrical Representation of the Integral  $\int dx \div \sqrt{(x+a)(x+b)(x+c)}$ ,” *Phil. Mag.*, April 1853, [113].

“Note on the Porism of the in-and-circumscribed Polygon,” *Phil. Mag.* August 1853, [115].

“Correction of two Theorems relating to the Porism of the in-and-circumscribed Polygon,” *Phil. Mag.* November 1853, [116].

“Developments of the Porism of the in-and-circumscribed Polygon,” *Phil. Mag.* May 1854, [128].

respectively, while those in the cases of the quadrangle, hexagon, octagon, &c., are

$$D = 0, \quad \begin{vmatrix} D, & E \\ E, & F \end{vmatrix} = 0, \quad \begin{vmatrix} D, & E, & F \\ E, & F, & G \\ F, & G, & H \end{vmatrix} = 0, \quad \&c.$$

respectively. The demonstration of this fundamental theorem is for greater completeness here reproduced; but the chief object of the memoir is to direct attention to a curious analytical theorem which is an easy *à priori* consequence of the Porism, and to obtain the relations for the several polygons up to the enneagon, in a new and simple form which puts in evidence *à posteriori* for these cases the analytical theorem just referred to. The analytical theorem rests upon the following considerations:—the relation for a hexagon ought to include that for a triangle; in fact a triangle with its sides in order twice over is a form of hexagon; the condition for an octagon should in like manner include that for a quadrangle; and so in other cases. Let the cubic function,  $\text{disct.}(U + \xi V)$ , be represented by  $1 + \alpha\beta\xi + \gamma\xi^2 + \delta\xi^3$ ; the coefficients  $A, B, C, D, E$ , &c. are functions of  $\beta, \gamma, \delta$ . Write

$$\begin{aligned} C &= (3), \quad D = (4), \\ \begin{vmatrix} C, & D \\ D, & E \end{vmatrix} &= (5), \quad \begin{vmatrix} D, & E \\ E, & F \end{vmatrix} = (6), \\ \begin{vmatrix} C, & D, & E \\ D, & E, & F \\ E, & F, & G \end{vmatrix} &= (7), \end{aligned}$$

then (3), (4), (5) are respectively prime functions of  $\beta, \gamma, \delta$ ; that is they cannot be decomposed into factors, rational functions of these quantities; and it is convenient to denote this by writing (3) = [3], (4) = [4], (5) = [5]. But by what precedes, (6) contains the factor (3), that is [3]; and if the other factor, which is prime, is denoted by [6], then we have (6) = [6][3]. The next term (7) is prime, that is we have (7) = [7]; but we have (8) = [8][4]; the term (9) gives (9) = [9][3], and so on. Thus we have (12) = [12][6][4][3], the numbers in ( ) being all the factors, so the number itself included, and as well composite as prime, of the number in ( ), the factors 2 and 1 being however excluded. To make this clearer, it may be remarked that the last-mentioned equation has the geometrical signification that the relation for a dodecagon is the aggregate of the relations for a proper dodecagon, a proper hexagon, a quadrangle, and a triangle; that is, the relation for a dodecagon implies one or other of the last-mentioned relations. The relations for the several polygons up to the

enneagon are in the memoir obtained in a form which puts in evidence the property in question, that is, the series of equations

$$\begin{aligned}(3) &= [3], \\(4) &= [4], \\(5) &= [5], \\(6) &= [6][3], \\(7) &= [7], \\(8) &= [8][4], \\(9) &= [9][3].\end{aligned}$$

To do this, the discriminant is represented, not as above in terms of the constants  $\beta, \gamma, \delta$ , but in a somewhat different form, by means of the constants  $b [= \beta]$ ,  $c$ ,  $d$ , the last two whereof are such that  $c = 0$  is the relation for the triangle,  $d = 0$  the relation for the quadrangle; thus  $[3] = c$ ,  $[4] = d$ , and for the particular cases considered, the analytical theorem consists herein, that  $c$  is a factor of (6), and of (9), and that  $d$  is a factor of (8). I have, for the sake of homogeneity, introduced into the formulæ the quantity  $a (= 1)$ , but this is a matter of form only.

The functions  $[3]$ ,  $[4]$ , &c. have been spoken of as prime; they are so, in fact, so far as they are calculated; and that they are so in general rests on the assumption that for a polygon of a given number of sides, there is but one form of relation: if, for instance, in the equation  $[12] = 0$ , which is the condition for a proper dodecagon, the function  $[12]$  could be decomposed into rational factors, then equating each of these factors to zero, we should have so many distinct forms of relation for a proper dodecagon. I believe that the assumption and reasoning are valid; but without entering further into this, I take it for granted that in the general case the functions  $[3]$ ,  $[4]$ , &c. are in fact prime. But the coefficients  $\beta, \gamma, \delta$ , or  $b, c, d$  instead of being so many independent arbitrary quantities, may be given as rational functions of other quantities (if, for instance, the two conics are circles, radii  $R, r$ , and distance between the centres  $a$ , then  $\beta, \gamma, \delta$  will be functions of  $R, r, a$ ): and it is in a case of this kind quite conceivable that the functions  $[3]$ ,  $[4]$ , &c., considered as functions of these new elements, should cease to be prime functions. In fact, in the case just referred to of the two circles (the original case of the Porism as considered by Fuss), the functions  $[4]$ ,  $[6]$ , &c., which correspond to a polygon of an even number of sides, appear to be each of them decomposable into two factors: the memoir contains some remarks tending to show *à priori* that in the case in question this decomposition takes place. I was led to examine the point by the elegant formulæ obtained in an essentially different manner by M. Mention, *Bull. de l'Acad. de St Pétersbourg*, t. 1. pp. 15, 30, and 507 (1860), in reference to the case of the two circles (it thereby appears that the decomposition takes place for the quadrangle and the hexagon); and these formulæ are reproduced in the memoir.

*Demonstration of the general Formula of the Relation between the two Conics.*

The equation of a conic passing through the points of intersection of the conics

$$U = 0, \quad V = 0,$$

is of the form

$$U + mV = 0,$$

where  $m$  is an arbitrary parameter. Suppose that the conic touches a given line, we have for the determination of  $m$  a quadratic equation; and conversely, if the roots of this quadratic equation are given, the line is also given; that is, the two roots may be considered as parameters which determine the particular line.

Let  $k$  be a given value of  $m$ ; the parameters of any tangent of the conic  $U + kV = 0$  are  $k, p$ , but as  $k$  is common to all the tangents, we may consider the particular tangent as determined by the single parameter  $p$ . And a point of the conic  $U + kV = 0$  may be considered as determined by the same parameter  $p$  which determines the tangent at that point.

As regards the conic  $V = 0$ , the common parameter for all its tangents is  $\infty$ , and we may consider any other tangent of this conic as determined by the parameter  $\theta$ , and a point of the conic as determined by the same parameter  $\theta$ .

Suppose, in the first instance, that the two conics are

$$\begin{aligned} U &= ax^2 + by^2 + cz^2 = 0, \\ V &= x^2 + y^2 + z^2 = 0; \end{aligned}$$

the equation of the tangent of  $U + kV = 0$ , the parameter whereof is  $p$  (in fact a common tangent of the conics  $U + kV = 0$ ,  $U + pV = 0$ ), is easily found to be

$$x\sqrt{b-c}\sqrt{a+k}\sqrt{a+p} + y\sqrt{c-a}\sqrt{b+k}\sqrt{b+p} + z\sqrt{a-b}\sqrt{c+k}\sqrt{c+p} = 0;$$

and if this meet the conic  $V = 0$  in the points  $P, P'$ , the parameters whereof are  $k, \theta$  and  $k, \theta'$ , or say  $\theta$  and  $\theta'$  respectively, then the coordinates of the point  $P$  are given by

$$x : y : z = \sqrt{b-c}\sqrt{a+\theta} : \sqrt{c-a}\sqrt{b+\theta} : \sqrt{a-b}\sqrt{c+\theta};$$

and substituting these values in the foregoing equation, we have

$$(b-c)\sqrt{a+k}\sqrt{a+p}\sqrt{a+\theta} + (c-a)\sqrt{b+k}\sqrt{b+p}\sqrt{b+\theta} + (a-b)\sqrt{c+k}\sqrt{c+p}\sqrt{c+\theta} = 0,$$

as an equation connecting the parameters  $p$  and  $\theta$ . This equation may be replaced by

$$\begin{aligned} \sqrt{(a+k)(a+p)(a+\theta)} &= \lambda + \mu a, \\ \sqrt{(b+k)(b+p)(b+\theta)} &= \lambda + \mu b, \\ \sqrt{(c+k)(c+p)(c+\theta)} &= \lambda + \mu c. \end{aligned}$$

from which  $\lambda, \mu$  are to be eliminated; and squaring and reducing, we have

$$\begin{aligned}\lambda^2 &= abc + kp\theta, \\ -2\lambda\mu &= bc + ca + ab - (p\theta + kp - k\theta),\end{aligned}$$

and thence

$$(bc + ca + ab - p\theta - kp - k\theta)^2 - 4(a + b + c + k + p + \theta)(abc + kp\theta) = 0$$

as the rational form of the original equation. But the same rational equation would, it is clear, be obtained from the system

$$\begin{aligned}\sqrt{(k+a)(k+b)(k+c)} &= L + Mk, \\ \sqrt{(p+a)(p+b)(p+c)} &= L + Mp, \\ \sqrt{(\theta+a)(\theta+b)(\theta+c)} &= L + M\theta,\end{aligned}$$

by the elimination of  $L$  and  $M$ . And it follows from Abel's theorem (but the result might be verified by means of Euler's fundamental theorem for the addition of elliptic functions), that if

$$\Pi\xi = \int \frac{d\xi}{\sqrt{(\xi+a)(\xi+b)(\xi+c)}},$$

then the last-mentioned system is equivalent to the transcendental equation

$$\Pi\theta = \Pi p \pm \Pi k,$$

in which the arbitrary constant which should have been inserted, and the sign of  $\Pi k$ , are determined by the consideration that for  $k = \infty$  (which gives  $\Pi k = 0$ ) we ought to have  $\theta = p$ , and therefore  $\Pi\theta = \Pi p$ .

There is of course a similar equation in  $\theta'$ , and the terms with  $\Pi k$  must be taken with opposite signs, and we have thence the theorem:

"If  $\theta, \theta'$  are the parameters of the points  $P, P'$  in which the conic  $V = 0$  is intersected by the tangent, the parameter whereof is  $p$ , of the conic  $U + kV = 0$ , then the equations

$$\begin{aligned}\Pi\theta &= \Pi p - \Pi k, \\ \Pi\theta' &= \Pi p + \Pi k,\end{aligned}$$

determine the parameters  $\theta, \theta'$  of the points in question." And again:

"If the two variable parameters  $\theta, \theta'$  are connected by the equation

$$\Pi\theta' - \Pi\theta = 2\Pi k,$$

then the line  $PP'$  will be a tangent of the conic  $U + kV = 0$ ."



The foregoing demonstration relates to the particular forms  $U = ax^2 + by^2 + cz^2$ ,  $V = x^2 + y^2 + z^2$ ; but observing that the function  $\sqrt{(\xi + a)(\xi + b)(\xi + c)}$ , which enters under the integral sign in the transcendental function  $\Pi\xi$ , is the square root of the discriminant of  $U + \xi V$ , the theory of covariants shows at once that the conclusions apply to any forms whatever of  $U, V$ , the expression for the transcendental function being

$$\Pi\xi = \int \frac{d\xi}{\sqrt{\text{discr.}(U + \xi V)}}.$$

Consider now a triangle inscribed in the conic  $V = 0$ , and with its sides touching the conics

$$U + kV = 0, \quad U + k'V = 0, \quad U + k''V = 0,$$

then if  $\theta, \theta', \theta''$  are the parameters of the angles, we have

$$\begin{aligned} \Pi\theta' - \Pi\theta &= 2\Pi k, \\ \Pi\theta'' - \Pi\theta' &= 2\Pi k', \\ \Pi\theta - \Pi\theta'' &= 2\Pi k'', \end{aligned}$$

and thence

$$\Pi k + \Pi k' + \Pi k'' = 0,$$

as the relation which must subsist between the parameters  $k, k', k''$  of the conics touched by the sides; and similarly for a polygon of  $n$  sides, the relation between the parameters is

$$\Pi k_1 + \Pi k_2 + \cdots + \Pi k_n = 0.$$

But by Abel's theorem, this transcendental equation is equivalent to an algebraical one.

In fact, calling the radical  $\sqrt{D\xi}$ , then if  $\varphi x, \chi x$  are rational and integral functions of  $x$  with arbitrary coefficients, and if

$$\varphi^2 x - y^2 \chi^2 x \cdot Dx = A(x - k_1)(x - k_2) \cdots (x - k_n),$$

(this implies that  $\varphi x$  is of a degree not exceeding  $n$  and  $\chi x$  of a degree not exceeding  $n - 3$ ; that is, for  $n$  even the degrees of  $\varphi, \chi$  are  $\frac{1}{2}n, \frac{1}{2}(n - 4)$ ; and for  $n$  odd they are  $\frac{1}{2}(n - 1), \frac{1}{2}(n - 3)$ ), then the algebraical equation is that obtained by the elimination of the arbitrary coefficients from the system of equations

$$\begin{aligned} \varphi k_1 + \chi k_1 \sqrt{Dk_1} &= 0, \\ \varphi k_2 + \chi k_2 \sqrt{Dk_2} &= 0, \\ &\vdots \\ \varphi k_n + \chi k_n \sqrt{Dk_n} &= 0; \end{aligned}$$

or, what is the same thing, for  $n$  odd,  $= 2p - 1$ , it is

$$\{1, \theta, \dots, \theta^{p-1}, \sqrt{D\theta}, \theta\sqrt{D\theta}, \dots, \theta^{p-2}\sqrt{D\theta}\} = 0,$$

and for  $n$  even,  $= 2p$ , it is

$$\{1, \theta, \dots, \theta^p, \sqrt{D\theta}, \theta\sqrt{D\theta}, \dots, \theta^{p-2}\sqrt{D\theta}\} = 0,$$

where the expressions in  $\{ \}$  denote respectively the determinants, of  $2p - 1$  lines, and  $2p$  lines, formed by substituting for  $\theta$  the values  $k_1, k_2, \dots, k_n$  respectively. Thus for  $n = 3$ , the equation is

$$\begin{vmatrix} 1, & k_1, & \sqrt{Dk_1} \\ 1, & k_2, & \sqrt{Dk_2} \\ 1, & k_3, & \sqrt{Dk_3} \end{vmatrix} = 0;$$

and for  $n = 4$ , it is

$$\begin{vmatrix} 1, & k_1, & k_1^2, & \sqrt{Dk_1} \\ 1, & k_2, & k_2^2, & \sqrt{Dk_2} \\ 1, & k_3, & k_3^2, & \sqrt{Dk_3} \\ 1, & k_4, & k_4^2, & \sqrt{Dk_4} \end{vmatrix} = 0,$$

and so on.

Suppose

$$\sqrt{D\xi} = A + B\xi + C\xi^2 + D\xi^3 + E\xi^4 + \dots;$$

then substituting the corresponding expressions for  $\sqrt{Dk_1}, \sqrt{Dk_2}$ , &c., the determinant will divide by  $\{1, \theta, \theta^2, \dots, \theta^{p-1}\}$ , and it may be seen without difficulty that the resulting equation on putting therein  $k_1 = k_2 = \dots = k_n = 0$ , will, according as  $n = 3, 4, 5, 6$ , &c., &c., be

$$C = 0, \quad D = 0, \quad \begin{vmatrix} C, & D \\ D, & E \end{vmatrix}, \quad \begin{vmatrix} D, & E \\ E, & F \end{vmatrix}, \quad \begin{vmatrix} C, & D, & E \\ D, & E, & F \\ E, & F, & G \end{vmatrix},$$

which is the theorem above referred to.

## II.

### *Application to the several Polygons up to the Enneagon.*

If the equation of the inscribed conic is  $U = 0$ , and that of the circumscribed conic is  $V = 0$ , and if the discriminant  $(U + \xi V)$  is in the first instance represented by  $1 + 4\beta\xi + 4\gamma\xi^2 + 4\delta\xi^3$ , then the square root of the discriminant is

$$1 + 2\beta\xi + 2(\gamma - \beta^2)\xi^2 + 2(\delta - 2\beta\gamma + 2\beta^3)\xi^3 + \&c.,$$

267]

ON THE PORISM OF THE IN-AND-CIRCUMSCRIBED POLYGON.

299

so that the condition for the triangle is

$$\gamma - \beta^2 = 0,$$

and that for the quadrangle is

$$\delta - 2\beta\gamma + 2\beta^3 = 0.$$

It is obviously convenient to introduce into the formulæ, in the place of  $\gamma$  and  $\delta$ , the quantities

$$c = \gamma - \beta^2, \quad d = \delta - 2\beta\gamma + 2\beta^3;$$

and writing also, for symmetry of notation,  $b$  in the place of  $\beta$ , we have

$$\beta = b, \quad \gamma = c + b^2, \quad \delta = d + 2bc,$$

so that the discriminant will be

$$= 1 + 4b\xi + 4(c + b^2)\xi^2 + 4(d + 2bc)\xi^3,$$

which is

$$= (1 + 2b\xi + 2c\xi^2)^2 + 4(d\xi^3 - c^2\xi^4).$$

But for homogeneity I introduce the quantity  $a = 1$ , and put the discriminant

$$= (1 + 2b\xi + 2ac\xi^2)^2 + 4a^2(d\xi^3 - c^2\xi^4).$$

The square root, divided by  $a^2$ , is

$$a^{-2}(1 + 2b\xi + 2ac\xi^2) \sqrt{1 + \{4a^2(d\xi^3 - c^2\xi^4)\} \div (1 + 2b\xi + 2ac\xi^2)^2};$$

or developing, this is

$$\begin{aligned} & a^{-2}(1 + 2b\xi + 2ac\xi^2) \\ & + 2(d\xi^3 - c^2\xi^4) \div (1 + 2b\xi + 2ac\xi^2) \\ & - 2a^2(d\xi^3 - c^2\xi^4)^2 \div (1 + 2b\xi + 2ac\xi^2)^3 \\ & + 4a^4(d\xi^3 - c^2\xi^4)^3 \div (1 + 2b\xi + 2ac\xi^2)^5 \\ & - 10a^6(d\xi^3 - c^2\xi^4)^4 \div (1 + 2b\xi + 2ac\xi^2)^7 \\ & + \&c.; \end{aligned}$$

or representing this by  $1 + 2B\xi + 2C\xi^2 + 2D\xi^3 + \&c.$ , we have

$$C\xi^2 + D\xi^3 + E\xi^4 + F\xi^5 + G\xi^6 + H\xi^7 + I\xi^8 + J\xi^9 + K\xi^{10} + L\xi^{11} + M\xi^{12} + \dots$$

[the expansion of the foregoing is a large array of monomial terms in  $a, b, c, d$  with signed integer coefficients, occupying the body of the page; representative entries include

$$\begin{aligned} & (1 - 2b\xi - 2ac\xi^2) \text{ contributes the leading terms,} \\ & \text{further terms include } -8b^3\xi^3, -24ab^2c\xi^4, +64a^3bc^3\xi^6, \\ & \text{up to } +16a^4c\xi^8 - 160a^4bc^4\xi^9 + \&c.], \end{aligned}$$

and the values of the coefficients  $C, D, E$ , &c. thus are:

the following table sets out the coefficients  $C, D, E, F, G, H, I, J, K, L, M$  as polynomials in  $a, b, c, d$  (with  $a = 1$ ):

|       |                                                                                                                                                                                  |               |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $C :$ | $-1 \cdot c,$                                                                                                                                                                    | leading term; |
| $D :$ | $+1 \cdot d,$                                                                                                                                                                    |               |
| $E :$ | $bd - 2, \quad c^2 - 1,$                                                                                                                                                         |               |
| $F :$ | $6abcd + 2, \quad ac^3 + 2,$                                                                                                                                                     |               |
| $G :$ | $a^2d^2 - 1, \quad a^2bd^2 + 6, \quad a^2bcd + 8, \quad ab^3d^2 - 24,$<br>$ab^2c^2 - 8, \quad b^4d - 32, \quad \&c.$                                                             |               |
| $H :$ | $a^3bd^2 + 6, \quad a^3cd^2 + 6, \quad a^3b^2cd^2 - 245,$<br>$a^2b^4cd - 360, \quad ab^4d + 168, \quad b^6 - 32, \quad \&c.$                                                     |               |
| $I :$ | $a^4d^3 - 1, \quad a^4bd^3 + 2, \quad a^3b^2cd^2 - 1320,$<br>$a^3c^2d^2 - 1320, \dots, \quad a^2b^4d^2 + 240, \quad a^2c^4d + 30,$<br>$b^4cd - 160, \quad b^6d - 64, \quad \&c.$ |               |
| $J :$ | $a^5bd^3 + 2, \quad a^4b^2cd^2 + 144, \quad a^4b^3d^3 + 360,$<br>$a^4bcd^3 + 1480, \quad a^3b^2c^2d^2 - 98000,$<br>$a^3b^3cd^2 - 672240, \quad a^3b^5d^2 + 560, \quad \&c.$      |               |
| $K :$ | $a^5b^2d^2 - 300, \quad a^5bc^2d^2 - 560, \quad a^4b^3cd^2 + 160,$<br>$a^4bc^3d + 11600, \quad \&c.$                                                                             |               |
| $L :$ | $a^5c^4 - 5, \quad a^5bc^3d + 1400, \quad \&c.$                                                                                                                                  |               |
| $M :$ | $\&c.;$                                                                                                                                                                          |               |

But in the sequel only the coefficients up to  $I$  are made use of: the expressions for  $J, K, L, M$  may however be useful, and they are given accordingly.

The sums of the numerical coefficients are given here and elsewhere, as they are very useful for verifications; thus, putting  $a = b = c = d = 1$ , we have, as should be,

$$\sqrt{1 + 4\xi + 8\xi^2 + 12\xi^3} = 1 + 2\xi + 2\xi^2 (1, +1, -3, +4, -3, +4, -19, +60, -124, +214, -455, \&c.)(1, \xi)^n.$$

Proceeding now to form the several terms of the matrix

$$(C, D, E, F, G, \dots),$$

$$(D, E, F, G, H, \dots),$$

267]

ON THE PORISM OF THE IN-AND-CIRCUMSCRIBED POLYGON.

301

&amp;c.,

which may be represented by 12, 13, &amp;c., viz.

$$12 = \begin{vmatrix} C & D \\ D & E \end{vmatrix}, \quad 13 = \begin{vmatrix} C & E \\ D & F \end{vmatrix},$$

we have, up to 45, which is all that is required:

$$\begin{aligned} 12 &= a^{-1} \times (ad^2 - 1, bcd - 2, c^3 - 1), \\ 13 &= a^{-1} \times (abd^2 + 2, ac^2d - 1, b^2cd + 4, bc^3 + 2), \\ 14 &= a^{-1} \times (a^2cd^2 + 1, ab^2d^2 - 4, abc^2d + 6, ac^4 + 2, b^3cd - 8, b^2c^3 - 4), \\ 15 &= a^{-1} \times (a^3d^3 + 1, a^2bcd^2 - 2, a^2c^2d + 4, ab^3d^2 + 8, ab^2c^2d - 20, abc^4 - 8, b^4cd + 16, b^3c^3 + 8), \\ 23 &= (acd^2 - 2, bc^2d - 2, c^4 - 1), \\ 24 &= (a^2d^3 - 1, abcd^2 + 4, b^2c^2d + 4, bc^4 + 2), \\ 25 &= (a^2bd^3 + 4, a^2c^2d^2 + 5, ab^2cd^2 - 8, abc^3d + 4, ac^5 + 2, b^3c^2d - 8, b^2c^4 - 4), \\ 34 &= (a^2bd^3 + 2, a^2c^2d^2 - 3, ab^2cd^2 - 4, ac^5 - 2), \\ 35 &= (a^3cd^3 - 2, a^2b^2d^3 - 8, a^2c^3d - 2, ab^3cd^2 + 8, abc^5 + 4), \\ 45 &= (a^4d^4 - 1, a^4bcd^3 + 4, a^3c^2d^2 - 8, a^3b^3d^3 + 8, a^3b^2c^2d^2 + 12, a^2bc^4d - 4, a^2c^6 - 4). \end{aligned}$$

Forming in like manner the determinants of the matrix

$$(C, D, E, F, \dots),$$

$$(D, E, F, G, \dots),$$

$$(E, F, G, H, \dots),$$

and representing these by 123, 124, &amp;c., viz.

$$123 = \begin{vmatrix} C & D & E \\ D & E & F \\ E & F & G \end{vmatrix}, \text{ \&c.,}$$

we have, up to 234,

$$\begin{aligned} 123 &= a^{-2} \times (a^3d^4 + 1, a^2bcd^3 + 2, a^2c^3d^2 - 1, abcd^4 - 2, ac^6 - 1), \\ 124 &= a^{-1} \times (a^2bd^4 - 4, a^2c^2d^3 - 3, a^2b^2cd^3 - 8, a^2bc^3d^2 - 4, a^2c^5d - 2, ab^3c^2d^2 + 4, abc^6 + 2), \\ 134 &= a^{-1} \times (a^4cd^4 + 1, a^3b^2d^4 + 4, a^3bc^2d^3 + 8, a^3c^4d^2 - 3, a^2b^3cd^3 + 8, a^2b^2c^3d^2 + 12, a^2c^7 - 2), \\ 234 &= (a^4d^5 - 1, a^3bcd^4 - 4, a^3c^3d^3 - 4, a^2b^2c^2d^3 - 4, a^2bc^4d^2 - 6, a^2c^6d - 2). \end{aligned}$$

and further, the determinants of the matrix

$$\begin{aligned} & (C, D, E, F, \dots), \\ & (D, E, F, G, \dots), \\ & (E, F, G, H, \dots), \\ & (F, G, H, I, \dots), \end{aligned}$$

in the present case, the single determinant

$$1234 = \begin{vmatrix} C, & D, & E, & F \\ D, & E, & F, & G \\ E, & F, & G, & H \\ F, & G, & H, & I \end{vmatrix},$$

we have

$$1234 = a^{-1} \times (a^6 cd^6 - 3, a^5 bc^2 d^5 - 12, a^5 c^4 d^4 - 4, a^4 b^2 c^3 d^4 - 16, a^4 bc^5 d^3 - 14, a^4 c^7 d^2 - 3, a^3 b^3 c^4 d^3 - 8, a^3 b^2 c^6 d^2 - 12, \dots)$$

The conditions for the triangle and the quadrangle are  $c = 0, d = 0$ ; those for the pentagon and hexagon are  $12 = 0, 23 = 0$ ; for the heptagon and octagon,  $123 = 0, 234 = 0$ ; and that for the enneagon is  $1234 = 0$ . The foregoing values show that 23 and 1234 (which belong to the hexagon and the enneagon) divide by  $c$  (which belongs to the triangle), and that 234 (which belongs to the octagon) divides by  $d$  (which belongs to the quadrangle). But I was not prepared for the destruction which will be observed in the several determinants, of the terms involving the lower powers of  $a$  (that is, the terms of the highest orders in  $b, c, d$ ), and which renders these expressions so much more simple than they would otherwise have been.

Representing the reduced equations for the several polygons, as before, by

$$\begin{aligned} [3] &= 0, \\ [4] &= 0, \\ [5] &= 0, \\ [6] &= 0, \\ [7] &= 0, \\ [8] &= 0, \\ [9] &= 0, \text{ \&c.}, \end{aligned}$$

then retaining the quantity  $a (= 1)$  for homogeneity, but rejecting the powers of  $a$  which divide out, and reversing in some cases the signs, the values of the functions [3], [4], &c. are

$$\begin{aligned} [3] &= c + 1, \\ [4] &= d + 1, \\ [5] &= ad^2 + 1, \quad bcd + 2, \quad c^3 + 1, \\ [6] &= ad^2 + 2, \quad bcd + 2, \quad c^3 + 1, \\ [7] &= a^2d^4 + 1, \quad abcd^3 + 2, \quad ac^3d^2 - 1, \quad bc^4d - 2, \quad c^6 - 1, \\ [8] &= a^2d^4 + 1, \quad abcd^3 + 4, \quad ac^3d^2 + 4, \quad b^2cd^2 + 4, \quad bc^4d + 6, \quad c^6 + 2, \\ [9] &= a^3cd^6 + 3, \quad a^2bc^2d^5 + 12, \quad a^2c^4d^4 + 4, \quad ab^2c^3d^4 + 16, \quad abc^5d^3 - 14, \quad ac^7d^2 + 3, \quad b^3c^4d^3 + 8, \quad b^2c^6d^2 + 12. \end{aligned}$$

The similarity of form for the relations corresponding to the pentagon and the hexagon, and for those corresponding to the heptagon and the octagon, is, I am inclined to think, accidental; the functions are homogeneous as regards degree and weight; and the degrees and weights of the two consecutive functions being identical, the literal parts must be similarly constituted.

### III.

#### *M. Mention's Formulæ for the Case of two Circles.*

In the case of two circles, if, as usual, the radii of the inscribed and circumscribed circles are put equal to  $r$  and  $R$  respectively, and the distance of their centres to  $a$ , then the equations of the inscribed and circumscribed circles respectively may be taken to be

$$\begin{aligned} x^2 + y^2 - r^2 &= 0, \\ (x - a)^2 + y^2 - R^2 &= 0; \end{aligned}$$

and if, in the notation of M. Mention, we put

$$\frac{R^2 - r^2}{r^2} = \nu, \quad \frac{1}{r^4}(r^4 + R^4 + a^4 - 2r^2R^2 - 2r^2a^2 - 2R^2a^2) = -\nu',$$

then the quadratic radical is

$$\sqrt{(1 + 4\nu\xi)\{[1 + (2 + 2\nu)\xi]^2 + 4\nu'\xi^3\}}.$$

THE COLLECTED MATHEMATICAL PAPERS  
of  
ARTHUR CAYLEY

Volume IV — pages 304–353

Paper 267 (conclusion), Papers 268, 269, 270, 271



and comparing this with

$$\sqrt{1 + 4b\xi + 4(c + b^2)\xi^2 + 4(d + 2bc)\xi^3},$$

we have

$$\begin{aligned} b &= i + 2, \\ c + b^2 &= i^2 + 6i + \nu + 5, \\ d + 2bc &= 4i(i^2 + 2i + \nu + 1), \end{aligned}$$

and thence

$$\begin{aligned} b &= i + 2, \\ c &= 2i + \nu + 1, \\ d &= -2i\nu - 2i; \end{aligned}$$

and by means of these values, or by effecting the development in a different manner, we find

$$\begin{aligned} B &= \begin{cases} i \\ + 2, \end{cases} \\ C &= \begin{cases} i \cdot 2 \\ + \nu + 1, \end{cases} \\ D &= 2\{i(-\nu - 1)\}, \\ E &= \begin{cases} i^2 \cdot 4\nu \\ + i \cdot 8(\nu + 1) \\ - (\nu + 1)^2, \end{cases} \\ F &= 2 \begin{cases} i^2 \cdot -4\nu \\ + i(\nu + 1)(3\nu - 5) \\ + 2(\nu + 1)^2, \end{cases} \\ G &= 2 \begin{cases} i^3 \cdot 8\nu \\ + i^2 \cdot 24\nu \\ + i \cdot 4\nu(-3\nu + 5) \\ + (\nu + 1)(\nu - 7), \end{cases} \\ H &= 4 \begin{cases} i^3 \cdot -8\nu \\ + i^2 \cdot -32\nu \\ + i^2 \cdot 4\nu(3\nu - 11) \\ + i^2 \cdot 16(3\nu - 1) \\ + i \cdot (\nu + 1)(\nu - 7)(-5\nu + 3) \\ - 4(\nu + 1)^2(\nu - 3). \end{cases} \end{aligned}$$

These values give

$$\begin{aligned}
 C &= \begin{cases} 2i \\ +\nu+1, \end{cases} = [4], \\
 &= [5], \\
 D &= 2\{i(-\nu-1)\}, \\
 CE - D^2 &= \begin{cases} i^2 \cdot 8\nu \\ +i \cdot 4(\nu+1) \\ -2(\nu+1)^2, \end{cases} = [6], \\
 \begin{vmatrix} C, & D, & E \\ D, & E, & F \\ E, & F, & G \end{vmatrix} &= \begin{cases} i^3 \cdot 64\nu^2 \\ +i^2 \cdot -32\nu(\nu+1)(\nu-1) \\ +i \cdot -16\nu(\nu+1)^2 \\ +i \cdot -8(3\nu-1)(\nu+1)^2 \\ +i \cdot -4(\nu+1)^3 \end{cases} = [4] [8]. \\
 \begin{vmatrix} D, & E \\ E, & F \\ F, & G \end{vmatrix} &= i(-\nu-1) \begin{cases} 16i^2\nu \\ +(\nu+1)^2 \end{cases}
 \end{aligned}$$

It will be remarked that  $[4] = i(-\nu-1)$  breaks up into the factors  $i$  and  $\nu+1$ ; and so  $[6] = (2i-\nu-1)\{4\nu i^2 + (\nu+1)^2\}$  breaks up into the factors  $2i-\nu-1$  and  $4\nu i^2 + (\nu+1)^2$ .

It may be added that the developed expression of  $[6]$  is

$$\begin{cases} i^2 \cdot 8\nu \\ +i \cdot 4(\nu+1) \\ -2(\nu+1)^2, \end{cases}$$

so that the difference between this and  $[5]$  is  $i^2 \cdot 4(\nu+1)^2$ , which is  $= d^2$ ; this agrees with the former result.

M. Mention has also given, but not in a developed form, the formulæ for the enneagon and the endecagon, and the following formula for the decagon, viz.

$$[16i^2\nu + (\nu+1)^2]^2 + 16i^2\nu(\nu+1)^2 \cdot [2i^2(1-\nu) - (\nu+1)^2]^2 = 0.$$

## IV.

*Considerations as to the form of relation, in the case of two circles, for Polygons of an odd and even number of sides respectively.*

The relation between the two conics, or condition for the existence of the polygon, is the same whatever point of the circumscribed conic is taken as an angle of the polygon. Take for an angle, a point of intersection of the two conics. Consider first the case of the triangle; if a point of intersection is taken as a vertex  $A$  of the triangle, then the sides  $AB$ ,  $AC$  coincide in direction with the tangent at  $A$  to the inscribed conic  $U$ , hence  $B$  and  $C$  coincide together at the point where this tangent meets the circumscribed conic  $V$ ;  $BC$  is therefore a tangent of  $V$ , but it is by hypothesis a tangent of  $U$ ; hence for the triangle the relation between the inscribed conic  $U$  and the circumscribed conic  $V$  may be stated as follows: viz. a common tangent of  $U$  and  $V$  meets  $V$  at a point of contact of a tangent of  $U$ .

In like manner for the quadrangle, if  $A$  be taken at a point of intersection, the sides  $AB$  and  $AD$  will coincide in direction with the tangent to  $U$  at this point, consequently  $B$  and  $D$  must coincide at the point where this tangent meets  $V$ ; hence also  $CB$ ,  $CD$ , the two tangents to  $U$  from the point  $C$ , must coincide, or  $C$  must be a point of intersection of the conics  $U$ ,  $V$ . In other words, the pole, with respect to the inscribed conic  $U$ , of a common chord  $AC$  of the two conics must lie on the circumscribed conic  $V$ ; this is therefore the condition for the quadrilateral.

In the ordinary mode of drawing the figures, with two conics which do not intersect, the points and lines employed in the foregoing constructions are imaginary, but the conics may be so drawn that these points and lines are all real.

In general, for a polygon of an odd number,  $2n + 1$ , of sides, then starting from a point of intersection, the sides will coincide in pairs, viz. the first and last, second and last but one, and so on, the middle or  $(n + 1)$ th side being a common tangent of the two conics. But for a figure of an even number,  $2n$ , of sides, then starting from a point of intersection of the two conics, the  $n$ th side will terminate at a second point of intersection, and then the same series of sides will be repeated in the reverse order, so that the sides will coincide in pairs, first and last, second and last but one,  $n$ th and  $(n + 1)$ th. For a figure of an odd number of sides, the relation involves only a single point of intersection, but for a figure of an even number of sides, it involves two points of intersection.

Now in the case of two circles, for a polygon of an odd number of sides, the same relation is obtained, whether we take as the point of intersection one of the actual points of intersection, or a circular point at infinity, and the relation  $[2n + 1] = 0$  does not break up into factors. And so for a polygon of an even number of sides, then taking for the two points of intersection, the two actual points of intersection, or the two circular points at infinity, we have one form of result; but taking for them an actual point of intersection and a circular point at infinity, we have a different form of result; and the equation  $[2n] = 0$  does break up into factors.

This is verified very simply in the case of the quadrangle. Taking for the two points of intersection the circular points at infinity, the line joining them is the line infinity, and its pole (with respect to the inscribed circle) is the centre of this circle; the relation therefore is that the centre of the inscribed circle lies on the circumscribed circle. But when this is the case, it is easy to see that the pole (with respect to the inscribed circle) of the radical axis, lies also on the circumscribed circle; this pole and the centre of the inscribed circle are in fact the extremities of a diameter of the circumscribed circle. The condition thus obtained is  $R^2 - a^2 = 0$  (which is M. Mention's condition  $i = 0$ ). We have next to find the analytical relation when the pole (with respect to the inscribed circle), of the line joining one of the actual points of intersection with a circular point at infinity is a point on the circumscribed circle. This I effect as follows :—taking  $z = 0$  as the equation of line infinity, if the origin be taken on the middle point of the radical axis, and if  $x = 0$  be the radical axis, then the equations of the two circles may be taken to be

$$\begin{aligned}\text{Inscribed circle, } & x^2 + y^2 - 2lxz - l'z^2 = 0, \\ \text{Circumscribed circle, } & x^2 + y^2 - 2Lxz - L'z^2 = 0,\end{aligned}$$

a circular point at infinity is

$$x : y : z = 1 : i : 0, \quad (i = \sqrt{-1}),$$

an actual point of intersection is

$$x : y : z = 0 : \sqrt{l'} : 1.$$

The line joining these is

$$ix - y + z\sqrt{l'} = 0,$$

its pole, with respect to the inscribed circle, is

$$x : y : z = -l\sqrt{l'} : i\sqrt{l'} - il : 1;$$

and if this be a point of the circumscribed circle

$$-l'(l' - 2il\sqrt{l'}) + 2Li\sqrt{l'} - L' = 0,$$

that is

$$2(L - l)i\sqrt{l'} = l'^2 + L',$$

or

$$(l'^2 + L')^2 + 4(L - l)^2 l' = 0,$$

which is the required relation : but to express it in terms of the ordinary data  $R, r, a$ , the equations of the circles, putting therein  $z = 1$ , become

$$\begin{aligned}(x - l)^2 + y^2 &= l' + l^2, \\ (x - L)^2 + y^2 &= L' + L^2,\end{aligned}$$

and therefore

$$\begin{aligned} a &= L - l, \\ R^2 &= L' + L^2; \end{aligned}$$

these equations give

$$\begin{aligned} r^2 &= l' + l^2, \\ a &= L - l; \end{aligned}$$

and thence

$$\begin{aligned} L &= \frac{R^2 - r^2 + a^2}{2a}, \\ l' &= R^2 - \left( \frac{R^2 - r^2 + a^2}{2a} \right)^2 = \frac{1}{4a^2} (2r^2 R^2 + 2r^2 a^2 + 2R^2 a^2 - r^4 - R^4 - a^4); \end{aligned}$$

if with M. Mention we write

$$\frac{1}{4a^2} (r^4 + R^4 + a^4 - 2r^2 R^2 - 2r^2 a^2 - 2R^2 a^2) = -\nu'.$$

The equation

$$(l'^2 + L')^2 + 4(L - l)^2 l' = 0$$

thus becomes

$$r^4 + 4a^2 \cdot \frac{-r^4}{4a^2} \nu' = 0,$$

that is, it becomes  $\nu' + 1 = 0$ , which is the other factor of the complete condition  $\nu'(\nu' + 1) = 0$ .

## 268.

ON A NEW AUXILIARY EQUATION IN THE THEORY OF  
EQUATIONS OF THE FIFTH ORDER.

*[From the Philosophical Transactions of the Royal Society of London, vol. CLI. (for the year 1861), pp. 263–276. Received February 20,—Read March 7, 1861.]*

CONSIDERING the equation of the fifth order, or quintic equation,

$$(* \mid v, 1)^5 = (v - x_1)(v - x_2)(v - x_3)(v - x_4)(v - x_5) = 0,$$

and putting as usual

$$f\omega = x_1 + \omega x_2 + \omega^2 x_3 + \omega^3 x_4 + \omega^4 x_5,$$

where  $\omega$  is an imaginary fifth root of unity, then, according to Lagrange's general theory for the solution of equations,  $f\omega$  is the root of an equation of the order 24, called the Resolvent Equation, but the solution whereof depends ultimately on an equation of the sixth order, viz.

$$(f\omega)^5, (f\omega^2)^5, (f\omega^3)^5, (f\omega^4)^5$$

are the roots of an equation of the fourth order, each coefficient whereof is determined by an equation of the sixth order; and moreover the other coefficients can be all of them rationally expressed in terms of any one coefficient assumed to be known; the solution thus depends on a single equation of the sixth order. In particular the last coefficient, or

$$(f\omega \cdot f\omega^2 \cdot f\omega^3 \cdot f\omega^4)^5,$$

is determined by an equation of the sixth order; and not only so, but its fifth root, or

$$f\omega \cdot f\omega^2 \cdot f\omega^3 \cdot f\omega^4,$$

(which is a rational function of the roots, and is the function called by Mr Cockle the Resolvent Product), is also determined by an equation of the sixth order : this

equation may be called the Resolvent-Product Equation. But the recent researches of Mr Cockle and Mr Harley<sup>1</sup> show that the solution of an equation of the fifth order may be made to depend on an equation of the sixth order, originating indeed in, and closely connected with, the resolvent-product equation, but of a far more simple form; this is the auxiliary equation referred to in the title of the present memoir. The connexion of the two equations, and the considerations which led to the new one, will be pointed out in the sequel; but I will here state synthetically the construction of the auxiliary equation. Representing for shortness the roots  $(x_1, x_2, x_3, x_4, x_5)$  of the given quintic equation by 1, 2, 3, 4, 5, and putting moreover

$$\overline{12345} = 12 + 23 + 34 + 45 + 51, \text{ \&c.}$$

(where on the right-hand side 12, 23, &c. stand for  $x_1x_2, x_2x_3, \text{ \&c.}$ ), then the auxiliary equation, say

$$(* \mid \phi, 1)^6 = 0,$$

has for its roots

$$\begin{aligned} \phi_1 &= \overline{12345} - \overline{24135}, & \phi_4 &= \overline{21435} - \overline{13245}, \\ \phi_2 &= \overline{13425} - \overline{32145}, & \phi_5 &= \overline{31245} - \overline{14325}, \\ \phi_3 &= \overline{14235} - \overline{43125}, & \phi_6 &= \overline{41325} - \overline{12435}, \end{aligned}$$

and, it follows therefrom, is of the form

$$(1, 0, C, 0, E, F, G \mid \phi, 1)^6$$

where  $C, E, G$  are rational and integral functions of the coefficients of the given equation, being in fact seminvariants, and  $F$  is a mere numerical multiple of the square root of the discriminant.

The roots of the given quintic equation are each of them rational functions of the roots of the auxiliary equation, so that the theory of the solution of an equation of the fifth order appears to be now carried to its extreme limit. We have in fact

$$\begin{aligned} \phi_1\phi_2 + \phi_3\phi_4 + \phi_5\phi_6 &= (* \mid x_2, 1)^4, \\ \phi_1\phi_5 + \phi_2\phi_3 + \phi_4\phi_6 &= (* \mid x_3, 1)^4, \\ \phi_1\phi_3 + \phi_2\phi_6 + \phi_3\phi_5 &= (* \mid x_4, 1)^4, \\ \phi_1\phi_4 + \phi_2\phi_5 + \phi_3\phi_6 &= (* \mid x_5, 1)^4, \end{aligned}$$

where  $(* \mid x_1, 1)^4, \text{ \&c.}$  are the values, corresponding to the roots  $x_1, \text{ \&c.}$  of the given equation, of a given quartic function. And combining these equations respectively with the quintic equations satisfied by the roots  $x_1, \text{ \&c.}$  respectively, it follows that, conversely, the roots  $x_1, x_2, \text{ \&c.}$  are rational functions of the combinations  $\phi_1\phi_2 + \phi_3\phi_4 + \phi_5\phi_6, \phi_1\phi_5 + \phi_2\phi_3 + \phi_4\phi_6, \text{ \&c.}$  respectively, of the roots of the auxiliary equation.

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1. Cockle, "Researches in the Higher Algebra," *Manchester Memoirs*, t. xv. pp. 131–142 (1858). Harley, "On the Method of Symmetric Products, and its Application to the Finite Algebraic Solution of Equations," *Manchester Memoirs*, t. xv. pp. 172–219 (1859). Harley, "On the Theory of Quintics," *Quart. Math. Journ.* t. iii. pp. 343–359 (1859).

It is proper to notice that, combining together in every possible manner the six roots of the auxiliary equation, there are in all fifteen combinations of the form  $\phi_1\phi_2 + \phi_3\phi_4 + \phi_5\phi_6$ . But the combinations occurring in the above-mentioned equations are a completely determinate set of five combinations : the equation of the order 15, whereon depend the combinations  $\phi_1\phi_2 + \phi_3\phi_4 + \phi_5\phi_6$ , is not rationally decomposable into three quintic equations, but only into a quintic equation having for its roots the above-mentioned five combinations, and into an equation of the tenth order, having for its roots the other ten combinations, and being an irreducible equation. Suppose that the auxiliary equation and its roots are known ; the method of ascertaining what combinations of roots correspond to the roots of the quintic equation would be to find the rational quintic factor of the equation of the fifth order, and observe what combinations of the roots of the auxiliary equation are also roots of this quintic factor. The direct calculation of the auxiliary equation by the method of symmetric functions would, I imagine, be very laborious. But the coefficients are seminvariants, and the process explained in my memoir on the Equation of Differences, [262], was therefore applicable, and by means of it, the equation, it will be seen, is readily obtained. The auxiliary equation gives rise to a corresponding covariant equation, which is given at the conclusion of the memoir.

1. I will commence by referring to some of the results obtained by Mr Cockle and Mr Harley.

In the paper "Researches on the Higher Algebra," Mr Cockle, dealing with the quintic equation

$$v^5 - 5Qv + E = 0,$$

obtains for the Resolvent Product  $\theta (= f\omega f\omega^2 f\omega^3 f\omega^4)$  the equation

$$\theta^6 + 2QE \cdot 5^2 \theta^4 + 2Q^4 \cdot 5^4 \theta^3 + Q^2 E^2 \cdot 5^{10} \theta^2 - (5^8 Q^5 - E^2) 5^4 \theta + 5^{15} Q^6 = 0;$$

and he remarks that this equation may be written

$$(\theta^3 + 5^2 QE\theta + 5^7 Q^4)^2 = 5^{10} (108Q^5 - E^2) \theta,$$

so that  $\sqrt[3]{-\theta}$  is determined by an equation of the sixth order, involving the quadratic radical  $\sqrt{E(E^2 - 108Q^5)}$ , which is in fact the square root of the discriminant of the quintic equation.

2. Mr Harley, in his paper "On the Symmetric Product &c.," makes use of the functions

$$\tau = x_1x_2 + x_2x_3 + x_3x_4 + x_4x_5 + x_5x_1 (= \overline{12345}),$$

$$\tau' = x_1x_3 + x_3x_5 + x_5x_2 + x_2x_4 + x_4x_1 (= \overline{24135}),$$

and he obtains for the form  $v^5 - 5Qv + E = 0$ , the relation  $\theta = 5\tau\tau'$ , which, since here  $\tau + \tau' = 0$ , gives  $\theta = -5\tau^2$ .



Hence  $\tau (= \sqrt{-\frac{1}{5}\theta})$  is the root of an equation of the sixth order involving the radical  $\sqrt{E(E^2 - 108Q^5)}$ , and which is in fact ( $t = \tau\sqrt{5} = \sqrt{-\theta}$ ), the equation

$$t^6 + 5QE t^2 + \sqrt{E(E^2 - 108Q^5)} t - 5Q^4 = 0,$$

given in Mr Harley's paper "On the Theory of Quintics."

3. And in the same paper there is given a system of equations

$$t_1 t_2 + t_3 t_4 + t_5 t_6 = x_1 (3Q - x_1^2), \text{ \&c.},$$

connecting the five roots of the given quintic equation with the combinations

$$t_1 t_2 + t_3 t_4 + t_5 t_6, \text{ \&c.}$$

of the roots of the equation in  $t$ .

4. I quote also, with a slight change of notation, the following results from the paper "On the Symmetric Product &c.," viz. considering the quintic equation under the form

$$(a, b, c, d, e, f \mid v, 1)^5 = 0,$$

we have

$$\begin{aligned} f\omega \cdot f\omega^4 &= \Sigma x^2 + \tau(\omega + \omega^4) + \tau'(\omega^2 + \omega^3), \\ f\omega^2 \cdot f\omega^3 &= \Sigma x^2 + \tau(\omega^2 + \omega^3) + \tau'(\omega + \omega^4), \end{aligned}$$

where

$$\Sigma x^2 = x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 = \frac{1}{a^2} (b^2 - 2ac),$$

and thence, observing also that  $\tau + \tau' = -\frac{c}{a}$ ,

$$a^4 e (= a^4 f\omega f\omega^2 f\omega^3 f\omega^4) = 5a^2 bd - 5ab^2 c + b^4 + 5a^2 \tau \tau',$$

or, as this equation may also be written,

$$4a^4 \theta = (5ac - 2b^2)^2 - 5a^2 (\tau - \tau')^2;$$

and hence the Resolvent Product  $\theta (= f\omega f\omega^2 f\omega^3 f\omega^4)$  being determined by an equation of the sixth order, this is also the case with the function  $(\tau - \tau')^2$ .

5. But the twelve functions  $\pm (\tau - \tau')$  can be divided into two sets of six functions each, so that each set is determined by an equation of the sixth order involving a single quadratic radical. This was in fact suggested to me by Mr Harley's equation in  $t$ ; for in the case considered  $\tau + \tau'$  was  $= 0$ , or  $2\tau = \tau - \tau'$ , and the equation in  $t$  was presumably the particular form of the equation for  $\frac{1}{2}(\tau - \tau')$  in the general case. But it will presently appear in what manner the conclusion should have been arrived at *à priori*.

268]

313

6. The preceding remarks show the connexion between the function  $\phi (= \tau - \tau')$  to which belongs the new auxiliary equation, and the Resolvent Product  $\theta (= f\omega f\omega^2 f\omega^3 f\omega^4)$ . The relation was given for the denumerate form of the quintic; but taking, instead, the standard form  $(a, b, c, d, e, f \mid v, 1)^5 = 0$ , it becomes

$$4a^4\theta = 2500(ac - b^2)^2 - 5a^2\phi^2.$$

7. The foregoing equation shows that  $\phi$  is a seminvariantive function of the roots. In fact

$$f\omega = x_1 - x_5 + \omega(x_2 - x_5) + \omega^2(x_3 - x_5) + \omega^3(x_4 - x_5),$$

is seminvariantive, and  $f\omega^2, f\omega^3, f\omega^4$ , being in like manner seminvariantive, the product  $\theta (= f\omega f\omega^2 f\omega^3 f\omega^4)$  is also seminvariantive;  $ac - b^2$  and  $a$  are seminvariants, and therefore  $\phi$  is a seminvariantive function.

8. But it is easy to show this directly. For representing, as before, the roots by 1, 2, 3, 4, 5, we have

$$(1-5)(2-5) + (2-5)(3-5) + (3-5)(4-5) = 12 + 23 + 34 - 5(1 + 2 \cdot 2 + 2 \cdot 3 + 4) + 3 \cdot 5^2,$$

$$(2-5)(4-5) + (4-5)(1-5) + (1-5)(3-5) = 24 + 41 + 13 - 5(2 + 2 \cdot 4 + 2 \cdot 1 + 3) + 3 \cdot 5^2;$$

and the difference of the right-hand sides is

$$12 + 23 + 34 - 24 - 41 - 13 + 5(2 + 4 + 3 - 4 - 1 - 3) - 0,$$

which is  $= \overline{12345} - \overline{24135}$ . So that  $\phi_1$ ,

$$= (1-5)(2-5) + (2-5)(3-5) + (3-5)(4-5) - [(2-5)(4-5) + (4-5)(1-5) + (1-5)(3-5)],$$

is a function of the differences of the roots, that is, it is a seminvariantive function.

9. To account for the division of the twelve values of  $\pm(\tau - \tau')$  into two sets as above, and to explain the formation of a set, consider the symbols 1, 2, 3, 4, 5 as belonging to five points. We may with these five points form in all  $(\frac{1}{2} \cdot 1 \cdot 2 \cdot 3 \cdot 4 =)$  12 pentagons, and the symbol 12345 of any pentagon may of course be read backwards or forwards from any point ( $12345 = 23451 = \&c. = 15432 = \&c.$ ) without alteration of its meaning. Now attaching to each arrangement of the five numbers a sign, + or -, according to the ordinary rule of signs, 12345 being as usual positive, the arrangements 12345, 23451, &c. ... 15432, &c., which belong to the same pentagon, have all of them the same sign; and we may consequently connect with each pentagon the sign + or -; there are, in fact, six pentagons with the sign + and six with the sign -; and to each positive pentagon there corresponds a negative pentagon, which is derived from it by stellation, viz. to the positive pentagon 12345 there corresponds the negative one 24135, and so for the other positive pentagons. The above-mentioned system of equations

$$\begin{aligned}\phi_1 &= \overline{12345} - \overline{24135}, & \phi_4 &= \overline{21435} - \overline{13245}, \\ \phi_2 &= \overline{13425} - \overline{32145}, & \phi_5 &= \overline{31245} - \overline{14325}, \\ \phi_3 &= \overline{14235} - \overline{43125}, & \phi_6 &= \overline{41325} - \overline{12435},\end{aligned}$$

in fact exhibits the six positive pentagons, each accompanied by its stellated negative pentagon, and the formation of the system of equations is thus completely explained; the order of arrangement of the pairs *inter se* (or, what comes to the same thing, the order of arrangement of the suffixes of the  $\phi$ 's) is wholly immaterial.

10. The six pairs of pentagons, or, what is the same thing, the  $\phi$ 's, correspond to each other in pairs in a fivefold manner, *quoad* the numbers 1, 2, 3, 4, 5 respectively; thus, *quoad* 5, the pairs are  $\phi_1$  and  $\phi_4$ ,  $\phi_2$  and  $\phi_5$ ,  $\phi_3$  and  $\phi_6$ , or say 1 and 4, 2 and 5, 3 and 6. The relation is best seen by means of the positive pentagons; thus, *quoad* 5, in the pentagons 12345 and 21435, the points adjacent to 5 in the one of them are the points 2, 3, and in the other of them the complementary points 1, 4; and so in the other cases. The fivefold correspondence is shown by the symbolical equations

$$\begin{aligned} 1 &= 16, 24, 35, \\ 2 &= 12, 34, 56, \\ 3 &= 15, 23, 46, \\ 4 &= 13, 26, 35, \\ 5 &= 14, 25, 36, \end{aligned}$$

which, in fact, indicate the combinations of the  $\phi$ 's which correspond to the several roots of the quintic.

11. It is proper to notice that the right-hand sides of the last-mentioned equations contain all the duads formed with the six numbers 1, 2, 3, 4, 5, 6, each duad once, and once only. There are in all six such synthemes of duads, viz.

|          |          |          |
|----------|----------|----------|
| 12.34.56 | 12.35.46 | 12.36.45 |
| 13.25.46 | 13.24.56 | 13.25.46 |
| 14.26.35 | 14.25.36 | 14.23.56 |
| 15.24.36 | 15.26.34 | 15.26.34 |
| 16.23.45 | 16.23.45 | 16.24.35 |
| 12.34.56 | 12.35.46 | 12.36.45 |
| 13.26.45 | 13.26.45 | 13.24.56 |
| 14.25.36 | 14.23.56 | 14.26.35 |
| 15.23.46 | 15.24.36 | 15.23.46 |
| 16.24.35 | 16.25.34 | 16.25.34 |

which is in fact the theorem whereon depends the existence, for six letters, of a 6-valued function not symmetrical in respect of five letters. There is not any peculiarity in the syntheme of duads which above presented itself; the occurrence of this particular syntheme, instead of any other, arises merely from the arbitrary selection of the suffixes of the  $\phi$ 's.

12. It is hardly necessary to remark that if the pentagon 12345 had been assumed negative instead of positive, the only difference would be that the  $\phi$ 's would have their signs reversed.

268]

315

13. I proceed now to the calculation of the Auxiliary Equation. As the working is rather easier for that form, I shall in the first instance take for the given quintic the denumerate form

$$(a, b, c, d, e, f \mid v, 1)^5 = 0.$$

Representing, as before, the roots  $x_1, x_2, x_3, x_4, x_5$  of this equation by 1, 2, 3, 4, 5, and writing

$$\overline{12345} = 12 + 23 + 34 + 45 + 51, \text{ \&c.}$$

(where on the right-hand side 12, 23, &c. stand for  $x_1x_2, x_2x_3$ , &c.), we have to find the equation

$$(* \mid \phi, 1)^6 = 0,$$

the roots whereof are

$$\begin{aligned} \phi_1 &= \overline{12345} - \overline{24135}, & \phi_4 &= \overline{21435} - \overline{13245}, \\ \phi_2 &= \overline{13425} - \overline{32145}, & \phi_5 &= \overline{31245} - \overline{14325}, \\ \phi_3 &= \overline{14235} - \overline{43125}, & \phi_6 &= \overline{41325} - \overline{12435}. \end{aligned}$$

As already remarked, the coefficients are seminvariants, and if the equation is in the first instance calculated for the particular case  $f = 0$ , the terms in  $f$  can be separately determined. But putting  $f = 0$ , one of the roots, say 5, becomes  $= 0$ , and the remaining roots 1, 2, 3, 4 are the roots of the quartic equation  $(a, b, c, d, e \mid v, 1)^4 = 0$ .

14. Writing for shortness

$$\overline{1234} = 12 + 23 + 34, \text{ \&c.,}$$

and putting also

$$\begin{aligned} A &= 12 + 34, \\ B &= 13 + 42, \\ C &= 14 + 23, \end{aligned}$$

then we have

$$\begin{aligned} \phi_1 &= \overline{1234} - \overline{2413} = 12 + 23 + 34 - 24 - 41 - 13 = A - B + 23 - 14, \\ \phi_2 &= \overline{1342} - \overline{3214} = 13 + 34 + 42 - 32 - 21 - 14 = B - C + 34 - 12, \\ \phi_3 &= \overline{1423} - \overline{4312} = 14 + 42 + 23 - 43 - 31 - 12 = C - A + 42 - 13, \\ \phi_4 &= \overline{2143} - \overline{1324} = 21 + 14 + 43 - 13 - 32 - 24 = A - B - 23 + 14, \\ \phi_5 &= \overline{3124} - \overline{1423} = 31 + 12 + 24 - 14 - 42 - 23 = B - C - 34 + 12, \\ \phi_6 &= \overline{4132} - \overline{1243} = 41 + 13 + 32 - 12 - 24 - 43 = C - A - 42 + 13. \end{aligned}$$

15. We have then

$$(\phi - \phi_1)(\phi - \phi_4) = (\phi - A + B)^2 - (14 - 23)^2,$$

316

ON A NEW AUXILIARY EQUATION IN

[268

where  $\overline{1234}$  denotes the product of the four roots; the functions  $A, B, C$ , and the product  $\overline{1234}$ , are each of the degree zero in the coefficients  $(a, b, c, d, e)$ ; and if we put

$$\begin{aligned} b &= -c', \\ c &= -4ae + bd, \end{aligned}$$

then we in fact have

$$\begin{aligned} d &= 4ace - ad^2 - b^2e, \\ a\Sigma A &= -b, \\ a^2\Sigma AB &= c, \\ a^3 ABC &= -d, \\ a^4 \cdot \overline{1234} &= e. \end{aligned}$$

But on the understanding that  $\phi$  is ultimately to be changed into  $a\phi$ , it is allowable, and it will be convenient to write

$$\begin{aligned} \Sigma A &= -b, \\ \Sigma AB &= c, \\ ABC &= -d, \\ \overline{1234} &= ae. \end{aligned}$$

16. I assume also

$$\begin{aligned} B + C - A &= \alpha, \\ C + A - B &= \beta, \\ A + B - C &= \gamma; \end{aligned}$$

we have thus

$$\begin{aligned} (\phi - \phi_1)(\phi - \phi_4) &= (\phi + \alpha)(\phi - \beta) + 4ae, \\ (\phi - \phi_2)(\phi - \phi_5) &= (\phi + \beta)(\phi - \gamma) + 4ae, \\ (\phi - \phi_3)(\phi - \phi_6) &= (\phi + \gamma)(\phi - \alpha) + 4ae, \end{aligned}$$

so that the equation in  $\phi$  is

$$[(\phi + \beta)(\phi - \gamma) + 4ae][(\phi + \gamma)(\phi - \alpha) + 4ae][(\phi + \alpha)(\phi - \beta) + 4ae] = 0.$$

17. To obtain the symmetrical functions of  $\alpha, \beta, \gamma$  it is only necessary to remark that if in the identical equation

$$(1, b, c, d \mid \theta, 1)^3 = (\theta - A)(\theta - B)(\theta - C),$$

we put  $\frac{1}{2}(x + A + B + C) = \frac{1}{2}(x - b)$ , in the place of  $\theta$ , the equation becomes

$$(1, b, c, d \mid \frac{1}{2}(x - b), 2)^3 = (x + \alpha)(x + \beta)(x + \gamma),$$

so that we have

$$\begin{aligned} \Sigma \alpha &= -b = -c, \\ \Sigma \alpha \beta &= -b^2 + 4c = -16ae + 4bd - c^2, \\ \alpha \beta \gamma &= b^3 - 4bc + 8d = 16ace - 4ad^2 - 8b^2e + 4bcd - c^3. \end{aligned}$$

268]

317

18. The developed expression for the equation in  $\phi$  is easily found to be

$$\begin{aligned} & \phi^6 \\ & + \phi^4 \cdot (-2 \Sigma \alpha^2 + 12ae) \\ & + \phi^3 \cdot (2 \Sigma \alpha^2 \beta^2 - 4ae (2 \Sigma \alpha^2 + \Sigma \alpha \beta) + 48a^2 e^2) \\ & + \phi \cdot (-4ae (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)) \\ & + (-\alpha^2 \beta^2 \gamma^2 + 4ae \alpha \beta \gamma \Sigma \alpha - 16a^2 e^2 \Sigma \alpha \beta + 64a^3 e^3) = 0. \end{aligned}$$

19. In this equation the coefficient of  $\phi$  is

$$\begin{aligned} & -4ae \Sigma (B - A)(C - B)(A - C) \\ & = 32ae (A - B)(B - C)(C - A); \end{aligned}$$

or, neglecting the multiplier  $a$ , it is

$$-32 \cdot 1 \cdot 2 \cdot 3 \cdot 4 (1 - 2)(1 - 3)(1 - 4)(2 - 3)(2 - 4)(3 - 4),$$

which is the value for  $a = 0$ , of

$$-32 (1 - 2)(1 - 3)(1 - 4)(1 - 5)(2 - 3)(2 - 4)(2 - 5)(3 - 4)(3 - 5)(4 - 5),$$

i.e. the coefficient in question is

$$-32 \cdot 25 \sqrt{5} \sqrt{a^2 f^2 + \&c.} = -800 \sqrt{5} \sqrt{a^2 f^2 + \&c.},$$

where  $a^2 f^2 + \&c.$  denotes the discriminant of the denumerate form

$$(a, b, c, d, e, f \mid v, 1)^5.$$

20. The remaining coefficients are rational functions of  $a, b, c, d, e$ , which have to be completed by the introduction of the terms in  $f$ . We have

$$\begin{aligned} \text{Coeff. } \phi^4 &= -(\Sigma \alpha)^2 + 2(-16ae + 4bd - c^2) \\ &+ 2 \Sigma \alpha \beta + 12ae. \end{aligned}$$

$$\begin{aligned} \text{Coeff. } \phi^3 &= (-16ae + 4bd - c^2)^2 \\ &- 2(a^2 \beta^2 \gamma^2) \Sigma \alpha - 2c(16ace - 8ad^2 - 8b^2e + 4bcd - c^3) \\ &- 4ae(2 \Sigma \alpha)^2 - 4ac^2e \\ &+ 4ae(2 \Sigma \alpha \beta) + 4ae(-16ae + 4bd - c^2) \\ &+ 48a^2 e^2 + 48a^2 e^2. \end{aligned}$$

$$\begin{aligned} \text{Coeff. } \phi^0 &= -(16ace - 8ad^2 - 8b^2e + 4bcd - c^3)^2 \\ &+ 4ace(16ace - 8ad^2 - 8b^2e + 4bcd - c^3) \\ &- \alpha^2 \beta^2 \gamma^2 - 16a^2 e^2 (-16ae + 4bd - c^2) \\ &+ 4ae \alpha \beta \gamma \Sigma \alpha + 64a^3 e^3 \\ &- 16a^2 e^2 \Sigma \alpha \beta \\ &+ 64a^3 e^3. \end{aligned}$$

21. Effecting the developments, these are

$$\begin{array}{rclcl}
 ae & - & 20 & a^2bf & + & 240 \\
 bd & + & 8 & a^2cef & - & 128 \\
 c^2 & - & 3 & a^2d^4 & - & 64 \\
 & & & abcd & + & 240 \\
 & & & abde & - & 128 \\
 & & & ac^2e & - & 128 \\
 & & & acd^2 & + & 16 \\
 & & & b^2ce & + & 16 \\
 & & & b^2d^2 & + & 16 \\
 & & & bc^2d & + & 16 \\
 & & & c^4 & + & 3
 \end{array}$$

the first of which is in fact complete; the others being completed, we obtain the equation in  $\phi$ , viz. :

22. For the denumerate form  $(a, b, c, d, e, f \mid v, 1)^5 = 0$ , the equation in  $\phi$  is

$$(a^6 X, -800\sqrt{5}a^5\sqrt{D}X, a^4 X, a^4 X, a^2 X, X)(\phi, 1)^6 = 0,$$

[the explicit columns of monomial-coefficients given in the original printed page 318 : column  $a^4$  produces the developed table including  $ae(-20)$ ,  $bd(+8)$ ,  $c^2(-3)$ , etc., column  $a^2$  produces the table for the cubic-in- $\phi$  coefficient, and the constant column produces the  $\phi^0$  table; these are reproduced verbatim in the printed text and amount to several score of monomials each — they are taken as given by Cayley without alteration], where  $D(= a^2f^2 + \&c.)$  denotes the discriminant for the denumerate form.

268]

319

I proceed now to form the expression for  $\phi_1\phi_4 + \phi_2\phi_5 + \phi_3\phi_6$ .

23. Writing for convenience  $x$  in the place of the root 5, we have

$$\begin{aligned}\phi_1 &= A - B + \{23 - 14 + x(1 + 4 - 2 - 3)\}, \\ \phi_4 &= A - B - \{23 - 14 + x(1 + 4 - 2 - 3)\},\end{aligned}$$

or

$$\phi_1\phi_4 = (A - B)^2 - \{23 - 14 + x(1 + 4 - 2 - 3)\}^2.$$

The terms without  $x$  are, as before,  $(A - B)^2 - C^2 + 4 \cdot \overline{1234}$ , or  $-\alpha\beta + 4 \cdot \overline{1234}$ , and we have

$$\begin{aligned}\phi_1\phi_4 &= -\alpha\beta + 4 \cdot 12 \cdot 34 \\ &\quad + 2x(1 + 4 - 2 - 3)(14 - 23) \\ &\quad - x^2(1 + 4 - 2 - 3)^2;\end{aligned}$$

and in like manner

$$\begin{aligned}\phi_2\phi_5 &= -\beta\gamma + 4 \cdot \overline{1234} \\ &\quad + 2x(1 + 2 - 3 - 4)(12 - 34) \\ &\quad - x^2(1 + 2 - 3 - 4)^2,\end{aligned}$$

and

$$\begin{aligned}\phi_3\phi_6 &= -\gamma^2 + 4 \cdot 12 \cdot 34 \\ &\quad + 2x(1 + 3 - 4 - 2)(13 - 42) \\ &\quad - x^2(1 + 3 - 4 - 2)^2.\end{aligned}$$

24. The roots 1, 2, 3, 4, contained in these expressions explicitly, and in  $\alpha, \beta, \gamma$ , are the roots of the equation  $\frac{1}{v-x}(a, b, c, d, e, f \mid v, 1)^5 = 0$ , or, what is the same thing,

$$(a', b', c', d', e' \mid v, 1)^4 = 0,$$

where

$$\begin{aligned}a' &= a, \\ b' &= ax + b, \\ c' &= ax^2 + bx + c, \\ d' &= ax^3 + bx^2 + cx + d, \\ e' &= ax^4 + bx^3 + cx^2 + dx + e.\end{aligned}$$

Omitting, as before, a power of  $a$ , which is ultimately restored, we have

$$\begin{aligned}\phi_1\phi_4 + \phi_2\phi_5 + \phi_3\phi_6 &= -\Sigma\alpha\beta + 12a'e' \\ &\quad + 2x\Sigma(1 + 4 - 2 - 3)(14 - 23) \\ &\quad - x^2\Sigma(1 + 4 - 2 - 3)^2,\end{aligned}$$

where the  $\Sigma$ 's in the second and third lines denote each of them the sum of the three terms obtained by the cyclical permutations of 2, 3, 4.



The first line is

$$\begin{aligned} & (16a'e' - 4b'd' + c'^2) + 12a'e' \\ & = 28a'e' - 4b'd' + c'^2. \end{aligned}$$

The second line is  $2x$  into  $2\Sigma 1 \cdot 2 - 3\Sigma 123$ , or it is

$$\begin{aligned} & = 2x [(-b'c' + 3a'd') + 3a'd'] \\ & = 2x (6a'd' - 2b'c'); \end{aligned}$$

and the third line is  $-x^2$  into  $3\Sigma 1^2 - 2\Sigma 12$ , or it is

$$\begin{aligned} & = 3(b'^2 - 2a'c') - 2a'c' \\ & = x^2 (8a'c' - 3b'^2). \end{aligned}$$

Hence, combining the three terms,

$$\begin{aligned} \phi_1\phi_4 + \phi_2\phi_5 + \phi_3\phi_6 & = 28a'e' - 4b'd' + c'^2 \\ & + x(12a'd' - 2b'c') \\ & + x^2(8a'c' - 3b'^2), \end{aligned}$$

or substituting for  $(a', b', c', d', e')$  their values, the right-hand side is

$$= (40a^2, 32ab, 28ac - 8b^2, 44ad - 8bc, 28ae - 4bd + c^2 \mid x, 1)^4,$$

where  $x$  stands for  $x_5$ , and on the left-hand side the factor  $a^2$  is to be restored.

25. Writing for shortness

$$(* \mid x, 1)^4 = (40a^2, 32ab, 28ac - 8b^2, 44ad - 8bc, 28ae - 4bd + c^2 \mid x, 1)^4,$$

the equation is

$$a^2 (\phi_1\phi_4 + \phi_2\phi_5 + \phi_3\phi_6) = (* \mid x_5, 1)^4;$$

and the system of equations to which this belongs is

$$\begin{aligned} a^2 (\phi_1\phi_2 + \phi_3\phi_4 + \phi_5\phi_6) & = (* \mid x_1, 1)^4, \\ a^2 (\phi_1\phi_5 + \phi_2\phi_3 + \phi_4\phi_6) & = (* \mid x_2, 1)^4, \\ a^2 (\phi_1\phi_3 + \phi_2\phi_6 + \phi_3\phi_5) & = (* \mid x_3, 1)^4, \\ a^2 (\phi_2\phi_4 + \phi_3\phi_6 + \phi_4\phi_5) & = (* \mid x_4, 1)^4, \\ a^2 (\phi_1\phi_4 + \phi_2\phi_5 + \phi_3\phi_6) & = (* \mid x_5, 1)^4; \end{aligned}$$

so that the roots  $x_1, x_2, x_3, x_4, x_5$  will be rational functions of the combinations  $\phi_1\phi_2 + \phi_3\phi_4 + \phi_5\phi_6$ , &c. respectively, of the equation in  $\phi$ .

**268]****321**

26. Passing now to the standard form  $(a, b, c, d, e, f \mid v, 1)^5 = 0$ , the equation in  $\phi$  is

$$(a^6, -100a^5\sqrt{5}\sqrt{\square}X, 2000a^4X, a^4X, a^2X, X, 40000X)(\phi, 1)^6 = 0,$$

[where again the column-tables of monomial coefficients are reproduced verbatim from page 321 of the original printed text; these tables enumerate, for each column, the monomials  $ae$ ,  $bd$ ,  $c^2$ ,  $a^2ef$ ,  $abcf$ ,  $abde$ ,  $ac^2e$ ,  $acd^2$ ,  $b^2ce$ ,  $b^2d^2$ ,  $bc^2d$ ,  $c^4$ , etc., with their integer coefficients  $+1, -2, +1, +1, -2, +14, -2, +8, -10, -40, -50, +15$ , &c., the full table being given on the original page], where  $\square (= a^2f^2 + \&c.)$  denotes the Discriminant for the Standard form.

27. And if we put

$$(* \mid x, 1)^4 = 20(2a^2, 8ab, 22ac - 10b^2, 18ad - 10bc, 7ae - 10bd + 5c^2 \mid x, 1)^4,$$

then we have

$$a^2(\phi_1\phi_2 + \phi_3\phi_4 + \phi_5\phi_6) = (* \mid x_1, 1)^4,$$

$$a^2(\phi_1\phi_5 + \phi_2\phi_3 + \phi_4\phi_6) = (* \mid x_2, 1)^4,$$

&c., which lead to rational expressions for the roots  $x_1, x_2, x_3, x_4, x_5$ , in terms of the combinations  $\phi_1\phi_2 + \phi_3\phi_4 + \phi_5\phi_6$ , &c. respectively.

C. IV.

41

28. Consider now the quintic function

$$U = (a, b, c, d, e, f \mid x, y)^5 = a(x - \alpha y)(x - \beta y)(x - \gamma y)(x - \delta y)(x - \epsilon y);$$

and treating the numbers 1, 2, 3, 4, 5 as corresponding to  $\alpha, \beta, \gamma, \delta, \epsilon$  respectively, write

$$\Phi = \widetilde{12345} - \widetilde{24135}, \text{ \&c.},$$

where

$$\widetilde{12345} = \widetilde{12} + \widetilde{23} + \widetilde{34} + \widetilde{45} + \widetilde{51},$$

in which  $\widetilde{12}$ , &c. denote respectively

$$\frac{1}{y^2} \cdot \frac{1}{x - \alpha y} \cdot \frac{1}{x - \beta y}, \text{ \&c.}$$

Then we have

$$\Phi = \frac{a}{U} \{\phi x - \chi y\},$$

where

$$\phi = \overline{12345} - \overline{24135},$$

and

$$\overline{12345} = 12 + 23 + 34 + 45 + 51 = \alpha\beta + \beta\gamma + \gamma\delta + \delta\epsilon + \epsilon\alpha,$$

and

$$\overline{24135} = 24 + 41 + 13 + 35 + 52 = \beta\delta + \delta\alpha + \alpha\gamma + \gamma\epsilon + \epsilon\beta,$$

and

$$\chi = (\overline{12345}) - (\overline{24135}),$$

$$(\overline{12345}) = 123 + 234 + 345 + 451 + 512 = \alpha\beta\gamma + \beta\gamma\delta + \gamma\delta\epsilon + \delta\epsilon\alpha + \epsilon\alpha\beta,$$

$$(\overline{24135}) = 241 + 413 + 135 + 352 + 524 = \beta\delta\alpha + \delta\alpha\gamma + \alpha\gamma\epsilon + \gamma\epsilon\beta + \epsilon\beta\delta.$$

29. In fact,

$$\Phi = \frac{a}{U} \{(\overline{12345}) - (\overline{24135})\},$$

where

$$(\overline{12345}) = 123 + 234 + 345 + 451 + 512,$$

$$(\overline{24135}) = 241 + 413 + 135 + 352 + 524,$$

where 123, &c. denote respectively

$$\frac{1}{y^3} (x - \alpha y)(x - \beta y)(x - \gamma y), \text{ \&c.},$$

and  $(\overline{12345}) - (\overline{24135})$  thus presents itself as a cubic function divided by  $y^3$ . But in this cubic function the coefficients of  $x^3$ ,  $x^2y$  vanish. For the coefficient of any power of  $x$  will be

$$123 + 234 + 345 + 451 + 512 - 241 - 413 - 135 - 352 - 524,$$

268]

323

where, first, for  $x^3$ , 123, &c. denote respectively unity; the coefficient of  $x^3$  therefore vanishes. Next, for  $x^2y$ , 123, &c. denote respectively  $-(1+2+3)$ , &c.  $(=\alpha+\beta+\gamma)$ , and the coefficient of  $x^2y$  also vanishes. But for  $xy^2$ , 123, &c. denote respectively  $12+23+31$   $(=\alpha\beta+\beta\gamma+\gamma\alpha)$ , &c. respectively; the positive terms are

$$(12+23+31) + (23+34+42) + (34+45+53) + (45+51+14) + (51+12+25),$$

which are

$$\begin{aligned} &= 2(12+23+34+45+51) + (24+41+13+35+52) \\ &= 2 \cdot \overline{12345} + \overline{24135}; \end{aligned}$$

and the negative terms, taken positively, are

$$(24+41+12) + (41+13+34) + (13+35+51) + (35+52+23) + (52+24+45),$$

which are

$$\begin{aligned} &= (12+23+34+45+51) + 2(24+41+13+35+52) \\ &= \overline{12345} + 2 \cdot \overline{24135}; \end{aligned}$$

so that the difference, or coefficient of  $xy^2$ , is

$$= \overline{12345} - \overline{24135},$$

which is  $=\phi$ .

And for  $y^3$ , 123, &c. denote respectively  $-123$   $(=-\alpha\beta\gamma)$ , &c., so that the coefficient of  $y^3$  is  $=\chi$ .

30. The cubic function is therefore  $=\phi xy^2 - \chi y^3$ ; and dividing by  $y^2$ , we have

$$\Phi = \frac{a}{U} (\phi x - \chi y).$$

$\Phi$  is thus a fractional covariantive function, the leading coefficient whereof is  $\phi$ , and the equation for the determination of  $\Phi$  is consequently that deduced from the equation for  $\phi$ , by replacing therein the seminvariants by the corresponding covariants.

The equation [denoting the covariants as in 141 and 143,  $A$  the quintic itself, &c.] is

$$\begin{aligned} &- 100A^5B, \\ &0, \\ &+ 2000A^4(6B^2 - 4H) \\ &- 800A^5\sqrt{B}\sqrt{\text{disct.}}, \\ &AJ - 25D^2, \quad |\Phi, 1)^6 = 0 \end{aligned}$$

where the coefficients are of the orders 30,  $-$ , 22,  $-$ , 14, 10, 6  
41-2

respectively. The last coefficient [now given in the form  $AJ - 25D^2$ ] being of the degree 6 in the coefficients  $(a, b, c, d, e, f)$ , is not given in the Tables; it is therefore merely indicated by  $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F} \mid x, y)^6$ , the leading coefficient  $\mathfrak{A}$  being of course the last coefficient in the equation for  $\phi$ , to the standard form.

I refrain from at present entering into the consideration of the values of the expressions  $\Phi_1\Phi_2 + \Phi_3\Phi_4 + \Phi_5\Phi_6$ , &c.

*Addition*, Nov. 10, 1862. [Originally printed in a later memoir "On Tschirnhausen's Transformation" post, 275.]

I take the opportunity of remarking, with reference to [the foregoing] memoir, that I recently discovered that the auxiliary equation there considered is in fact due to Jacobi, who, in his paper, "Observatiunculæ ad theoriæ aequationum pertinentes," *Crelle*, t. XIII. (1835), pp. 340–352, under the heading "Observatio de aequatione sexti gradus ad quam aequationes quinti gradus revocari possunt," gives the theory, and observes that the equation is of the form

$$\phi^6 + a_2\phi^4 + a_4\phi^2 + a_6 = 32\sqrt{D}\phi,$$

and mentions that the value of  $a_2$  is easily found to be (I adapt his notation to the denumerate form  $(a, b, c, d, e, f \mid v, 1)^5 = 0$ )

$$= 40ae - 16bd + 6c^2$$

(this ought, however, to be divided by  $-2$ ), but that the values of  $a_4$ ,  $a_6$  "paullo ampliores calculos poscunt."

The value of the coefficient in question is correctly obtained (page 270 of my memoir [as printed in the *Philosophical Transactions*]) in the form

$$\begin{aligned} &+ 2(-16ae + 4bd - c^2) \\ &+ 12ae; \end{aligned}$$

but the reduced value is given in two places (page 271) as equal to

$$\begin{aligned} &- 32ae, \quad \text{this should be } - 20ae, \\ &+ 8db, \quad \quad \quad + 8bd, \\ &- 3c^2, \quad \quad \quad + 3c^2. \end{aligned}$$

The last-mentioned correct value was used in obtaining the coefficient for the standard form, which coefficient is given correctly, page 274. [The correction here indicated  $-20$  in place of  $-32$ , is made *ante* p. 318.]

## A SEVENTH MEMOIR ON QUANTICS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLI. (for the year 1861), pp. 277–292. Received February 28,—Read March 14, 1861.]

THE present memoir relates chiefly to the theory of ternary cubics. Since the date of my Third Memoir on Quantics, [144], M. Aronhold has published the continuation of his researches on ternary cubics, in the memoir “Theorie der homogenen Functionen dritten Grades von drei Veränderlichen,” *Crelle*, t. LV. pp. 97–191 (1858). He there considers two derived contravariants, linear functions of the fundamental ones, and which occupy therein the position which the fundamental contravariants  $PU$ ,  $QU$  do in my Third Memoir; in the notation of the present memoir these derived contravariants are

$$\begin{aligned} YU &= 3T \cdot PU - 4S \cdot QU, \\ ZU &= -48S^2 \cdot PU + T \cdot QU; \end{aligned}$$

and for the canonical form  $x^3 + y^3 + z^3 + 6lxyz$ , they acquire respectively the factor  $(1 + 8l^3)^2$ , viz. in this case

$$\begin{aligned} YU &= (1 + 8l^3)^2 \{l(\xi^3 + \eta^3 + \zeta^3) - 3l^2 \xi \eta \zeta\}, \\ ZU &= (1 + 8l^3)^2 \{(1 + 2l^3)(\xi^3 + \eta^3 + \zeta^3) + 18l^2 \xi \eta \zeta\}. \end{aligned}$$

The derived contravariants have with the covariants  $U$ ,  $HU$ , even a more intimate connexion than have the contravariants  $PU$ ,  $QU$ ; and the advantage of the employment of  $YU$ ,  $ZU$  fully appears by M. Aronhold’s memoir.

But the conclusion is, not that the contravariants  $PU$ ,  $QU$  are to be rejected, but that the system is to be completed by the addition thereto of two derived covariants, linear functions of  $U$ ,  $HU$ ; these derived covariants, suggested to me by M. Aronhold’s memoir, are in the present memoir called  $CU$ ,  $DU$ ; their values are

$$\begin{aligned} CU &= -T \cdot U + 24S \cdot HU, \\ DU &= 8S^2 \cdot U - 3T \cdot HU; \end{aligned}$$

and for the canonical form  $x^3 + y^3 + z^3 + 6lxyz$ , they acquire respectively, not indeed  $(1 + 8l^3)^2$ , but the simple power  $(1 + 8l^3)$ , as a factor, viz. in this case

$$\begin{aligned} CU &= (1 + 8l^3) \{(-1 + 4l^3)(x^3 + y^3 + z^3) + 18l^2xyz\}, \\ DU &= (1 + 8l^3) \{l^2(5 + 4l^3)(x^3 + y^3 + z^3) + 3(1 - 10l^3)xyz\}; \end{aligned}$$

it was in fact by means of this condition as to the factor  $(1 + 8l^3)$ , that the foregoing expressions for  $CU$ ,  $DU$  were obtained.<sup>2</sup>

The formulæ of my Third Memoir and those of M. Aronhold are by this means brought into harmony and made parts of a whole; instead of the two intermediates

$$\alpha U + 6\beta HU, \quad 6\alpha PU + \beta QU,$$

in Tables 68 and 69 of my Third Memoir, or of the intermediates

$$\alpha U + 6\beta HU, \quad -2\alpha YU + 2\beta ZU,$$

of M. Aronhold's theory, we have the four intermediates

$$\alpha U + 6\beta HU, \quad -2\alpha YU + 2\beta ZU, \quad 2\alpha CU - 2\beta DU, \quad 6\alpha PU + \beta QU,$$

in Tables 74, 75, 76, and 77 of the present memoir. These four Tables embrace the former results, and the new ones which relate to the covariants  $CU$ ,  $DU$ ; and they are what is most important in the present memoir. I have, however, excluded from the Tables, and I do not in the memoir consider (otherwise than incidentally) the covariant of the sixth order  $\Theta U$ , or the contravariant (reciprocant)  $FU$ .

I have given in the memoir a comparison of my notation with that of M. Aronhold. A short part of the memoir relates to the binary cubic and the binary quartic, viz. each of these quantics has a covariant of its own order, forming with it an intermediate  $\alpha U + \beta W$ , the covariants whereof contain quantics in  $(\alpha, \beta)$ , the coefficients of which are invariants of the original quantic. The formulæ which relate to these cases are in fact given in my Fifth Memoir, [156], but they are reproduced here in order to show the relations between the quantics in  $(\alpha, \beta)$  contained in the formulæ. As regards the binary quartic, these results are required for the discussion of the like question in regard to the ternary cubic, viz. that of finding the relations between the different quantics in  $(\alpha, \beta)$  contained in the formulæ relating to the ternary cubic. Some of these relations have been obtained by M. Hermite in the memoir "Sur les formes cubiques à trois indéterminées," *Liouville*, t. III. pp. 37–40 (1858), and in that "Sur la Résolution des Equations du quatrième degré," *Comptes Rendus*, XLVI. p. 715 (1858), and by M. Aronhold in his memoir already referred to; and in particular I reproduce and demonstrate some of the results in the last-mentioned memoir of M. Hermite. But the relations in question are in the present memoir exhibited in a more complete and systematic form.

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2. M. Aronhold, in a letter dated Berlin, 17 June 1861, has pointed out to me that the covariants  $CU$ ,  $DU$  are in his notation  $P_\Theta$ ,  $P_\Psi$ , and that they belong to the forms called Conjugate Forms, § 27 of his memoir. But the explicit development of the properties of these covariants is not on this account the less interesting. Added 20 Sept. 1861.—A.C.

The paragraphs and Tables of the present memoir are numbered consecutively with those of my former memoirs on Quantics.

231. For the binary cubic  $(a, b, c, d \mid x, y)^3$ , if  $U$  be the cubic itself,  $HU$  the Hessian,  $\Phi U$  the cubicovariant, and  $D$  the discriminant (see Fifth Memoir, Nos. 115, 118), then

Covariant and other Tables, No. 71.

$$D(\alpha U + \beta \Phi U) = (\alpha^2 - \beta^2 D) U,$$

so that the quantics in  $(\alpha, \beta)$  all of them depend on  $\alpha^2 - \beta^2 D$ .

232. For the binary quartic  $(a, b, c, d, e \mid x, y)^4$ , if  $U$  be the quartic itself,  $HU$  the Hessian,  $\Phi U$  the cubicovariant,  $I, J$ , the quadrinvariant and the cubinvariant, and  $D (= I^3 - 27J^2)$  the discriminant (see Fifth Memoir, Nos. 128, 134), then

Table No. 72.

$$\begin{aligned}\Phi(\alpha U + 6\beta HU) &= (1, 0, -9I, -54J \mid \alpha, \beta)^3 \cdot U, \\ H(\alpha U + 6\beta HU) &= -\frac{1}{12} \frac{d}{d\beta} (1, 0, -9I, -54J \mid \alpha, \beta)^3 \cdot U \\ &\quad + \frac{1}{3} \frac{d}{d\alpha} (1, 0, -9I, -54J \mid \alpha, \beta)^3 \cdot HU, \\ I(\alpha U + 6\beta HU) &= (I, 18J, 3I^2 \mid \alpha, \beta)^2, \\ J(\alpha U + 6\beta HU) &= (J, I^2, 9IJ, -I^3 + 54J^2 \mid \alpha, \beta)^3, \\ D(\alpha U + 6\beta HU) &= (1, 0, -18I, -108J, 81I^2, 972IJ, 2916J^2 \mid \alpha, \beta)^6 \cdot D^3 \\ &= [(1, 0, -9I, -54J \mid \alpha, \beta)^3]^2 \cdot D.\end{aligned}$$

233. Writing for the moment

$$G = (1, 0, -9I, -54J \mid \alpha, \beta)^3,$$

then the Hessian, cubicovariant, and discriminant of this cubic function of  $(\alpha, \beta)$  are respectively

$$\begin{aligned}HG &= -3(I, 18J, 3I^2 \mid \alpha, \beta)^2, \\ \Phi G &= 54(J, I^2, 9IJ, -I^3 + 54J^2 \mid \alpha, \beta)^3, \\ DG &= -108D;\end{aligned}$$

so that the covariants of the intermediate  $\alpha U + 6\beta HU$  are all of them expressible by means of the cubic function  $G$ .

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3. The coefficient  $2916J^2$  is in the Fifth Memoir erroneously given as  $-2916J^2$ . [This correction should have been made, vol. ii. p. 549.]



It may be noticed that  $G$  is what the left-hand side of the equation

$$I(HU)^2 - J \cdot HU \cdot U^2 + JI^3 = -(\Phi U)^2$$

(see Fifth Memoir, No. 128) becomes on writing therein  $\alpha, -6\beta$ , for  $U, HU$  respectively, and throwing out the factor 4.

234. I take the opportunity of remarking with respect to a binary quartic  $U = (a, b, c, d \mid x, y)^4$ , that the Hessian of the cubicovariant, to fix the numerical factor, say  $\frac{1}{144}(\partial_x^2 \Phi U \cdot \partial_y^2 \Phi U - (\partial_x \partial_y \Phi U)^2)$ , is

$$I^3 U^2 - 36J \cdot U \cdot HU + 12I(HU)^2,$$

which is

$$= (IU - 18JHU)^2 + 4(I^3 - 27J^2)(HU)^2;$$

or if  $I^3 - 27J^2 = 0$ , that is if the quartic has a pair of equal factors, the Hessian of the cubicovariant is a perfect square.

235. Coming now to the ternary cubic  $U = (a, b, c, f, g, h, i, j, k, l \mid x, y, z)^3$ , I give in the first place the following comparison of my notation with that of M. Aronhold.

**Aronhold. Cayley.**

|            |                                        |
|------------|----------------------------------------|
| $f$        | $-CHU$                                 |
| $\Delta f$ | $-6HU$                                 |
| $S$        | $4S$                                   |
| $T$        | $-T$                                   |
| $A_8 /$    | $R$                                    |
| $T_f$      | $6PU$                                  |
| $A_5 /$    | $-2YU$                                 |
| $J$        | $2ZU$                                  |
|            | $CU$                                   |
|            | $DU$                                   |
| $0$        |                                        |
| $H$        | $-FU$                                  |
| $F$        | $2(\Theta_w U - TU^2 + 4SU \cdot HU),$ |

where the notations  $YU, ZU$  (to correspond to M. Aronhold's  $P_f, Q_f$ ) and the notations  $CU, DU$  are first employed in the present memoir. I remark in regard to  $P_f (= -2YU)$ , where, as already mentioned,

$$YU = (1 + 8l^3)^2 \{l(\xi^3 + \eta^3 + \zeta^3) - 3l^2 \xi \eta \zeta\},$$

that in my Memoir on Curves of the Third Order (*Phil. Trans.*, t. CXLVII. (1857), see p. 427), [146], I was led incidentally to the curve

$$l(\xi^3 + \eta^3 + \zeta^3) - 3l^2 \xi \eta \zeta = 0,$$

and that I there gave the equation

$$3T \cdot PU - 4S \cdot QU = (1 + 8l^3)^2 \{l(\xi^3 + \eta^3 + \zeta^3) - 3l^2 \xi \eta \zeta\}.$$

But the curve

$$(1 + 2l^3)(\xi^3 + \eta^3 + \zeta^3) + 18l^2 \xi \eta \zeta = 0,$$

which corresponds to ( $Q_f = 2ZU$ ), does not occur in that memoir.

236. I remark, further, in regard to M. Aronhold's  $\Theta, H$ , that these are what he calls "Zwischenformen," viz. they are covariants of the cubic and of the adjoint linear form  $\xi x + \eta y + \zeta z$ , or as they might be termed Contracovariants. For the canonical form  $U = x^3 + y^3 + z^3 + 6lxyz$ , the value of  $\Theta$  is

$$(yz - l^2 x^2, zx - l^2 y^2, xy - l^2 z^2, l^2 yz - lx^2, l^2 zx - ly^2, l^2 xy - lz^2 \mid \xi, \eta, \zeta)^2,$$

which is a form which occurs incidentally in my memoir last referred to (see p. 427). The value of  $H$  in the same case is

$$(-2l(1 + 2l^3)x^2 - 6lyz, \dots, -(1 + 4l^3)x^2 + 2l(1 + 2l^3)yz, \dots \mid \xi, \eta, \zeta)^2,$$

which does not occur in that memoir. In my Third Memoir on Quantics I purposely abstained from the consideration of any such forms.

237. My covariants  $\Theta U$  and  $\psi U$  involved unsymmetrically the cubic and its Hessian, and it did not occur to me how a similar covariant, such as M. Aronhold's  $\psi$ , which involves the two functions symmetrically, was to be formed. Let  $(A, B, C)$  be the first derived functions,  $(a, b, c, f, g, h)$  the second derived functions of the cubic  $U$ , and  $(A', B', C')$  the first derived functions,  $(a', b', c', f', g', h')$  the second derived functions of the Hessian  $HU$ , then disregarding numerical factors, we have

$$\Theta U = (bc - f^2, \dots, gh - af, \dots \mid A', B', C')^2,$$

and

$$\Theta_1 U = (b'c' - f'^2, \dots, g'h' - a'f', \dots \mid A, B, C)^2,$$

$$\psi = (bc' + b'c - 2ff', \dots, gh' + g'h - af' - a'f, \dots \mid A, B, C \mid A', B', C');$$

and considering  $U = 0$  as the equation of a curve of the third order, the equations  $\Theta U = 0$ ,  $\Theta_1 U = 0$ ,  $\psi = 0$  have the following significations, viz.  $\Theta U = 0$  is the locus of a point, such that its second or line polar with respect to the Hessian touches its first or conic polar with respect to the cubic :  $\Theta_1 U$  is the locus of a point such that its second or line polar with respect to the cubic touches its first or conic polar with respect to the Hessian : and  $\psi = 0$  is the locus of a point such that its second or line polar with respect to the cubic, and its second or line polar with respect to the Hessian are reciprocals (that is, each passes through the pole of the other of them) with respect to the conic which is the envelope of a line cutting the first or conic polar of the point with respect to the cubic, and the first or conic polar of the point with respect to the Hessian, in two pairs of points which are harmonically related to each other : such being in fact the immediate interpretation of the analytical formula. But this in passing.

238. The formulæ (Tables 68 and 69 of my Third Memoir) for the discriminants of the intermediates  $\alpha U + 6\beta HU$  and  $6\alpha PU + \beta QU$  respectively are

$$\begin{aligned} R(\alpha U + 6\beta HU) &= [(1, 0, -24S, -8T, -48S^2 \mid \alpha, \beta)^4]^3 R, \\ R(6\alpha PU + \beta QU) &= [(48S, 8T, -96S^2, -24TS, -T^2 - 16S^3 \mid \alpha, \beta)^4]^3 R. \end{aligned}$$

In M. Hermite's paper in the *Comptes Rendus*, already referred to, there are given between these quantics in  $(\alpha, \beta)$  certain relations which (although less simple than the relations that will afterwards be obtained) I now proceed to investigate. Putting in the first formula  $\alpha \div \beta = p$ , and in the second formula  $\alpha \div \beta = \theta$ , we have

$$\begin{aligned} R(pU + 6HU) &= 0, \quad \text{if } (1, 0, -24S, -8T, -48S^2 \mid p, 1)^4 = 0, \\ R(6\theta PU + QU) &= 0, \quad \text{if } (48S, 8T, -96S^2, -24TS, -T^2 - 16S^3 \mid \theta, 1)^4 = 0, \end{aligned}$$

which equations in  $p, \theta$ , are about to be considered in place of the quantics from which they respectively arise.

239. It is convenient to write <sup>4</sup>

$$B = \frac{1}{64} T^2 - 64S^3$$

(so that  $T^2 = A^2 + B^2$  and, for the canonical form,

$$A = -4l + 4l^4, \quad B = 1 + 8l^3).$$

Making this change, and joining to the equation in  $p$  that derived from it by writing  $q$  for  $p$ , and interchanging  $A, B$ , we have the three equations

$$\begin{aligned} (1, 0, -6A, -8T, -3A^2 \mid p, 1)^4 &= 0, \\ (1, 0, -6B, -8T, -3B^2 \mid q, 1)^4 &= 0, \\ (12A, 8T, -6A^2, -6TA, -T^2 - 7A^3 \mid \theta, 1)^4 &= 0. \end{aligned}$$

240. The signification of the equation in  $q$  is as follows, viz. if the quantic

$$U = (* \mid x, y, z)^3$$

is transformed into the canonical form

$$X^3 + Y^3 + Z^3 + 6LXYZ$$

by means of the linear equations

$$(X, Y, Z) = (\Delta \mid x, y, z),$$

---

4. A slight inaccuracy in three of Hermite's formulæ, which should be  $\beta, B$ , is here corrected. Hermite's  $S_3, B$  is Hermite's  $S_4$ , and  $p, q, \theta, \Delta$  are Hermite's  $J, S_1, \Delta, \theta$ : there is corresponding to formula in the present memoir.

269]

331

where  $\Delta$  is a matrix, then using the same letter  $\Delta$  to represent the determinant formed out of this matrix, or determinant of substitution, we have

$$q = \frac{3}{\Delta^3},$$

so that the equation in  $q$  is one that presents itself in the question of the reduction of the cubic to its canonical form.

In fact the linear transformation gives

$$S \Delta^4 = -l + l^4,$$

and thence

$$\begin{aligned} T \Delta^6 &= 1 - 20l^3 - 8l^6, \\ (T^2 - 64S^3) \Delta^{12} &= (1 + 8l^3)^2, \end{aligned}$$

which, writing  $B^2$  in the place of  $T^2 - 64S^3$ , becomes

$$B^2 \Delta^{12} = (1 + 8l^3)^2,$$

whence also

$$\begin{aligned} B \Delta^6 &= 1 + 8l^3, \quad \text{or} \quad 8l^3 = B \Delta^6 - 1, \\ 8T \Delta^6 &= 8 - 20(B \Delta^6 - 1) - (B \Delta^6 - 1)^2 \\ &= 27 - 18B \Delta^6 - B^2 \Delta^{12} \end{aligned}$$

or, as this may be written,

$$\frac{81}{\Delta^{12}} - \frac{54B}{\Delta^6} - \frac{24T}{\Delta^4} - 3B^2 = 0,$$

which, putting therein  $q = \frac{3}{\Delta^3}$ , becomes

$$(1, 0, -6B, -8T, -3B^2 \mid q, 1)^4 = 0,$$

the above-mentioned equation in  $q$ .

241. The relation between  $\theta$  and  $q$  is

$$\theta = -\frac{1}{2} \frac{T}{A} + \frac{B^2}{2Aq},$$

as may be verified without difficulty. That between  $\theta$  and  $p$  is

$$\theta = -\frac{1}{2} \left( p - \frac{A}{p} \right),$$

as appears by the identical equation

$$\begin{aligned} &(12A, 8T, -6A^2, -6TA, -T^2 - 7A^3 \mid \tfrac{1}{2}(p + \tfrac{A}{p}), 1)^4 \\ &= \frac{1}{64p^4} (8A, 8T, -64A^2, -72TA, -4(6A^3 - 64T^2), -72TA^2, -12A^4, 8TA^3, 8A^4 \mid p, 1)^8 \\ &= \frac{1}{64p^4} (1, 0, -6A, -8T, -3A^2 \mid p, 1)^4 \cdot (8A, 8T, 6A^2, 0, -A^3 \mid p, 1)^4 \end{aligned}$$

where the second factor of the product on the right-hand side is

$$-\frac{A}{p} \cdot 2(1, 0, -6A, -8T, -3A^2 \mid p, 1)^4.$$

The relation between  $p$  and  $q$  is then at once found to be

$$p = \frac{\frac{2A^2}{B}}{q + \frac{A}{B} + \frac{2T}{B}},$$

or (since  $p, q$  and  $A, B$  may be simultaneously interchanged)

$$p = \frac{\frac{2A^2}{B}}{q + \frac{B}{q} + \frac{2T}{B}}.$$

242. Let the equations in  $p, q$  be represented by  $\phi p = 0, \psi q = 0$  respectively; then we have

$$\phi p = p^4 - 6Ap^2 - 8Tp - 3A^2,$$

and therefore

$$\frac{1}{4}\phi'p = p^3 - 3Ap - 2T,$$

whence

$$\begin{aligned} \frac{1}{4}p\phi'p &= p^4 - 3Ap^2 - 2Tp \\ &= 3Ap^2 + 6Tp + 3A^2, \end{aligned}$$

and therefore

$$q = \frac{2B^2p}{\frac{1}{4}p\phi'p - \phi p} = \frac{24B^2}{\phi'p},$$

with a like formula for  $p$ , that is we have

$$q = \frac{24B^2}{\phi'p}, \quad p = \frac{24A^2}{\psi'q},$$

which with the equation

$$\theta = \frac{1}{2} \left( p - \frac{A}{p} \right),$$

are the system of equations connecting  $\theta, p, q$ .

243. As already remarked, we have to consider the two derived covariants

$$\begin{aligned} CU &= -T \cdot U + 24S \cdot HU, \\ DU &= 8S^2 \cdot U - 2T \cdot HU, \end{aligned}$$

and the two derived contravariants

$$\begin{aligned} YU &= 3T \cdot PU - 4S \cdot QU, \\ ZU &= -48S^2 \cdot PU + T \cdot QU, \end{aligned}$$

which for the canonical form  $x^3 + y^3 + z^3 + 6lxyz$  are as follows :

Table No. 70 (addition to).

$$\begin{aligned} CU &= (1 + 8l^3) [(-1 + 4l^3)(x^3 + y^3 + z^3) + 18l^2 xyz], \\ DU &= (1 + 8l^3) [l^2(5 + 4l^3)(x^3 + y^3 + z^3) + 3(1 - 10l^3)xyz], \\ YU &= (1 + 8l^3)^2 [l(\xi^3 + \eta^3 + \zeta^3) - 3l^2 \xi\eta\zeta], \\ ZU &= (1 + 8l^3)^2 [(1 + 2l^3)(\xi^3 + \eta^3 + \zeta^3) + 18l^2 \xi\eta\zeta]. \end{aligned}$$

244. We have conversely

$$\begin{aligned} 3R \cdot U &= 3T \cdot CU + 24S \cdot DU, \\ 3R \cdot HU &= 8S^2 \cdot CU + T \cdot DU, \\ -3R \cdot PU &= T \cdot YU + 4S \cdot QU, \\ -3R \cdot QU &= 48S^2 \cdot YU + 3T \cdot PU, \end{aligned}$$

and also the following formulæ, viz. if

$$2\alpha CU - 2\beta DU = \alpha' U + 6\beta' HU;$$

which give, conversely,

$$\begin{aligned} \alpha' &= -2T\alpha - 16S^2\beta, \\ \beta' &= 8S\alpha + T\beta, \\ \alpha &= \frac{1}{2R}(T\alpha' + 16S^2\beta'), \\ \beta &= \frac{1}{2R}(-8S\alpha' - 2T\beta'); \end{aligned}$$

and moreover, if

$$-2\alpha YU + 2\beta ZU = 6\alpha' PU + \beta' QU;$$

which give, conversely,

$$\begin{aligned} \alpha' &= -(T\alpha + 16S^2\beta), \\ \beta &= -(-8S\alpha - 2T\beta), \\ \alpha &= -\frac{1}{2R}(-2T\alpha' - 16S^2\beta'), \\ \beta &= -\frac{1}{2R}(8S\alpha' + T\beta'); \end{aligned}$$

so that the relation between  $(\alpha, \beta)$  and  $(\alpha', \beta')$  in the present case is similar to that between  $(\alpha', \beta')$  and  $(\alpha, \beta)$  in the former case. It may be noticed that in all these systems of linear equations, the determinant of transformation is a multiple of  $64S^3 - T^2 (= R)$ .

245. It will be convenient, before giving the Tables for the covariants of

$$\alpha U + 6\beta HU, \quad 2\alpha CU - 2\beta DU, \quad 6\alpha PU + \beta QU, \quad 2\alpha YU - 2\beta ZU,$$

which replace Tables 68 and 69 of my Third Memoir, to give the following separate Table of the quantics in  $(\alpha, \beta)$  which enter into the expressions of the invariants in Tables 68 and 69, and in these new Tables.

Table No. 73.

$$\begin{aligned} &(1, 0, -24S, -8T, -48S^2 \mid \alpha, \beta)^4, \\ &(S, T, 24S^2, 4TS, T^2 - 48S^3 \mid \alpha, \beta)^4, \\ &(T, 96S^2, 60TS, 20T^2, 240TS^2, -48T^2S + 4608S^4, -8T^3 + 576T^2S^2 \mid \alpha, \beta)^6. \\ &(48S, 8T, -96S^2, -24TS, -T^2 - 16S^3 \mid \alpha, \beta)^4, \\ &(T^2 + 192S^3, 128TS^2, 18T^2S + 384S^4, 6T^2, 6TS \mid \alpha, \beta)^4, \\ &(T^3 + 64TS^3, 5T^2S^2 - 64S^5, -8T^3 + 4608TS^4, 1920T^2S^3 + 73728S^6, \\ &\quad 360T^3S + 38400TS^4, 20T^4 + 8960T^2S^3, \\ &\quad 840T^3S^2 + 7680TS^5, \\ &\quad 36T^4S + 384T^2S^4 + 24576S^7, \\ &\quad T^5 - 40T^3S^3 + 2560TS^6 \mid \alpha, \beta)^6, \end{aligned}$$

where the first part of the Table contains the quantics in  $(\alpha, \beta)$  which relate to the forms  $\alpha U + 6\beta HU$  and  $-2\alpha YU + 2\beta ZU$ , and the second part of the Table contains the quantics in  $(\alpha, \beta)$  which relate to the forms  $6\alpha PU + \beta QU$  and  $-2\alpha CU + 2\beta DU$ .

The quantics in  $(\alpha, \beta)$  contained in the foregoing Table are in the sequel indicated by means of their leading coefficients; as thus,

$$(1, 0, -24S, \dots \mid \alpha, \beta)^4, \quad (S, T, \dots \mid \alpha, \beta)^4, \quad (T^2 + 192S^3, \dots \mid \alpha, \beta)^4, \quad \&c.$$

246. It is easy to see what transformations must be performed on the results in Tables 68 and 69, in order to obtain the new Tables. Thus, in the formation of Table 74, Table 68 gives  $\alpha U + 6\beta HU$  and  $H(\alpha U + 6\beta HU)$ , and from these  $C(\alpha U + 6\beta HU)$ ,

$D(\alpha U + 6\beta HU)$  have to be found : the same Table gives also  $P(\alpha U + 6\beta HU)$ ,  $Q(\alpha U + 6\beta HU)$ , but the expressions of these quantities  $YU$ ,  $ZU$  have to be introduced in the place of  $PU$ ,  $QU$ ; and from the expressions so transformed are deduced also the expressions for  $Y(\alpha U + 6\beta HU)$ ,  $Z(\alpha U + 6\beta HU)$ . Table 75 is to be deduced from Table 69 by writing therein  $(\alpha, \beta')$ , for  $(\alpha, \beta)$ , and then putting  $6\alpha'PU + \beta'QU = -2\alpha YU + 2\beta ZU$  and  $\alpha', \beta'$  but, as is seen above, if for  $\alpha', \beta'$  as functions of  $\alpha, \beta$  and of the invariants which is in some of the formulæ,  $YU$ ,  $ZU$ , have to be introduced in the place of  $PU$ ,  $QU$ . And so for the Tables 76 and 77. The actual effectuation of the transformations would, it is almost needless to remark, be very laborious, but the forms of the results are easily foreseen, and the results can then be verified by means of one or two coefficients only. The new Tables are

Table No. 74.

$$\begin{aligned}
 R(\alpha U + 6\beta HU) &= R \times [(1, 0, -24S, \dots | \alpha, \beta)^4]^3, \\
 S(\alpha U + 6\beta HU) &= (S, T, \dots | \alpha, \beta)^4, \\
 T(\alpha U + 6\beta HU) &= [(T, 96S^2, \dots | \alpha, \beta)^4]^2. \\
 (\alpha U + 6\beta HU) &= \alpha U + 6\beta HU, \\
 H(\alpha U + 6\beta HU) &= -\frac{1}{2} \times \begin{cases} \partial_\beta (1, 0, -24S, \dots | \alpha, \beta)^4 \cdot U \\ -6\partial_\alpha (1, 0, -24S, \dots | \alpha, \beta)^4 \cdot HU, \end{cases} \\
 C(\alpha U + 6\beta HU) &= (1, 0, -24S, \dots | \alpha, \beta)^4 \times \\
 &\quad \begin{cases} \partial_\beta (S, T, \dots | \alpha, \beta)^4 \cdot U \\ -6\partial_\alpha (S, T, \dots | \alpha, \beta)^4 \cdot HU, \end{cases} \\
 D(\alpha U + 6\beta HU) &= \frac{1}{12} (1, 0, -24S, \dots | \alpha, \beta)^4 \times \\
 &\quad \begin{cases} \partial_\beta (T, 96S^2, \dots | \alpha, \beta)^4 \cdot U \\ +\partial_\alpha (T, 96S^2, \dots | \alpha, \beta)^4 \cdot HU, \end{cases} \\
 P(\alpha U + 6\beta HU) &= -\frac{1}{3R} \times \begin{cases} \partial_\beta (S, T, \dots | \alpha, \beta)^4 \cdot YU \\ +\partial_\alpha (S, T, \dots | \alpha, \beta)^4 \cdot ZU, \end{cases} \\
 Q(\alpha U + 6\beta HU) &= -\frac{2}{3R} \times \begin{cases} \partial_\beta (T, 96S^2, \dots | \alpha, \beta)^4 \cdot YU \\ +\partial_\alpha (T, 96S^2, \dots | \alpha, \beta)^4 \cdot ZU, \end{cases} \\
 Y(\alpha U + 6\beta HU) &= [(1, 0, -24S, \dots | \alpha, \beta)^4]^2 \times (-2\alpha YU + 2\beta ZU), \\
 Z(\alpha U + 6\beta HU) &= \frac{1}{2} [(1, 0, -24S, \dots | \alpha, \beta)^4]^2 \times \\
 &\quad \begin{cases} \partial_\beta (1, 0, -24S, \dots | \alpha, \beta)^4 \cdot YU \\ +\partial_\alpha (1, 0, -24S, \dots | \alpha, \beta)^4 \cdot ZU. \end{cases}
 \end{aligned}$$



No. 75.

$$\begin{aligned}
R(-2\alpha YU + 2\beta ZU) &= -4096R^4 \times [(S, T, \dots | \alpha, \beta)^4]^3, \\
S(-2\alpha YU + 2\beta ZU) &= -R^2 \times (1, 0, -24S, \dots | \alpha, \beta)^4, \\
T(-2\alpha YU + 2\beta ZU) &= -8R^3 \times (T, 96S^2, \dots | \alpha, \beta)^4. \\
-2\alpha YU + 2\beta ZU &= -2\alpha YU + 2\beta ZU, \\
H(-2\alpha YU + 2\beta ZU) &= -\frac{1}{2}R \times \begin{cases} \partial_\beta (S, T, \dots | \alpha, \beta)^4 \cdot YU \\ + \partial_\alpha (S, T, \dots | \alpha, \beta)^4 \cdot ZU, \end{cases} \\
C(-2\alpha YU + 2\beta ZU) &= -16R^2 (S, T, \dots | \alpha, \beta)^4 \times \\
&\quad \begin{cases} \partial_\beta (1, 0, -24S, \dots | \alpha, \beta)^4 \cdot YU \\ + \partial_\alpha (1, 0, -24S, \dots | \alpha, \beta)^4 \cdot ZU, \end{cases} \\
D(-2\alpha YU + 2\beta ZU) &= -\frac{4}{3}R^2 (S, T, \dots | \alpha, \beta)^4 \times \\
&\quad \begin{cases} \partial_\beta (T, 96S^2, \dots | \alpha, \beta)^4 \cdot YU \\ + \partial_\alpha (T, 96S^2, \dots | \alpha, \beta)^4 \cdot ZU, \end{cases} \\
P(-2\alpha YU + 2\beta ZU) &= \frac{1}{3}R^2 \times \begin{cases} \partial_\beta (1, 0, -24S, \dots | \alpha, \beta)^4 \cdot U \\ - 6\partial_\alpha (1, 0, -24S, \dots | \alpha, \beta)^4 \cdot HU, \end{cases} \\
Q(-2\alpha YU + 2\beta ZU) &= -\frac{2}{3}R^2 \times \begin{cases} \partial_\beta (T, 96S^2, \dots | \alpha, \beta)^4 \cdot U \\ - 6\partial_\alpha (T, 96S^2, \dots | \alpha, \beta)^4 \cdot HU, \end{cases} \\
Y(-2\alpha YU + 2\beta ZU) &= 256R^4 [(S, T, \dots | \alpha, \beta)^4]^2 \times (\alpha U + 6\beta HU), \\
Z(-2\alpha YU + 2\beta ZU) &= -512R^4 [(S, T, \dots | \alpha, \beta)^4]^2 \times \\
&\quad \begin{cases} \partial_\beta (S, T, \dots | \alpha, \beta)^4 \cdot U \\ - 6\partial_\alpha (S, T, \dots | \alpha, \beta)^4 \cdot HU. \end{cases}
\end{aligned}$$

No. 76.

$$R(2\alpha CU - 2\beta DU) = -4096R^4 \times [(T^2 + 192S^3, \dots | \alpha, \beta)^4]^3,$$

$$S(2\alpha CU - 2\beta DU) = -8R^2 \times (48S, 8T, \dots | \alpha, \beta)^4,$$

$$T(2\alpha CU - 2\beta DU) = -8R^3 \times (-8T^3 + 4608TS^4, \dots | \alpha, \beta)^4.$$

$$2\alpha CU - 2\beta DU = 2\alpha CU - 2\beta DU,$$

$$H(2\alpha CU - 2\beta DU) = \frac{1}{2} \times \left\{ \begin{array}{l} \partial_\beta (T^2 + 192S^3, \dots | \alpha, \beta)^4 \cdot CU \\ + \partial_\alpha (T^2 + 192S^3, \dots | \alpha, \beta)^4 \cdot DU, \end{array} \right.$$

$$C(2\alpha CU - 2\beta DU) = -16R^2 (T^2 + 192S^3, \dots | \alpha, \beta)^4 \times \left\{ \begin{array}{l} \partial_\beta (48S, 8T, \dots | \alpha, \beta)^4 \cdot CU \\ + \partial_\alpha (48S, 8T, \dots | \alpha, \beta)^4 \cdot DU, \end{array} \right.$$

$$D(2\alpha CU - 2\beta DU) = \frac{4}{3} R^2 (T^2 + 192S^3, \dots | \alpha, \beta)^4 \times \left\{ \begin{array}{l} \partial_\beta (-8T^3 + 4608TS^4, \dots | \alpha, \beta)^4 \cdot CU \\ + \partial_\alpha (-8T^3 + 4608TS^4, \dots | \alpha, \beta)^4 \cdot DU, \end{array} \right.$$

$$P(2\alpha CU - 2\beta DU) = \frac{1}{6R} \times \left\{ \begin{array}{l} 6\partial_\beta (48S, 8T, \dots | \alpha, \beta)^4 \cdot PU \\ + \partial_\alpha (48S, 8T, \dots | \alpha, \beta)^4 \cdot QU, \end{array} \right.$$

$$Q(2\alpha CU - 2\beta DU) = \frac{4}{R} \times \left\{ \begin{array}{l} 6\partial_\beta (-8T^3 + 4608TS^4, \dots | \alpha, \beta)^4 \cdot PU \\ - \partial_\alpha (-8T^3 + 4608TS^4, \dots | \alpha, \beta)^4 \cdot QU, \end{array} \right.$$

$$Y(2\alpha CU - 2\beta DU) = 216R^2 [(T^2 + 192S^3, \dots | \alpha, \beta)^4]^2 \times (6\alpha PU + \beta QU),$$

$$Z(2\alpha CU - 2\beta DU) = -512R^4 [(T^2 + 192S^3, \dots | \alpha, \beta)^4]^2 \times \left\{ \begin{array}{l} 6\partial_\beta (T^2 + 192S^3, \dots | \alpha, \beta)^4 \cdot PU \\ - \partial_\alpha (T^2 + 192S^3, \dots | \alpha, \beta)^4 \cdot QU. \end{array} \right.$$

C. IV.

No. 77.

$$R(6\alpha PU + \beta QU) = R^2 \times [(48S, 8T, \dots | \alpha, \beta)^4]^3,$$

$$S(6\alpha PU + \beta QU) = (T^2 + 192S^3, \dots | \alpha, \beta)^4,$$

$$T(6\alpha PU + \beta QU) = (-8T^3 + 4608TS^4, \dots | \alpha, \beta)^4.$$

$$(6\alpha PU + \beta QU) = 6\alpha PU + \beta QU,$$

$$H(6\alpha PU + \beta QU) = -\frac{1}{R} \times \begin{cases} -6\partial_\beta (48S, 8T, \dots | \alpha, \beta)^4 \cdot PU \\ -\partial_\alpha (48S, 8T, \dots | \alpha, \beta)^4 \cdot QU, \end{cases}$$

$$C(6\alpha PU + \beta QU) = -(48S, 8T, \dots | \alpha, \beta)^4 \times \begin{cases} -6\partial_\beta (T^2 + 192S^3, \dots | \alpha, \beta)^4 \cdot PU \\ -\partial_\alpha (T^2 + 192S^3, \dots | \alpha, \beta)^4 \cdot QU, \end{cases}$$

$$D(6\alpha PU + \beta QU) = \frac{1}{12} (48S, 8T, \dots | \alpha, \beta)^4 \times \{ \partial_\beta (-8T^3 + 4608TS^4, \dots | \alpha, \beta)^4 \cdot QU,$$

$$P(6\alpha PU + \beta QU) = -\frac{1}{R} \times \begin{cases} \partial_\beta (T^2 + 192S^3, \dots | \alpha, \beta)^4 \cdot CU \\ -\partial_\alpha (T^2 + 192S^3, \dots | \alpha, \beta)^4 \cdot DU, \end{cases}$$

$$Y(6\alpha PU + \beta QU) = -\frac{1}{2} R [(48S, 8T, \dots | \alpha, \beta)^4]^2 \times (-2\alpha CU + 2\beta DU),$$

$$Z(6\alpha PU + \beta QU) = -\frac{1}{4} R [(48S, 8T, \dots | \alpha, \beta)^4]^2 \times \begin{cases} \partial_\beta (48S, 8T, \dots | \alpha, \beta)^4 \cdot CU \\ + \partial_\alpha (48S, 8T, \dots | \alpha, \beta)^4 \cdot DU. \end{cases}$$

247. It will be noticed how Tables 74 and 75 form a system involving only the quantics in  $(\alpha, \beta)$  contained in the first part of Table 73, and how, in like manner, Tables 76 and 77 form a system involving only the quantics in  $(\alpha, \beta)$  contained in the second part of Table 73; and, moreover, how in each pair of Tables the covariants, &c. correspond to each other as follows, viz.

$$\begin{array}{l} R, S, T, 1, H, C, D, P, Q, Y, Z \quad \text{to} \\ S, R, T, Y, P, Z, Q, H, D, 1, C. \end{array}$$

Thus in Table 74,—the formula for  $H(\alpha U + 6\beta HU)$ , and in Table 75,—the formula for  $P(2\alpha YU - 2\beta ZU)$ , each of them involve the same factor

$$\begin{cases} \partial_\beta (1, 0, -24S, \dots | \alpha, \beta)^4 \cdot U \\ -6\partial_\alpha (1, 0, -24S, \dots | \alpha, \beta)^4 \cdot HU, \end{cases}$$

and so in all the other cases.

248. The quantics in  $(\alpha, \beta)$  in each part of the foregoing Table 73 are covariantly connected together. In fact, considering the function  $(1, 0, -24S, \dots | \alpha, \beta)^4$ , which for shortness I call  $G$ , we have

$$\begin{aligned} G &= (1, 0, -24S, \dots | \alpha, \beta)^4, \\ IG &= 0, \\ JG &= 4(64S^3 - T^2) = 4R, \\ DG &= (IG)^3 - 27(JG)^2 = -432R^2, \\ HG &= -\frac{1}{3}(S, T, \dots | \alpha, \beta)^4, \\ \Phi G &= 2(T, 96S^2, \dots | \alpha, \beta)^6. \end{aligned}$$

The last-mentioned formulæ, by the aid of Table 72, give rise to the following more general system in which they are themselves included.

Table No. 78.

$$\begin{aligned} \lambda G + 6\mu HG &= \lambda G + 6\mu HG, \\ H(\lambda G + 6\mu HG) &= 36R\mu^2 G + \lambda^2 HG, \\ \Phi(\lambda G + 6\mu HG) &= (\lambda^3 - 216R\mu^3) \cdot 2(T, 96S^2, \dots | \alpha, \beta)^6, \\ I(\lambda G + 6\mu HG) &= 72R\mu^2, \\ J(\lambda G + 6\mu HG) &= 4R(\lambda^3 + 216R\mu^3), \\ D(\lambda G + 6\mu HG) &= -432R^2(\lambda^3 - 216R\mu^3)^2. \end{aligned}$$

The expression for  $H(\lambda G + 6\mu HG)$ , putting therein  $\lambda = 0$ , shows that, to a numerical factor *près*,  $H \cdot HG$  is equal to  $G$ , and hence, disregarding numerical factors, we may say that each of the quartics  $(1, 0, -24S, \dots | \alpha, \beta)^4$ ,  $(S, T, \dots | \alpha, \beta)^4$  is the Hessian of the other of them, and that the sextic  $(T, 96S^2, \dots | \alpha, \beta)^6$  is the cubicovariant of each of them.

249. Similarly, if the function  $(48S, 8T, \dots | \alpha, \beta)^4$  is for shortness called  $G$ , then we have

$$\begin{aligned} IG &= 0, \\ JG &= 4(64S^3 - T^2)^2 = 4R, \\ DG &= (IG)^3 - 27(JG)^2 = -432R^2, \\ HG &= -\frac{1}{3}(T^2 + 192S^3, \dots | \alpha, \beta)^4, \\ \Phi G &= -2(-8T^3 + 4608TS^4, \dots | \alpha, \beta)^6. \end{aligned}$$

The last-mentioned formulæ, by the aid of the same Table 72, give rise to the more general system in which they are themselves included.

Table No. 79.

$$\begin{aligned} \lambda G + 6\mu HG &= \lambda G + 6\mu HG, \\ H(\lambda G + 6\mu HG) &= 36R^2\mu^2G + \lambda^2HG, \\ \Phi(\lambda G + 6\mu HG) &= (\lambda^3 - 216R^2\mu^3) \times -2(-8T^3 + 4608TS^4, \dots | \alpha, \beta)^6, \\ I(\lambda G + 6\mu HG) &= 72R^2\mu^2, \\ J(\lambda G + 6\mu HG) &= 4R(\lambda^3 + 216R^2\mu^3), \\ D(\lambda G + 6\mu HG) &= -432R^2(\lambda^3 - 216R^2\mu^3)^2. \end{aligned}$$

The expression for  $H(\lambda G + 6\mu HG)$ , putting therein  $\lambda = 0$ , shows that, to a numerical factor *près*,  $H \cdot HG$  is equal to  $G$ ; so that, disregarding numerical factors, we may say that each of the quartics  $(48S, T, \dots | \alpha, \beta)^4$ ,  $(T^2 + 192S^3, \dots | \alpha, \beta)^4$  is the Hessian of the other of them, and that the sextic  $(-8T^3 + 4608TS^4, \dots | \alpha, \beta)^6$  is the cubicovariant of each of them.

250. But besides this, the quantics in  $(\alpha, \beta)$  in the two parts of the Table 73 are linearly connected together : the linear relations in question are in fact the equations whereon depend the expressions for the invariants in Tables 76 and 77 as deduced from those in Tables 74 and 75 ; and in the order of proof, they precede the formulæ in these four Tables. The linear relations are

Table No. 80.

$$\begin{aligned}
(1, 0, -24S, \dots | -2T\alpha - 16S^2\beta, 8S\alpha + T\beta)^4 &= -16R(T^2 + 192S^3, \dots | \alpha, \beta)^4, \\
(S, T, \dots | -2T\alpha - 16S^2\beta, 8S\alpha + T\beta)^4 &= R^2(48S, 8T, \dots | \alpha, \beta)^4, \\
(T, 96S^2, \dots | -2T\alpha - 16S^2\beta, 8S\alpha + T\beta)^6 &= -8R^2(-8T^3 + 4608TS^4, \dots | \alpha, \beta)^6. \\
(48S, 8T, \dots | T\alpha + 16S^2\beta, -8S\alpha - 2T\beta)^4 &= -16R^2(S, T, \dots | \alpha, \beta)^4, \\
(T^2 + 192S^3, \dots | T\alpha + 16S^2\beta, -8S\alpha - 2T\beta)^4 &= -R^2(1, 0, -24S, \dots | \alpha, \beta)^4, \\
(-8T^3 + 4608TS^4, \dots | T\alpha + 16S^2\beta, -8S\alpha - 2T\beta)^6 &= -8R^3(T, 96S^2, \dots | \alpha, \beta)^6.
\end{aligned}$$

Hence, attending to the remarks on the Tables 78 and 79, we may say that the quartics

$$(1, 0, -24S, \dots | \alpha, \beta)^4, \quad (48S, T, \dots | \alpha, \beta)^4,$$

which belong to the two parts respectively of Table 73, and which are related, the first of them to the discriminants of  $\alpha U + 6\beta HU$  and  $-2\alpha YU + 2\beta ZU$ , and the second to the discriminants of  $6\alpha PU + \beta QU$ ,  $-2\alpha CU + 2\beta DU$ , have these relations to each other, viz. each is a linear transformation of the Hessian of the other of them, and the cubicovariant of each is a linear transformation of the cubicovariant of the other.

## 270.

## ON THE DOUBLE TANGENTS OF A CURVE OF THE FOURTH ORDER.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLI. (for the year 1861), pp. 357–362. Received May 30,—Read June 20, 1861.]

THE present memoir is intended to be supplementary to that “On the Double Tangents of a Plane Curve” (*Phil. Trans.*, vol. CXLIX. (1859), pp. 193–212) [260]. I take the opportunity of correcting an error which I have there fallen into, and which is rather a misleading one, viz. the emanants  $U_1, U_2, \dots$  were numerically determined in such manner as to become equal to  $U$  on putting  $(x_1, y_1, z_1)$  equal to  $(x, y, z)$ ; the numerical determination should have been (and in the latter part of the memoir is assumed to be) such as to render  $H_1, H_2$ , &c. equal to  $H$ , on making the substitution in question; that is, in the place of the formulæ

$$U_1 = \frac{1}{n} (x_1 \partial_x + y_1 \partial_y + z_1 \partial_z) U,$$

$$U_2 = \frac{1}{n(n-1)} (x_1 \partial_x + y_1 \partial_y + z_1 \partial_z)^2 U, \text{ \&c.},$$

there ought to have been

$$U_1 = \frac{1}{(n-2)} (x_1 \partial_x + y_1 \partial_y + z_1 \partial_z) U,$$

$$U_2 = \frac{1}{(n-2)(n-3)} (x_1 \partial_x + y_1 \partial_y + z_1 \partial_z)^2 U, \text{ \&c.}$$

[this error is corrected *ante* p. 189].

The points of contact of the double tangents of the curve of the fourth order or quartic  $U = 0$ , are given as the intersections of the curve with a curve of the fourteenth order  $\Pi = 0$ ; the last-mentioned curve is not absolutely determinate, since instead of  $\Pi = 0$ , we may, it is clear, write  $\Pi + MU = 0$ , where  $M$  is an arbitrary

function of the tenth order. I have in the memoir spoken of Hesse's original form (say  $\Pi_1 = 0$ ) of the curve of the fourteenth order obtained by him in 1850, and of his transformed form (say  $\Pi_2 = 0$ ) obtained in 1856. The method in the memoir itself (Mr Salmon's method) gives, in the case in question of a quartic curve, a third form, say  $\Pi_3 = 0$ . It appears by his paper "On the Determination of the Points of Contact of Double Tangents to an Algebraic Curve" (*Quart. Math. Journ.* vol. III. p. 317 (1859)), that Mr Salmon has verified by algebraic transformations the equivalence of the last-mentioned form with those of Hesse; but the process is not given. The object of the present memoir is to demonstrate the equivalence in question, viz. that of the equation  $\Pi_3 = 0$  with the one or other of the equations  $\Pi_1 = 0$ ,  $\Pi_2 = 0$ , in virtue of the equation  $U = 0$ . The transformation depends, 1st, on a theorem used by Hesse for the deduction of his second form  $\Pi_2 = 0$  from the original form  $\Pi_1 = 0$ , which theorem is given in his paper "Transformation der Gleichung der Curven 14ten Grades welche eine gegebene Curve 4ten Grades in den Berührungspunkten ihrer Doppeltangenten schneiden," *Crelle*, t. LII. pp. 97–103 (1856), containing the transformation in question; I prove this theorem in a different and (as it appears to me) more simple manner; 2nd, on a theorem relating to a cubic curve proved incidentally in my memoir "On the Conic of Five-pointic Contact at any point of a Plane Curve" (*Phil. Trans.*, vol. CXLIX. (1859), see p. 385 [261]), the cubic curve being in the present case any first emanent of the given quartic curve: the demonstration occupies only a single paragraph, and it is here reproduced; and I reproduce also Hesse's demonstration of the equivalence of the two forms  $\Pi_1 = 0$  and  $\Pi_2 = 0$ .

Let  $U = (* | x, y, z)^4$  be a quartic function of  $(x, y, z)$ ;  $(A, B, C, F, G, H)$  the reciprocal system

$$(bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch);$$

and let  $H$  be the Hessian of  $U$ , or determinant  $abc - af^2 - bg^2 - ch^2 + 2fgh$  ( $H$  is of course a sextic function of  $x, y, z$ );  $(a', b', c', f', g', h')$  the second differential coefficients of  $H$ ;  $(A', B', C', F', G', H')$  the reciprocal system

$$(b'c' - f'^2, c'a' - g'^2, a'b' - h'^2, g'h' - a'f', h'f' - b'g', f'g' - c'h').$$

Then  $U = 0$  being the equation of a quartic curve, the equation of the curve of the fourteenth order which by its intersections determines the points of contact of the double tangents of the quartic curve, may be taken to be (Hesse's original form)

$$\Pi_1 = (A, B, C, F, G, H | \partial_x H, \partial_y H, \partial_z H)^2 - 3H (A, B, C, F, G, H | \partial_x, \partial_y, \partial_z)^2 H = 0.^5$$

Or it may be taken to be (Hesse's transformed form)

$$\Pi_2 = 5 (A, B, C, F, G, H | \partial_x H, \partial_y H, \partial_z H)^2 - 3 (A', B', C', F', G', H' | \partial_x U, \partial_y U, \partial_z U)^2 = 0.$$

And moreover, if  $U_1 = \frac{1}{2} (x_1 \partial_x + y_1 \partial_y + z_1 \partial_z) U$ , and if  $H_1$  be the Hessian of  $U_1$ , and  $(a'', b'', c'', f'', g'', h'')$  the second differential coefficients of  $H - 3H_1$ , where in the differentiations  $(x_1, y_1, z_1)$  are treated as constants but after the differentiations are

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5. In quoting this formula in my former memoir, the numerical factor 3 is by mistake omitted. [This correction should have been made *ante* p. 187.]



effected they are replaced by  $(x, y, z)$ , and if  $(A'', B'', C'', F'', G'', H'')$  be the reciprocal system

$$(b''c'' - f''^2, c''a'' - g''^2, a''b'' - h''^2, g''h'' - a''f'', h''f'' - b''g'', f''g'' - c''h''),$$

then the equations of the curve of the fourteenth order may be taken to be (Salmon's form)

$$\Pi_3 = (A'', B'', C'', F'', G'', H'' \mid \partial_x U, \partial_y U, \partial_z U)^2 = 0.$$

I have preferred to write the three equations in the foregoing forms; but it is clear that the terms

$$(A, B, C, F, G, H \mid \partial_x, \partial_y, \partial_z)^2 H; \quad (A', B', C', F', G', H' \mid \partial_x, \partial_y, \partial_z)^2 U$$

might also have been written

$$(A, B, C, F, G, H \mid a', b', c', 2f', 2g', 2h'); \quad (A', B', C', F', G', H' \mid a, b, c, 2f, 2g, 2h).$$

As already noticed, it has been shown by Hesse (and his demonstration is to be here reproduced) that the two forms  $\Pi_1 = 0$  and  $\Pi_2 = 0$  are equivalent to each other. And the object of the memoir is to show that the third form  $\Pi_3 = 0$  is equivalent to the other two. The equivalences in question subsist in virtue of the equation  $U = 0$ , that is, the functions  $\Pi_1, \Pi_2, \Pi_3$  are not identical, but differ from each other by multiples of  $U$ .

*Demonstration of Hesse's Theorem.*

Let  $(a, b, c, f, g, h)$ ,  $(a', b', c', f', g', h')$  be any systems of coefficients of a ternary quadratic function;  $(A, B, C, F, G, H)$ ,  $(A', B', C', F', G', H')$  the reciprocal systems as above,  $(x, y, z)$  arbitrary quantities. Consider the function

$$\begin{aligned} \Omega = & (a, b, c, f, g, h \mid x, y, z)^2 (A', B', C', F', G', H' \mid a, b, c, 2f, 2g, 2h) \\ & - (A', B', C', F', G', H' \mid ax + hy + gz, \quad hx + by + fz, \quad gx + fy + cz)^2. \end{aligned}$$

The term involving  $A'$  is

$$a(a, b, c, f, g, h \mid x, y, z)^2 - (ax + hy + gz)^2,$$

which is

$$\begin{aligned} & = (ab - h^2)y^2 + (ac - g^2)z^2 + 2(af - gh)yz, \\ & = Cy^2 + Bz^2 - 2Fyz; \end{aligned}$$

and the term involving  $2F'$  is

$$f(a, b, c, f, g, h \mid x, y, z)^2 - (hx + by + fz)(gx + fy + cz),$$

which is

$$\begin{aligned} & = (af - gh)x^2 + (f^2 - bc)yz + (fg - ch)zx + (hf - bg)xy, \\ & = -Fx^2 - Ayz + Hzx + Gxy; \end{aligned}$$

and the entire expression for  $\Omega$  is thus

$$\begin{aligned} & A' (Cy^2 + Bz^2 - 2Fyz) \\ & + B' (Az^2 + Cx^2 - 2Gzx) \\ & + C' (Bx^2 + Ay^2 - 2Hxy) \\ & + 2F' (-Fx^2 - Ayz + Hxz + Gxy) \\ & + 2G' (-Gy^2 - Bzx + Fxy + Hyz) \\ & + 2H' (-Hz^2 - Cxy + Gyz + Fzx); \end{aligned}$$

or, what is the same thing,

$$\Omega = (BC' + B'C - 2FF', CA' + C'A - 2GG', AB' + A'B - 2HH', GH' + G'H - AF' - A'F, HF' + H'F - BG' - B'G,$$

which is really the fundamental theorem. It is however used as follows; viz. the right-hand side being symmetrical in regard to the two systems

$$(a, b, c, f, g, h), \quad (a', b', c', f', g', h'),$$

the left-hand side, which is not in form symmetrical as regards the two systems, must be so in reality; or if  $\Omega'$  is what  $\Omega$  becomes by interchanging the two systems, then  $\Omega' = \Omega$ ; or substituting for  $\Omega$  and  $\Omega'$  their values, we have

$$\begin{aligned} & (a, b, c, f, g, h \mid x, y, z)^2 \cdot (A', B', C', F', G', H' \mid a, b, c, 2f, 2g, 2h) \\ & - (A', B', C', F', G', H' \mid ax + hy + gz, hx + by + fz, gx + fy + cz)^2 \\ & = (a', b', c', f', g', h' \mid x, y, z)^2 \cdot (A, B, C, F, G, H \mid a', b', c', 2f', 2g', 2h') \\ & - (A, B, C, F, G, H \mid a'x + h'y + g'z, h'x + b'y + f'z, g'x + f'y + c'z)^2, \end{aligned}$$

which is Hesse's theorem.

If in particular  $(a, b, c, f, g, h)$  are the second differential coefficients of a function  $u = (* \mid x, y, z)^p$ , and  $(a', b', c', f', g', h')$  the second differential coefficients of a function  $u' = (* \mid x, y, z)^{p'}$ , then the equation becomes

$$\begin{aligned} & p(p-1)u \cdot (A', B', C', F', G', H' \mid \partial_x, \partial_y, \partial_z)^2 u - (p-1)^2 (A', B', C', F', G', H' \mid \partial_x u, \partial_y u, \partial_z u)^2 \\ & = p'(p'-1)u' \cdot (A, B, C, F, G, H \mid \partial_x, \partial_y, \partial_z)^2 u' - (p'-1)^2 (A, B, C, F, G, H \mid \partial_x u', \partial_y u', \partial_z u')^2; \end{aligned}$$

and if for  $u, u'$  we take the quartic function  $U$  and the sextic function  $H$ , its Hessian, we have

$$\begin{aligned} & 12U \cdot (A', B', C', F', G', H' \mid \partial_x, \partial_y, \partial_z)^2 U - 9(A', B', C', F', G', H' \mid \partial_x U, \partial_y U, \partial_z U)^2 \\ & = 30H \cdot (A, B, C, F, G, H \mid \partial_x, \partial_y, \partial_z)^2 H - 25(A, B, C, F, G, H \mid \partial_x H, \partial_y H, \partial_z H)^2; \end{aligned}$$

and if in this identical equation we write  $U = 0$ , then from the resulting equation and the equation

$$\Pi_1 = -3H(A, B, C, F, G, H \mid \partial_x, \partial_y, \partial_z)^2 H + (A, B, C, F, G, H \mid \partial_x H, \partial_y H, \partial_z H)^2$$

we may eliminate any one of the three terms

$$\begin{aligned} & (A', B', C', F', G', H' \mid \partial_x U, \partial_y U, \partial_z U)^2, \\ & H \cdot (A, B, C, F, G, H \mid \partial_x, \partial_y, \partial_z)^2 H, \\ & (A, B, C, F, G, H \mid \partial_x H, \partial_y H, \partial_z H)^2; \end{aligned}$$

and in particular if the second term be eliminated, we obtain the equation

$$\Pi_2 = 5 (A, B, C, F, G, H \mid \partial_x H, \partial_y H, \partial_z H)^2 - 3 (A', B', C', F', G', H' \mid \partial_x U, \partial_y U, \partial_z U)^2,$$

and the equivalence of the two forms  $\Pi_1 = 0$  and  $\Pi_2 = 0$  is thus established.

But Hesse's theorem leads also to the demonstration of the equivalence of the third form  $\Pi_3 = 0$ . To use it for this purpose, I remark that if  $(a'', b'', c'', f'', g'', h'')$  are the second differential coefficients of  $H - 3H_1$ , where after the differentiations  $(x_1, y_1, z_1)$  are to be replaced by  $(x, y, z)$ , then the theorem gives

$$\begin{aligned} & 12 U \cdot (A'', B'', C'', F'', G'', H'' \mid \partial_x, \partial_y, \partial_z)^2 U - 9 (A'', B'', C'', F'', G'', H'' \mid \partial_x U, \partial_y U, \partial_z U)^2 \\ & = (a'', b'', c'', f'', g'', h'' \mid x, y, z)^2 \cdot (A, B, C, F, G, H \mid \partial_x, \partial_y, \partial_z)^2 (H - 3H_1) \\ & \quad - (A, B, C, F, G, H \mid a''x + h''y + g''z, h''x + b''y + f''z, g''x + f''y + c''z)^2. \end{aligned}$$

But on putting  $(x, y, z)$  for  $(x_1, y_1, z_1)$  we have (since  $H$  is a homogeneous function of the order 6, and  $H_1$  before the change is a homogeneous function of the order 3 in  $(x, y, z)$ ) :  $a''x + h''y + g''z = 5\partial_x H - 3 \cdot 2\partial_x H_1 = 5\partial_x H - 3\partial_x H$  (since, on making the substitution,  $H_1 = H$ , but  $\partial_x H_1 = \frac{1}{2}\partial_x H$ )  $= 2\partial_x H$ ; and thus

$$(a''x + h''y + g''z, h''x + b''y + f''z, g''x + f''y + c''z) = (2\partial_x H, 2\partial_y H, 2\partial_z H);$$

and similarly, on making the substitution,

$$(a'', b'', c'', f'', g'', h'' \mid x, y, z)^2 = 6 \cdot 5H - 3 \cdot 3 \cdot 2H_1 = (30 - 18) H = 12 H.$$

Hence writing therein  $U = 0$ , the foregoing equation becomes

$$\begin{aligned} -9 (A'', B'', C'', F'', G'', H'' \mid \partial_x U, \partial_y U, \partial_z U)^2 &= 12 H \cdot (A, B, C, F, G, H \mid \partial_x, \partial_y, \partial_z)^2 (H - 3H_1) \\ &\quad - 4 (A, B, C, F, G, H \mid \partial_x H, \partial_y H, \partial_z H)^2, \end{aligned}$$

which may also be written

$$\begin{aligned} -9 (A'', B'', C'', F'', G'', H'' \mid \partial_x U, \partial_y U, \partial_z U)^2 &= 12 H \cdot (A, B, C, F, G, H \mid \partial_x, \partial_y, \partial_z)^2 H \\ &\quad - 36 H \cdot (A, B, C, F, G, H \mid \partial_x, \partial_y, \partial_z)^2 H_1 \\ &\quad - 4 (A, B, C, F, G, H \mid \partial_x H, \partial_y H, \partial_z H)^2, \end{aligned}$$

where  $(x_1, y_1, z_1)$  are ultimately to be replaced by  $(x, y, z)$ . The second line in fact vanishes, which I show as follows :

*Demonstration of my Theorem for a Cubic Curve.*

Let  $U = (* | x, y, z)^3$  be a cubic function; it may by a linear transformation of the coordinates be reduced to the canonical form  $x^3 + y^3 + z^3 + 6lxyz$ , and we then have

$$\begin{aligned}
 (A, B, C, F, G, H | \partial_x, \partial_y, \partial_z)^2 H \div 6^2 &= (yz - l^2 x^2) \cdot -6l^2 x \\
 &+ (zx - l^2 y^2) \cdot -6l^2 y \\
 &+ (xy - l^2 z^2) \cdot -6l^2 z \\
 &+ 2(l^2 yz - lx^2) \cdot (1 + 2l^3) x \\
 &+ 2(l^2 zx - ly^2) \cdot (1 + 2l^3) y \\
 &+ 2(l^2 xy - lz^2) \cdot (1 + 2l^3) z \\
 &= -18l^2 xyz + 6l^4 (x^3 + y^3 + z^3) \\
 &+ 6l^2 (1 + 2l^3) xyz - 2l (1 + 2l^3) (x^3 + y^3 + z^3) \\
 &= (-12l^2 + 12l^5) xyz + (-2l + 2l^4) (x^3 + y^3 + z^3) \\
 &= 2(-l + l^4) (x^3 + y^3 + z^3 + 6lxyz);
 \end{aligned}$$

or since  $-l + l^4$  is equal to the quartinvariant  $S$ , and the equation is an invariantive one, we have for any cubic function whatever

$$(A, B, C, F, G, H | \partial_x, \partial_y, \partial_z)^2 H \div 6^2 = 2S \cdot U,$$

which is the theorem in question. There is a difference of notation, and consequently a different numerical factor, in the theorem as stated in the memoir on the conic of five-pointic contact, referred to above.

If  $U$  is as above,  $U$  is a quartic function  $(* | x, y, z)^4$ , and  $U_1 = \frac{1}{2}(x_1 \partial_x + y_1 \partial_y + z_1 \partial_z) U$  is a cubic function, and we have

$$(A_1, B_1, C_1, F_1, G_1, H_1 | \partial_x, \partial_y, \partial_z)^2 H_1 \div 6^2 = 2S_1 \cdot U_1,$$

where it is to be noticed that  $S_1$  denotes a quartic function in the coefficients of  $U_1$ , and consequently a quartic function in  $(x_1, y_1, z_1)$ , the coefficients being quartic functions of the coefficients of  $U$ . On writing  $(x, y, z)$  in the place of  $(x_1, y_1, z_1)$ ,  $S_1$  becomes a quartic function of  $(x, y, z)$ , which is in fact a quarticovariant quartic of  $U$ .

If in the foregoing equation we write  $(x, y, z)$  in the place of  $(x_1, y_1, z_1)$ , then  $U_1$  becomes equal to  $U$ , and consequently, if  $U = 0$ , the right-hand side of the equation vanishes. Moreover,  $(a_1, b_1, c_1, f_1, g_1, h_1)$ , the second differential coefficients of  $U_1$ , become

equal to  $(a, b, c, f, g, h)$ , and consequently the coefficients  $(A_1, B_1, C_1, F_1, G_1, H_1)$  become equal to  $(A, B, C, F, G, H)$ . Hence, assuming always that  $U = 0$ , the equation becomes

$$(A, B, C, F, G, H \mid \partial_x, \partial_y, \partial_z)^2 H_1 = 0,$$

where after the differentiations  $(x_1, y_1, z_1)$  are replaced by  $(x, y, z)$ . This is the form which is required for the present purpose.

Now returning to the foregoing expression of  $-9(A'', B'', C'', F'', G'', H'' \mid \partial_x U, \partial_y U, \partial_z U)^2$ , this now becomes

$$\begin{aligned} -9\Pi_3 &= -9(A'', B'', C'', F'', G'', H'' \mid \partial_x U, \partial_y U, \partial_z U)^2 \\ &= 4\{3H \cdot (A, B, C, F, G, H \mid \partial_x, \partial_y, \partial_z)^2 H - (A, B, C, F, G, H \mid \partial_x U, \partial_y U, \partial_z U)^2\}, \end{aligned}$$

so that the equation  $\Pi_3 = 0$  gives

$$\Pi_1 = (A, B, C, F, G, H \mid \partial_x U, \partial_y U, \partial_z U)^2 - 3H \cdot (A, B, C, F, G, H \mid \partial_x, \partial_y, \partial_z)^2 H = 0,$$

and the equivalence of the equations  $\Pi_1 = 0$  and  $\Pi_3 = 0$  is thus established.

## 271.

SUR L'INVARIANT LE PLUS SIMPLE D'UNE FONCTION QUAD-  
RATIQUE BI-TERNAIRE, ET SUR LE RÉSULTANT DE TROIS  
FONCTIONS QUADRATIQUES TERNAIRES.

[From the *Journal für die reine und angewandte Mathematik (Crelle)*, tom. LVII. (1860),  
pp. 139–148.]

M. SYLVESTER a trouvé depuis longtemps, pour le résultant des trois fonctions quadratiques ternaires

$$\begin{aligned} &(a, b, c, f, g, h)(x, y, z)^2, \\ &(a', b', c', f', g', h')(x, y, z)^2, \\ &(a'', b'', c'', f'', g'', h'')(x, y, z)^2, \end{aligned}$$

une expression remarquable

$$12R = 16C_{12} - C_6^2,$$

où  $C_{12}, C_6$  représentent des fonctions données des coefficients qui sont non seulement des invariants mais aussi des combinants des trois fonctions quadratiques (Voir le mémoire de M. Sylvester, "On the Calculus of forms otherwise the theory of Invariants," § VII., *Camb. et Dub. Math. Journ.*, t. VIII., de la, pp. 256–269 de l'année 1853, Équation (A.), p. 267).

Pour expliquer la formation du fonction  $C_6$ , il convient de considérer la fonction quadratique bi-ternaire formée par les coefficients  $a, b, c, f, g, h; a', b', c', f', g', h'; a'', b'', c'', f'', g'', h''$  et par les coefficients du quadratique réciproque

$$A, B, C, F, G, H, \quad A', B', C', F', G', H', \quad A'', B'', C'', F'', G'', H'';$$

savoir je considère

$$U = \begin{vmatrix} a, & h, & g, & F, & G, & H \\ h, & b, & f, & F', & G', & H' \\ g, & f, & c, & F'', & G'', & H'' \\ F, & F', & F'', & a', & h', & g' \\ G, & G', & G'', & h', & b', & f' \\ H, & H', & H'', & g', & f', & c' \end{vmatrix} \cdot (x, y, z, \xi, \eta, \zeta)^2$$

cette notation représentant la fonction

$$\begin{aligned}
 & (a, b, c, f, g, h) (x, y, z)^2 \cdot \xi^2 \\
 & + (a', b', c', f', g', h') (x, y, z)^2 \cdot \eta^2 \\
 & + (a'', b'', c'', f'', g'', h'') (x, y, z)^2 \cdot \zeta^2 \\
 & + (A, B, C, F, G, H) (x, y, z)^2 \cdot 2\eta\zeta \\
 & + (A', B', C', F', G', H') (x, y, z)^2 \cdot 2\zeta\xi \\
 & + (A'', B'', C'', F'', G'', H'') (x, y, z)^2 \cdot 2\xi\eta,
 \end{aligned}$$

ou, ce qui est la même chose,

$$\begin{aligned}
 & (a x^2 + b y^2 + c z^2 + 2f yz + 2g zx + 2h xy) \xi^2 \\
 & + (a' x^2 + b' y^2 + c' z^2 + 2f' yz + 2g' zx + 2h' xy) \eta^2 \\
 & + (a'' x^2 + b'' y^2 + c'' z^2 + 2f'' yz + 2g'' zx + 2h'' xy) \zeta^2 \\
 & + (A x^2 + B y^2 + C z^2 + 2F yz + 2G zx + 2H xy) 2\eta\zeta \\
 & + (A' x^2 + B' y^2 + C' z^2 + 2F' yz + 2G' zx + 2H' xy) 2\zeta\xi \\
 & + (A'' x^2 + B'' y^2 + C'' z^2 + 2F'' yz + 2G'' zx + 2H'' xy) 2\xi\eta.
 \end{aligned}$$

Soit 123 le déterminant

$$\begin{vmatrix} \partial_{\xi_1} & \partial_{\eta_1} & \partial_{\zeta_1} \\ \partial_{\xi_2} & \partial_{\eta_2} & \partial_{\zeta_2} \\ \partial_{\xi_3} & \partial_{\eta_3} & \partial_{\zeta_3} \end{vmatrix}$$

et 1'2'3' le déterminant

$$\begin{vmatrix} \partial_{x_1} & \partial_{y_1} & \partial_{z_1} \\ \partial_{x_2} & \partial_{y_2} & \partial_{z_2} \\ \partial_{x_3} & \partial_{y_3} & \partial_{z_3} \end{vmatrix}$$

et soient  $U_1, U_2, U_3$  ce que devient  $U$  lorsqu'on y écrit  $x_1, y_1, z_1, \xi_1, \eta_1, \zeta_1; x_2, y_2, z_2, \xi_2, \eta_2, \zeta_2; x_3, y_3, z_3, \xi_3, \eta_3, \zeta_3$  au lieu de  $x, y, z, \xi, \eta, \zeta$ ; on a alors cet invariant (analogue à l'invariant quadratique d'une fonction binaire d'ordre pair), savoir :

$$\frac{1}{384} \overline{123}^3 \cdot \overline{1'2'3'}^3 U_1 U_2 U_3,$$

où, comme à l'ordinaire les valeurs  $x, y, z, \xi, \eta, \zeta$  sont rétablies après les différentiations au lieu de  $x_1, y_1, z_1, \xi_1, \eta_1, \zeta_1$ , etc.

Je représente cet invariant par

$$\left\{ \begin{array}{cccccc} a, & b, & c, & f, & g, & h \\ a', & b', & c', & f', & g', & h' \\ a'', & b'', & c'', & f'', & g'', & h'' \\ A, & B, & C, & F, & G, & H \\ A', & B', & C', & F', & G', & H' \\ A'', & B'', & C'', & F'', & G'', & H'' \end{array} \right\}.$$

on voit sans peine que l'invariant peut être développé, et qu'alors il prend la forme

$$\begin{aligned} & \left\{ \begin{array}{l} A, B, C, F, G, H \\ a, b, c, f, g, h \\ a', b', c', f', g', h' \\ a'', b'', c'', f'', g'', h'' \end{array} \right\} \\ +2 & \left\{ \begin{array}{l} A', B', C', F', G', H' \\ a', b', c', f', g', h' \\ a, b, c, f, g, h \end{array} \right\} - \left\{ \begin{array}{l} A, B, C, F, G, H \\ a', b', c', f', g', h' \\ a', b', c', f', g', h' \end{array} \right\} \\ & - \left\{ \begin{array}{l} A'', B'', C'', F'', G'', H'' \\ a'', b'', c'', f'', g'', h'' \\ a, b, c, f, g, h \end{array} \right\} \end{aligned}$$

où le premier terme

$$\left\{ \begin{array}{l} A, B, C, F, G, H \\ a, b, c, f, g, h \\ a', b', c', f', g', h' \\ a'', b'', c'', f'', g'', h'' \end{array} \right\}$$

dénote la fonction

$$\begin{aligned} & ab'c'' + ab''c' + a'b''c + a'bc'' + a''bc' + a''b'c \\ & - 2af'f'' - 2a'f'f - 2a''ff' \\ & - 2bg'g'' - 2b'g''g - 2b''gg' \\ & - 2ch'h'' - 2c'h''h - 2c''hh' \\ & + 2fg'h'' + 2fg''h' + 2f'g''h + 2f'gh'' + 2f''gh' + 2f''g'h \end{aligned}$$

et de même pour les autres termes. L'invariant contient les termes

$$ab'c'' + 2AB'G'' + 2fg'h'' + 2FG'H'' - 2af'f'' - 2aBG + 2aF^2 - 4AFF'' + 4fAF - 4fGH,$$



et on déduit de là son expression complète en y écrivant

$$ab'c'' + ab''c' + a'b''c + a'bc'' + a''bc' + a''b'c$$

au lieu de  $ab'c''$  ;

$$af'f'' + a'f''f + a''ff' + bg'g'' + b'g''g + b''gg' + ch'h'' + c'h''h + c''hh'$$

au lieu de  $af'f''$ , et ainsi de suite ; la somme des coefficients est  $6 + 2 \cdot 6 + 2 \cdot 6 + 4 \cdot 6 + 2 \cdot 9 + 2 \cdot 9 + 2 \cdot 9 + 4 \cdot 9 + 4 \cdot 9 + 4 \cdot 9 = 216$ , ce qui est exact, car il n'y a pas de termes qui se détruisent, et dans l'expression  $\frac{1}{384} \overline{123}^3 \cdot \overline{1'2'3'}^3 U_1 U_2 U_3$ , le nombre des termes qui composent le produit des deux déterminants carrés est  $36 \times 36 = 1296$ , chacun de ces termes est multiplié par le facteur  $8 \times 8 = 64$  introduit par la différentiation, et l'on a

$$\frac{1}{384} \times 64 \times 1296 = 216.$$

L'invariant qui vient d'être donné est lié avec l'invariant  $T$  de M. Aronhold (lequel se rapporte à une fonction ternaire cubique). C'est pour cela que je le désigne par la même lettre  $T$ , et pour fixer sa valeur j'écris

$$2T = ab'c'' + \text{etc.}$$

Cela posé, je forme les deux combinants  $C_6$  et  $C_{12}$  de la manière suivante.

*Combinant du sixième ordre  $C_6$ .* Soient  $U, V, W$  les trois fonctions quadratiques ternaires ; considérez la fonction syzygétique  $\lambda U + \mu V + \nu W$ , où  $\lambda, \mu, \nu$  sont des quantités arbitraires ; formez avec les variables  $\xi, \eta, \zeta$  le réciproquant<sup>6</sup> (multiplié par 2) de cette fonction, le résultat sera de l'ordre 2 par rapport aux coefficients de  $U, V, W$ , de l'ordre 2 par rapport à  $\lambda, \mu, \nu$ , et de l'ordre 2 par rapport à  $\xi, \eta, \zeta$  ; c. à d. ce résultat sera une fonction quadratique bi-ternaire de  $\lambda, \mu, \nu$  et  $\xi, \eta, \zeta$  dont l'invariant  $2T$  (lequel sera par conséquent de l'ordre 6 par rapport aux coefficients de  $U, V, W$ ) est le combinant  $C_6$  qu'il s'agissait de trouver.

*Combinant du douzième ordre  $C_{12}$ .* Considérez comme auparavant la fonction syzygétique  $\lambda U + \mu V + \nu W$ , formez le discriminant (multiplié par 6) de cette fonction ; le résultat sera de l'ordre 3 par rapport aux coefficients de  $U, V, W$ , et de l'ordre 3 par rapport à  $\lambda, \mu, \nu$  ; c. à d. il sera une fonction cubique ternaire de  $\lambda, \mu, \nu$ , dont

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6. Le réciproquant de  $(a, b, c, f, g, h)(x, y, z)^2$  est

$$\begin{vmatrix} \xi & \eta & \zeta \\ a & h & g & \xi \\ h & b & f & \eta \\ g & f & c & \zeta \end{vmatrix} = -(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})(\xi, \eta, \zeta)^2.$$

271]

SUR L'INVARIANT LE PLUS SIMPLE D'UNE FONCTION &amp;C.

353

l'invariant  $S$  de M. Aronhold<sup>7</sup> (lequel sera de l'ordre 12 par rapport aux coefficients de  $U, V, W$ ) est le combinant  $C_{12}$  qu'il s'agissait de trouver.

On peut exprimer ces résultats d'une manière abrégée en disant que  $C_6$  est l'invariant  $2T$  de la fonction bi-ternaire  $2R$  et que  $C_{12}$  est l'invariant  $S$  de la fonction cubique ternaire  $6D$ , en désignant par  $R$  et  $D$  le réciproquant et le discriminant de la fonction  $\lambda U + \mu V + \nu W$ .

Pour démontrer l'équation

$$12R = 16C_{12} - C_6^2$$

il suffit de faire voir que cette équation est vérifiée lorsqu'au lieu des trois fonctions données  $U, V, W$  on substitue trois fonctions de la forme  $lU + mV + nW$ , les coefficients  $l, m, n$  ayant été choisis de manière à simplifier autant que possible les expressions des trois fonctions. Il est facile de voir qu'il est permis de prendre pour les trois fonctions les formes dont s'est servi M. Sylvester dans son mémoire, savoir

$$\begin{aligned} p(\xi^2 - \zeta^2), \\ q(\eta^2 - \zeta^2), \\ y^2 + 2fyz + 2gzx + 2hxy. \end{aligned}$$

Mais il suit des recherches de M. Hesse, et M. Sylvester aussi a depuis reconnu, qu'on peut prendre pour les trois fonctions les formes encore plus simples

$$\begin{aligned} x^2 + 2lyz, \\ y^2 + 2lzx, \\ z^2 + 2lxy, \end{aligned}$$

qui se réduisent aux fonctions dérivées de la fonction cubique  $x^3 + y^3 + z^3 + 6lxyz$ . Or, en prenant ces valeurs de  $U, V, W$ , on obtient

$$\lambda U + \mu V + \nu W = (\lambda, \mu, \nu, \lambda l, \mu l, \nu l)(x, y, z)^2.$$

C. IV.

45

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7. Pour la fonction cubique ternaire  $(a, b, c, i, j, k, i_1, j_1, k_1, l)(x, y, z)^3$ , l'invariant  $S$  est

$$\begin{aligned} & -2l^2(kj_1 + k_1j + i_1j_1) \\ & + 3l(ijk + i_1j_1k_1) \\ & + l(aj_1j + bk_1i_1 + ci_1j_1) \\ & - labc \\ & - (aij_1k_1 + bj_1k_1i_1 + ck_1i_1j_1) \\ & + (j^2k_1^2 + k^2i_1^2 + i^2j_1^2) \\ & + (bcj_1k_1 + cak_1i_1 + abi_1j_1) \\ & - (aj_1^2j + bk_1^2k + ci_1^2i + ai^2k_1 + bj^2i_1 + ck^2j_1). \end{aligned}$$

THE COLLECTED MATHEMATICAL PAPERS  
of  
ARTHUR CAYLEY

Volume IV — pages 354–378

Paper 271 (conclusion), Papers 272, 273, 274, 275 (start)

Le réciproquant (multiplié par 2) est

$$= -2 \begin{vmatrix} \xi, & \eta, & \zeta \\ \xi, & \lambda, & \nu l, & \mu l \\ \eta, & \nu l, & \mu, & \lambda l \\ \zeta, & \mu l, & \lambda l, & \nu \end{vmatrix},$$

lequel, en forme développée, donne le résultat

$$\begin{aligned} & (-2l^2\nu \dots\dots\dots + 2\mu\nu \dots\dots\dots) \xi^2 \\ & + (\dots - 2l^2\nu \dots\dots\dots + 2\nu\lambda \dots) \eta^2 \\ & + (\dots\dots - 2l^2\nu \dots\dots\dots + 2\lambda\mu) \zeta^2 \\ & + (-2l\lambda^2 \dots\dots\dots + 2l\mu\nu \dots) 2\eta\zeta \\ & + (\dots - 2l\mu^2 \dots\dots\dots + 2l^2\nu\lambda \dots) 2\zeta\xi \\ & + (\dots\dots - 2l\nu^2 \dots\dots\dots + 2l^2\lambda\mu) 2\xi\eta, \end{aligned}$$

que l'on peut aussi exprimer comme fonction quadratique bi-ternaire, savoir:

$$\begin{pmatrix} -2l^2 & \cdot & \cdot & 1 & \cdot & \cdot \\ \cdot & -2l^2 & \cdot & \cdot & 1 & \cdot \\ \cdot & \cdot & -2l^2 & \cdot & \cdot & 1 \\ -2l & \cdot & \cdot & l^2 & \cdot & \cdot \\ \cdot & -2l & \cdot & \cdot & l^2 & \cdot \\ \cdot & \cdot & -2l & \cdot & \cdot & l^2 \end{pmatrix} (\lambda, \mu, \nu)^2 (\xi, \eta, \zeta)^2.$$

En représentant l'invariant  $2T$  par le développement incomplet donné ci-dessus, on obtient dans le cas dont il s'agit

$$\begin{aligned} 2T = & \begin{vmatrix} -2l^2 & -2l^2 & -2l^2 \\ \cdot & -2l^2 & \cdot \\ \cdot & \cdot & -2l^2 \end{vmatrix} - \begin{vmatrix} -2l & \cdot & \cdot \\ \cdot & -2l & \cdot \\ \cdot & \cdot & -2l \end{vmatrix} + \begin{vmatrix} l^2 & \cdot & \cdot \\ \cdot & l^2 & \cdot \\ \cdot & \cdot & l^2 \end{vmatrix} - \text{etc. etc.} \\ & + 2 \begin{vmatrix} -2l & \cdot & \cdot \\ \cdot & -2l & \cdot \\ \cdot & \cdot & -2l \end{vmatrix}, \end{aligned}$$

271]

SUR L'INVARIANT LE PLUS SIMPLE D'UNE FONCTION &C.

355

ce qui se réduit à la valeur finale

$$\begin{aligned} & -8l^6 + 2 \\ & - (4l^6 + 4l^3 + 4l^3) \\ & - (4l^3 + 4l^3 + 4l^3) \\ & - (l^6 + l^6 + l^6) \\ & + 2(-8l^3 + 2l^3) \\ & = 2(1 - 20l^3 - 8l^6). \end{aligned}$$

Les invariants  $S$  et  $T$  de la fonction cubique  $x^3 + y^3 + z^3 + 6lxyz$  sont

$$\begin{aligned} S &= -l + l^4, \\ T &= 1 - 20l^3 - 8l^6; \end{aligned}$$

l'invariant  $2T$  de la fonction quadratique bi-ternaire est donc précisément égal à  $2T$ , où  $T$  est l'invariant qui vient d'être donné, et on a

$$C_6 = 2T.$$

Le discriminant (multiplié par 6) de la fonction  $\lambda U + \mu V + \nu W$  est

$$= 6 \begin{vmatrix} \lambda, & l\nu, & l\mu \\ l\nu, & \mu, & l\lambda \\ l\mu, & l\nu, & \nu \end{vmatrix} = 6[-l^2\lambda^3 - l^2\mu^3 - l^2\nu^3 + (1 + 2l^3)\lambda\mu\nu],$$

où la fonction en parenthèses est ce que devient le Hessien de la fonction cubique  $x^3 + y^3 + z^3 + 6lxyz$ , lorsqu'on y substitue  $\lambda, \mu, \nu$  au lieu de  $x, y, z$ . Son invariant  $S$  est

$$\begin{aligned} &= (1 + 2l^3)^2 216l^2 + (1 + 2l^3)^4 \\ &= 1 + 8l^3 + 240l^6 + 464l^9 + 16l^{12}. \end{aligned}$$

Cette quantité est en même temps un invariant de la fonction cubique

$$x^3 + y^3 + z^3 + 6lxyz,$$

et comme telle elle doit s'exprimer en fonction des deux invariants  $S$  et  $T$  de cette dernière: en effet elle se réduit à

$$(1 - 20l^3 - 8l^6)^2 - 48(-l + l^4)^3 = T^2 - 48S^3.$$

Nous avons donc

$$C_{12} = T^2 - 48S^3,$$

et comme nous venons de trouver

$$C_6 = 2T,$$

l'équation  $12R = 16C_{12} - C_6^2$  se réduit à

$$\begin{aligned} 12R &= 16(T^2 - 48S^3) - 4T^2 \\ &= 12T^2 - 768S^3, \end{aligned}$$

ou enfin à

$$R = T^2 - 64S^3,$$

équation qui fait voir que le résultant  $R$  dont il s'agit est effectivement égal à  $R = (1 + 8l^3)^3$ , c. à d. au discriminant de la fonction cubique

$$x^3 + y^3 + z^3 + 6lxyz,$$

ce qui donne la vérification du théorème général. Je m'étais servi d'abord d'une analyse moins simple, en considérant le cas dans lequel on prend pour les trois fonctions les formes

$$x^2 + y^2 + z^2 + 2lyz + 2mzx + 2nxy.$$

Il vaut la peine, je crois, de donner les expressions correspondantes de  $C_6$ ,  $C_{12}$ ,  $R$ . On a ici à considérer la fonction

$$\lambda(x^2 - y^2) + \mu(x^2 + y^2 + z^2 + 2lyz + 2mzx + 2nxy) + \nu(z^2 - y^2),$$

laquelle peut s'écrire comme suit,

$$(\mu + \lambda, \mu - \lambda - \nu, \mu + \nu, l\mu, m\mu, n\mu)(x, y, z)^2.$$

Le réciproquant (multiplié par 2) est

$$= -2 \begin{vmatrix} \xi, & \eta, & \zeta \\ \xi, & \mu + \lambda, & n\mu, & m\mu \\ \eta, & n\mu, & \mu - \lambda - \nu, & l\mu \\ \zeta, & m\mu, & l\mu, & \mu + \nu \end{vmatrix}$$

et en écrivant  $a, b, c, f, g, h$  au lieu de  $1 - l^2, 1 - m^2, 1 - n^2, mn - l, nl - m, lm - n$ , cette fonction est

$$\begin{aligned} &= (2a\mu^2 - 2\lambda\mu - 2\nu\lambda - 2l^2)\xi^2 \\ &\quad + (2b\mu^2 + 2l^2 + 2\mu^2 + 2\nu\lambda)\eta^2 \\ &\quad + (2c\mu^2 - 2\mu\lambda - 2\nu\lambda - 2\lambda^2)\zeta^2 \\ &\quad + (2f\mu^2 - 2l\lambda\mu)2\eta\zeta \\ &\quad + (2g\mu^2 + 2m\lambda\mu + 2ml\nu\mu)2\zeta\xi \\ &\quad + (2h\mu^2 - 2n\nu\mu)2\xi\eta, \end{aligned}$$

ou, écrite en forme de fonction quadratique bi-ternaire,

$$= \begin{pmatrix} 0, & 2a, & -2, & 0, & 1, & -1 \\ 0, & 2b, & 0, & 1, & 1, & 1 \\ -2, & 2c, & 0, & -1, & -1, & 0 \\ 0, & 2f, & 0, & 0, & 0, & -l \\ 0, & 2g, & 0, & m, & 0, & l \\ 0, & 2h, & 0, & -n, & 0, & 0 \end{pmatrix} (\lambda, \mu, \nu)^2 (\xi, \eta, \zeta)^2.$$

271]

SUR L'INVARIANT LE PLUS SIMPLE D'UNE FONCTION &C.

357

En exprimant comme auparavant l'invariant  $2T$  par une somme de cinq termes, on trouve sans peine que ces termes sont respectivement

$$18 - 4l^2 - 4m^2 - 4n^2, \quad -4l^2, \quad -4m^2, \quad -4n^2, \quad 0;$$

l'invariant sera donc  $-8(l^2 + m^2 + n^2) + 18$ ; et l'on a pour le combinant du sixième ordre

$$C_6 = -8(l^2 + m^2 + n^2) + 18.$$

Le discriminant (multiplié par 6) de la fonction  $\lambda U + \mu V + \nu W$  est

$$\begin{aligned} &= 6 \begin{vmatrix} \mu + \lambda, & n\mu, & m\mu \\ n\mu, & \mu - \lambda - \nu, & l\mu \\ m\mu, & l\mu, & \mu + \nu \end{vmatrix} \\ &= 6(1 - l^2 - m^2 - n^2 + 2lmn)\mu^3 + 0 \cdot \nu^3 \\ &\quad + 6(m^2 - n^2)\mu^2\nu - 6\lambda^2\mu - 6\lambda\nu^2 - 6\mu^2\lambda - 6\nu\lambda^2 + 6(m^2 - l^2)\lambda\mu^2 - 6\lambda\mu\nu \\ &= (0, 6(1 - l^2 - m^2 - n^2 + 2lmn), 0, 2(m^2 - n^2), -2, -2, -2, -2, 2(m^2 - l^2), -1)(\lambda, \mu, \nu)^3 \\ &= (a, b, c, i, j, k, l_1, j_1, h, 1)(\lambda, \mu, \nu)^3, \end{aligned}$$

où l'on a posé  $a = 0$ ,  $b = 6(1 - l^2 - m^2 - n^2 + 2lmn)$ , etc. L'invariant  $S$  de cette dernière fonction est

$$\begin{aligned} &= 1 \\ &\quad - 2(-4(m^2 - n^2) + 4 - 4(m^2 - n^2)) \\ &\quad - 3(8(m^2 - n^2) + 8(m^2 - l^2)) \\ &\quad - 4b \\ &\quad + 16(m^2 - n^2) - 16(m^2 - n^2)(m^2 - l^2) + 16(m^2 - l^2) \\ &\quad + 16(m^2 - n^2)^2 + 16 + 16(m^2 - l^2) \\ &\quad + 16b \\ &= 9 + 12b + 16(l^4 + m^4 + n^4 - m^2n^2 - n^2l^2 - l^2m^2), \end{aligned}$$

ou, si l'on substitue la valeur de  $b$ ,

$$C_{12} = 16(l^4 + m^4 + n^4 - m^2n^2 - n^2l^2 - l^2m^2) + 144lmn - 72(l^2 + m^2 + n^2) + 81.$$

On vient de trouver

$$C_6 = -8(l^2 + m^2 + n^2) + 18,$$

équation qui donne

$$C_6^2 = 64(l^4 + m^4 + n^4 + 2m^2n^2 + 2n^2l^2 + 2l^2m^2) - 288(l^2 + m^2 + n^2) + 324.$$

De ces résultats combinés on conclut

$$16C_{12} - C_6^2 = 192(l^4 + m^4 + n^4 - 2m^2n^2 - 2n^2l^2 - 2l^2m^2) + 2304lmn - 864(l^2 + m^2 + n^2) + 972.$$

Évidemment l'expression du résultant est, à un facteur numérique près,

$$R = (3 + 2l + 2m + 2n)(3 + 2l - 2m - 2n)(3 - 2l + 2m - 2n)(3 - 2l - 2m + 2n),$$

ou en réduisant

$$R = 16(l^4 + m^4 + n^4 - 2m^2n^2 - 2n^2l^2 - 2l^2m^2) + 192lmn - 72(l^2 + m^2 + n^2) + 81.$$

On retrouve donc

$$12R = 16C_{12} - C_6^2,$$

équation qu'il s'agissait de vérifier.

En terminant je remarque que la plus grande partie de ce mémoire est tirée d'un manuscrit daté "Machynlleth, 13 Août 1853."

Londres, 22 Mars 1859.



## 272.

## DÉMONSTRATION D'UN THÉORÈME DE JACOBI PAR RAPPORT AU PROBLÈME DE PFAFF.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LVII. (1860), pp. 273–277.]

Dans le mémoire de Jacobi “Theoria novi multiplicatoris etc.” (t. XXIX. de ce Journal, 1845) on trouve p. 253 le passage que voici: “Methodum ad solvendum problema Pfaffianum ab ipso auctore adhibitam, data occasione observo per plures et altiores procedere integrationes quam methodus vera et genuina poscat. Quam novam methodum pro exemplo simplice explicabo. Ad aequationem differentialem

$$X_1 dx_1 + X_2 dx_2 + X_3 dx_3 + X_4 dx_4 = 0$$

per duas aequationes integrandam poscit Pfaffiana methodus. . . integrationem completam systematis trium aequationum differentialium primi ordinis inter quatuor variables ac deinde unius aequationis differentialis primi ordinis inter duas variables. Illius igitur systematis integrali uno invento, secundum illam methodum restat integratio completa duarum aequationum differentialium primi ordinis inter tres variables sive unius aequationis differentialis secundi ordinis inter duas variables ac deinde aequationis differentialis primi ordinis inter duas variables. At observo si integrali illo invento exprimat  $x_4$  per  $x_1, x_2, x_3$ , aequationem differentialem propositam abire in aliam linearem primi ordinis inter tres variables conditioni integrabilitatis satisfaciendam; cujus integrationem vidimus (voir p. 246) absolvi posse per integrationes separatas duarum aequationum differentialium primi ordinis inter duas variables. Unde in locum aequationis differentialis secundi ordinis, tantum integrandae sunt duae aequationes differentiales separatae primi ordinis, quae est reductio maxime insignis; integrationi autem aequationis differentialis primi ordinis postremo praestandae omnino supersedetur. Tractatio hujus rei gravissimae completa ac generalis alii commentationi reservanda

est.” Le théorème dont il s’agit est énoncé dans les mots “conditioni integrabilitatis satisfacientem,” mais j’ai cité le passage en entier pour montrer l’importance qu’à bon droit Jacobi attachait à la théorie dont ce théorème fait partie. On sait que le problème de la solution d’une équation à différences partielles du premier ordre n’est qu’un cas particulier du problème de Pfaff et que la solution des équations différentielles de la Dynamique et celle d’autres systèmes est intimement liée avec la question de la solution de ce cas particulier: le problème de Pfaff est ainsi de la plus haute importance dans l’analyse. Jacobi a parlé autre part d’un mémoire sur la mécanique analytique dont il s’est beaucoup occupé, lequel aurait naturellement des rapports avec celui-là; malheureusement ni l’un ni l’autre de ces mémoires n’ont paru.

Pour démontrer le théorème, considérons l’expression différentielle

$$\nabla = X_1 dx_1 + X_2 dx_2 + X_3 dx_3 + X_4 dx_4.$$

En supposant que  $a = \phi(x_1, x_2, x_3, x_4)$  soit une intégrale des équations différentielles de Pfaff, si au moyen de cette équation on exprime  $x_4$  en fonction de  $(x_1, x_2, x_3, a)$  l’expression  $\nabla$  prendra la forme

$$\nabla_1 = Y_1 dx_1 + Y_2 dx_2 + Y_3 dx_3 + A da$$

et il s’agit de faire voir que l’équation

$$Y_1 dx_1 + Y_2 dx_2 + Y_3 dx_3 = 0$$

à laquelle  $\nabla_1 = 0$  se réduit, lorsqu’on fait  $a = \text{constante}$ , sera intégrable au moyen d’un facteur, ou autrement dit, que l’expression  $Y_1 dx_1 + Y_2 dx_2 + Y_3 dx_3$  sera de la forme  $Udu$ , c. à d. qu’il existe une quantité  $u$ , fonction de  $x_1, x_2, x_3, a$ , telle qu’en traitant  $a$  comme une constante on a

$$Y_1 dx_1 + Y_2 dx_2 + Y_3 dx_3 = Udu.$$

Ainsi dans cette formule  $du$  denote  $\frac{du}{dx_1} dx_1 + \frac{du}{dx_2} dx_2 + \frac{du}{dx_3} dx_3$ , et si l’on veut regarder  $a$  comme variable il faudrait y écrire  $du - \frac{du}{da} da$  au lieu de  $du$ ; on aurait ainsi

$$\nabla_1 = U \left( du - \frac{du}{da} da \right) + A da,$$

ou en changeant la signification de  $A$ ,

$$\nabla = X_1 dx_1 + X_2 dx_2 + X_3 dx_3 + X_4 dx_4 = Udu + Ada,$$

où  $u, a$ , et de même  $U, A$ , seront des fonctions données de  $x_1, x_2, x_3, x_4$ ; c’est en effet la forme à laquelle dans le problème de Pfaff il s’agit de réduire l’expression différentielle  $\nabla$ .

Or les équations différentielles de Pfaff sont le système que voici, savoir en écrivant

$$12 = \frac{dX_2}{dx_1} - \frac{dX_1}{dx_2}, \text{ etc.},$$

272]

PAR RAPPORT AU PROBLÈME DE PFAFF.

361

ce qui implique  $12 = -21$ ,  $11 = 0$ , etc., et en introduisant la variable auxiliaire  $t$ , qui doit être éliminée, le système est

$$\begin{aligned} 0 &= X_1 dx_1 + X_2 dx_2 + X_3 dx_3 + X_4 dx_4, \\ X_1 dt &= \quad * \quad 12 dx_2 + 13 dx_3 + 14 dx_4, \\ X_2 dt &= 21 dx_1 \quad * \quad + 23 dx_3 + 24 dx_4, \\ X_3 dt &= 31 dx_1 + 32 dx_2 \quad * \quad + 34 dx_4, \\ X_4 dt &= 41 dx_1 + 42 dx_2 + 43 dx_3 \quad *, \end{aligned}$$

lequel ne contient que quatre équations indépendantes. On donne à ce système une forme plus symétrique en écrivant  $x_0$  au lieu de  $t$ , et en y mettant de plus

$$X_1 = 01 = -10, \text{ etc.}$$

les équations deviennent par ce moyen

$$\begin{aligned} 0 &= \quad 01 dx_1 + 02 dx_2 + 03 dx_3 + 04 dx_4, \\ 0 &= 10 dx_0 \quad * \quad + 12 dx_2 + 13 dx_3 + 14 dx_4, \\ 0 &= 20 dx_0 + 21 dx_1 \quad * \quad + 23 dx_3 + 24 dx_4, \\ 0 &= 30 dx_0 + 31 dx_1 + 32 dx_2 \quad * \quad + 34 dx_4, \\ 0 &= 40 dx_0 + 41 dx_1 + 42 dx_2 + 43 dx_3 \quad *, \end{aligned}$$

et on déduit de là

$$dx_0 : dx_1 : dx_2 : dx_3 : dx_4 = 1234. : 2340. : 3401. : 4012. : 0123.,$$

où 1234., etc. sont les fonctions de Jacobi que j'ai nommées "Pfaffiens," savoir on a

$$\begin{aligned} 12.34 &= 12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23, \\ 2340. &= 23 \cdot 40 + 24 \cdot 03 + 20 \cdot 34, \quad = \quad * - X_2 34 - X_3 42 - X_4 23, \\ 3401. &= 34 \cdot 01 + 30 \cdot 14 + 31 \cdot 40, \quad = X_1 34 \quad * + X_3 41 + X_4 13, \\ 4012. &= 40 \cdot 12 + 41 \cdot 20 + 42 \cdot 01, \quad = -X_1 24 - X_2 41 \quad * - X_4 12, \\ 0123. &= 01 \cdot 23 + 02 \cdot 31 + 03 \cdot 12, \quad = X_1 23 + X_2 31 + X_3 12 \quad *. \end{aligned}$$

Cette transformation se trouve en effet dans le mémoire de Jacobi "Ueber die Pfaffsche Methode etc." (t. II. de ce Journal p. 347 et suivantes, 1827), où cependant Jacobi ne s'est pas servi de l'algorithme  $X_1 = 01 = -10$  etc. au moyen duquel la forme de la solution devient si simple.

Faisons attention à présent à ce que l'équation

$$a = \phi(x_1, x_2, x_3, x_4)$$

est supposée être une intégrale des équations différentielles. On en déduit d'abord  $x_4 = \text{fonc.}(x_1, x_2, x_3, a)$ , et de là en traitant  $a$  comme constante

$$dx_4 = \frac{dx_4}{dx_1} dx_1 + \frac{dx_4}{dx_2} dx_2 + \frac{dx_4}{dx_3} dx_3,$$

équation qui doit être satisfaite par les valeurs qu'on vient de trouver pour  $dx_0 : dx_1 : dx_2 : dx_3 : dx_4$ , et qui par conséquent donne:

$$0 = 0123. - \frac{dx_4}{dx_1} 2340. - \frac{dx_4}{dx_2} 3401. - \frac{dx_4}{dx_3} 4012.$$

En y substituant pour 0123., etc. leurs valeurs, modifiées en sorte que tous les termes deviennent positifs (ce qui se fait à l'aide des relations  $12 = -21$  etc.), on trouve

$$\begin{aligned} & \frac{dx_4}{dx_1} ( \quad * \quad X_2 34 + X_3 42 + X_4 23 ) \\ & + \frac{dx_4}{dx_2} ( X_1 43 \quad * \quad + X_3 14 + X_4 31 ) \\ & + \frac{dx_4}{dx_3} ( X_1 24 + X_2 41 \quad * \quad + X_4 12 ) \\ & + ( X_1 32 + X_2 13 + X_3 21 \quad * \quad ) = 0, \end{aligned}$$

équation aux différences partielles à laquelle satisfait la variable  $x_4$ , lorsqu'au moyen de l'équation  $a = \phi(x_1, x_2, x_3, x_4)$  (qui est une intégrale des équations différentielles de Pfaff) elle est donnée en fonction de  $x_1, x_2, x_3$ . Nous allons voir que c'est cette dernière équation, dans laquelle est contenu le théorème dont la démonstration fait l'objet de cette note. En effet l'expression différentielle

$$\nabla = X_1 dx_1 + X_2 dx_2 + X_3 dx_3 + X_4 dx_4$$

ayant été transformée en

$$\nabla_1 = Y_1 dx_1 + Y_2 dx_2 + Y_3 dx_3 + A da$$

par la substitution de  $a$  au lieu de  $x_4$ , il s'en suit qu'on a les relations:

$$\begin{aligned} Y_1 &= X_1 + X_4 \frac{dx_4}{dx_1}, \\ Y_2 &= X_2 + X_4 \frac{dx_4}{dx_2}, \\ Y_3 &= X_3 + X_4 \frac{dx_4}{dx_3}, \\ A &= X_4 \frac{dx_4}{da}, \end{aligned}$$

la variable  $x_4$  contenue dans les quantités  $X_1, X_2, X_3, X_4$  doit être remplacée par son expression en fonction de  $x_1, x_2, x_3, a$ .

Cela posé, la condition pour que l'équation

$$Y_1 dx_1 + Y_2 dx_2 + Y_3 dx_3 = 0$$

soit intégrable par un facteur étant désignée par

$$Z = 0,$$

272]

PAR RAPPORT AU PROBLÈME DE PFAFF.

363

on sait que

$$Z = Y_1 \left( \frac{dY_2}{dx_3} - \frac{dY_3}{dx_2} \right) + Y_2 \left( \frac{dY_3}{dx_1} - \frac{dY_1}{dx_3} \right) + Y_3 \left( \frac{dY_1}{dx_2} - \frac{dY_2}{dx_1} \right).$$

Mais d'après les relations que l'on vient d'établir on a

$$\begin{aligned} \frac{dY_2}{dx_3} &= \frac{dX_2}{dx_3} + \frac{dX_2}{dx_4} \frac{dx_4}{dx_3} + \left( \frac{dX_4}{dx_3} + \frac{dX_4}{dx_4} \frac{dx_4}{dx_3} \right) \frac{dx_4}{dx_2} + X_4 \frac{d^2 x_4}{dx_2 dx_3}, \\ \frac{dY_3}{dx_2} &= \frac{dX_3}{dx_2} + \frac{dX_3}{dx_4} \frac{dx_4}{dx_2} + \left( \frac{dX_4}{dx_2} + \frac{dX_4}{dx_4} \frac{dx_4}{dx_2} \right) \frac{dx_4}{dx_3} + X_4 \frac{d^2 x_4}{dx_2 dx_3}, \end{aligned}$$

d'où l'on tire

$$\frac{dY_2}{dx_3} - \frac{dY_3}{dx_2} = 23 + 24 \frac{dx_4}{dx_3} + 43 \frac{dx_4}{dx_2},$$

et on trouve de même

$$\begin{aligned} \frac{dY_3}{dx_1} - \frac{dY_1}{dx_3} &= 31 + 34 \frac{dx_4}{dx_1} + 41 \frac{dx_4}{dx_3}, \\ \frac{dY_1}{dx_2} - \frac{dY_2}{dx_1} &= 12 + 14 \frac{dx_4}{dx_2} + 42 \frac{dx_4}{dx_1}; \end{aligned}$$

donc en ajoutant ces équations après les avoir multipliées par  $X_1 + X_4 \frac{dx_4}{dx_1}$ ,  $X_2 + X_4 \frac{dx_4}{dx_2}$ ,  $X_3 + X_4 \frac{dx_4}{dx_3}$ , les termes qui contiennent les produits  $\frac{dx_4}{dx_1} \frac{dx_4}{dx_2}$ , etc. se détruisent, et l'on obtient l'équation finale

$$\begin{aligned} Z &= \frac{dx_4}{dx_1} ( * X_2 34 + X_3 42 + X_4 23) \\ &\quad + \frac{dx_4}{dx_2} (X_1 43 * + X_3 14 + X_4 31) \\ &\quad + \frac{dx_4}{dx_3} (X_1 24 + X_2 41 * + X_4 12) \\ &\quad + (X_1 32 + X_2 13 + X_3 21 *), \end{aligned}$$

dont la seconde partie s'évanouit comme il a été démontré ci-dessus. Par conséquent la condition d'intégrabilité  $Z = 0$  se trouve remplie, ce qu'il s'agissait de prouver.

Londres, 3 Septembre 1859.

46–2

## 273.

## NOTE SUR LA TRANSFORMATION DE TSCHIRNHAUSEN.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LVIII. (1861), pp. 259–262.]

On trouve dans le mémoire de M. Hermite “Sur quelques théorèmes d’algèbre et la résolution de l’équation du quatrième degré” (*Comptes Rendus*, t. XLVI. p. 961, 1858) un théorème très-important relatif à la transformation de Tschirnhausen, à l’aide de laquelle une équation  $f(x) = 0$  est transformée en une autre du même degré en  $y$  par la substitution  $y = \phi x$ , où  $\phi x$  désigne une fonction rationnelle. En considérant pour fixer les idées une équation du quatrième degré

$$(a, b, c, d, e \mid x, 1)^4 = 0,$$

M. Hermite donne à l’équation  $y = \phi x$  la forme que voici,

$$y = aT + (ax + 4b)T_0 + (ax^2 + 4bx + 6c)T_1 + (ax^3 + 4bx^2 + 6cx + 4d)T_2,$$

et il fait voir que les coefficients de la transformée en  $y$  ont la propriété suivante: toute fonction de ces coefficients, laquelle exprimée en fonction de  $a, b, c, d, e, T, T_0, T_1, T_2$  ne contient pas  $T$ , est un invariant, c. à d. un invariant des deux fonctions

$$(a, b, c, d, e \mid \xi, \eta)^4, \quad (T_0, T_1, T_2 \mid \eta, -\xi)^2.$$

Cela revient à dire qu’en supposant la valeur de  $T$  déterminée de manière à faire évanouir dans l’équation en  $y$  le coefficient du second terme (de  $y^3$ ), les autres coefficients seront des invariants, de sorte que, si dans le polynome en  $y$  qui est égal à zéro on considère  $y$  comme une constante absolue, le polynome tout entier sera un invariant des deux fonctions ci-dessus mentionnées. On trouve sans peine la valeur que doit avoir  $T$ , elle est donnée par l’équation

$$0 = aT + 3bT_0 + 3cT_1 + dT_2,$$

ce qui conduit pour  $y$  à la valeur

$$y = (ax + b)T_0 + (ax^2 + 4bx + 3c)T_1 + (ax^3 + 4bx^2 + 6cx + 3d)T_2,$$

et en même temps la transformée en  $y$  aura la propriété dont il s'agit.

Par rapport à la forme de l'expression que l'on vient de trouver pour  $y$  il est bon de remarquer que les coefficients numériques qu'on y rencontre, hormis ceux du terme en  $x^0$ , ou de  $bT_0 + 3cT_1 + 3dT_2$ , sont les coefficients de la puissance  $(1, 1)^4$ , tandis que les coefficients qui ont été désignés comme faisant exception sont ceux de la puissance  $(1, 1)^3$ . Une remarque pareille s'applique au cas général. Pour démontrer le théorème énoncé, je représente l'équation qui vient d'être écrite par  $y = V$ , la transformée en  $y$  sera donc

$$(y - V_1)(y - V_2)(y - V_3)(y - V_4) = 0,$$

où  $V_1, V_2, V_3, V_4$  sont ce que devient  $V$  lorsqu'on y substitue successivement pour  $x$  les quatre racines de l'équation  $(a, b, c, d, e \mid x, 1)^4 = 0$ . Or, en considérant  $y$  comme une constante, pour que l'expression qui forme la première partie de l'équation que l'on vient d'écrire soit un invariant, les conditions à remplir consistent en ce que cette expression se réduise à zéro au moyen de l'un et l'autre des opérateurs

$$a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e - (T_1\partial_{T_0} + 2T_2\partial_{T_1}),$$

$$4b\partial_a + 3c\partial_b + 2d\partial_c + e\partial_d - (2T_0\partial_{T_1} + T_1\partial_{T_2}).$$

Ces conditions seront satisfaites si chacun des facteurs  $y - V_1$ , etc. est doué de cette même propriété, c. à d. si  $y - V$ , ou plus simplement si  $V$ , en y faisant  $x$  égal à l'une des racines de l'équation en  $x$ , se réduit à zéro au moyen de l'un et l'autre des deux opérateurs ci-dessus écrits. Je considère le premier des deux opérateurs et pour abrégé je le désigne par

$$\Delta - (T_1\partial_{T_0} + 2T_2\partial_{T_1}).$$

Pour avoir  $\Delta V$ , il faut d'abord former la valeur de  $\Delta x$ . On l'obtient en opérant avec  $\Delta$  sur l'équation  $(a, b, c, d, e \mid x, 1)^4 = 0$ , ce qui donne

$$(a, b, c, d \mid x, 1)^3 \Delta x + (a, b, c, d \mid x, 1)^3 = 0, \quad \text{ou } \Delta x = -1.$$

La partie de  $\Delta V$  qui dépend de la variation de  $x$  est par conséquent

$$-aT_1 + (-2ax - 4b)T_1 + (-3ax^2 - 8bx - 6c)T_2.$$

Pour l'autre partie de  $\Delta V$  on trouve aisément

$$aT_0 + (4ax + 6b)T_1 + (4ax^2 + 12bx + 9c)T_2,$$

et de là en ajoutant

$$\Delta V = 2(ax + b)T_1 + (ax^2 + 4bx + 3c)T_2,$$

ce qui est précisément égal à

$$(T_1\partial_{T_0} + 2T_2\partial_{T_1})V.$$

Donc  $V$  se réduit à zéro par l'opérateur

$$\Delta - (T_1 \partial_{T_0} + 2T_2 \partial_{T_1}).$$

De même en représentant le second opérateur par

$$\nabla - (2T_0 \partial_{T_1} + T_1 \partial_{T_2})$$

on trouve d'abord

$$(a, b, c, d \mid x, 1)^3 \nabla x + x (b, c, d, e \mid x, 1)^3 = 0,$$

mais en ayant égard à l'équation  $(a, b, c, d, e \mid x, 1)^4 = 0$  la valeur de  $\nabla x$  se réduit à  $\nabla x = x^2$ . La partie de  $\nabla V$  due à la variation de  $x$  est par conséquent

$$ax^2 T_0 + (2ax^3 + 4bx^2) T_1 + (3ax^4 + 8bx^3 + 6cx^2) T_2.$$

L'autre partie de  $\nabla V$  est

$$(4bx + 3c) T_0 + (4bx^2 + 12cx + 6d) T_1 + (4bx^3 + 12cx^2 + 12dx + 3e) T_2.$$

En les ajoutant, le coefficient de  $T_2$  s'évanouit en vertu de l'équation en  $x$ , et l'on trouve

$$\nabla V = (ax^2 + 4bx + 3c) T_0 + 2(ax^3 + 4bx^2 + 6cx + 3d) T_1,$$

ce qui est précisément égal à

$$(2T_0 \partial_{T_1} + T_1 \partial_{T_2}) V.$$

$V$  se réduit donc à zéro au moyen de l'opérateur

$$\nabla - (2T_0 \partial_{T_1} + T_1 \partial_{T_2}),$$

ce qui achève la démonstration dont il s'agissait. Il va sans dire que la démonstration serait conduite d'une manière semblable pour une équation de degré quelconque. Pour avoir l'exemple le plus simple, je prends les équations

$$(a, b, c, d \mid x, 1)^3 = 0,$$

$$y = (ax + b) T_0 + (ax^2 + 3bx + 2c) T_1,$$

et pour effectuer l'élimination j'écris

$$yx = (ax^2 + bx) T_0 + (-cx - d) T_1,$$

$$yx^2 = (-2bx^2 - 3cx - d) T_0 + (-cx^2 - dx) T_1.$$

Maintenant on a les trois équations

$$0 = bT_0 + 2cT_1 - y + x(aT_0 + 3bT_1) + x^2 \cdot aT_0,$$

$$0 = -dT_1 + x(bT_0 - cT_1 - y) + x^2 \cdot aT_0,$$

$$+ x(-3cT_0 - dT_1)$$

$$0 = -dT_0 + x^2(-2bT_0 - cT_1 - y),$$

donc l'équation en  $y$  est

$$\begin{vmatrix} y - bT_0 - 2cT_1, & -aT_0 - 3bT_1, & -aT_0 \\ dT_1, & y - bT_0 + cT_1, & -aT_0 \\ dT_0, & 3cT_0 + dT_1, & y + 2bT_0 + cT_1 \end{vmatrix} = 0.$$



En ordonnant ce déterminant suivant les puissances de  $y$ , les coefficients de  $y^3$ ,  $y^2$ ,  $y$ ,  $y^0$  deviennent des formes binaires en  $T_0, T_1$  des ordres 0, 1, 2, 3. En calculant les valeurs de ces quatre coefficients on les trouve respectivement

$$\begin{aligned} &= 1, \\ &= 0, \\ &= 3(ac - b^2, ad - bc, bd - c^2 \mid T_0, T_1)^2, \\ &= (a^2d - 3abc + 2b^3, 3abd - 6ac^2 + 3b^2c, -3acd + 6b^2d - 3bc^2, -ad^2 + 3bcd - 2c^3 \mid T_0, T_1)^3, \end{aligned}$$

c'est-à-dire que l'équation en  $y$  est celle-ci:

$$y^3 + 3y(ac - b^2, ad - bc, bd - c^2 \mid T_0, T_1)^2 + (a^2d - 3abc + 2b^3, 3abd - 6ac^2 + 3b^2c, -3acd + 6b^2d - 3bc^2, -ad^2 + 3bcd - 2c^3 \mid T_0, T_1)^3 = 0,$$

équation qui remplit en effet la condition ci-dessus mentionnée, d'avoir pour coefficients des invariants des deux formes:

$$(a, b, c, d \mid \xi, \eta)^3, (T_0, T_1 \mid \eta, -\xi).$$

Dans le cas particulier dont il s'agit, la fonction  $(T_0, T_1 \mid \eta, -\xi)$  est linéaire, et l'on peut même dire que les coefficients sont des covariants de la seule fonction  $(a, b, c, d \mid T_0, T_1)^3$  en y considérant  $T_0, T_1$  comme les variables.

J'ai cru qu'il y avait de l'intérêt de donner cette vérification. D'ailleurs je remarque qu'au moyen du théorème même on aurait pu trouver tout de suite l'équation en  $y$ , en écrivant d'abord  $T_1 = 0$ , ce qui donne le système

$$\begin{aligned} (a, b, c, d \mid x, 1)^3 &= 0, \\ y &= (ax + b)T_0, \end{aligned}$$

et de là

$$a^2(a, b, c, d \mid y - bT_0, aT_0)^3 = 0$$

ou enfin

$$y^3 + 3y(b^2 - ac)T_0^2 + (a^2d - 3abc + 2b^3)T_0^3 = 0.$$

Les valeurs des coefficients peuvent être complétées, eu égard à ce qu'ils doivent être des invariants de  $(a, b, c, d \mid \xi, \eta)^3, (T_0, T_1 \mid \eta, -\xi)$  (ou, comme on vient de le dire, des covariants de  $(a, b, c, d \mid T_0, T_1)^3$ ). Mais ce qu'on peut réduire la seconde équation des coefficients  $T_0, T_1$  sont au nombre de deux, ce n'est que dans le cas particulier, où il l'on aurait à une équation linéaire.

Londres, 18 Avril 1860.

# 274.

## DEUXIÈME NOTE SUR LA TRANSFORMATION DE TSCHIRNHAUSEN.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LVIII. (1861), pp. 263–269.]

A la fin de ma première note sur ce sujet j'ai appliqué la transformation de Tschirnhausen à l'équation du troisième degré mise sous la forme

$$ax^3 + 3bx^2 + 3cx + d = (a, b, c, d \mid x, 1)^3 = 0.$$

En y substituant  $\frac{1}{3}b, \frac{1}{3}c$  au lieu de  $b, c$ , cette équation se change en

$$ax^3 + bx^2 + cx + d = (a, b, c, d \mid x, 1)^3 = 0,$$

et en même temps le résultat obtenu dans ma première note s'énonce de la manière suivante:

En calculant pour l'équation

$$(a, b, c, d \mid x, 1)^3 = 0,$$

la transformée en

$$y = (ax + \frac{1}{3}b) T_0 + (ax^2 + bx + \frac{1}{3}c) T_1,$$

on obtient

$$\begin{aligned} & y^3 \\ & + \frac{1}{3}y \left\{ \begin{aligned} & T_0^2 (3ac - b^2) \\ & + T_0 T_1 (9ad - bc) \\ & + T_1^2 (3bd - c^2) \end{aligned} \right\} \\ & + \frac{1}{27} \left\{ \begin{aligned} & T_0^3 (27a^2d - 9abc + 2b^3) \\ & + T_0^2 T_1 (27abd - 18ac^2 + 3b^2c) \\ & + T_0 T_1^2 (-27acd + 18b^2d - 3bc^2) \\ & + T_1^3 (-27ad^2 + 9bcd - 2c^3) \end{aligned} \right\} = 0. \end{aligned}$$

274]

DEUXIÈME NOTE SUR LA TRANSFORMATION DE TSCHIRNHAUSEN.

369

Je vais me servir de cette formule, pour en déduire l'équation qui d'une manière analogue est la transformée de l'équation du quatrième ordre

$$(a, b, c, d, e \mid x, 1)^4 = 0.$$

J'écris d'abord

$$(a, b, c, d, e \mid x, 1)^4 = 0,$$

$$y = (ax + \frac{1}{4}b) T_0 + (ax^2 + bx + \frac{1}{2}c) T_1 + (ax^3 + bx^2 + cx + \frac{1}{4}d) T_2,$$

et je remarque qu'en faisant  $e = 0$ , le système proposé se partage en deux, dont le premier est

$$x = 0,$$

$$y = \frac{1}{4}(bT_0 + 2cT_1 + 3dT_2),$$

et le second:

$$(a, b, c, d \mid x, 1)^3 = 0,$$

$$y = (ax + \frac{1}{4}b) T_0 + (ax^2 + bx + \frac{1}{2}c) T_1 - \frac{1}{4}dT_2;$$

ou, ce qui est la même chose,

$$(a, b, c, d \mid x, 1)^3 = 0,$$

$$y + \frac{1}{4}(bT_0 + 2cT_1 + 3dT_2) = (ax + \frac{1}{4}b) T_0 + (ax^2 + bx + \frac{1}{2}c) T_1.$$

Une circonstance analogue a lieu dans l'équation en  $y$ , résultat de l'élimination du système proposé. Pour  $e = 0$  son premier membre se résout de même en deux facteurs qui égaux à zéro sont les résultats de l'élimination du premier et du second système ci-dessus écrits. Le premier de ces deux facteurs est donc

$$y - \frac{1}{4}(bT_0 + 2cT_1 + 3dT_2),$$

et le second (en vertu de la formule donnée antérieurement)

$$\{y + \frac{1}{4}(bT_0 + 2cT_1 + 3dT_2)\}^3$$

$$+ \frac{1}{3}[(3ac - b^2) T_0^2 + \text{etc.}] \{y + \frac{1}{4}(bT_0 + 2cT_1 + 3dT_2)\}$$

$$+ \frac{1}{27}[(27a^2d - 9abc + 2b^3) T_0^3 + \text{etc.}];$$

donc en multipliant les deux facteurs, et en égalant à zéro leur produit, on a la transformée en  $y$  de la forme  $(a, b, c, d, 0 \mid x, 1)^4$ . Or dans le cas général, où  $e$  est différent de zéro, les coefficients de la transformée en  $y$  sont des invariants des deux formes

$$(a, b, c, d, e \mid \xi, \eta)^4, \quad (T_0, T_1, T_2 \mid \eta, -\xi)^2.$$

Cette propriété permet de déduire leurs valeurs générales des valeurs particulières qu'ils ont pour  $e = 0$ . Je formerai de cette manière la transformée en  $y$  pour la forme  $(a, b, c, d, 0 \mid x, 1)^4$ , je passerai de là à la forme  $(a, b, c, d, 0 \mid x, 1)^4$  (ce qui se fait en écrivant  $4b, 6c, 4d$  au lieu de  $b, c, d$ ), et enfin je compléterai les valeurs des coefficients en  $y$  introduisant  $e$  au moyen de la propriété que doivent posséder les coefficients d'être des invariants des deux formes

$$(a, b, c, d, e \mid \xi, \eta)^4, \quad (T_0, T_1, T_2 \mid \eta, -\xi)^2.$$

On obtient d'abord l'équation en  $y$  sous la forme

$$(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} \mid y, 1)^4 = 0,$$

où

$$\begin{aligned} 8\mathfrak{C} = & T_0^2 (8ac - 3b^2) \\ & + T_0 T_1 (24ad - 4bc) \\ & + T_1^2 (8bd - 4c^2) \\ & + T_1 T_2 (-2bd) \\ & + T_2 T_3 (-4cd) \end{aligned}$$

$$\begin{aligned} 8\mathfrak{D} = & T_0^3 (8a^2d - 4abc + b^3) \\ & + T_0^2 T_1 (4abd - 3ac^2 + 2b^2c) \\ & + T_0^2 T_2 (-2acd + 4b^2d - 4ad^2) \\ & + T_0 T_1^2 (-12ad^2 + 16bc^2) \\ & + T_0 T_1 T_2 (-4bd^2) \\ & + T_0 T_2^2 (-7bd^2) \\ & + T_1^2 T_2 (-2cd^2) \\ & + T_2^3 (-1d^3). \end{aligned}$$

$$\begin{aligned} 256\mathfrak{E} = & T_0^4 (64a^2bd + 16ab^2c - 3b^4) \\ & + T_0^3 T_1 (-128a^2cd - 80ab^2d + 64abc^2 - 8b^3c) \\ & + (T_0^3 T_2 \text{ and other powers, including:}) \\ & T_0^2 T_1^2 (-128abcd + 64ac^3 - 48b^3d + 8b^2c^2) \\ & + T_0^2 T_1 T_2 (64abd^2 + 64ac^2d - 128bc^2d + 32bc^3) \\ & + T_0^2 T_2^2 (128acd^2 - 64bc^2d + 16c^4) \\ & + T_0 T_1^3 (-192a^2d^2 + 32abcd - 4b^2d^2) \\ & + T_0 T_1^2 T_2 (-288abd^2 + 144ac^2d + 8b^2cd) \\ & + (\text{further terms}) \\ & - 160b^2d^2 + 48bc^2d \\ & + T_0 T_1 T_2^2 (+192ad^3 - 128bcd^2 + 32c^3d) \\ & + T_0 T_2^3 (-48acd^2 + 14b^2d^2) \\ & - 144acd^2 + 8bcd^3 \\ & + T_1^4 (-48bd^3 + 8c^3d^2) \\ & + (\text{remaining terms}) \\ & + (-16d^4) \\ & + T_2^4 (-8d^4). \end{aligned}$$

[Note: The coefficient table for  $256\mathfrak{E}$  is densely packed in the original printing; the OCR layout shows several columns of monomial coefficients in  $T_0, T_1, T_2$  which have been rendered above in approximate column-by-column form. The reader is referred to the scan for the precise tabular layout.]

Ce calcul achevé et substituant la forme  $(a, b, c, d, 0 \mid x, 1)^4$  au lieu de  $(a, b, c, d, 0 \mid x, 1)^4$  (où  $4b, 6c, 4d$  au lieu de  $b, c, d$ ) on obtient tous les termes de l'équation cherchée, hormis ceux qui contiennent  $e$ : et ces derniers s'obtiennent au moyen de ce que les coefficients des différentes puissances de  $y$  se réduisent à zéro par l'opérateur

$$a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e - T_1\partial_{T_0} - 2T_2\partial_{T_1}.$$

274]

DEUXIÈME NOTE SUR LA TRANSFORMATION DE TSCHIRNHAUSEN.

371

Cela ne présente pas de difficulté, je supprime donc les calculs intermédiaires et je donne le résultat final que voici: les équations

$$(a, b, c, d, e \mid x, 1)^4 = 0,$$

$$y = (ax + b)T_0 + (ax^2 + 4bx + 3c)T_1 + (ax^3 + 4bx^2 + 6cx + 3d)T_2,$$

conduisent à la transformée:

$$(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} \mid y, 1)^4 = 0,$$

où l'on a (table dense en monomes  $T_0^i T_1^j T_2^k$  avec coefficients en  $a, b, c, d, e$ ):

[Figure: large multi-column coefficient table for  $\mathfrak{C}$ ,  $\mathfrak{D}$ ,  $\mathfrak{E}$  in monomes of  $T_0, T_1, T_2$  with polynomial entries in  $a, b, c, d, e$ . The table spans most of pp. 371–372 of the original printing.]

The table for  $\mathfrak{C}$  has the form (column entries are coefficients of  $T_0^2, T_0T_1, T_0T_2, T_1^2, T_1T_2, T_2^2$  respectively):

$$\mathfrak{C} = 2 \begin{pmatrix} ac + 8 \\ b^2 - 3 \end{pmatrix} T_0^2 + \begin{pmatrix} ad + 24 \\ bc - 8 \end{pmatrix} T_0T_1 + \begin{pmatrix} ae + 32 \\ bd - 2 \\ c^2 - 6 \end{pmatrix} T_0T_2 + \begin{pmatrix} ae + 16 \\ bd + 8 \\ c^2 - 6 \end{pmatrix} T_1^2 + \begin{pmatrix} be + 24 \\ cd - 4 \end{pmatrix} T_1T_2 + \begin{pmatrix} ce + 8 \\ d^2 - 3 \end{pmatrix} T_2^2.$$

Une table semblable s'applique à  $\mathfrak{D}$  et à  $\mathfrak{E}$ , dont les expressions complètes sont données aux pp. 371–372 de l'édition originale. [Voir scan pour la disposition détaillée.]

J'écris

$$\begin{aligned} U' &= aT_0^2 + 4bT_0T_1 + c(2T_0T_2 + 4T_1^2) + 4dT_1T_2 + eT_2^2, \\ H' &= (ac - b^2)T_0^2 + 2(ad - bc)T_0T_1 + (ae - 2bd + c^2)T_0T_2 + 4(bd - c^2)T_1^2 \\ &\quad + 2(be - cd)T_1T_2 + (ce - d^2)T_2^2, \end{aligned}$$

47–2

372

DEUXIÈME NOTE SUR LA TRANSFORMATION DE TSCHIRNHAUSEN.

[274

et je représente par  $4\Phi'$  la valeur qui vient d'être trouvée pour  $\mathfrak{D}$ . Ces expressions  $U'$ ,  $H'$ ,  $\Phi'$  sont des invariants des deux formes  $(a, b, c, d, e \mid \xi, \eta)^4$ ,  $(T_0, T_1, T_2 \mid \eta, -\xi)^2$ .

On a de plus les invariants

$$ae - 4bd + 3c^2, \quad ace - ad^2 - b^2e + 2bcd - c^3,$$

que je représente comme à l'ordinaire par  $I, J$ , et l'invariant  $T_0T_2 - T_1^2$  que je représente par  $\Theta'$ . Cela posé on a

$$\mathfrak{C} = 6H' - 2I\Theta',$$

$$\mathfrak{D} = 4\Phi',$$

$$\mathfrak{E} = IU'^2 - 3H'^2 + I^2\Theta'^2 + 12J\Theta'U' + 2I\Theta'H'.$$

La dernière de ces équations peut être vérifiée aisément, pour cela on a seulement besoin de remarquer qu'en posant  $a = e = 1$ ,  $b = d = 0$ ,  $c = \theta$ , elle devient

$$\begin{aligned} & (1 + 3\theta^2) \{T_0^2 + \theta(2T_0T_2 + 4T_1^2) + T_2^2\}^2 \\ & - 3 \{\theta T_0^2 + (1 + \theta^2)T_0T_2 - 4\theta^2T_1^2 + \theta T_2^2\}^2 \\ & + (1 + 3\theta^2)^2 (T_0T_2 - T_1^2)^2 \\ & + 12(\theta - \theta^3)(T_0T_2 - T_1^2)(T_0^2 + \theta(2T_0T_2 + 4T_1^2) + T_2^2) \\ & + 2(1 + 3\theta^2)(T_0T_2 - T_1^2)(\theta T_0^2 + (1 + \theta^2)T_0T_2 - 4\theta^2T_1^2 + \theta T_2^2) \\ & = T_0^4 \\ & \quad + T_0^3T_1(12\theta) \\ & \quad + T_0^2T_1^2(-6\theta^2 + 54\theta^4) \\ & \quad + T_0^2T_2^2(2 + 36\theta^2) \\ & \quad + T_0T_1^2T_2(-4 + 36\theta^2) \\ & \quad + T_1^4(-18\theta^2 + 81\theta^4) \\ & \quad + T_2^2T_1^2(-6\theta^2 + 54\theta^4) \\ & \quad + T_0T_2^3(12\theta) \end{aligned}$$

équation qui est identique. L'expression de l'invariant  $I$  (quadrinvariant) de la fonction  $(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} \mid y, 1)^4$  est  $\mathfrak{E} + 3\mathfrak{C}^2$  ou  $\mathfrak{E} + 3(H' - \frac{1}{3}I\Theta')^2$ , c'est-à-dire

$$IU'^2 - 3H'^2 + I^2\Theta'^2 + 12J\Theta'U' + 2I\Theta'H' + 3H'^2 + \frac{1}{3}I^2\Theta'^2 - 2I\Theta'H',$$

ou enfin

$$IU'^2 + \frac{4}{3}I^2\Theta'^2 + 12J\Theta'U',$$

274]

DEUXIÈME NOTE SUR LA TRANSFORMATION DE TSCHIRNHAUSEN.

373

ce qui est égal à

$$\frac{1}{I} [(IU' + 6J\Theta')^2 + \frac{4}{3}(I^3 - 27J^2)\Theta'^2].$$

La condition à remplir pour que cet invariant se réduise à zéro peut donc être présentée sous la forme

$$IU' + [6J \pm 2\sqrt{-\frac{1}{3}(I^3 - 27J^2)}] \Theta' = 0,$$

ce qui s'accorde avec un résultat trouvé par M. Hermite.

Il doit y avoir, ce me semble, une équation identique de la forme

$$JU'^3 - IU'^2H' + I^2\Theta'U' + MU' = -\Phi'^2$$

qui servirait à exprimer le carré de  $\Phi'$  au moyen des autres invariants  $U'$ ,  $H'$ ,  $\Theta'$ ,  $I$ ,  $J$ , mais en supposant que cette équation existe, la forme du facteur  $M$ , que je n'ai pas encore cherchée, reste à déterminer; l'invariant  $J$  (cubinvariant) de la forme  $(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} \mid y, 1)^4$  contient  $\Phi'^2$  et il faudrait employer l'identité dont je viens de parler pour réduire à sa forme la plus simple cet invariant; dans l'état actuel de la question je ne m'occupe donc pas de l'expression du cubinvariant de  $(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} \mid y, 1)^4$ .

Pour passer au cas d'une équation du cinquième ordre, on devra faire usage de la formule qui se rapporte à la forme  $(a, b, c, d, e \mid x, 1)^4$ . En faisant la substitution nécessaire on arrive à ce résultat que pour l'équation

$$(a, b, c, d, e \mid x, 1)^4 = 0$$

la transformée en

$$y = (ax + \frac{1}{4}b)T_0 + (ax^2 + bx + \frac{1}{2}c)T_1 + (ax^3 + bx^2 + cx + \frac{1}{4}d)T_2$$

est la suivante

$$(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} \mid y, 1)^4 = 0.$$

[Figure: large multi-column tables giving  $\mathfrak{C}$ ,  $\mathfrak{D}$ ,  $\mathfrak{E}$  as explicit polynomials in  $T_0, T_1, T_2$  with coefficients formed from monomes of  $a, b, c, d, e$  such as  $ac + 8$ ,  $b^2 - 3$ ,  $a^2bd + 8$ ,  $abce - 60$ , etc. The tables are densely printed on pp. 373–374 of the original and contain hundreds of monomial entries grouped column-by-column under each  $T_0^i T_1^j T_2^k$  heading. The reader is referred to the original scan for the full layout.]

The leading entries (samples from the columns) are:

$$\mathfrak{C} = \frac{1}{4} \begin{bmatrix} T_0^2 : & ac + 8, & b^2 - 3; \\ T_0 T_1 : & ad + 24, & bc - 4; \\ T_0 T_2 : & ae + 32, & bd - 2, & c^2 - 6; \\ T_1^2 : & ae + 16, & bd + 8, & c^2 - 6; \\ T_1 T_2 : & be + 24, & cd - 4; \\ T_2^2 : & ce + 8, & d^2 - 3. \end{bmatrix}$$

[Figure: similar columnar tables for  $\mathfrak{D}$  and  $\mathfrak{E}$  with respective leading entries  $T_0^3 : a^2d + 8$ ,  $abc - 4$ ,  $b^3 + 1$ , etc., and  $T_0^4 : a^3e + 256$ ,  $a^2bd - 64$ ,  $ab^2c + 16$ ,  $b^4 - 3$ , etc. The complete tables are reproduced in the scanned original.]

374

DEUXIÈME NOTE SUR LA TRANSFORMATION DE TSCHIRNHAUSEN.

[274

[Figure: continuation of the multi-column coefficient tables for  $\mathfrak{E}$ . Each column carries a header of the form  $T_0^i T_1^j T_2^k$  (with  $i + j + k = 4$ ), and the column entries are integer-weighted monomes of  $a, b, c, d, e$  summing to the coefficient of that monome in  $256\mathfrak{E}$ . Sample headers include  $T_0^4$ ,  $T_0^3 T_1$ ,  $T_0^3 T_2$ ,  $T_0^2 T_1^2$ ,  $T_0^2 T_1 T_2$ ,  $T_0^2 T_2^2$ ,  $T_0 T_1^3$ ,  $T_0 T_1^2 T_2$ ,  $T_0 T_1 T_2^2$ ,  $T_0 T_2^3$ ,  $T_1^4$ ,  $T_1^3 T_2$ ,  $T_1^2 T_2^2$ ,  $T_1 T_2^3$ ,  $T_2^4$ .]

En m'appuyant sur ce résultat j'espère être à même de trouver la formule pour l'équation du cinquième ordre.

Londres, 11 Mai 1860.



# 275.

## ON TSCHIRNHAUSEN'S TRANSFORMATION.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLI. (for the year 1861), pp. 561–578.  
Received November 7,—Read December 5, 1861.]

The memoir of M. Hermite, “Sur quelques théorèmes d’algèbre et la résolution de l’équation du quatrième degré,” *Comptes Rendus*, t. XLVI. p. 961 (1858), contains a very important theorem in relation to Tschirnhausen’s Transformation of an equation  $f(x) = 0$  into another of the same degree in  $y$ , by means of the substitution  $y = \phi x$ , where  $\phi$  is a rational and integral function of  $x$ . In fact, considering for greater simplicity a quartic equation,

$$(a, b, c, d, e \mid x, 1)^4 = 0,$$

M. Hermite gives to the equation  $y = \phi x$  the following form,

$$y = aT + (ax + 4b) B + (ax^2 + 4bx + 6c) C + (ax^3 + 4bx^2 + 6cx + 4d) D,$$

(I write  $B, C, D$  in the place of his  $T_0, T_1, T_2$ ), and he shows that the transformed equation in  $y$  has the following property: viz., every function of the coefficients which, expressed as a function of  $a, b, c, d, e, T, B, C, D$ , does not contain  $T$ , is an invariant, that is, an invariant of the two quantics

$$(a, b, c, d, e \mid X, Y)^4, \quad (B, C, D \mid Y, -X)^2.$$

This comes to saying that if  $T$  be so determined that in the equation for  $y$  the coefficient of the second term ( $y^3$ ) shall vanish, the other coefficients will be invariants; or if, in the function of  $y$  which is equated to zero, we consider  $y$  as an absolute constant, the function of  $y$  will be an invariant of the two quantics. It is easy to find the value of  $T$ ; this is in fact given by the equation

$$0 = aT + 3bB + 3cC + dD;$$

and we have thence for the value of  $y$ ,

$$y = (ax + b)B + (ax^2 + 4bx + 3c)C + (ax^3 + 4bx^2 + 6cx + 3d)D;$$

so that for this value of  $y$  the function of  $y$  which equated to zero gives the transformed equation will be an invariant of the two quantics. It is proper to notice that in the last-mentioned expression for  $y$ , all the coefficients except those of the term in  $x^0$ , or  $bB + 3cC + 3dD$ , are those of the binomial  $(1, 1)^4$ , whereas the excepted coefficients are those of the binomial  $(1, 1)^3$ ; this suffices to show what the expression for  $y$  is in the general case.

I have in the two papers, "Note sur la Transformation de Tschirnhausen," and "Deuxième Note sur la Transformation de Tschirnhausen," Crelle, t. LVIII. pp. 259 and 263 (1861), [273 and 274], obtained the transformed equations for the cubic and quartic equations; and by means of a grant from the Government Grant Fund, I have been enabled to procure the calculation, by Messrs Davis and Otter, under my superintendence, of the transformed equation for the quintic equation. The several results are given in the present memoir; and for greater completeness, I reproduce the demonstration which I have given in the former of the above-mentioned two Notes, of the general property, that the function of  $y$  is an invariant. At the end of the memoir I consider the problem of the reduction of the general quintic equation to Mr Jerrard's form  $x^5 + ax + b = 0$ .

Considering for simplicity the foregoing two equations

$$(a, b, c, d, e \mid x, 1)^4 = 0,$$

$$y = (ax + b)B + (ax^2 + 4bx + 3c)C + (ax^3 + 4bx^2 + 6cx + 3d)D;$$

let the second of these be represented by  $y = V$ , the transformed equation in  $y$  is

$$(y - V_1)(y - V_2)(y - V_3)(y - V_4) = 0,$$

where  $V_1, V_2, V_3, V_4$  are what  $V$  becomes upon substituting therein for  $x$  the roots  $x_1, x_2, x_3, x_4$  of the quartic equation respectively. Considering  $y$  as a constant, the conditions to be satisfied in order that the function in  $y$  may be an invariant are that this function shall be reduced to zero by each of the two operators

$$a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e - (B\partial_C + 2C\partial_D),$$

$$4b\partial_a + 3c\partial_b + 2d\partial_c + e\partial_d - (2C\partial_B + B\partial_C).$$

These conditions will be satisfied if each of the factors  $y - V_1$ , &c. has the property in question; that is, if  $y - V$ , or (what is the same thing) if  $V$ , supposing that  $x$  denotes therein a root of the quartic equation, is reduced to zero by each of the two operators. Considering the first operator, which for shortness I represent by

$$\Delta - (D\partial_C + 2C\partial_B),$$

in order to obtain  $\Delta V$  we require the value of  $\Delta x$ . To find it, operating with  $\Delta$  on the quartic equation, we have

$$(a, b, c, d \mid x, 1)^3 \Delta x + (a, b, c, d \mid x, 1)^3 = 0,$$

275]

ON TSCHIRNHAUSEN'S TRANSFORMATION.

377

or  $\Delta x = -1$ . In  $\Delta V$ , the part which depends on the variation of  $\Delta x$  then is

$$-aB + (-2ax - 4b)C + (-3ax^2 - 8bx - 6c)D,$$

and the other part of  $\Delta V$  is at once found to be

$$+aB + (4ax + 6b)C + (4ax^2 + 12bx + 9c)D;$$

whence, adding,

$$\Delta V = 2(ax + b)C + (ax^2 + 4bx + 3c)D,$$

and this is precisely equal to

$$(D\partial_C + 2C\partial_B)V;$$

so that  $V$  is reduced to zero by the operator  $\Delta - (D\partial_C + 2C\partial_B)$ .

Similarly, if the second operator is represented by

$$\nabla - (2C\partial_D + B\partial_C),$$

then we have

$$(a, b, c, d \mid x, 1)^3 \nabla x + x(b, c, d, e \mid x, 1)^3 = 0,$$

which by means of the equation

$$(a, b, c, d, e \mid x, 1)^4 = 0$$

is reduced to  $\nabla x = x^2$ . Hence in  $\nabla V$  the part depending on the variation of  $x$  is

$$(ax^2)B + (2ax^3 + 4bx^2)C + (3ax^4 + 8bx^3 + 6cx^2)D,$$

and the other part of  $\nabla V$  is at once found to be

$$(4bx + 3c)B + (4bx^2 + 12cx + 6d)C + (4bx^3 + 12cx^2 + 12dx + 3e)D;$$

and, adding, the coefficient of  $D$  vanishes on account of the quartic equation, and we have

$$\nabla V = (ax^2 + 4bx + 3c)B + 2(ax^3 + 4bx^2 + 6cx + 3d)C,$$

which is precisely equal to

$$(2C\partial_D + B\partial_C)V,$$

so that  $V$  is reduced to zero by the operator

$$\nabla - (2C\partial_D + B\partial_C),$$

which completes the demonstration; and the demonstration in the general case is precisely similar.

In the case of the cubic equation we have

$$(a, b, c, d \mid x, 1)^3 = 0,$$

$$y = (ax + b)B + (ax^2 + 3bx + 2c)C;$$

C, IV.

and writing the second equation in the form

$$(y - bB - 2cC) + x(-aB - 3bC) + x^2(-aC) = 0,$$

multiplying by  $x$  and reducing by the cubic equation, we have

$$dC + x(y - bB + cC) + x^2(-aB) = 0,$$

and repeating the process,

$$dB + x(3cB + dC) + x^2(y + 2bB + cC) = 0;$$

or, what is the same thing, we have the system of equations

$$\begin{vmatrix} y - bB - 2cC, & -aB - 3bC, & -aC \\ dC, & y - bB + cC, & -aB \\ dB, & 3cB + dC, & y + 2bB + cC \end{vmatrix} (1, x, x^2) = 0,$$

and the resulting equation in  $y$  is of course that formed by equating to zero the determinant formed out of the matrix in this equation. The developed expression is

$$(1, 0, \mathfrak{C}, \mathfrak{D} \mid y, 1)^3 = 0,$$

where The coefficients  $\mathfrak{C}$  and  $\mathfrak{D}$  are given column-wise in the original printing as follows:

$$\mathfrak{C} = \frac{1}{3} [(ac - b^2) B^2 + (ad - 3bc) BC + (-3bd + c^2) C^2] \quad (\text{column sums } -2, -3, -2),$$

$$\mathfrak{D} = \frac{1}{27} [(a^2d - 3abc + 2b^3) B^3 + (3abd + 3b^2c) B^2C + (6b^2d + 6acd) BC^2 + (ad^2 + 3bcd + 3) C^3].$$

[Figure: Cayley's column-table for the coefficients  $\mathfrak{C}$  and  $\mathfrak{D}$ . The columns are headed  $B^2, BC, C^2$  (for  $\mathfrak{C}$ ) and  $B^3, B^2C, BC^2, C^3$  (for  $\mathfrak{D}$ ); within each column the entries are monomes in  $a, b, c, d$  with their numerical coefficients, the column sums being zero as a check (the original adds annotations like  $+1, -1, +3$  etc., giving the positive and negative subtotals for each column).]

The sum of the coefficients in each column should here and elsewhere in the present memoir be equal to zero, and I have by way of verification annexed to each column the sums ( $\pm$  a number) of the positive and negative coefficients. The coefficients  $\mathfrak{C}, \mathfrak{D}$ , and therefore the function in  $y$ , are invariants of the two forms,

$$(a, b, c, d \mid X, Y)^3, \quad (B, C \mid Y, -X)^2;$$

or in the present case, where there are only two coefficients  $B, C$ , the coefficients  $\mathfrak{C}, \mathfrak{D}$ , and therefore also the function in  $y$ , are covariants of the single form  $(a, b, c, d \mid B, C)^3$ , considering therein  $(B, C)$  as the facients.

It may be remarked that in the present case, assuming the invariance of the function in  $y$ , we may obtain the transformed equation in a very simple manner by writing in the first instance  $C = 0$ ; this gives

$$(a, b, c, d \mid x, 1)^3 = 0, \quad y = (ax + b)B,$$

and thence

$$\frac{1}{a}(a, b, c, d \mid y - bB, aB)^3 = 0,$$

or developing,

$$y^3 + 3y(ac - b^2)B^2 + (a^2d - 3abc + 2b^3)B^3 = 0,$$

where the expressions for the coefficients are to be completed by the consideration that these coefficients are covariants of the form  $(a, b, c, d \mid B, C)^3$ . But it is only in the case in hand of a cubic equation that the transformed equation can be obtained in this manner.

In the case of a quartic equation, we have

$$(a, b, c, d, e \mid x, 1)^4 = 0,$$

$$y = (ax + b)B + (ax^2 + 2bx + 3c)C + (ax^3 + 3bx^2 + 6cx + 4d)D,$$

and these give the system of equations

$$(y - bB - 3cC - 4dD, -aB - 4bC - 6cD, -aC - 4bD, -aD \mid 1, x, x^2, x^3) = 0,$$

$$\begin{vmatrix} y - bB - 3cC - 4dD & -aB - 4bC - 6cD & -aC - 4bD & -aD \\ eD & y - bB - 3cC + dD & -aB - 4bC & -aC \\ eC & 4dC + eD & y - bB + 3cC + dD & -aB \\ eB & 4dB + eC & 6cB + 4dC + eD & y + 3bB + 3cC + dD \end{vmatrix} = 0,$$

and the transformed equation is therefore found by equating to zero the determinant formed out of the matrix contained in this equation.

The developed result, which was obtained by a different process in the “Deuxième Note” above referred to, is

$$(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} \mid y, 1)^4 = 0,$$

where the coefficients  $\mathfrak{C}, \mathfrak{D}, \mathfrak{E}$  are quartic, sextic, and octic functions of the form  $(a, b, c, d, e \mid B, C, D)^k$  in the two sets of variables; the detailed expressions, given in the original on this and the following pages as a two-dimensional array of monomial coefficients in  $(a, b, c, d, e)$  keyed against the monomial products of  $(B, C, D)$ , are reproduced below:

[Large multivariate coefficient table for  $\frac{1}{4}\mathfrak{C}$ , expressed as a sum of monomial products  $\text{monom}(a, b, c, d, e) \cdot \text{monom}(B, C, D)$  with small integer coefficients (typical entries:  $+3, -2, +6$ ); the table spans the lower half of printed page 379 of the original. The OCR of this dense table is not trustworthy and the table is here represented as a placeholder; the cleanly typeset expressions for the binary forms  $U', H', \Phi'$  on page 380, together with the identities for  $\mathfrak{C}, \mathfrak{D}, \mathfrak{E}$  given there in terms of  $U', H', \Theta'$ , suffice for all further use.]

[The remainder of the expression for  $\frac{1}{4}\mathfrak{C}$  continues at the top of page 380 in the original, together with the analogous (much larger) multivariate coefficient tables for  $\frac{1}{8}\mathfrak{D}$  and  $\frac{1}{16}\mathfrak{E}$ . These are large two-dimensional arrays of small integer coefficients indexed by the monomials of  $a, b, c, d, e$  and of  $B, C, D$ ; the OCR of these tables is not faithful. Cayley's purpose in displaying them is to enable the next identity (below), which expresses the same three quantities compactly in terms of invariants  $U', H', \Phi', \Theta'$  of the auxiliary quartic. We retain the prose, the new notation, and the identity, all of which are independently legible.]

I write

$$U' = aB^3 + 4bB^2C + c(2BD + 4C^2) + 4dCD + eD^2,$$

$H' = (ac - b^2)B^2 + 2(ad - bc)BC + (ae - 2bd + c^2)BD + 4(bd - c^2)C^2 + 2(be - cd)CD + (ce - d^2)D^2$ , and I represent by  $\Phi'$  the expression which has just been found for  $\mathfrak{E}$ . These functions  $U', H', \Phi'$  are invariants of the two forms

$$(a, b, c, d, e \mid X, Y)^4, \quad (B, C, D \mid Y, -X)^2;$$

we have, moreover, the invariants

$$I = ae - 4bd + 3c^2, \quad J = ace - ad^2 - b^2e + 2bcd - c^3$$

275]

ON TSCHIRNHAUSEN'S TRANSFORMATION.

381

which I represent as usual by  $I, J$ , and the invariant  $BD - C^2$ , which I represent by  $\Theta'$ . This being so, we have

$$\begin{aligned} \mathfrak{E} &= 6H' - 2I\Theta', \\ \mathfrak{D} &= 4\Phi', \\ \mathfrak{E} &= IU'^2 - 3H'^2 + I^2\Theta'^2 + 12J\Theta'U' + 2I\Theta'H', \end{aligned}$$

the last of which may be verified as follows:—viz., writing  $a = e = 1$ ,  $b = d = 0$ ,  $c = \theta$ , it becomes

$$\begin{aligned} &(1 + 3\theta^2)[B^2 + \theta(2BD + 4C^2) + D^2]^2 \\ &\quad - 3[\theta B^2 + (1 + \theta^2)BD - 4\theta^2C^2 + \theta D^2]^2 \\ &\quad + (1 + 3\theta^2)^2(BD - C^2)^2 \\ &\quad + 12(\theta - \theta^3)(BD - C^2)[B^2 + \theta(2BD + 4C^2) + D^2] \\ &\quad + 2(1 + 3\theta^2)(BD - C^2)[\theta B^2 + (1 + \theta^2)BD - 4\theta^2C^2 + \theta D^2] \\ &= B^4 \\ &\quad + B^3D (12\theta) \\ &\quad + B^2C^2 (-6\theta + 54\theta^3) \\ &\quad + B^2D^2 (2 + 36\theta^2) \\ &\quad + BC^2D (-4 + 36\theta^2) \\ &\quad + C^4 (1 - 18\theta^2 + 81\theta^4) \\ &\quad + CD^2 (-6\theta + 54\theta^3) \\ &\quad + BD^3 (12\theta) \\ &\quad + D^4, \end{aligned}$$

which is an identical equation. [This is in effect the identity of the “Deuxième Note,” see ante p. 372, but the left-hand side was in the original printed incorrectly, with four lines only instead of five.]

The expression for the invariant  $I$  (quadrinvariant) of the function  $(1, 0, \mathfrak{E}, \mathfrak{D}, \mathfrak{E} \mid y, 1)^4$  is  $\mathfrak{E} + 3\mathfrak{E}^2$ , or  $\Phi' + 3(H' - \frac{1}{3}I\Theta')^2$ , viz. it is

$$IU'^2 - 3H'^2 + I^2\Theta'^2 + 12J\Theta'U' + 2I\Theta'H'$$

or, finally, it is

$$IU'^2 + 9H'^2 + \frac{1}{3}I^2\Theta'^2 + 12J\Theta'U' - 2I\Theta'H',$$

which is equal to

$$\begin{aligned} &IU'^2 + \frac{1}{3}I^2\Theta'^2 + 12J\Theta'U', \\ &= \frac{1}{I}[(IU' + 6J\Theta')^2 + \frac{1}{3}(I^3 - 27J^2)\Theta'^2]; \end{aligned}$$

so that the condition in order that this invariant may be equal to zero is

$$IU' + \left[6J \pm 2\sqrt{-\frac{1}{3}(I^3 - 27J^2)}\right]\Theta' = 0,$$

which agrees with a result of M. Hermite's.

There should, I think, be an identical equation of the form

$$JU'^3 - IU'^2H' + 4H'^3 + M\Theta' = -\Phi'^2,$$

which would serve to express the square of the invariant  $\Phi'$  in terms of the other invariants  $U', H', \Theta', I, J$ ; but assuming that such an equation exists, the form of the factor  $M$  remains to be ascertained. The invariant  $J$  (cubinvariant) of the form  $(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} | y, 1)^4$  contains  $\Phi'^2$ , and it would be necessary to make use of the identical equation just referred to in order to reduce it to its simplest form; and (this being so) I have not sought for the expression of the cubinvariant of  $(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} | y, 1)^4$ .

For the quintic we have the equations

$$(a, b, c, d, e, f | x, 1)^5 = 0,$$

$$\begin{aligned} y &= (ax + b) B \\ &+ (ax^2 + 2bx + 3c) C \\ &+ (ax^3 + 3bx^2 + 6cx + 4d) D \\ &+ (ax^4 + 4bx^3 + 10cx^2 + 10dx + 5e) E, \end{aligned}$$

and this leads to the system of equations

$$\left| \begin{array}{cccccc} y - bB - 4cC - 6dD - 5eE & -aB - 5bC - 10cD - 10dE & -aC - 5bD - 10cE & -aD - 5bE & -aE \\ fE & y - bB - 4cC - 6dD + eE & -aB - 5bC - 10cD & -aC - 5bD & -aD \\ fD & 5eD + fE & y - bB - 4cC + 4dD + eE & -aB - 5bC & -aC \\ fC & 5eC + fD & 10dC + 5eD + fE & y - bB + 6cC + 4dD + eE & -aB \\ fB & 5eB + fC & 10dB + 5eC + fD & 10cB + 10dC + 5eD + fE & y + 4bB + 6cC + 4dD + eE \end{array} \right| = 0,$$

and the transformed equation is obtained by equating to zero the determinant formed out of the matrix contained in this equation.

The determinant in question was calculated by the formula

$$\begin{array}{rcl} & & \text{Div.} \\ & -12 \cdot 345 & 1 \\ & +13 \cdot 245 & 2 \\ & -14 \cdot 235 & 3 \\ & +15 \cdot 234 & 4 \\ \square = & -23 \cdot 145 & 5 \\ & +24 \cdot 135 & 6 \\ & -25 \cdot 134 & 7 \\ & -34 \cdot 125 & 8 \\ & +35 \cdot 124 & 9 \\ & -45 \cdot 123 & 10, \end{array}$$

where the duadic symbols refer to the first and fifth columns, viz. 12 is the determinant formed out of the lines 1 and 2 of these columns, and so for the other like symbols; and the triadic symbols refer to the second, third, and fourth columns, viz. 345 is the determinant formed out of the lines 3, 4, 5 of these columns, and so for the other like symbols.

The ten divisions were separately calculated. It is to be noticed that these divisions other than 4 and 6 correspond to each other in pairs, while each of the divisions 4 and 6 corresponds to itself, as thus:

$$\begin{array}{rcl} \text{Div. } 1, & -10 \\ & 2, & -9 \\ & 3, & -7 \\ & 5, & -8 \\ & 4, & -4 \\ & 6, & -6, \end{array}$$

viz. if in the place of

$$y; \ a, b, c, d, e, f; \ B, C, D, E,$$

we write

$$-y; \ f, e, d, c, b, a; \ E, D, C, B,$$

then division 1 becomes division 10 with its sign reversed, and so for divisions 2 and 9, 3 and 7, 5 and 8; while each of the divisions 4 and 6 is unaltered, except that the sign is reversed. But the corresponding divisions were each of them calculated, and the property in question was used as a verification. Another very convenient verification, which was employed for the several divisions, was obtained by putting

$$a = b = c = d = e = f = B = C = D = E = 1,$$

384

ON TSCHIRNHAUSEN'S TRANSFORMATION.

[275]

upon which supposition the determinant becomes

$$\begin{vmatrix} y-26, & -16, & -6, & -1 & \\ 1, & y-10, & -16, & -6, & -1 \\ 1, & 6, & y, & -6, & -1 \\ 1, & 6, & 16, & y+10, & -1 \\ 1, & 6, & 16, & 16, & y+15 \end{vmatrix},$$

and the values of the ten divisions respectively are

|        |        |        |        |       |
|--------|--------|--------|--------|-------|
| $y^3,$ | $y^2,$ | $y,$   | 1      | Div.  |
| 26,    | −544,  | +4608, | −16384 | 1     |
| 96,    | −540,  | +6144, | −24576 | 2     |
| 36,    | +672,  | −3584, | +24576 | +15 3 |
| 1,     | 0,     | 0,     | 0      | 4     |
| 26,    | 0,     | 0,     | 0      | 5     |
| 6,     | 0,     | 0,     | 0      | 6     |
| 16,    | +544,  | +3584, | +24576 | 7     |
| +576,  |        |        |        | 8     |
| +288,  | +6144, | +16384 |        | 9     |
| +4608, | +24576 |        |        | 10    |
| 1,     | 0,     | 0,     | 0,     | 0     |

[The exact integer entries in this table are reproduced from the scan with approximate fidelity only: the original printing on this page is broken in two places and the OCR is correspondingly unreliable; consult the IA scan, p. 384, for the authoritative listing. The sum of the ten divisions equals  $y^5$ , in agreement with the verification  $a = b = \cdots = E = 1$  stated below.]

A verification similar to this was in fact employed at each step of the calculation of a division: viz. in forming a product such as  $(\lambda X + \mu Y + \&c.)(\lambda' X + \mu' Y + \&c.)$ , where  $\lambda, \mu, \&c., \lambda', \mu', \dots$  are numerical coefficients, and  $X, Y, \&c.$  are monomial products of  $a, b, c, d, e, f$  and  $B, C, D, E$ , the sum of the numerical coefficients of the product is  $(\lambda + \mu + \&c.)(\lambda' + \mu' + \&c.)$ .

It was of course necessary to employ such verifications, as a test of the correctness of the several divisions, before proceeding to collect them together, but the collection itself affords an exceedingly good ultimate verification. The following is an exemplification: the terms in  $y$  which involve the product  $BCDE$  are obtained by the collection of the corresponding terms in the ten divisions, as follows:

[Multi-column tabular collection by monomial in  $(a, b, \dots, f)$ ; OCR illegible. Net coefficient of  $BCDE$  verifies to +1228; cross-sum check yields 0.]

275]

ON TSCHIRNHAUSEN'S TRANSFORMATION.

385

where it may be remarked that the greater part, but not all, of the component coefficients are divisible by 5. I soon observed in the process of summing the ten divisions that all the resulting coefficients should be divisible by 5 (the only exception is as to the terms in  $y^4$  which contain  $B^4, C^4, D^4$ , and  $E^4$



respectively), and the circumstance that they are so in each particular instance is as far as it goes a verification, which, however, only applies to those of the component coefficients which are not themselves divisible by 5. But it was known *à priori* (I will presently show how this is so) that the sum of the resulting coefficients should be equal to zero, and that they are so in fact is a verification as to all the coefficients. The foregoing specimen term  $BCDE$  is one which remains unaltered when  $B, C, D, E$  are changed into  $E, D, C, B$ ; and on making the further change  $a, b, c, d, e, f$  into  $f, e, d, c, b, a$ , the coefficient of  $BCDE$  remains, as it should do, unaltered; this is a verification of the coefficients of the terms  $ace^2, b^2df; ad^2e, b^2cf$ , which are respectively interchanged by the substitution in question, but not of the other terms  $c^2f, abef, acdf, b^2e^2, bcd f$ , which are respectively unaltered by the substitution. I did not employ what would have been another convenient verification of the several divisions, viz. the comparison of their values on putting therein  $a = b = c = d = e = f = 1$ , with the corresponding values as calculated independently from the determinant

$$\begin{vmatrix} y - B - 4C - 6D - 5E & -B - 5C - 10D - 10E & -C - 5D - 10E & -D - 5E & -E \\ E & y - B - 4C - 6D + E & -B - 5C - 10D & -C - 5D & -D \\ D & 5D + E & y - B - 4C + 4D + E & -B - 5C & -C \\ C & 5C + D & 10C + 5D + E & y - B + 6C + 4D + E & -B \\ B & 5B + C & 10B + 5C + D & 10B + 10C + 5D + E & y + 4B + 6C + 4D + E \end{vmatrix}$$

the calculation of the ten divisions of this determinant would however itself have been a work of some labour.

The last-mentioned determinant is  $= y^5$ ; in fact, equating it to zero, we have the transformed equation corresponding to the system of equations

$$(1, 1, 1, 1, 1, 1 \mid x, 1)^5 = 0,$$

$$y = (x + 1)B + (x^2 + 5x + 4)C + (x^3 + 5x^2 + 10x + 6)D + (x^4 + 5x^3 + 10x^2 + 10x + 4)E.$$

But the first of these equations is  $(x + 1)^5 = 0$ , and the second is

$$y = (x + 1)[B + (x + 4)C + (x^2 + 4x + 6)D + (x^3 + 4x^2 + 6x + 4)E],$$

so that for each of the five equal roots  $x = -1$ , we have  $y = 0$ , or the transformed equation in  $y$  is  $y^5 = 0$ .

386

# ON TSCHIRNHAUSEN'S TRANSFORMATION.

[275

And since upon writing  $a = b = c = d = e = f = 1$  the transformed equation becomes  $y^5 = 0$ , it is clear that in the coefficient of any monomial product of  $B, C, D, E$ , the sum of the numerical coefficients of the several monomial products of  $a, b, c, d, e, f$  must be  $= 0$ , which is the property above referred to as affording a verification of the calculated expression of the transformed equation.

The final result is that the equations

$$(a, b, c, d, e, f \mid x, 1)^5 = 0,$$

$$\begin{aligned} y &= (ax + b) B \\ &+ (ax^2 + 2bx + 3c) C \\ &+ (ax^3 + 3bx^2 + 6cx + 4d) D \\ &+ (ax^4 + 4bx^3 + 10cx^2 + 10dx + 5e) E \end{aligned}$$

give for the transformed equation in  $y$

$$(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F} \mid y, 1)^5 = 0,$$

where  $\mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}$  are functions of the coefficients  $(a, b, c, d, e, f)$  and  $(B, C, D, E)$  of the degrees 2, 3, 4, 5 respectively in the latter, and each homogeneous of degree 2, 3, 4, 5 in the former. The full developed expressions are displayed on pages 386–390 of the original as four multi-column tables of monomial coefficients, viz. one for each of  $\mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}$ , each table indexed in one direction by the monomials of  $(B, C, D, E)$

of the appropriate weight, and in the other direction by the monomials of  $(a, b, c, d, e, f)$ . We reproduce these tables in the form of placeholders, since the OCR of these dense tabular arrays is not faithful and the surrounding text suffices to identify their content:

[Table for  $\frac{1}{4}\mathfrak{C}$  (degree 2 in  $B, C, D, E$ ): header monomials  $B^2, BC, BD, C^2, BE, CD, CE, DE, D^2, E^2$ ; row monomials of degree 2 in  $a, b, c, d, e, f$  ( $ac, b^2, ad, bc, \dots, e^2$  etc.). Coefficients are small integers ranging over  $\{-16, \dots, +12\}$ .]

275] ON TSCHIRNHAUSEN'S TRANSFORMATION. 387

[Continuation of the table for  $\frac{1}{4}\mathfrak{C}$ , and beginning of the table for  $\frac{1}{8}\mathfrak{D}$  (degree 3 in  $B, C, D, E$ ): header monomials  $B^3, B^2C, BC^2, C^3, B^2D, BCD, B^2E, \dots, DE^2, E^3$ ; row monomials of degree 3 in  $a, b, c, d, e, f$ . Coefficients are small integers, generally divisible by 5 except in the four leading positions  $B^3, C^3, D^3, E^3$ .]

388 ON TSCHIRNHAUSEN'S TRANSFORMATION. [275

[Table for  $\frac{1}{16}\mathfrak{E}$  (degree 4 in  $B, C, D, E$ ): header monomials  $B^4, B^3C, B^3D, B^2C^2, B^3E, B^2CD, \dots, DE^3, E^4$ ; row monomials of degree 4 in  $a, b, c, d, e, f$  ( $a^2ce, a^2d^2, a^2cf, \dots$ ). Coefficients are integers, generally larger in magnitude than for  $\mathfrak{C}$  or  $\mathfrak{D}$ , e.g.  $\pm 70, \pm 195, \pm 1000$ .]

275] ON TSCHIRNHAUSEN'S TRANSFORMATION. 389

[Continuation of the table for  $\frac{1}{16}\mathfrak{E}$ , and beginning of the table for  $\frac{1}{32}\mathfrak{F}$  (degree 5 in  $B, C, D, E$ ): header monomials  $B^5, B^4C, B^4D, B^3C^2, B^4E, B^3CD, \dots, DE^4, E^5$ ; row monomials of degree 5 in  $a, b, c, d, e, f$ . Coefficients are larger still, e.g. entries with values such as  $\pm 245, \pm 5070, \pm 23010, \pm 25475$  appear in the column-sum rows.]

390 ON TSCHIRNHAUSEN'S TRANSFORMATION. [275

[Conclusion of the table for  $\frac{1}{32}\mathfrak{F}$ , with the final column sums that verify (after multiplication by  $\frac{1}{5}$ ) the divisibility property stated above.]

Upon writing  $(B, C, D, E) = (x^4, xy^3, xy^2, y^4)$ , the foregoing values of  $\mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}$  become covariants of the quintic  $(a, b, c, d, e, f \mid x, y)^5$ . In fact [using for the covariants of the quintic the notation of §§ 141 and 143,  $A$  the quintic itself,

$$\begin{aligned}\frac{1}{4}\mathfrak{C} &= 2C, \\ \frac{1}{8}\mathfrak{D} &= 2F, \\ \frac{1}{16}\mathfrak{E} &= A^2E - 2C^2, \\ \frac{1}{32}\mathfrak{F} &= A^2F - 2CF.\end{aligned}$$

275] ON TSCHIRNHAUSEN'S TRANSFORMATION. 391

This [the circumstance that the values thus become covariants of the quintic] is a further verification of the foregoing result.

I will conclude by showing how the formula may be applied to the reduction of the general quintic equation to Mr Jerrard's form  $x^5 + ax + b = 0$ . It was long ago remarked by Professor Sylvester that Tschirnhausen's Transformation, in its original form, gave the means of effecting this reduction. In fact, if the transforming equation be

$$y = \alpha + \beta x + \gamma x^2 + \delta x^3 + \varepsilon x^4,$$

then the equation in  $y$  is of the form

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F} \mid y, 1)^5 = 0,$$

where  $\mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}$  are, in regard to  $\alpha, \beta, \gamma, \delta, \varepsilon$ , of the degrees 1, 2, 3, 4, 5 respectively. And by assuming, say  $\alpha$  a linear function of  $\beta, \gamma, \delta, \varepsilon$ , we may make  $\mathfrak{B} = 0$ , and we have then  $\mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}$  functions of the degrees 2, 3, 4, 5 respectively of the quantities  $\beta, \gamma, \delta, \varepsilon$ : and these can be determined by means of a quadric

equation and a cubic equation in such manner that  $\mathfrak{C} = 0, \mathfrak{D} = 0$ , in which case the equation in  $y$  will be of the required form. For considering  $\beta, \gamma, \delta, \varepsilon$  as the coordinates of a point in space, the equations  $\mathfrak{C} = 0, \mathfrak{D} = 0$  will be the equations of a quadric surface and a cubic surface respectively, and if  $\beta, \gamma, \delta, \varepsilon$  be the coordinates of a point on the curve of intersection, the required conditions will be satisfied. And by combining with the equation of the quadric surface, the equation of any tangent plane thereto (or by the different process which is made use of in the sequel), we may, by means of a quadric equation, find a generating line of the quadric surface, and then, by means of a cubic equation, a point of intersection of this line with the cubic surface, i.e. a point the coordinates whereof give the required values  $\beta, \gamma, \delta, \varepsilon$ . And similarly for the new form of Tschirnhausen's Transformation; the only difference being that, starting with an equation in  $y$  which contains the four arbitrary quantities  $B, C, D, E$ , we obtain in the first instance an equation

$$(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F} \mid y, 1)^5 = 0$$

wanting the second term. And then  $B, C, D, E$  are to be so determined that  $\mathfrak{C} = 0, \mathfrak{D} = 0$ .

To proceed with the reduction, I write the foregoing value of  $\mathfrak{C}$  in the form

$$\frac{1}{4}\mathfrak{C} = \left( \begin{array}{cccc} 4ac - 4b^2 & 6ad - 6bc & 4ae - 4bd & af - be \\ 6ad - 6bc & 4ae + 20bd - 24c^2 & af + 15be - 16cd & 4bf - 4ce \\ 4ae - 4bd & af + 15be - 16cd & 4bf + 20ce - 24d^2 & 6cf - 6de \\ af - be & 4bf - 4ce & 6cf - 6de & 4df - 4e^2 \end{array} \middle| B, C, D, E \right)^2,$$

which for shortness may be represented by

$$\frac{1}{4}\mathfrak{C} = \left( \begin{array}{cccc} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & p \end{array} \middle| B, C, D, E \right)^2,$$

392

ON TSCHIRNHAUSEN'S TRANSFORMATION.

[275

or say

$$\frac{1}{4}\mathfrak{C} = (\Omega \mid B, C, D, E)^2.$$

Now, by a formula in my memoir "On the Automorphic Linear Transformation of a Bipartite Quadric Function," *Phil. Trans.*, vol. CXLVIII. (1858), see p. 44, [153], if  $T$  denote any skew symmetric matrix of the order 4, then if

$$(B, C, D, E) = (\Omega^{-1}(\Omega - T)(\Omega + T)^{-1}\Omega \mid B', C', D', E'),$$

in which formula  $\Omega^{-1}, \Omega - T, (\Omega + T)^{-1}, \Omega$  are all matrices which are to be compounded together into a single matrix, we have identically

$$(\Omega \mid B, C, D, E)^2 = (\Omega \mid B', C', D', E')^2.$$

Let  $Q$  denote the determinant  $|\Omega + T|$ , then the terms of the matrix  $(\Omega + T)^{-1}$  are respectively divided by  $Q$ , and we may write

$$(\Omega + T)^{-1} = \frac{1}{Q} Q(\Omega + T)^{-1},$$

where  $Q(\Omega + T)^{-1}$  is the matrix obtained from the matrix  $(\Omega + T)^{-1}$  by multiplying each term by  $Q$ , the terms of  $Q(\Omega + T)^{-1}$  being thus rational and integral functions of the terms of the matrix  $(\Omega + T)$ . Hence if, instead of the before-mentioned relation between  $(B, C, D, E)$  and  $(B', C', D', E')$ , we assume

$$(B, C, D, E) = (\Omega^{-1}(\Omega - T)Q(\Omega + T)^{-1}\Omega \mid B', C', D', E'),$$

we find

$$(\Omega \mid B, C, D, E)^2 = Q^2(\Omega \mid B', C', D', E')^2.$$

If here the matrix  $T$  is such that we have  $Q = 0$ , i.e.  $\text{Det.}(\Omega + T) = 0$  (which is a quadric relation between the terms of the skew matrix, that is, each term is contained therein in the first and second powers only), then the equation becomes

$$(\Omega \mid B, C, D, E)^2 = 0.$$

It is clear that this can only be the case in consequence of the coefficients of trans- formation in the equation

$$(B, C, D, E) = (\Omega^{-1}(\Omega - T)Q(\Omega + T)^{-1}\Omega \mid B', C', D', E')$$

being such that there shall exist at least two linear relations between the quantities  $(B, C, D, E)$ , and I assume (without stopping to prove it) that they are such that the number of such linear relations is in fact two. That is, the last-mentioned equation establishes between the quantities  $(B, C, D, E)$  two linear relations, in virtue whereof  $\mathfrak{C} = 0$ . And this being so, we may, without loss of generality, write  $D' = 0$ ,  $E' = 0$ , or put

$$(B, C, D, E) = (\Omega^{-1}(\Omega - T)Q(\Omega + T)^{-1}\Omega \mid B', C', 0, 0);$$

275]

ON TSCHIRNHAUSEN'S TRANSFORMATION.

393

so that  $B, C, D, E$  are linear functions of  $B', C'$ , such that  $\mathfrak{C} = 0$ . And then substituting these values for  $(B, C, D, E)$ , we find  $\mathfrak{D}$  a cubic function of  $B', C'$ ; so that, putting  $\mathfrak{D} = 0$ , we have a cubic equation to determine the ratio  $B' : C'$ .

The foregoing reasoning presents no real difficulty, but it is expressed by means of a very condensed notation, and it may be proper to illustrate it by the case of the quadric function  $x^2 + y^2 + z^2$ . Considering the equations

$$\begin{aligned} x &= (1 + \lambda^2 - \mu^2 - \nu^2)x' + 2(\lambda\mu - \nu)y' + 2(\lambda\nu + \mu)z', \\ y &= 2(\lambda\mu + \nu)x' + (1 - \lambda^2 + \mu^2 - \nu^2)y' + 2(\mu\nu - \lambda)z', \\ z &= 2(\nu\lambda - \mu)x' + 2(\mu\nu + \lambda)y' + (1 - \lambda^2 - \mu^2 + \nu^2)z', \end{aligned}$$

these equations, if the expressions for  $x, y, z$  had been divided by  $1 + \lambda^2 + \mu^2 + \nu^2$ , would have given

$$x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2.$$

Hence they actually do give

$$x^2 + y^2 + z^2 = (1 + \lambda^2 + \mu^2 + \nu^2)^2(x'^2 + y'^2 + z'^2);$$

or if

$$1 + \lambda^2 + \mu^2 + \nu^2 = 0,$$

they give

$$x^2 + y^2 + z^2 = 0.$$

But if

$$1 + \lambda^2 + \mu^2 + \nu^2 = 0,$$

then

$$\begin{vmatrix} 1 + \lambda^2 - \mu^2 - \nu^2 & 2(\lambda\mu - \nu) & 2(\lambda\nu + \mu) \\ 2(\lambda\mu + \nu) & 1 - \lambda^2 + \mu^2 - \nu^2 & 2(\mu\nu - \lambda) \\ 2(\nu\lambda - \mu) & 2(\mu\nu + \lambda) & 1 - \lambda^2 - \mu^2 + \nu^2 \end{vmatrix} = 0,$$

so that we have

$$x : y : z = 1 + \lambda^2 - \mu^2 - \nu^2 : 2(\lambda\mu + \nu) : 2(\nu\lambda - \mu),$$

which is the same result as would have been found by writing  $y' = z' = 0$ , and which comes to saying that  $x, y, z$  are not independent, but are connected by two linear relations.

The equation  $\text{Det.}(\Omega + T) = 0$ , written at length, will be

$$\begin{vmatrix} a, & h - \tau, & g + \sigma, & l + \lambda \\ h + \tau, & b, & f - \rho, & m + \mu \\ g - \sigma, & f + \rho, & c, & n + \nu \\ l - \lambda, & m - \mu, & n - \nu, & p \end{vmatrix} = 0,$$

394

ON TSCHIRNHAUSEN'S TRANSFORMATION.

[275

where  $\lambda, \mu, \nu, \rho, \sigma, \tau$  are the arbitrary constituents of the skew matrix; or developing, this is

$$\begin{vmatrix} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & p \end{vmatrix}$$

terms quadratic in  $\lambda, \mu, \nu, \rho, \sigma, \tau$  with coefficients  
+  $\left( \begin{array}{l} \text{bilinear in the entries } a, b, c, f, g, h, l, m, n, p \text{ of } \Omega, \\ \text{viz. the developed expansion as displayed in the original.} \end{array} \right) + (\lambda\rho + \mu\sigma + \nu\tau)^2 = 0,$

the first term whereof, substituting for  $(a, b, c, f, g, h, l, m, n, p)$  their values, is in fact equal to the discriminant  $a^4f^4 + \&c.$  of the quintic  $(a, b, c, d, e, f \mid x, y)^5$ . There is no loss of generality in putting all but two of the quantities  $(\lambda, \mu, \nu, \rho, \sigma, \tau)$  equal to zero; in fact this leaves in the formulae a single arbitrary quantity, which is the right number, since the ratios  $B : C : D : E$  have to satisfy only the two conditions  $\mathfrak{C} = 0, \mathfrak{D} = 0$ .

[An Addition of a half page, dated Nov. 10, 1862, and referring to the Memoir “On a New Auxiliary Equation in the Theory of Equations of the Fifth Order”, 268, is printed at the end of that paper, ante p. 324.]

276.

ON THE ANALYTICAL THEORY OF THE CONIC.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLII. (for the year 1862), pp. 639–662.  
Received May 8,—Read May 15, 1862.]

THE decomposition into its linear factors of a decomposable quadric function cannot be effected in a symmetrical manner otherwise than by formulae containing super- numerary arbitrary quantities; thus, for a binary quadric (which of course is always decomposable) we have

$$(a, b, c \mid x, y)^2 = \frac{1}{(a, b, c \mid x', y')^2} \{ (a, b, c \mid x, y \mid x', y') \pm \sqrt{ac - b^2} (xy' - x'y) \}^2$$

or the expression for a linear factor is

$$\frac{1}{\sqrt{(a,b,c|x',y')^2}} \{ (a,b,c | x,y | x',y') \pm \sqrt{ac-b^2} (xy' - x'y) \},$$

which involves the arbitrary quantities  $(x', y')$ . And this appears to be the reason why, in the analytical theory of the conic, the questions which involve the decomposition of a decomposable ternary quadric have been little or scarcely at all considered: thus, for instance, the expressions for the coordinates of the points of intersection of a conic by a line (or say the line-equations of the two ineunts), and the equations for the tangents (separate each from the other) drawn from a given point not on the conic, do not appear to have been obtained. These questions depend on the decomposition of a decomposable ternary quadric, which decomposition itself depends on that for the simplest case, when the quadric is a perfect square. Or we may say that in the first instance they depend on the transformation of a given quadric function  $U = (* | x, y, z)^2$  into the form  $W^2 + V$ , where  $W$  is a linear function, given save as to a constant factor (that is,  $W = 0$  is the equation of a given line), and  $V$  is a decomposable quadric function, which is ultimately decomposed into its linear factors,  $= QR$ , so that we have  $U = W^2 + QR$ .

396

ON THE ANALYTICAL THEORY OF THE CONIC.

[276

The formula for this purpose, which is exhibited in the eight different forms I, II, III, IV, I (bis), II (bis), III (bis), IV (bis), is the analytical basis of the whole theory; and the greater part of the memoir relates to the establishment of these forms.

The solution of the geometrical questions above referred to is (as shown in the memoir) involved in and given immediately by these forms. It is also shown that the formulae are greatly simplified in the case e.g. of tangents drawn to a conic from a point in a conic having double contact with the first-mentioned conic, and that in this case they lead to the linear Automorphic Transformation of the ternary quadric. The memoir concludes with some formulae relating to the case of two conics, which however is treated of in only a cursory manner.

*Article Nos. 1–17, relating to a single conic.*

1. The point-equation of the conic is

$$(a, b, c, f, g, h | x, y, z)^2 = 0,$$

which expresses that the point  $(x, y, z)$  is an ineunt of the conic.

The line-equation of the same conic is

$$\begin{vmatrix} \xi, & \eta, & \zeta, & 0 \\ \xi, & a, & h, & g \\ \eta, & h, & b, & f \\ \zeta, & g, & f, & c \end{vmatrix} = 0,$$

or putting

$$(A, B, C, F, G, H) = (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch),$$

the line-equation is

$$(A, B, C, F, G, H | \xi, \eta, \zeta)^2 = 0,$$

which expresses that the line  $(\xi, \eta, \zeta)$  (that is, the line the point-equation whereof is  $\xi x + \eta y + \zeta z = 0$ ) is a tangent of the conic. We are thus in the analytical theory of the conic concerned with the quadrics  $(a, b, c, f, g, h | x, y, z)^2$  and  $(A, B, C, F, G, H | \xi, \eta, \zeta)^2$ , which are the characteristics or nilfactums of these equations respectively.

2. I put also

$$K = \begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix},$$

or, what is the same thing,

$$K = abc - af^2 - bg^2 - ch^2 + 2fgh.$$

3. It may be convenient to notice that when  $(a, \dots | x, y, z)^2$  breaks up into factors, the conic the equation whereof is  $(a, \dots | x, y, z)^2 = 0$  becomes a pair of lines; and that, when  $(a, \dots | x, y, z)^2$  is a perfect square, the conic becomes a pair of coincident lines, or say a twofold line. But a pair of lines, distinct or coincident, cannot be represented by a line-equation. The analytical formulae presently given show that in the former case  $(A, \dots | \xi, \eta, \zeta)^2$  is the square of a linear function, which equated to zero gives the line-equation of the point of intersection of the two lines, or node of the conic; and the equation  $(A, \dots | \xi, \eta, \zeta)^2 = 0$  accordingly represents such point considered as a pair of coincident points, or say a twofold point. But in the latter case, where the conic is a twofold line,  $(A, \dots | \xi, \eta, \zeta)^2$  is identically equal to zero, and the line-equation  $(A, \dots | \xi, \eta, \zeta)^2 = 0$  is a mere identity  $0 = 0$ , thus ceasing to have any signification at all. And the like remarks apply to the conic as represented by the line-equation  $(A, \dots | \xi, \eta, \zeta)^2 = 0$ , the conic here breaking up into a pair of distinct or coincident points, &c.

4. It is proper to remark also that

$$(a, \dots | x', y', z' | x, y, z) = 0$$

is the equation of the polar of the point  $(x', y', z')$  in regard to the conic, and that

$$(A, \dots | \xi', \eta', \zeta' | \xi, \eta, \zeta) = 0$$

is the line-equation of the pole of the line  $(\xi', \eta', \zeta')$ ; or, what is the same thing, the point-coordinates of the pole are

$$A\xi' + H\eta' + G\zeta' : H\xi' + B\eta' + F\zeta' : G\xi' + F\eta' + C\zeta'.$$

5. The inverse matrix is

$$\begin{pmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{pmatrix}^{-1} = \frac{1}{K} \begin{pmatrix} A, & H, & G \\ H, & B, & F \\ G, & F, & C \end{pmatrix},$$

but it is convenient to disregard the factor  $\frac{1}{K}$ , and speak of  $(A, B, C, F, G, H)$  as the inverse or reciprocal coefficients. The equation just written down implies the relations  $Aa + Hh + Gg = K$ ,  $Ah + Hb + Gf = 0$ , &c., which may be arranged in two different ways as a system of nine equations.

6. We have also

$$(BC - F^2, CA - G^2, AB - H^2, GH - AF, HF - BG, FG - CH) = K(a, b, c, f, g, h),$$

and

$$ABC - AF^2 - BG^2 - CH^2 + 2FGH = K^2,$$

which are well-known theorems.

7. I notice also the theorem

$$(a, \dots | x, y, z)^2 \cdot (a, \dots | x', y', z')^2 - [(a, \dots | x, y, z | x', y', z')]^2 = (A, \dots | yz' - y'z, zx' - z'x, xy' - x'y)^2,$$

which is much used in the sequel: it may be mentioned, in passing, that this is included in the more general theorem

$$\begin{vmatrix} (a, \dots | x, y, z | l, m, n) & (a, \dots | x', y', z' | l, m, n) \\ (a, \dots | x, y, z | l', m', n') & (a, \dots | x', y', z' | l', m', n') \end{vmatrix} \\ = (A, \dots | yz' - y'z, zx' - z'x, xy' - x'y | mn' - m'n, nl' - n'l, lm' - l'm),$$

which is at once deducible from

$$\begin{vmatrix} Ll + Mm + Nn & L'l + M'm + N'n \\ Ll' + Mm' + Nn' & L'l' + M'm' + N'n' \end{vmatrix} = (MN' - M'N)(mn' - m'n) + (NL' - N'L)(nl' - n'l) + (LM' - L'M)(lm' - l'm),$$

by writing therein

$$(L, M, N) = (ax + hy + gz, hx + by + fz, gx + fy + cz),$$

$$(L', M', N') = (ax' + hy' + gz', hx' + by' + fz', gx' + fy' + cz').$$

8. Suppose now that

$$(a, b, c, f, g, h \mid x, y, z)^2$$

breaks up into factors, or say that we have

$$(a, b, c, f, g, h \mid x, y, z)^2 = 2(\alpha x + \beta y + \gamma z)(\alpha' x + \beta' y + \gamma' z),$$

the values of the coefficients  $(a, \dots)$  then are

$$(a, b, c, f, g, h) = (2\alpha\alpha', 2\beta\beta', 2\gamma\gamma', \beta\gamma' + \beta'\gamma, \gamma\alpha' + \gamma'\alpha, \alpha\beta' + \alpha'\beta),$$

and forming from these the inverse coefficients  $(A, \dots)$  and the discriminant  $K$ , we find

$$(A, B, C, F, G, H) = -(\beta\gamma' - \beta'\gamma, \gamma\alpha' - \gamma'\alpha, \alpha\beta' - \alpha'\beta)^2,$$

$$K = 0.$$

9. The last-mentioned equation,  $K = 0$ , is the condition in order that  $(a, \dots \mid x, y, z)^2$  may break up into factors; and when it does so, we have

$$(A, \dots \mid \xi, \eta, \zeta)^2 = -[(\beta\gamma' - \beta'\gamma, \gamma\alpha' - \gamma'\alpha, \alpha\beta' - \alpha'\beta \mid \xi, \eta, \zeta)]^2,$$

that is,  $(a, \dots \mid x, y, z)^2$  breaking up into factors,  $(A, \dots \mid \xi, \eta, \zeta)^2$  is a perfect square; and equating it to zero, we have

$$[(\beta\gamma' - \beta'\gamma, \gamma\alpha' - \gamma'\alpha, \alpha\beta' - \alpha'\beta \mid \xi, \eta, \zeta)]^2 = 0;$$

which,  $(\xi, \eta, \zeta)$  being line-coordinates, gives (as a two-fold point) the point of intersection of the lines  $(\alpha, \beta, \gamma), (\alpha', \beta', \gamma')$ , that is, the lines  $\alpha x + \beta y + \gamma z = 0, \alpha' x + \beta' y + \gamma' z = 0$ .

276]

# ON THE ANALYTICAL THEORY OF THE CONIC.

399

10. If  $(a, \dots \mid x, y, z)^2$  is a perfect square, then  $\alpha' : \beta' : \gamma' = \alpha : \beta : \gamma$ ; whence not only, as before,  $K = 0$ , but the coefficients  $(A, B, C, F, G, H)$  all vanish (this implies the first-mentioned condition,  $K = 0$ ); and the line-equation  $(A, \dots \mid \xi, \eta, \zeta)^2 = 0$  becomes the mere identity  $0 = 0$ .

11. Conversely if  $K = 0$ , then  $(a, \dots \mid x, y, z)^2$  breaks up into factors; and if  $(A, B, C, F, G, H)$  all vanish, then  $(a, \dots \mid x, y, z)^2$  is a perfect square. The conclusions stated *ante*, No. 3, are thus sustained.

12. I assume, first, that  $(a, \dots \mid x, y, z)^2$  is a perfect square (No. 13); and secondly, that it breaks up into factors (No. 14); and I proceed to inquire how in the one case the root, and in the other case the factors, can be determined in a symmetrical form.

13. Considering the before-mentioned identical equation

$$(a, \dots \mid x, y, z)^2 \cdot (a, \dots \mid x', y', z')^2 - [(a, \dots \mid x, y, z \mid x', y', z')]^2 = (A, \dots \mid yz' - y'z, zx' - z'x, xy' - x'y)^2,$$

if  $(a, \dots \mid x, y, z)^2$  is a perfect square, then by what precedes, the right-hand side of the equation vanishes, and we have

$$(a, \dots \mid x, y, z)^2 = \frac{[(a, \dots \mid x, y, z \mid x', y', z')]^2}{(a, \dots \mid x', y', z')^2},$$

and the root of  $(a, \dots \mid x, y, z)^2$  is thus seen to be

$$\pm \frac{(a, \dots \mid x, y, z \mid x', y', z')}{\sqrt{(a, \dots \mid x', y', z')^2}},$$

an expression which involves the quantities  $(x', y', z')$ , the values whereof may be assumed at pleasure without altering the value of the expression. For instance, assuming for  $(x', y', z')$  the values  $(1, 0, 0), (0, 1, 0), (0, 0, 1)$  successively, the different values of the expression are

$$\frac{ax + hy + gz}{\sqrt{a}}, \quad \frac{hx + by + fz}{\sqrt{b}}, \quad \frac{gx + fy + cz}{\sqrt{c}}.$$



But if, as assumed,  $(a, \dots | x, y, z)^2$  be a perfect square  $= (\alpha x + \beta y + \gamma z)^2$ , then

$$(a, b, c, f, g, h) = (\alpha^2, \beta^2, \gamma^2, \beta\gamma, \gamma\alpha, \alpha\beta),$$

and each of the foregoing values becomes equal to the root  $\alpha x + \beta y + \gamma z$ . It is somewhat singular that it is not possible to obtain symmetrical formulae without employing in this manner supernumerary arbitrary quantities such as  $(x', y', z')$ .

14. Next, if  $(a, \dots | x, y, z)^2$ , instead of being a perfect square, only breaks up into factors, then in the foregoing identical equation the right-hand side is a perfect square, and by the formula just obtained its value is

$$\frac{[(A, \dots | yz' - y'z, zx' - z'x, xy' - x'y | X, Y, Z)]^2}{(A, \dots | X, Y, Z)^2}$$

400

ON THE ANALYTICAL THEORY OF THE CONIC.

[276

where  $(X, Y, Z)$  are supernumerary arbitrary quantities. The identical equation then gives

$$(a, \dots | x, y, z)^2 = \frac{1}{(a, \dots | x', y', z')^2} \left[ [(a, \dots | x, y, z | x', y', z')]^2 + \frac{[(A, \dots | yz' - y'z, \dots | X, Y, Z)]^2}{(A, \dots | X, Y, Z)^2} \right]$$

and consequently

$$(a, \dots | x, y, z)^2 = \frac{1}{(a, \dots | x', y', z')^2} \text{Product of} \\ \left\{ (a, \dots | x, y, z | x', y', z') \pm \frac{(A, \dots | X, Y, Z | yz' - y'z, zx' - z'x, xy' - x'y)}{\sqrt{(A, \dots | X, Y, Z)^2}} \right\},$$

a formula which exhibits the decomposition of  $(a, \dots | x, y, z)^2$  assumed to be a function which breaks up into factors; the formula contains the two sets of supernumerary arbitrary quantities  $(x', y', z')$  and  $(X, Y, Z)$ . It will be remembered that  $(A, \dots)$  denotes the system of inverse or reciprocal coefficients  $(bc - f^2, \dots)$ .

15. Consider the formula

$$(a, b, c, f, g, h | \eta\zeta' - \eta'\zeta, \zeta\xi' - \zeta'\xi, \xi\eta' - \xi'\eta)^2 = (a, b, c, f, g, h | \xi', \eta', \zeta')^2,$$

which gives

$$\begin{aligned} a &= c\eta^2 + b\zeta^2 - 2f\eta\zeta, \\ b &= a\zeta^2 + c\xi^2 - 2g\zeta\xi, \\ c &= b\xi^2 + a\eta^2 - 2h\xi\eta, \\ f &= -a\eta\zeta - f\xi^2 + g\xi\eta + h\zeta\xi, \end{aligned}$$

and from these we deduce

$$\begin{aligned} g &= -b\zeta\xi + f\xi\eta - g\eta^2 + h\eta\zeta, \\ h &= -c\xi\eta + f\zeta\xi + g\eta\zeta - h\zeta^2; \end{aligned}$$

viz.

$$\begin{pmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{pmatrix} \begin{pmatrix} \xi \\ \eta \\ \zeta \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

viz.  $a\xi + h\eta + g\zeta = 0$ , &c.

Also

$$(bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch) = (\xi', \eta', \zeta')^2 (A, B, C, F, G, H | \xi, \eta, \zeta)^2,$$

that is

$$bc - f^2 = \xi^2(A, B, C, F, G, H | \xi, \eta, \zeta)^2, \quad \&c.$$

Hence also

$$(bc - f^2, \dots | l, m, n)^2 = (l\xi + m\eta + n\zeta)^2 (A, \dots | \xi, \eta, \zeta)^2,$$

$$(bc - f^2, \dots | l, m, n | l', m', n') = (l\xi + m\eta + n\zeta)(l'\xi + m'\eta + n'\zeta) (A, \dots | \xi, \eta, \zeta)^2;$$

and moreover

$$abc - af^2 - bg^2 - ch^2 + 2fgh = 0.$$

276]

ON THE ANALYTICAL THEORY OF THE CONIC.

401

16. The last equation shows that  $(a, \dots | \eta\zeta' - \eta'\zeta, \zeta\xi' - \zeta'\xi, \xi\eta' - \xi'\eta)^2$ , considered as a function of  $(\xi', \eta', \zeta')$ , breaks up into factors. Or since the expression is not altered by interchanging  $(\xi', \eta', \zeta')$  and  $(\xi, \eta, \zeta)$ , the same expression, considered as a function of  $(\xi, \eta, \zeta)$ , breaks up into factors. It is in fact easy to see that any quantic whatever,  $(* | \eta\zeta' - \eta'\zeta, \zeta\xi' - \zeta'\xi, \xi\eta' - \xi'\eta)^2$ , considered as a function of  $(\xi, \eta, \zeta)$ , breaks up into linear factors; for in virtue of the equation  $\xi'(\eta\zeta' - \eta'\zeta) + \eta'(\zeta\xi' - \zeta'\xi) + \zeta'(\xi\eta' - \xi'\eta) = 0$ , any one of the quantities  $\eta\zeta' - \eta'\zeta, \zeta\xi' - \zeta'\xi, \xi\eta' - \xi'\eta$  can be expressed as a linear function of the other two; so that the quantic can be expressed as a linear function of any two of the three quantities; and *quel* homogeneous function of two quantities, it of course breaks up into factors, linear functions of these two quantities.

We may in all the formulae interchange  $(x', y', z')$  and  $(x, y, z)$ , writing  $(a', b', c', f', g', h')$  in the place of  $(a, b, c, f, g, h)$ .

17. Putting, in like manner,

$$(A, B, C, F, G, H | yz' - y'z, zx' - z'x, xy' - x'y)^2 = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} | x, y, z)^2,$$

we obtain

$$\begin{aligned}\mathfrak{A} &= Cy^2 + Bz^2 - 2Fyz, \\ \mathfrak{B} &= Az^2 + Cx^2 - 2Gzx, \\ \mathfrak{C} &= Bx^2 + Ay^2 - 2Hxy, \\ \mathfrak{F} &= -Ayz - Fx^2 + Gxy + Hzx, \\ \mathfrak{G} &= -Bzx + Fxy - Gy^2 + Hyz, \\ \mathfrak{H} &= -Cxy + Fzx + Gyz - Hz^2,\end{aligned}$$

viz.

$$\begin{pmatrix} \mathfrak{A}, & \mathfrak{H}, & \mathfrak{G} \\ \mathfrak{H}, & \mathfrak{B}, & \mathfrak{F} \\ \mathfrak{G}, & \mathfrak{F}, & \mathfrak{C} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

viz.  $\mathfrak{A}x + \mathfrak{H}y + \mathfrak{G}z = 0$ , &c.

Also

$$(\mathfrak{B}\mathfrak{C} - \mathfrak{F}^2, \mathfrak{C}\mathfrak{A} - \mathfrak{G}^2, \mathfrak{A}\mathfrak{B} - \mathfrak{H}^2, \mathfrak{G}\mathfrak{H} - \mathfrak{A}\mathfrak{F}, \mathfrak{H}\mathfrak{F} - \mathfrak{B}\mathfrak{G}, \mathfrak{F}\mathfrak{G} - \mathfrak{C}\mathfrak{H}) = (x, y, z)^2 K(a, b, c, f, g, h | x, y, z)^2;$$

that is

$$\mathfrak{B}\mathfrak{C} - \mathfrak{F}^2 = x^2 K(a, b, c, f, g, h | x, y, z)^2, \quad \&c.;$$

whence also

$$(\mathfrak{B}\mathfrak{C} - \mathfrak{F}^2, \dots | \lambda, \mu, \nu)^2 = (\lambda x + \mu y + \nu z)^2 K(a, \dots | x, y, z)^2,$$

$$(\mathfrak{B}\mathfrak{C} - \mathfrak{F}^2, \dots | \lambda, \mu, \nu | \lambda', \mu', \nu') = (\lambda x + \mu y + \nu z)(\lambda' x + \mu' y + \nu' z) K(a, \dots | x, y, z)^2;$$

and moreover

$$\mathfrak{A}\mathfrak{B}\mathfrak{C} - \mathfrak{A}\mathfrak{F}^2 - \mathfrak{B}\mathfrak{G}^2 - \mathfrak{C}\mathfrak{H}^2 + 2\mathfrak{F}\mathfrak{G}\mathfrak{H} = 0.$$

402

ON THE ANALYTICAL THEORY OF THE CONIC.

[276

The last equation shows that  $(A, \dots | yz' - y'z, zx' - z'x, xy' - x'y)^2$ , considered as a function of  $(x', y', z')$ , breaks up into factors, or (what is the same thing) this expression, considered as a function of  $(x, y, z)$ , breaks up into factors; we may in all the formulae interchange  $(x, y, z)$  and  $(x', y', z')$ , writing  $(\mathfrak{A}', \mathfrak{B}', \mathfrak{C}', \mathfrak{F}', \mathfrak{G}', \mathfrak{H}')$  in the place of  $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$ .

*Article Nos. 18 to 28, relating to a single conic in connexion with a point or line.*

18. I apply the decomposition formula to the function  $(A, \dots | yz' - y'z, \dots)^2$ , which, considered as a function of  $(x, y, z)$ , breaks up into factors. We have

$$(A, \dots | yz' - y'z, \dots)^2 = (\mathfrak{A}', \dots | x, y, z)^2 \\ = \frac{1}{(\mathfrak{A}', \dots | x, y, z)^2 \cdot x'^2 - y'^2 \cdot z'^2 (\mathfrak{B}'\mathfrak{C}' - \mathfrak{F}'^2, \dots | \lambda, \mu, \nu)^2}.$$

But we have

$$(\mathfrak{A}', \dots | l, m, n)^2 = (A, \dots | mz' - ny', \dots)^2, \\ (\mathfrak{A}', \dots | l, m, n | x, y, z) = (A, \dots | mz' - ny', \dots | yz' - y'z, \dots), \\ (\mathfrak{B}'\mathfrak{C}' - \mathfrak{F}'^2, \dots | mz - ny, \dots | \lambda, \mu, \nu) = [x'(mz - ny) + y'(nx - lz) + z'(ly - mx)](\lambda x' + \mu y' + \nu z') K(a, \dots | x', y', z')^2, \\ (\mathfrak{B}'\mathfrak{C}' - \mathfrak{F}'^2, \dots | \lambda, \mu, \nu)^2 = (\lambda x' + \mu y' + \nu z')^2 K(a, \dots | x', y', z')^2,$$

and thence

$$\frac{(\mathfrak{B}'\mathfrak{C}' - \mathfrak{F}'^2, \dots | mz - ny, \dots | \lambda, \mu, \nu)}{\sqrt{(\mathfrak{B}'\mathfrak{C}' - \mathfrak{F}'^2, \dots | \lambda, \mu, \nu)^2}} = \begin{vmatrix} x' & y' & z' \\ x & y & z \\ l & m & n \end{vmatrix} \sqrt{K(a, \dots | x', y', z')^2};$$

whence we have

$$(A, \dots | yz' - y'z, \dots)^2 = \frac{1}{(\mathfrak{A}', \dots | x, y, z)^2} \text{Product of} \\ \left\{ (A, \dots | mz' - ny', \dots | yz' - y'z, \dots) \pm \begin{vmatrix} x' & y' & z' \\ x & y & z \\ l & m & n \end{vmatrix} \sqrt{K(a, \dots | x', y', z')^2} \right\};$$

and the identical equation

$$(a, \dots | x, y, z)^2 \cdot (a, \dots | x', y', z')^2 - [(a, \dots | x, y, z | x', y', z')]^2 = (A, \dots | yz' - y'z, \dots)^2$$

276]

ON THE ANALYTICAL THEORY OF THE CONIC.

403

now gives

$$(a, \dots | x, y, z)^2 = \text{Quotient by } (a, \dots | x', y', z')^2 \text{ of} \\ \left\{ \begin{array}{l} [(a, \dots | x, y, z | x', y', z')]^2 \\ \text{I.} \quad + \text{Quotient by } (A, \dots | mz' - ny', \dots)^2 \text{ of Product} \\ \left\{ (A, \dots | mz' - ny', \dots | yz' - y'z, \dots) \pm \begin{vmatrix} x' & y' & z' \\ x & y & z \\ l & m & n \end{vmatrix} \sqrt{K(a, \dots | x', y', z')^2} \right\} \end{array} \right.$$

where the Product part may also be written

$$(a, \dots | l, m, n | x', y', z') \cdot (a, \dots | x, y, z | x', y', z') \\ - (a, \dots | x', y', z')^2 \cdot (a, \dots | x, y, z | l, m, n) \\ \pm \sqrt{K(a, \dots | x', y', z')^2} \begin{vmatrix} x' & y' & z' \\ x & y & z \\ l & m & n \end{vmatrix}.$$

19. Writing in the formula I.

$$(a, h, g, f, h, b, c, f, g | x', y', z') = (\xi', \eta', \zeta'),$$

we have

$$\begin{pmatrix} a, h, g \\ h, b, f \\ g, f, c \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \xi' \\ \eta' \\ \zeta' \end{pmatrix}, \quad \begin{pmatrix} A, H, G \\ H, B, F \\ G, F, C \end{pmatrix} \begin{pmatrix} \xi' \\ \eta' \\ \zeta' \end{pmatrix} = K \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix},$$

and thence

$$K(a, \dots | x', y', z')^2 = (A, \dots | \xi', \eta', \zeta')^2.$$

Assume

$$(a, \dots | x, y, z | x', y', z') = \xi'x + \eta'y + \zeta'z.$$

$$(l, m, n) = (\nu\eta' - \mu\zeta', \lambda\zeta' - \nu\xi', \mu\xi' - \lambda\eta');$$

then from the foregoing values of  $(x', y', z')$

$$\begin{aligned} mz' - ny' &= \frac{1}{K} \left\{ (\lambda\zeta' - \nu\xi')(G\xi' + F\eta' + C\zeta') - (\mu\xi' - \lambda\eta')(H\xi' + B\eta' + F\zeta') \right\} \\ &= \frac{1}{K} \left\{ \lambda[\xi'(A\xi' + H\eta' + G\zeta') + \eta'(H\xi' + B\eta' + F\zeta') + \zeta'(G\xi' + F\eta' + C\zeta')] \right. \\ &\quad \left. - \lambda\xi'(A\xi' + H\eta' + G\zeta') - \mu\xi'(H\xi' + B\eta' + F\zeta') - \nu\xi'(G\xi' + F\eta' + C\zeta') \right\}. \end{aligned}$$

[continued p. 404]

that is

$$mz' - ny' = \frac{1}{K} \left\{ \lambda(A, \dots | \xi', \eta', \zeta')^2 - \xi'(A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\},$$

and similarly

$$nx' - lz' = \frac{1}{K} \left\{ \mu(A, \dots | \xi', \eta', \zeta')^2 - \eta'(A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\},$$

$$ly' - mx' = \frac{1}{K} \left\{ \nu(A, \dots | \xi', \eta', \zeta')^2 - \zeta'(A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\};$$

and thence

$$\begin{aligned} (A, H, G | mz' - ny', \dots) \\ = \frac{1}{K} \left\{ (A, H, G | \lambda, \mu, \nu) \cdot (A, \dots | \xi', \eta', \zeta')^2 \right. \\ \left. - (A, H, G | \xi', \eta', \zeta') \cdot (A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\} \end{aligned}$$

with the like equations, writing  $H, B, F$  and  $G, F, C$  in the place of  $A, H, G$  successively: and we then have

$$(A, \dots | mz' - ny', \dots)^2 = \frac{1}{K} \left\{ (A, \dots | \xi', \eta', \zeta' | mz' - ny', \dots) \cdot (A, \dots | \xi', \eta', \zeta')^2 - (A, \dots | \xi', \eta', \zeta')^2 \cdot (A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\}.$$

But the foregoing values of  $mz' - ny'$ ,  $nx' - lz'$ ,  $ly' - mx'$  give also

$$(A, \dots | \xi', \eta', \zeta' | mz' - ny', \dots) = \frac{1}{K} \left\{ (A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \cdot (A, \dots | \xi', \eta', \zeta')^2 - (A, \dots | \xi', \eta', \zeta')^2 \cdot (A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\} = 0;$$

so that

$$(A, \dots | mz' - ny', \dots)^2 = -\frac{1}{K} (A, \dots | \xi', \eta', \zeta')^2 \cdot (a, \dots | \lambda x - \mu \xi', \dots)^2.$$

Similarly,

$$(A, \dots | mz' - ny', \dots | yz' - y'z, \dots) = \frac{1}{K} \left\{ (A, \dots | \xi', \eta', \zeta' | yz' - y'z, \dots) \cdot (A, \dots | \xi', \eta', \zeta')^2 - (A, \dots | \xi', \eta', \zeta')^2 \cdot (A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\}.$$

But

$$\begin{aligned} (A, \dots | \xi', \eta', \zeta' | yz' - y'z, \dots) &= (A\lambda + H\mu + G\nu)(yz' - y'z) \\ &\quad + (H\lambda + B\mu + F\nu)(zx' - z'x) \\ &\quad + (G\lambda + F\mu + C\nu)(xy' - x'y), \\ &= x[y'(G\lambda + F\mu + C\nu) - z'(H\lambda + B\mu + F\nu)] \\ &\quad + y[z'(A\lambda + H\mu + G\nu) - x'(G\lambda + F\mu + C\nu)] \\ &\quad + z[x'(H\lambda + B\mu + F\nu) - y'(A\lambda + H\mu + G\nu)], \end{aligned}$$

which, substituting for  $x', y', z'$  their values

$$(x', y', z') = \frac{1}{K} \begin{pmatrix} A, H, G \\ H, B, F \\ G, F, C \end{pmatrix} (\xi', \eta', \zeta'),$$

becomes

$$\begin{aligned} &= \frac{1}{K} \left\{ x[(BC - F^2)(\nu\eta' - \mu\zeta') + (FG - CH)(\lambda\zeta' - \nu\xi') + (HF - BG)(\mu\xi' - \lambda\eta')] \right. \\ &\quad + y[(FG - CH)(\nu\eta' - \mu\zeta') + (CA - G^2)(\lambda\zeta' - \nu\xi') + (GH - AF)(\mu\xi' - \lambda\eta')] \\ &\quad \left. + z[(HF - BG)(\nu\eta' - \mu\zeta') + (GH - AF)(\lambda\zeta' - \nu\xi') + (AB - H^2)(\mu\xi' - \lambda\eta')] \right\}, \end{aligned}$$

which is

$$= (a, \dots | x, y, z | \nu\eta' - \mu\zeta', \dots);$$

and by merely writing  $(\xi', \eta', \zeta')$  in the place of  $(\lambda, \mu, \nu)$ , we have

$$(A, \dots | \xi', \eta', \zeta' | yz' - y'z, \dots) = 0;$$

so that we find

$$(A, \dots | mz' - ny', \dots | yz' - y'z, \dots) = \frac{1}{K} (A, \dots | \xi', \eta', \zeta')^2 \cdot (a, \dots | x, y, z | \nu\eta' - \mu\zeta', \lambda\zeta' - \nu\xi', \mu\xi' - \lambda\eta').$$



$$-\frac{1}{K}(A, \dots | \xi', \eta', \zeta')^2 \cdot (A, \dots | \lambda, \mu, \nu | \xi, \eta, \zeta) \pm \sqrt{-(A, \dots | \xi', \eta', \zeta')^2} \begin{vmatrix} \xi, \eta, \zeta \\ \xi', \eta', \zeta' \\ \lambda, \mu, \nu \end{vmatrix}.$$

21. And also

$$(A, \dots | \lambda, \mu, \nu)^2 = \text{Quotient by } (a, \dots | x', y', z')^2 \text{ of}$$

$$\text{IV. } \begin{cases} K(\xi x' + \eta y' + \zeta z')^2 \\ + \text{Quotient by } K(A, \dots | ny' - mz', \dots)^2 \text{ of Product} \\ [K\Phi \pm \sqrt{-K}(a, \dots | x', y', z')^2 (A, \dots | ny' - mz', \dots | \lambda, \mu, \nu)], \end{cases}$$

where

$$\Phi = \begin{vmatrix} \xi & \eta & \zeta \\ ax' + hy' + gz', & hx' + by' + fz', & gx' + fy' + cz' \\ ny' - mz', & lz' - nx', & mx' - ly' \end{vmatrix},$$

which may also be written

$$= (a, \dots | x', y', z' | l, m, n)(x'\xi + y'\eta + z'\zeta) - (a, \dots | x', y', z')^2 \cdot (l\xi + m\eta + n\zeta).$$

22. The geometrical signification is obvious. The formulae I. and II. each of them show that the equation

$$(a, \dots | x, y, z)^2 = 0$$

408

ON THE ANALYTICAL THEORY OF THE CONIC.

[276

of the conic may be written in the form

$$W^2 + \frac{1}{M}QR = 0,$$

where  $Q = 0, R = 0$  are any two tangents of the conic, and  $W = 0$  is the line joining the points of contact, or chord of contact corresponding to the two tangents; viz., in the formula I. we have

$$W = (a, \dots | x', y', z' | x, y, z),$$

$$\left. \begin{matrix} Q \\ R \end{matrix} \right\} = (A, \dots | mz' - ny', \dots | yz' - y'z, \dots) \pm \sqrt{-K(a, \dots | x, y, z)^2} \begin{vmatrix} x, y, z \\ x', y', z' \\ l, m, n \end{vmatrix},$$

(or for a different form of  $Q, R$  see the formula). The quantities  $(x', y', z')$  are the coordinates of the point of intersection of the two tangents, or pole of the chord of contact;  $(l, m, n)$  are supernumerary arbitrary quantities, the values whereof do not affect the result<sup>1</sup>. And in the formula II. we have

$$W = \xi'x + \eta'y + \zeta'z,$$

$$\left. \begin{matrix} Q \\ R \end{matrix} \right\} = \Pi \pm \sqrt{-(A, \dots | \xi', \eta', \zeta')^2} (a, \dots | \nu\eta' - \mu\zeta', \dots | x, y, z),$$

(for the value of  $\Pi$  see the formula). The quantities  $(\xi', \eta', \zeta')$  are the line-coordinates of the chord of contact (viz. the point-equation of this line is  $\xi'x + \eta'y + \zeta'z = 0$ );  $(\lambda, \mu, \nu)$  are supernumerary arbitrary quantities.

23. In the like manner the formulae III. and IV. each of them show that the line-equation

$$(A, \dots | \xi, \eta, \zeta)^2 = 0$$

of the conic may be written in the form

$$W^2 + \frac{1}{M}QR = 0,$$

<sup>1</sup>In a different point of view, viz. if we consider the formula I. as a transformation of the function  $(a, \dots | x, y, z)^2$ , then  $(x', y', z')$  and  $(l, m, n)$  would be each of them supernumerary arbitrary quantities; and so in the other like cases.

where  $Q = 0, R = 0$  are any two ineunts of the conic, and  $W = 0$  is the point of intersection of the corresponding tangents; viz. in the formula III. we have

$$W = (a, \dots | \nu\eta' - \mu\zeta', \dots | \xi, \eta, \zeta),$$

$$\left. \begin{matrix} Q \\ R \end{matrix} \right\} = (a, \dots | \nu\eta' - \mu\zeta', \dots | \lambda\zeta' - \nu\xi', \dots) \pm \sqrt{-(A, \dots | \xi', \eta', \zeta')^2} \begin{vmatrix} \xi, \eta, \zeta \\ \xi', \eta', \zeta' \\ \lambda, \mu, \nu \end{vmatrix},$$

(for another form of  $Q, R$  see the formula).

276]

ON THE ANALYTICAL THEORY OF THE CONIC.

409

The quantities  $\xi', \eta', \zeta'$  are the line-coordinates of the line through the two ineunts, or chord of contact;  $(\lambda, \mu, \nu)$  are supernumerary arbitrary quantities; and so in the formula IV. we have

$$W = x'\xi + y'\eta + z'\zeta,$$

$$\left. \begin{matrix} Q \\ R \end{matrix} \right\} = K\Phi \pm \sqrt{-K(a, \dots | x', y', z')^2} (A, \dots | ny' - mz', \dots | \xi, \eta, \zeta),$$

(for the value of  $\Phi$  see the formula), where  $x', y', z'$  are the point-coordinates of the intersection of tangents at the two ineunts, or pole of the chord of contact;  $(l, m, n)$  are supernumerary arbitrary quantities.

24. We may, instead of the supernumerary arbitrary quantities  $(l, m, n)$  of the formula I., introduce the quantities  $(\lambda, \mu, \nu)$ , where

$$(l, m, n) = \frac{1}{K} \begin{pmatrix} A, H, G \\ H, B, F \\ G, F, C \end{pmatrix} (\lambda, \mu, \nu).$$

This gives

$$\begin{aligned} (A, H, G | mz' - ny', \dots) &= A(mz' - ny') + H(nx' - lz') + G(ly' - mx') \\ &= x'(Hn - Gm) + y'(Gl - An) + z'(Am - Hl) \\ &= \frac{1}{K} \left\{ x' [H(G\lambda + F\mu + C\nu) - G(H\lambda + B\mu + F\nu)] \right. \\ &\quad + y' [G(A\lambda + H\mu + G\nu) - A(G\lambda + F\mu + C\nu)] \\ &\quad \left. + z' [A(H\lambda + B\mu + F\nu) - H(A\lambda + H\mu + G\nu)] \right\} \\ &= x'(g\mu - h\nu) + y'(f\mu - b\nu) + z'(c\mu - f\nu) \\ &= \mu(gx' + fy' + cz') - \nu(hx' + by' + fz'); \end{aligned}$$

we have thus the system

$$\begin{aligned} (A, H, G | mz' - ny', \dots) &= \mu(gx' + fy' + cz') - \nu(hx' + by' + fz'), \\ (H, B, F | mz' - ny', \dots) &= \nu(ax' + hy' + gz') - \lambda(gx' + fy' + cz'), \\ (G, F, C | mz' - ny', \dots) &= \lambda(hx' + by' + fz') - \mu(ax' + hy' + gz'). \end{aligned}$$

and thence

$$\begin{aligned} &(A, \dots | mz' - ny', \dots | yz' - y'z, \dots) \\ &= \begin{vmatrix} yz' - y'z, zx' - z'x, xy' - x'y \\ ax' + hy' + gz', hx' + by' + fz', gx' + fy' + cz' \\ \lambda, \mu, \nu \end{vmatrix} \end{aligned}$$

or observing that the term in  $\lambda$  is

$$-(zx' - z'x)(gx' + fy' + cz') + (xy' - x'y)(hx' + by' + fz'),$$

52

410

ON THE ANALYTICAL THEORY OF THE CONIC.

[276



which is

$$\begin{aligned}
 &= x' \{ x'(ax' + hy' + gz') + y'(hx' + by' + fz') + z'(gx' + fy' + cz') \} \\
 &\quad - x \cdot x'(ax' + hy' + gz') \\
 &\quad - y \cdot x'(hx' + by' + fz') \\
 &\quad - z \cdot x'(gx' + fy' + cz') \\
 &= -x'(a, \dots | x, y, z | x', y', z') + x(a, \dots | x', y', z')^2,
 \end{aligned}$$

with similar expressions for the terms in  $\mu, \nu$ , we have

$$(A, \dots | mz' - ny', \dots | yz' - y'z, \dots) = -(\lambda x' + \mu y' + \nu z') \cdot (a, \dots | x, y, z | x', y', z') + (\lambda x + \mu y + \nu z) \cdot (a, \dots | x', y', z')^2,$$

$$(A, \dots | mz' - ny', \dots)^2 = -(\lambda x' + \mu y' + \nu z') \cdot (a, \dots | l, m, n | x', y', z') + (\lambda l + \mu m + \nu n) \cdot (a, \dots | x', y', z')^2,$$

where

$$(a, \dots | l, m, n | x', y', z') = \lambda x' + \mu y' + \nu z',$$

so that

$$\begin{aligned}
 \lambda l + \mu m + \nu n &= \frac{1}{K}(A, \dots | \lambda, \mu, \nu)^2, \\
 (A, \dots | mz' - ny', \dots)^2 &= -(\lambda x' + \mu y' + \nu z')^2 + \frac{1}{K}(A, \dots | \lambda, \mu, \nu)^2 \cdot (a, \dots | x', y', z')^2.
 \end{aligned}$$

Moreover,

$$\Pi = \begin{vmatrix} x, & y, & z \\ x', & y', & z' \\ l, & m, & n \end{vmatrix} = -\frac{1}{K}[(A\lambda + H\mu + G\nu)(yz' - y'z) + (H\lambda + B\mu + F\nu)(zx' - z'x) + (G\lambda + F\mu + C\nu)(xy' - x'y)],$$

which is

$$= -\frac{1}{K}(A, \dots | \lambda, \mu, \nu | yz' - y'z, \dots),$$

and hence instead of the formula I. we have

$$(a, \dots | x, y, z)^2 = \text{Quotient by } (a, \dots | x', y', z')^2 \text{ of}$$

$$\text{I. (bis) } \left\{ \begin{aligned} &[(a, \dots | x, y, z | x', y', z')]^2 \\ &+ \text{Quotient by } (A, \dots | \lambda, \mu, \nu)^2(a, \dots | x', y', z')^2 - K(\lambda x' + \mu y' + \nu z')^2 \text{ of } K \text{ into Product} \\ &\{(\lambda x' + \mu y' + \nu z') \cdot (a, \dots | x, y, z | x', y', z') - (\lambda x + \mu y + \nu z) \cdot (a, \dots | x', y', z')^2\} \\ &\pm \sqrt{-K(a, \dots | x', y', z')^2} (A, \dots | \lambda, \mu, \nu | yz' - y'z, \dots). \end{aligned} \right.$$

25. If, in like manner, in the formula II. we introduce, instead of  $(\lambda, \mu, \nu)$ , the new quantities  $(l, m, n)$ , where

$$(\lambda, \mu, \nu) = \begin{pmatrix} a, h, g \\ h, b, f \\ g, f, c \end{pmatrix} (l, m, n),$$

276]

ON THE ANALYTICAL THEORY OF THE CONIC.

411

or, what is the same thing,

$$(l, m, n) = \frac{1}{K} \begin{pmatrix} A, H, G \\ H, B, F \\ G, F, C \end{pmatrix} (\lambda, \mu, \nu),$$

then we have

$$\begin{aligned}
 (a, h, g | \nu\eta' - \mu\zeta', \dots) &= n(H\xi' + B\eta' + F\zeta') - m(G\xi' + F\eta' + C\zeta'), \\
 (h, b, f | \nu\eta' - \mu\zeta', \dots) &= l(G\xi' + F\eta' + C\zeta') - n(A\xi' + H\eta' + G\zeta'), \\
 (g, f, c | \nu\eta' - \mu\zeta', \dots) &= m(A\xi' + H\eta' + G\zeta') - l(H\xi' + B\eta' + F\zeta');
 \end{aligned}$$

and thence

$$(a, \dots | \nu\eta' - \mu\zeta', \dots | x, y, z) = \begin{vmatrix} & x & & y & z \\ A\xi' + H\eta' + G\zeta', & H\xi' + B\eta' + F\zeta', & G\xi' + F\eta' + C\zeta' & & \\ & l, & m, & n & \end{vmatrix}$$

$$\begin{aligned}
 &= (A, \dots | mz - ny, \dots | \xi', \eta', \zeta'), \\
 (a, \dots | \nu\eta' - \mu\zeta', \dots)^2 &= \frac{1}{K} \left[ (A, \dots | \lambda, \mu, \nu)^2 \cdot (A, \dots | \xi', \eta', \zeta')^2 - (A, \dots | \lambda, \mu, \nu | \xi', \eta', \zeta')^2 \right] \\
 &= (a, \dots | l, m, n)^2 \cdot (A, \dots | \xi', \eta', \zeta')^2 - K(l\xi' + m\eta' + n\zeta')^2; \\
 \Pi &= \begin{vmatrix} & x & y & z \\ A\xi' + H\eta' + G\zeta', & H\xi' + B\eta' + F\zeta', & G\xi' + F\eta' + C\zeta' & \end{vmatrix} \\
 &= (\eta z' - \zeta y')(G\xi' + F\eta' + C\zeta') - [(H\xi' + B\eta' + F\zeta')] \dots \quad [\text{bracketed determinant expansion as printed}] \\
 &= \lambda \{ (\xi'x + \eta'y + \zeta'z)(A\xi' + H\eta' + G\zeta') - x(A, \dots | \xi', \eta', \zeta')^2 \} \\
 &\quad + \mu \{ (\xi'x + \eta'y + \zeta'z)(H\xi' + B\eta' + F\zeta') - y(A, \dots | \xi', \eta', \zeta')^2 \} \\
 &\quad + \nu \{ (\xi'x + \eta'y + \zeta'z)(G\xi' + F\eta' + C\zeta') - z(A, \dots | \xi', \eta', \zeta')^2 \},
 \end{aligned}$$

and the formula II. thus becomes

$$\begin{aligned}
 (a, \dots | x, y, z)^2 &= \text{Quotient by } (A, \dots | \xi', \eta', \zeta')^2 \text{ of} \\
 \text{II. (bis)} \quad &\begin{cases} K(\xi'x + \eta'y + \zeta'z)^2 \\ + \text{Quotient by } (a, \dots | l, m, n)^2 \cdot (A, \dots | \xi', \eta', \zeta')^2 - K(l\xi' + m\eta' + n\zeta')^2 \text{ of Product} \\ \{ K(l\xi' + m\eta' + n\zeta') \cdot (\xi'x + \eta'y + \zeta'z) - K(a, \dots | x, y, z | \xi', \eta', \zeta') \cdot (a, \dots | \xi', \eta', \zeta') \} \\ \pm \sqrt{-(A, \dots | \xi', \eta', \zeta')^2} (A, \dots | mz - ny, \dots | \xi', \eta', \zeta'). \end{cases}
 \end{aligned}$$

52–2

412

# ON THE ANALYTICAL THEORY OF THE CONIC.

[276

26. And from these we at once deduce the inverse or reciprocal formulae

$$\begin{aligned}
 (A, \dots | \xi, \eta, \zeta)^2 &= \text{Quotient by } (A, \dots | \xi', \eta', \zeta')^2 \text{ of} \\
 \text{III. (bis)} \quad &\begin{cases} [(A, \dots | \xi, \eta, \zeta | \xi', \eta', \zeta')]^2 \\ + \text{Quotient by } (a, \dots | l, m, n)^2 \cdot (A, \dots | \xi', \eta', \zeta')^2 - K(l\xi' + m\eta' + n\zeta')^2 \text{ of } K \text{ into Product} \\ \{ (l\xi' + m\eta' + n\zeta') \cdot (A, \dots | \xi, \eta, \zeta | \xi', \eta', \zeta') - (l\xi + m\eta + n\zeta) \cdot (A, \dots | \xi', \eta', \zeta')^2 \} \\ \pm \sqrt{-(A, \dots | \xi', \eta', \zeta')^2} (a, \dots | l, m, n | \eta\zeta' - \zeta\eta', \dots). \end{cases}
 \end{aligned}$$

27. And

$$\begin{aligned}
 (A, \dots | \xi, \eta, \zeta)^2 &= \text{Quotient by } (a, \dots | x', y', z')^2 \text{ of} \\
 \text{IV. (bis)} \quad &\begin{cases} K(\xi x' + \eta y' + \zeta z')^2 \\ + \text{Quotient by } (A, \dots | \lambda, \mu, \nu)^2 \cdot (a, \dots | x', y', z')^2 - K(\lambda x' + \mu y' + \nu z')^2 \text{ of Product} \\ \pm \sqrt{-K(a, \dots | x', y', z')^2} (a, \dots | x', y', z' | \mu\zeta - \nu\eta, \dots). \end{cases}
 \end{aligned}$$

These four formulae have the same geometrical significations with the original four formulae to which they correspond respectively.

28. The eight formulae become all of them the same or very similar for the quadric form  $(a, \dots | x, y, z)^2 = x^2 + y^2 + z^2$ , which of course implies  $(A, \dots | \xi, \eta, \zeta)^2 = \xi^2 + \eta^2 + \zeta^2$ . Thus selecting any one of them at pleasure, e.g. the formula II. (bis), this becomes

$$\begin{aligned}
 &\{ (x^2 + y^2 + z^2)(\xi'^2 + \eta'^2 + \zeta'^2) - (\xi'x + \eta'y + \zeta'z)^2 \} \\
 &\times \{ (l^2 + m^2 + n^2)(\xi'^2 + \eta'^2 + \zeta'^2) - (l\xi' + m\eta' + n\zeta')^2 \} \\
 &= [(l\xi' + m\eta' + n\zeta')(x\xi' + m\eta' + n\zeta') - (lx + my + nz)(\xi'^2 + \eta'^2 + \zeta'^2)]^2 \\
 &\quad + (\xi'^2 + \eta'^2 + \zeta'^2) \begin{vmatrix} \xi' & \eta' & \zeta' \\ x & y & z \\ l & m & n \end{vmatrix}^2,
 \end{aligned}$$

where the terms independent of  $\xi'^2 + \eta'^2 + \zeta'^2$  destroy each other. Omitting these terms, and dividing by  $\xi'^2 + \eta'^2 + \zeta'^2$ , the resulting equation is found to be

$$\begin{vmatrix} \xi'^2 + \eta'^2 + \zeta'^2, & \xi'x + \eta'y + \zeta'z, & \xi'l + \eta'm + \zeta'n \\ \xi'x + \eta'y + \zeta'z, & x^2 + y^2 + z^2, & lx + my + nz \\ \xi'l + \eta'm + \zeta'n, & lx + my + nz, & l^2 + m^2 + n^2 \end{vmatrix} = 0,$$

which is a well-known identical equation.

276]

ON THE ANALYTICAL THEORY OF THE CONIC.

413

*Article Nos. 29 to 33, relating to a single conic in connexion with an ineunt or a tangent of a conic of double contact.*

29. The formulae assume a very simple form when the point of intersection of the two tangents, or the line of junction of the two ineunts of the conic, is an ineunt or a tangent of a conic having double contact with the first-mentioned conic. Thus, if to the conic

$$(a, \dots | x, y, z)^2 = 0$$

tangents are drawn from a point  $(x', y', z')$  of the conic

$$(a, \dots | x, y, z)^2 + (\xi'x + \eta'y + \zeta'z)^2 = 0,$$

then we have

$$(a, \dots | x', y', z')^2 = -(\xi'x' + \eta'y' + \zeta'z')^2;$$

and using the form I. (bis), and putting therein  $(\xi', \eta', \zeta')$  in the place of the arbitrary quantities  $(\lambda, \mu, \nu)$ , the equation of the tangent divides out by  $\xi'x' + \eta'y' + \zeta'z'$ , and owing to this fact it becomes

$$\begin{aligned} & (a, \dots | x', y', z' | x, y, z) + (\xi'x' + \eta'y' + \zeta'z')(\xi'x + \eta'y + \zeta'z) \\ & \pm \frac{1}{\sqrt{K}}(A, \dots | \xi', \eta', \zeta' | yz' - y'z, zx' - z'x, xy' - x'y) = 0, \end{aligned}$$

which is of the form

$$(\alpha, \beta, \gamma | x', y', z' | x, y, z) = 0,$$

where the matrix  $(\alpha, \beta, \gamma)$  is =

$$\begin{pmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{pmatrix} = \begin{pmatrix} a + \xi'^2, & h + \xi'\eta' + \frac{1}{\sqrt{K}}(G\xi' + F\eta' + C\zeta'), & g + \xi'\zeta' - \frac{1}{\sqrt{K}}(H\xi' + B\eta' + F\zeta') \\ h + \xi'\eta' - \frac{1}{\sqrt{K}}(G\xi' + F\eta' + C\zeta'), & b + \eta'^2, & f + \eta'\zeta' + \frac{1}{\sqrt{K}}(A\xi' + H\eta' + G\zeta') \\ g + \xi'\zeta' + \frac{1}{\sqrt{K}}(H\xi' + B\eta' + F\zeta'), & f + \eta'\zeta' - \frac{1}{\sqrt{K}}(A\xi' + H\eta' + G\zeta'), & c + \zeta'^2 \end{pmatrix}.$$

30. But instead of further developing these formulae, I prefer to consider the formulae which give the points of contact of the tangents in question, viz. the ineunts of the conic  $(a, \dots | x, y, z)^2 = 0$ , or the tangents through the point  $(x', y', z')$  of the conic  $(a, \dots | x, y, z)^2 + (\xi'x + \eta'y + \zeta'z)^2 = 0$ .

414

ON THE ANALYTICAL THEORY OF THE CONIC.

[276

We have as before

$$(a, \dots | x', y', z')^2 = -(\xi'x' + \eta'y' + \zeta'z')^2,$$

and using the formula IV. (bis) and writing therein  $(\xi', \eta', \zeta')$  in the place of the arbitrary quantities  $(\lambda, \mu, \nu)$ , the equation contains the factor  $\xi'x' + \eta'y' + \zeta'z'$ , and dividing by this factor, and by  $K$ , the line-equation of the ineunt is

$$x'\xi + y'\eta + z'\zeta + \frac{1}{K}(A\xi' + H\eta' + G\zeta')(\xi'x' + \eta'y' + \zeta'z') \cdot (A, \dots | \xi', \eta', \zeta' | \xi, \eta, \zeta)$$

$$\pm \frac{1}{\sqrt{K}}(a, \dots | x', y', z' | \xi' \zeta - \zeta' \xi, \dots) = 0.$$

Selecting the positive sign, the coordinates of the corresponding ineunt are

$$\begin{aligned} x' + \frac{1}{K}(A\xi' + H\eta' + G\zeta')(\xi'x' + \eta'y' + \zeta'z') + \frac{1}{\sqrt{K}}\{\eta'(gx' + fy' + cz') - \zeta'(hx' + by' + fz')\}, \\ y' + \frac{1}{K}(H\xi' + B\eta' + F\zeta')(\xi'x' + \eta'y' + \zeta'z') + \frac{1}{\sqrt{K}}\{\zeta'(ax' + hy' + gz') - \xi'(gx' + fy' + cz')\}, \\ z' + \frac{1}{K}(G\xi' + F\eta' + C\zeta')(\xi'x' + \eta'y' + \zeta'z') + \frac{1}{\sqrt{K}}\{\xi'(hx' + by' + fz') - \eta'(ax' + hy' + gz')\}, \end{aligned}$$

and taking  $(X, Y, Z)$  for the coordinates of the ineunt in question, and putting for shortness

$$\begin{aligned} \alpha &= -\frac{1}{\sqrt{K}}(g\eta' - h\zeta'), \quad \beta = -\frac{1}{\sqrt{K}}(f\eta' - b\zeta'), \quad \gamma = -\frac{1}{\sqrt{K}}(c\eta' - f\zeta'), \\ \alpha' &= -\frac{1}{\sqrt{K}}(a\zeta' - g\xi'), \quad \beta' = 1 - \frac{1}{\sqrt{K}}(h\zeta' - f\xi'), \quad \gamma' = -\frac{1}{\sqrt{K}}(g\zeta' - c\xi'), \\ \alpha'' &= -\frac{1}{\sqrt{K}}(h\xi' - a\eta'), \quad \beta'' = -\frac{1}{\sqrt{K}}(b\xi' - h\eta'), \quad \gamma'' = 1 - \frac{1}{\sqrt{K}}(f\xi' - g\eta'), \end{aligned}$$

we may write

$$\begin{aligned} (1 + P)X &= (2 - \alpha)x' - \beta y' - \gamma z' + \frac{1}{K}(A\xi' + H\eta' + G\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ (1 + P)Y &= -\alpha' x' + (2 - \beta')y' - \gamma' z' + \frac{1}{K}(H\xi' + B\eta' + F\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ (1 + P)Z &= -\alpha'' x' - \beta'' y' + (2 - \gamma'')z' + \frac{1}{K}(G\xi' + F\eta' + C\zeta')(\xi'x' + \eta'y' + \zeta'z'), \end{aligned}$$

where  $P$ , which is arbitrary, may be put

$$= -\frac{1}{K}(A, \dots | \xi', \eta', \zeta')^2.$$

276]

ON THE ANALYTICAL THEORY OF THE CONIC.

415

31. These equations then give

$$(x', y', z') = \begin{pmatrix} \alpha, \beta, \gamma \\ \alpha', \beta', \gamma' \\ \alpha'', \beta'', \gamma'' \end{pmatrix} (X, Y, Z),$$

which can be verified without difficulty by reversing the process; and we have thus the coordinates  $(X, Y, Z)$  in terms of  $(x', y', z')$ , and reciprocally.

32. If  $(X_1, Y_1, Z_1)$  are the coordinates of the other ineunt, we have, it is clear,

$$(x', y', z') = \begin{pmatrix} 2 - \alpha, -\beta, -\gamma \\ -\alpha', 2 - \beta', -\gamma' \\ -\alpha'', -\beta'', 2 - \gamma'' \end{pmatrix} (X_1, Y_1, Z_1);$$

or substituting for  $(x', y', z')$  their values in terms of  $(X, Y, Z)$ ,

$$(2X_1, 2Y_1, 2Z_1) = \begin{pmatrix} \alpha, \beta, \gamma \\ \alpha', \beta', \gamma' \\ \alpha'', \beta'', \gamma'' \end{pmatrix} (X + X_1, Y + Y_1, Z + Z_1),$$

so that  $(X + X_1, Y + Y_1, Z + Z_1)$  are the same linear functions of  $2X_1, 2Y_1, 2Z_1$  that  $(X, Y, Z)$  are of  $(x', y', z')$ ; that is, we have

$$\begin{aligned} \frac{1}{2}(1 + P)(X + X_1) &= (2 - \alpha)X_1 - \beta Y_1 - \gamma Z_1 + \frac{1}{K}(A\xi' + H\eta' + G\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ \frac{1}{2}(1 + P)(Y + Y_1) &= -\alpha' X_1 + (2 - \beta')Y_1 - \gamma' Z_1 + \frac{1}{K}(H\xi' + B\eta' + F\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ \frac{1}{2}(1 + P)(Z + Z_1) &= -\alpha'' X_1 - \beta'' Y_1 + (2 - \gamma'')Z_1 + \frac{1}{K}(G\xi' + F\eta' + C\zeta')(\xi'x' + \eta'y' + \zeta'z'), \end{aligned}$$

which equations may be written

$$(1 + P)(X, Y, Z) = \begin{pmatrix} \mathfrak{a}, \mathfrak{b}, \mathfrak{c} \\ \mathfrak{a}', \mathfrak{b}', \mathfrak{c}' \\ \mathfrak{a}'', \mathfrak{b}'', \mathfrak{c}'' \end{pmatrix} (X_1, Y_1, Z_1),$$

where the values of the coefficients are

$$\begin{aligned} \mathfrak{a} &= (a\eta' - h\xi') + \frac{1}{K}\xi'(A\xi' + H\eta' + G\zeta'), & \mathfrak{b} &= -\frac{1}{\sqrt{K}}(h\eta' - b\xi') + \frac{1}{K}\xi'(H\xi' + B\eta' + F\zeta'), \\ \mathfrak{c} &= -\frac{1}{\sqrt{K}}(g\eta' - f\xi') + \frac{1}{K}\xi'(G\xi' + F\eta' + C\zeta'); & \mathfrak{a}' &= 1 - \frac{1}{\sqrt{K}}(A\xi' - h\eta') + \frac{1}{K}\eta'(A\xi' + H\eta' + G\zeta'), \\ \mathfrak{b}' &= -\frac{1}{\sqrt{K}}(B\eta' - h\zeta') + \frac{1}{K}\eta'(H\xi' + B\eta' + F\zeta'), & \mathfrak{c}' &= -\frac{1}{\sqrt{K}}(F\eta' - c\xi') + \frac{1}{K}\eta'(G\xi' + F\eta' + C\zeta'); \\ \mathfrak{a}'' &= -\frac{1}{\sqrt{K}}(g\xi' - c\eta') + \frac{1}{K}\zeta'(A\xi' + H\eta' + G\zeta'), & \mathfrak{b}'' &= -\frac{1}{\sqrt{K}}(b\xi' - h\zeta') + \frac{1}{K}\zeta'(H\xi' + B\eta' + F\zeta'), \\ \mathfrak{c}'' &= 1 - \frac{1}{\sqrt{K}}(f\xi' - g\eta') + \frac{1}{K}\zeta'(G\xi' + F\eta' + C\zeta'). \end{aligned}$$

416

ON THE ANALYTICAL THEORY OF THE CONIC.

[276

and considering  $(X, Y, Z)$  and  $(X_1, Y_1, Z_1)$  as quantities connected by the foregoing linear relations, we have identically

$$(a, \dots | X, Y, Z)^2 = (a, \dots | X_1, Y_1, Z_1)^2.$$

The investigation leads thus to the automorphic transformation of the quadric function, a transformation first effected by M. Hermite<sup>2</sup>.

33. It is to be remarked that the foregoing formulae show that  $(x', y', z')$  being the coordinates of a point on the conic  $(a, \dots | x, y, z)^2 + (\xi'x + \eta'y + \zeta'z)^2 = 0$ , from which point tangents are drawn to the conic  $(a, \dots | x, y, z)^2 = 0$ , the ineunts (or points of contact), enter linearly into the equations of the tangents, the ineunts (or coordinates of contact), and the polar. And it may be added that the equation of the conic enveloped by the polar (that is, the polar conic of  $(a, \dots | x, y, z)^2 + (\xi'x + \eta'y + \zeta'z)^2 = 0$ ) has for its equation

$$\{K + (A, \dots | \xi', \eta', \zeta')^2\}(a, \dots | x, y, z)^2 - K(\xi'x + \eta'y + \zeta'z)^2 = 0,$$

and that the coordinates of the point of contact of the polar with this conic are

$$\begin{aligned} x' &+ \frac{1}{K}(A\xi' + H\eta' + G\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ y' &+ \frac{1}{K}(H\xi' + B\eta' + F\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ z' &+ \frac{1}{K}(G\xi' + F\eta' + C\zeta')(\xi'x' + \eta'y' + \zeta'z'); \end{aligned}$$

so that  $(x', y', z')$  also enter linearly into the expressions for the coordinates of the last-mentioned point.

*Article Nos. 34 to 37, relating to two conics.*

34. Considering now the two conics

$$U = (a, b, c, f, g, h | x, y, z)^2 = 0,$$

$$U' = (a', b', c', f', g', h' | x, y, z)^2 = 0;$$

suppose that the conic

$$\theta U + \theta' U' = (\theta a + \theta' a', \dots | x, y, z)^2 = 0$$

represents a pair of lines.

The condition for this is

$$\text{Disc.}(\theta a + \theta' a', \dots | x, y, z)^2 = 0,$$

<sup>2</sup>See my "Memoir on the Automorphic Transformation of a Bipartite Quadric Function," *Phil. Trans.*, t. cxlviii. (1858), pp. 39–46, [115].

which is

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D} \mid \theta, \theta')^3 = 0,$$

where

$$\begin{aligned}\mathfrak{A} &= K, & \mathfrak{D} &= K', \\ \mathfrak{B} &= Aa' + Bb' + Cc' + 2Ff' + 2Gg' + 2Hh', \\ \mathfrak{C} &= A'a + B'b + C'c + 2F'f + 2G'g + 2H'h,\end{aligned}$$

276]

ON THE ANALYTICAL THEORY OF THE CONIC.

417

(the significations of  $K', A', B', C', F', G', H'$  being of course analogous to those of  $K, A, B, C, F, G, H$ ). The three roots  $\theta : \theta'$  correspond, it is clear, to the three pairs of lines which can be drawn through the intersections of the two conics.

35. The equation

$$\text{Disc.}(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D} \mid \theta, \theta')^3 = 0,$$

which is of the fourth order in  $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$ , and of the sixth order as regards  $(a, b, c, f, g, h)$  and  $(a', b', c', f', g', h')$  respectively, is the condition, in order that the two conics may touch each other. Assuming that it is satisfied, the cubic equation in  $\theta : \theta'$  has a pair of equal roots; or say there is a twofold root and a onefold root; the twofold root gives the pair of lines drawn from the point of contact to the other two points of intersection, the onefold root gives the pair made up of the common tangent and the line joining the other two points of intersection.

36. In particular, suppose that the two conics are

$$\begin{aligned}2(\rho x + \sigma y + \tau z)(\rho' x + \sigma' y + \tau' z) &= 0, \\ 2(\lambda x + \mu y + \nu z)(\lambda' x + \mu' y + \nu' z) &= 0;\end{aligned}$$

so that

$$\begin{aligned}(a, b, c, f, g, h) &= (2\rho\rho', 2\sigma\sigma', 2\tau\tau', \sigma\tau' + \sigma'\tau, \tau\rho' + \tau'\rho, \rho\sigma' + \rho'\sigma), \\ (a', b', c', f', g', h') &= (2\lambda\lambda', 2\mu\mu', 2\nu\nu', \mu\nu' + \mu'\nu, \nu\lambda' + \nu'\lambda, \lambda\mu' + \lambda'\mu), \\ (A, B, C, F, G, H) &= -(\sigma\tau' - \sigma'\tau, \tau\rho' - \tau'\rho, \rho\sigma' - \rho'\sigma, \dots)^2, \\ (A', B', C', F', G', H') &= -(\mu\nu' - \mu'\nu, \nu\lambda' - \nu'\lambda, \lambda\mu' - \lambda'\mu, \dots)^2;\end{aligned}$$

and thence also

$$\begin{aligned}\mathfrak{A} &= K = 0, & \mathfrak{D} &= K' = 0, \\ \mathfrak{B} &= Aa' + \&c. = -2 \begin{vmatrix} \lambda, & \mu, & \nu \\ \lambda', & \mu', & \nu' \\ \sigma\tau', & \tau\rho', & \rho\sigma' \end{vmatrix} \cdot \begin{vmatrix} \lambda, & \mu, & \nu \\ \lambda', & \mu', & \nu' \\ \sigma'\tau, & \tau'\rho, & \rho'\sigma \end{vmatrix}, \\ \mathfrak{C} &= A'a + \&c. = -2 \begin{vmatrix} \rho, & \sigma, & \tau \\ \rho', & \sigma', & \tau' \\ \mu\nu', & \nu\lambda', & \lambda\mu' \end{vmatrix} \cdot \begin{vmatrix} \rho, & \sigma, & \tau \\ \rho', & \sigma', & \tau' \\ \mu'\nu, & \nu'\lambda, & \lambda'\mu \end{vmatrix};\end{aligned}$$

and the equation in  $(\theta, \theta')$  is

$$\mathfrak{B}\theta^2 + \mathfrak{C}\theta'^2 = 0;$$

hence writing  $\theta = \mathfrak{C}$ ,  $\theta' = -\mathfrak{B}$ , the equation of the pair of lines is

$$\begin{vmatrix} \rho, & \sigma, & \tau \\ \rho', & \sigma', & \tau' \\ \mu\nu', & \nu\lambda', & \lambda\mu' \end{vmatrix} \begin{vmatrix} \rho, & \sigma, & \tau \\ \rho', & \sigma', & \tau' \\ \mu'\nu, & \nu'\lambda, & \lambda'\mu \end{vmatrix} (\rho x + \sigma y + \tau z)(\rho' x + \sigma' y + \tau' z) - \begin{vmatrix} \lambda, & \mu, & \nu \\ \lambda', & \mu', & \nu' \\ \sigma\tau', & \tau\rho', & \rho\sigma' \end{vmatrix} \begin{vmatrix} \lambda, & \mu, & \nu \\ \lambda', & \mu', & \nu' \\ \sigma'\tau, & \tau'\rho, & \rho'\sigma \end{vmatrix} (\lambda x + \mu y + \nu z)(\lambda' x + \mu' y + \nu' z) = 0;$$

53

418

ON THE ANALYTICAL THEORY OF THE CONIC.

[276

and it is easy to see that the left-hand side does in fact break up into factors, and that the equation is

$$\left| \begin{array}{ccc} x, y, z \\ \mu\tau' - \nu\sigma', \nu\rho' - \lambda\tau', \lambda\sigma' - \mu\rho' \\ \sigma'\nu - \tau'\mu, \tau'\lambda - \rho'\nu, \rho'\mu - \sigma'\lambda \end{array} \right| \cdot \left| \begin{array}{ccc} x, y, z \\ \mu'\tau - \nu'\sigma, \nu'\rho - \lambda'\tau, \lambda'\sigma - \mu'\rho \\ \sigma\nu' - \tau\mu', \tau\lambda' - \rho\nu', \rho\mu' - \sigma\lambda' \end{array} \right| = 0,$$

which of course might have been obtained at once by means of the four points which are the intersection of each component line of the first conic by each component line of the second conic.

37. Suppose that the first conic is

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0,$$

while the second conic is the pair of lines

$$2(\lambda x + \mu y + \nu z)(\lambda'x + \mu'y + \nu'z) = 0;$$

then, putting as before,

$$(\theta a + \theta' \cdot 2\lambda\lambda', \dots \mid x, y, z)^2 = 0,$$

we have

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D} \mid \theta, \theta')^3 = 0,$$

where

$$\mathfrak{A} = K,$$

$$\mathfrak{B} = 2(A, B, C, F, G, H \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu'),$$

$$\mathfrak{C} = -(a, b, c, f, g, h \mid \mu\nu' - \mu'\nu, \nu\lambda' - \nu'\lambda, \lambda\mu' - \lambda'\mu)^2,$$

$$\mathfrak{D} = 0;$$

and the equation in  $(\theta, \theta')$  is

$$K\theta^3 + 2(A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu')\theta^2\theta' - (a, \dots \mid \mu\nu' - \mu'\nu, \dots)^2\theta\theta'^2 = 0,$$

which may be written

$$[K\theta + (A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu')\theta']^2 = \left[ [(A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu')]^2 + K(a, \dots \mid \mu\nu' - \mu'\nu, \dots)^2 \right] \theta'^2,$$

that is

$$= (A, \dots \mid \lambda, \mu, \nu)^2 \cdot (A, \dots \mid \lambda', \mu', \nu')^2 \cdot \theta'^2,$$

$$K\theta = [\pm\sqrt{(A, \dots \mid \lambda, \mu, \nu)^2} \sqrt{(A, \dots \mid \lambda', \mu', \nu')^2} - (A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu')]\theta';$$

we may assume

$$\theta = \pm\sqrt{(A, \dots \mid \lambda, \mu, \nu)^2} \sqrt{(A, \dots \mid \lambda', \mu', \nu')^2} - (A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu'), \quad \theta' = K,$$

so that the conic

$$\begin{aligned} & \{ \pm\sqrt{(A, \dots \mid \lambda, \mu, \nu)^2} \sqrt{(A, \dots \mid \lambda', \mu', \nu')^2} - (A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu') \} (a, \dots \mid x, y, z)^2 \\ & + 2K(\lambda x + \mu y + \nu z)(\lambda'x + \mu'y + \nu'z) = 0 \end{aligned}$$

breaks up into a pair of lines.

Putting for shortness

$$\pm\sqrt{(A, \dots \mid \lambda, \mu, \nu)^2} \sqrt{(A, \dots \mid \lambda', \mu', \nu')^2} - (A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu') = \Omega,$$

the coefficients on the left-hand side of the equation are

$$(\Omega a + 2K\lambda\lambda', \dots, \Omega f + K(\mu\nu' + \mu'\nu), \dots),$$

whence, after all reductions, the inverse function is

$$\{(A, \dots | \lambda, \mu, \nu | \xi, \eta, \zeta) \sqrt{(A, \dots | \lambda', \mu', \nu')^2} + (A, \dots | \lambda', \mu', \nu' | \xi, \eta, \zeta) \sqrt{(A, \dots | \lambda, \mu, \nu)^2}\}^2,$$

and the remainder of the process of decomposition is effected without difficulty.

*Addition, 18 December, 1862.*

The formulae II. and II. (bis) each of them give the tangents of the conic  $(a, \dots | x, y, z)^2 = 0$  at the ineunts of intersection with the line  $\xi'x + \eta'y + \zeta'z = 0$ . A very elegant formula for these ineunts themselves was communicated to me by Mr Spottiswoode, and I have since found that the same or an equivalent formula is made use of by M. Aronhold in his recent valuable memoir, “Ueber eine neue algebraische Behandlungsweise der Integrale irrationaler Differentiale, &c.,” *Crelle*, t. LXI. pp. 95–145 (1862). The formula is as follows, viz. for the conic and line,

$$(a, b, c, f, g, h | x, y, z)^2 = 0,$$

$$\xi'x + \eta'y + \zeta'z = 0,$$

then

$$\begin{aligned} x : y : z = \\ (l\xi' + m\eta' + n\zeta') \frac{d\phi}{d\xi'} + \eta'(gl + fm + cn) - \zeta'(hl + bm + fn) + l\sqrt{\phi}, \\ : (l\xi' + m\eta' + n\zeta') \frac{d\phi}{d\eta'} + \zeta'(al + hm + gn) - \xi'(gl + fm + cn) + m\sqrt{\phi}, \\ : (l\xi' + m\eta' + n\zeta') \frac{d\phi}{d\zeta'} + \xi'(hl + bm + fn) - \eta'(al + hm + gn) + n\sqrt{\phi}; \end{aligned}$$

where

$$\phi = \begin{vmatrix} \xi' & \eta' & \zeta' \\ \xi' & a & h & g \\ \eta' & h & b & f \\ \zeta' & g & f & c \end{vmatrix} = -(A, B, C, F, G, H | \xi', \eta', \zeta')^2,$$

so that  $\frac{1}{2} \frac{d\phi}{d\xi'}$ ,  $\frac{1}{2} \frac{d\phi}{d\eta'}$ ,  $\frac{1}{2} \frac{d\phi}{d\zeta'}$  are respectively

$$= -(A\xi' + H\eta' + G\zeta'), \quad -(H\xi' + B\eta' + F\zeta'), \quad -(G\xi' + F\eta' + C\zeta'),$$

and where  $l, m, n$  are supernumerary arbitrary quantities.

53–2

420

[277

## 277.

### ON THE WAVE SURFACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 16–22.]

SOME very beautiful results in relation to the Wave Surface have been recently obtained by Herr Zech, in the Memoirs “Die Eigenschaften der Wellenflächen der zweiaxigen Krystalle mittels der höhern Geometrie abgeleitet,” *Crelle*, t. LII. pp. 243–254 (1856), and “Die Krümmungslinien der Wellenfläche zweiaxiger Krystalle,” *Crelle*, t. LIV. pp. 72–77 (1857). According to the former of Fresnel’s two modes of generation, the wave surface is the envelope of a plane whose perpendicular distance  $v$  is a certain given function of the direction cosines  $l, m, n$ . For the same system of direction cosines, there are in fact two values of the perpendicular distance: call these  $v$  and  $w$ , and let the corresponding planes (parallel of course to each other) be called  $P$  and  $Q$ . Then the entire system of the planes  $P$  and  $Q$  envelope the wave surface, viz. the planes  $P$  may be considered as enveloping one sheet and the planes  $Q$  the other sheet of the surface.



But if instead of considering the entire system of planes we consider only the planes  $P$  and the parallel planes  $Q$ , for which the perpendicular on the plane  $P$  has a given constant value  $v$ , then the planes  $P$  will envelope a developable  $F$ , and the planes  $Q$  will envelope a developable  $G$ , these two developables being, it is to be observed, distinct surfaces, not sheets of one and the same surface. Or, what is the same thing, the planes  $P$  for which the perpendicular distance has a given constant value  $v$  will envelope a developable  $F$ , and the planes  $P$  for which the perpendicular distance of the parallel planes  $Q$  has a given constant value  $w$ , will envelope a developable  $G$ . And it is obvious that the developables  $F$  and  $G$  touch the wave surface along curves. The equation of the developable  $F$  contains of course the arbitrary parameter  $v$ , and the equation of the developable  $G$  contains in like manner the arbitrary parameter  $w$ , so that we in fact have two series of developables

277]

ON THE WAVE SURFACE.

421

$F$  and  $G$  respectively touching the wave surface along two series of curves. And it is shown in the second of the Memoirs above referred to that these curves are the curves of curvature of the wave surface.

The developable  $F$  is obtained as the envelope of the plane  $P$ , whose equation is

$$lx + my + nz = v,$$

where  $v$  has a given constant value and  $l, m, n$  are parameters which vary, subject to the two conditions

$$l^2 + m^2 + n^2 = 1,$$

$$\frac{l^2}{a^2 - v^2} + \frac{m^2}{b^2 - v^2} + \frac{n^2}{c^2 - v^2} = 0.$$

And in like manner the developable  $G$  is obtained as the envelope of the plane  $Q$ , whose equation is

$$lx + my + nz = \frac{abc}{w} \sqrt{\frac{l^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2}},$$

where  $w$  has a given constant value and  $l, m, n$  are parameters which vary, subject to the two conditions

$$l^2 + m^2 + n^2 = 1,$$

$$\frac{l^2}{a^2 - w^2} + \frac{m^2}{b^2 - w^2} + \frac{n^2}{c^2 - w^2} = 0.$$

{It is hardly necessary to remark that if in the last-mentioned system of equations, the parameter  $w$  is also treated as variable, we obtain the wave surface: in fact  $v^2, w^2$  being the roots of the equation

$$\frac{l^2}{a^2 - \theta} + \frac{m^2}{b^2 - \theta} + \frac{n^2}{c^2 - \theta} = 0,$$

we have, attending to the condition  $l^2 + m^2 + n^2 = 1$ , the identical equation

$$l^2(b^2 - \theta)(c^2 - \theta) + m^2(c^2 - \theta)(a^2 - \theta) + n^2(a^2 - \theta)(b^2 - \theta) = (v^2 - \theta)(w^2 - \theta),$$

and thence

$$vw = \frac{abc}{w} \sqrt{\frac{l^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2}},$$

which shows that the system of equations for the plane  $Q$  ( $w$  being treated as variable) is in fact identical with the system of equations for the plane  $P$ .}

The form of the equations of the planes  $P$  and  $Q$  respectively shows that each of these planes is parallel to a tangent plane of the cone

$$\frac{l^2}{a^2 - v^2} + \frac{m^2}{b^2 - v^2} + \frac{n^2}{c^2 - v^2} = 0.$$

422

ON THE WAVE SURFACE.

[277

or in other words, that the planes  $P$  and  $Q$  are respectively tangents to the conic or infinitely thin surface of the second order, which is the section of the last-mentioned cone by the plane at infinity. Moreover it is obvious from the same equations that the plane  $P$  is a tangent plane of the sphere whose equation is

$$x^2 + y^2 + z^2 = v^2,$$

and that the plane  $Q$  is a tangent plane of the ellipsoid whose equation is

$$a^2x^2 + b^2y^2 + c^2z^2 = \frac{a^2b^2c^2}{w^2}.$$

Hence each of the developables  $F$  and  $G$  is the envelope of a plane which is the common tangent plane of two surfaces of the second order; such developables are in general of the eighth order, see my paper “On the Developable Surfaces which arise from Two Surfaces of the Second Order,” *Camb. and Dubl. Math. Journ.*, t. v. pp. 46–57 (1850), [84], and it will be presently seen that this is in fact the order of the developables  $F$  and  $G$  respectively.

The before-mentioned cone

$$\frac{l^2}{a^2 - v^2} + \frac{m^2}{b^2 - v^2} + \frac{n^2}{c^2 - v^2} = 0,$$

(Zech’s cone  $K$ ) is a cone having for its focal lines the optic axes (or normals to the circular sections) of the ellipsoid  $a^2x^2 + b^2y^2 + c^2z^2 = 1$  (Zech’s ellipsoid  $E$ ) which is used in the theory of the Wave Surface in the place of Fresnel’s Surface of Elasticity. The complementary cone

$$(a^2 - v^2)x^2 + (b^2 - v^2)y^2 + (c^2 - v^2)z^2 = 0,$$

(Zech’s cone  $C$ ) meets the last-mentioned ellipsoid in a curve lying on the sphere whose equation is  $x^2 + y^2 + z^2 = \frac{1}{v^2}$ , a property which may be considered as affording the geometrical construction of the magnitude  $v$ , by means of the cone  $K$ . The ellipsoid  $a^2x^2 + b^2y^2 + c^2z^2 = \frac{a^2b^2c^2}{w^2}$  is obviously an ellipsoid similar to the ellipsoid  $E$ , and the value of  $v$  being determined as above, the sphere  $x^2 + y^2 + z^2 = \frac{1}{v^2}$  and the ellipsoid  $a^2x^2 + b^2y^2 + c^2z^2 = \frac{a^2b^2c^2}{w^2}$  (which are used in the preceding geometrical construction of the developables  $F$  and  $G$  respectively) may be considered as given by construction.

To find the equation of the developable  $F$ , we have, from the equations

$$lx + my + nz = v,$$

$$l^2 + m^2 + n^2 = 1,$$

$$\frac{l^2}{a^2 - v^2} + \frac{m^2}{b^2 - v^2} + \frac{n^2}{c^2 - v^2} = 0,$$

277]

ON THE WAVE SURFACE.

423

treating them by the method of arbitrary multipliers in the usual manner, we obtain

$$x + \left( \rho + \frac{\theta}{a^2 - v^2} \right) l = 0,$$

$$y + \left( \rho + \frac{\theta}{b^2 - v^2} \right) m = 0,$$

$$z + \left( \rho + \frac{\theta}{c^2 - v^2} \right) n = 0,$$

equations which give in the first instance  $v + \rho = 0$  or  $\rho = -v$ , and then substituting this value for  $\rho$ ,

$$l = \frac{x(v^2 - a^2)}{v(v^2 - a^2) + \theta}, \quad m = \frac{y(v^2 - b^2)}{v(v^2 - b^2) + \theta}, \quad n = \frac{z(v^2 - c^2)}{v(v^2 - c^2) + \theta},$$

and thence

$$\frac{x^2(v^2 - a^2)}{v(v^2 - a^2) + \theta} + \frac{y^2(v^2 - b^2)}{v(v^2 - b^2) + \theta} + \frac{z^2(v^2 - c^2)}{v(v^2 - c^2) + \theta} = v,$$

$$\frac{x^2(v^2 - a^2)}{[v(v^2 - a^2) + \theta]^2} + \frac{y^2(v^2 - b^2)}{[v(v^2 - b^2) + \theta]^2} + \frac{z^2(v^2 - c^2)}{[v(v^2 - c^2) + \theta]^2} = 0,$$

the latter of which is the derived equation of the former with respect to the parameter  $\theta$ , hence writing the former equation under the form

$$\frac{x^2(v^2 - a^2)}{\theta + v(v^2 - a^2)} + \frac{y^2(v^2 - b^2)}{\theta + v(v^2 - b^2)} + \frac{z^2(v^2 - c^2)}{\theta + v(v^2 - c^2)} = v,$$

or what is the same thing,

$$(A, B, C, D \mid \theta, 1)^3 = 0,$$

where

$$A = 3v^2,$$

$$B = (v^2 - x^2)(v^2 - a^2) + (v^2 - y^2)(v^2 - b^2) + (v^2 - z^2)(v^2 - c^2),$$

$$C = (v^2 - y^2 - z^2)(v^2 - b^2)(v^2 - c^2) + (v^2 - z^2 - x^2)(v^2 - c^2)(v^2 - a^2) + (v^2 - x^2 - y^2)(v^2 - a^2)(v^2 - b^2),$$

the equation of the developable  $F$  is

$$(AD - BC)^2 - 4(AC - B^2)(BD - C^2) = 0.$$

The investigation for the developable  $G$  is very similar to the preceding. Write for shortness

$$\frac{abc}{w} \sqrt{\frac{l^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2}} = \Lambda,$$

then the system of equations is

$$lx + my + nz - \Lambda = 0,$$

$$l^2 + m^2 + n^2 = 1,$$

$$\frac{l^2}{a^2 - w^2} + \frac{m^2}{b^2 - w^2} + \frac{n^2}{c^2 - w^2} = 0,$$

424

ON THE WAVE SURFACE.

[277

and we then have

$$x - \frac{abc}{wa^2\Lambda} l + \left( \rho + \frac{\theta}{a^2 - w^2} \right) l = 0,$$

$$y - \frac{abc}{wb^2\Lambda} m + \left( \rho + \frac{\theta}{b^2 - w^2} \right) m = 0,$$

$$z - \frac{abc}{wc^2\Lambda} n + \left( \rho + \frac{\theta}{c^2 - w^2} \right) n = 0,$$

equations which give  $\rho = 0$ , and substituting this value, we obtain

$$l = \frac{wa^2\Lambda x}{abc - wa^2\Lambda \frac{\theta}{a^2 - w^2}} \quad \dots$$

Substituting these values in the equations

$$\frac{l^2}{a^2 - w^2} + \frac{m^2}{b^2 - w^2} + \frac{n^2}{c^2 - w^2} = 0,$$

$$\frac{l^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2} = \Lambda^2,$$

we find

$$\frac{x^2}{\left\{ \frac{abc}{wa^2\Lambda} (a^2 - w^2) - \theta \right\}^2} (a^2 - w^2) + \dots = 0,$$

$$\frac{x^2\left(\frac{abc}{wa^2} - \theta\right)^2}{\left\{\frac{abc}{wa^2\Lambda}(a^2 - w^2) - \theta\right\}^2} + \dots = \Lambda^2,$$

and multiplying the first equation by  $-\theta$  and the second equation by  $\Lambda^2$  and adding, we obtain

$$\frac{x^2(w^2 - a^2)}{\frac{abc}{wa^2}(a^2 - w^2) - \theta} + \frac{y^2(w^2 - b^2)}{\frac{abc}{wb^2}(b^2 - w^2) - \theta} + \frac{z^2(w^2 - c^2)}{\frac{abc}{wc^2}(c^2 - w^2) - \theta} = \frac{abc}{w},$$

of which equation the latter of the foregoing two equations is the derived equation with respect to the parameter  $\Lambda^2$ .

277]

ON THE WAVE SURFACE.

425

Comparing this with the foregoing equation

$$\frac{x^2(v^2 - a^2)}{v(v^2 - a^2) + \theta} + \frac{y^2(v^2 - b^2)}{v(v^2 - b^2) + \theta} + \frac{z^2(v^2 - c^2)}{v(v^2 - c^2) + \theta} = v,$$

we see that it is deduced from it by writing

$$a^2x^2, b^2y^2, c^2z^2, \frac{w^2}{a^2} - 1, \frac{w^2}{b^2} - 1, \frac{w^2}{c^2} - 1, \frac{abc}{w}, \Lambda\theta$$

in the place of

$$x^2, y^2, z^2, v^2 - a^2, v^2 - b^2, v^2 - c^2, v, \theta$$

respectively, and the equation of the developable  $G$  is therefore

$$(A'D' - B'C')^2 - 4(A'C' - B'^2)(B'D' - C'^2) = 0,$$

where  $A', B', C', D'$  are what  $A, B, C, D$  become when

$$x^2, y^2, z^2, v^2 - a^2, v^2 - b^2, v^2 - c^2, v, \theta$$

are replaced by

$$a^2x^2, b^2y^2, c^2z^2, \frac{w^2}{a^2} - 1, \frac{w^2}{b^2} - 1, \frac{w^2}{c^2} - 1, \frac{abc}{w}, \Lambda\theta.$$

The equations of the developables  $F$  and  $G$ , although radically distinct from each other, are consequently similar in form, and each is at once deducible from the other.

H

2, Stone Buildings, W.C., 9th March, 1858.

426

[278

## 278.

### NOTE ON THE SINGULAR SOLUTIONS OF DIFFERENTIAL EQUATIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 36, 37.]

THE following investigation (which has been in my possession for a good many years) affords I think a simple explanation of the theory of the singular solutions of differential equations.

Let the primitive equation be

$$c^n + Pc^{n-1} + Qc^{n-2} + \dots = 0,$$

where  $c$  is the arbitrary constant and  $P, Q \dots$  are any functions of  $x, y$ ; then the differential equation is obtained by eliminating  $c$  from the foregoing equation and the derived equation

$$P'c^{n-1} + Q'c^{n-2} + \dots = 0,$$

and the result may be represented by

$$F(P, Q, \dots, P', Q', \dots) = 0.$$

Assume now

$$c^n + Pc^{n-1} + Qc^{n-2} + \dots = (c + X)(c + Y)(c + Z) \dots,$$

then we have

$$\begin{aligned} P \text{ \&c.} &= X + Y + Z + \text{\&c.}, \\ Q &= XY + XZ + YZ + \text{\&c.}, \end{aligned}$$

and consequently

$$\begin{aligned} P' \text{ \&c.} &= X' + Y' + Z' + \text{\&c.}, \\ Q' &= (Y + Z + \text{\&c.})X' + \text{\&c.}, \end{aligned}$$

278]                      NOTE ON THE SINGULAR SOLUTIONS OF DIFFERENTIAL EQUATIONS.                      427

and substituting these values in the function  $F(P, Q, \dots, P', Q', \dots)$  it is clear that we shall have  $F(P, Q, \dots, P', Q', \dots) = UX'Y'Z' \dots$  where  $U$  is a symmetrical function of  $X, Y, Z, \text{\&c.}$ , and therefore a function of  $P, Q, \dots$ ; and this equation will be identically true whatever values we attribute to  $X', Y', Z', \dots$ , hence putting these quantities respectively equal to unity, we have

$$\begin{aligned} P' &= n, \\ Q' &= (n - 1)P, \\ R' \text{ \&c.} &= (n - 2)Q, \end{aligned}$$

and with these values

$$U = F(P, Q, \dots, P', Q', \dots),$$

that is  $U = 0$  is the result obtained by eliminating  $c$  from the primitive equation and the equation

$$nc^{n-1} + (n - 1)Pc^{n-2} + \dots = 0,$$

which is the equation obtained by differentiating the primitive equation with respect to the arbitrary constant  $c$ : that is,  $U = 0$  being the singular solution, the differential equation is

$$UX'Y'Z' \dots = 0.$$

It is to be remarked that ( $P, Q, \text{\&c.}$  being rational and integral functions) then if the roots  $X, Y, Z, \text{\&c.}$  are also rational and integral functions, the differential equation contains the differential equation does not really contain the rational and integral factor  $U$ , but  $X'Y'Z' \dots$  is here a rational fraction containing  $U$  in the denominator and  $UX'Y'Z' \dots$  is an indecomposable rational and integral function. This is easily verified *à posteriori* for a quadratic equation.

2, Stone Buildings, W.C., 28th January, 1858.

54–2

428

[279

## 279.

### ON A THEOREM RELATING TO SPHERICAL CONICS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), p. 53.]

THE following theorem was given by Prof. MacCullagh: “If three lines at right angles to each other pass through a fixed point  $O$  so that two of them are confined to given planes: the third line traces out a cone of the second order whose sections parallel to the given planes are circles, and the plane containing

the other two lines envelopes a cone of the second order whose sections by planes parallel to the given planes are parabolas.”

Referring the figure to the sphere we have a trirectangular triangle  $XYZ$ , of which two angles  $X, Y$  lie on fixed arcs  $A, B$ . The angle  $Z$  generates a spherical conic  $U'$  having  $A, B$  for its cyclic arcs. The side  $XY$  envelopes a spherical conic  $U$  touched by the arcs  $A, B$ . The conic  $U'$  is evidently the supplementary conic of  $U$ , hence the poles of  $A, B$  are the foci of  $U$ . We may drop altogether the consideration of the triangle  $XYZ$  and consider only the side  $XY$ , we have then the theorem:

If a quadrantal arc  $XY$  slides between the two fixed arcs  $A, B$ , the envelope of  $XY$  is a spherical conic  $U$  touched by the fixed arcs  $A, B$ , and which has for its foci the poles of these same arcs  $A, B$ .

It is worth while to notice the great reduction of order which takes place in consequence of the arc  $XY$  being a quadrant. If  $XY$  had been an arc of a given magnitude  $\theta$ , the envelope would have been a spherical curve of an order certainly higher than 6. For considering the corresponding problem *in plano*, the envelope in the particular case where the fixed lines  $A, B$  are at right angles to each other is a curve of the sixth order, and in the general case where the two fixed lines are not at right angles the order is higher: the problem *in plano* corresponds of course, not to the general problem on the sphere, but to that in which  $\theta$  is indefinitely small.

[continued p. 429]

280.

ON THE CONICS WHICH TOUCH FOUR GIVEN LINES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 94–96.]

There are considerable practical difficulties in drawing a figure of the system of conics which touch four given lines, but a notion of the figure of the system may be obtained as follows:

Figure 20 may be taken to represent any quadrilateral whatever, having all its sides real; and if we attend only to the unshaded spaces, it will be seen that there are five regions which are called the inner, upper, lower, right-hand, and left-hand regions respectively. The inner diagonals are  $AC$  the vertical diagonal and  $BD$  the horizontal diagonal; the former of these traverses the inner, upper, and lower regions; the latter the inner, right-hand, and left-hand regions; the outer diagonal is  $EF$  which traverses the upper region and the right-hand and left-hand regions. The inner or vertical and horizontal diagonals meet in the inner region, the vertical and outer diagonals meet in the upper region, the horizontal and outer diagonals meet in

[Figure 20: a complete quadrilateral  $ABCD$  with extended sides meeting at the diagonal points  $E, F$ ; the five regions (inner, upper, lower, right-hand, left-hand) are distinguished by shading.]

the left-hand region. No conic touching the four lines lies wholly or in part in the shaded regions, and every conic touching the four lines lies wholly in the inner region or wholly in the upper region, or partly in the upper and partly in the lower region, or partly in the right-hand and partly in the left-hand region. It will be convenient to consider, 1°. the conics which lie in the inner region; 2°. the conics which lie in the upper and lower regions, or in the upper region only; 3°. the conics which lie in the right-hand and left-hand regions.

1°. The conics in the inner region are obviously ellipses; an extreme term is the finite right line  $BD$ , considered as an indefinitely thin ellipse; this gradually broadens out and there is (as a mean term) an ellipse which touches the four lines in the points in which they are intersected two and two by the lines joining the points  $E, F$  with the point of intersection of the diagonals  $AC$  and  $BD$ , the ellipse then narrows in the transverse direction and at length reduces itself to the finite line  $AC$  considered as an indefinitely thin ellipse.

2°. Considering first the conics which lie in the upper and lower regions, these are of course hyperbolas, an extreme term is the infinite portions  $A\infty$  and  $C\infty$  of the line  $AC$ , considered as an indefinitely thin hyperbola: we have then hyperbolas such that for each of them, one branch lies in the lower region and touches the two lines through  $A$ , while the other branch lies in the upper region and touches the two lines through  $C$ ; the points of contact of the lower branch with the lines through  $A$  gradually recede from  $A$ , but the point of contact on the line  $AD$ , which adjoins the left-hand region, recedes with the greater rapidity, and it at last becomes infinite while the point of contact with the line  $AB$  which adjoins the right-hand region remains finite: we have thus a hyperbola having the line  $AD$  for its asymptote; the point at infinity of this line belongs of course indifferently to the upper and lower regions; and we may therefore consider one branch as lying in the lower region and touching  $AB$  and (at infinity)  $AD$ ; the other branch as lying in the upper region and touching the two lines through  $C$ , and besides (at infinity) the line  $AD$ . We have next a series of hyperbolas such that for each of them, one branch lies in the lower region and touches only the line  $AB$ , the other branch lies in the upper region and touches the two lines through  $C$  and besides the line  $AD$ . We arrive again at a limiting case when the point of contact on the line  $AB$  passes off to infinity, or  $AB$  becomes an asymptote; we have here in the lower region a branch touching (at infinity)  $AB$  and in the upper region a branch touching the two lines through  $C$ , the line  $AD$ , and besides (at infinity) the line  $AB$ . To this succeeds a series of hyperbolas such that for each of them, one branch lies in the lower region but does not touch either of the lines through  $A$ , the other branch lies in the upper region and touches as well the two lines through  $C$  as also the two lines through  $A$ . Finally the branch in the lower region passes off to infinity, or the conic becomes a parabola lying wholly in the upper region.

We have thus arrived at the conics which lie wholly in the upper region, the extreme term being a parabola: this passes into an ellipse touching of course the four lines, and gradually reducing itself to the finite line  $EF$  considered as an indefinitely thin ellipse.

280]

# ON THE CONICS WHICH TOUCH FOUR GIVEN LINES.

431

3°. We come now to the conics which lie in the right-hand and left-hand regions; an extreme term is the conic composed of the infinite portions  $E\infty$  and  $F\infty$  of the line  $EF$ , considered as an indefinitely thin hyperbola; we have then a series of hyperbolas such that for each of them the branch in the right-hand region touches the two lines through  $E$ , and the branch in the left-hand region the two lines through  $F$ ; we then arrive at a limiting case where the branch in the right-hand region touches the line  $EC$  the upper boundary of the right-hand region at  $\infty$ , or where  $EC$  is an asymptote; we may say that the branch in the right-hand region touches the line  $EB$  and (at infinity) the line  $EC$ , while the branch in the left-hand region touches not only the two lines through  $F$  but also (at infinity) the line  $EC$ ; we have next a series of hyperbolas such that for each of them the branch in the right-hand region touches only the line  $EB$  while the branch in the left-hand region touches as well the two lines through  $F$  as the line  $EC$ ; we have then again a limiting case, viz. the branch in the right-hand region touches the line  $BC$  at infinity, or  $BC$  is an asymptote; we may here say that the branch in the right-hand region touches the line  $BE$  and (at infinity) the line  $BC$ , while the branch in the left-hand region touches the lines through  $D$  and (at infinity) the line  $BC$ ; and to this succeeds a series of hyperbolas such that for each of them the branch in the right-hand region touches the two lines through  $B$  while the branch in the left-hand region touches the two lines through  $D$ ; the hyperbola finally becomes the infinite portions  $B\infty$  and  $D\infty$  of the line  $BD$ , considered as an indefinitely thin hyperbola.

It is to be noticed that the entire system commences with the finite line  $BD$  considered as an indefinitely thin ellipse and passes on continuously to the infinite portions  $B\infty$  and  $D\infty$  of the same line, considered as an indefinitely thin hyperbola: it is therefore a cyclical series, and any term whatever might have been taken for the commencement of the series.

432

[281

## 281.

### NOTE ON THE WAVE SURFACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 142–144.]

In the paper “On the Wave Surface” in the last Number of the *Journal*, [277], I stated that it was shown in the second of Zech’s Memoirs that the curves of contact of the developables  $F$  and  $G$  with the wave surface were the curves of curvature of the wave surface. I had not examined the demonstration of this theorem, and I had overlooked the author’s note “Die Krümmungslinien der Wellenfläche zweiaxiger Krystalle, &c.,” *Crelle*, t. LIV. p. 94 (Feb. 1858), where he points out that in a phrase which he quotes, he had assumed without demonstration a theorem which was in fact erroneous, and that he retracted all that he had said in No. 11 of his Memoir (the portion which contains the theorem as to the curves of curvature of the wave surface). M. Bertrand in a note in the *Comptes Rendus*, t. XLVII. pp. 817–819 (Nov. 1858), after referring to my paper, remarks that the theorem as to the curves of curvature of the wave surface appeared to him so remarkable that he hastened to investigate a proof of it, but that he very soon discovered that the theorem was unfortunately erroneous; and he proceeds to show why the theorem cannot be true. M. Bertrand’s demonstration is as follows:

THEOREM I. If from a point  $O$ , we let fall perpendiculars on the tangent planes of a surface, the locus of their feet is a new surface. Let  $P$  be a point of this surface corresponding to the point  $M$  of the first surface, the normal at  $P$  passes through the middle point of  $OM$ .

THEOREM II. If the curve of curvature of a surface is such that the tangent planes at the several points of the curve are equidistant from a point  $O$ , the curve of curvature is situate on a sphere having the point  $O$  for its centre.

For suppose that we let fall from the point  $O$  perpendiculars on the tangent planes to the surface at the several points of the curve of curvature in question. The



locus of the feet will be a spherical curve, a normal at any point  $P$  of the curve will it is clear pass through the point  $O$ ; besides, in virtue of the first theorem, another normal will pass through the middle point of the radius  $OM$  drawn to the corresponding point  $M$  of the surface; the tangent to the curve which is the locus of the points  $P$  is therefore perpendicular to the plane  $MOP$ , and consequently to the line  $MP$ . But when a developable surface is circumscribed about a sphere, the perpendiculars let fall from the centre of the sphere on the tangent planes of the developable surface have their feet on the generating lines; and consequently the curve which is the locus of the points  $P$  is situate on the developable surface (viz., the developable surface enveloped by the tangent planes at the several points of the curve of curvature of the given surface) and cuts the generating lines at right angles. But the curve, the locus of the point  $M$ , being by hypothesis a curve of curvature of the given surface, also cuts at right angles the generating lines of the developable surface, the two curves are therefore equidistant curves on the developable surface, viz.  $MP$  is constant, and since by hypothesis  $OP$  is constant,  $OM$  is also constant, which proves the second theorem.

This being premised, it is to be recollected that the wave surface may be generated in two different ways; 1°. it is the locus of the extremities of the central perpendiculars to the diametral sections of an ellipsoid  $E$ , equal to the axes of these sections; 2°. it is the envelope of planes parallel to the diametral sections of a second ellipsoid  $E'$  at distances inversely proportional to the axes of the sections. To obtain all the tangent planes of the wave surface situate at a distance  $h$  from the centre, it is necessary to find in the ellipsoid the diametral sections which have an axis equal to  $\frac{1}{h}$ ; for this, we may cut the ellipsoid by a concentric sphere of the radius  $\frac{1}{h}$ , and draw tangent planes to the cone having for its vertex the centre of the ellipsoid and passing through the curve of intersection with the sphere. The tangent planes of the wave surface respectively parallel to the tangent planes of the cone will be at the distance  $h$  from the centre, and, if they touched the wave surface along a curve of curvature, it would follow from the foregoing theorem II. that their points of contact would be all at the same distance from the centre; but the distance of the centre of the wave surface from the point of contact of a tangent plane is inversely proportional to the perpendicular let fall from the centre of the ellipsoid  $E'$  on the tangent plane at the corresponding point of this ellipsoid: it follows that the curve of intersection of an ellipsoid with a concentric sphere is not such that the tangent planes of the ellipsoid at the several points of the curve are at the same distance from the centre, and consequently the tangent planes which were under consideration do not determine by their points of contact a curve of curvature of the wave surface.

I remark in relation to M. Bertrand's theorem I. that it is immediately connected with a well-known theorem which occurs in optics—viz., if rays proceeding from a point are reflected at a surface, and if from the radiant point to the surface and thence along the reflected ray, we measure off a constant distance, the surface which is the locus of the point so obtained (the secondary focal surface of Dandelin and Quetelet)

C. IV.

55

is an orthogonal trajectory of the reflected rays. In fact, if  $O$  be the radiant point, and  $OM'$  the incident ray, and if from  $M'$  we measure off on the reflected ray a distance  $= -OM'$ , that is, on the reflected ray produced backwards, a distance  $M'P = OM'$ , then the whole distance from  $O$  is  $OM' - OM' = 0$ , and the surface which is the locus of the point  $P$  is consequently an orthogonal trajectory to the reflected rays. But the point  $P$  may, it is clear, be constructed as follows: viz., on the tangent plane at  $M'$  let fall the perpendicular  $OP'$ , and produce it to a point  $P$  such that  $OP' = P'P$ . And if we produce  $OM'$  to  $M$  so that  $OM' = M'M$ , then it is clear that the locus of  $M$  is a surface similar and similarly situated with the original surface, but of double the magnitude, and that  $OP$  is the perpendicular from  $O$  upon the tangent plane at  $M$  of the last-mentioned surface. And, by what precedes, the line  $PM'$  from  $P$  to the middle point of  $OM$  is a normal of the surface which is the locus of  $P$ : the corresponding theorem *in plano* is in fact actually given by Dandelin.

282.

ON A PARTICULAR CASE OF CASTILLON'S PROBLEM.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 157–164.]

The problem referred to as Castillon's problem, being itself a particular case of the problem of the in-and-circumscribed triangle, is as follows: viz. "In a circle to inscribe a triangle the sides of which pass through three given points," and the reciprocal problem of course presents itself "about a circle to circumscribe a triangle, the angles of which lie in given lines." If in Castillon's problem the three given points are the angles of a triangle circumscribed about the circle, or if in the reciprocal problem the three given lines are the sides of a triangle circumscribed about the circle, we have a circle and an inscribed and circumscribed triangle, such that the sides of the inscribed triangle pass through the angles of the circumscribed triangle, and the problem arises "given one of the triangles to determine the other triangle." The problem, so far as I am aware, was first proposed by Clausen, who has given, *Crelle*, t. IV. (1829), p. 391, a very elegant solution, which I propose to reproduce here.

Let the angle of a point be defined as the inclination to a fixed radius, of the line from the centre through the given point.

Let  $\alpha, \beta, \gamma$  be the angles of the points of contact of the sides of the circumscribed triangle.

$\alpha', \beta', \gamma'$ , the angles of the angular points of the circumscribed triangle.

$a, b, c$ , the angles of the angular points of the inscribed triangle.

$a', b', c'$ , the angles of the intersections of the perpendiculars from the centre on the sides of the inscribed triangle.

Therefore

$$2a' = b + c, \quad 2\alpha' = \beta + \gamma,$$

$$2b' = c + a, \quad 2\beta' = \gamma + \alpha,$$

$$2c' = a + b, \quad 2\gamma' = \alpha + \beta.$$

55–2

ON A PARTICULAR CASE OF CASTILLON'S PROBLEM.

Hence observing that the distance of one of the angles of the circumscribed triangle is  $\sec \frac{1}{2}(\beta - \gamma)$ , and that the projection of this line perpendicular to the corresponding side of the inscribed triangle is equal to  $\sec \frac{1}{2}(\beta - \gamma) \cos(\alpha' - a')$ , which is equal to the projection of the radius perpendicular to the same side, or to  $\cos \frac{1}{2}(b - c)$ , we have

$$\cos(\alpha' - a') = \cos \frac{1}{2}(\beta - \gamma) \cos \frac{1}{2}(b - c),$$

$$\cos(\beta' - b') = \cos \frac{1}{2}(\gamma - \alpha) \cos \frac{1}{2}(c - a),$$

$$\cos(\gamma' - c') = \cos \frac{1}{2}(\alpha - \beta) \cos \frac{1}{2}(a - b).$$

Write

$$b - c = 4f, \quad b + c - 2\alpha = 4x,$$

$$c - a = 4g, \quad c + a - 2\beta = 4y,$$

$$a - b = 4h, \quad a + b - 2\gamma = 4z;$$

therefore

$$\cos(y + z + g - h) = \cos 2f \cos(f + y - z),$$

$$\cos(z + x + h - f) = \cos 2g \cos(g + z - x),$$

$$\cos(x + y + f - g) = \cos 2h \cos(h + x - y),$$

or, since  $f + g + h = 0$ ,

$$\cos\{(y - h) + (z + g)\} = \cos 2f \cos\{(y - h) - (z + g)\},$$

$$\begin{aligned}\cos\{(z-f)+(x+h)\} &= \cos 2g \cos\{(z-f)-(x+h)\}, \\ \cos\{(x-g)+(y+f)\} &= \cos 2h \cos\{(x-g)-(y+f)\},\end{aligned}$$

or

$$\begin{aligned}\tan^2 f &= \tan(y-h) \tan(z+g), \\ \tan^2 g &= \tan(z-f) \tan(x+h), \\ \tan^2 h &= \tan(x-g) \tan(y+f).\end{aligned}$$

Put

$$\begin{aligned}y-h &= \zeta, & z+g &= \zeta_1, \\ z-f &= \eta, & x+h &= \eta_1, \\ x-g &= \xi, & y+f &= \xi_1, \\ \tan f &= l, & \tan \xi &= x, & \tan \xi_1 &= x_1, \\ \tan g &= m, & \tan \eta &= y, & \tan \eta_1 &= y_1, \\ \tan h &= n, & \tan \zeta &= z, & \tan \zeta_1 &= z_1.\end{aligned}$$

We have then

$$\begin{aligned}\tan^2 f &= \tan \zeta \tan \zeta_1, \\ \tan^2 g &= \tan \eta \tan \eta_1, \\ \tan^2 h &= \tan \xi \tan \xi_1,\end{aligned}$$

or, what is the same thing,

$$l^2 = xx_1, \quad m^2 = yy_1, \quad n^2 = zz_1.$$

282]

# ON A PARTICULAR CASE OF CASTILLON'S PROBLEM.

437

But to obtain equations involving only one of the sets  $(x, y, z), (x_1, y_1, z_1)$ , it is proper to write

$$\begin{aligned}\tan^2 f &= \tan(\zeta+g) \tan \zeta = \tan \zeta_1 \tan(\zeta_1-h), \\ \tan^2 g &= \tan(\eta+h) \tan \eta = \tan \eta_1 \tan(\eta_1-f), \\ \tan^2 h &= \tan(\xi+f) \tan \xi = \tan \xi_1 \tan(\xi_1-g);\end{aligned}$$

taking the first set of equations, we have

$$l^2 = \frac{x(z+m)}{1-mz} \cdot \frac{z(1-nx)}{x+n},$$

therefore

$$z = \frac{l^2 - mx}{x + l^2 n} = \frac{m^2(1-nx)}{x+n};$$

therefore

$$y+l : 1-ly = nl^2 + ln + (1-m^2n)x : n-lm^2 + (1+lm^2n)x;$$

therefore

$$\begin{aligned}m^2\{n-lm^2+(1+lm^2n)x\}(l^2n+x) &= (l^2-mx)\{m^2+ln+(1-m^2n)x\}; \\ & (l^4m^2+l^2n-l^2m^4n^2+l^2m^4n^2) \\ & +x(-n^2+lm^2n^2-l^2mn^2-l^2m^4-m^4-lmn+l^2-l^2m^2n) \\ & +x^2(-lm+m^4n-n^3-lm^2n^2) = 0.\end{aligned}$$

Now  $l+m+n=lmn$ , and by means of this relation we find

$$\frac{1}{l^2} \text{Coef. } x^0 = l\{lm^2+l^2n-n^2(l+m+n)+m(l+m+n)^2\},$$

$$\text{Coef. } x = -n^2+mn(l+m+n)-ln(l+m+n)-(l+m+n)^2-m^2-lmn+l^2-l^2m^2n(l+m+n),$$

$$\frac{1}{m^2} \text{Coef. } x^2 = \{-l^2m + m^2(l + m + n) - ln^2 - n(l + m + n)^2\};$$

or, reducing,

$$\begin{aligned} \frac{1}{l^2} \text{coef. } x^0 &= (m^3 + 2m^2n - n^3) + l(3m^2 + 2mn - n^2) + l^2(m + n) \\ &= (m + n)\{(m^2 + mn - n^2) + l(3m - n) + l^2\} \\ &= (m + n)\{(l + m)^2 + (l + m)(l + n) - (l + n)^2\}, \\ -\text{coef. } x &= m^3 + m^2n + mn^2 + n^3 + 2l(m^2 + 2mn + n^2) + 2l^2(m + n) \\ &= (m + n)\{m^2 + n^2 + 2l(m + n) + 2l^2\} \\ &= (m + n)\{(l + m)^2 + (l + n)^2\}, \\ \frac{1}{m^2} \text{coef. } x^2 &= (m^3 - 2m^2n - n^3) + l(n^2 - 2mn - 3n^2) - l^2(m + n) \\ &= (m + n)\{(m^2 - mn - n^2) + l(m - 3n) - l^2\} \\ &= (m + n)\{(l + m)^2 - (l + m)(l + n) - (l + n)^2\}; \end{aligned}$$

438

ON A PARTICULAR CASE OF CASTILLON'S PROBLEM.

[282

and the equation becomes

$$\begin{aligned} &l^2\{(l + m)^2 + (l + m)(l + n) - (l + n)^2\} \\ &\quad - 2lx\{(l + m)^2 + (l + n)^2\} \\ &\quad + x^2\{(l + m)^2 - (l + m)(l + n) - (l + n)^2\} = 0, \end{aligned}$$

which may be written

$$[l\{(l + m)^2 - (l + m)(l + n) - (l + n)^2\} - x\{(l + m)^2 + (l + n)^2\}]^2 = 5(l + m)^2(l + n)^2,$$

and we have therefore

$$\frac{x}{l} = \frac{(l + m)^2 + \sqrt{(5)}(l + m)(l + n) + (l + n)^2}{(l + m)^2 - (l + m)(l + n) - (l + n)^2},$$

or, reducing and observing also that  $l^2 = xx_1$ , we have

$$\frac{x}{l} = \frac{x_1}{l} = \frac{l + m + \frac{1}{2}\{1 + \sqrt{(5)}\}(l + n)}{(l + m) - \frac{1}{2}\{1 + \sqrt{(5)}\}(l + n)}.$$

The values of  $(l, m, n)$  are

$$l = \tan \frac{1}{2}(b - c), \quad m = \tan \frac{1}{2}(c - a), \quad n = \tan \frac{1}{2}(a - b),$$

and those of  $(x, y, z), (x_1, y_1, z_1)$  are

$$\begin{aligned} x &= \tan \frac{1}{2}(b + c - 2\gamma), \quad y = \tan \frac{1}{2}(c + a - 2\alpha), \quad z = \tan \frac{1}{2}(a + b - 2\beta), \\ x_1 &= \tan \frac{1}{2}(b + c - 2\beta), \quad y_1 = \tan \frac{1}{2}(c + a - 2\beta), \quad z_1 = \tan \frac{1}{2}(a + b - 2\alpha), \end{aligned}$$

and the foregoing result shows that the values of

$$\frac{x}{l} \text{ or } \frac{x_1}{l}; \quad \frac{y}{m} \text{ or } \frac{y_1}{m}; \quad \frac{z}{n} \text{ or } \frac{z_1}{n}$$

are

$$\begin{aligned} &\frac{\sin \frac{1}{2}(a - b) + \frac{1}{2}\{1 + \sqrt{(5)}\} \sin \frac{1}{2}(c - a)}{\sin \frac{1}{2}(a - b) - \frac{1}{2}\{1 + \sqrt{(5)}\} \sin \frac{1}{2}(c - a)}, \\ &\frac{\sin \frac{1}{2}(b - c) + \frac{1}{2}\{1 + \sqrt{(5)}\} \sin \frac{1}{2}(a - b)}{\sin \frac{1}{2}(b - c) - \frac{1}{2}\{1 + \sqrt{(5)}\} \sin \frac{1}{2}(a - b)}, \end{aligned}$$

$$\frac{\sin \frac{1}{2}(c-a) + \frac{1}{2}\{1 + \sqrt{(5)}\} \sin \frac{1}{2}(b-c)}{\sin \frac{1}{2}(c-a) - \frac{1}{2}\{1 + \sqrt{(5)}\} \sin \frac{1}{2}(b-c)}.$$

Write for a moment  $\frac{x}{l} = \frac{x_1}{l} = k$ ; therefore

$$\frac{k-1}{1+k} \cdot \frac{1}{xx_1} \cdot \frac{l(k-\frac{1}{k})}{1+k} = -l \frac{P-Q}{P+Q} \text{ if } k = \frac{P}{Q};$$

and taking  $P, Q$  for the numerator and the denominator respectively of the foregoing fractional expression for  $k$ , we find

282]

ON A PARTICULAR CASE OF CASTILLON'S PROBLEM.

439

also

$$\begin{aligned} P^2 - Q^2 &= 2\{1 + \sqrt{(5)}\} \sin \frac{1}{2}(a-b) \sin \frac{1}{2}(c-a), \\ 2PQ &= 2[\sin^2 \frac{1}{2}(a-b) - \frac{1}{4}\{1 + \sqrt{(5)}\}^2 \sin^2 \frac{1}{2}(c-a)] \\ &= -\{1 + \sqrt{(5)}\}[\frac{1}{2}\{1 - \sqrt{(5)}\} \sin^2 \frac{1}{2}(a-b) + \frac{1}{2}\{1 + \sqrt{(5)}\} \sin^2 \frac{1}{2}(c-a)]; \end{aligned}$$

also

$$\frac{z-x}{1+xx_1} = \tan \frac{1}{2}(\beta - \gamma),$$

we have therefore

$$\begin{aligned} \tan \frac{1}{2}(\beta - \gamma) &= \frac{-2 \sin \frac{1}{2}(b-c) \sin \frac{1}{2}(c-a) \sin \frac{1}{2}(a-b)}{\frac{1}{2}\{1 - \sqrt{(5)}\} \sin^2 \frac{1}{2}(a-b) + \frac{1}{2}\{1 + \sqrt{(5)}\} \sin^2 \frac{1}{2}(c-a)}, \\ \tan \frac{1}{2}(\gamma - \alpha) &= \frac{-2 \sin \frac{1}{2}(b-c) \sin \frac{1}{2}(c-a) \sin \frac{1}{2}(a-b)}{\frac{1}{2}\{1 - \sqrt{(5)}\} \sin^2 \frac{1}{2}(b-c) + \frac{1}{2}\{1 + \sqrt{(5)}\} \sin^2 \frac{1}{2}(a-b)}, \\ \tan \frac{1}{2}(\alpha - \beta) &= \frac{-2 \sin \frac{1}{2}(b-c) \sin \frac{1}{2}(c-a) \sin \frac{1}{2}(a-b)}{\frac{1}{2}\{1 - \sqrt{(5)}\} \sin^2 \frac{1}{2}(c-a) + \frac{1}{2}\{1 + \sqrt{(5)}\} \sin^2 \frac{1}{2}(b-c)}, \end{aligned}$$

equations which determine the circumscribed triangle when the inscribed triangle is given.

The more general problem, for a conic and an inscribed and circumscribed triangle such that the sides of the inscribed triangle pass through the angles of the circumscribed triangle, “given one of the triangles to determine the other” is solved by Möbius, *Crelle*, t. v. (1830), p. 103, by means of his Barycentric Calculus, which is in fact the method of trilinear coordinates. The solution is in effect as follows:

Let  $\xi = 0, \eta = 0, \zeta = 0$  be the equations of the sides of the inscribed triangle,  $\xi, \eta, \zeta$  may be considered as containing each of them an implicit constant factor, and the equation of the conic may be taken to be

$$\eta\zeta + \zeta\xi + \xi\eta = 0;$$

moreover, if  $x = 0, y = 0, z = 0$  be the equations of the sides of the circumscribed triangle, then considering  $x, y, z$  as also containing each of them an implicit constant factor, the equation of the conic may be taken to be

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0.$$

Suppose now the sides of the inscribed triangle pass through the angles of the circumscribed triangle, we have equations such as

$$\xi = b'y + cz, \quad \eta = c'z + ax, \quad \zeta = a'x + by;$$

substituting these values we must have identically

$$\eta\zeta + \zeta\xi + \xi\eta + m(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy) = 0,$$

$$aa' + m = 0, \quad bc + bc' + b'c = 2m,$$

$$bb' + m = 0, \quad ca + ca' + c'a = 2m,$$

$$cc' + m = 0, \quad ab + ab' + a'b = 2m,$$

and, substituting for  $a', b', c'$  their values,

$$(m - bc)^2 = mb^2, \quad (m - ca)^2 = mc^2, \quad (m - ab)^2 = ma^2,$$

or, if  $m = n^2$ ,

$$n^2 - bc = nb, \quad n^2 - ca = nc, \quad n^2 - ab = na,$$

[the signs on the second side must be all + or all - and the - sign may be omitted without loss of generality].

Hence

$$a = b = c = -\frac{1}{2}\{1 + \sqrt{(5)}\}n, \quad \sqrt{(5)} \text{ being written for } +\sqrt{(5)}.$$

$$a' = b' = c' = -\frac{1}{2}\{1 - \sqrt{(5)}\}n = -va,$$

if for shortness

$$v = \frac{1 - \sqrt{(5)}}{1 + \sqrt{(5)}}, \text{ i.e. } v = \frac{\sqrt{(5)} - 1}{\sqrt{(5)} + 1}, \text{ or } v^2 - 3v + 1 = 0,$$

and  $v$  having this value, the equations give

$$\xi = vy - z, \quad \eta = vz - x, \quad \zeta = vx - y,$$

whence also

$$4(2v - 1)x = (3v - 1)\xi + v\eta + \zeta,$$

$$4(2v - 1)y = \xi + (3v - 1)\eta + v\zeta,$$

$$4(2v - 1)z = v\xi + \eta + (3v - 1)\zeta.$$

Each side of the circumscribed triangle has on it four points, viz. two angles of the circumscribed triangle, a point of contact with the conic, and an intersection with the corresponding side of the inscribed triangle. Thus for the side  $x = 0$ , the four points are given as the intersections of  $x = 0$ , with

$$y = 0, \quad z = 0, \quad y - z = 0, \quad vy + z = 0,$$

and the anharmonic ratio of the four points is therefore a given quantity.\*

Again, each side of the inscribed triangle has on it four points, viz. two angles of the inscribed triangle, a point of intersection with the tangent at the opposite angle of the inscribed conic, and a point of intersection with the corresponding side of the circumscribed triangle.

Thus for the side  $\xi = 0$ , the four points are given as the points of intersection of  $\xi = 0$  with the lines

$$\eta = 0, \quad \zeta = 0, \quad \eta + \zeta = 0, \quad \eta + (3v - 1)\zeta = 0,$$

and the anharmonic ratio of the four points is therefore a given quantity.\*

If we draw tangents at the angles of the inscribed triangle, we have a new triangle, the sides of which are  $\eta + \zeta = 0$ ,  $\zeta + \xi = 0$ ,  $\xi + \eta = 0$ , and joining the angles of this triangle with the points of contact of the opposite sides (i.e. the angles of the

inscribed triangle), we have three lines  $\eta - \zeta = 0$ ,  $\zeta - \xi = 0$ ,  $\xi - \eta = 0$  meeting in a point  $\xi = \eta = \zeta$ , which is obviously the same as the point  $x = y = z$ , which is the point of intersection of the lines joining the angles of the circumscribed triangle with the points of contact of the opposite sides.\*

The coordinates of the points of contact of the sides of the circumscribed triangle are given by  $(x = 0, y - z = 0)$ ,  $(y = 0, z - x = 0)$ ,  $(z = 0, x - y = 0)$ , these points form therefore an inscribed triangle the sides of which are

$$y + z - x = 0, \quad z + x - y = 0, \quad x + y - z = 0.$$

Again, the tangents at the angles of the inscribed triangle form a circumscribed triangle the sides of which are

$$\eta + \zeta = 0, \quad \zeta + \xi = 0, \quad \xi + \eta = 0;$$

therefore

$$(\xi + \eta) - v(\zeta + \eta) = (1 - v)\eta - v\zeta + \xi = 2vx - (1 - v + v^2)y - (1 - v + v^2)z = -2v(y + z - x),$$

and we thus have

$$\begin{aligned}\xi + \eta - v(\zeta + \eta) &= -2v(y + z - x), \\ \eta + \zeta - v(\eta + \xi) &= -2v(z + x - y), \\ \zeta + \xi - v(\xi + \eta) &= -2v(x + y - z),\end{aligned}$$

equations which show that the sides of the second inscribed triangle pass through the angles of the second circumscribed triangle, and that the two systems are consequently reciprocal.\*

The four theorems marked \* are all of them contained in the paper by Möbius.

C. IV.

56

442

[283

### 283.

#### ON A THEOREM RELATING TO HOMOGRAPHIC FIGURES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 177–180.]

The following theorem is not new, but I do not remember where to find it:

Given any two homographic figures; there exists in the first figure a point  $S$ , which is the focus of an (ellipse or hyperbola, say an) ellipse  $\Sigma$ ; and in the second figure a point  $S'$ , which is the focus of a (hyperbola or ellipse, say a) hyperbola  $\Sigma'$ , such that the points  $S, S'$  and the conics  $\Sigma, \Sigma'$  correspond to each other, and the conics  $\Sigma, \Sigma'$  are so related that the foci and vertices of the one may be superimposed upon the vertices and foci of the other. Moreover the perpendiculars from  $S, S'$  upon corresponding tangents of  $\Sigma, \Sigma'$  will be equal.

Write

$$x : y : 1 = \alpha\xi + \beta\eta + \gamma : \alpha'\xi + \beta'\eta + \gamma' : \alpha''\xi + \beta''\eta + \gamma'',$$

then  $l, m, \lambda, \mu$  may be so determined that  $(x - l) + i(y - m)$  shall vanish with  $(\xi - \lambda) + i(\eta - \mu)$ , for if we write

$$\begin{aligned}J &= \alpha - l\alpha'', & J' &= \alpha' - m\alpha'', \\ K &= \beta - l\beta'', & K' &= \beta' - m\beta'', \\ L &= \gamma - l\gamma'', & L' &= \gamma' - m\gamma'',\end{aligned}$$

we have

$$(x - l) + i(y - m) = \frac{(J + iJ')\xi + (K + iK')\eta + (L + iL')}{\alpha''\xi + \beta''\eta + \gamma''},$$

and therefore

$$\begin{aligned}K + iK' &= i(J + iJ'), \\ L + iL' &= (J + iJ')(\lambda + i\mu),\end{aligned}$$

283]

#### ON A THEOREM RELATING TO HOMOGRAPHIC FIGURES.

443

that is

$$K = -J', \quad K' = J,$$

or

$$\begin{aligned}l\beta'' + m\alpha'' &= \alpha' + \beta, \\ -l\alpha'' + m\beta'' &= -\alpha + \beta',\end{aligned}$$

whence also

$$l(\alpha''^2 + \beta''^2) = \alpha'\beta'' - \alpha''\beta' + \alpha\alpha'' + \beta\beta'',$$

$$m(\alpha''^2 + \beta''^2) = \alpha''\beta - \alpha\beta'' + \alpha\alpha' + \beta'\beta'',$$

which determine  $l, m$ ; and then  $\lambda, \mu$  are given by

$$\lambda + i\mu = \frac{L + iL'}{J + iJ'},$$

but more simply as follows, viz. remarking that if  $c = \beta'\gamma'' - \beta''\gamma'$ , &c., so that

$$\xi : \eta : 1 = ax + a'y + a'' : bx + b'y + b'' : cx + c'y + c'',$$

then we have

$$\lambda(c^2 + c'^2) = bc' - b'c + ac + a'c',$$

$$\mu(c^2 + c'^2) = ca' - c'a + bc + b'c';$$

the equations  $x = l, y = m$  give the point  $S$  and  $\xi = \lambda, \eta = \mu$  the point  $S'$ . The values of  $J, J'$  are

$$J = \frac{-c\alpha'' - c'\beta''}{\alpha''^2 + \beta''^2}, \quad J' = \frac{-c'\alpha'' + c\beta''}{\alpha''^2 + \beta''^2},$$

and therefore

$$J^2 + J'^2 = \frac{c^2 + c'^2}{\alpha''^2 + \beta''^2} = \frac{k^2}{K^2};$$

if

$$k = \sqrt{c^2 + c'^2}, \quad K = \sqrt{(\alpha''^2 + \beta''^2)};$$

we have therefore

$$k = \sqrt{c^2 + c'^2}, \quad K = \sqrt{(\alpha''^2 + \beta''^2)}.$$

Consider now the expression

$$(x - l) \cos \vartheta + (y - m) \sin \vartheta - \varpi,$$

which made equal to 0 would be the equation of a line the perpendicular distance of which from  $S$  is  $\varpi$ , and which distance is inclined at an angle  $\vartheta$  to the axis of  $x$ ; then

$$\begin{aligned} (x - l) \cos \vartheta + (y - m) \sin \vartheta &= \frac{1}{2}[(\cos \vartheta - i \sin \vartheta)\{(x - l) + i(y - m)\} \\ &\quad + (\cos \vartheta + i \sin \vartheta)\{(x - l) - i(y - m)\}] \\ &= \frac{1}{2(\alpha''\xi + \beta''\eta + \gamma'')}[(\cos \vartheta - i \sin \vartheta)\{(\xi - \lambda) + i(\eta - \mu)\}(J + iJ') \\ &\quad + (\cos \vartheta + i \sin \vartheta)\{(\xi - \lambda) - i(\eta - \mu)\}(J - iJ')] \end{aligned}$$

56-2

444

ON A THEOREM RELATING TO HOMOGRAPHIC FIGURES.

[283

$$\begin{aligned} &= \frac{k}{2(\alpha''\xi + \beta''\eta + \gamma'')K} [\{\cos(\vartheta - \vartheta_0) - i \sin(\vartheta - \vartheta_0)\}\{(\xi - \lambda) + i(\eta - \mu)\} + \{\cos(\vartheta - \vartheta_0) + i \sin(\vartheta - \vartheta_0)\}\{(\xi - \lambda) - i(\eta - \mu)\}] \\ &= \frac{k}{(\alpha''\xi + \beta''\eta + \gamma'')K} [(\xi - \lambda) \cos(\vartheta - \vartheta_0) + (\eta - \mu) \sin(\vartheta - \vartheta_0)], \end{aligned}$$

where  $J, J'$  have been replaced by

$$J = \frac{k}{K} \cos \vartheta_0, \quad J' = -\frac{k}{K} \sin \vartheta_0.$$

And putting besides

$$\vartheta - \vartheta_0 = \theta,$$



we have more simply

$$(x-l)\cos\vartheta + (y-m)\sin\vartheta = \frac{k}{(\alpha''\xi + \beta''\eta + \gamma'')K} [(\xi - \lambda)\cos\theta + (\eta - \mu)\sin\theta],$$

whence

$$\begin{aligned} & (x-l)\cos\vartheta + (y-m)\sin\vartheta - \varpi \\ &= \frac{k}{(\alpha''\xi + \beta''\eta + \gamma'')K} \left\{ (\xi - \lambda)\cos\theta + (\eta - \mu)\sin\theta - \frac{\varpi K(\alpha''\xi + \beta''\eta + \gamma'')}{k} \right\}. \end{aligned}$$

Write now

$$\alpha'' = K \cos j, \quad \beta'' = K \sin j; \quad a = \frac{k}{K}, \quad b = \frac{k}{K}(\alpha''\lambda + \beta''\mu + \gamma'');$$

then

$$\begin{aligned} & (x-l)\cos\vartheta + (y-m)\sin\vartheta - \varpi \\ &= \sqrt{(\alpha''^2 + \beta''^2)} a [(\xi - \lambda)\cos\phi + (\eta - \mu)\sin(\phi - \varpi)], \end{aligned}$$

where it will be noticed that

$$(x-l)\cos\vartheta + (y-m)\sin\vartheta - \varpi = 0,$$

and

$$(\xi - \lambda)\cos\phi + (\eta - \mu)\sin\phi - \varpi = 0,$$

283]

# ON A THEOREM RELATING TO HOMOGRAPHIC FIGURES.

445

are lines in the first figure and in the second figure, corresponding to each other and at the same distance  $\varpi$  from the points  $S, S'$  respectively. Call these lines  $T$  and  $T'$ .

But the relations between  $\theta, \phi, \varpi$  are given by

$$\varpi \cos j = a \cos \theta - b \cos \phi,$$

$$\varpi \sin j = a \sin \theta - b \sin \phi,$$

or, by changing the fixed axes from which  $\theta, \phi$  are respectively measured,

$$\varpi = a \cos \theta - b \cos \phi,$$

$$0 = a \sin \theta - b \sin \phi.$$

Now in these equations  $\varpi$  is the perpendicular distance from  $S$  upon the line  $(x-l)\cos\vartheta + (y-m)\sin\vartheta - \varpi = 0$  and  $\theta$  (which only differs from  $\vartheta$  by a constant angle) is the inclination of this perpendicular to a certain fixed line;  $\varpi$  is also the perpendicular distance of the line  $(\xi - \lambda)\cos\phi + (\eta - \mu)\sin\phi - \varpi = 0$  from the point  $S'$ , and  $\phi$  is the inclination of this perpendicular to a fixed line. Eliminating successively  $\phi$  and  $\theta$ , we have

$$\varpi = a \cos \theta - \sqrt{(b^2 - a^2 \sin^2 \theta)},$$

$$\varpi = -b \cos \phi + \sqrt{(a^2 - b^2 \sin^2 \phi)},$$

or, as these equations may also be written,

$$\varpi^2 - 2\varpi a \cos \theta + a^2 - b^2 = 0,$$

$$\varpi^2 + 2\varpi b \cos \phi + b^2 - a^2 = 0.$$

Suppose  $a > b$ , the former equation shows that the line  $T$  is a tangent to a certain hyperbola, and the latter equation shows that the line  $T'$  is a tangent to a certain ellipse, and it is easily seen that, taking for the transverse axes the lines from which the angles  $\theta$  and  $\phi$  are respectively measured, the equation of the hyperbola is

$$\frac{x^2}{b^2} - \frac{y^2}{a^2 - b^2} = 1,$$

and that of the ellipse is

$$\frac{\xi^2}{a^2} + \frac{\eta^2}{a^2 - b^2} = 1,$$

which are the conics  $\Sigma, \Sigma'$  referred to in the enunciation. And if the second conic is superimposed upon the first in such manner that the coordinates  $\xi, \eta$  may belong to the same axes with  $x, y$ , then the two conics will have the assumed relation, viz. the foci of either conic will coincide with the vertices of the other conic.

446

[284

## 284.

### ON A NEW ANALYTICAL REPRESENTATION OF CURVES IN SPACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 225–236.]

The ordinary analytical representation of a curve in space by the equations of two surfaces passing through the curve is, even in the case where the curve is the complete intersection of the two surfaces, inappropriate as involving the consideration of surfaces which are extraneous to the curve; and the objection becomes more serious when the curve is not the complete intersection of any two surfaces; for in this case the curve can only be represented in conjunction with a curve or curves extraneous to itself. A curve in space can be in a mode represented by means of the equation of the developable surface having the curve for its edge of regression; but this corresponds to the representation of a plane curve by means of its equation in line coordinates; a representation which is very useful in addition to, but which is not to be substituted for, the equation in point coordinates. It occurred to me some years ago that it might be advantageous to represent a curve in space by means of the cone passing through the curve and having for its vertex an arbitrary point<sup>1</sup>; and although I have not advanced beyond the first steps of the theory, the results which I have obtained may, I think, be interesting to geometers. The conclusion is that a curve in space may be represented by a homogeneous equation  $V = 0$  between six coordinates  $(p, q, r, s, t, u)$  such that  $ps + qt + ru = 0$ ; the function  $V$  is moreover such that in virtue of this relation between the coordinates, and of the equation  $V = 0$  itself, we have

$$d_p V \cdot d_s V + d_q V \cdot d_t V + d_r V \cdot d_u V = 0,$$

284]

### ON A NEW ANALYTICAL REPRESENTATION OF CURVES IN SPACE.

447

or what is the same thing we have identically

$$d_p V \cdot d_s V + d_q V \cdot d_t V + d_r V \cdot d_u V = LV + M(ps + qt + ru),$$

where  $L$  and  $M$  are functions of  $p, q, r, s, t, u$ . But the converse proposition, viz. any equation whatever  $V = 0$ , where  $V$  satisfies the condition just referred to represents a curve in space, is not true; it would obviously be an important point in the theory to ascertain what further conditions must be satisfied by the function  $V$ .

The establishment of the foregoing results is very easy; in fact if  $x, y, z, w$  are current coordinates of the ordinary kind (point-coordinates) and  $\alpha, \beta, \gamma, \delta$  the coordinates of an arbitrary point; the equation of any cone whatever having for its vertex the point  $(\alpha, \beta, \gamma, \delta)$  may be represented by a homogeneous equation between the six determinants of the matrix

$$\begin{pmatrix} x & y & z & w \\ \alpha & \beta & \gamma & \delta \end{pmatrix}$$

or if we write

$$\begin{aligned} p &= \gamma y - \beta z, & s &= \delta x - \alpha w, \\ q &= \alpha z - \gamma x, & t &= \delta y - \beta w, \end{aligned}$$

<sup>1</sup>See my paper "On the cones which pass through a curve of the third order in space," *Phil. Mag.*, t. XII. (1856), p. 20, [200], where a curve of the third order in space is in effect so represented.

$$r = \beta x - \alpha y, \quad u = \delta z - \gamma w,$$

values which it is well known give identically

$$ps + qt + ru = 0,$$

then the cone will be represented by a homogeneous equation

$$V = 0$$

between the six coordinates  $(p, q, r, s, t, u)$ . It remains to find the conditions in order that all the cones so represented, viz. the cones obtained by giving any values whatever to the arbitrary quantities  $\alpha, \beta, \gamma, \delta$  which enter implicitly into the coordinates  $p, q, r, s, t, u$ , pass through one and the same curve; for when this is the case, the equation  $V = 0$  may be properly considered as the equation of the curve.

Assume then that all the cones pass through the same curve; if we give to one of the arbitrary quantities  $\alpha, \beta, \gamma, \delta$ , say  $\alpha$ , the infinitesimal variation  $d\alpha$ , then the function  $V$  becomes  $V + d_\alpha V \cdot d\alpha$ , and each of the equations  $V = 0$ ,  $V + d_\alpha V \cdot d\alpha = 0$  belongs to a cone passing through the curve; the equation  $d_\alpha V = 0$  is therefore the equation of a surface passing through the curve; and in like manner the four equations

$$d_\alpha V = 0, \quad d_\beta V = 0, \quad d_\gamma V = 0, \quad d_\delta V = 0$$

are each of them the equation of a surface passing through the curve, or these equations must be simultaneously satisfied for all the points of the curve, they must consequently reduce themselves to two independent relations. But  $V$  is given as a

448                      ON A NEW ANALYTICAL REPRESENTATION OF CURVES IN SPACE.                      [284

function of  $p, q, r, s, t, u$ , through which quantities it is a function of  $\alpha, \beta, \gamma, \delta$ ; the last-mentioned four equations become therefore

$$\begin{aligned} 0 &= -d_r V \cdot y + d_q V \cdot z - d_t V \cdot w, \\ 0 &= d_r V \cdot x - d_p V \cdot z - d_u V \cdot w, \\ 0 &= -d_q V \cdot x + d_p V \cdot y - d_u V \cdot w, \\ 0 &= d_s V \cdot x + d_t V \cdot y + d_u V \cdot z, \end{aligned}$$

and from the first, second, and third equation, or any other combination of three equations, we obtain at once the condition

$$d_p V \cdot d_s V + d_q V \cdot d_t V + d_r V \cdot d_u V = 0,$$

so that if this condition is satisfied, the four equations do in fact reduce themselves to two independent equations; the condition in question is thus shown to be necessary. But the condition only implies that all the cones having their vertices in the neighbourhood of the point  $(\alpha, \beta, \gamma, \delta)$  pass through one and the same curve; this will be the case for instance for a series of cones all of them circumscribed about one and the same surface, those having their vertices in the neighbourhood of the point  $(\alpha, \beta, \gamma, \delta)$  will all pass through the curve of contact with the surface of the cone having for its vertex the point in question. But the curve of contact is not a fixed curve for all positions of the vertex, and the condition before referred to is consequently insufficient.

It may be noticed that the systems  $p, q, r$  and  $s, t, u$  are not similar to each other and that the six coordinates cannot be in any way divided into two systems which are similar to each other: the symmetry of the coordinates is in fact that of the vertices (or sides) of a complete quadrilateral (or quadrangle); thus we may divide the coordinates into two sets in a fourfold manner as follows:

$$\begin{aligned} u, t, p; & \quad r, q, s, \\ s, q, u; & \quad p, t, r, \\ r, t, s; & \quad u, q, p, \\ p, q, r; & \quad s, t, u, \end{aligned}$$

where each left-hand set corresponds to three vertices forming a triangle and each right-hand set to the remaining three vertices *in lineo*. It may be noticed also that if in the equation  $V = 0$  of any curve in space

we substitute for  $p, q, r, s, t, u$ , their values, and equate to zero the coefficients of the different powers and products of  $\alpha, \beta, \gamma, \delta$ , each of the equations so obtained will belong to a surface passing through the curve, and the entire system of these equations will be equivalent to two relations only between the coordinates  $x, y, z, w$ . But any two of these surfaces will not in general intersect only in the curve, i.e. the curve will not be the complete intersection of any two of the surfaces. It may be added that the equation of any other surface whatever through the curve will be obtained by equating to zero a syzygetic function of the functions which equated to zero give the surfaces first referred to.

284] ON A NEW ANALYTICAL REPRESENTATION OF CURVES IN SPACE. 449

As an example of the theory, the equation, in the new coordinates, of a line in space will be

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

where the constants are such that

$$AF + BG + CH = 0.$$

This may be verified as follows: suppose that the line in question is given as the line of junction of the points  $(\alpha', \beta', \gamma', \delta')$  and  $(\alpha'', \beta'', \gamma'', \delta'')$ ; then the cone through the line and the arbitrary point  $(\alpha, \beta, \gamma, \delta)$  is nothing else than the plane through the three points; its equation is therefore

$$\begin{vmatrix} x & y & z & w \\ \alpha & \beta & \gamma & \delta \\ \alpha' & \beta' & \gamma' & \delta' \\ \alpha'' & \beta'' & \gamma'' & \delta'' \end{vmatrix} = 0,$$

which may be written

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

the values of  $A$ , &c. being

$$\begin{aligned} A &= \alpha'\delta'' - \alpha''\delta', & F &= \beta'\gamma'' - \beta''\gamma', \\ B &= \beta'\delta'' - \beta''\delta', & G &= \gamma'\alpha'' - \gamma''\alpha', \\ C &= \gamma'\delta'' - \gamma''\delta', & H &= \alpha'\beta'' - \alpha''\beta', \end{aligned}$$

which in fact satisfy the relation

$$AF + BG + CH = 0.$$

So again if the line is given as the intersection of the planes

$$ax + by + cz + dw = 0,$$

$$a'x + b'y + c'z + d'w = 0,$$

then the equation of the cone (plane) through this line and the arbitrary point  $(\alpha, \beta, \gamma, \delta)$  is

$$(ax + by + cz + dw)(a'\alpha + b'\beta + c'\gamma + d'\delta) - (a'x + b'y + c'z + d'w)(a\alpha + b\beta + c\gamma + d\delta) = 0,$$

which is

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

the values of  $A$ , &c. being

$$\begin{aligned} A &= bc' - b'c, & F &= ad' - a'd, \\ B &= ca' - c'a, & G &= bd' - b'd, \\ C &= ab' - a'b, & H &= cd' - c'd, \end{aligned}$$

which also satisfy the relation

$$AF + BG + CH = 0.$$

I annex the following further investigation: let  $p', q', r', s', t', u'$  and  $p'', q'', r'', s'', t'', u''$  be what  $p, q, r, s, t, u$  become when  $\alpha, \beta, \gamma, \delta$  are changed into  $\alpha', \beta', \gamma', \delta'$  and  $\alpha'', \beta'', \gamma'', \delta''$  respectively: the line represented by the equation

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

is also represented by the equations

$$Ap' + Bq' + Cr' + Fs' + Gt' + Hu' = 0,$$

$$Ap'' + Bq'' + Cr'' + Fs'' + Gt'' + Hu'' = 0,$$

which, reverting to the coordinates  $x, y, z, w$ , may be written

$$\mathfrak{A}'x + \mathfrak{B}'y + \mathfrak{C}'z + \mathfrak{D}'w = 0,$$

$$\mathfrak{A}''x + \mathfrak{B}''y + \mathfrak{C}''z + \mathfrak{D}''w = 0,$$

where

$$\begin{aligned} \mathfrak{A}' &= C\beta' - B\gamma' + F\delta', & \mathfrak{A}'' &= C\beta'' - B\gamma'' + F\delta'', \\ \mathfrak{B}' &= -C\alpha' + A\gamma' + G\delta', & \mathfrak{B}'' &= -C\alpha'' + A\gamma'' + G\delta'', \\ \mathfrak{C}' &= B\alpha' - A\beta' + H\delta', & \mathfrak{C}'' &= B\alpha'' - A\beta'' + H\delta'', \\ \mathfrak{D}' &= -F\alpha' - G\beta' - H\gamma', & \mathfrak{D}'' &= -F\alpha'' - G\beta'' - H\gamma'', \end{aligned}$$

in which form the equations represent two planes each of them through the given line: the equation of the cone (plane) through the given line and the arbitrary point  $(\alpha, \beta, \gamma, \delta)$  is

$$\begin{aligned} &(\mathfrak{A}'x + \mathfrak{B}'y + \mathfrak{C}'z + \mathfrak{D}'w)(\mathfrak{A}''\alpha + \mathfrak{B}''\beta + \mathfrak{C}''\gamma + \mathfrak{D}''\delta) \\ & - (\mathfrak{A}'\alpha + \mathfrak{B}'\beta + \mathfrak{C}'\gamma + \mathfrak{D}'\delta)(\mathfrak{A}''x + \mathfrak{B}''y + \mathfrak{C}''z + \mathfrak{D}''w) = 0, \end{aligned}$$

or, developing,

$$(\mathfrak{B}'\mathfrak{C}'' - \mathfrak{B}''\mathfrak{C}')p + (\mathfrak{C}'\mathfrak{A}'' - \mathfrak{C}''\mathfrak{A}')q + (\mathfrak{A}'\mathfrak{B}'' - \mathfrak{A}''\mathfrak{B}')r + (\mathfrak{A}'\mathfrak{D}'' - \mathfrak{A}''\mathfrak{D}')s + (\mathfrak{B}'\mathfrak{D}'' - \mathfrak{B}''\mathfrak{D}')t + (\mathfrak{C}'\mathfrak{D}'' - \mathfrak{C}''\mathfrak{D}')u = 0.$$

Now

$$\begin{aligned} \mathfrak{B}'\mathfrak{C}'' - \mathfrak{B}''\mathfrak{C}' &= (-C\alpha' + A\gamma' + G\delta')(B\alpha'' - A\beta'' + H\delta'') \\ & - (-C\alpha'' + A\gamma'' + G\delta'')(B\alpha' - A\beta' + H\delta'), \end{aligned}$$

which putting

$$\begin{aligned} \beta'\gamma'' - \beta''\gamma' &= p_1, & \alpha'\delta'' - \alpha''\delta' &= s_1, \\ \gamma'\alpha'' - \gamma''\alpha' &= q_1, & \beta'\delta'' - \beta''\delta' &= t_1, \\ \alpha'\beta'' - \alpha''\beta' &= r_1, & \gamma'\delta'' - \gamma''\delta' &= u_1, \end{aligned}$$

become

$$A^2p_1 + ABq_1 + ACr_1 - (BG + CH)s_1 + AGt_1 + AHu_1,$$

or as it may be written

$$A(Ap_1 + Bq_1 + Cr_1 + Fs_1 + Gt_1 + Hu_1) - (AF + BG + CH)s_1.$$

Instead of writing at once  $AF + BG + CH = 0$ , I denote it by  $\nabla$  and I write also

$$Ap_1 + Bq_1 + Cr_1 + Fs_1 + Gt_1 + Hu_1 = \Omega.$$

The series of coefficients  $\mathfrak{B}'\mathfrak{C}'' - \mathfrak{B}''\mathfrak{C}'$ , &c. is

$$\begin{aligned} A\Omega - \nabla s_1, \quad B\Omega - \nabla t_1, \quad C\Omega - \nabla u_1, \\ F\Omega - \nabla p_1, \quad G\Omega - \nabla q_1, \quad H\Omega - \nabla r_1, \end{aligned}$$

and the required equation, restoring for  $\Omega, \nabla$  their values, is

$$(Ap + Bq + Cr + Fs + Gt + Hu)(Ap_1 + Bq_1 + Cr_1 + Fs_1 + Gt_1 + Hu_1) - (AF + BG + CH)(ps_1 + p_1s + qt_1 + q_1t + ru_1 + r_1u) = 0,$$

which in virtue of  $AF + BG + CH = 0$  becomes

$$(Ap + Bq + Cr + Fs + Gt + Hu)(Ap_1 + Bq_1 + Cr_1 + Fs_1 + Gt_1 + Hu_1) = 0.$$

The equation

$$Ap_1 + Bq_1 + Cr_1 + Fs_1 + Gt_1 + Hu_1 = 0$$

would imply that the two points  $(\alpha', \beta', \gamma', \delta')$ ,  $(\alpha'', \beta'', \gamma'', \delta'')$  are in the same plane

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

(or, what is the same thing, that this line is intersected by the line through the two points); in this exceptional case, the planes determined by the given line and the two points respectively are one and the same plane, and they do not by their intersection determine the given line. But in every other case the factor  $Ap_1 + Bq_1 + Cr_1 + Fs_1 + Gt_1 + Hu_1$  is not equal to zero, and the foregoing equation becomes

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

which is as it should be.

In what precedes it has been shown that the equation of the line through the points  $(\alpha', \beta', \gamma', \delta')$  and  $(\alpha'', \beta'', \gamma'', \delta'')$  is

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

where

$$\begin{aligned} A &= \alpha'\delta'' - \alpha''\delta', & F &= \beta'\gamma'' - \beta''\gamma', \\ B &= \beta'\delta'' - \beta''\delta', & G &= \gamma'\alpha'' - \gamma''\alpha', \\ C &= \gamma'\delta'' - \gamma''\delta', & H &= \alpha'\beta'' - \alpha''\beta', \end{aligned}$$

and that the equation of the line of intersection of the planes  $ax + by + cz + dw = 0$ ,  $a'x + b'y + c'z + d'w = 0$  is

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

where

$$\begin{aligned} A &= bc' - b'c, & F &= ad' - a'd, \\ B &= ca' - c'a, & G &= bd' - b'd, \\ C &= ab' - a'b, & H &= cd' - c'd. \end{aligned}$$

57-2

452

ON A NEW ANALYTICAL REPRESENTATION OF CURVES IN SPACE.

[284

It may be added that the condition of intersection of the two lines

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

$$A'p + B'q + C'r + F's + G't + H'u = 0,$$

is

$$AF' + A'F + BG' + B'G + CH' + C'H = 0,$$

and any other problems in relation to the line, for instance to find the condition that a line may pass through a given point or lie in a given plane, &c. may also be solved by means of the new coordinates.

A curve of the second order in space is either a pair of lines or a plane conic, each of which species may degenerate into a pair of intersecting lines. If we write

$$W = Ap + Bq + Cr + Fs + Gt + Hu,$$

$$W' = A'p + B'q + C'r + F's + G't + H'u,$$

then the equation of the pair of lines will be  $V = WW' = 0$ , and it is worth while to show that this value of  $V$  satisfies the fundamental equation  $d_p V \cdot d_s V + d_q V \cdot d_t V + d_r V \cdot d_u V = 0$ ; we have in fact  $d_p V = AW' + A'W$ , &c. and thence the left-hand side is

$$W'^2(AF + BG + CH) + WW'(AF' + A'F + BG' + B'G + CH' + C'H) + W^2(A'F' + B'G' + C'H'),$$

which in fact vanishes in virtue of the relations

$$AF + BG + CH = 0, \quad A'F' + B'G' + C'H' = 0, \quad WW' = 0.$$

The like proof applies to any curve made up of two or more curves.

Consider in the next place the plane conic given by the equations

$$ax + by + cz + dw = 0,$$

$$x^2 + y^2 + z^2 + w^2 = 0.$$

To find the equation of the cone having for its vertex the point  $(\alpha, \beta, \gamma, \delta)$  and passing through the conic we must, according to Joachimsthal's method, substitute in the two equations for  $x, y, z, w$  the values  $\lambda x + \mu \alpha$ ,  $\lambda y + \mu \beta$ ,  $\lambda z + \mu \gamma$ ,  $\lambda w + \mu \delta$ , and from the resulting equations eliminate  $\lambda, \mu$ . The two equations become

$$\lambda^2(x^2 + y^2 + z^2 + w^2) + 2\lambda\mu(\alpha x + \beta y + \gamma z + \delta w) + \mu^2(\alpha^2 + \beta^2 + \gamma^2 + \delta^2) = 0,$$

$$\lambda(ax + by + cz + dw) + \mu(a\alpha + b\beta + c\gamma + d\delta) = 0,$$

and thence eliminating  $\lambda, \mu$  we find

$$\begin{aligned} & (a\alpha + b\beta + c\gamma + d\delta)^2(x^2 + y^2 + z^2 + w^2) \\ & - 2(a\alpha + b\beta + c\gamma + d\delta)(ax + by + cz + dw)(\alpha x + \beta y + \gamma z + \delta w) \\ & + (ax + by + cz + dw)^2(\alpha^2 + \beta^2 + \gamma^2 + \delta^2) = 0, \end{aligned}$$

284] ON A NEW ANALYTICAL REPRESENTATION OF CURVES IN SPACE. 453

where the left-hand side is as it should be a function of  $p, q, r, s, t, u$ ; the equation may in fact be written

$$\begin{aligned} & a^2(q^2 + r^2 + s^2) + b^2(p^2 + r^2 + t^2) + c^2(p^2 + q^2 + u^2) + d^2(p^2 + q^2 + r^2) \\ & + 2bc(-qr + tu) + 2ca(-rp + us) + 2ab(-pq + st) \\ & + 2ad(qu - rt) + 2bd(rs - pu) + 2cd(pt - qs) = 0 \end{aligned}$$

which treated as a quadric in  $(a, b, c, d)$  may be written

$$\begin{pmatrix} q^2 + r^2 + s^2 & -pq + st & -rp + us & qu - rt \\ -pq + st & p^2 + r^2 + t^2 & -qr + tu & rs - pu \\ -rp + us & -qr + tu & p^2 + q^2 + u^2 & pt - qs \\ qu - rt & rs - pu & pt - qs & p^2 + q^2 + r^2 \end{pmatrix} (a, b, c, d)^2 = 0,$$

or treated as a quadric in  $(p, q, r, s, t, u)$ , in the form

$$\begin{pmatrix} b^2 + c^2 + d^2 & -ab & -ac & \cdot & -cd & bd \\ -ab & c^2 + a^2 + d^2 & -bc & cd & \cdot & -ad \\ -ac & -bc & a^2 + b^2 + d^2 & -bd & ad & \cdot \\ \cdot & cd & -bd & a^2 + b^2 & ab & ac \\ -cd & \cdot & ad & ab & a^2 + c^2 & bc \\ bd & -ad & \cdot & ac & bc & a^2 + d^2 \end{pmatrix} (p, q, r, s, t, u)^2 = 0.$$

Or again, in a form which is one of a system of four forms,

$$\begin{aligned} & (b^2 + c^2, c^2 + a^2, a^2 + b^2, -bc, -ca, -ab)(p, q, r)^2 \\ & + (b^2 + d^2, c^2 + d^2, c^2 + d^2, bc, ca, ab)(s, t, u)^2 \\ & + 2 \begin{pmatrix} \cdot & -cd & bd \\ cd & \cdot & -ad \\ -bd & ad & \cdot \end{pmatrix} (p, q, r)(s, t, u) = 0. \end{aligned}$$



Representing the equation by  $V = 0$ , the function  $V$  should verify the fundamental equation  $d_p V \cdot d_s V + d_q V \cdot d_t V + d_r V \cdot d_u V = 0$ , and we in fact have

$$\begin{aligned} d_p V &= (b^2 + c^2)p - abq - acr & + cdt - bdu, \\ d_s V &= & - cdq + bdr + (a^2 + d^2)s + abt + acu, \end{aligned}$$

and thence forming the product  $d_p V \cdot d_s V$ , and from it the two analogous products, we find

|       | $d_p V \cdot d_s V =$     | $d_q V \cdot d_t V =$     | $d_r V \cdot d_u V =$     |
|-------|---------------------------|---------------------------|---------------------------|
| $p^2$ | 0                         | $-abcd$                   | $+abcd$                   |
| $q^2$ | $+abcd$                   | 0                         | $-abcd$                   |
| $r^2$ | $-abcd$                   | $+abcd$                   | 0                         |
| $s^2$ | $+abcd$                   | 0                         | $-abcd$                   |
| $t^2$ | $-abcd$                   | $+abcd$                   | 0                         |
| $u^2$ | 0                         | $-abcd$                   | $+abcd$                   |
| $pq$  | $-cd(b^2 + c^2)$          | $+abcd$                   | $-cd(a^2 - b^2)$          |
| $qr$  | $-ad(b^2 - c^2)$          | $+bd(a^2 + c^2)$          | $-bd(a^2 - c^2)$          |
| $rp$  | $+cd(a^2 + d^2)$          | $-ad(b^2 + d^2)$          | $+ad(c^2 - d^2)$          |
| $st$  | $+bd(a^2 + d^2)$          | $-ab(c^2 + d^2)$          | $-ab(c^2 - d^2)$          |
| $tu$  | $-ad(b^2 - c^2)$          | $-ad(b^2 + d^2)$          | $+ad(b^2 - d^2)$          |
| $us$  | $+cd(a^2 + d^2)$          | $-bc(a^2 - d^2)$          | $-ab(c^2 - d^2)$          |
| $ps$  | $-a^2b^2 - c^2d^2$        | $-ab(c^2 + d^2)$          | $-ab(c^2 - d^2)$          |
| $qt$  | $+(a^2 + d^2)(b^2 + c^2)$ | $+ab(c^2 + d^2)$          | $-ac(c^2 + d^2)$          |
| $ru$  | $+a^2c^2 + b^2d^2$        | $+(b^2 + d^2)(c^2 + a^2)$ | $-ab(c^2 - d^2)$          |
| $pt$  | $+ac(b^2 + c^2)$          | $+bc(c^2 + d^2)$          | $+ab(c^2 + d^2)$          |
| $qu$  | $+ab(b^2 + c^2)$          | $-b^2c^2 - a^2d^2$        | $+bc(a^2 + b^2)$          |
| $rs$  | $-bc(b^2 + c^2)$          | $-ad(b^2 - c^2)$          | $+(c^2 + d^2)(d^2 - b^2)$ |
| $pu$  | $-bc(a^2 + d^2)$          | $+bc(b^2 - d^2)$          | .                         |

by which the equation is verified.

I conclude with some remarks relating to a different part of the subject. It is shown by M. E. de Jonquières in his "Essai sur la génération des courbes géométriques, &c." *Mém. Prés. à l'Acad. de Paris*, t. XVI. (1858), that the equation  $U = 0$  of any geometrical plane curve can be presented in a variety of different ways in the form

$$\begin{vmatrix} P, & Q \\ P', & Q' \end{vmatrix} = 0,$$

this being in fact the fundamental theorem of his very beautiful investigations. I am not aware that it has ever been considered whether the equation  $U = 0$  of a plane curve can in general be represented in the form

$$\begin{vmatrix} P, & Q, & R \\ P', & Q', & R' \\ P'', & Q'', & R'' \end{vmatrix} = 0$$

or in the analogous forms where the left-hand side is a determinant of a higher order. As regards surfaces, the equation  $U = 0$  of a geometrical surface cannot in general be presented in the form

$$\begin{vmatrix} P, & Q \\ P', & Q' \end{vmatrix} = 0,$$

nor in the similar forms where the left-hand side is a determinant of a higher order. We may consequently classify surfaces of a given order according to the forms of this nature by which they can be represented, or as I propose to term it, according to their “frangibility.” It is obvious that this question is immediately connected with that of the representation of curves in space by means of the ordinary coordinates of analytical geometry; for instance if we have a surface  $U = 0$ , which can be represented in the form

$$\begin{vmatrix} P, & Q, & R \\ P', & Q', & R' \\ P'', & Q'', & R'' \end{vmatrix} = 0,$$

then we can describe upon the surface curves such as

$$\begin{vmatrix} P', & Q', & R' \\ P'', & Q'', & R'' \end{vmatrix} = 0,$$

viz. the curve so represented is the curve which in conjunction with the curve ( $R' = 0, R'' = 0$ ) makes up the complete intersection of the two surfaces  $P'R'' - P''R' = 0, Q'R'' - Q''R' = 0$ : the curve in question is not in general the complete intersection of any two surfaces. If the surface  $U = 0$  can be represented in the form  $\begin{vmatrix} P, & Q \\ P', & Q' \end{vmatrix} = 0$ , then we can describe upon the surface, curves such as the curve ( $P = 0, Q = 0$ ) which, although it is the complete intersection of two surfaces, is not the complete intersection of the given surface  $U = 0$  by any other surface. But if the surface  $U = 0$  cannot be represented in any such form, or as we may express it, if the surface is infrangible, then it would appear that the only curves which can be described upon the surface are those which are the complete intersection of the given surface by some other surface. The question is, I think, an interesting one in the theory of surfaces, but I doubt whether much will be done in this manner as regards the theory of curves in space, and it appears to me that there is more to be hoped for from the theory previously explained in the present paper.

*2, Stone Buildings, W.C., June 2, 1859.*

285.

ON THE SYSTEM OF CONICS HAVING DOUBLE CONTACT WITH EACH OTHER.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 246–250.]

CONSIDER the conics which pass through four points coinciding two and two together; the two points of each pair of coincident points are to be regarded as lying in a line which will be a tangent to the conic, and the system is thus a system of conics touching the same two lines at the same two points: or if we replace the two lines by a conic of the system, it is the system of conics having double contact with a given conic at the same two points. The lines may be spoken of as the tangents, and the points as the ineunts of the system; the line joining the two points is the axis, and the point of intersection of the two lines, the pole. It is assumed that the system contains real conics; the pole and axis must consequently be real, but the tangents and ineunts may be either all of them real or else all of them imaginary—we may for shortness say that in the former case the double contact is real, and that in the latter case it is imaginary. Consider first the case of a real double contact, the system of conics commences with an ellipse differing only slightly from the finite portion of the axis included between the ineunts, this ellipse being the first of a series of ellipses the last of which only slightly differs from a parabola, we have then a parabola, then a hyperbola differing only slightly from the parabola, and which is the first of a series of hyperbolas of which the last differs only slightly from the two tangents: all the before-mentioned conics are included within the angles at the pole or vertex of the triangle, viz. the ellipses and parabola within the angle towards the axis or base, and the hyperbolas, one branch of each of them within this angle and the other branch within the opposite angle. And it will be convenient to term these conics the “Multi- form Series.” Passing over the intermediate case, we come to a hyperbola differing only slightly from the two tangents, the first of a series of hyperbolas of which the

285]

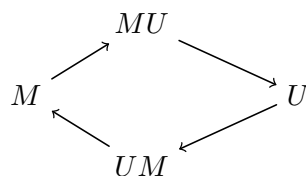
ON THE SYSTEM OF CONICS HAVING DOUBLE CONTACT &C.

457

last differs only slightly from the infinite portions, not included between the ineunts, of the axis; all these hyperbolas lying without the angles at the pole or vertex of the triangle. The last-mentioned hyperbolas may be termed the “Uniform Series.” Passing over another intermediate case we return to the multiform series, the entire system of conics forming in fact a complete cycle. The intermediate case forming the transition from the multiform to the uniform series is a conic which considered as generated by a point is the pair of tangents, but which considered as enveloped by a line is a pair of points coincident with the pole. The intermediate case forming the transition from the uniform series to the multiform series is a conic which considered as generated by a point is a pair of lines coincident with the axis, but considered as enveloped by a line is the pair of ineunts.

When the double contact is imaginary, the conics of the system may still be considered as forming a multiform and a uniform series, but the uniform series consists entirely of imaginary conics. The multiform series commences with an ellipse differing only slightly from the pole; this is the first of a series of ellipses of which the last differs only slightly from a parabola, we have next the parabola, and then a series of hyperbolas, the first of which differs only slightly from the parabola, and the last of which differs only slightly from the axis. The ellipses and parabolas lie all of them on the same side of the axis; the hyperbolas lie, one branch of each on the one side of the axis, and the other branch on the other side. The uniform series (being imaginary) of course does not admit of description. The intermediate or transition cases are the same as for a real double contact, with only the variation occasioned by the tangents and ineunts being imaginary. The conics of the entire system are considered (as in the case of a real double contact) to form a cycle, but here part of the cycle, viz. the uniform series, is imaginary.

Suppose that  $M$  and  $U$  stand for the multiform and uniform series,  $MU$  for the transition form from the multiform to the uniform series, and  $UM$  for the transition form from the uniform to the multiform series. The cycle may be represented as in the figure.



Call either of the conics  $MU$ ,  $UM$  the centre of the cycle, and say that the cycle is arranged line-wise when  $MU$  is considered as the centre, and point-wise when  $UM$  is considered as the centre. We may pass from one conic of the cycle to another incentrically or excentrically, i.e. by passing towards the centre or away from the centre. The intermediates of two conics are the conics passed through in going from the first to the second incentrically, the extramediates are those passed through in going from the first to the second excentrically. In speaking of the extramediates or the intermediates of a single conic, such conic is considered as the first conic and the second conic is understood to be the centre. The nearer extramediates are those passed through previous to reaching an extremity of the cycle, the further extramediates are those passed through subsequent to reaching an extremity of the cycle. It is clear that the intermediates and the nearer extramediates make up the series multiform or uniform to which the conic belongs, and that the further extramediates are the other series; and moreover that if the cycle be arranged line-wise and point-wise successively, the intermediates of the one arrangement are the nearer extramediates of the other arrangement and vice versa; the further extramediates of each arrangement being the same.

Two points of a conic may be said to be conjunctive with respect to a given line when it is possible to pass from the one to the other along the curve without crossing the line and disjunctive in the contrary case—it being understood that both the parabola and the hyperbola are to be treated as closed curves, viz. the points at infinity of the parabola are to be considered as one and the same point, and so the points at each extremity of either asymptote of the hyperbola are to be considered as one and the same point—and in like manner two tangents of a conic are said to be conjunctive with respect to a given point, when it is possible by the revolution of the one tangent to arrive at the other tangent without sweeping through the point in question, and disjunctive in the contrary case.

Consider now any conic of the series, and call this simply the conic. Take upon the conic any two points  $a$ ,  $a'$  and joining these with a variable point in the axis, we obtain the lines  $ab'$  and  $a'b$  meeting the conic in the points  $b$ ,  $b'$  (so that  $ab'$  and  $a'b$  meet on the axis). We have then the conic in a series of points  $a$ ,  $b$ ,  $c$ , ... and a second series  $a'$ ,  $b'$ ,  $c'$ , ... which are homographically related to each other, and which possess the property that taking any two points  $b$ ,  $c$  of the first series and the corresponding points  $b'$ ,  $c'$  of the second series, the lines  $bc'$ ,  $b'c$  meet on the axis. It is immaterial how the point  $a$  is chosen, but  $a$  being chosen at pleasure, the system will obviously depend on the way in which the point  $a'$  is chosen; so that the points of a conic may be considered as homographically related to each other in an infinity of different ways. The reciprocal construction of course applies to the tangents of a conic, so that the tangents of a conic may be considered as homographically related to each other in an infinity of different ways: and not only this, but it is clear that the tangents at points homographically related to each other are also homographically related to each other, and vice versa.

The line joining corresponding points envelopes a conic, one of the conics of the system, and which may for shortness be spoken of simply as the envelope. The point of intersection of corresponding tangents generates a conic, one of the conics of a system, and which may in like manner be spoken of simply as the locus.

We may now enunciate the following theorem: Let the system be arranged line-wise, the envelope is an extramediate of the conic; viz. if a pair of corresponding points of the conic (all pairs have in this respect the same property) be conjunctive with respect to the axis, a nearer extramediate, but if disjunctive, then a further extramediate.

Let the system be arranged point-wise, the locus is an extramediate of the conic; viz. if a pair of corresponding tangents of the conic (all pairs have in this respect the same property) be conjunctive with respect to the pole, a nearer extramediate, but if disjunctive, then a further extramediate.

It is however proper to remark that when the double contact is imaginary, the pair of corresponding points or tangents (in fact any pair of points or tangents) is necessarily conjunctive, so that in this case the second alternatives in the two theorems have no application, and the envelope or locus is always a nearer extramediate. When the double contact is real, the pair of corresponding points or tangents may be either conjunctive or disjunctive, and the character (nearer or further) of the extramediate is determined accordingly.

The double contact being either real or imaginary, we may as a limiting case assume that the corresponding points or tangents coincide, the envelope or locus is in this case the conic itself. Suppose that the double contact is real, we have here two other limiting cases, viz. first, one of a pair (and therefore of each pair) of corresponding points or tangents may be situate in the axis or pass through the pole; this is the limit between the two cases of the pair of corresponding points or tangents being conjunctive and disjunctive, and therefore the envelope or locus must be the conic which is the limit between the nearer extramediates and the further extramediates, i.e. the envelope is the pair of ineunts and the locus is the pair of tangents.

Secondly, the line through a pair of corresponding points may pass through the pole, or the point of intersection of a pair of corresponding tangents may lie in the axis; the envelope or locus is here the furthest of the extramediates, viz. the envelope is the pair of points coincident with the pole, the locus is the pair of lines coincident with the axis. If considering one of the pair of corresponding points or lines as fixed, the other point or line passes through the last-mentioned limiting position, the envelope or locus returns back through the series of further extramediates.

In the statement of the preceding theorems, the envelope and locus have been considered separately, but we may if we please consider the locus as generated by the point of intersection of the tangents at the corresponding points: the locus and envelope are in this case reciprocal polars with respect to the conic; it should be noticed that the pair of corresponding points and the pair of corresponding tangents are either both conjunctive, or else both disjunctive.

*2, Stone Buildings, W.C., 3rd June, 1859.*

286.

NOTE ON THE VALUE OF CERTAIN DETERMINANTS, THE TERMS  
OF WHICH ARE THE SQUARED DISTANCES OF POINTS IN  
A PLANE OR IN SPACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 275–277.]

THE values of the several determinants mentioned in my paper “On a Certain Theorem in the Geometry of Position,” *Cambridge Mathematical Journal*, Old Series, t. II. (1842), p. 267, [1], are as follows:

$$(1) \begin{vmatrix} 0, & \overline{12}^2, & \overline{13}^2, & 1 \\ \overline{21}^2, & 0, & \overline{23}^2, & 1 \\ \overline{31}^2, & \overline{32}^2, & 0, & 1 \\ 1, & 1, & 1, & 0 \end{vmatrix} = \Sigma \overline{12}^2 \overline{21}^2 - \Sigma \overline{12}^2 \overline{23}^2,$$

where the  $\Sigma$ ,  $\Sigma$  contain 3 and 6 terms respectively.

$$(2) \begin{vmatrix} 0, & \overline{12}^2, & \overline{13}^2, & \overline{14}^2, & 1 \\ \overline{21}^2, & 0, & \overline{23}^2, & \overline{24}^2, & 1 \\ \overline{31}^2, & \overline{32}^2, & 0, & \overline{34}^2, & 1 \\ \overline{41}^2, & \overline{42}^2, & \overline{43}^2, & 0, & 1 \\ 1, & 1, & 1, & 1, & 0 \end{vmatrix} = \Sigma \overline{12}^2 \overline{23}^2 \overline{34}^2 - \Sigma \overline{12}^2 \overline{34}^2 \overline{43}^2 - \Sigma \overline{12}^2 \overline{23}^2 \overline{31}^2,$$

where the  $\Sigma$ ,  $\Sigma$ ,  $\Sigma$  contain 24, 12 and 8 terms respectively.

286]

NOTE ON THE VALUE OF CERTAIN DETERMINANTS &C.

461

$$(3) \begin{vmatrix} 0, & \overline{12}^2, & \overline{13}^2, & \overline{14}^2, & \overline{15}^2, & 1 \\ \overline{21}^2, & 0, & \overline{23}^2, & \overline{24}^2, & \overline{25}^2, & 1 \\ \overline{31}^2, & \overline{32}^2, & 0, & \overline{34}^2, & \overline{35}^2, & 1 \\ \overline{41}^2, & \overline{42}^2, & \overline{43}^2, & 0, & \overline{45}^2, & 1 \\ \overline{51}^2, & \overline{52}^2, & \overline{53}^2, & \overline{54}^2, & 0, & 1 \\ 1, & 1, & 1, & 1, & 1, & 0 \end{vmatrix} = -\Sigma \overline{12}^2 \overline{23}^2 \overline{34}^2 \overline{45}^2 \\ - \Sigma \overline{12}^2 \overline{23}^2 \overline{34}^2 \overline{43}^2 \\ + \Sigma \overline{12}^2 \overline{23}^2 \overline{45}^2 \overline{54}^2 \\ + \Sigma \overline{12}^2 \overline{23}^2 \overline{34}^2 \overline{41}^2 \\ + \Sigma \overline{12}^2 \overline{23}^2 \overline{31}^2 \overline{45}^2$$

where the  $\Sigma$ ,  $\Sigma$ ,  $\Sigma$ ,  $\Sigma$ ,  $\Sigma$  contain 120, 15, 60, 30, and 40 terms respectively.

$$(4) \begin{vmatrix} 0, & \overline{12}^2, & \overline{13}^2, & \overline{14}^2 \\ \overline{21}^2, & 0, & \overline{23}^2, & \overline{24}^2 \\ \overline{31}^2, & \overline{32}^2, & 0, & \overline{34}^2 \\ \overline{41}^2, & \overline{42}^2, & \overline{43}^2, & 0 \end{vmatrix} = \Sigma \overline{12}^2 \overline{21}^2 \overline{34}^2 \overline{43}^2 - \Sigma \overline{12}^2 \overline{23}^2 \overline{34}^2 \overline{41}^2,$$

where the  $\Sigma$ ,  $\Sigma$  contain 3 and 6 terms respectively.

$$(5) \begin{vmatrix} 0, & \overline{12}^2, & \overline{13}^2, & \overline{14}^2, & \overline{15}^2 \\ \overline{21}^2, & 0, & \overline{23}^2, & \overline{24}^2, & \overline{25}^2 \\ \overline{31}^2, & \overline{32}^2, & 0, & \overline{34}^2, & \overline{35}^2 \\ \overline{41}^2, & \overline{42}^2, & \overline{43}^2, & 0, & \overline{45}^2 \\ \overline{51}^2, & \overline{52}^2, & \overline{53}^2, & \overline{54}^2, & 0 \end{vmatrix} = \Sigma \overline{12}^2 \overline{23}^2 \overline{34}^2 \overline{45}^2 \overline{51}^2 \\ - \Sigma \overline{12}^2 \overline{23}^2 \overline{31}^2 \overline{45}^2 \overline{54}^2$$

where the  $\Sigma$ ,  $\Sigma$  contain 24 and 20 terms respectively.

And it is proper to remark that it is not in the preceding formulæ (as in the memoir above referred to in which  $\overline{12}$  denotes a distance between two points 1 and 2) assumed that  $\overline{12}$  and  $\overline{21}$  are equal.

The formula (1) gives, if  $a, b, c$  represent the sides, the well-known expression for the area of a triangle

$$(\text{area})^2 = \frac{1}{16}(2b^2c^2 + 2c^2a^2 + 2a^2b^2 - a^4 - b^4 - c^4).$$

Similarly the formula (2), if  $a, b, c, f, g, h$  represent the edges, viz.  $\overline{23} = a, \overline{31} = b, \overline{12} = c, \overline{14} = f, \overline{24} = g, \overline{34} = h$ , gives for the volume of a tetrahedron

$$\begin{aligned} (\text{volume})^2 &= \frac{1}{144} \{ b^2c^2(g^2 + h^2) + c^2a^2(h^2 + f^2) + a^2b^2(f^2 + g^2) \\ &\quad + g^2h^2(b^2 + c^2) + h^2f^2(c^2 + a^2) + f^2g^2(a^2 + b^2) \\ &\quad - a^2f^2(a^2 + f^2) - b^2g^2(b^2 + g^2) - c^2h^2(c^2 + h^2) \\ &\quad - a^2g^2h^2 - b^2h^2f^2 - c^2f^2g^2 - a^2b^2c^2 \}, \\ &= \frac{1}{144} W \quad \text{suppose.} \end{aligned}$$

462

NOTE ON THE VALUE OF CERTAIN DETERMINANTS &C.

[286

Now

$$\begin{aligned} 4 \times \text{surface} &= \sqrt{2g^2h^2 + 2h^2a^2 + 2a^2g^2 - a^4 - h^4 - g^4} \\ &\quad + \sqrt{2f^2h^2 + 2h^2b^2 + 2b^2f^2 - h^4 - b^4 - f^4} \\ &\quad + \sqrt{2g^2f^2 + 2f^2c^2 + 2c^2g^2 - g^4 - f^4 - c^4} \\ &\quad + \sqrt{2b^2c^2 + 2c^2a^2 + 2a^2b^2 - a^4 - b^4 - c^4}, \\ &= x + y + z + w \quad \text{suppose,} \end{aligned}$$

and the norm of this is

$$(x^4 + y^4 + z^4 + w^4 - 2y^2z^2 - 2z^2w^2 - 2w^2x^2 - 2x^2y^2 - 2y^2w^2 - 2x^2z^2)^2 - 64x^2y^2z^2w^2,$$

which is of the sixteenth order, and must be of the form  $WQ$  where  $Q$  is a function of  $a, b, c, f, g, h$  of the tenth order. The expression of this function is given by Prof. Sylvester in his paper "On the Relation between the Volume of a Tetrahedron &c.," *Camb. and Dubl. Math. Jour.*, t. VIII. (1853), pp. 171–178, viz. the value is

$$\begin{aligned} Q &= a^2b^2c^2\{f^4 + g^4 + h^4 + g^2h^2 + h^2f^2 + f^2g^2 + b^2c^2 + c^2a^2 + a^2b^2 - (f^2 + g^2 + h^2)(a^2 + b^2 + c^2)\} \\ &\quad + a^2g^2h^2\{f^4 + b^4 + c^4 + b^2c^2 + c^2f^2 + f^2b^2 + g^2h^2 + h^2f^2 + f^2g^2 - (f^2 + b^2 + c^2)(a^2 + g^2 + h^2)\} \\ &\quad + b^2h^2f^2\{g^4 + c^4 + a^4 + c^2a^2 + a^2g^2 + g^2c^2 + h^2f^2 + f^2h^2 + h^2g^2 - (g^2 + c^2 + a^2)(b^2 + h^2 + f^2)\} \\ &\quad + c^2f^2g^2\{h^4 + a^4 + b^4 + a^2b^2 + b^2h^2 + h^2a^2 + f^2g^2 + g^2c^2 + c^2f^2 - (h^2 + a^2 + b^2)(c^2 + f^2 + g^2)\}, \end{aligned}$$

and, as there remarked, the equation  $Q = 0$  expresses the condition that the radius of the inscribed sphere may be infinite.

*2, Stone Buildings, W.C., June 10th, 1859.*

287.

NOTE ON THE EQUATION FOR THE SQUARED DIFFERENCES  
OF THE ROOTS OF A CUBIC EQUATION.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 307–309.]

THE question of finding the equation for the squared differences of the roots, presents, in the case of a cubic equation, a peculiarity which does not occur for equations of a higher order, viz. we may in the first instance form the equation for the differences of the roots taken in a given cyclical order, and thence deduce the equation for the squared differences of the roots. Let the cubic equation be

$$U = (a, b, c, d \mid x, 1)^3 = a(x - \alpha)(x - \beta)(x - \gamma) = 0,$$

the function

$$\Pi\{\theta - (\beta - \gamma)\},$$

which equated to zero gives for  $\theta$  the values  $\beta - \gamma$ ,  $\gamma - \alpha$ ,  $\alpha - \beta$ , which are the differences of the roots taken in a given circular order, has for any interchanges whatever of the roots, two values only, viz. that just written down, and the value  $\Pi\{\theta + (\beta - \gamma)\}$  [which is  $\Pi\{\theta - (\gamma - \beta)\}$ ], which may be deduced therefrom by changing first the sign of  $\theta$  and then the sign of the entire expression (or what is the same thing, by changing the signs of the terms containing the even powers of  $\theta$ ); we may consequently write

$$\Pi\{\theta - (\beta - \gamma)\} = P - Q\sqrt{\square(\alpha, \beta, \gamma)},$$

which  $P$ ,  $Q$  are symmetrical functions of the roots, and  $\sqrt{\square(\alpha, \beta, \gamma)}$  or  $(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$  is a function the square of which is a symmetrical function of the roots, and such

464

NOTE ON THE EQUATION FOR THE SQUARED DIFFERENCES

[287

symmetrical functions of the roots can of course be expressed as functions of the coefficients. We have in fact

$$P = \theta^3 + \theta(2\beta\gamma - \Sigma\alpha^2) = a^{-3}\{a^3\theta^3 + 9(ac - b^2)\theta\},$$

and

$$Q = 1,$$

where  $\square$  is the discriminant of the cubic function,

$$= a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2.$$

Consequently

$$\Pi\{\theta - (\beta - \gamma)\} = a^{-3}\{a^3\theta^3 + 9(ac - b^2)\theta - \sqrt{-27\square}\},$$

and forming the similar equation

$$\Pi\{\theta + (\beta - \gamma)\} = a^{-3}\{a^3\theta^3 + 9(ac - b^2)\theta + \sqrt{-27\square}\},$$

multiplying the two equations together and writing  $u$  in the place of  $\theta^2$ , we find

$$\Pi\{u - (\beta - \gamma)^2\} = a^{-6}\{[a^3u + 9(ac - b^2)]^2u + 27\square\},$$

and the equation for the squared differences of the roots is thus seen to be

$$\{a^3u + 9(ac - b^2)\}^2u + 27\square = 0,$$

or what is the same thing

$$a^6u^3 + 18a^3(ac - b^2)u^2 + 81(ac - b^2)^2u + 27\square = 0.$$



I remark that if  $\omega$  is an imaginary cube root of unity (so that  $(\omega - \omega^2)^2 = -3$ ,  $\omega - \omega^2$  being thus only another form of  $\sqrt{-3}$ ) then if in the expression for  $\Pi\{\theta - (\beta - \gamma)\}$  we write  $\frac{\theta}{(\omega - \omega^2)a}$  in the place of  $\theta$ , the equation assumes the more simple form

$$\Pi\{\theta - \frac{1}{3}a(\omega - \omega^2)(\beta - \gamma)\} = \theta^3 - 3(ac - b^2)\theta - a^3\sqrt{\square},$$

which if  $U$  be the cubic function,  $H$  the Hessian  $= (ac - b^2, ad - bc, bd - c^2 \mid x, y)^2$ , and  $\square$  the discriminant as before, is a particular case (obtained by writing  $x = 1, y = 0$ ) of the equation

$$\Pi\{\theta - \frac{1}{3}a(\omega - \omega^2)(x - \alpha y)\} = \theta^3 - 3H\theta - U\sqrt{\square},$$

which equation can be at once obtained from the equation (where  $\Phi$  is the cubi covariant of the cubic function)

$$\sqrt[3]{\frac{1}{2}(\Phi + U^2\sqrt{\square})} - \sqrt[3]{\frac{1}{2}(\Phi - U^2\sqrt{\square})} = \frac{1}{3}a(\omega - \omega^2)(\beta - \gamma)(x - \alpha y),$$

287]

# OF THE ROOTS OF A CUBIC EQUATION.

465

given in my Fifth Memoir on Quantics, *Phil. Trans.*, t. CXLVIII. (1858), [156]. For writing for a moment

$$\theta = \sqrt[3]{X} - \sqrt[3]{Y},$$

we find

$$\theta^3 = X - Y - 3\sqrt[3]{XY}\theta,$$

or

$$\theta^3 + 3\sqrt[3]{XY}\theta - (X - Y) = 0,$$

where  $\sqrt[3]{XY} = \sqrt[3]{\frac{1}{4}(\Phi^2 - U^4\square)}$ , which by the equation

$$\Phi^2 - U^4\square = -4H^3$$

(given in the Memoir) is  $= -H$ , and  $(X - Y)$  is  $= U^2\sqrt{\square}$ , so that the equation in  $\theta$  is, as above,  $\theta^3 - 3H\theta - U^2\sqrt{\square} = 0$ , an equation which is satisfied by  $\theta = \frac{1}{3}a(\omega - \omega^2)(\beta - \gamma)(x - \alpha y)$ ; and the other two roots being of course of the like form, the cubic function in  $\theta$  is equal to  $\Pi\{\theta - \frac{1}{3}a(\omega - \omega^2)(\beta - \gamma)(x - \alpha y)\}$  which proves the theorem.

2, Stone Buildings, W.C., Nov. 3rd, 1859.

288.

NOTE ON THE CURVATURE OF A PLANE CURVE AT A DOUBLE POINT, AND ON THE CURVATURE OF SURFACES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 322–326.]

THE radius of curvature of a plane curve at a double point is most readily determined as follows: viz. if  $x, y$  be the coordinates of the double point,  $\theta$  the inclination to the axis of  $x$  of the tangent to the branch under consideration, and  $u, v$  the coordinates measured from the double point as origin, parallel and perpendicular to the tangent, of the consecutive point on such branch, then the radius of curvature for the branch in question is

$$\frac{u^2}{2v};$$

the coordinates of the consecutive point are

$$x + u \cos \theta - v \sin \theta,$$

$$y + u \sin \theta + v \cos \theta,$$

and the value of  $u^2 \div 2v$  is to be found by substituting these expressions for  $x, y$  in the equation of the curve. Let  $U = 0$  be the equation of the curve; then at the double point,  $U$  and the differential coefficients of the first order vanish, let those of the second order be  $(a, b, c)$  and those of the third order  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d})$ , then substituting we have

$$\begin{aligned} & \frac{1}{2}(a, b, c)(u \cos \theta - v \sin \theta, u \sin \theta + v \cos \theta)^2 \\ & + \frac{1}{6}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d})(u \cos \theta - v \sin \theta, u \sin \theta + v \cos \theta)^3 = 0. \end{aligned}$$

The coefficient of  $u^2$  is

$$\frac{1}{2}(a, b, c)(\cos \theta, \sin \theta)^2,$$

288] NOTE ON THE CURVATURE OF A PLANE CURVE AT A DOUBLE POINT &C. 467

which vanishes, since it is by putting this function equal to zero that we find the direction of the tangents at the double point. For the present purpose we are concerned with one of the two branches only, and in all that follows the ratio  $\cos \theta : \sin \theta$  will denote a determinate root of the quadratic equation; viz. the root which corresponds to the branch in question.

The equation takes therefore the form

$$Buv + \frac{1}{2}Cv^2 + \frac{1}{6}Du^3 + \&c. = 0,$$

which might be satisfied by assuming  $Buv + \frac{1}{2}Cv^2 = 0$ , but the values so obtained belong to the branch which does not touch the tangent; the proper solution is

$$Buv + \frac{1}{6}Du^3 = 0,$$

and thence

$$\frac{u^2}{2v} = -\frac{3B}{D},$$

or substituting for  $B, D$  their values, the radius of curvature is

$$= -\frac{3(a, b, c)(\cos \theta, \sin \theta)(-\sin \theta, \cos \theta)}{(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d})(\cos \theta, \sin \theta)^3},$$

where the expression in the numerator is

$$= (-a + c) \sin \theta \cos \theta + b(\cos^2 \theta - \sin^2 \theta),$$

$$= \frac{1}{2}[(c - a) \sin 2\theta + 2b \cos 2\theta].$$

The curvature, at the point of contact, of the curve in which any surface is intersected by a tangent plane cannot be found by the ordinary theory of the curvature of surfaces, being (by reason that the point of contact is a double point on the curve) a case of exception from that theory. The foregoing method may be applied to the solution of the question as follows: Let  $U = 0$  be the equation of the surface, and suppose that  $(x, y, z)$  refer to the point of contact of the tangent plane. Let  $L, M, N$  be the first differential coefficients of  $U$  at this point,  $(a, b, c, f, g, h)$  the second differential coefficients,  $(\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota)$  the third differential coefficients. Let  $\lambda, \mu, \nu$  be proportional to the cosines of the inclinations to the axis of the tangent to one of the two branches through the double point; the ratios  $\lambda : \mu : \nu$  are determined by

$$(L, M, N)(\lambda, \mu, \nu) = 0,$$

$$(a, b, c, f, g, h)(\lambda, \mu, \nu)^2 = 0.$$

Let  $u, v$  be proportional to the coordinates of a consecutive point measured, from the point of contact as origin, in the direction of the tangent and in the perpendicular direction in the tangent plane. The cosines of the inclinations of  $u$  to the axes are as  $\lambda : \mu : \nu$ ; those for the inclinations of the normal to the axes are as  $L : M : N$ ; hence for the coordinate  $v$  which is perpendicular to the plane of the last-mentioned

468 NOTE ON THE CURVATURE OF A PLANE CURVE AT A DOUBLE POINT, [288

two lines, the cosines of the inclinations to the axes are as  $N\mu - M\nu : L\nu - N\lambda : M\lambda - L\mu$ , and the coordinates of the consecutive point may be taken to be

$$x + \lambda u + (N\mu - M\nu)v,$$

$$y + \mu u + (L\nu - N\lambda)v,$$

$$z + \nu u + (M\lambda - L\mu)v.$$

Substituting these values in the equation of the surface, the terms involving the first powers of  $u, v$  vanish, and the term involving  $u^2$  also vanishes in virtue of the relation  $(a, b, c, f, g, h)(\lambda, \mu, \nu)^2 = 0$ . The equation consequently becomes

$$Buv + \frac{1}{2}Cv^2 + \frac{1}{6}Du^3 + \dots = 0,$$

and we have as before for the branch in question,

$$\frac{u^2}{2v} = -\frac{3B}{D}.$$

In the present case  $u, v$  have been taken, not as before equal, but only proportional, to the coordinates of the consecutive point measured from the point of contact parallel and perpendicular to the tangent, the values of the coordinates are in fact

$$\sqrt{\lambda^2 + \mu^2 + \nu^2} u, \quad \sqrt{L^2 + M^2 + N^2} \sqrt{\lambda^2 + \mu^2 + \nu^2} v,$$

and the expression for the radius of curvature is

$$= \frac{u^2}{2v} \cdot \frac{\sqrt{\lambda^2 + \mu^2 + \nu^2}}{\sqrt{L^2 + M^2 + N^2} \sqrt{\lambda^2 + \mu^2 + \nu^2}} \cdot \sqrt{\lambda^2 + \mu^2 + \nu^2},$$

or substituting for  $\frac{u^2}{2v}$  the value  $-\frac{3B}{D}$  and for  $B, D$  their values, the radius of curvature is

$$= -\frac{3\sqrt{\lambda^2 + \mu^2 + \nu^2}}{\sqrt{L^2 + M^2 + N^2}} \cdot \frac{(a, b, c, f, g, h)(\lambda, \mu, \nu)(N\mu - M\nu, L\nu - N\lambda, M\lambda - L\mu)}{(\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota)(\lambda, \mu, \nu)^3},$$

where, as already noticed, the ratios  $\lambda : \mu : \nu$  are determined by the equations

$$(L, M, N)(\lambda, \mu, \nu) = 0,$$

$$(a, b, c, f, g, h)(\lambda, \mu, \nu)^2 = 0,$$

the system of roots selected being that which corresponds to the branch under consideration. It may be noticed that these two equations give

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})(L, M, N)^2 \cdot (a, b, c, f, g, h)(\lambda, \mu, \nu)^2 - K \cdot [(L, M, N)(\lambda, \mu, \nu)]^2 = 0,$$

where as usual

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) = (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch),$$

$$K = abc - af^2 - bg^2 - ch^2 + 2fgh,$$

288]

AND ON THE CURVATURE OF SURFACES.

469

and the expression on the left-hand side considered as a function of  $(\lambda, \mu, \nu)$  is decomposable into a pair of factors. Selecting the proper factor and equating it to zero, we have in addition to the linear equation

$$(L, M, N)(\lambda, \mu, \nu) = 0,$$

a new linear equation; these two equations determine the ratios  $\lambda : \mu : \nu$ .

P.S.—I remark that the process adopted for finding the radius of curvature at a double point of a plane curve is the most simple one for the case of an ordinary point on the curve. In fact let  $U = 0$  be the curve,  $(x, y)$  the coordinates of the point in question,  $L, M$  the corresponding values of the first differential coefficients,  $(a, b, c)$  those of the second differential coefficients. The coordinates of the consecutive point are

$$x + u \cos \theta - v \sin \theta,$$

$$y + u \sin \theta + v \cos \theta.$$

Substituting these in the equation of the curve, we have

$$L(u \cos \theta - v \sin \theta) + M(u \sin \theta + v \cos \theta) + \frac{1}{2}(a, b, c)(u \cos \theta - v \sin \theta, u \sin \theta + v \cos \theta)^2 = 0,$$

the coefficient of  $u$  must be zero, or we have

$$L \cos \theta + M \sin \theta = 0,$$

giving

$$\sin \theta = \frac{L}{\sqrt{L^2 + M^2}}, \quad \cos \theta = -\frac{M}{\sqrt{L^2 + M^2}},$$

and the equation may be reduced to

$$\sqrt{L^2 + M^2} v + \frac{1}{2}(a, b, c)(\cos \theta, \sin \theta)^2 u^2 = 0,$$

or what is the same thing

$$(L^2 + M^2)^{\frac{3}{2}} v + \frac{1}{2}(a, b, c)(M, -L)^2 u^2 = 0,$$

whence the radius of curvature

$$= -\frac{u^2}{2v} = \frac{(L^2 + M^2)^{\frac{3}{2}}}{(a, b, c)(M, -L)^2},$$

which is the ordinary form for the radius of curvature at any point of a curve represented by the equation  $U = 0$ .

2, *Stone Buildings, W.C., October 2nd, 1859.*

289.

ON SOME NUMERICAL EXPANSIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 366–369.]

THERE are in the theory of series and the calculus of finite differences and the differential calculus, various sets of numbers, such for instance as Bernoulli's Numbers, or the successive differences of  $0^m$ , which from their more or less frequent occurrence and the complexity of their law it is desirable to tabulate. To these belong, I think, the coefficients of the successive powers of  $x$  in the expansion of a given power of  $\frac{\log(1+x)}{x}$ , or when the index of the power of the function is an indeterminate quantity  $r$ , then the coefficients of the several terms of the rational and integral functions of  $r$  which form the coefficients of the successive powers of  $x$  in the expansion in question. We have

$$\log \frac{\log(1+x)}{x} = -\frac{1}{2}x + \frac{5}{24}x^2 - \frac{1}{8}x^3 + \frac{251}{2880}x^4 - \frac{19}{288}x^5 + \frac{19087}{362880}x^6 - \&c.,$$

which is most easily verified by differentiating and multiplying by  $\log(1+x)$ , which gives on the left-hand side  $\frac{1}{1+x}$ , and on the right-hand side the expansion  $1 - x + x^2 - \&c$ . And from the above, multiplying each side by  $r$ , and taking the exponential, we find

$$\begin{aligned} \left\{ \frac{\log(1+x)}{x} \right\}^r &= 1 - \left(\frac{r}{2}\right)x \\ &\quad + (A_2r^2 + A'_2r)x^2 \\ &\quad - (A_3r^3 + A'_3r^2 + A''_3r)x^3 \\ &\quad + \&c. \end{aligned}$$

which may be verified by writing  $r = 1$ .

289]

ON SOME NUMERICAL EXPANSIONS.

471

As an instance of the use of the preceding table, let it be required to find expressions for the combinations without repetitions of the series of natural numbers  $1, 2, 3, \dots (k-1)$ , or what is the same thing for the coefficients of the powers of  $k$  in  $k(k-1)(k-2)\dots(k-n+1)$ . We have

$$k(k-1)(k-2)\dots(k-n+1) = A_1^n k^n - A_2^n k^{n-1} + A_3^n k^{n-2} - \&c.$$

whence

$$\begin{aligned} &= \Pi n \cdot \text{coefficient } x^n \text{ in } (1+x)^k; \\ (-)^r A_r^n &= \Pi n \cdot \text{coefficient } x^n k^{n-r} \text{ in } (1+x)^k \\ &= \Pi n \cdot \text{coefficient } x^n k^{n-r} \text{ in } e^{k \log(1+x)} \\ &= \frac{\Pi n}{\Pi(n-r)} \cdot \text{coefficient } x^n \text{ in } \{\log(1+x)\}^{n-r} \\ &= \frac{\Pi n}{\Pi(n-r)} r \cdot \text{coefficient of } x^r \text{ in } \left\{ \frac{\log(1+x)}{x} \right\}^{n-r}. \end{aligned}$$

Thus

$$\begin{aligned} A_2^n &= n\{\tfrac{1}{2}(n-1)\}, \\ A_3^n &= n(n-1)\{\tfrac{1}{6}(n-2)^2 + \tfrac{1}{4}(n-2)\}, \\ A_4^n &= n(n-1)(n-2)\{\tfrac{1}{24}(n-3)^3 + \tfrac{1}{8}(n-3)^2 + \tfrac{1}{8}(n-3)\}, \end{aligned}$$

and so on, as far as the expansion has been effected. It may be remarked that the general expression for the algebraical transcendent  $A_r^n$  is given in Dr Schläfli's paper, "Sur les coefficients du développement du produit  $(1+x)(1+2x)\dots[1+(n-1)x]$  suivant les puissances ascendantes de  $x$ ." *Crelle*, t. XLIII. [1852], pp. 1–22, but the law is a very complicated one.

We have

$$\log(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5 - \frac{1}{6}x^6 + \frac{1}{7}x^7 - \&c.$$

and dividing by  $x$ , and taking the logarithm, we have as before

$$\log\left(\frac{1}{x}\log(1+x)\right) = -\frac{1}{2}x + \frac{5}{24}x^2 - \frac{1}{8}x^3 + \frac{251}{2880}x^4 - \frac{19}{288}x^5 + \frac{19087}{362880}x^6 - \&c.,$$

which may be considered as the first of a series of logarithmic derivatives, viz., dividing by  $-\frac{1}{2}x$  and taking the logarithm we have

$$\log\left(-\frac{2}{x}\log\left(\frac{1}{x}\log(1+x)\right)\right) = -\frac{5}{12}x + \frac{49}{720}x^2 - \frac{959}{30240}x^3 + \frac{6515}{169344}x^4 - \&c.,$$

and by the like process

$$\log\left(-\frac{12}{5x}\log\left(-\frac{2}{x}\log\left(\frac{1}{x}\log(1+x)\right)\right)\right) = -\frac{49}{120}x + \frac{34471}{604800}x^2 - \frac{2787221}{101606400}x^3 + \&c.,$$

and so on.

472

# ON SOME NUMERICAL EXPANSIONS.

[289

Suppose in general that

$$\phi x = x + B_1x^2 + C_1x^3 + \dots,$$

and let it be required to find the  $r^{\text{th}}$  function  $\phi^r x$ . It is easy to see that the successive coefficients are rational and integral functions of  $r$  of the degrees 1, 2, 3, &c. respectively; we have in fact

$$\begin{aligned}\phi^r x &= \phi^0 x \\ &+ \frac{r}{1} (\phi^1 x - \phi^0 x) \\ &+ \frac{r \cdot r-1}{1 \cdot 2} (\phi^2 x - 2\phi^1 x + \phi^0 x) \\ &+ \frac{r \cdot r-1 \cdot r-2}{1 \cdot 2 \cdot 3} (\phi^3 x - 3\phi^2 x + 3\phi^1 x - \phi^0 x), \\ &+ \&c.,\end{aligned}$$

and by successive substitutions,

$$\begin{aligned}\phi^0 x &= x, \\ \phi^1 x &= x + B_1x^2 + C_1x^3 + D_1x^4 + \dots, \\ \phi^2 x &= x + 2B_1x^2 + (2B_1^2 + 2C_1)x^3 + (B_1^3 + 5B_1C_1 + 2D_1)x^4 + \dots, \\ \phi^3 x &= x + 3B_1x^2 + (6B_1^2 + 3C_1)x^3 + (9B_1^3 + 15B_1C_1 + 3D_1)x^4 + \dots.\end{aligned}$$

Whence

$$\begin{aligned}\phi^r x &= x \\ &+ rB_1x^2 \\ &+ \{(r^2 - r)B_1^2 + rC_1\}x^3 \\ &+ \{(\frac{1}{6}r^3 - \frac{1}{2}r^2 + \frac{1}{3}r)B_1^3 + (\frac{5}{2}r^2 - \frac{5}{2}r)B_1C_1 + rD_1\}x^4 \\ &+ \&c.\end{aligned}$$

It would, I think, be worth while to continue the expansion some steps further.

2, Stone Buildings, W.C., Oct. 2nd, 1859.

290.

A DISCUSSION OF THE STURMIAN CONSTANTS FOR CUBIC  
AND QUARTIC EQUATIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. IV. (1861), pp. 7–12.]

FOR the cubic equation

$$(a, b, c, d \mid x, 1)^3 = 0,$$

the Sturmiian Constants (or leading coefficients of the Sturmiian functions) are

$$a, \quad a, \quad b^2 - ac, \quad -a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2.$$

| If the signs of the constants,<br>that is, of the functions for<br>$+\infty$ , are | then the signs<br>of the functions for $-\infty$<br>are |                                                           |
|------------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------|
| $++++$                                                                             | $-+-+$                                                  | three real roots.<br>case cannot occur.<br>one real root. |
| $++++$                                                                             | $-+++$                                                  |                                                           |
| $++-+$                                                                             | $----+$                                                 |                                                           |
| $+++-$                                                                             | $--+-$                                                  |                                                           |
| $++--$                                                                             | $--++$                                                  |                                                           |

The second case would give a loss of variations of sign in passing from  $+\infty$  to  $-\infty$ , which is inconsistent with Sturm's theorem. To show à posteriori that the case cannot occur, we may form the identical equation

$$(a^2d - 3abc + 2b^3)^2 = -a^2(-a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2) + 4(b^2 - ac)^3,$$

and, this being so, then in the case in question, the right-hand side would consist of two terms, each of them negative, while the left-hand side is essentially positive.

In the particular case where the third constant vanishes, or  $b^2 - ac = 0$ , we have

$$\begin{aligned} -a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2 &= -(ad - bc)^2 + 4(b^2 - ac)(c^2 - bd) \\ &= -(ad - bc)^2, \text{ is negative;} \end{aligned}$$

hence, regarding the evanescent term as being at pleasure positive or negative, we have in each case a combination of signs corresponding to one real root.

The general result (which is well known) is, that there are three real roots or one real root according as

$$-a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2$$

is positive or negative.

For the quartic equation  $(a, b, c, d, e \mid x, 1)^4 = 0$ , the Sturmiian constants are

$$a, \quad a, \quad b^2 - ac, \quad 3aJ + 2(b^2 - ac)I, \quad I^3 - 27J^2,$$

if, as usual,

$$\begin{aligned} I &= ae - 4bd + 3c^2, \\ J &= ace - ad^2 - b^2e + 2bcd - c^3. \end{aligned}$$

| If signs of constants,<br>i.e. of functions for $+\infty$ , are | then signs of functions<br>for $-\infty$ are |                                                    |
|-----------------------------------------------------------------|----------------------------------------------|----------------------------------------------------|
| $+++++$                                                         | $+- - ++$                                    | 4 real roots.<br>no real root.                     |
| $++-++$                                                         | $+- +++$                                     |                                                    |
| $+++ - +$                                                       | $+++ - +$                                    |                                                    |
| $++-++$                                                         |                                              |                                                    |
| $++++ -$                                                        | $+- + - -$                                   | 2 real roots.<br>2, cannot occur.<br>2 real roots. |
| $++-+-$                                                         | $+++ -$                                      |                                                    |
| $+++ - +$                                                       | $+- + -$                                     |                                                    |
| $++- - -$                                                       |                                              |                                                    |

The non-existing combination of signs is

$$\begin{aligned} I^3 - 27J^2 &= -, \\ 3aJ + 2(b^2 - ac)I &= +, \\ b^2 - ac &= -. \end{aligned}$$

To show à posteriori that this case cannot occur, write

$$\Psi = a^2d - 3abc + 2b^3,$$

$$X = 3aJ + 2(b^2 - ac)I,$$

290]

FOR CUBIC AND QUARTIC EQUATIONS.

475

then we have identically

$$9(3a^2J^2 + X^2)\Psi^2 = -4a^2X^3 + 36(b^2 - ac)^2X^2 - 4a^2(b^2 - ac)^3(I^3 - 27J^2),$$

which is impossible under the given combination of signs, since the left-hand side would be positive, and the right-hand side negative.

To prove the above identity—the relation  $JU^4 - IU^2H + 4H^3 + \Phi^2 = 0$ , between the covariants of the quartic, gives

$$a^2J + a^2(b^2 - ac)I - 4(b^2 - ac)^3 + \Phi^2 = 0,$$

or, what is the same thing,

$$\Phi^2 = -a^2J - a^2(b^2 - ac)I + 4(b^2 - ac)^3.$$

But

$$X = 3aJ + 2(b^2 - ac)I,$$

and thence

$$3\Phi^2 + a^2X = -a^2(b^2 - ac)I + 12(b^2 - ac)^3,$$

$$3\Phi^2 = -a^2X - a^2(b^2 - ac)I + 12(b^2 - ac)^3,$$

and the identity will be true, if

$$\begin{aligned} (3X^2 + 9a^2J^2) \left[ -X - (b^2 - ac)I + 12\frac{(b^2 - ac)^3}{a^2} \right] \\ = -4X^3 + 36\frac{(b^2 - ac)^2}{a^2}X^2 - 4(b^2 - ac)^3(I^3 - 27J^2). \end{aligned}$$

This gives

$$(3X^2 + 9a^2J^2)[-X - (b^2 - ac)I] = -4X^3 - 4(b^2 - ac)^3I^3,$$

or, what is the same thing,

$$(3X^2 + 9a^2J^2)\{X + (b^2 - ac)I\} = 4\{X^3 + (b^2 - ac)^3I^3\},$$



or, dividing by  $X + (b^2 - ac)I$ ,

$$3X^2 + 9a^2J^2 = 4\{X^2 - X(b^2 - ac)I + (b^2 - ac)^2I^2\},$$

and reducing

$$X^2 - 4X(b^2 - ac)I - 9a^2J^2 + 4(b^2 - ac)^2I^2 = 0,$$

or finally

$$\{X - 3aJ - 2(b^2 - ac)I\}\{X + 3aJ - 2(b^2 - ac)I\} = 0,$$

which is true in virtue of

$$X = 3aJ + 2(b^2 - ac)I,$$

and the identity is thus proved.

476

# A DISCUSSION OF THE STURMIAN CONSTANTS

[290

The general conclusion is,

if  $I^3 - 27J^2$  is positive, the four roots are all real or all imaginary,

viz., all real if  $b^2 - ac$  and  $3aJ + 2(b^2 - ac)I$  are both positive, imaginary if otherwise. But if  $I^3 - 27J^2$  is negative, the roots are two of them real, and the other two imaginary.

The following special cases may be noticed,

1°.  $b^2 - ac = 0$ , here

$$9(3a^2J^2 + X^2)\Psi^2 = -4a^2X^3, \text{ or } X = 3aJ + 2(b^2 - ac)I = 3aJ, \text{ is negative,}$$

so that,

if  $I^3 - 27J^2$  is +, the roots are all imaginary;

if  $I^3 - 27J^2$  is -, the roots are two real and two imaginary.

2°.  $X = 3aJ + 2(b^2 - ac)I = 0$ , here

$$27a^2J^2\Psi^2 = -4a^2(b^2 - ac)^3(I^3 - 27J^2),$$

or  $b^2 - ac$ ,  $I^3 - 27J^2$  are of opposite signs, and if

$b^2 - ac = -$ ,  $I^3 - 27J^2 = +$ , the roots are all imaginary,

$b^2 - ac = +$ ,  $I^3 - 27J^2 = -$ , the roots are two real and two imaginary.

3°.  $b^2 - ac = 0$ ,  $X = 3aJ + 2(b^2 - ac)I = 0$ , here  $J = 0$ , that is,

$$2bcd - ad^2 - c^3 = 0, \text{ or } (ad - bc)^2 + c^2(ac - b^2) = 0, \text{ or } ad - bc = 0, \text{ and } I^3 - 27J^2 = I^3,$$

$$I = ae - 4bd + 3c^2 = ae - 4\frac{b^4}{a^2} + 3\frac{b^4}{a^2} = ae - \frac{b^4}{a^2} = \frac{1}{a^2}(a^3e - b^4),$$

whence

$I = +$ , the roots are all imaginary.

$I = -$ , the roots are two real and two imaginary.

This is easily verified, in fact  $ac - b^2 = 0$ ,  $ad - bc = 0$ , give  $c = \frac{b^2}{a}$ ,  $d = \frac{bc}{a} = \frac{b^3}{a^2}$ , and the equation becomes

$$ax^4 + 4bx^3 + 6\frac{b^2}{a}x^2 + 4\frac{b^3}{a^2}x + e = 0,$$

or, which is the same thing,

$$\frac{1}{a^3}\{(ax + b)^4 + (a^3e - b^4)\} = 0,$$

so that the roots are all imaginary, or two real and two imaginary, according to the sign of  $a^3e - b^4$  as above.

290]

FOR CUBIC AND QUARTIC EQUATIONS.

477

$$(a, b, c, d, e, f \mid x, 1)^5,$$
$$a, \quad a, \quad C, \quad D, \quad E, \quad F,$$
 $C, D, E, F$ 
$$\left. \begin{array}{l} - + + + \\ + - + + \\ + + - + \\ + + + - \end{array} \right\} 1 \text{ real root, 4 imaginary roots.}$$
$$\left. \begin{array}{l} + + - - \\ + - - - \\ - + + - \\ - - + - \end{array} \right\} \text{3 real roots, 2 imaginary roots.}$$
$$\left. \begin{array}{l} - + + - \\ + - - + \\ - + - \end{array} \right\} \text{cases which do not occur.}$$

2, *Stone Buildings, W.C., September 29, 1859.*

**291.**

ON THE DEMONSTRATION OF A THEOREM RELATING TO THE  
MOMENTS OF INERTIA OF A SOLID BODY.

*[From the Quarterly Journal of Pure and Applied Mathematics, vol. IV. (1861), pp. 25–27.]*

CONSIDERING in the first instance the analogous question in plano, let  $a = \int x^2 dm$ ,  $b = \int y^2 dm$ ,  $h = \int xy dm$ , where the integration extends over any closed figure whatever, then it is to be shown that the equations

$$(a, h, b \mid p, q)^2 = 1, \quad \text{and} \quad (a + b)(p^2 + q^2) - (a, h, b \mid p, q)^2 = 1,$$

represent respectively ellipses.

If in the first case, by a transformation of coordinates,

$$(a, h, b \mid p, q)^2 = a_1 p_1^2 + b_1 q_1^2,$$

then  $a_1, b_1$  are the roots of the quadratic equations,

$$\begin{vmatrix} a - \rho, & h \\ h, & b - \rho \end{vmatrix} = 0,$$

and if in the second case,

$$(a + b)(p^2 + q^2) - (a, h, b \mid p, q)^2 = a_1 p_1^2 + b_1 q_1^2,$$

then  $a_1, b_1$  are the roots of

$$\begin{vmatrix} b - \rho, & -h \\ -h, & a - \rho \end{vmatrix} = 0,$$

the two equations being in fact the same equation,

$$\rho^2 - (a + b)\rho + ab - h^2 = 0,$$

and the conditions that the curve may be an ellipse, are

$$a + b = +, \quad ab - h^2 = +,$$

the former of which requires no demonstration; to prove the latter, changing merely the variables under the integral sign, I write

$$a' = \int x'^2 dm', \quad b' = \int y'^2 dm', \quad h' = \int x'y' dm',$$

these quantities being of course respectively equal to  $a, b, h$ , we have then

$$ab' + a'b - 2hh' = \iint (xy' - x'y)^2 dm dm' = 2(ab - h^2),$$

or since the quantity under the integral sign is a square,  $ab - h^2$  is positive.

For the analogous problem *in solido*, we have

$$a = \int x^2 dm, \quad b = \int y^2 dm, \quad c = \int z^2 dm, \quad f = \int yz dm, \quad g = \int zx dm, \quad h = \int xy dm,$$

and it is to be shown that the equations

$$(a, b, c, f, g, h \mid p, q, r)^2 = 1,$$

$$(a + b + c)(p^2 + q^2 + r^2) - (a, b, c, f, g, h \mid p, q, r)^2 = 1,$$

represent respectively ellipsoids.

The conditions in the first problem are

$$\begin{aligned} a + b + c &= +, \\ bc + ca + ab - f^2 - g^2 - h^2 &= +, \\ abc - af^2 - bg^2 - ch^2 + 2fgh &= +, \end{aligned}$$

the first of which is obviously true: as regards the second, the theorem *in plano* shows that each of the quantities  $bc - f^2, ca - g^2, ab - h^2$  is positive, or merely reproducing the investigation, we find

$$2(bc + ca + ab - f^2 - g^2 - h^2) = \iint [(yz' - y'z)^2 + (zx' - z'x)^2 + (xy' - x'y)^2] dm dm',$$

which proves the theorem, and where it is to be observed that the integral may also be written

$$\iint [(x^2 + y^2 + z^2)(x'^2 + y'^2 + z'^2) - (xx' + yy' + zz')^2] dm dm';$$

and for the third, we find in a precisely similar manner,

$$6(abc - af^2 - bg^2 - ch^2 + 2fgh) = \iint \begin{vmatrix} x & y & z \\ x' & y' & z' \\ x'' & y'' & z'' \end{vmatrix}^2 dm dm',$$

which proves the theorem. The integral may also be written

$$\iint \begin{vmatrix} x^2 + y^2 + z^2 & xx' + yy' + zz' & xx'' + yy'' + zz'' \\ x'x + y'y + z'z & x'^2 + y'^2 + z'^2 & x'x'' + y'y'' + z'z'' \\ x''x + y''y + z''z & x''x' + y''y' + z''z' & x''^2 + y''^2 + z''^2 \end{vmatrix} dm dm'.$$

The conditions in the second problem are

$$\begin{aligned} (b+c) + (c+a) + (a+b) &= +, \\ (c+a)(a+b) + (a+b)(b+c) + (b+c)(c+a) - f^2 - g^2 - h^2 &= +, \\ (a+b)(b+c)(c+a) - (b+c)f^2 - (c+a)g^2 - (a+b)h^2 - 2fgh &= +, \end{aligned}$$

the first and second of which are respectively equivalent to

$$\begin{aligned} a+b+c &= +, \\ (a+b+c)^2 + bc + ca + ab - f^2 - g^2 - h^2 &= +, \end{aligned}$$

which are already proved. The last may be written

$$(a+b+c)(bc+ca+ab-f^2-g^2-h^2) - (abc-af^2-bg^2-ch^2+2fgh) = +,$$

which, putting for shortness,

$$\begin{aligned} A &= x^2 + y^2 + z^2, & B &= x'^2 + y'^2 + z'^2, & C &= x''^2 + y''^2 + z''^2, \\ F &= x'x'' + y'y'' + z'z'', & G &= x''x + y''y + z''z, & H &= xx' + yy' + zz', \end{aligned}$$

is by what precedes expressible in the form

$$\begin{aligned} \frac{1}{6} \iint \{A(BC-F^2) + B(CA-G^2) + C(AB-H^2) - (ABC - AF^2 - BG^2 - CH^2 + 2FGH)\} dm dm' \\ = \frac{1}{3} \iint (ABC - FGH) dm dm', \end{aligned}$$

or, since  $\sqrt{BC} > F$ ,  $\sqrt{CA} > G$ ,  $\sqrt{AB} > H$ , we have  $ABC > FGH$ , or  $ABC - FGH = +$ , and therefore the value of the integral is also positive.

2, Stone Buildings, W.C., 6th March, 1860.

## 292.

### A THEOREM IN CONICS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. IV. (1861), pp. 131–133.]

The following theorem is given in Todhunter's *Conic Sections*, [Ed. 7, p. 304], "If ellipses be inscribed in a triangle, each with one focus in a fixed straight line, the locus of the other focus is a conic section through the angular points of the triangle." A focus is the intersection of tangents to the conic from the circular points at infinity; and instead of the circular points at infinity we may substitute any two points whatever. This being so, let the equations of the sides of the triangle be  $x = 0$ ,  $y = 0$ ,  $z = 0$ , and let a pair of tangents to the curve from the points  $(\alpha, \beta, \gamma)$ ,  $(\alpha', \beta', \gamma')$  meet in the point  $(\xi, \eta, \zeta)$ , and the other pair of tangents from the same two points meet in the point  $(X, Y, Z)$ . I find that we have the very simple relation

$$X\xi : Y\eta : Z\zeta = \alpha\alpha' : \beta\beta' : \gamma\gamma',$$

and consequently, when the locus of the point  $(\xi, \eta, \zeta)$  is given, that of the point  $(X, Y, Z)$  is at once determined by substituting in the equation of the first-mentioned locus, in the place of  $\xi, \eta, \zeta$ , the values  $\frac{\alpha\alpha'}{\xi}, \frac{\beta\beta'}{\eta}, \frac{\gamma\gamma'}{\zeta}$ , or as we may express it, the second locus is derived from the first by the method of reciprocal trilinear substitutions. And, in particular, when the first locus is a line, the second locus is a conic through the angular points of the triangle, which is Mr Todhunter's theorem. I have considered some of the properties of this substitution in a Memoir "Sur quelques transmutations des Courbes," *Liouville*, t. xiv. (1849), pp. 40–46 and t. xv. (1850), pp. 351–356, [80 and 81].

To demonstrate the theorem, I take for the equation of the conic

$$\sqrt{(lx)} + \sqrt{(my)} + \sqrt{(nz)} = 0,$$

and I write for shortness

$$\begin{aligned} \beta\zeta - \gamma\eta, \quad \gamma\xi - \alpha\zeta, \quad \alpha\eta - \beta\xi &= A, B, C, \\ \beta'\zeta - \gamma'\eta, \quad \gamma'\xi - \alpha'\zeta, \quad \alpha'\eta - \beta'\xi &= A', B', C', \\ \xi(\beta\gamma' - \beta'\gamma) + \eta(\gamma\alpha' - \gamma'\alpha) + \zeta(\alpha\beta' - \alpha'\beta) &= \Delta, \\ \xi\alpha\alpha'(\beta\gamma' - \beta'\gamma) + \eta\beta\beta'(\gamma\alpha' - \gamma'\alpha) + \zeta\gamma\gamma'(\alpha\beta' - \alpha'\beta) &= \square, \end{aligned}$$

so that in fact

$$\Delta = \begin{vmatrix} \xi & \eta & \zeta \\ \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \end{vmatrix}, \quad \square = \alpha\beta\gamma\alpha'\beta'\gamma' \begin{vmatrix} \frac{1}{\alpha} & \frac{1}{\beta} & \frac{1}{\gamma} \\ \frac{1}{\alpha'} & \frac{1}{\beta'} & \frac{1}{\gamma'} \\ \frac{\xi}{\alpha} & \frac{\eta}{\beta} & \frac{\zeta}{\gamma} \end{vmatrix},$$

and we have

$$\begin{aligned} BC' - B'C &= \xi\Delta, \quad CA' - C'A = \eta\Delta, \quad AB' - A'B = \zeta\Delta, \\ \beta\gamma'BC' - \beta'\gamma B'C &= \gamma\alpha'CA' - \gamma'\alpha C'A = \alpha\beta'AB' - \alpha'\beta A'B = \square. \end{aligned}$$

The conditions in order that the conic

$$\sqrt{(lx)} + \sqrt{(my)} + \sqrt{(nz)} = 0$$

may touch the line through  $(\alpha, \beta, \gamma)$  and  $(\xi, \eta, \zeta)$  is

$$\frac{l}{A} + \frac{m}{B} + \frac{n}{C} = 0,$$

and the condition in order that it may touch the line through  $(\alpha', \beta', \gamma')$  and  $(\xi, \eta, \zeta)$  is

$$\frac{l}{A'} + \frac{m}{B'} + \frac{n}{C'} = 0,$$

and we thus have

$$l : m : n = \frac{1}{\frac{1}{BC'} - \frac{1}{B'C}} : \frac{1}{\frac{1}{CA'} - \frac{1}{C'A}} : \frac{1}{\frac{1}{AB'} - \frac{1}{A'B}},$$

or, what is the same thing,

$$l : m : n = AA'\xi : BB'\eta : CC'\zeta,$$

which determine the constants in the equation of the conic.

Consider now the tangents to the conic from the point  $(\alpha, \beta, \gamma)$ ; if the equation of the tangent is assumed to be

$$px + qy + rz = 0,$$

then we have

$$\begin{aligned} p\alpha + q\beta + r\gamma &= 0, \\ \frac{l}{p} + \frac{m}{q} + \frac{n}{r} &= 0, \end{aligned}$$

and these equations are of course satisfied by  $p : q : r = A : B : C$ , since the line through  $(\xi, \eta, \zeta)$  is a tangent. They are also satisfied by

$$p : q : r = \frac{l}{A} : \frac{m}{B} : \frac{n}{C},$$

as is obvious by substitution, we have therefore

$$\frac{l}{A}x + \frac{m}{B}y + \frac{n}{C}z = 0,$$

or more simply

$$\frac{A'\xi}{\alpha}x + \frac{B'\eta}{\beta}y + \frac{C'\zeta}{\gamma}z = 0,$$

for the equation of the other tangent through  $(\alpha, \beta, \gamma)$ , and we have in like manner

$$\frac{A\xi}{\alpha'}x + \frac{B\eta}{\beta'}y + \frac{C\zeta}{\gamma'}z = 0,$$

for the equation of the other tangent through  $(\alpha', \beta', \gamma')$ ; the last-mentioned two lines intersect in the point  $X, Y, Z$ , that is we have

$$X : Y : Z = \frac{BC'}{\beta\gamma'} - \frac{B'C}{\beta'\gamma} : \frac{CA'}{\gamma\alpha'} - \frac{C'A}{\gamma'\alpha} : \frac{AB'}{\alpha\beta'} - \frac{A'B}{\alpha'\beta},$$

or attending to an above-mentioned equation, we have

$$X\xi : Y\eta : Z\zeta = \alpha\alpha' : \beta\beta' : \gamma\gamma',$$

which is the property in question. In the particular case, where the points  $(\alpha, \beta, \gamma), (\alpha', \beta', \gamma')$  are the foci, the theorem is an immediate consequence of the well-known proposition that the product of the perpendiculars let fall from the two foci upon any tangent of the conic is a constant.

2, Stone Buildings, W.C., *17th March*, 1860.



293.

ON A CERTAIN SYSTEM OF FUNCTIONAL SYMBOLS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. IV. (1861), pp. 225–230.]

M. Hermite, in the Memoir “Sur la Résolution de l’équation du cinquième degré,” *Comptes Rendus*, t. xlv., (1858), p. 508, has given for equations of the fifth degree a solution by means of elliptic functions, analogous to the well-known trigonometrical solution of (the irreducible case of) the cubic equation. He has for this purpose to consider the effect of transforming certain functions  $\phi\omega$ ,  $\psi\omega$ , connected with the elliptic functions, by replacing therein the argument  $\omega$  by a linear fraction of the form  $\frac{c+d\omega}{a+b\omega}$ , where  $a$ ,  $b$ ,  $c$ ,  $d$  are integer numbers satisfying the condition

$$ad - bc = 1,$$

and he finds for  $\phi\left(\frac{c+d\omega}{a+b\omega}\right)$  (and the like would apply to  $\psi\left(\frac{c+d\omega}{a+b\omega}\right)$ ), six different expressions according to the evenness, or values in relation to the modulus 2, of the coefficients  $a$ ,  $b$ ,  $c$ ,  $d$ , viz., the congruence

$$ad - bc \equiv 1 \pmod{2}$$

admits of the six solutions shown by the table

|      | $a$ | $b$ | $c$ | $d$ |
|------|-----|-----|-----|-----|
| I.   | 1   | 0   | 0   | 1   |
| II.  | 0   | 1   | 1   | 0   |
| III. | 1   | 1   | 0   | 1   |
| IV.  | 1   | 1   | 1   | 0   |
| V.   | 1   | 0   | 1   | 1   |
| VI.  | 0   | 1   | 1   | 1   |

(where 0 or 1 denotes that the coefficient to which it refers is even or odd), and to each solution there corresponds a distinct value of  $\phi\left(\frac{c+d\omega}{a+b\omega}\right)$  or  $\psi\left(\frac{c+d\omega}{a+b\omega}\right)$ .

It appeared to me that the formulae in question were not only of the highest importance in the theory of elliptic functions, but that they were very remarkable in and for themselves, and that the question might be looked at in the following manner.

Suppose that  $A$ ,  $B$  are functional symbols, or, so to term them, substitutions, the laws of operation being

$$A(X, Y) = \left(\theta \frac{X}{Y}, \frac{1}{Y}\right),$$

$$B(X, Y) = (Y, X),$$

where  $\theta$  is a constant. The equations mean that the effect of  $A$  is to change  $X$  into  $\theta \frac{X}{Y}$ , and  $Y$  into  $\frac{1}{Y}$ , and the effect of  $B$  to change  $X$  into  $Y$ , and  $Y$  into  $X$ . We have

$$A^2X = \theta \frac{\theta X/Y}{1/Y} = \theta^2 X, \quad A^2Y = \frac{1}{1/Y} = Y, \quad \text{or} \quad A^2(X, Y) = (\theta^2 X, Y),$$

$$B^2X = X, \quad B^2Y = Y, \quad \text{or} \quad B^2(X, Y) = (X, Y);$$

thus  $B$  is periodic of the second order; and in the particular case  $\theta = 1$ ,  $A$  is also periodic of the second order.

This being so, let  $K$  denote any one of the compound symbols  $A^p$ ,  $BA^p$ ,  $A^qB$ ,  $A^qBA^p$ , &c.; I say that the species of  $K$  is determined by connecting with each of the symbols in question, a certain continued fraction as follows:

|                 |                                                      |
|-----------------|------------------------------------------------------|
| $A^p$ ,         | $p + \omega$ ,                                       |
| $BA^p$ ,        | $p - \frac{1}{\omega}$ ,                             |
| $A^qB$ ,        | $-\frac{1}{q + \omega}$ ,                            |
| $BA^qB$ ,       | $-\frac{1}{q - \frac{1}{\omega}}$ ,                  |
| $A^qBA^p$ ,     | $p - \frac{1}{q + \omega}$ ,                         |
| $BA^qBA^p$ ,    | $p - \frac{1}{q - \frac{1}{\omega}}$ ,               |
| $A^rBA^qBA^p$ , | $p - \frac{1}{q - \frac{1}{r + \omega}}$ ,           |
| &c.,            | $p - \frac{1}{q - \frac{1}{r - \frac{1}{\omega}}}$ , |

and when the continued fraction is reduced to the form  $\frac{c + d\omega}{a + b\omega}$ , where, of course,  $ad - bc = 1$ , then, that the corresponding symbol  $K$  is of the species I. if  $a, b, c, d \equiv 1, 0, 0, 1 \pmod{2}$ , and in like manner, that the species is II., III., IV., V., or VI., according as  $a, b, c, d$  are in regard to the modulus 2 of the form specified in the second, third, fourth, fifth, or sixth line of the foregoing table.

We have then a system of equations which I write in the form

$$K(X, Y) = \begin{cases} (\frac{X}{Y}, Y) & \text{for } K \text{ of the species I.} \\ (\frac{Y}{X}, X) & \text{II.} \\ (\frac{1}{Y}, \frac{X}{Y}) & \text{III.} \\ (\frac{1}{X}, \frac{Y}{X}) & \text{IV.} \\ (\frac{X}{Y}, \frac{1}{Y}) & \text{V.} \\ (\frac{Y}{X}, \frac{1}{X}) & \text{VI.} \end{cases}$$

viz., the first line denotes that  $K(X, Y) = (\theta^x X, \theta^y Y)$ , and so for the other lines, the values of the indices  $x, y$  being different in the several cases, but I have written the system in the above form, to put in evidence more distinctly the theorem as applied to the most simple case, where  $\theta = 1$ .

It will be sufficient to indicate how the theorem is proved in this particular case: I will suppose that for

$$K = BA^q BA^p \dots,$$

we have

$$K(X, Y) = \begin{cases} (\frac{X}{Y}, Y) & \text{for } K \text{ of the species I.} \\ (\frac{Y}{X}, X) & \text{II.} \\ (\frac{1}{Y}, \frac{X}{Y}) & \text{III.} \\ (\frac{1}{X}, \frac{Y}{X}) & \text{IV.} \\ (\frac{X}{Y}, \frac{1}{Y}) & \text{V.} \\ (\frac{Y}{X}, \frac{1}{X}) & \text{VI.} \end{cases}$$

and let it be inquired what is the effect of the symbol  $K' = A^r BA^q BA^p \dots$ , which, since  $A^r(X, Y) = (\theta^r X, Y)$ , is equivalent to  $AK$  or  $K$ , according as  $r$  is odd or even.

Let  $\Theta = \frac{c+d\omega}{a+b\omega}$  be the continued fraction which corresponds to  $K$ , and  $\Theta' = \frac{c'+d'\omega}{a'+b'\omega}$  the continued fraction which corresponds to  $K'$ , then  $\Theta'$  is deduced from  $\Theta$  by putting therein  $r + \omega$  for  $\omega$ , or we have

$$\frac{c+d(\omega+r)}{a+b(\omega+r)} = \frac{c'+d'\omega}{a'+b'\omega},$$

or

$$c' = c + dr, \quad d' = d, \quad a' = a + br, \quad b' = b,$$

and therefore when  $r$  is even, or  $\equiv 0 \pmod{2}$ , then

$$c' = c, \quad d' = d, \quad a' = a, \quad b' = b, \quad (\text{mod } 2),$$

or  $K'$  is of the same species as  $K$ . But in this case since  $A^r(X, Y) = (X, Y)$  and  $r$  is even, we ought to have  $K'$  equivalent to  $K$ ; and hence  $r$  being even, the theorem, assumed to be true for  $K$ , is also true for  $K'$ .

But if  $r$  be odd, or  $\equiv 1 \pmod{2}$ , then

$$c' = c + 1, \quad d' = d, \quad a' = a + 1, \quad b' = b, \quad (\text{mod } 2),$$

whence, according as

$K$  is of species I., II., III., IV., V., VI., so

$K'$  V., IV., VI., II., I., III.

But if from  $K(X, Y)$  we derive  $AK(X, Y)$ , we find that of whatever species  $K$  may be, the species of  $AK$  is identical with the corresponding species of  $K'$ , and in the case in question, where  $r$  is odd, it is clear that we ought to have  $K'$  equivalent to  $AK$ . Hence in this case also, and therefore generally, if the theorem be true for  $K$  it will be also true for  $K'$ .

In a very similar manner it may be shown that if the theorem is true for  $K' = A^r B A^q B A^p \dots$ , it will be true for  $B A^r B A^q B A^p \dots$ , and the foregoing cases include the case where the symbol is of the form  $B A^r B A^q B$  or  $A^r B A^q B$  (terminating in a  $B$ ), for it is only necessary to assume that the last index of  $A$  is zero; hence the theorem, if true in any case, as it obviously is for  $A^0(X, Y) = (X, Y)$ , is true universally.

When  $\theta$  is not equal 1, the only difference is that the terms of  $K(X, Y)$  are respectively multiplied by certain powers  $\theta^x$ ,  $\theta^y$  of  $\theta$ , and it is not difficult to see that for  $K = A^r B A^q B A^p \dots$  the indices  $x, y$  are of the forms

$$x = cr + bq + ap + \dots,$$

$$y = c'r + b'q + a'p + \dots,$$

where  $\mathfrak{c}, \mathfrak{b}, \mathfrak{a}, \dots, \mathfrak{c}', \mathfrak{b}', \mathfrak{a}', \dots$  have only the values 0, +1, -1, the values of

$\mathfrak{c}, \mathfrak{c}'$  depending on the species of  $BA^qBA^p\dots$ ,

$\mathfrak{b}, \mathfrak{b}'$   $BA^p\dots$ ,

&c.

in a manner which can easily be determined. In the particular case, where  $\theta^{16} = 1$  (as it is in M. Hermite's problem), it is only necessary to know the values of  $x, y$  in regard to the modulus 16, and these values can be expressed in terms of the numbers  $(a, b, c, d)$  which correspond to the symbol  $K$  (this is in itself a remark-able theorem), and I find that (to modulus 16) for  $K$  of the species

$$x = \begin{cases} d(c+d) - 1 & \text{I.} \\ -ab & \text{II.} \\ c(c-d) - 1 & \text{III.} \\ d(d-c) - 1 & \text{V.} \\ c(d-c) + 1 & \text{IV.} \\ cd & \text{VI.} \end{cases}, \quad y = \begin{cases} a(a-b) - 1 & \text{I.} \\ b(a+b) - 1 & \text{II.} \\ ab & \text{III.} \\ -cd & \text{V.} \\ a(b-a) + 1 & \text{IV.} \\ -b(b-a) - 1 & \text{VI.} \end{cases}$$

This may be shown by induction in like manner as the theorem in regard to the parts independent of  $\theta$ . Thus  $K, K'$  as before, let  $K$  be of the species I., and assume that  $K(X, Y) = (\theta^x X, \theta^y Y)$  where

$$x = d(c+d) - 1, \quad y = a(a-b) - 1 \pmod{16},$$

then if  $r$  be even, since  $A^r(X, Y) = (\theta^r X, Y)$ , we ought to have  $K'(X, Y) = (\theta^{x+r} X, \theta^y Y)$ . But in this case  $K'$  is of the same species with  $K$ , viz., of the species I., and by the theorem,  $K'(X, Y) = (\theta^{x'} X, \theta^{y'} Y)$  where

$$x' = d'(c' + d') - 1, \quad y' = a'(a' - b') - 1 \pmod{16};$$

and we ought to have

$$x' = x + r, \quad y' = y \pmod{16},$$

that is

$$d'(c' + d') - 1 = d(c+d) - 1 + r, \quad \pmod{16},$$

$$a'(a' - b') - 1 = a(a-b) - 1,$$

or substituting for  $a', b', c', d'$  their values  $a + br, b, c + dr, d$ , these become

$$d(c+d+dr) = d(c+d) + r, \quad \pmod{16},$$

$$(a+br)(a-b+br) = a(a-b),$$

or

$$(d^2 - 1)r = 0,$$

$$\pmod{16}.$$

But  $r$  being even, these become

$$d^2 - 1 \equiv 0, \quad 2ab - b^2 + b^2r \equiv 0 \pmod{8};$$

or, since in the case under consideration,  $a, b, c, d \equiv 1, 0, 0, 1 \pmod{2}$ , and therefore  $b^2r \equiv 0 \pmod{8}$ , the two congruences become

$$d^2 - 1 \equiv 0, \quad a^2 - (a - b)^2 \equiv 0 \pmod{8},$$

which are satisfied since  $d, a$ , and  $a - b$  are all odd. And in a similar way the theorem is proved for all the other cases.

In M. Hermite's problem,  $X, Y$  are replaced by  $\phi\omega, \psi\omega$ ,  $AX, AY$  are  $\phi(\omega + 1), \psi(\omega + 1)$ , and  $BX, BY$  are  $\phi\left(-\frac{1}{\omega}\right), \psi\left(-\frac{1}{\omega}\right)$ , the properties of the functions  $\phi, \psi$  being such that, as in the preceding investigation,

$$A(X, Y) = \left(\theta \frac{X}{Y}, \frac{1}{Y}\right), \quad B(X, Y) = (Y, X),$$

where  $\theta (= e^{\frac{1}{8}\pi i})$  is a sixteenth root of unity, and  $KX, KY$  are respectively equal to  $\phi\left(\frac{c + d\omega}{a + b\omega}\right), \psi\left(\frac{c + d\omega}{a + b\omega}\right)$ , and the formulae are precisely similar to those of the present paper.

2, Stone Buildings, W.C., 5 May, 1860.

294.

ON A NEW ANALYTICAL REPRESENTATION OF CURVES IN SPACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. V. (1862), pp. 81–86.]

The employment of a new kind of coordinates for the analytical representation of curves in space is suggested in my former paper under the same title *Journal*, t. III., pp. 225–236 (1859). The idea was as follows: viz. if  $(x, y, z, w)$  are current coordinates of a point in space (ordinary point coordinates), and  $(\alpha, \beta, \gamma, \delta)$  the coordinates of a particular point, then taking  $(p, q, r, s, t, u)$  to represent the minor determinants formed out of the matrix

$$\begin{pmatrix} x & y & z & w \\ \alpha & \beta & \gamma & \delta \end{pmatrix},$$

viz.

$$\begin{aligned} p &= \gamma y - \beta z, & s &= \delta x - \alpha w, \\ q &= \alpha z - \gamma x, & t &= \delta y - \beta w, \\ r &= \beta x - \alpha y, & u &= \delta z - \gamma w, \end{aligned}$$

values which satisfy identically

$$ps + qt + ru = 0,$$

then the equation of a cone passing through a given curve and having for its vertex the arbitrary point  $(\alpha, \beta, \gamma, \delta)$ , is of the form

$$V = 0,$$

$V$  being a homogeneous function of the six new coordinates  $(p, q, r, s, t, u)$ . And it was proposed to consider  $V = 0$  as the equation of the curve.

But as remarked in the paper, it is not every function  $V$  of the coordinates  $(p, q, r, s, t, u)$  which equated to zero, does in fact represent a curve. In order that

the equation  $V = 0$  may represent a curve, it is necessary, that when any infinitesimal variations whatever are given to the constants  $(\alpha, \beta, \gamma, \delta)$ , thus converting the equation into  $V + \delta V = 0$ , the two equations  $V = 0$ ,  $\delta V = 0$  (considered as equations in ordinary point coordinates) shall represent one and the same curve, whatever the system of infinitesimal variations attributed to  $\alpha, \beta, \gamma, \delta$  may be. Let  $P, Q, R, S, T, U$  denote the differential coefficients of  $V$  in regard to  $p, q, r, s, t, u$  respectively, then the equation  $\delta V = 0$ , breaks up into the equations

$$\begin{aligned} -Ry + Qz - Sw &= 0, \\ Rx &\quad -Pz - Tw = 0, \\ -Qx + Py &\quad -Uw = 0, \\ Sx + Ty + Uz &\quad = 0, \end{aligned}$$

and the system composed of these four equations and the equation  $V = 0$  (considered as equations in ordinary point coordinates) must belong to one and the same curve.

The four equations gave

$$PS + QT + RU = 0,$$

a relation between the differential coefficients of  $V$  which must be satisfied either identically or in virtue of the equation  $V = 0$ . And this relation existing, any two of the four equations lead to the other two. Attending exclusively to the coordinates  $(p, q, r, s, t, u)$  and considering  $(x, y, z, w)$  as mere arbitrary multipliers, the above equation

$$PS + QT + RU = 0$$

is the only relation between the differential coefficients of  $V$  which is deducible from the four equations.

But it was noticed that the equation  $V = 0$ , even when  $V$  is a function such that we have (identically or in virtue of the equation  $V = 0$ ) the equation  $PS + QT + RU = 0$ , does not of necessity represent a curve. Some further relation or relations between the differential coefficients of  $V$  must therefore exist, either identically or in virtue of the equation  $V = 0$ ; and such relations can be found by resorting to the second differential  $\delta^2 V$  of the function  $V$ . In fact not only the equation  $\delta V = 0$  but the entire series of relations  $\delta^2 V = 0$ ,  $\delta^3 V = 0, \dots$  should be satisfied by the coordinates of any point of the curve. I find by means of the equation  $\delta^2 V = 0$  a plexus of equations, which are consequently necessary, and I am inclined to believe sufficient, in order that the equation  $V = 0$  may in fact represent a curve; the equations of the plexus are, it will be seen, very numerous, and certainly only a small number of them are independent, but this is a question which I have not as yet investigated.

Attending to the expressions for  $p, q, r, s, t, u$ , we have

$$\begin{aligned} d\alpha &= \quad -yd_r + zd_q - wd_s = (1), \\ d\beta &= xd_r \quad -zd_p - wd_t = (2), \\ d\gamma &= -xd_q + yd_p \quad -wd_u = (3), \\ d\delta &= xd_s + yd_t + zd_u \quad = (4), \end{aligned}$$



and writing for convenience  $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$  instead of  $d_\alpha, d_\beta, d_\gamma, d_\delta$ , we have

$$d = (1)\mathbf{a} + (2)\mathbf{b} + (3)\mathbf{c} + (4)\mathbf{d}.$$

It was in effect by operating on  $V$  with this symbol and equating to zero the coefficients of  $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ , that the before-mentioned equations

$$\begin{aligned} -Ry + Qz - Sw &= 0, \\ Rx - Pz - Tw &= 0, \\ -Qx + Py - Uw &= 0, \\ Sx + Ty + Uz &= 0, \end{aligned}$$

were found.

If to these equations we join the equation

$$Ax + By + Cz + Dw = 0,$$

where  $A, B, C, D$  are arbitrary multipliers, we can express  $x, y, z, w$  in terms of  $A, B, C, D$  in such manner as to satisfy the four equations, viz. we have

$$\begin{aligned} x &= BU - CT + DP, \\ y &= -AU + CS + DQ, \\ z &= AT - BS + DR, \\ w &= -AP - BQ - CR. \end{aligned}$$

and if in the expressions for (1), (2), (3), (4) we substitute for  $x, y, z, w$  these values, and form therewith the value of  $d$ , which value I will for distinction call  $\mathfrak{D}$ , we have

$$\begin{aligned} \mathfrak{D} = & (Ud_r + Td_q + Pd_s, Qd_s - Sd_q, Rd_s - Sd_r, Rd_q - Qd_r) \\ & (Pd_t - Td_p, Ud_r + Qd_t + Sd_p, Rd_t - Td_r, Pd_r - Rd_p) \\ & (Pd_u - Ud_p, Qd_u - Ud_q, Rd_u + Sd_p + Td_q, Qd_p - Pd_q) \\ & (Td_u - Ud_t, Ud_s - Sd_u, Sd_t - Td_s, Pd_s + Qd_t + Rd_u) (A, B, C, D)(\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}), \end{aligned}$$

viz.  $\mathfrak{D}$  is a lineo-linear function of the two sets of indeterminate quantities  $(A, B, C, D)$ ,  $(\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d})$ , the coefficients thereof being the operators

$$Ud_r + Td_q + Pd_s, Qd_s - Sd_q, \text{ \&c.}$$

It may be remarked that we have identically

$$\mathfrak{D}V = (PS + QT + RU)(A\mathbf{a} + B\mathbf{b} + C\mathbf{c} + D\mathbf{d}),$$

since obviously each term such as  $(Qd_s - Sd_q)V$ , which is equal to  $QS - SQ$ , vanishes identically. The equation  $\mathfrak{D}V = 0$  gives therefore only the before-mentioned equation  $PS + QT + RU = 0$ , which is as it should be.

The equation  $\mathfrak{D}^2V = 0$ , is then to be satisfied independently of the values of  $(A, B, C, D)$  and  $(a, b, c, d)$ , and as  $\mathfrak{D}$  contains 16 distinct terms,  $\mathfrak{D}^2$  will contain in all  $\frac{1}{2} \cdot 16 \cdot 17$  or 136 distinct terms. The equation  $\mathfrak{D}^2V = 0$  gives therefore a plexus of 136 equations, and the equations in each succeeding plexus, involved in  $\mathfrak{D}^3V = 0$ ,  $\mathfrak{D}^4V = 0$ , &c. will, of course, be still be more numerous.

If  $V = 0$  be the plane conic which is the intersection of the surfaces

$$x^2 + y^2 + z^2 + w^2 = 0,$$

$$ax + by + cz + dw = 0,$$

then we have

$$V = (b^2 + c^2, -ab, -ac, \cdot, cd, -bd, c^2 + a^2, -bc, -cd, \cdot, ad, a^2 + b^2, bd, -ad, \cdot, b^2 + d^2, ab, ac, a^2 + d^2, bc, c^2 + d^2)(p, q, r, s, t, u)^2.$$

The values of  $P, Q, R, S, T, U$  (omitting a common factor 2) are

$$P = (b^2 + c^2, -ab, -ac, \cdot, +cd, -bd \mid p, q, r, s, t, u),$$

&c.,

and if we proceed to form a term in  $\mathfrak{D}^2V$ , say the coefficient of  $A^2a^2$ , this is  $(Ud_r + Td_q + Pd_s)^2V$ , or

$$(a^2 + b^2, c^2 + a^2, a^2 + d^2, -cd, +bd, -cb \mid U, T, P)^2.$$

The coefficient therein of  $\theta$  is

$$(a^2 + b^2, c^2 + a^2, a^2 + d^2, -cd, +bd, -cb \mid -bd, cd, b^2 + d^2)^2,$$

that is, it is

$$\begin{aligned} & (a^2 + b^2)b^2d^2 - 2cd \cdot cd(b^2 + d^2) \\ & + (c^2 + a^2)c^2d^2 + 2bd \cdot -bd(b^2 + c^2) \\ & + (a^2 + d^2)(b^2 + c^2)^2 - 2cb \cdot -bd \cdot cd, \end{aligned}$$

where the terms in which  $(b^2 + c^2)$  does not appear as a factor are together equal to

$$a^2d^2(b^2 + c^2) + d^2(b^4 + c^4),$$

the entire expression thus divides by  $b^2 + c^2$  the quotient being

$$(a^2 + d^2)(b^2 + c^2) - 4c^2d^2 - 2b^2d^2 + a^2d^2 + d^2(b^2 + c^2),$$

which is equal to  $a^2(b^2 + c^2 + d^2)$ , or restoring the factor  $b^2 + c^2$ , we see that in  $\mathfrak{D}^2V$  the coefficient of  $A^2a^2$  is

$$a^2(b^2 + c^2 + d^2)(b^2 + c^2)V + \&c.$$

The complete value must, it is clear, be of the form

$$a^2(b^2 + c^2 + d^2)V + k(ps + qt + ru),$$

vanishing in virtue of the equations  $V = 0$ ,  $ps + qt + ru = 0$ , and this being so, observing that  $V$  contains no term in  $ps$ , we have  $k =$  coefficient  $ps$  in

$$(a^2 + b^2, c^2 + a^2, a^2 + d^2, -cd, +bd, -cb \mid U, T, P)^2,$$

that is

$$k = 2(a^2 + b^2, c^2 + a^2, a^2 + d^2, -cd, +bd, -cb \mid -bd, cd, b^2 + d^2)(ca, ba, 0),$$

or

$$\begin{aligned} \frac{1}{2}k &= (a^2 + b^2) \cdot -bd \cdot ca - cd[cd \cdot 0 + (b^2 + d^2)ba] \\ &\quad + (c^2 + a^2) \cdot cd \cdot ba + bd[(b^2 + d^2)ca - bd \cdot 0] \\ &\quad + (a^2 + d^2) \cdot 0 - cb[-bd \cdot ba + cd \cdot ca], \end{aligned}$$

which is

$$= abcd \left[ -(a^2 + b^2) - (b^2 + c^2) + (d^2 + a^2) + (b^2 + d^2) + (b^2 - c^2) \right] = 0.$$

The coefficient  $k$  consequently vanishes, and therefore in  $\mathfrak{D}^2V$  the coefficient of  $A^2\mathfrak{a}^2$  is  $a^2(b^2 + c^2 + d^2)V$ , but I have not worked out the coefficients of the other terms.

2, Stone Buildings, W.C., 30th October, 1860.

## 295.

## ON THE CONSTRUCTION OF THE NINTH POINT OF INTER-SECTION OF THE CUBICS WHICH PASS THROUGH EIGHT GIVEN POINTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. v. (1862), pp. 222–233.]

I REPRODUCE with additional developments the solution which has been given of this interesting problem.

The equation of a given cubic may be written

$$qU - pV = 0,$$

where  $U = 0$ ,  $V = 0$  are any two conics meeting the cubic in the same four points;  $p = 0$  is the line joining the remaining two points of intersection of the cubic with the conic  $U = 0$ ; and  $q = 0$  is the line joining the remaining two points of intersection of the cubic with the conic  $V = 0$ ; the relation between the arbitrary constant factors implicitly contained in the functions  $qU$  and  $pV$  is assumed to be properly determined.

The form is employed by Plücker, *Theorie der algebraischen Curven*, p. 56 (1839), and in connexion therewith he gives some geometrical considerations which, he remarks, contain implicitly the solution of the above-mentioned problem.

The form is also the analytical basis of the investigations of M. Chasles, “Construction de la courbe du troisième ordre par neuf points,” *Comptes Rendus*, t. xxxvi. (1853), pp. 942–952. In fact calling to mind the theorem that in the pencil of conics  $U - aV = 0$  (where  $a$  is an arbitrary multiplier) the anharmonic ratio of the multipliers  $a$  is equal to the anharmonic ratio of the tangents at any one of the points of intersection, or (what is the same thing) to the anharmonic ratio of

the polars of any point whatever in regard to the conics, and recollecting that the anharmonic ratio in question is said to be the anharmonic ratio of the conics them-selves; then since the equation  $qU - pV = 0$  is satisfied by the system

$$U - aV = 0, \quad p - aq = 0,$$

(where  $a$  is arbitrary) we have at once the theorem, p. 949, viz. that if there be a pencil of lines through a point, and corresponding anharmonically thereto, a pencil of conics through the same four points; the locus of the intersections of a line by the corresponding conic is a cubic through the five points: and, conversely, that a given cubic may be so generated, the point of the pencil of lines, and the four points of the pencil of conics being any five points whatever of the cubic.

This gives at once the construction (M. Chasles' first construction) for the cubic through nine given points. In fact if the points are called 1, 2, 3, 4, 5, 6, 7, 8, 9; then grouping the points in any manner, it is only necessary to find a point  $x$  such that

$$x(1, 2, 3, 4, 5) = 6789(1, 2, 3, 4, 5),$$

that is, such that the pencil of lines  $x1, x2, x3, x4, x5$  shall correspond anharmonically to the pencil of conics 67891, 67892, 67893, 67894, 67895. The foregoing notation is that employed in M. de Jonquières' "Essai sur la génération des Courbes géométriques, &c.," Mém. Sav. Strang., t. xvi. (1858), which I take the opportunity of referring to. In fact, if  $x$  satisfies the foregoing condition, then taking through the point  $x$  any other line, and corresponding anharmonically thereto a conic through the points 6, 7, 8, 9, the locus of the intersections of the line and conic will be a cubic through the nine points. But the condition in question gives

$$x(1, 2, 3, 4) = 6789(1, 2, 3, 4),$$

$$x(1, 2, 3, 5) = 6789(1, 2, 3, 5),$$

which (by the anharmonic property of the points of a conic) show, the first that  $x$  is in a certain conic passing through 1, 2, 3, 4, and the second that  $x$  is in a certain conic passing through 1, 2, 3, 5; the two conics intersect in the points 1, 2, 3, and in a fourth point which is the required point  $x$ . Or we may say that  $x$  is given by the condition that the pencils  $x(1, 2, 3, 4)$  and  $x(1, 2, 3, 5)$  shall have given anharmonic ratios. It will presently be seen how  $x$  can be determined by the ruler alone.

Suppose now that the points 1, 2, 3, 4, 5, 6, 7, 8, 9 are the points of intersection of two cubics; the construction should become indeterminate; this is only the case when the two conics which by their intersection should determine  $x$  become one and the same conic. This implies that the conic  $x(1, 2, 3, 4)$  passes through 5, or that  $x, 1, 2, 3, 4, 5$  are points of the same conic. And then since by the anharmonic property of the points of a conic  $x(1, 2, 3, 4) = 5(1, 2, 3, 4)$ , we have

$$5(1, 2, 3, 4) = 6789(1, 2, 3, 4).$$

The grouping of the nine points is altogether arbitrary, hence there are in all  $(9 \times 70 =) 630$  such equations, which are really equivalent to only two equations, and which when eight of the points are given, determine the ninth point. Supposing that the given points are 1, 2, 3, 4, 5, 6, 7, 8, the equations for the determination of the remaining point 9 may be taken to be

$$9(5, 6, 7, 8) = 1234(5, 6, 7, 8),$$

$$9(4, 6, 7, 8) = 1235(4, 6, 7, 8),$$

which (it is to be remarked) determine 9 in a similar way to that in which  $x$  is given in the construction of the cubic through nine points; viz. 9 is the fourth intersection of two conics which pass through the points 5, 6, 7, 8 and the points 4, 6, 7, 8 respectively. Or we may say that 9 is given by the conditions that the pencils  $9(5, 6, 7, 8)$  and  $9(4, 6, 7, 8)$  shall have given anharmonic ratios.

The foregoing equations

$$9(5, 6, 7, 8) = 1234(5, 6, 7, 8),$$

$$9(4, 6, 7, 8) = 1235(4, 6, 7, 8),$$

are equivalent to and constitute the geometrical interpretation of the equations obtained (previous to M. Chasles' Memoir) by Weddle in the paper "On the construction of the ninth point of intersection of two curves of the third degree when the other eight points are given," *Cambridge and Dublin Mathematical Journal*, t. vi., pp. 83–86 (1851). In fact, reproducing his analysis with only a slight change of notation, let 012 denote the determinant

$$\begin{vmatrix} x_0 & y_0 & z_0 \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix}$$

so that  $012 = 0$  is the equation of the line through the points 1 and 2; and in like manner let 012345 denote the determinant

$$\begin{vmatrix} x_0^2 & y_0^2 & z_0^2 & y_0 z_0 & z_0 x_0 & x_0 y_0 \\ x_1^2 & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{vmatrix} \text{ \&c.}$$

so that  $012345 = 0$  is the equation of the conic through the points 1, 2, 3, 4, 5. Of course 123, 123456, &c. will denote given functions of the coordinates of the points 1, 2, 3, the points 1, 2, 3, 4, 5, 6, &c. This being so

$$012345 \cdot 078 = \lambda \cdot 012347 \cdot 058$$

is the equation of a particular cubic passing through the points 1, 2, 3, 4, 5, 7, 8, and which if we properly determine  $\lambda$ , viz. if we write

$$\lambda = \frac{612345 \cdot 678}{612347 \cdot 658},$$

will also pass through the point 6.

And similarly

$$012345 \cdot 076 = \mu \cdot 012347 \cdot 056$$

is the equation of a particular cubic curve passing through the points 1, 2, 3, 4, 5, 6, 7 and which if we properly determine  $\mu$ , viz. if we write

$$\mu = \frac{812345 \cdot 876}{812347 \cdot 856},$$

will also pass through the point 8. Hence the two curves, each of them passing through the points 1, 2, 3, 4, 5, 6, 7, 8 will intersect in the remaining point 9; and writing 9 for 0, and combining the two equations, we have

$$\frac{978 \cdot 956}{958 \cdot 976} = \frac{\lambda}{\mu} = \frac{612345}{612347} \cdot \frac{812347}{812345},$$

or, what is the same thing,

$$\frac{956 \cdot 978}{958 \cdot 967} = \frac{123456 \cdot 123478}{123458 \cdot 123467},$$

which is Weddle's equation, and is equivalent to the above-mentioned equation

$$9(5, 6, 7, 8) = 1234(5, 6, 7, 8).$$

To prove this I remark that we have identically

$$012 \cdot 034 \cdot 514 \cdot 523 - 014 \cdot 023 \cdot 512 \cdot 534 = 012345.$$

In fact the left-hand side equated to zero is the equation of the conic through 1, 2, 3, 4, 5, and such left-hand side must therefore, save to a mere numerical factor, be equal to 012345. And to determine this factor it is to be observed that 012345 contains the term

$$+x_0^2 \cdot y_1^2 \cdot z_2^2 \cdot y_3 z_3 \cdot z_4 x_4 \cdot x_5 y_5,$$

but that there is no such term in  $012 \cdot 034 \cdot 514 \cdot 523$ , and that there is in  $-014 \cdot 023 \cdot 512 \cdot 534$  the equivalent term

$$+x_0^2 \cdot y_1^2 \cdot z_2^2 \cdot y_3 z_3 \cdot z_4 x_4 \cdot x_5 y_5,$$

so that the numerical factor is rightly determined.

The foregoing identity written under the form

$$\frac{012 \cdot 034}{014 \cdot 023} - \frac{512 \cdot 534}{514 \cdot 523} = \frac{012345}{014 \cdot 023 \cdot 514 \cdot 523}$$

shows that, when  $012345 = 0$ , i.e. if 0 be a point of the conic through 1, 2, 3, 4, 5, then we have

$$0(1, 2, 3, 4) = 5(1, 2, 3, 4),$$

which is in fact the anharmonic property of the points of a conic. And observing that  $012345 = 051234$ , and substituting 5, 6, for 0, 5 respectively, the identity becomes

$$\frac{512 \cdot 534}{514 \cdot 523} - \frac{612 \cdot 634}{614 \cdot 623} = \frac{561234}{514 \cdot 523 \cdot 614 \cdot 623}.$$

The equation  $012345 = 0$  may be written

$$\frac{012 \cdot 034}{014 \cdot 023} - \frac{512 \cdot 534}{514 \cdot 523} = 0,$$

and hence the anharmonic ratio of the conics

$$012345 = 0, \quad 012346 = 0, \quad 012347 = 0, \quad 012348 = 0$$

is equal to that of the quantities

$$\frac{512 \cdot 534}{514 \cdot 523}, \quad \frac{612 \cdot 634}{614 \cdot 623}, \quad \frac{712 \cdot 734}{714 \cdot 723}, \quad \frac{812 \cdot 834}{814 \cdot 823},$$

or calling these quantities for a moment  $\alpha, \beta, \gamma, \delta$ , it is

$$\frac{(\alpha - \beta)(\gamma - \delta)}{(\alpha - \delta)(\beta - \gamma)},$$

where

$$\alpha - \beta = \frac{512 \cdot 534}{514 \cdot 523} - \frac{612 \cdot 634}{614 \cdot 623} = \frac{561234}{514 \cdot 523 \cdot 614 \cdot 623},$$

and forming in this manner the expressions of each of the four factors  $\alpha - \beta, \gamma - \delta, \alpha - \delta, \beta - \gamma$ , we have

$$\frac{(\alpha - \beta)(\gamma - \delta)}{(\alpha - \delta)(\beta - \gamma)} = \frac{561234 \cdot 781234}{581234 \cdot 671234},$$

so that in the equation

$$\frac{956 \cdot 978}{958 \cdot 967} = \frac{561234 \cdot 781234}{581234 \cdot 671234},$$

the right-hand side is

$$= 1234(5, 6, 7, 8),$$

and since by what precedes the left-hand side is  $= 9(5, 6, 7, 8)$ , the equation is

$$9(5, 6, 7, 8) = 1234(5, 6, 7, 8),$$

which is the transformation in question.

Now resuming the two equations

$$9(5, 6, 7, 8) = 1234(5, 6, 7, 8),$$

$$9(4, 6, 7, 8) = 1235(4, 6, 7, 8),$$

the right-hand sides are given anharmonic ratios, and as we have seen the question is to find 9 so that the anharmonic ratios  $9(5, 6, 7, 8)$ ,  $9(4, 6, 7, 8)$  shall have given values. But for the geometrical solution by the ruler alone, we have the preliminary question, from the given eight points, without the assistance of the before-mentioned conics, to construct the given anharmonic ratios  $1234(5, 6, 7, 8)$  and  $1235(4, 6, 7, 8)$ . The solution of both questions is given in Dr Hart's paper, "Construction by the ruler alone to determine the ninth point of intersection of two curves of the third degree," *Cambridge and Dublin Mathematical Journal*, t. vi., pp. 181, 182 (1851).



*The Preliminary Question.* The anharmonic ratio  $1234(5, 6, 7, 8)$  is equal to that of the polars of an arbitrary point  $X$  in regard to the conics  $12345$ ,  $12346$ ,  $12347$ ,  $12348$  respectively (these polars all pass through one and the same point). Now to construct the polars of  $X$  in regard to these conics, and first in regard to the conic  $12345$ : the fourth harmonics of  $X$  in regard to the lines  $12$ ,  $34$ , in regard to the lines  $13$ ,  $42$ , and in regard to the lines  $14$ ,  $23$ , meet in a point; and considering the several combinations  $1234$ ,  $1235$ ,  $1245$ ,  $1345$ ,  $2345$  we have thus five points; these lie on a line which is the required polar of  $X$  in regard to the conic  $12345$ . The polars in regard to the other conics are obtained in the same manner; and it is clear that the first above-mentioned point (viz. that deduced from the points  $1, 2, 3, 4$ ) is in fact the point of intersection of the four polars, or point of the pencil formed by the polars.

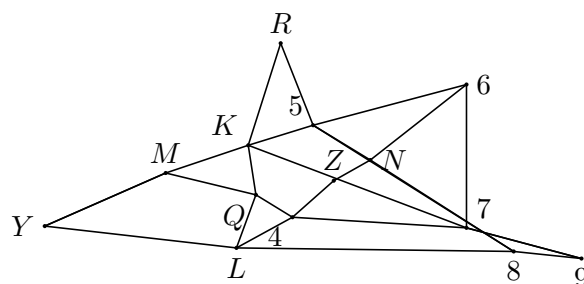
*The Principal Question* then is, given the points  $4, 5, 6, 7, 8$  to find the point  $9$ , such that

$$9(5, 6, 7, 8) = 1234(5, 6, 7, 8),$$

$$9(4, 6, 7, 8) = 1235(4, 6, 7, 8),$$

where the right-hand sides represent given anharmonic ratios.

For this, let  $65, 74$  (see the figure) meet in  $M$ , and on  $74$  find a point  $Q$  such



that the anharmonic ratio of the points  $(4, M, 7, Q)$  may be equal to the given ratio  $1235(4, 6, 7, 8)$ , say

$$(4, M, 7, Q) = 1235(4, 6, 7, 8),$$

and say let  $64, 85$  meet in  $N$  and on  $85$  find a point  $R$ , such that the anharmonic ratio of the points  $5, N, R, 8$  is equal to the given anharmonic ratio  $1234(5, 6, 7, 8)$ ,

$$(5, N, R, 8) = 1234(5, 6, 7, 8).$$

Join  $QR$  meeting  $65$  in  $K$  and  $64$  in  $L$ ; then  $7K$  and  $8L$  will meet in the required point  $9$ .

In fact taking  $Y$  as the intersection of the lines  $65KM$  and  $8L$ , and  $Z$  as the intersection of the lines  $64NL$  and  $7K$ ; then, as is clear from the figure, first, the

anharmonic ratio  $9(4, 6, 7, 8)$  is equal to that of the points  $(4, 6, Z, L)$  on the line  $46ZL$ , that is

$$9(4, 6, 7, 8) = (4, 6, Z, L),$$

which is

$$= (4, M, 7, Q),$$

since the lines  $6M, ZL, LQ$  meet in the point  $K$ ; but by the construction  $(4, M, 7, Q) = 1235(4, 6, 7, 8)$ , that is, we have

$$9(4, 6, 7, 8) = 1235(4, 6, 7, 8);$$

and, secondly, the anharmonic ratio  $9(5, 6, 7, 8)$  is equal to that of the points  $(5, 6, K, Y)$  on the line  $56KY$ , that is

$$9(5, 6, 7, 8) = (5, 6, K, Y),$$

which is

$$= (5, N, R, 8),$$

since the lines  $6N, KR, Y8$  meet in the same point  $L$ ; but by the construction  $(5, N, R, 8) = 1234(5, 6, 7, 8)$ , that is, we have

$$9(5, 6, 7, 8) = 1234(5, 6, 7, 8),$$

so that the point 9 satisfies the required conditions.

It has been already remarked that the point  $x$  in M. Chasles' theorem for the construction of the cubic through nine points is determined by precisely similar conditions to those which determine the ninth intersection of the two cubics; that is, the foregoing construction by the ruler alone is applicable to the determination of the point  $x$ ; and when this is once obtained, the remainder of the construction, giving the points of the cubic through the nine given points, can obviously be performed by the ruler alone. The construction for the cubic through nine points gives implicitly the relation between ten points of the cubic and such relation is accordingly expressed by the equation

$$x(1, 2, 3, 4, 5, 10) = 6789(1, 2, 3, 4, 5, 10),$$

which is one out of 210 similar forms. But it is possible that some more convenient form of the relation between the ten points may yet be found.

I proceed to further develop the analytical theory. Writing for convenience  $\omega$  in the place of 10, we have

$$x(1, 2, 3, 4, 5, 6) = 789\omega(1, 2, 3, 4, 5, 6),$$

or, what is the same thing,

$$x(1, 2, 3, 4) = 789\omega(1, 2, 3, 4),$$

$$x(1, 2, 3, 5) = 789\omega(1, 2, 3, 5),$$

$$x(1, 2, 3, 6) = 789\omega(1, 2, 3, 6),$$

502 ON THE CONSTRUCTION OF THE NINTH POINT OF INTERSECTION OF THE [295

which belong to three conics each of them passing through 1, 2, 3, and which must have a remaining fourth point of intersection.

The equation of the first conic is

$$012 \cdot 034 = \frac{\lambda}{\mu} \cdot 014 \cdot 023,$$

where

$$\lambda = 789\omega 12 \cdot 789\omega 34,$$

$$\mu = 789\omega 14 \cdot 789\omega 23,$$

and thence, in virtue of an identical equation already referred to,

$$-\lambda - \mu = 789\omega 13 \cdot 789\omega 24.$$

But we have identically

$$123 \cdot 0pq = 1pq \cdot 023 + 2pq \cdot 031 + 3pq \cdot 012,$$

and thence in particular

$$123 \cdot 034 = 134 \cdot 023 + 234 \cdot 031,$$

$$123 \cdot 014 = 214 \cdot 031 + 314 \cdot 012,$$

and the equation of the conic may therefore be written

$$012(134 \cdot 023 + 234 \cdot 031) \mu - 023(214 \cdot 031 + 314 \cdot 012) \lambda = 0,$$

that is

$$031 \cdot 012 \cdot 234 \cdot \mu + 012 \cdot 023 \cdot 314 (-\lambda - \mu) + 023 \cdot 031 \cdot 124 \cdot \lambda = 0,$$

or, substituting for  $\mu$ ,  $-\lambda - \mu$ ,  $\lambda$ , their values, this is

$$\begin{aligned} & 031 \cdot 012 \cdot 234 \cdot 789\omega 14 \cdot 789\omega 23 \\ & + 012 \cdot 023 \cdot 314 \cdot 789\omega 13 \cdot 789\omega 42 \\ & + 023 \cdot 031 \cdot 124 \cdot 789\omega 12 \cdot 789\omega 34 = 0, \end{aligned}$$

or, what is the same thing,

$$\frac{234 \cdot 789\omega 14 \cdot 789\omega 23}{023} + \frac{314 \cdot 789\omega 13 \cdot 789\omega 24}{031} + \frac{124 \cdot 789\omega 12 \cdot 789\omega 34}{012} = 0,$$

or, making a slight change of form, the equation of the conic is

$$\frac{423 \cdot 789\omega 41 \cdot 789\omega 23}{023} + \frac{431 \cdot 789\omega 42 \cdot 789\omega 31}{031} + \frac{412 \cdot 789\omega 43 \cdot 789\omega 12}{012} = 0.$$

The equations of the other two conics are deduced by writing successively 5 and 6 in the place of 4; and the condition in order that the conics may have a remaining fourth point of intersection is

$$\begin{vmatrix} 423 \cdot 789\omega 41 & 431 \cdot 789\omega 42 & 412 \cdot 789\omega 43 \\ 523 \cdot 789\omega 51 & 531 \cdot 789\omega 52 & 512 \cdot 789\omega 53 \\ 623 \cdot 789\omega 61 & 631 \cdot 789\omega 62 & 612 \cdot 789\omega 63 \end{vmatrix} = 0.$$

This equation, say  $D = 0$ , expresses the relation between the coordinates of the ten points 1, 2, 3, 4, 5, 6, 7, 8, 9,  $\omega$ , of the cubic. Hence if  $123456789\omega$  denote the determinant

$$\begin{vmatrix} x_1^3 & y_1^3 & z_1^3 & y_1^2 z_1 & z_1^2 x_1 & x_1^2 y_1 & y_1 z_1^2 & z_1 x_1^2 & x_1 y_1^2 & x_1 y_1 z_1 \\ x_2^3 & & & & & & & & & \\ & \&c. & & & & & & & \end{vmatrix},$$

$123456789\omega$  must be a factor of  $D$ , and a little consideration shows that the other factor which is of the order one as regards the coordinates of each of the points 1, 2, 3, and of the order three as regards the coordinates of each of the points 7, 8, 9,  $\omega$ , must be of the form  $123 \cdot 789 \cdot 78\omega \cdot 79\omega \cdot 89\omega$ . We must therefore have

$$D = \epsilon \cdot 123 \cdot 789 \cdot 78\omega \cdot 79\omega \cdot 89\omega \cdot 123456789\omega,$$

where the merely numerical factor  $\epsilon$  is, I believe, equal to +1 or else to -1.

In order to verify the factor  $123 \cdot 789 \cdot 78\omega \cdot 79\omega \cdot 89\omega$ , observing that the points 7, 8, 9,  $\omega$  enter symmetrically, it will be sufficient to show that 123, 789 are each of them factors of  $D$ , or, what is the same thing, that if  $123 = 0$ , or if  $789 = 0$ , then in either case  $D = 0$ .

First, if  $123 = 0$ , we may write

$$x_3 = \lambda x_1 + \mu x_2,$$

$$y_3 = \lambda y_1 + \mu y_2,$$

$$z_3 = \lambda z_1 + \mu z_2,$$

equations which give

$$423 = \lambda \cdot 421, \quad 431 = \lambda \cdot 421, \quad \&c.,$$

and the equation  $D = 0$ , thus becomes

$$\begin{vmatrix} 789\omega 41 & 789\omega 42 & 789\omega 43 \\ 789\omega 51 & 789\omega 52 & 789\omega 53 \\ 789\omega 61 & 789\omega 62 & 789\omega 63 \end{vmatrix} = 0,$$

or, what is the same thing,

$$\begin{vmatrix} 789\omega 14 & 789\omega 24 & 789\omega 34 \\ 789\omega 15 & 789\omega 25 & 789\omega 35 \\ 789\omega 16 & 789\omega 26 & 789\omega 36 \end{vmatrix} = 0.$$

Now if the terms in the same column are multiplied by  $789\omega 56$ ,  $789\omega 64$ ,  $789\omega 45$  respectively and added, then for the first column the sum is

$$789\omega 14 \cdot 789\omega 56 + 789\omega 15 \cdot 789\omega 64 + 789\omega 16 \cdot 789\omega 45,$$

which is  $= 0$ , and the sums for the second and third columns are each  $= 0$  in the same manner: wherefore the determinant vanishes as it should do.

Next, if  $789 = 0$ , we have identically

$$789\omega_{41} = 789 \cdot 741 \cdot \omega_{81} \cdot \omega_{94} + 781 \cdot 794 \cdot \omega_{89} \cdot \omega_{41},$$

which when  $789 = 0$  gives

$$789\omega_{41} = 781 \cdot 794 \cdot \omega_{89} \cdot \omega_{41},$$

and in like manner,

$$789\omega_{42} = 782 \cdot 794 \cdot \omega_{89} \cdot \omega_{42},$$

$$789\omega_{43} = 783 \cdot 794 \cdot \omega_{89} \cdot \omega_{43},$$

in which three equations 4 may be changed into 5 and 6 successively. The equation  $\square = 0$  thus becomes

$$\begin{vmatrix} 423 \cdot \omega_{41}, & 431 \cdot \omega_{42}, & 412 \cdot \omega_{43} \\ 523 \cdot \omega_{51}, & 531 \cdot \omega_{52}, & 512 \cdot \omega_{53} \\ 623 \cdot \omega_{61}, & 631 \cdot \omega_{62}, & 612 \cdot \omega_{63} \end{vmatrix} = 0,$$

or, what is the same thing,

$$\begin{vmatrix} 423 \cdot 41\omega, & 421 \cdot 4\omega_3, & 42\omega \cdot 431 \\ 523 \cdot 51\omega, & 521 \cdot 5\omega_3, & 52\omega \cdot 531 \\ 623 \cdot 61\omega, & 621 \cdot 6\omega_3, & 62\omega \cdot 631 \end{vmatrix} = 0,$$

and since the sum of the terms in each line of the determinant is  $= 0$ , the determinant is as it should be  $= 0$ .

The foregoing equation

$$\begin{vmatrix} 412 \cdot 789\omega_{43}, & 423 \cdot 789\omega_{41}, & 431 \cdot 789\omega_{42} \\ 512 \cdot 789\omega_{53}, & 523 \cdot 789\omega_{51}, & 531 \cdot 789\omega_{52} \\ 612 \cdot 789\omega_{63}, & 623 \cdot 789\omega_{61}, & 631 \cdot 789\omega_{62} \end{vmatrix} \\ = e \cdot 123 \cdot 789 \cdot 78\omega \cdot 79\omega \cdot 89\omega \cdot 123456789\omega,$$

(since  $412, 789\omega_{43}$ , &c. are interpretable functions of the coordinates) affords a geometrical interpretation of the equation

$$123456789\omega = 0$$

between the coordinates of the ten points of the cubic; but it would be more satisfactory if a similar identical equation could be found, having on the right-hand side the function  $123456789\omega$  without the irrelevant factor

$$123 \cdot 789 \cdot 78\omega \cdot 79\omega \cdot 89\omega.$$

*2, Stone Buildings, W.C., 6th March, 1862.*

296.

ON THE CONICS WHICH PASS THROUGH THE FOUR FOCI  
OF A GIVEN CONIC.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. v. (1862), pp. 275–280.]

The foci of a conic are the points of intersection of the tangents through the circular points at infinity; the pair of tangents through each of the circular points at infinity is a conic through the four foci; and we have thus two conics  $P = 0$ ,  $Q = 0$  passing through the four foci; the equation of any other conic through the four foci is of course  $P + \lambda Q = 0$ ; and in particular if  $\lambda$  be suitably determined this equation gives the axes of the conic.

I was led to develop the solution, in seeking to obtain the elegant formulae given in Mr P. J. Hensley's paper "Determination of the foci of the conic section expressed by trilinear coordinates," *Journal*, t. v., pp. 177–183, (March, 1862).

I take the coordinates to be proportional to the perpendicular distances of the point from the sides of the fundamental triangle, each distance divided by the perpendicular distance of the side from the opposite angle. This being so, the equation of the line infinity is

$$\frac{\alpha}{x} + \frac{\beta}{y} + \frac{\gamma}{z} = 0,$$

and,  $\alpha, \beta, \gamma$  denoting the sides of the fundamental triangle, the equation of the circle circumscribed about the triangle is

$$\frac{\alpha^2}{x} + \frac{\beta^2}{y} + \frac{\gamma^2}{z} = 0.$$

The foregoing two equations determine the circular points at infinity; and if  $(x_1, y_1, z_1)$  are the coordinates, there is no difficulty in obtaining the system of values

$$\begin{aligned} x_1 : y_1 : z_1 &= -\alpha : \beta(\cos C + i \sin C) : \gamma(\cos B + i \sin B) \\ &= \alpha(\cos C - i \sin C) : -\beta : \gamma(\cos A + i \sin A) \\ &= \alpha(\cos B + i \sin B) : \beta(\cos A - i \sin A) : -\gamma, \end{aligned}$$

where as usual  $i = \sqrt{(-1)}$ , and where  $A, B, C$  denote the angles of the triangle, so that the cosines and sines of these angles denote given functions of  $\alpha, \beta, \gamma$ . The coordinates  $(x_2, y_2, z_2)$  of the other circular point at infinity are of course obtained by merely writing  $-i$  for  $i$ . We find also

$$\begin{aligned} x_1x_2 : y_1y_2 : z_1z_2 : y_1z_2 + y_2z_1 : z_1x_2 + z_2x_1 : x_1y_2 + x_2y_1 \\ = -\alpha^2 : -\beta^2 : -\gamma^2 : \beta^2 + \gamma^2 - \alpha^2 : \gamma^2 + \alpha^2 - \beta^2 : \alpha^2 + \beta^2 - \gamma^2, \end{aligned}$$

which are the formulae chiefly made use of in the sequel.

Suppose now that the equation of the conic is

$$U = (a, b, c, f, g, h \mid x, y, z)^2 = 0;$$

then putting for a moment

$$\begin{aligned} U_1 &= (a, b, c, f, g, h \mid x_1, y_1, z_1)^2, \\ V_1 &= (a, b, c, f, g, h \mid x, y, z \mid x_1, y_1, z_1), \end{aligned}$$

and the like as regards  $U_2$  and  $W_2$ ; the tangents from the points  $(x_1, y_1, z_1)$  and  $(x_2, y_2, z_2)$  respectively are

$$U U_1 - V_1^2 = 0,$$

$$U U_2 - W_2^2 = 0,$$

which are respectively pairs of lines intersecting in the four foci. And it is moreover clear that the equation of the axes is

$$U_1 W_2^2 - U_2 V_1^2 = 0.$$

The foregoing equations may be written

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid yz_1 - y_1z, zx_1 - z_1x, xy_1 - x_1y)^2 = 0,$$

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid yz_2 - y_2z, zx_2 - z_2x, xy_2 - x_2y)^2 = 0,$$

where  $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$  are the inverse system of coefficients.

These may be written

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h} \mid x_1, y_1, z_1)^2 = 0,$$

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h} \mid x_2, y_2, z_2)^2 = 0,$$

where  $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}$  are quadric functions of  $x, y, z$ , viz.

$$\begin{aligned} \mathfrak{a} &= \mathfrak{B}z^2 + \mathfrak{C}y^2 - 2\mathfrak{F}yz, \\ \mathfrak{b} &= \mathfrak{C}x^2 + \mathfrak{A}z^2 - 2\mathfrak{G}zx, \\ \mathfrak{c} &= \mathfrak{A}y^2 + \mathfrak{B}x^2 - 2\mathfrak{H}xy, \\ \mathfrak{f} &= -\mathfrak{A}yz - \mathfrak{F}x^2 + \mathfrak{G}xy + \mathfrak{H}xz, \\ \mathfrak{g} &= -\mathfrak{B}zx - \mathfrak{G}y^2 + \mathfrak{H}yz + \mathfrak{F}yx, \\ \mathfrak{h} &= -\mathfrak{C}xy - \mathfrak{H}z^2 + \mathfrak{F}zx + \mathfrak{G}zy, \end{aligned}$$

and this being so, I combine the two equations as follows:

$$\begin{aligned} x_2^2(\mathfrak{a}, \dots | x_1, y_1, z_1)^2 + x_1^2(\mathfrak{a}, \dots | x_2, y_2, z_2)^2 &= 0, \\ y_2^2 &+ y_1^2 = 0, \\ z_2^2 &+ z_1^2 = 0, \\ y_2 z_2(\mathfrak{a}, \dots | x_1, y_1, z_1)^2 + y_1 z_1(\mathfrak{a}, \dots | x_2, y_2, z_2)^2 &= 0, \\ z_2 x_2 &+ z_1 x_1 = 0, \\ x_2 y_2 &+ x_1 y_1 = 0, \end{aligned}$$

any one of which is the equation of a conic passing through the four foci; the current coordinates being always  $(x, y, z)$ .

The first of these equations is

$$\begin{aligned} &\mathfrak{a}(-2x_1^2 x_2^2) + \mathfrak{b}(x_1^2 y_2^2 + x_2^2 y_1^2) + \mathfrak{c}(x_1^2 z_2^2 + x_2^2 z_1^2) \\ &+ 2\mathfrak{f}(x_1^2 y_2 z_2 + x_2^2 y_1 z_1) + 2\mathfrak{g}(x_1^2 z_2 x_1 + x_2^2 z_1 x_2) + 2\mathfrak{h}(x_2^2 x_1 y_1 + x_1^2 x_2 y_2) = 0, \end{aligned}$$

where the quantities multiplied by  $\mathfrak{a}$ ,  $\mathfrak{b}$ , &c. are all of them easily expressible in terms of  $x_1 x_2$ ,  $y_1 y_2$ ,  $z_1 z_2$ ,  $y_1 z_2 + y_2 z_1$ ,  $z_1 x_2 + z_2 x_1$ ,  $x_1 y_2 + x_2 y_1$ , which are respectively proportional to given functions of  $(\alpha, \beta, \gamma)$ ; and replacing for  $\mathfrak{a}$ ,  $\mathfrak{b}$ , &c. their values, the equation is

$$\begin{aligned} &(\mathfrak{B}z^2 + \mathfrak{C}y^2 - 2\mathfrak{F}yz) \cdot 2\alpha^4 \\ &+ (\mathfrak{C}x^2 + \mathfrak{A}z^2 - 2\mathfrak{G}zx) \cdot \{(\alpha^2 + \beta^2 - \gamma^2)^2 - 2\alpha^2 \beta^2\} \\ &+ (\mathfrak{A}y^2 + \mathfrak{B}x^2 - 2\mathfrak{H}xy) \cdot \{(\gamma^2 + \alpha^2 - \beta^2)^2 - 2\gamma^2 \alpha^2\} \\ &+ 2(-\mathfrak{A}yz - \mathfrak{F}x^2 + \mathfrak{G}xy + \mathfrak{H}xz) \cdot \alpha^2(\beta^2 + \gamma^2 - \alpha^2) \\ &+ 2(-\mathfrak{B}zx - \mathfrak{G}y^2 + \mathfrak{H}yz + \mathfrak{F}yz) \cdot -\alpha^2(\gamma^2 + \alpha^2 - \beta^2) \\ &+ 2(-\mathfrak{C}xy - \mathfrak{H}z^2 + \mathfrak{F}zx + \mathfrak{G}zy) \cdot -\alpha^2(\alpha^2 + \beta^2 - \gamma^2) = 0. \end{aligned}$$

Now putting for shortness

$$\square = \alpha^4 + \beta^4 + \gamma^4 - 2\beta^2 \gamma^2 - 2\gamma^2 \alpha^2 - 2\alpha^2 \beta^2,$$

so that  $-\square$  is equal to sixteen times the square of the area of the fundamental triangle, the coefficient of  $x^2$  is

$$= \mathfrak{B}(\square + 2\alpha^2 \gamma^2) + \mathfrak{C}(\square + 2\alpha^2 \beta^2) - 2\mathfrak{H}\{-\square - \alpha^2(\beta^2 + \gamma^2 - \alpha^2)\},$$

which is  $= (\mathfrak{B} + \mathfrak{C} + 2\mathfrak{H})\square + 2\alpha^2\{\mathfrak{B}\gamma^2 + \mathfrak{C}\beta^2 + \mathfrak{H}(\beta^2 + \gamma^2 - \alpha^2)\}$ .

And reducing in a similar manner the other coefficients, the equation is

$$\square\{(\mathfrak{B} + \mathfrak{C} + 2\mathfrak{H})x^2 + \mathfrak{A}(y + z)^2 - 2(\mathfrak{F} + \mathfrak{G})x(y + z)\} + 2\alpha^2 \Theta = 0,$$

where for shortness

$$\begin{aligned} \Theta &= x^2 \cdot \{\mathfrak{B}\gamma^2 + \mathfrak{C}\beta^2 + \mathfrak{H}(\beta^2 + \gamma^2 - \alpha^2)\} \\ &+ y^2 \cdot \{\mathfrak{C}\alpha^2 + \mathfrak{A}\gamma^2 + \mathfrak{G}(\gamma^2 + \alpha^2 - \beta^2)\} \\ &+ z^2 \cdot \{\mathfrak{A}\beta^2 + \mathfrak{B}\alpha^2 + \mathfrak{F}(\alpha^2 + \beta^2 - \gamma^2)\} \\ &+ yz \cdot \{-2\mathfrak{F}\alpha^2 + \mathfrak{A}(\beta^2 + \gamma^2 - \alpha^2) - \mathfrak{G}(\gamma^2 + \alpha^2 - \beta^2) - \mathfrak{H}(\alpha^2 + \beta^2 - \gamma^2)\} \\ &+ zx \cdot \{-2\mathfrak{G}\beta^2 - \mathfrak{F}(\beta^2 + \gamma^2 - \alpha^2) + \mathfrak{B}(\gamma^2 + \alpha^2 - \beta^2) - \mathfrak{H}(\alpha^2 + \beta^2 - \gamma^2)\} \\ &+ xy \cdot \{-2\mathfrak{H}\gamma^2 - \mathfrak{F}(\beta^2 + \gamma^2 - \alpha^2) - \mathfrak{G}(\gamma^2 + \alpha^2 - \beta^2) + \mathfrak{C}(\alpha^2 + \beta^2 - \gamma^2)\}, \end{aligned}$$



or, what is the same thing, the equation is

$$\square\{(\mathfrak{A} + \mathfrak{B} + \mathfrak{C} + 2\mathfrak{F} + 2\mathfrak{G} + 2\mathfrak{H})x^2 - 2(\mathfrak{A} + \mathfrak{F} + \mathfrak{G})x(x + y + z) + \mathfrak{A}(x + y + z)^2\} + 2\alpha^2\Theta = 0,$$

or putting for shortness

$$\begin{aligned}\mathfrak{A} + \mathfrak{B} + \mathfrak{C} + 2\mathfrak{F} + 2\mathfrak{G} + 2\mathfrak{H} &= \mathfrak{K}, \\ \mathfrak{A} + \mathfrak{F} + \mathfrak{G} &= \mathfrak{L}, \\ \mathfrak{F} + \mathfrak{B} + \mathfrak{H} &= \mathfrak{M}, \\ \mathfrak{G} + \mathfrak{H} + \mathfrak{C} &= \mathfrak{N},\end{aligned}$$

so that in fact

$$\mathfrak{L} + \mathfrak{M} + \mathfrak{N} = \mathfrak{K},$$

the equation is

$$\square\{\mathfrak{K}x^2 - 2\mathfrak{L}x(x + y + z) + \mathfrak{A}(x + y + z)^2\} + 2\alpha^2\Theta = 0.$$

The equation with  $y_2z_2$ ,  $y_1z_1$  is in a similar manner found to be

$$\mathfrak{a}\{\mathfrak{K}x^2 - [(\mathfrak{G} + \mathfrak{C})y^2 - (\mathfrak{H} + \mathfrak{B})z^2 + (\mathfrak{A} + 2\mathfrak{F} + \mathfrak{G} + \mathfrak{H})yz + (-\mathfrak{H} - \mathfrak{F} + \mathfrak{G})zx + (-\mathfrak{G} - \mathfrak{F} + \mathfrak{H})xy]\}$$

or, what is the same thing,

$$\square[(\mathfrak{A} + \mathfrak{B} + \mathfrak{C} + 2\mathfrak{F} + 2\mathfrak{G} + 2\mathfrak{H})yz - \{(\mathfrak{C} + \mathfrak{G} + \mathfrak{H})y + (\mathfrak{F} + \mathfrak{B} + \mathfrak{H})z\}(x + y + z) + \mathfrak{F}(x + y + z)^2] - (\beta^2 + \gamma^2 - \alpha^2)\Theta = 0,$$

or, finally,

$$\square\{\mathfrak{K}yz - (\mathfrak{N}y + \mathfrak{M}z)(x + y + z) + \mathfrak{F}(x + y + z)^2\} - (\beta^2 + \gamma^2 - \alpha^2)\Theta = 0.$$

Hence the entire system of equations is

$$\begin{aligned}\square\{\mathfrak{K}x^2 - 2\mathfrak{L}x(x + y + z) + \mathfrak{A}(x + y + z)^2\} + 2\alpha^2\Theta &= 0, \\ \square\{\mathfrak{K}y^2 - 2\mathfrak{M}y(x + y + z) + \mathfrak{B}(x + y + z)^2\} + 2\beta^2\Theta &= 0, \\ \square\{\mathfrak{K}z^2 - 2\mathfrak{N}z(x + y + z) + \mathfrak{C}(x + y + z)^2\} + 2\gamma^2\Theta &= 0, \\ \square\{\mathfrak{K}yz - (\mathfrak{N}y + \mathfrak{M}z)(x + y + z) + \mathfrak{F}(x + y + z)^2\} - (\beta^2 + \gamma^2 - \alpha^2)\Theta &= 0, \\ \square\{\mathfrak{K}zx - (\mathfrak{L}z + \mathfrak{N}x)(x + y + z) + \mathfrak{G}(x + y + z)^2\} - (\gamma^2 + \alpha^2 - \beta^2)\Theta &= 0, \\ \square\{\mathfrak{K}xy - (\mathfrak{M}x + \mathfrak{L}y)(x + y + z) + \mathfrak{H}(x + y + z)^2\} - (\alpha^2 + \beta^2 - \gamma^2)\Theta &= 0,\end{aligned}$$

which are the equations of six conics, each of them passing through the four foci.

From the first three of these, we have

$$\begin{aligned}&\frac{1}{\alpha^2}\{\mathfrak{K}x^2 - 2\mathfrak{L}x(x + y + z) + \mathfrak{A}(x + y + z)^2\} \\ &= \frac{1}{\beta^2}\{\mathfrak{K}y^2 - 2\mathfrak{M}y(x + y + z) + \mathfrak{B}(x + y + z)^2\} \\ &= \frac{1}{\gamma^2}\{\mathfrak{K}z^2 - 2\mathfrak{N}z(x + y + z) + \mathfrak{C}(x + y + z)^2\},\end{aligned}$$

which, allowing for the difference of notation, are Mr Hensley's equations: it appears by his investigation that their geometrical signification is as follows; viz. if for shortness we denote the equations by

$$A = B = C,$$

then if we consider the tangents parallel to the  $x$ -side of the fundamental triangle, and the tangents parallel to the  $y$ -side of the fundamental triangle, the equation  $A = B$  is the locus of a point such that the feet of the perpendiculars let fall from it on the four tangents lie in a circle. And similarly for the equations  $A = C$ ,  $B = C$ .

If we multiply the six equations by 1, 1, 1, 2, 2, 2 and add, we obtain the identical equation  $0 = 0$ ; if we multiply them by  $a, b, c, 2f, 2g, 2h$  and add, then after some easy reductions, we obtain for the equation of a new conic passing through the four foci

$$\square\{\Re U + K(x + y + z)^2\} + 2S\Theta = 0,$$

where

$$U = (a, b, c, f, g, h \mid x, y, z)^2,$$

$K$  is the discriminant  $abc - af^2 - bg^2 - ch^2 + 2fgh$ , and

$$S = (a\alpha^2 + b\beta^2 + c\gamma^2 - f(\beta^2 + \gamma^2 - \alpha^2) - g(\gamma^2 + \alpha^2 - \beta^2) - h(\alpha^2 + \beta^2 - \gamma^2),$$

or, what is the same thing,

$$S = (f + a - h - g)\alpha^2 + (g - h + b - f)\beta^2 + (h - g - f + c)\gamma^2.$$

It would be interesting to ascertain the geometrical signification of the six conics and of the last-mentioned new conic.

*2, Stone Buildings, W.C., March 13th, 1862.*

297.

ON SOME FORMULAE RELATING TO THE DISTANCES OF A  
POINT FROM THE VERTICES OF A TRIANGLE, AND TO THE  
PROBLEM OF TACTIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. v. (1862), pp. 381–384.]

The relation between the distances of four points 1, 2, 3, 4 in a plane is

$$\begin{vmatrix} 0 & 12^2 & 13^2 & 14^2 & 1 \\ 21^2 & 0 & 23^2 & 24^2 & 1 \\ 31^2 & 32^2 & 0 & 34^2 & 1 \\ 41^2 & 42^2 & 43^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{vmatrix} = 0,$$

where, see my paper “Note on the value of certain Determinants the terms of which are the squared distances of Points in a plane or in space,” *Quarterly Journal of Mathematics*, t. iii., p. 275 (1859), [286], the determinant is

$$= \Sigma 12^2 \cdot 23^2 \cdot 34^2 - \Sigma 12^2 \cdot 34^2 \cdot 43^2 - \Sigma 12^2 \cdot 23^2 \cdot 31^2,$$

an identity which subsists without the aid of the relations  $12 = 21$ , &c., and in which the  $\Sigma$ ,  $\Sigma$ ,  $\Sigma$  contain 24, 12, and 8 terms respectively.

Writing  $23 = f$ ,  $31 = g$ ,  $12 = h$ ,  $14 = a$ ,  $24 = b$ ,  $34 = c$ , the determinant is

$$\begin{aligned} &= 2 \{ g^2 h^2 (b^2 + c^2) + h^2 f^2 (c^2 + a^2) + f^2 g^2 (a^2 + b^2) \\ &\quad + b^2 c^2 (g^2 + h^2) + c^2 a^2 (h^2 + f^2) + a^2 b^2 (f^2 + g^2) \\ &\quad - a^2 f^2 (a^2 + f^2) - b^2 g^2 (b^2 + g^2) - c^2 h^2 (c^2 + h^2) \\ &\quad - b^2 c^2 f^2 - c^2 a^2 g^2 - a^2 b^2 h^2 - f^2 g^2 h^2 \} \\ &= -2D, \end{aligned}$$

if  $\square$  denote the function in  $\{ \}$  with the signs reversed. The function  $\square$  may be expressed in the form

$$\begin{aligned}\square &= a^4 f^2 + b^4 g^2 + c^4 h^2 + f^2 g^2 h^2 \\ &\quad + (a^2 f^2 + b^2 c^2)(f^2 - g^2 - h^2) \\ &\quad + (b^2 g^2 + c^2 a^2)(g^2 - h^2 - f^2) \\ &\quad + (c^2 h^2 + a^2 b^2)(h^2 - f^2 - g^2),\end{aligned}$$

and also in the form

$$\square = U^2 + (f + g + h)V,$$

if for shortness

$$\begin{aligned}U &= a^2 f + b^2 g + c^2 h + fgh, \\ V &= (a^2 f^2 + b^2 c^2)(f - g - h) \\ &\quad + (b^2 g^2 + c^2 a^2)(g - h - f) \\ &\quad + (c^2 h^2 + a^2 b^2)(h - f - g);\end{aligned}$$

and it may be remarked that since  $\square$  is an even function of  $f, g, h$ , we may in this last formula change at pleasure the signs of these quantities; we thus obtain in all four similar forms of the function  $\square$ .

It is clear that considering a triangle, and any point in the plane of the triangle,  $f, g, h$  may be taken to denote the sides of the triangle, and  $a, b, c$  the distances of the point from the vertices: and the equation  $\square = 0$  is the relation connecting the sides and distances.

The equation  $f + g + h = 0$  denotes that the vertices are *in lined*, and when this equation is satisfied we have

$$U = a^2 f + b^2 g + c^2 h + fgh = 0,$$

which is in fact, as it is easy to see, the relation connecting the distances of a point from any three points *in lined*.

For  $a, b, c$  write  $a + x, b + x, c + x$ ;  $x$  will be the radius of a circle touching the circles, radii  $a, b, c$ , described about the vertices as centres. The equation  $\square = 0$  becomes after all reductions

$$\begin{aligned}&U^2 - (f + g + h)V \\ &\quad + x[4U(af + bg + ch) \\ &\quad - 2(f + g + h)\{(af^2 + bc(b + c))(f - g - h) \\ &\quad + (bg^2 + ca(c + a))(g - h - f) \\ &\quad + (ch^2 + ab(a + b))(h - f - g)\}] \\ &\quad + x^2[f^2\{-4a^2 + 6a(b + c) - 6bc\} \\ &\quad + g^2\{-4b^2 + 6b(c + a) - 6ca\} \\ &\quad + h^2\{-4c^2 + 6c(a + b) - 6ab\}] = 0,\end{aligned}$$

which is a quadratic equation only: the two circles thus obtained are those which touch the given circles all three externally or all three internally. But by changing in every possible manner the signs of  $a, b, c$  we obtain in all four equations giving the eight tangent circles. It may be noticed that if as before  $f + g + h = 0$ ,  $U = 0$ ,

then not only the constant term vanishes, but the coefficient of  $x$  also vanishes or the equation becomes simply  $x^2 = 0$ .

In particular, suppose  $f = b + c$ ,  $g = c + a$ ,  $h = a + b$ ; developing this *de novo*, and putting for shortness

$$\begin{aligned} a + b + c &= p, \\ bc + ca + ab &= q, \\ abc &= r, \end{aligned}$$

we find

$$\begin{aligned} U &= 2\{px^2 + 2qx + pq - 2r\}, \\ V &= 2\{px^4 + 4qx^3 + (2pq + 12r)x^2 + 4q^2x + pq^2 - 4qr\}, \end{aligned}$$

and then the equation  $\square = U^2 - 2pV = 0$  gives

$$\begin{aligned} \frac{1}{4}\square &= (px^2 + 2qx + pq - 2r)^2 - p\{px^4 + 4qx^3 + (2pq + 12r)x^2 + 4q^2x + pq^2 - 4qr\} \\ &= 4\{(q^2 - 4pr)x^2 - 2qrx + r^2\}, \end{aligned}$$

so that we have

$$\frac{1}{16}\square = (q^2 - 4pr)x^2 - 2qrx + r^2 = (qx - r)^2 - 4prx^2 = 0,$$

and thence

$$qx - r = \pm x\sqrt{pr}, \quad \text{or} \quad x = \frac{r}{q \mp \sqrt{pr}},$$

which gives the radii of the circles inscribed in and circumscribed about the three circles radii  $a$ ,  $b$ ,  $c$ , whereof each touches the two others: a formula given by Descartes, *Epistolae* (Ed. 2, Franc. 1792), Pars iii., p. 261, in a letter to the Princess Elizabeth, viz. Descartes has

$$(d^2e^2 + d^2f^2 + e^2f^2 - 2def^2 - 2d^2ef - 2de^2f)x^2 - 2(de^2f^2 + d^2ef^2 + d^2e^2f)x + d^2e^2f^2 = 0,$$

which putting  $a$ ,  $b$ ,  $c$  for his  $d$ ,  $e$ ,  $f$ , becomes *ut supra*

$$x^2(q^2 - 4pr) - 2qrx + r^2 = 0.$$

In conclusion I notice the following formula which is obtained without difficulty, viz. if as before we have a triangle the sides whereof are  $f$ ,  $g$ ,  $h$ , and if  $a$ ,  $b$ ,  $c$  are the distances of a point from the vertices (so that as before  $\square = 0$ ) then the perpendicular distances of the point from the sides, each perpendicular distance divided by the perpendicular distance of the opposite vertex from the same side, are as follows: viz. the quotient for the side  $f$  is

$$= \frac{1}{16\Delta^2} [(b^2 - c^2)(g^2 - h^2) + f^2(b^2 + c^2 + g^2 + h^2 - 2a^2) - f^4],$$

where  $\Delta$  is the area of the triangle. It is clear that we ought to have

$$\sum_{\text{cyc}} \{(b^2 - c^2)(g^2 - h^2) + f^2(b^2 + c^2 + g^2 + h^2 - 2a^2) - f^4\} = 16\Delta^2,$$

and this equation in fact reduces itself to

$$2g^2h^2 + 2h^2f^2 + 2f^2g^2 - f^4 - g^4 - h^4 = 16\Delta^2,$$

which is right.

2, *Stone Buildings*, W.C., 11th September, 1862.

## 298.

**REPORT ON THE PROGRESS OF THE SOLUTION OF CERTAIN  
SPECIAL PROBLEMS OF DYNAMICS.**

[From the *Report of the British Association for the Advancement of Science*, 1862, pp. 184–252.]

My “Report on the Recent Progress of Theoretical Dynamics” was published in the Report of the British Association for the year 1857, [195]. The present Report (which is in some measure supplemental thereto) relates to the Special Problems of Dynamics: to give a general idea of the contents, I will at once mention the heads under which these problems are considered; viz., relating to the motion of a particle or system of particles, we have

- Rectilinear Motion;
- Central Forces, and in particular
- Elliptic Motion;
- The Problem of two Centres;
- The Spherical Pendulum;
- Motion as affected by the Rotation of the Earth, and Relative Motion in general;
- Miscellaneous Problems;
- The Problem of three bodies.

And relating to the motion of a solid body, we have

- The Transformation of Coordinates;
- Principal Axes, and Moments of Inertia;
- Rotation of a Solid Body;
- Kinematics of a Solid Body;
- Miscellaneous Problems.

As regards the first division of the subject, I remark that the lunar and planetary theories, as usually treated, do not (properly speaking) relate to the problem of three bodies, but to that of disturbed elliptic motion—a problem which is not considered in the present Report. The problem of the spherical pendulum is that of a particle moving on a spherical surface; but, with this exception, I do not much consider the motion of a particle on a given curve or surface, nor the motion in a resisting medium; what is said on these subjects is included under the head Miscellaneous Problems. The first six heads relate exclusively, and the head Miscellaneous Problems relates principally, to the motion of a single particle. As regards the second division of the subject, I will only remark that, from its intimate connexion with the theory of the motion of a solid body, I have been induced to make a separate head of the geometrical subject, “Transformation of Coordinates,” and to treat of it in considerable detail.

I have inserted at the end of the present Report a list of the memoirs and works referred to, arranged (not, as in the former Report, in chronological order, but) alphabetically according to the authors’ names: those referred to in the former Report formed for the purpose thereof a single series, which is not here the case. The dates specified are for the most part those on the title-page of the volume, being intended to show approximately the date of the researches to which they refer, but in some instances a more particular specification is made.

I take the opportunity of noticing a serious omission in my former Report, viz., I have not made mention of the elaborate memoir, Ostrogradsky “Mémoire sur les équations différentielles relatives au problème des Isopérimètres,” *Mém. de St Pét.*, t. iv. (6 sér.) pp. 385–517, 1850, which among other researches contains, and that in the most general form, the transformation of the equations of motion from the Lagrangian to the Hamiltonian form, and indeed the transformation of the general isoperimetric system (that is, the system arising from any problem in the calculus of variations) to the Hamiltonian form. I remark also, as regards the memoir of Cauchy referred to in the note p. 12 as an unpublished memoir of 1831, there is an “Extrait du Mémoire présenté à l’Académie de Turin le 11 Oct. 1831,” published in lithograph under the date Turin, 1832, with an addition dated 6 Mar. 1833. The Extract begins thus:—“§ I. Variation des Constantes Arbitraires. Soient données entre la variable  $t$ , ...  $n$  fonctions de  $t$  désignées par  $x, y, z$  ... et  $n$  autres fonctions de  $t$  désignées par  $u, v, w$ , ...  $2n$  équations différentielles du premier ordre et de la forme

$$\begin{aligned} \frac{dx}{dt} &= \frac{d\Omega}{du}, & \frac{dy}{dt} &= \frac{d\Omega}{dv}, & \frac{dz}{dt} &= \frac{d\Omega}{dw}, \dots \\ \frac{du}{dt} &= -\frac{d\Omega}{dx}, & \frac{dv}{dt} &= -\frac{d\Omega}{dy}, & \frac{dw}{dt} &= -\frac{d\Omega}{dz}, \dots \text{''} \end{aligned}$$

without any explanation as to the origin of these equations; and the formulae are then given for the variations of the constants in  $\frac{dt}{dz}$ , the integrals of the foregoing system; this seems sufficient to establish that Cauchy in the year 1831 was familiar with the Hamiltonian form of the equations of motion.

Bour's "Mémoire sur l'intégration des équations différentielles de la Mécanique," as published, Mém. prés. à l'Inst., t. xiv. pp. 792–821, is substantially the same as the extract thereof in Liouville's *Journal*, referred to in my former Report; but since the date of that Report there have been published in the *Comptes Rendus*, 1861 and 1862, several short papers by the same author; also Jacobi's great memoir, see list, Jacobi, *Nova Methodus* &c., 1862, edited after his decease by Clebsch; some valuable memoirs by Natani and Clebsch (*Crelle*, 1861 and 1862) relating to the Pfaffian system of equations (which includes those of Dynamics), and Boole "On Simultaneous Differential Equations of the First Order, in which the number of the Variables exceeds by more than one the number of the Equations," *Phil. Trans.*, t. CLII. (1862), pp. 437–454.

*Rectilinear Motion. Article Nos. 1 to 5.*

1. The determination of the motion of a falling body, which is the case of a constant force, is due to Galileo.

2. A variable force, assumed to be a force depending only on the position of the particle, may be considered as a function of the distance from any point in the line, selected at pleasure as a centre of force; but if, as usual, the force is given as a function of the distance from a certain point, it is natural to take that point for the centre of force. The problem thus becomes a particular case of that of central forces; and it is so treated in the *Principia*, Book I. § 7; the method has the advantage of explaining the paradoxical result which presents itself in the case  $\text{Force} \propto (\text{Dist.})^{-2}$ , and in some other cases where the force becomes infinite. According to theory, the velocity becomes infinite at the centre, but the direction of the motion is there abruptly reversed; so that the body in its motion does not pass through the centre, but on arriving there, forthwith returns towards its original position; of course such a motion cannot occur in nature, where neither a force nor a velocity ever is actually infinite.

3. Analytically the problem may be treated separately by means of the equation

$$\frac{d^2x}{dt^2} = X,$$

which is at once integrable in the form  $\left(\frac{dx}{dt}\right)^2 = C + 2 \int X dx$ .

4. The following cases may be mentioned:

Force  $\propto$  Dist. The law of motion is well known, being in fact the same as for the cycloidal pendulum.

Force  $\propto (\text{Dist.})^{-2}$ ,  $= \frac{\mu}{x^2}$ , which is the case above alluded to.

Assuming that the body falls from rest at a distance  $a$ , we have

$$x = a(1 - \cos \phi),$$

where, if  $n = \frac{\sqrt{\mu}}{a^{\frac{3}{2}}}$ ,  $\phi$  is given in terms of the time by means of the equation

$$nt = \phi - \sin \phi.$$



If the body had initially a small transverse velocity, the motion would be in a very eccentric ellipse, and the formulae are in fact the limiting form of those for elliptic motion.

5. There are various laws of force for which the motion may be determined. In particular it can be determined by means of Elliptic Integrals, in the case of a body attracted to two centres, force  $\propto (\text{dist.})^{-2}$ : see Legendre, *Exercices de Cal. Intég.*, t. ii. pp. 502–512, and *Théorie des Fonct. Ellip.*, t. i. pp. 531–538.

*Central Forces, Article Nos. 6 to 26.*

6. The theory of the motion of a body under the action of a given central force was first established in the *Principia*, Book I. §§ 2 and 3: viz. Prop. I. the areas are proportional to the times, that is (using the ordinary analytical notation),  $r^2 d\theta = h dt$ , and Prop. VI. Cor. 3,  $P \propto \frac{1}{SP^2 \cdot PV}$ ,  $= h^2 u^2 \left( \frac{d^2 u}{d\theta^2} + u \right)$ , so that

$$\frac{d^2 u}{d\theta^2} + u = \frac{P}{h^2 u^2}.$$

7. It is to be noticed that, given the orbit, the law of force is at once determined; and § 2 contains several instances of such determination; thus,

Prop. VII. If a body revolve in a circle, the law of force to a point  $S$  is force  $\propto \frac{1}{SP^2 \cdot PV^3}$  ( $P$  the body,  $PV$  the chord through  $S$ ).

Prop. IX. If a body move in a logarithmic spiral, force  $\propto (\text{dist.})^{-3}$ .

Prop. X. If a body move in an ellipse, force to centre  $\propto \text{dist.}$ , and as a particular case, if the body move in a parabola under the action of a force parallel to the axis, the force is constant. The particular case of motion in a parabola had been obtained by Galileo.

And § 3, Props. XI. XII. XIII. If a body move in an ellipse, hyperbola, or parabola under the action of a force tending to the focus, force  $\propto (\text{dist.})^{-2}$ .

8. But Newton had no direct method of solving the inverse problem (which depends on the solution of the differential equation), “Given the force to find the orbit.” Thus force  $\propto (\text{dist.})^{-2}$ , after it has been shown that an ellipse, a hyperbola, and a parabola may each of them be described under the action of such a force, the remainder of the solution consists in showing that, given the initial circumstances of the motion, a conic section (ellipse, parabola, or hyperbola, as the case may be) can be constructed, passing through the point of projection, having its tangent in the direction of the initial motion, and such that the velocity of the body describing the conic section under the action of the given central force is equal to the velocity of projection; which being so, the orbit will be the conic section so constructed. This is what is done, Prop. XVII.; it may be observed that the latus rectum is constructed not very elegantly by means of the latus rectum of an auxiliary orbit.

9. A more elegant construction was obtained by Cotes (see the *Harmonia Mensurarum*, pp. 103–105, and demonstration from the author's papers in the Notes by R. Smith, pp. 124, 125), depending on the position of a point  $C$ , such that the velocity acquired in falling under the action of the central force from  $C$  directly or through infinity<sup>1</sup> to  $P$  the point of projection, is equal to the given velocity of projection.

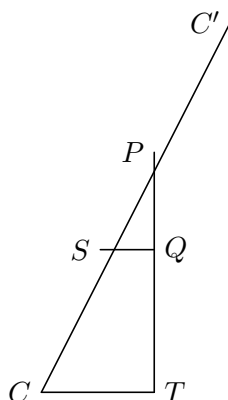
10. But Newton's original construction is now usually replaced by a construction which employs the space due to the velocity of projection considered as produced by a constant force equal to the central force at the point of projection.

11. § 9 of Book I. relates to revolving orbits, viz., it is shown that a body may be made to move in an orbit revolving round the centre of force, by adding to the central force required to make the body move in the same orbit at rest, a force  $\propto (\text{dist.})^{-3}$ . This appears very readily by means of the differential equation (*ante*, No. 6), viz. writing therein  $P + cu^3$  for  $P$ , and then  $\theta'$ ,  $h'$  in the place of  $\theta\sqrt{1 - \frac{c}{h^2}}$ ,  $h\sqrt{1 - \frac{c}{h^2}}$  respectively, the equation retains its original form, with  $\theta'$ ,  $h'$  in the place of  $\theta$ ,  $h$  respectively.

12. It may be remarked that when the original central force vanishes, the fixed orbit is a right line (not passing through the centre of force). It thus appears by § 9 that the curve  $u = A \cos(n\theta + B)$  may be described under the action of a force  $\propto (\text{dist.})^{-3}$ . A proposition in § 2, already referred to, shows that a logarithmic spiral may be described under the action of such a force.

13. But the case of a force  $\propto (\text{dist.})^{-3}$  was first completely discussed by Cotes in the *Harmonia Mensurarum*, pp. 31–35, 98–104, and Notes, pp. 117–173. There are in all five cases, according as the velocity of projection is

1. Less than that acquired in falling from infinity, or say equal to that acquired in falling from a point  $C$  to  $P$ , the point of projection.



<sup>1</sup>In the second case  $C$  lies on the radius vector produced beyond the centre, and the body is supposed to fall from rest at  $C$  (under the action of the central force considered as repulsive) to infinity, and then from the opposite infinity (with an initial velocity equal to the velocity so acquired) to  $P$ .

2. Equal to that acquired in falling from infinity.

3, 4, 5. Greater than that acquired in falling from infinity, or say equal to that acquired in falling from a point  $C$ , *through* infinity, to  $P$ ; viz.  $PQ$  being the direction of projection, and  $SQ$ ,  $CT$  perpendiculars thereon from  $S$  and  $C$  respectively,

3.  $SQ < TQ$ ;

4.  $SQ = TQ$ ;

5.  $SQ > TQ$ ;

the equations of the orbits being

1.  $u = Ae^{m\theta} + Be^{-m\theta}$ ,  $A$  and  $B$  same sign, so that rad. vector is never infinite.

2.  $u = Ae^{m\theta}$  or  $Be^{-m\theta}$ , logarithmic spiral.

3.  $u = Ae^{m\theta} + Be^{-m\theta}$ ,  $A$  and  $B$  opposite signs, so that rad. vector becomes infinite.

4.  $u = A\theta + B$ ,  $m = 0$ , reciprocal spiral.

5.  $u = A \cos(m\theta + B)$ ,  $m = n\sqrt{-1}$ .

14. The before-mentioned equation,

$$\frac{d^2u}{d\theta^2} + u = \frac{P}{h^2u^2},$$

is in effect given (but the equation is encumbered with a tangential force) in Clairaut's *Théorie de la Lune*, 1765. It is given in its actual form, and extensively used (in particular for obtaining the above-mentioned equations for Cotes' spirals) in Whewell's *Dynamics*, 1823. The equation appears to be but little known to continental writers, and (under the form  $u'' + u - aXu^{-3} = 0$ ) it is given as new by Schellbach as late as 1853. The formulae used in place of it are those which give  $t$  and  $\theta$  each of them in terms of  $r$ ; viz.

$$dt = \frac{r dr}{\{-h^2 + r^2(C - 2 \int P dr)\}^{\frac{1}{2}}},$$

$$d\theta = \frac{h dr}{r\{-h^2 + r^2(C - 2 \int P dr)\}^{\frac{1}{2}}},$$

which, however, assume that  $P$  is a function of  $r$  only.

15. Force  $\propto (\text{dist.})^{-2}$ . The law of motion in the conic sections is implicitly given by Newton's theorem for the equable description of the areas. For the parabola, if  $a$  denote the pericentric distance, and  $f$  the angle from pericentre or true anomaly, we have

$$t = \frac{a^{\frac{3}{2}}}{\sqrt{\mu}} \left( \tan \frac{1}{2}f + \frac{1}{3} \tan^3 \frac{1}{2}f \right).$$

For the ellipse we have an angle  $g$ , the mean anomaly varying directly as the time ( $g = nt$  if  $n = \frac{\sqrt{\mu}}{a^{\frac{3}{2}}}$ ); an auxiliary angle  $u$ , the eccentric anomaly, connected with  $g$  by the equation

$$g = u - e \sin u;$$

and then the radius vector  $r$  and the true anomaly  $f$  are given in terms of  $u$  by the equations  $r = a(1 - e \cos u)$ , and

$$\cos f = \frac{\cos u - e}{1 - e \cos u}, \quad \sin f = \frac{\sqrt{1 - e^2} \sin u}{1 - e \cos u}, \quad \text{and therefore} \quad \tan \frac{1}{2}f = \sqrt{\frac{1 + e}{1 - e}} \tan \frac{1}{2}u.$$

16. It is very convenient to have a notation for  $\frac{r}{a}$  and  $f$  considered as functions of  $e$ ,  $g$ , and I have elsewhere proposed to write

$$r = a \text{elqr}(e, g), \quad f = \text{elta}(e, g),$$

read elqr elliptic quotient radius, and elta elliptic true anomaly.

17. The formulae for the hyperbola correspond to those for the ellipse, but they contain exponential in the place of circular functions (see post, Elliptic Motion).

18. Euler, in the memoir “*Determinatio Orbitae Cometæ Anni 1742*,” (1743), p. 16 *et seq.*, obtained an expression for the time of describing a parabolic arc in terms of the radius vectors and the chord; viz. these being  $f$ ,  $g$ , and  $k$ , the expression is

$$t\sqrt{\mu} = \frac{1}{6}\sqrt{2}\{(f + g + k)^{\frac{3}{2}} - (f + g - k)^{\frac{3}{2}}\},$$

which, however, as remarked by Lagrange, *Méc. Anal.*, t. ii. (3rd edit. p. 28), is deducible from Lemma X. of the third book of the *Principia*. But the theorem in its actual form is due to Euler.

19. Lambert, in the *Proprietates Insigniores* &c. (1761), Theorem VII. Cor. 2, obtained the same theorem, and in section 4 he obtained the corresponding theorem for elliptic motion; viz. the expression for the time is

$$t = \frac{a^{\frac{3}{2}}}{\sqrt{\mu}}\{\phi - \phi' - (\sin \phi - \sin \phi')\},$$

if

$$\sin \frac{1}{2}\phi = \frac{1}{2}\sqrt{\frac{f + g + k}{a}}, \quad \sin \frac{1}{2}\phi' = \frac{1}{2}\sqrt{\frac{f + g - k}{a}}.$$

The form of the formula is, it will be observed, similar to that for motion in a straight line (*ante*, No. 4), and in fact the motion in the ellipse is, by an ingenious geometrical transformation, made to depend upon that in the straight line. The geometrical theorems upon which the transformation depends are stated, Cayley “On Lambert’s Theorem &c.” (1861).

20. The theorem was also obtained by Lagrange in the memoir “Recherches &c.” (1767) as a corollary to his solution of the problem of two centres; viz. upon making the attractive force of one of the centres equal to zero, and assuming that such centre is situate on the curve, the expression for the time presents itself in the form given by Lambert’s theorem.

21. Two other demonstrations of the theorem are given by Lagrange in the memoir “Sur une manière particulière d’exprimer le temps &c.” (1778), reproduced in Note V. of the second volume of the last edition (Bertrand’s) of the *Mécanique Analytique*. As M. Bertrand remarks, these demonstrations are very complete, very elegant, and very natural, assuming that the theorem is known beforehand.

Demonstrations were also given by Gauss, “Theoria Motus” (1809), p. 119 *et seq.*; Pagani, “Démonstration d’un théorème &c.” (1834); and (in connexion with Hamilton’s Principal Function) by Sir W. R. Hamilton, “On a General Method &c.” (1834), p. 282; Jacobi, “Zur Theorie &c.” (1837), p. 122; Cayley, “Note on the Theory of Elliptic Motion” (1856).

22. Connected with the problem of central forces, we have Sir W. R. Hamilton’s “Hodograph,” which in the paper (*Proc. R. Irish Acad.* 1847) is defined, and the fundamental properties thereof are stated; viz. if in an orbit round a centre of force there be taken on the perpendicular from the centre on the tangent at each point, a length equal to the velocity at that point of the orbit, the extremities of these lengths will trace out a curve which is the hodograph. As the product of the velocity into the perpendicular on the tangent is equal to twice the area swept out in a unit of time ( $vp = h$ ), the hodograph is the reciprocal polar of the orbit with respect to a circle described about the centre of force, radius  $= \sqrt{h}$ . Whence also the tangent at any point of the hodograph is perpendicular to the radius vector through the corresponding point of the orbit, and the product of the perpendicular on the tangent into the corresponding radius vector is  $= h$ .

If force  $\propto (\text{dist.})^{-2}$ , the hodograph, *quâ* reciprocal polar of a conic section with respect to a circle described about the focus, is a circle.

23. The following theorem is also given without demonstration; viz. if two circular hodographs, which have a common chord passing or tending through a common centre of force, be both cut at right angles by a third circle, the times of hodographically describing the intercepted arcs (that is, the times of describing the corresponding elliptic arcs) will be equal.

24. Droop, “On the Isochronism &c.” (1856), shows geometrically that the last-mentioned property is equivalent to Lambert’s theorem; and an analytical demonstration is also given, Cayley, “A demonstration of Sir W. R. Hamilton’s Theorem &c.” (1857). See also Sir W. R. Hamilton’s *Lectures on Quaternions* (1853), p. 614.

25. The laws of central force which have been thus far referred to are force  $\propto r$ ,  $\propto \frac{1}{r^2}$ ,  $\propto \frac{1}{r^3}$ ; and it has been seen that the case of a force  $P + \frac{C}{r^3}$  depends upon that

of a force  $P$ , so that the motions for the forces  $Ar + \frac{C}{r^3}$  and  $\frac{B}{r^2} + \frac{C}{r^3}$  are deducible from those for the forces  $Ar$  and  $\frac{B}{r^2}$  respectively. Some other laws of force, e.g.  $\frac{A}{r^2} \pm Br$ ,  $\frac{A}{r^2} + \frac{B}{r^3} + \frac{C}{r^4} + \frac{D}{r^5}$ , are considered by Legendre, “Théorie des Fonctions Elliptiques” (1825), being such as lead to results expressible by elliptic integrals, and also the law  $\frac{M}{r^2}$ , for which the result involves a peculiar logarithmic integral. But the most elaborate examination of the different cases in which the solution can be worked out by elliptic integrals or otherwise is given in Stader’s memoir “De Orbitis &c.” (1852), where the investigation is conducted by means of the formulae which give  $t$  and  $\theta$  in terms of  $r$  (*ante*, No. 14).

26. In speaking of a central force, it is for the most part implied that the force is a function of the distance: for some problems in which this is not the case, see post, Miscellaneous Problems, Nos. 86 and 87.

It is to be noticed that, although the problem of central forces may be (as it has so far been) considered as a problem *in plano* (viz. the plane of the motion has been made the plane of reference), yet that it is also interesting to consider it as a problem in space; in fact, in this case the integrals, though of course involved in those which belong to the plane problem, present themselves under very distinct forms, and afford interesting applications of the theory of canonical integrals, of the derivation of the successive integrals by Poisson’s method, and of other general dynamical theories. Moreover, in the lunar and planetary theories, the problem must of necessity be so treated. Without going into any details on this point, I will refer to Bertrand’s memoir, “Sur les équations différentielles de la Mécanique” (1852), Donkin’s memoir “On a Class of Differential Equations &c.” (1855), and Jacobi’s posthumous memoir, “Nova Methodus &c.” (1862).

*Elliptic Motion, Article Nos. 27—40.*

27. The question of the development of the true anomaly in terms of the mean anomaly (Kepler’s problem), and of the other developments which present themselves in the theory of elliptic motion, is one that has very much occupied the attention of geometers. The formulae on which it depends are mentioned *ante*, No. 15; they involve as an auxiliary quantity the eccentric anomaly  $u$ .

28. Consider first the equation

$$g = u - e \sin u,$$

which connects the mean anomaly  $g$  with the eccentric anomaly  $u$ .

Any function of  $u$ , and in particular  $u$  itself, and the functions  $\frac{\cos}{\sin} nu$  may be expanded in terms of  $g$  by means of Lagrange’s theorem (Lagrange, *Mém. de Berlin*, 1768–1769, “Théorie des Fonctions,” chap. 16, and “Traité de la Résolution des équations Numériques,” Note 11).

29. Considering next the equation

$$\tan \frac{1}{2}f = \sqrt{\frac{1+e}{1-e}} \tan \frac{1}{2}u,$$

which gives the true anomaly in terms of the eccentric anomaly, then, by replacing the circular functions by their exponential values (a process employed by Lagrange, *Mém. de Berlin*, 1776),  $f$  can be expressed in terms of  $u$ ; viz. the result is

$$f = u + 2\lambda \sin u + 2\lambda^2 \cdot \frac{1}{2} \sin 2u + 2\lambda^3 \cdot \frac{1}{3} \sin 3u + \&c.,$$

where  $\lambda = \frac{1-\sqrt{1-e^2}}{e} \left( = \frac{e}{1+\sqrt{1-e^2}} \right)$ . Hence if  $u, \sin u, \sin 2u, \&c.$  are expressed in terms of the mean anomaly,  $f$  will be obtained in the form  $f = g +$  a series of multiple sines of  $g$ , the coefficients of the different terms being given in the first instance as functions of  $e$  and  $\lambda$ ; and to complete the development  $\lambda$  and its powers have to be developed in powers of  $e$ . The solution is carried thus far in the *Mécanique Analytique* (1788), and in the *Mécanique Céleste* (1799).

30. We have next Bessel's investigations in the Berlin Memoirs for 1816, 1818, and 1824, and which are carried on mainly by means of the integral

$$2\pi I_h^k = \int_0^{2\pi} \cos(hu - k \sin u) du,$$

and various properties are there obtained and applications made of this important transcendant.

31. Relating to this integral we have Jacobi's memoir, "Formulae transformationis &c." (1836), Liouville, "Sur l'intégrale  $\int_0^{2\pi} \cos i(u - x \sin u) du$ " (1841), and Hansen's "Ermittelung der absoluten Störungen" (1843); the researches of Poisson in the *Connaissance des Temps* for 1825 and 1836 are closely connected with those of Bessel.

32. A very elegant formula, giving the actual expression of the coefficients considered as functions of  $e$  and  $\lambda$ , is given by Greatheed in the paper "Investigation of the General Term &c." (1838); viz. this is

$$f = g + 2\Sigma \lambda^r \frac{e^{\frac{1}{2}r(\lambda+\lambda^{-1})} + \lambda^{-r} e^{-\frac{1}{2}r(\lambda+\lambda^{-1})}}{r} \sin rg,$$

where, after developing in powers of  $\lambda$ , the negative powers of  $\lambda$  must be rejected, and the term independent of  $\lambda$  divided by 2. This result is extended to other functions of  $f$ , Cayley "On certain Expansions &c." (1842).

33. An expression for the coefficient of the general term as a function of  $e$  only is obtained, Lefort, "Expression Numérique &c." (1846). The expression, which, from the nature of the case, is a very complicated one, is obtained by means of Bessel's integral. This is an indirect process which really comes to the combination of the developments of  $f$  in terms of  $u$ , and  $u$  in terms of  $g$ ; and an equivalent result is obtained directly in this manner, Greedy, "General and Practical Solution &c." (1855).

34. We have also on the subject of these developments the very valuable and interesting researches of Hansen, contained in his *Fundamenta Nova* &c. (1838), in the memoir “Ermittelung der absoluten Störungen &c.” (1845), and in particular in the memoir “Entwicklung des Products &c.” (1853).

35. But the expression for the coefficient of the general term  $\frac{\cos}{\sin} r g$  in any of these expansions is so complicated that it was desirable to have for the coefficients corresponding to the values  $r = 0, 1, 2, 3, \dots$  the finally reduced expressions in which the coefficient of each power of  $e$  is given as a numerical fraction. Such formulae for  $\left(\frac{r}{a}\right)^n \frac{\cos}{\sin} j f$ , where  $j$  is a general symbol, the expansion being carried as far as  $e^6$ , were given, Leverrier, *Annales de l'Observatoire de Paris*, t. i. (1855).

36. And starting from these I deduced the results given in my “Tables of the Developments &c.” (1861); viz. these tables give  $(x = \frac{r}{a})$ ,

$$\left(\frac{r}{a}\right)^n \frac{\cos}{\sin} j f, \quad (n = -5, \dots, 5; j = 1, \dots, 7),$$

all carried to  $e^7$ .

37. The true anomaly  $f$  has been repeatedly calculated to a much greater extent, in particular by Schubert (*Ast. Théorique*, St Pet. 1822), as far as  $e^{20}$ . The expression for  $\frac{r}{a}$  as far as  $e^{20}$  is given in the same work, and that for  $\log \frac{r}{a}$  as far as  $e^{20}$  was calculated by Oriani, see Introd. to Delambre's *Tables du Soleil*, Paris (1806).

38. It may be remarked that when the motion of a body is referred to a plane which is not the plane of the elliptic orbit, then we have questions of development similar in some measure to those which regard the motion in the orbit; if, for instance,  $z$  be the distance from node,  $\phi$  the inclination, and  $x$  the reduced distance from node, then  $\cos \phi = \cos z \cos x$ , from which we may derive  $z = x +$  series of multiple sines of  $x$ . And there are, moreover, the questions connected with the development of the reciprocal distance of two particles—say  $(a^2 + a'^2 - 2aa' \cos \theta)^{-\frac{1}{2}}$ —which present themselves in the planetary theory; but this last is a wide subject, which I do not here enter upon. I will, however, just refer to Hansen's memoir, “Ueber die Entwicklung der negativen und ungeraden Potenzen &c.” (1854).